US 20250276017A1

(19) United States
(12) Patent Application Publication
Oliveira et al.
(10) Pub. No.: US 2025/0276017 A1
(43) Pub. Date: Sep. 4, 2025

(54) COMPOSITIONS AND METHODS FOR GENOMIC EDITING

(71) Applicant: **Intellia Therapeutics, Inc.**, Cambridge, MA (US)

(72) Inventors: **Ruan Oliveira**, Boston, MA (US); **Sung Hee Choi**, Brookline, MA (US); **Biao Liu**, Cambridge, MA (US); **Radhika Barve**, Cambridge, MA (US)

(73) Assignee: **Intellia Therapeutics, Inc.**, Cambridge, MA (US)

(21) Appl. No.: **18/980,502**

(22) Filed: **Dec. 13, 2024**

Related U.S. Application Data

(63) Continuation of application No. PCT/US23/68499, filed on Jun. 15, 2023.

(60) Provisional application No. 63/352,990, filed on Jun. 16, 2022.

Publication Classification

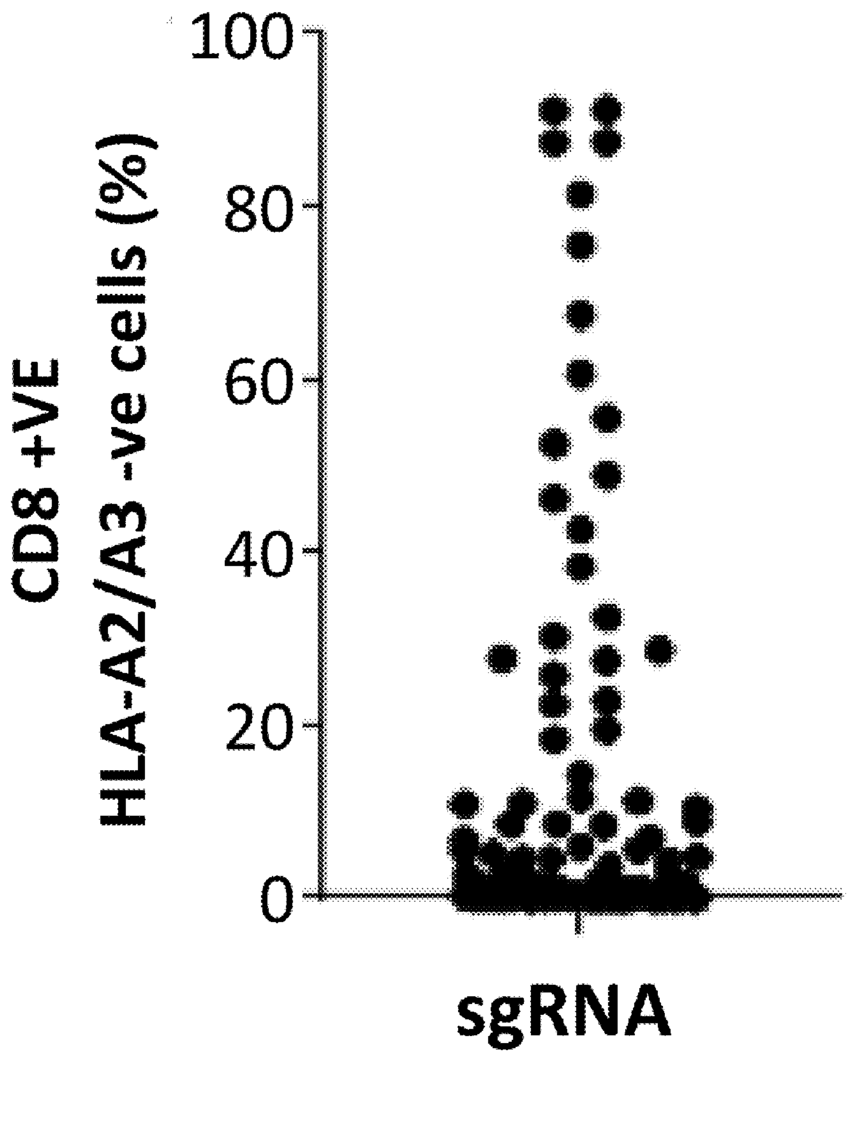
(51) **Int. Cl.**
A61K 35/17 (2025.01)
C12N 5/0783 (2010.01)
C12N 9/22 (2006.01)
C12N 15/11 (2006.01)

(52) **U.S. Cl.**
CPC ***A61K 35/17*** (2013.01); ***C12N 5/0636*** (2013.01); ***C12N 9/226*** (2025.05); ***C12N 15/111*** (2013.01); *C12N 2310/20* (2017.05); *C12N 2510/00* (2013.01)

(57) **ABSTRACT**

Compositions and methods for ex vivo genomic editing using *Neisseria meningitidis* (Nme) CRISPR/Cas9 systems are disclosed. The present disclosure provides for engineered cells comprising a genetical modification for use e.g., in adoptive cell transfer therapies.

Specification includes a Sequence Listing.



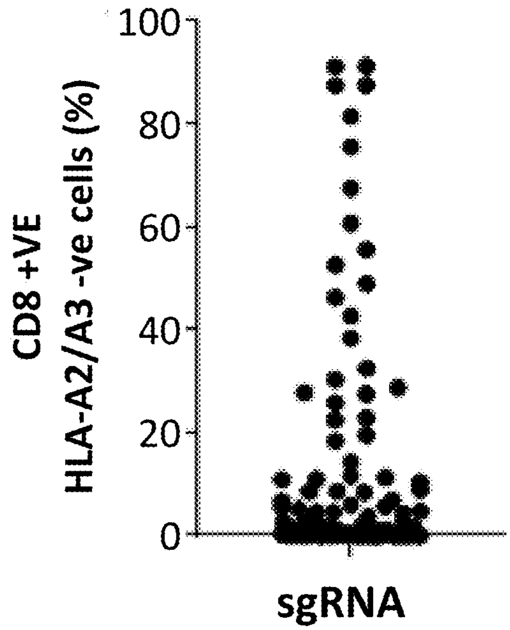

100
80
60
40
20
0
CD8 +VE
HLA-A2/A3 -ve cells (%)
sgRNA

Fig. 1A

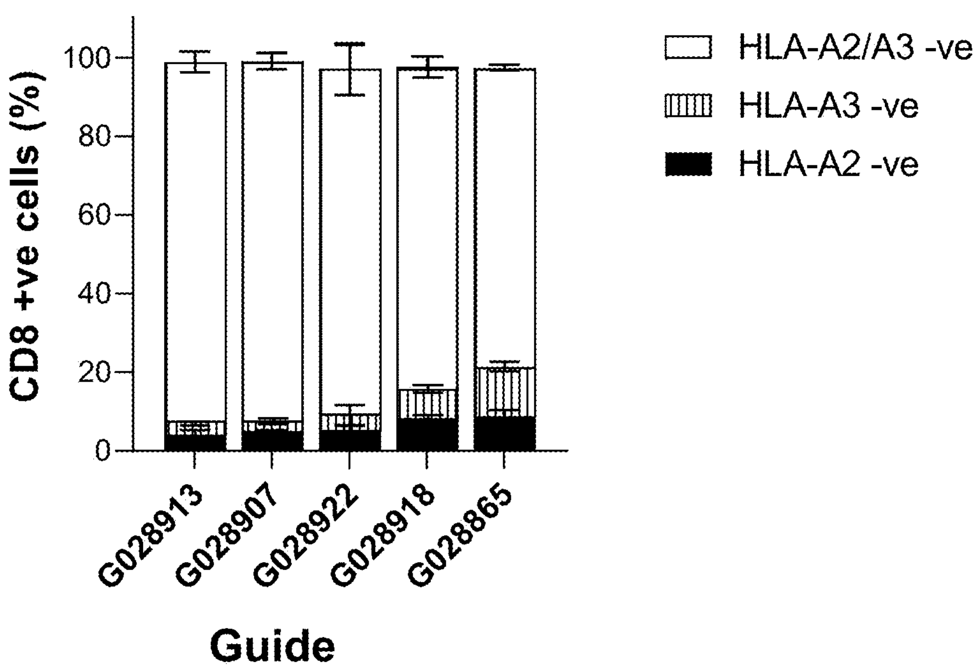

100
80
60
40
20
0
CD8 +ve cells (%)
G028913
G028907
G028922
G028918
G028865
Guide
HLA-A2/A3 -ve
HLA-A3 -ve
HLA-A2 -ve

Fig. 1B

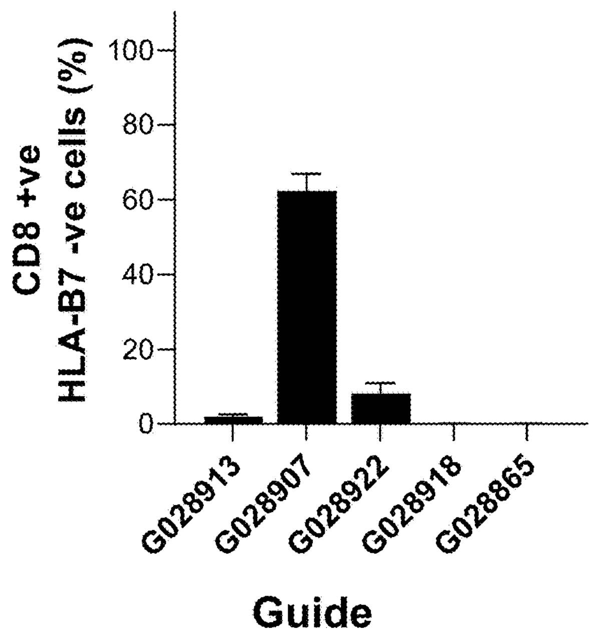

CD8 +ve
HLA-B7 -ve cells (%)
100
80
60
40
20
0
G028913
G028907
G028922
G028918
G028865
Guide

Fig. 1C

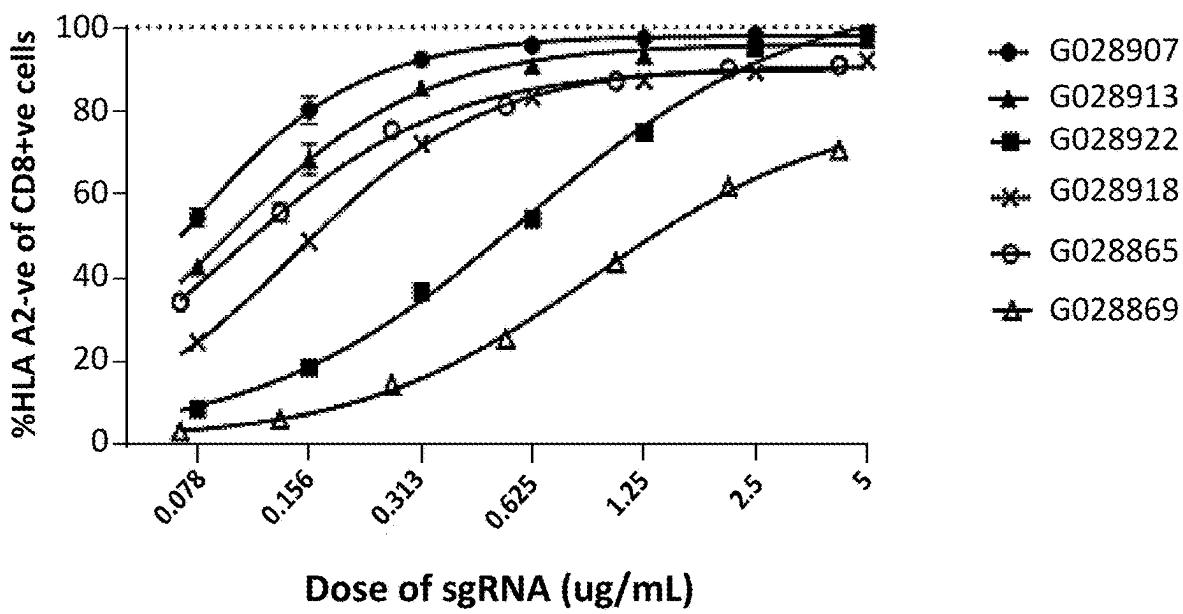

%HLA A2-ve of CD8+ve cells
100
80
60
40
20
0
0.078
0.156
0.313
0.625
1.25
2.5
5
Dose of sgRNA (ug/mL)
G028907
G028913
G028922
G028918
G028865
G028869

Fig. 2

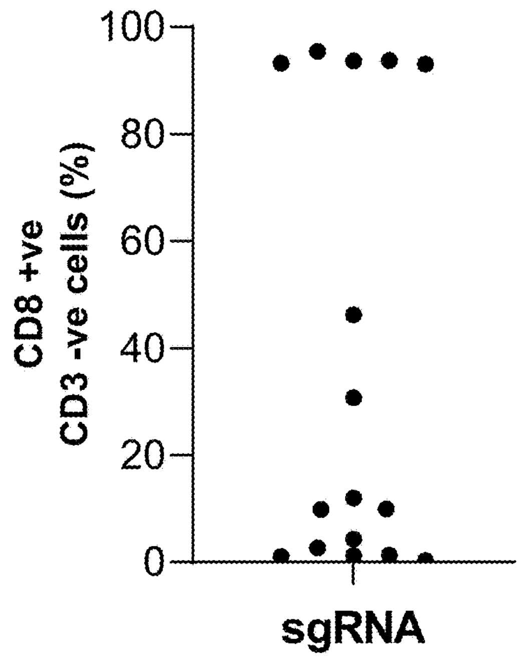

100
80
60
40
20
0
CD8 +ve
CD3 -ve cells (%)
sgRNA

Fig. 3A

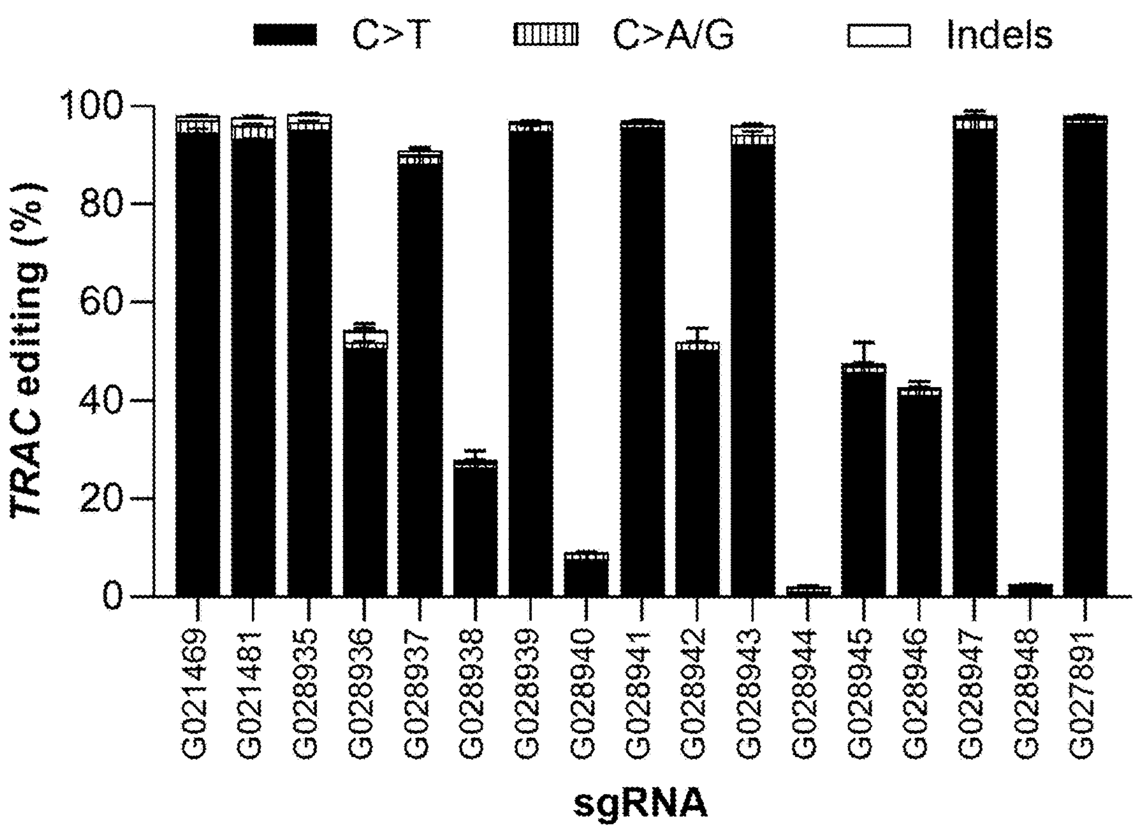
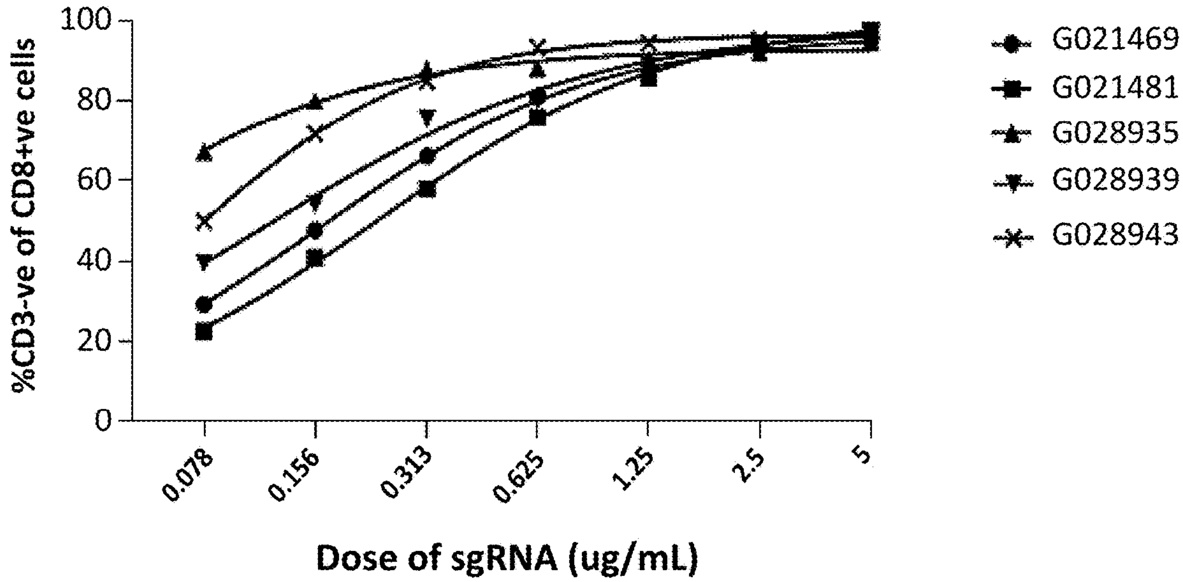
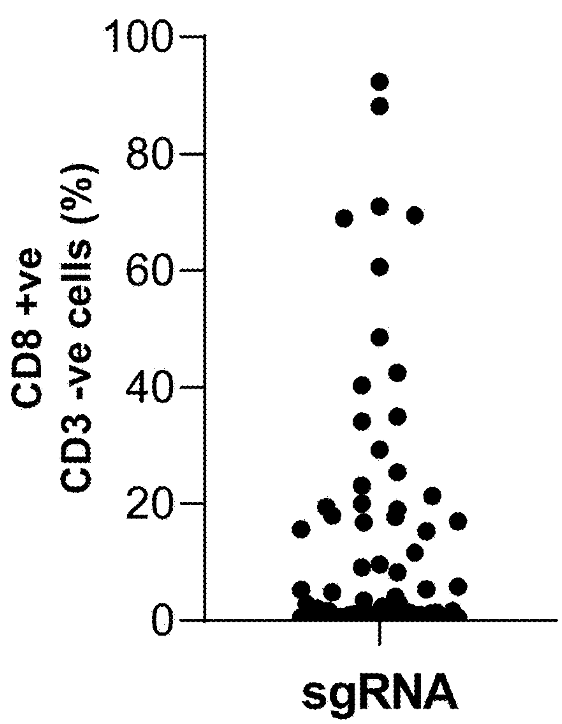

C>T
C>A/G
Indels
100
80
60
40
20
0
TRAC editing (%)
G021469
G021481
G028935
G028936
G028937
G028938
G028939
G028940
G028941
G028942
G028943
G028944
G028945
G028946
G028947
G028948
G027891
sgRNA

Fig. 3B



100
80
60
40
20
0
%CD3-ve of CD8+ve cells
0.078
0.156
0.313
0.625
1.25
2.5
5
Dose of sgRNA (ug/mL)
G021469
G021481
G028935
G028939
G028943

Fig. 4



100
80
60
40
20
0
CD8 +ve
CD3 -ve cells (%)
sgRNA

Fig. 5A

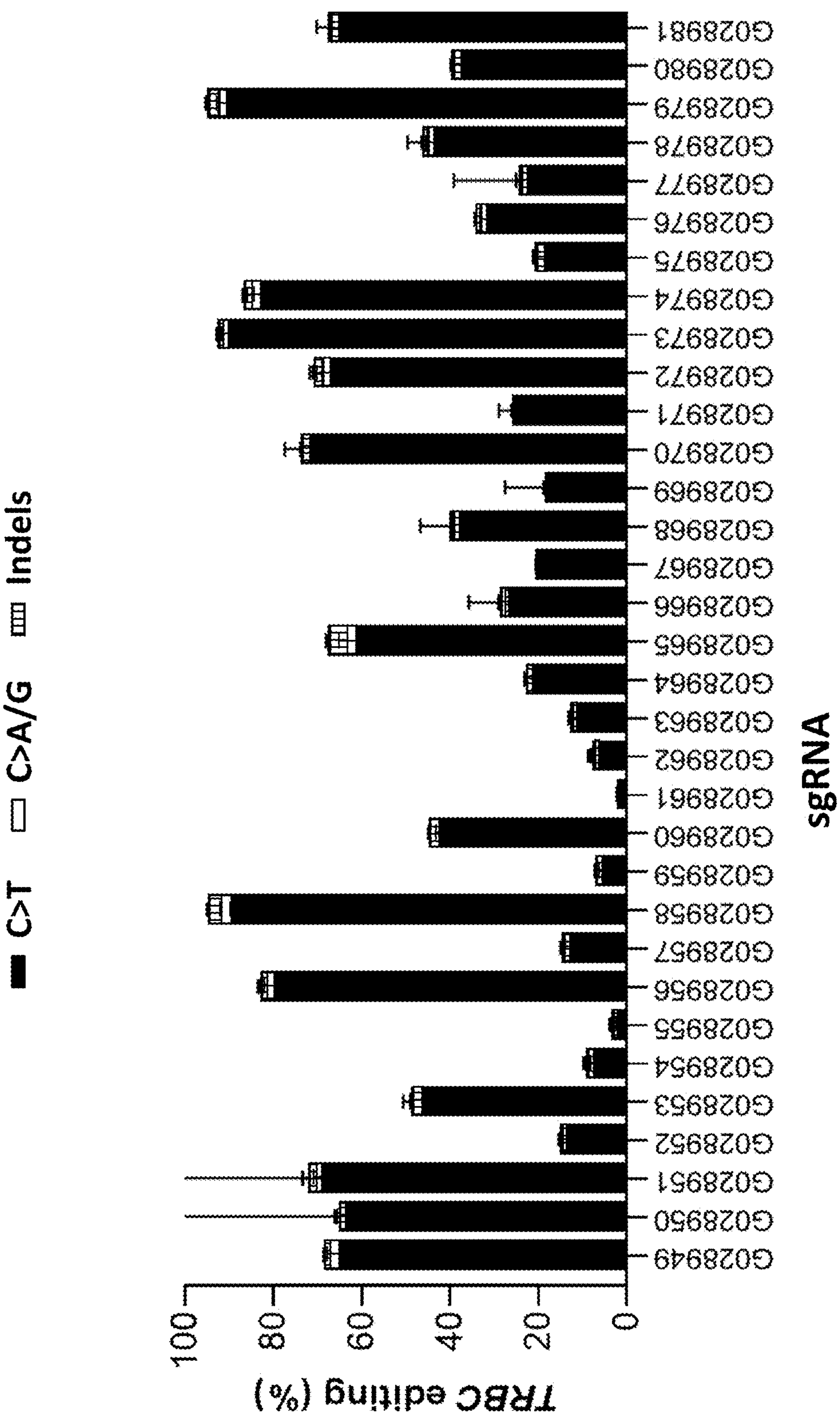

C>T
C>A/G
Indels
TRBC editing (%)
100
80
60
40
20
0
G028949
G028950
G028951
G028952
G028953
G028954
G028955
G028956
G028957
G028958
G028959
G028960
G028961
G028962
G028963
G028964
G028965
G028966
G028967
G028968
G028969
G028970
G028971
G028972
G028973
G028974
G028975
G028976
G028977
G028978
G028979
G028980
G028981
sgRNA

Fig. 5B

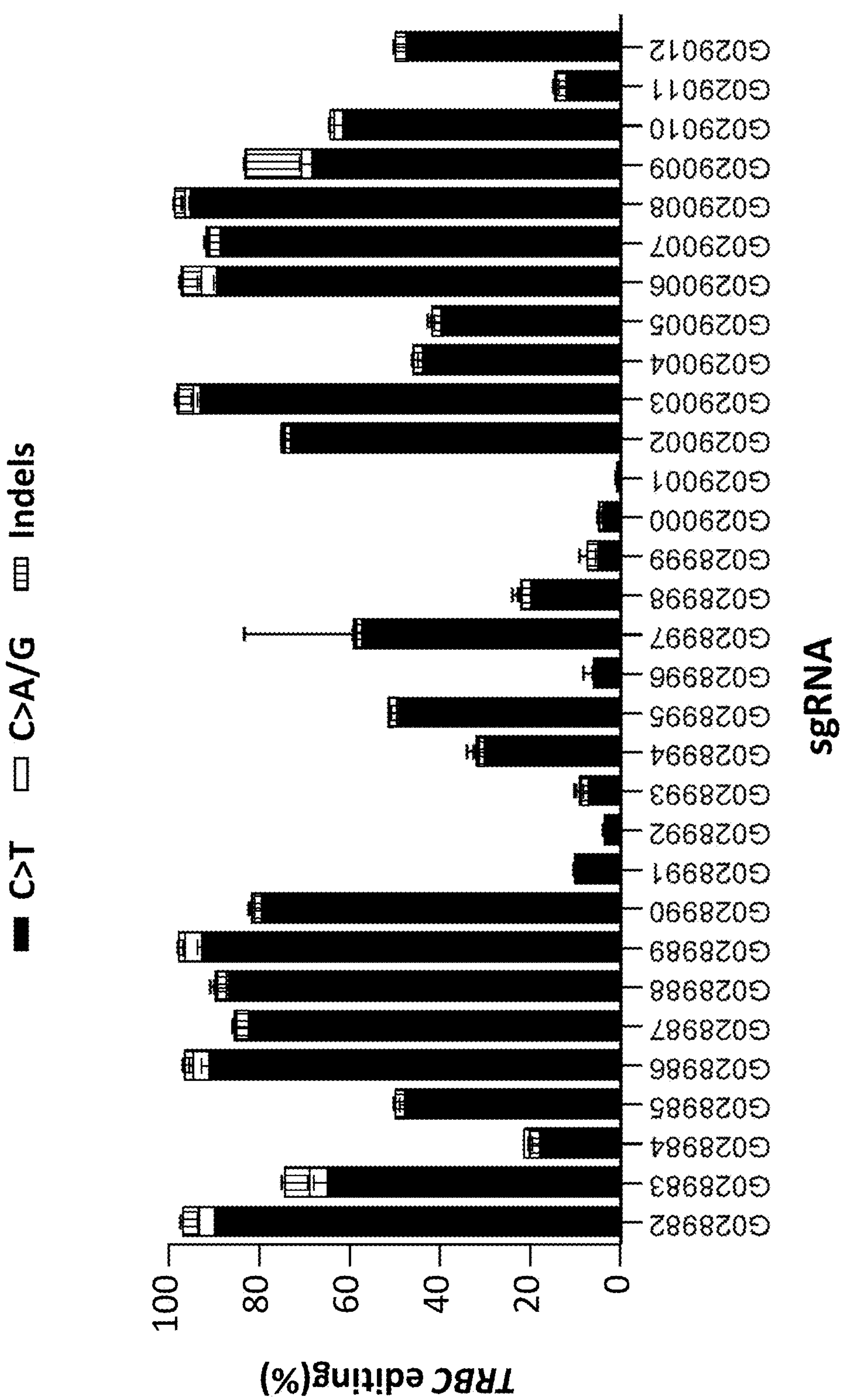

C>T
C>A/G
Indels
TRBC editing(%)
0
20
40
60
80
100
G028982
G028983
G028984
G028985
G028986
G028987
G028988
G028989
G028990
G028991
G028992
G028993
G028994
G028995
G028996
G028997
G028998
G028999
G029000
G029001
G029002
G029003
G029004
G029005
G029006
G029007
G029008
G029009
G029010
G029011
G029012
sgRNA

Fig. 5B (cont.)

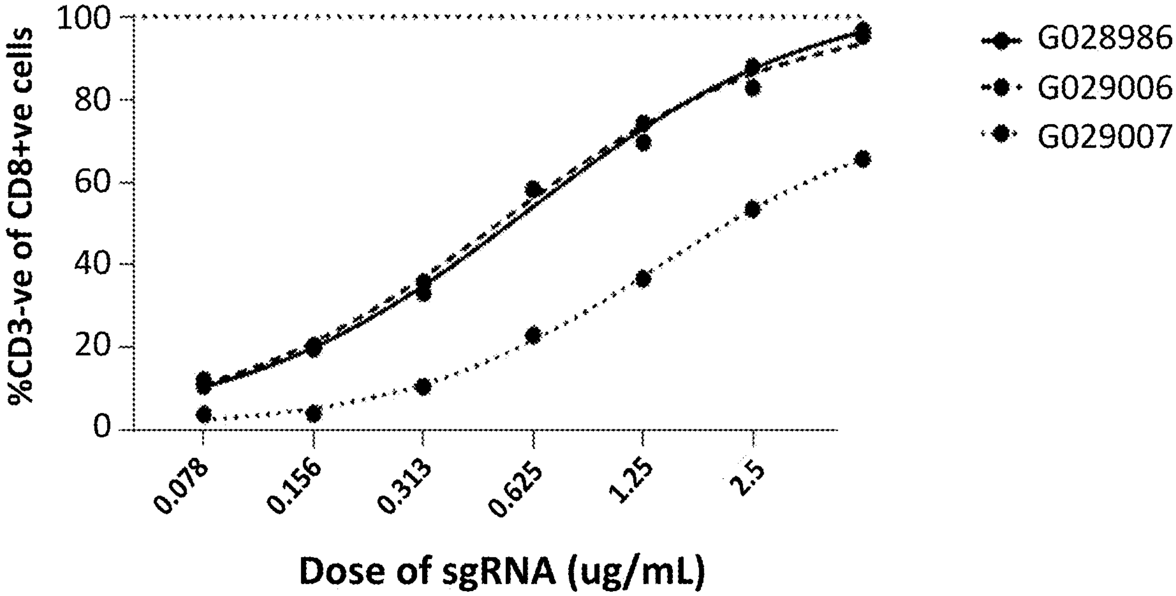

100
80
60
40
20
0
%CD3-ve of CD8+ve cells
0.078
0.156
0.313
0.625
1.25
2.5
Dose of sgRNA (ug/mL)
G028986
G029006
G029007

Fig. 6

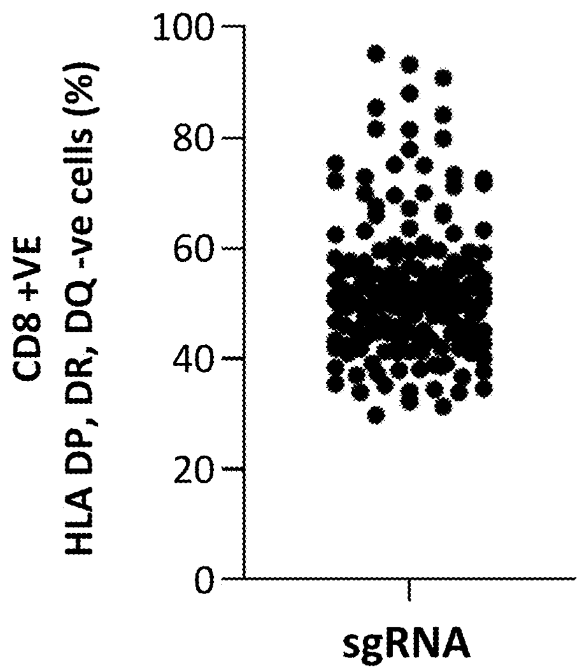

100
80
60
40
20
0
CD8 +VE
HLA DP, DR, DQ -ve cells (%)
sgRNA

Fig. 7A

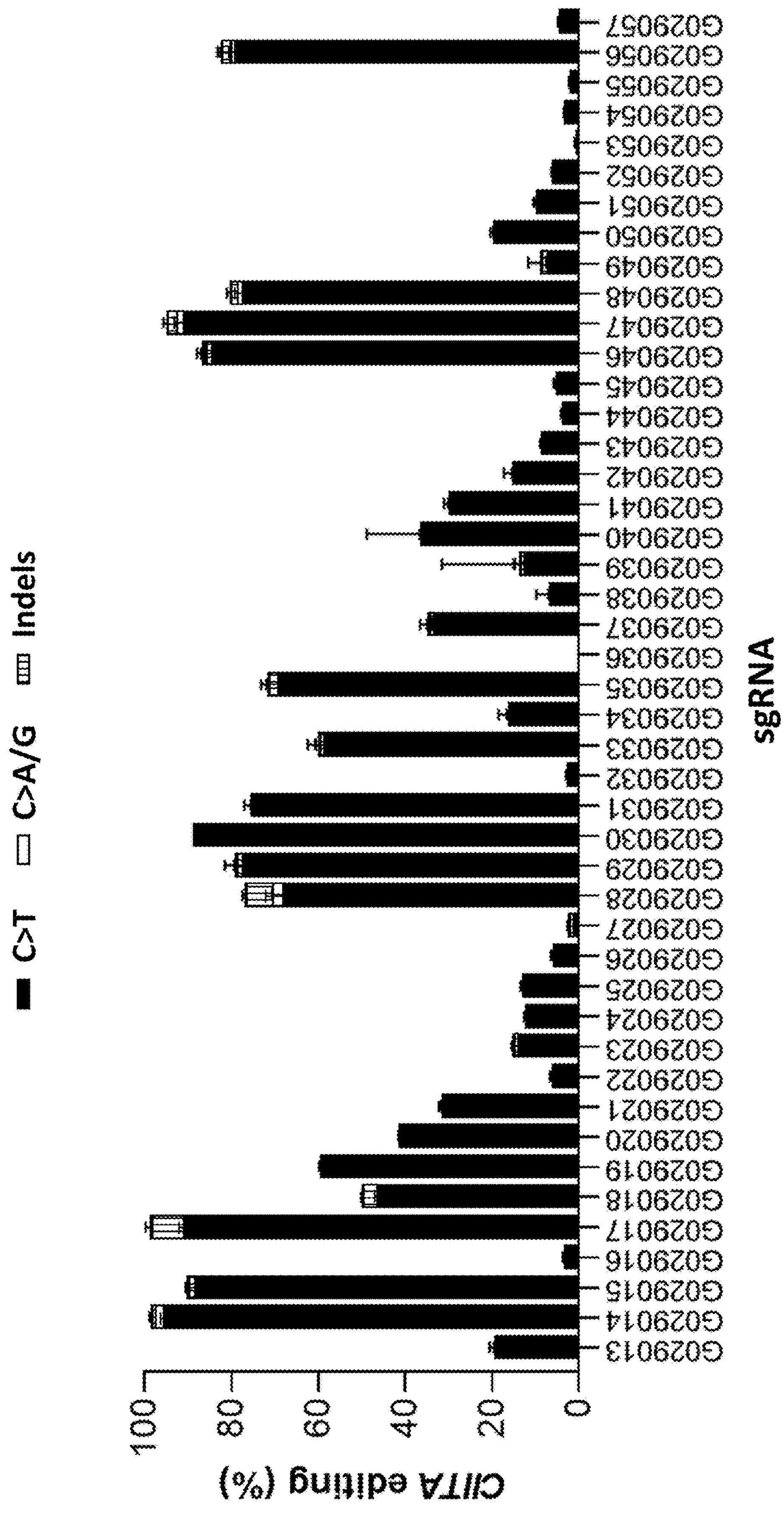

C>T
C>A/G
Indels
CIITA editing (%)
0
20
40
60
80
100
sgRNA
G029013
G029014
G029015
G029016
G029017
G029018
G029019
G029020
G029021
G029022
G029023
G029024
G029025
G029026
G029027
G029028
G029029
G029030
G029031
G029032
G029033
G029034
G029035
G029036
G029037
G029038
G029039
G029040
G029041
G029042
G029043
G029044
G029045
G029046
G029047
G029048
G029049
G029050
G029051
G029052
G029053
G029054
G029055
G029056
G029057

Fig. 7B

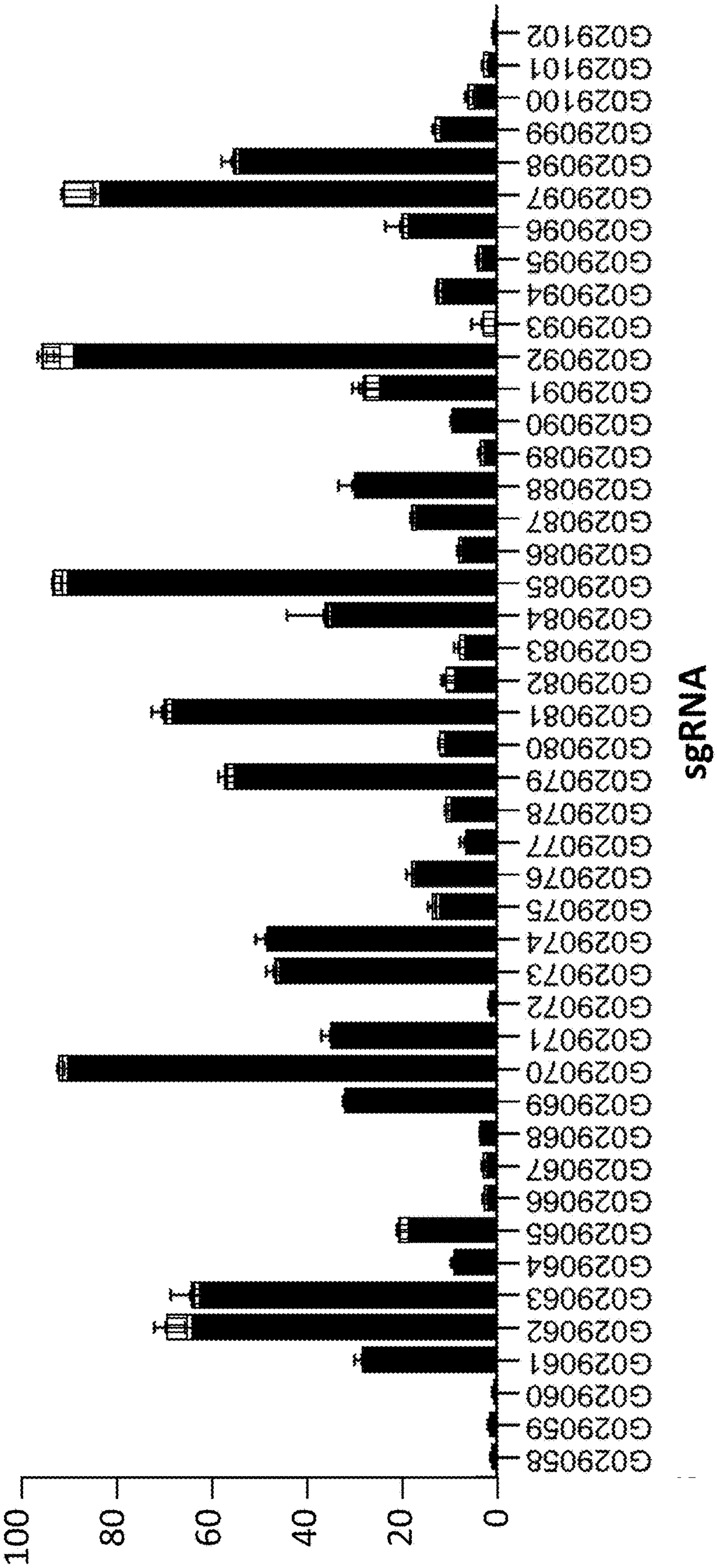

C>T
C>A/G
Indels
CIITA editing (%)
100
80
60
40
20
0
sgRNA
G029058
G029059
G029060
G029061
G029062
G029063
G029064
G029065
G029066
G029067
G029068
G029069
G029070
G029071
G029072
G029073
G029074
G029075
G029076
G029077
G029078
G029079
G029080
G029081
G029082
G029083
G029084
G029085
G029086
G029087
G029088
G029089
G029090
G029091
G029092
G029093
G029094
G029095
G029096
G029097
G029098
G029099
G029100
G029101
G029102

Fig. 7B (cont.)

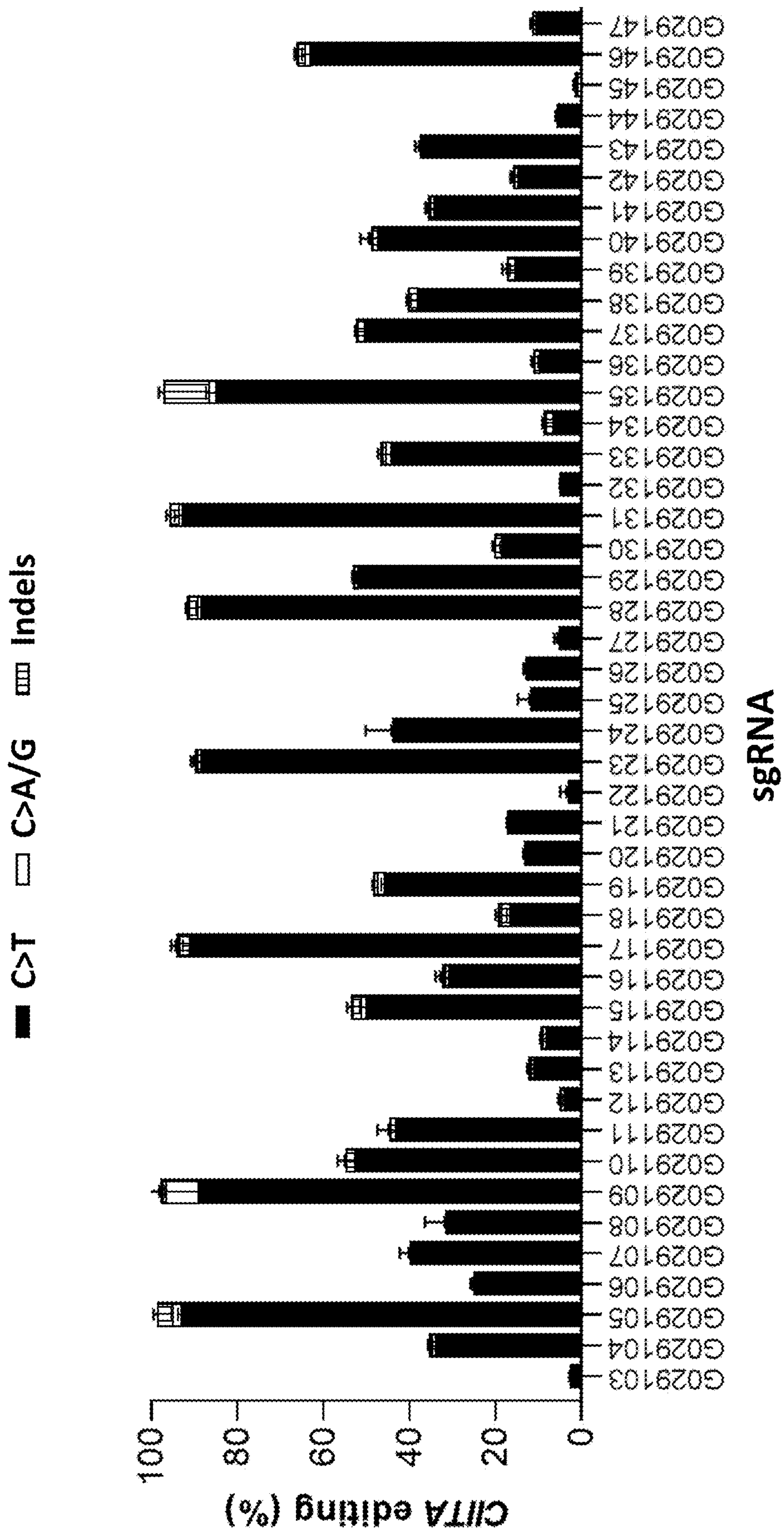

C>T
C>A/G
Indels
CIITA editing (%)
0
20
40
60
80
100
sgRNA
G029103
G029104
G029105
G029106
G029107
G029108
G029109
G029110
G029111
G029112
G029113
G029114
G029115
G029116
G029117
G029118
G029119
G029120
G029121
G029122
G029123
G029124
G029125
G029126
G029127
G029128
G029129
G029130
G029131
G029132
G029133
G029134
G029135
G029136
G029137
G029138
G029139
G029140
G029141
G029142
G029143
G029144
G029145
G029146
G029147

Fig. 7B (cont.)

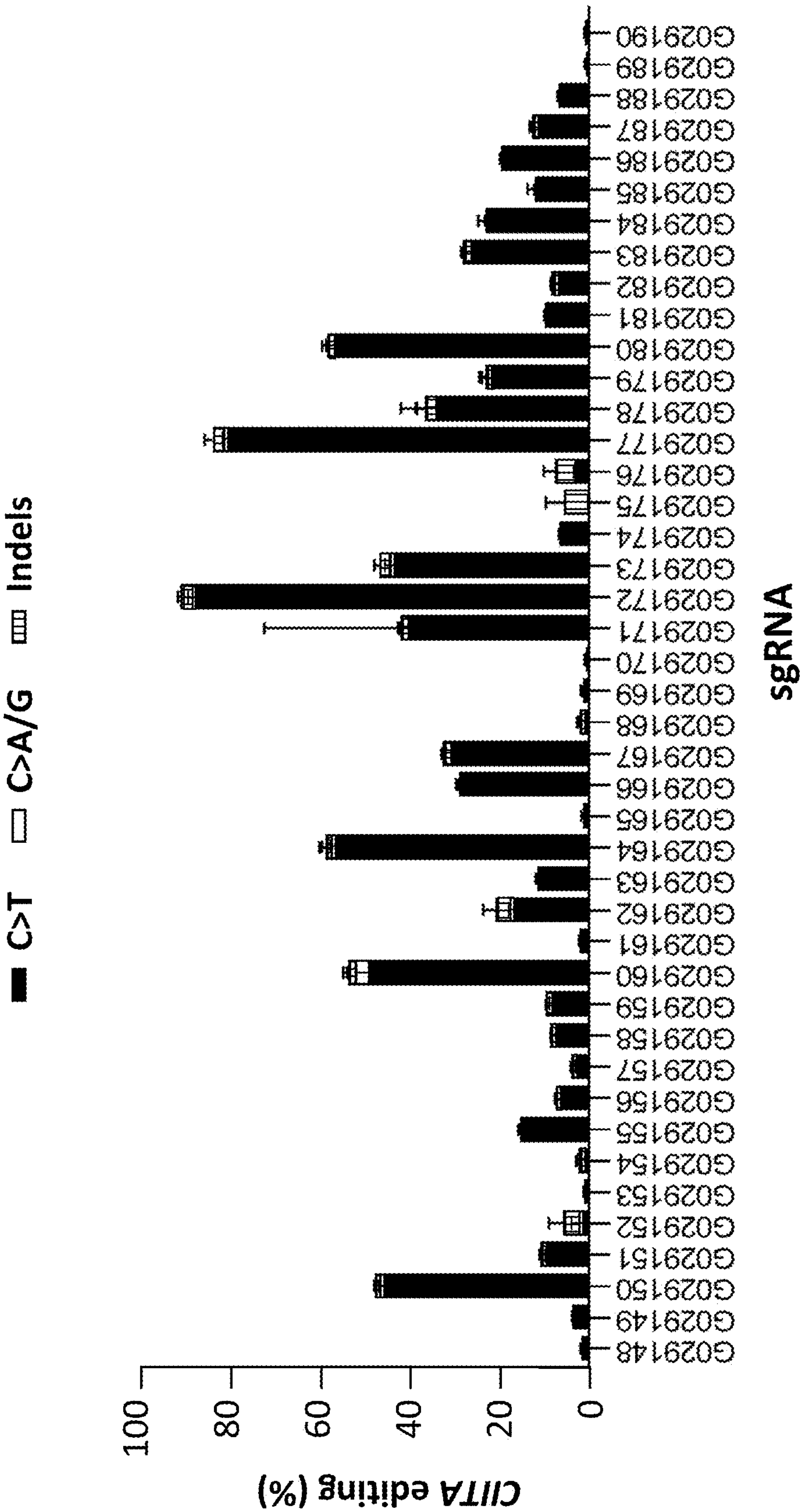
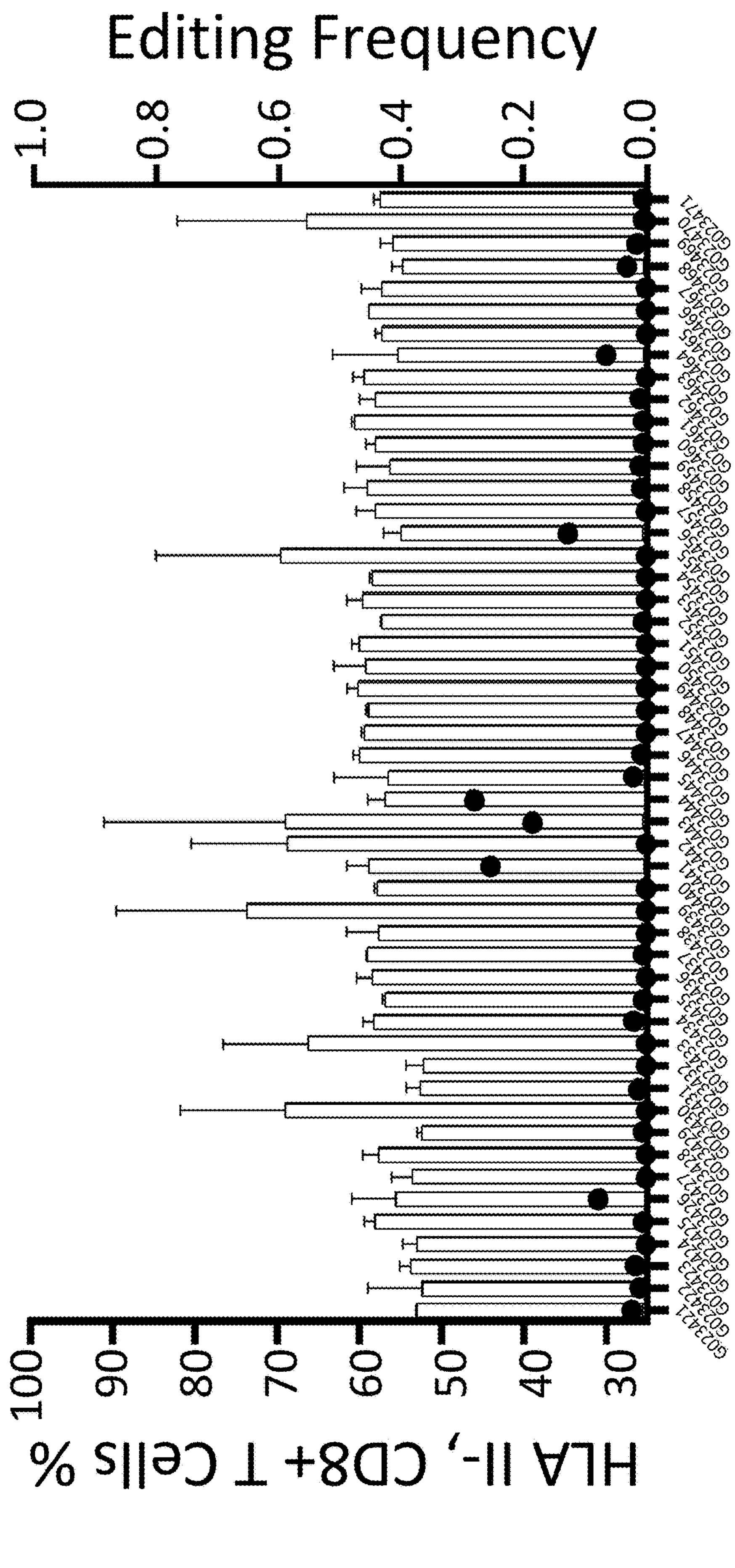

C>T
C>A/G
Indels
CIITA editing (%)
100
80
60
40
20
0
G029148
G029149
G029150
G029151
G029152
G029153
G029154
G029155
G029156
G029157
G029158
G029159
G029160
G029161
G029162
G029163
G029164
G029165
G029166
G029167
G029168
G029169
G029170
G029171
G029172
G029173
G029174
G029175
G029176
G029177
G029178
G029179
G029180
G029181
G029182
G029183
G029184
G029185
G029186
G029187
G029188
G029189
G029190
sgRNA

Fig. 7B (cont.)



Editing Frequency
1.0
0.8
0.6
0.4
0.2
0.0
HLA II-, CD8+ T Cells %
100
90
80
70
60
50
40
30

Fig. 8

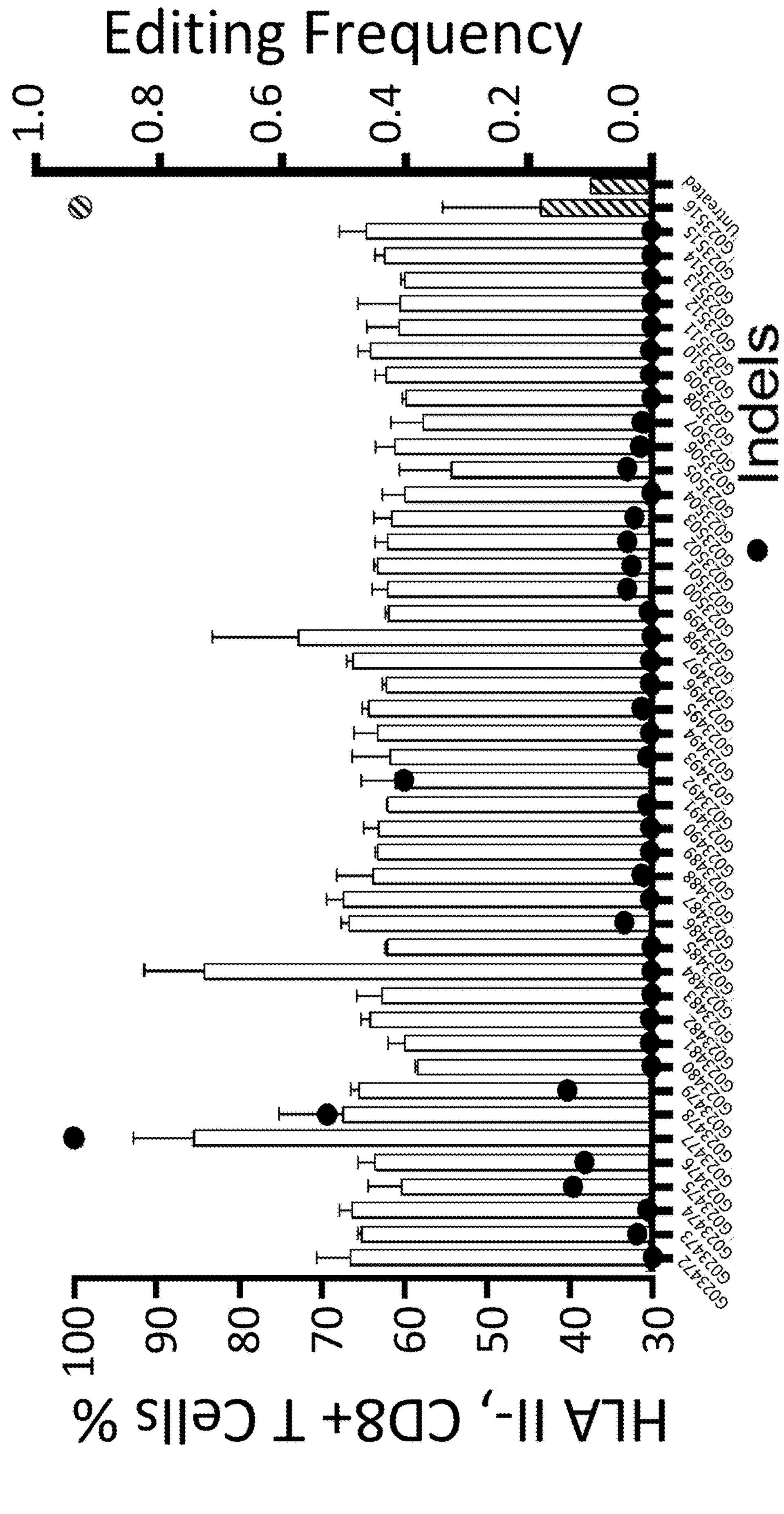

Editing Frequency
1.0
0.8
0.6
0.4
0.2
0.0
Indels
HLA II-, CD8+ T Cells %
100
90
80
70
60
50
40
30
G023472
G023473
G023474
G023475
G023476
G023477
G023478
G023479
G023480
G023481
G023482
G023483
G023484
G023485
G023486
G023487
G023488
G023489
G023490
G023491
G023492
G023493
G023494
G023495
G023496
G023497
G023498
G023499
G023500
G023501
G023502
G023503
G023504
G023505
G023506
G023507
G023508
G023509
G023510
G023511
G023512
G023513
G023514
G023515
G023516
Untreated

Fig. 8 (cont.)

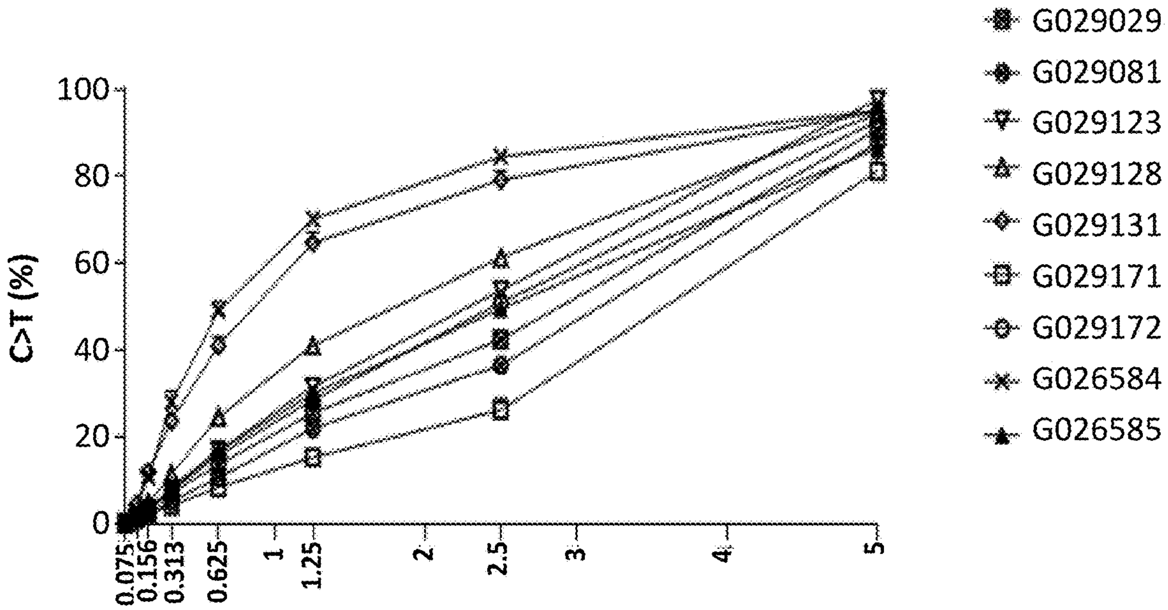

G029029
G029081
G029123
G029128
G029131
G029171
G029172
G026584
G026585
C>T (%)
100
80
60
40
20
0
0.075
0.156
0.313
0.625
1
1.25
2
2.5
3
4
5

Fig. 9A

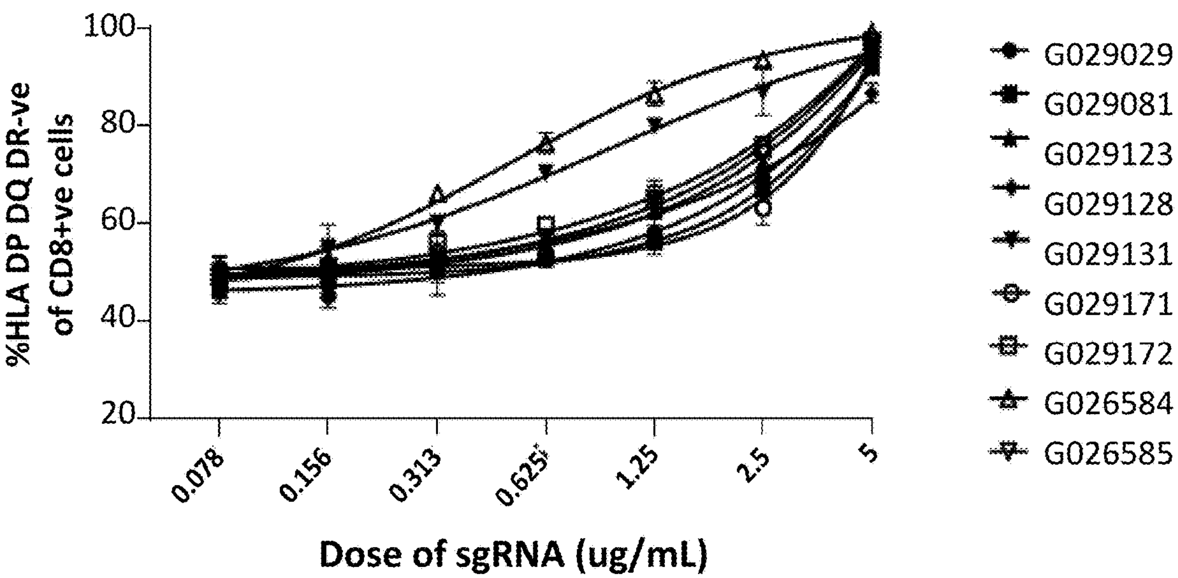

%HLA DP DQ DR-ve of CD8+ve cells
100
80
60
40
20
0.078
0.156
0.313
0.625
1.25
2.5
5
Dose of sgRNA (ug/mL)
G029029
G029081
G029123
G029128
G029131
G029171
G029172
G026584
G026585

Fig. 9B

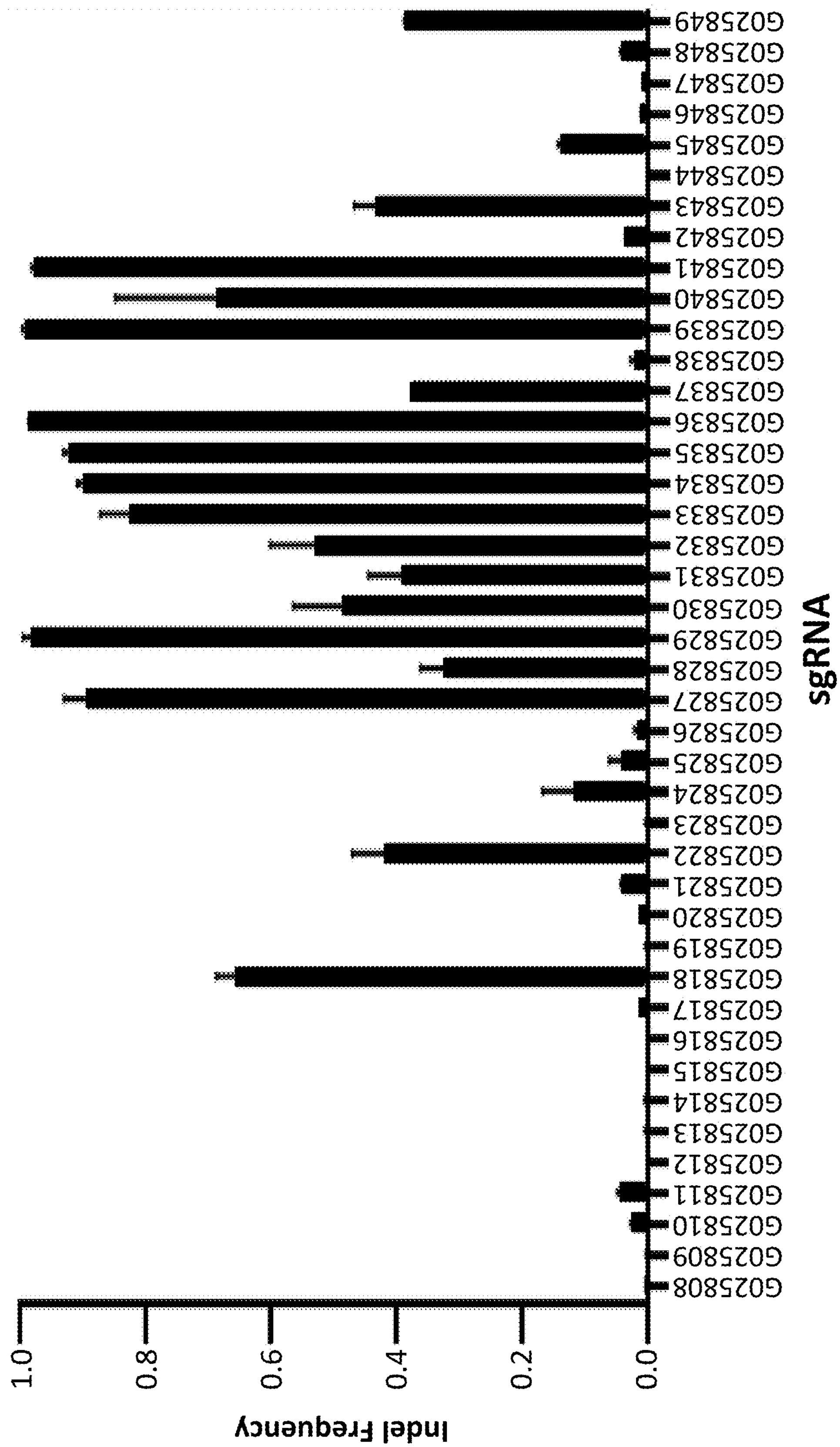

Indel Frequency
1.0
0.8
0.6
0.4
0.2
0.0
sgRNA
G025808
G025809
G025810
G025811
G025812
G025813
G025814
G025815
G025816
G025817
G025818
G025819
G025820
G025821
G025822
G025823
G025824
G025825
G025826
G025827
G025828
G025829
G025830
G025831
G025832
G025833
G025834
G025835
G025836
G025837
G025838
G025839
G025840
G025841
G025842
G025843
G025844
G025845
G025846
G025847
G025848
G025849

Fig. 10A

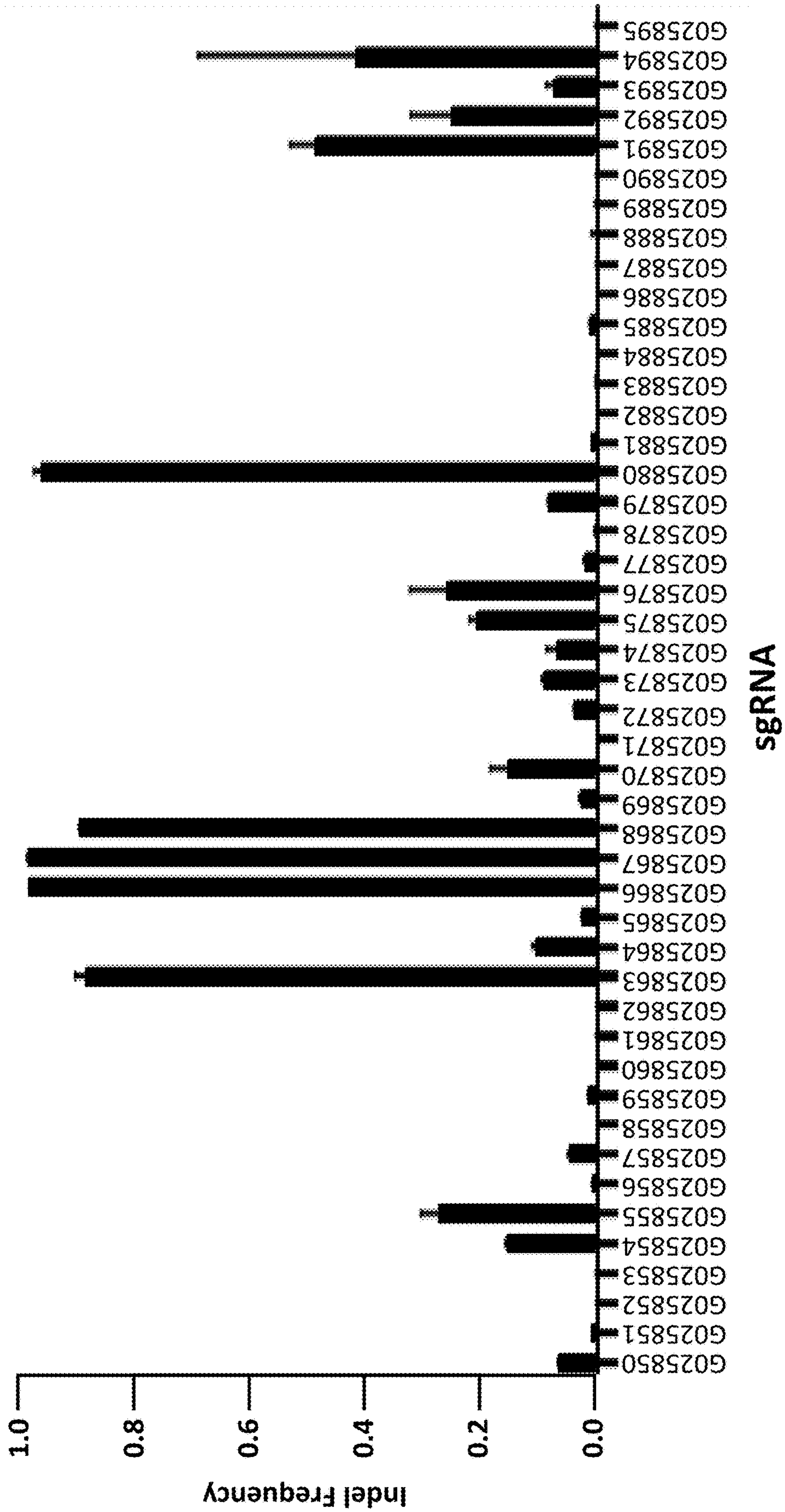

Indel Frequency
1.0
0.8
0.6
0.4
0.2
0.0
sgRNA
G025850
G025851
G025852
G025853
G025854
G025855
G025856
G025857
G025858
G025859
G025860
G025861
G025862
G025863
G025864
G025865
G025866
G025867
G025868
G025869
G025870
G025871
G025872
G025873
G025874
G025875
G025876
G025877
G025878
G025879
G025880
G025881
G025882
G025883
G025884
G025885
G025886
G025887
G025888
G025889
G025890
G025891
G025892
G025893
G025894
G025895

Fig. 10A (cont.)

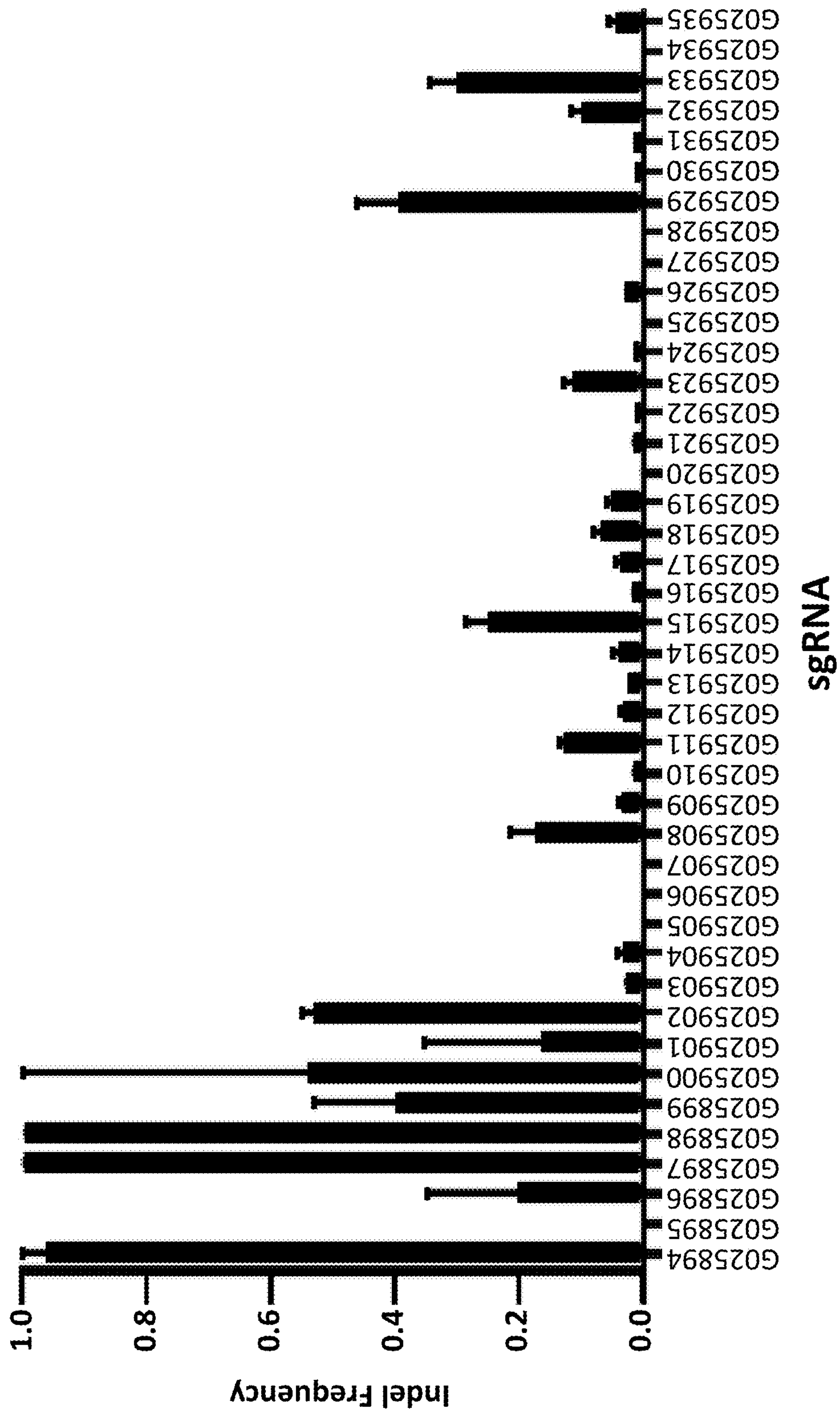
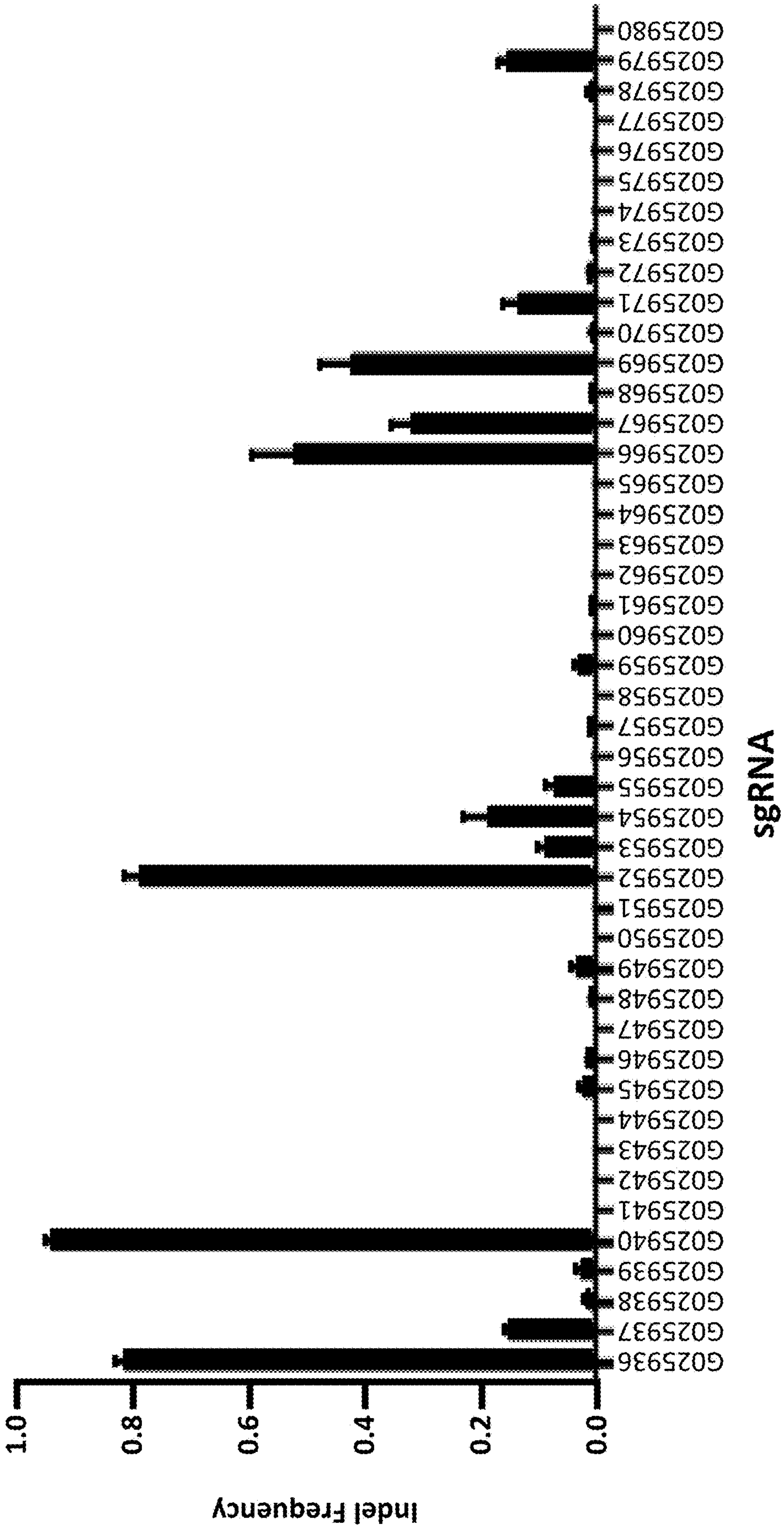

Indel Frequency
1.0
0.8
0.6
0.4
0.2
0.0
sgRNA
G025894
G025895
G025896
G025897
G025898
G025899
G025900
G025901
G025902
G025903
G025904
G025905
G025906
G025907
G025908
G025909
G025910
G025911
G025912
G025913
G025914
G025915
G025916
G025917
G025918
G025919
G025920
G025921
G025922
G025923
G025924
G025925
G025926
G025927
G025928
G025929
G025930
G025931
G025932
G025933
G025934
G025935

Fig. 10B



Indel Frequency
1.0
0.8
0.6
0.4
0.2
0.0
sgRNA
G025936
G025937
G025938
G025939
G025940
G025941
G025942
G025943
G025944
G025945
G025946
G025947
G025948
G025949
G025950
G025951
G025952
G025953
G025954
G025955
G025956
G025957
G025958
G025959
G025960
G025961
G025962
G025963
G025964
G025965
G025966
G025967
G025968
G025969
G025970
G025971
G025972
G025973
G025974
G025975
G025976
G025977
G025978
G025979
G025980

Fig. 10B (cont.)

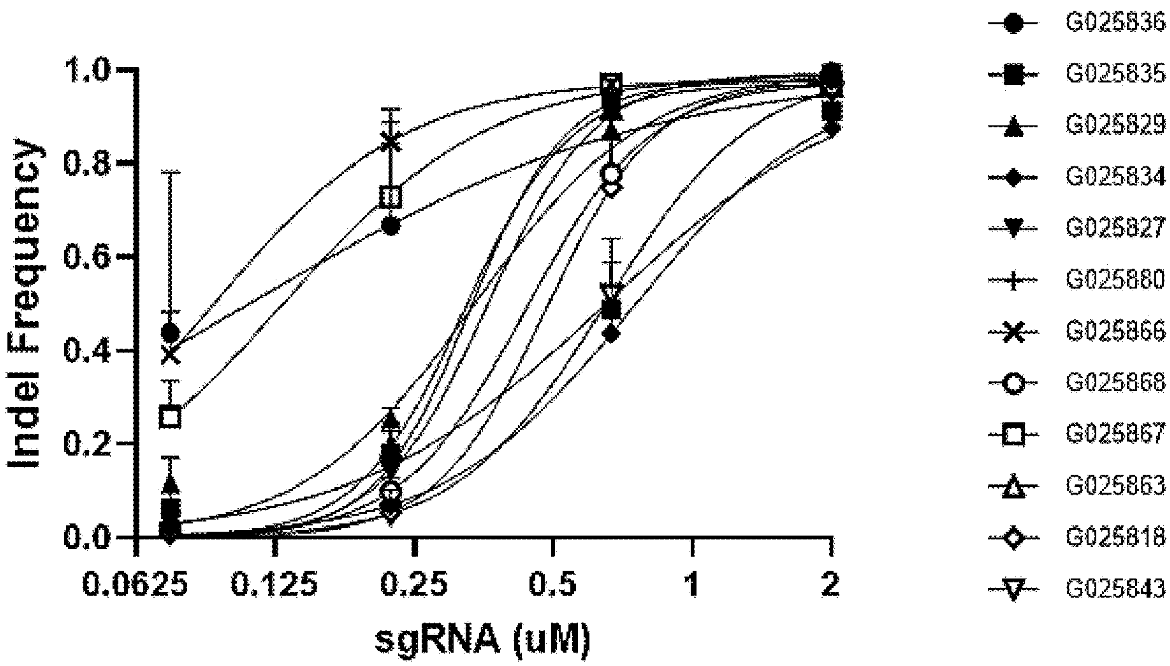


Fig. 11A

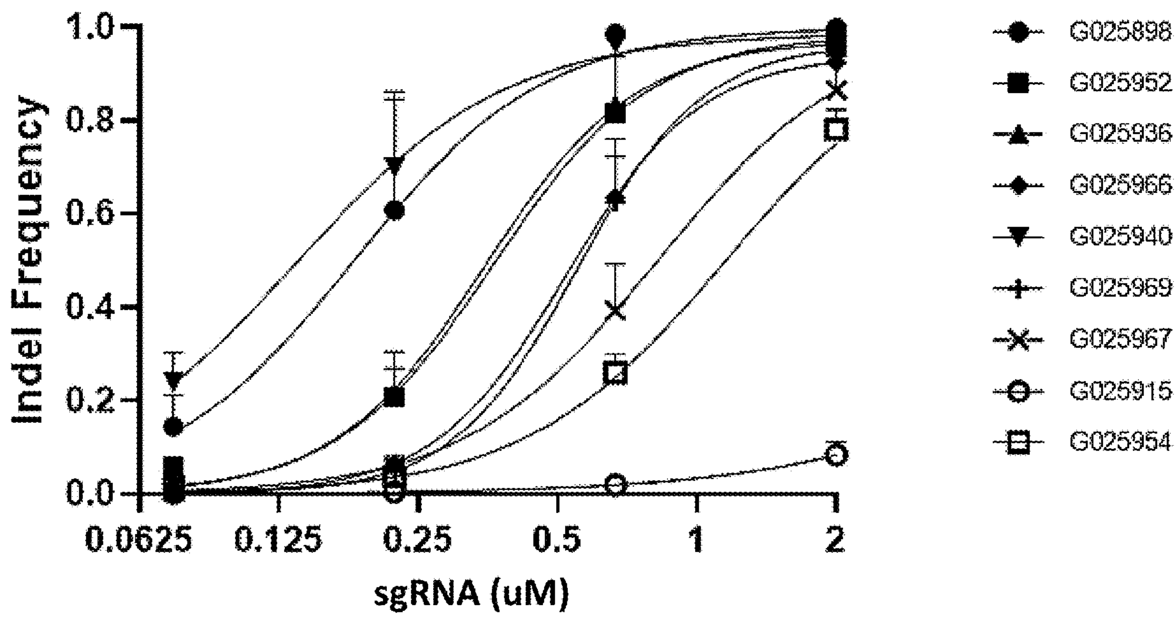
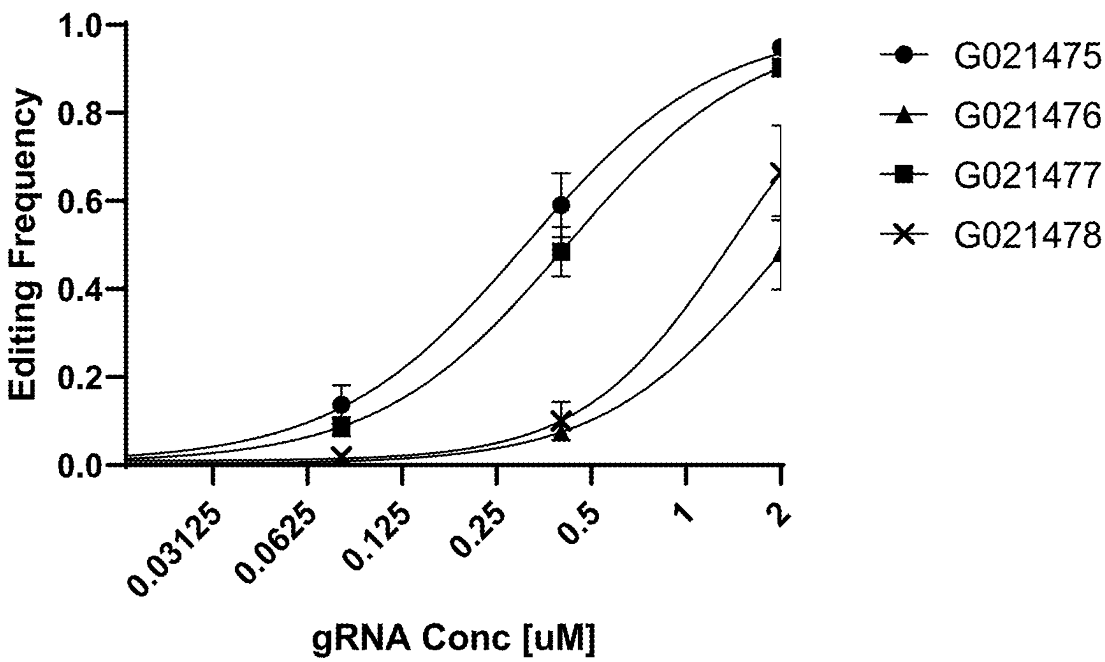
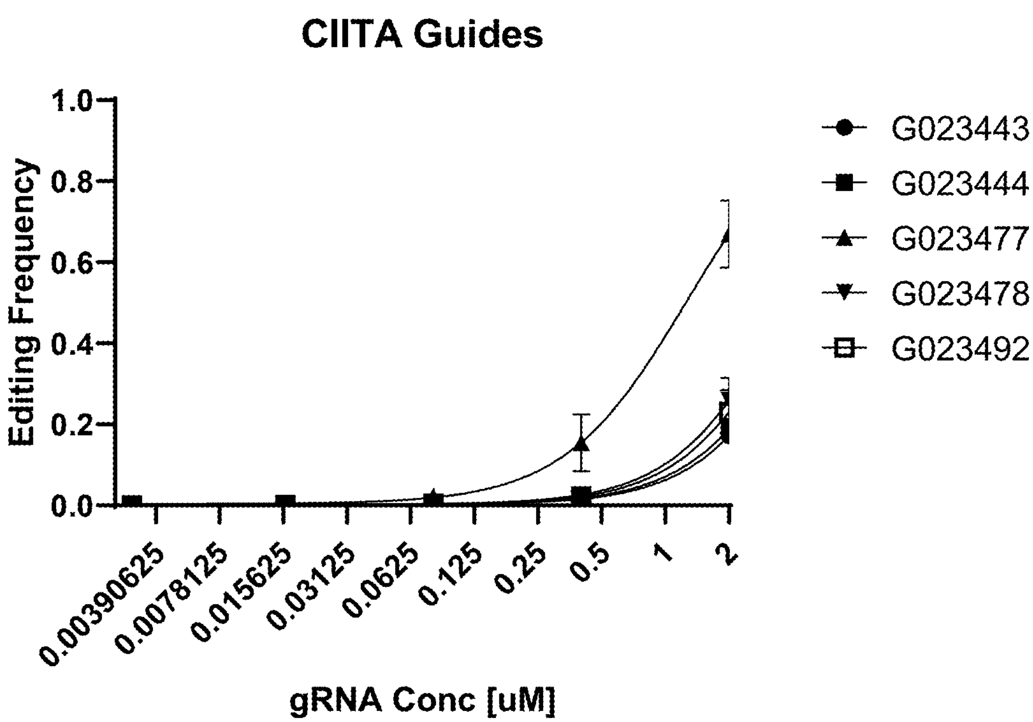


Fig. 11B



1.0
0.8
0.6
0.4
0.2
0.0
Editing Frequency
0.03125
0.0625
0.125
0.25
0.5
1
2
gRNA Conc [uM]
G021475
G021476
G021477
G021478

Fig. 12



CIITA Guides
1.0
0.8
0.6
0.4
0.2
0.0
Editing Frequency
0.00390625
0.0078125
0.015625
0.03125
0.0625
0.125
0.25
0.5
1
2
gRNA Conc [uM]
G023443
G023444
G023477
G023478
G023492

Fig. 13

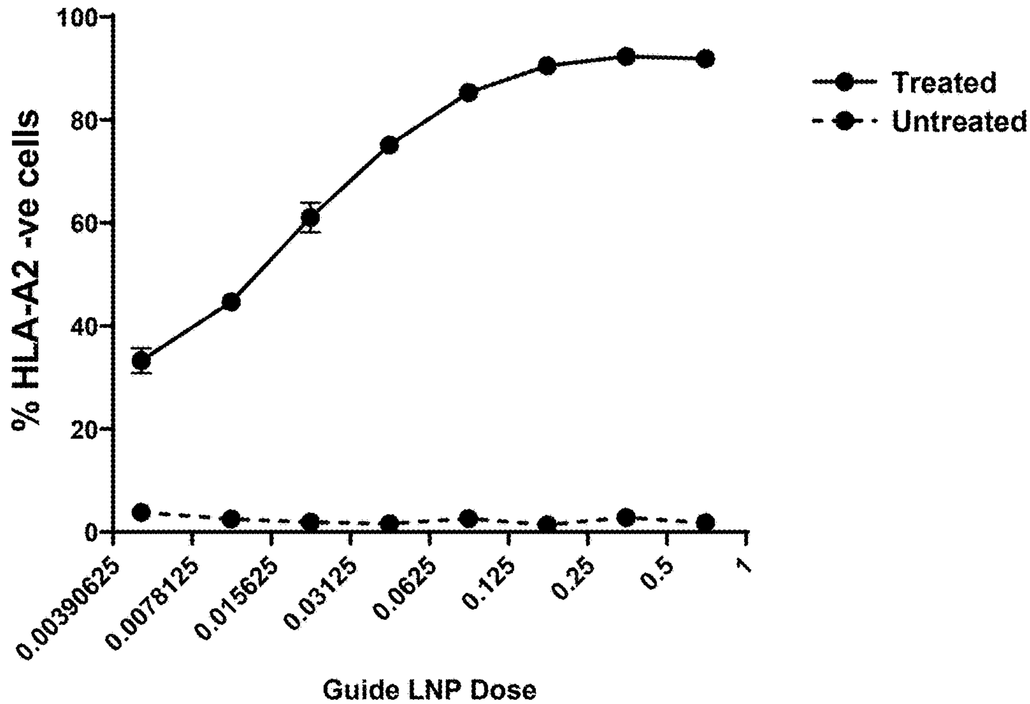

100
80
60
40
20
0
% HLA-A2 -ve cells
Treated
Untreated
0.00390625
0.0078125
0.015625
0.03125
0.0625
0.125
0.25
0.5
1
Guide LNP Dose

Fig. 14

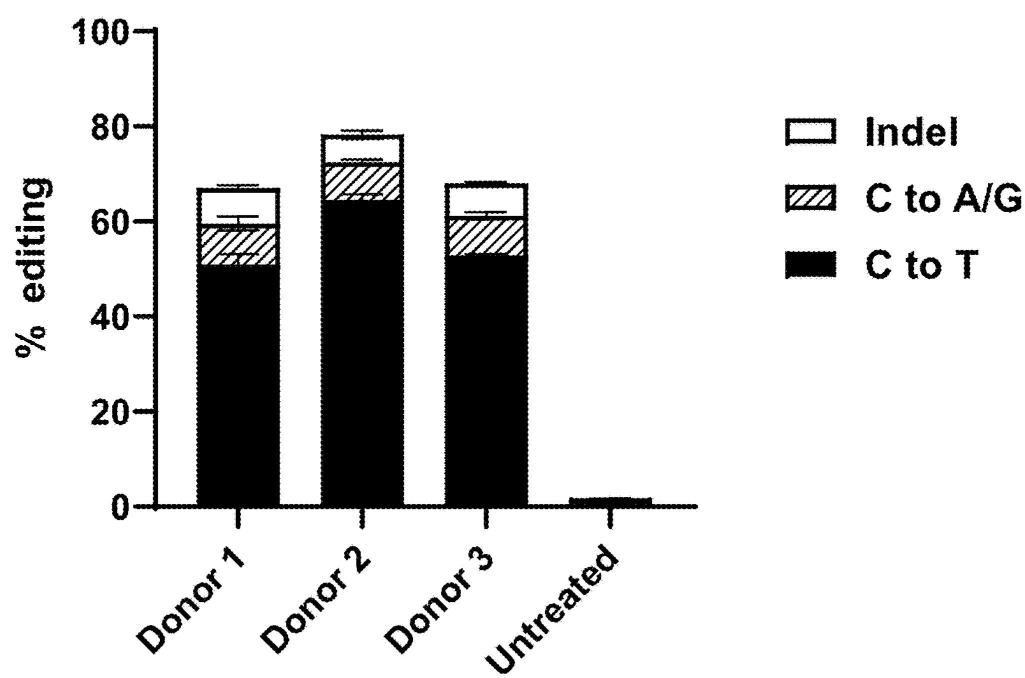

100
80
60
40
20
0
% editing
Indel
C to A/G
C to T
Donor 1
Donor 2
Donor 3
Untreated

Fig. 15

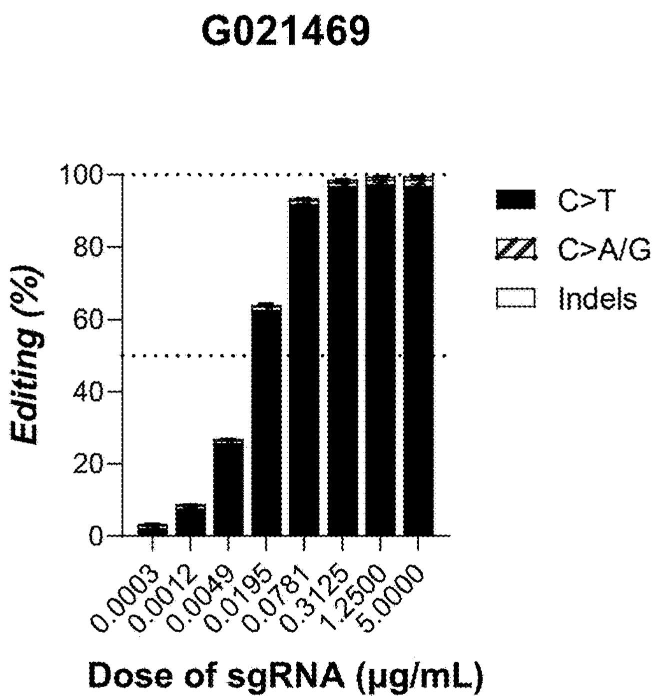


Fig. 16A

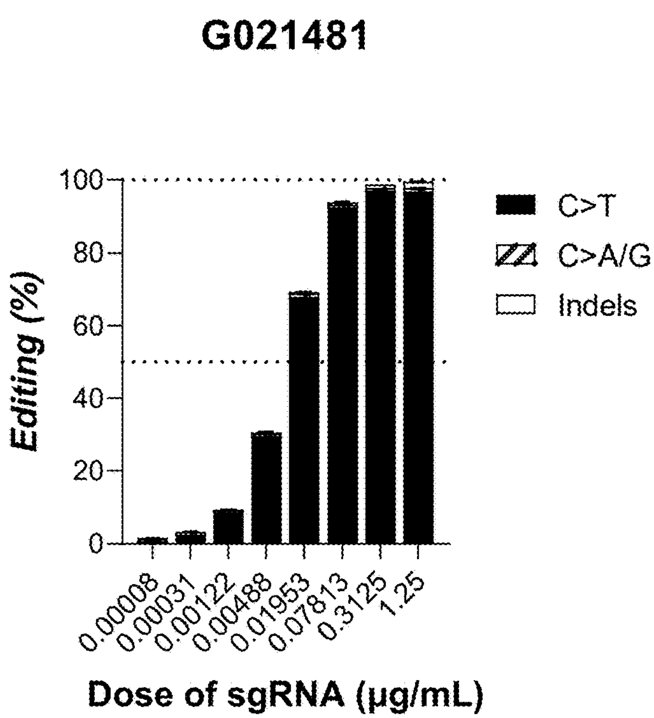


Fig. 16B

G028935

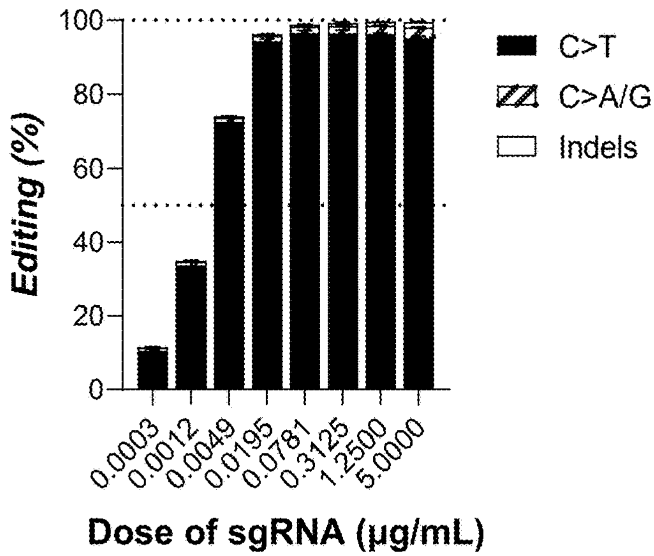


Fig. 16C

G028939

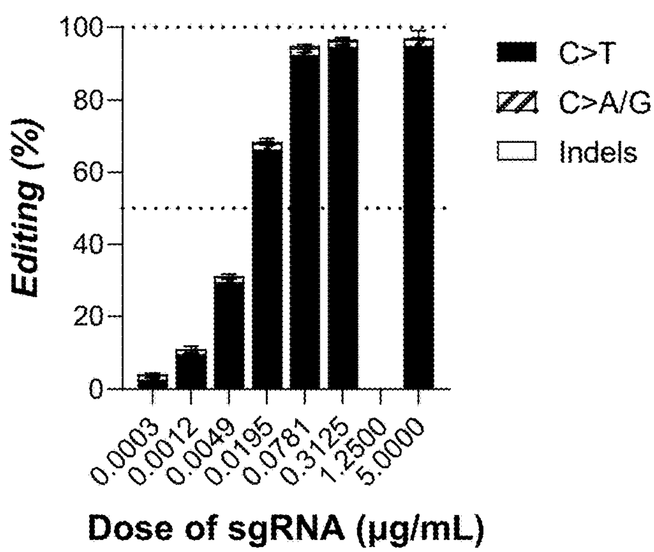


Fig. 16D

G028943

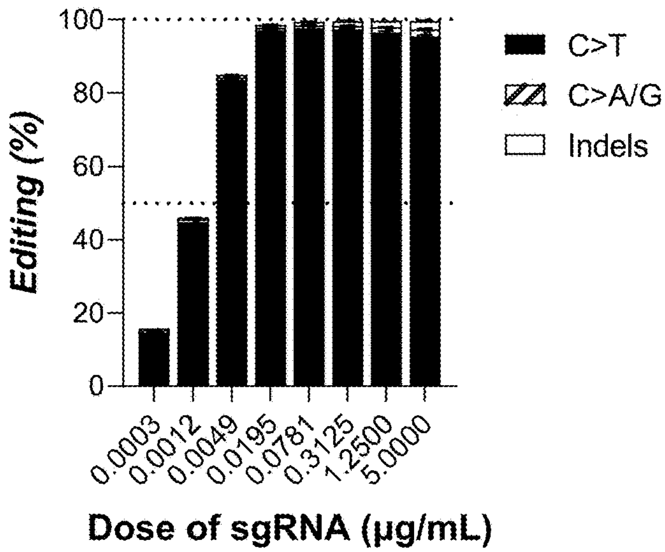


Fig. 16E

G028986

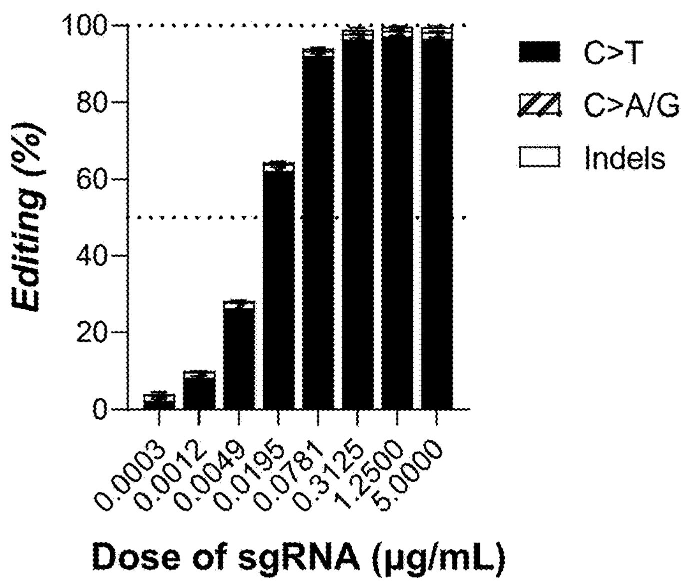


Fig. 16F

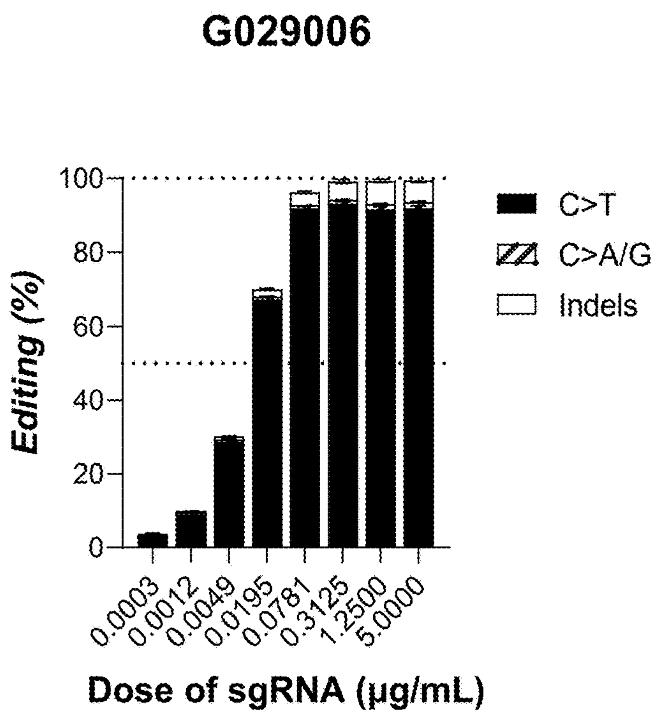


Fig. 16G

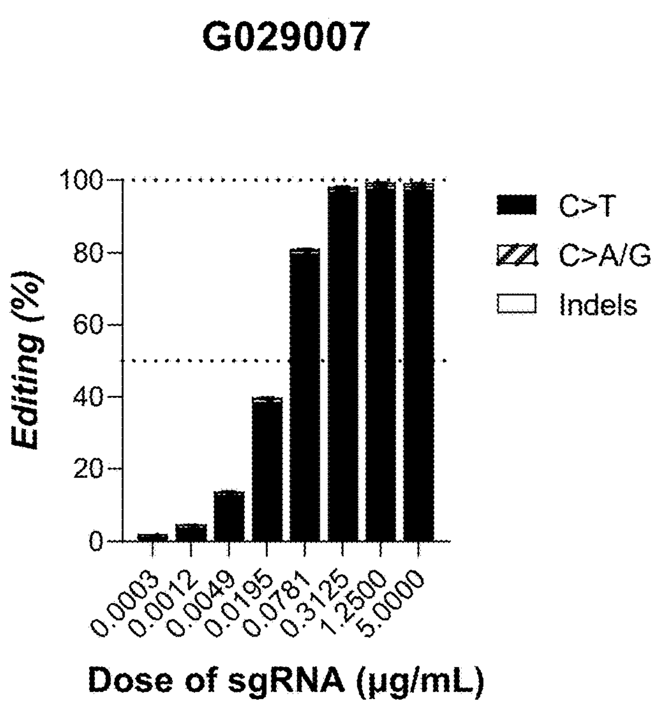


Fig. 16H

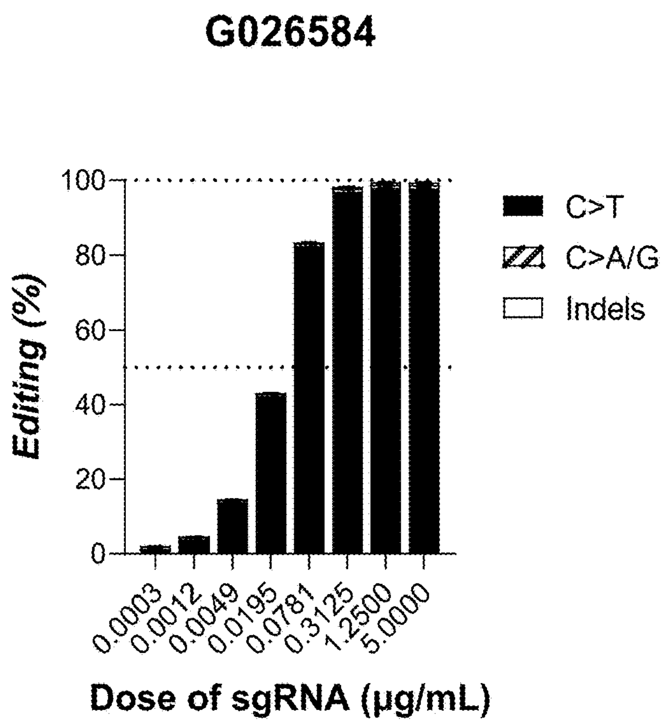


Fig. 16I

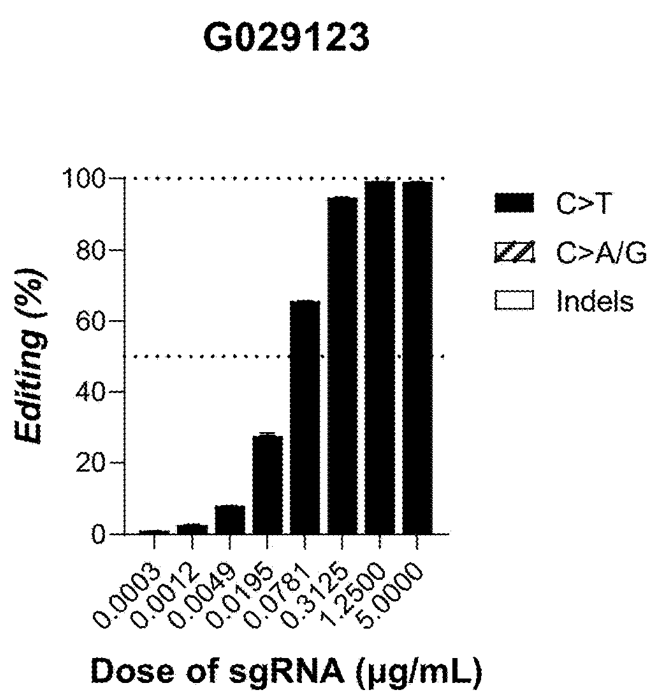


Fig. 16J

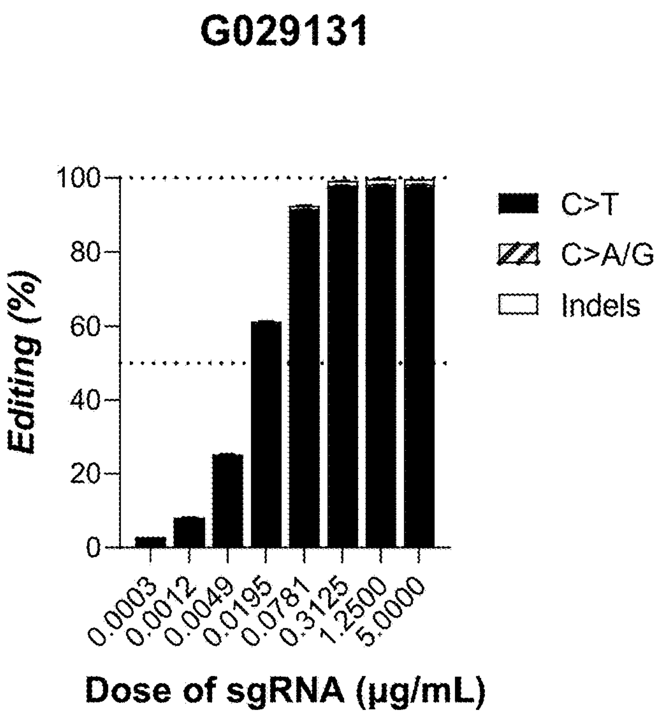


Fig. 16K

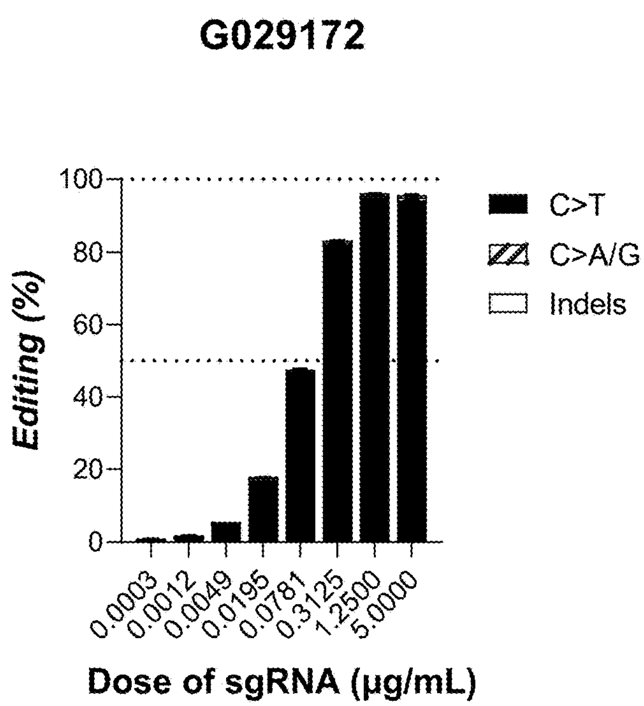


Fig. 16L

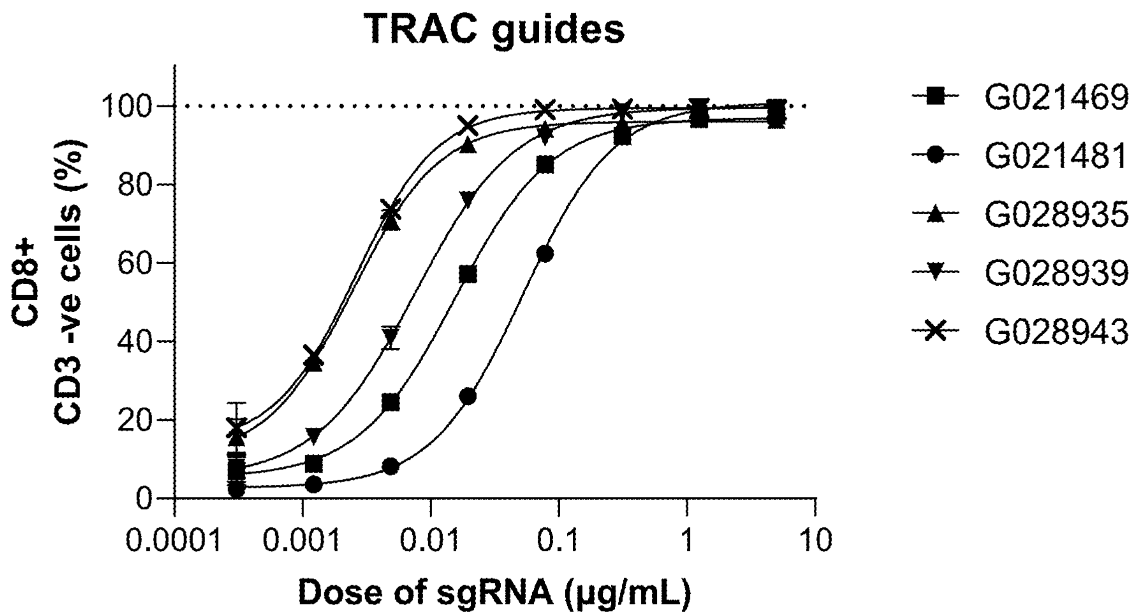


Fig. 17A

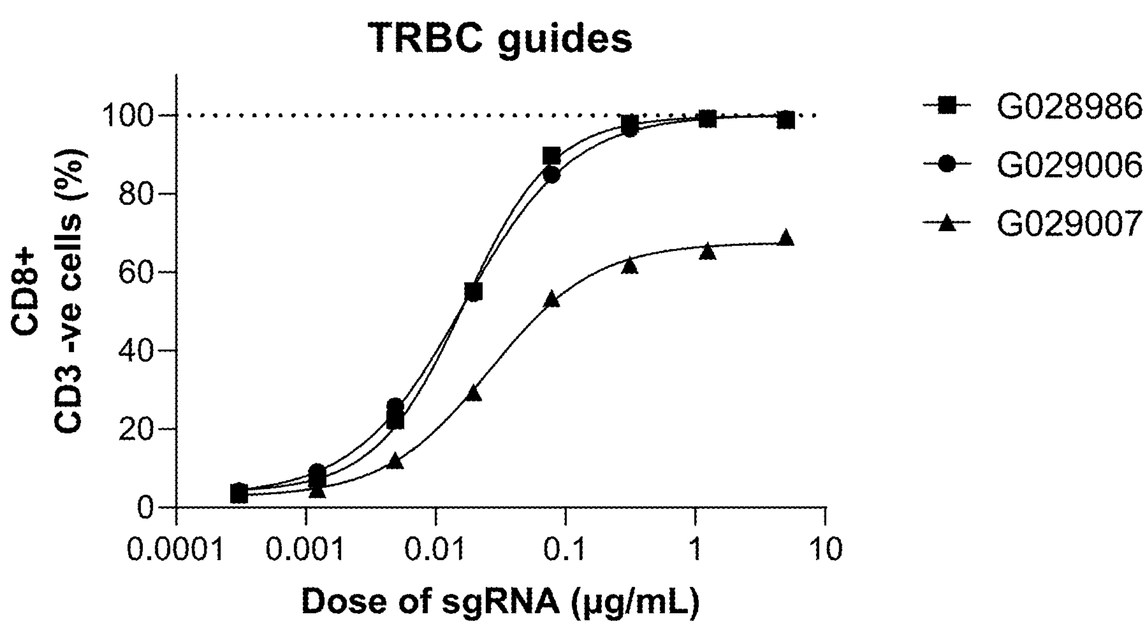


Fig. 17B

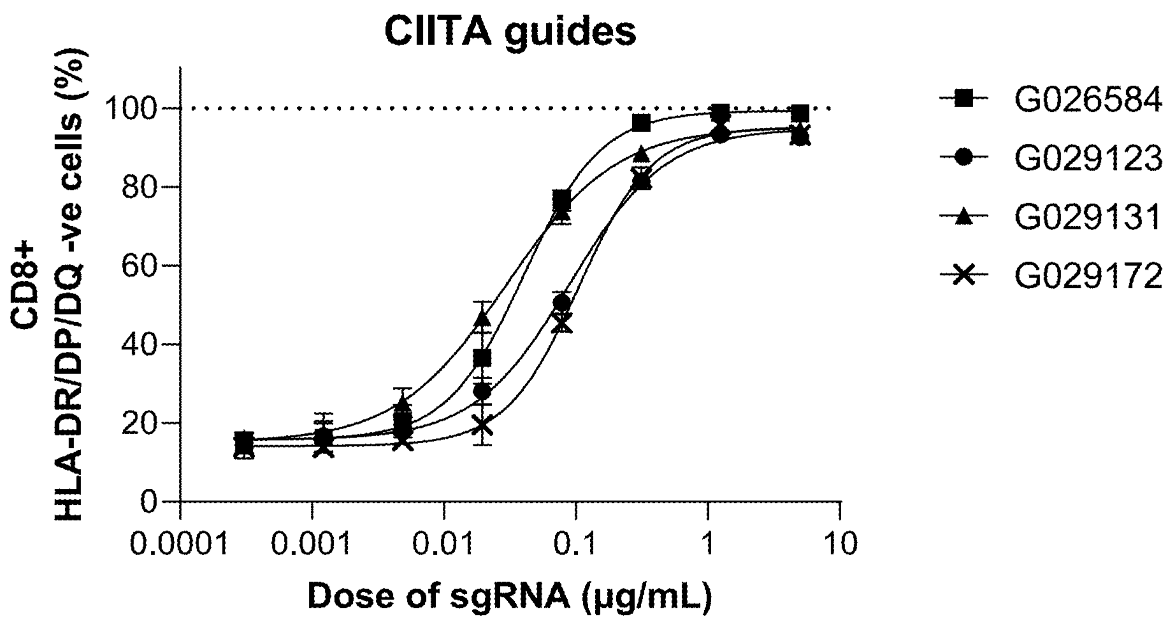


Fig. 17C

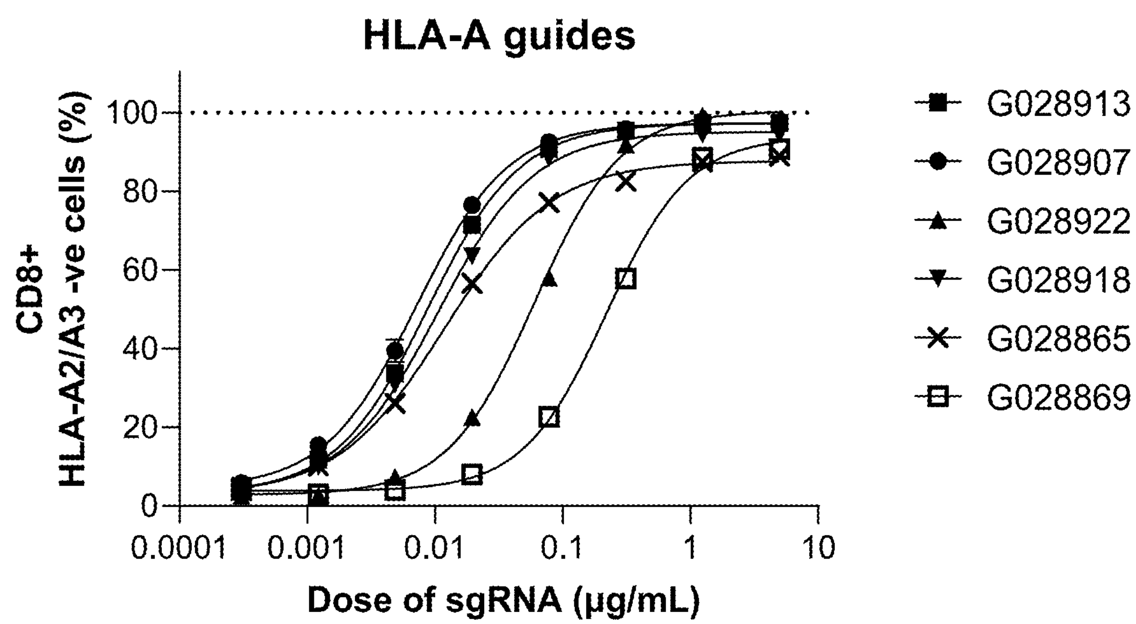


Fig. 17D

COMPOSITIONS AND METHODS FOR GENOMIC EDITING

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is a continuation of International Application No. PCT/US2023/068499, filed Jun. 15, 2023, which claims the benefit under 35 USC 119(e) of U.S. Provisional Application No. 63/352,990, filed Jun. 16, 2022, the content of each of which is herein incorporated by reference in its entirety.

SEQUENCE LISTING

[0002] This application contains a sequence listing, which has been submitted electronically in XML file format and is hereby incorporated by reference in its entirety. Said XML file, created on Jun. 13, 2023, is named "01155-0055-00PCT_SL.xml" and is 20,586,099 bytes in size.

INTRODUCTION AND SUMMARY

[0003] The present disclosure relates to genomic editing using *Neisseria meningitidis* CRISPR/Cas9 systems.

[0004] The ability to downregulate endogenous T-cell receptor (TCR) alpha and beta units, MHC class I, and MHC class II loci is critical for many in vivo and ex vivo utilities, e.g., when using allogeneic cells (originating from a donor) for transplantation or e.g., for creating a cell population in vitro that does not activate T cells. In particular, the transfer of allogeneic cells into a subject is of great interest to the field of cell therapy. The use of allogeneic cells has been limited due to the problem of rejection by the recipient subject's immune cells, which recognize the transplanted cells as foreign and mount an attack. To avoid the problem of immune rejection, cell-based therapies have focused on autologous approaches that use a subject's own cells as the cell source for therapy, an approach that is time-consuming and costly.

[0005] Typically, immune rejection of allogeneic cells results from a mismatching of major histocompatibility complex (MHC) molecules between the donor and recipient. Within the human population, MHC molecules exist in various forms, including e.g., numerous genetic variants of any given MHC gene, i.e., alleles, encoding different forms of MHC protein. The primary classes of MHC molecules are referred to as MHC class I and MHC class II. MHC class I molecules (e.g., HLA-A, HLA-B, and HLA-C in humans) are expressed on all nucleated cells and present antigens to activate cytotoxic T cells (CD8+ T cells or CTLs). MHC class II molecules (e.g., HLA-DP, HLA-DQ, and HLA-DR in humans) are expressed on only certain cell types (e.g., B cells, dendritic cells, and macrophages) and present antigens to activate helper T cells (CD4+ T cells or Th cells), which in turn provide signals to B cells to produce antibodies.

[0006] Slight differences, e.g., in MHC alleles between individuals can cause the T cells in a recipient to become activated. During T cell development, an individual's T cell repertoire is tolerized to one's own MHC molecules, but T cells that recognize another individual's MHC molecules may persist in circulation and are referred to as alloreactive T cells. Alloreactive T cells can become activated e.g., by the presence of another individual's cells expressing MHC molecules in the body, causing e.g., graft versus host disease and transplant rejection.

[0007] Methods and compositions for reducing the susceptibility of an allogeneic cell to rejection are of interest, including e.g., reducing the cell's expression of MHC protein to avoid recipient T cell responses. In practice, the ability to genetically modify an allogeneic cell for transplantation into a subject has been hampered by the requirement for multiple genomic edits to reduce all MHC protein expression, while at the same time, avoiding other harmful recipient immune responses. For example, while strategies to deplete MHC class I protein may reduce activation of CTLs, cells that lack MHC class I on their surface are susceptible to lysis by natural killer (NK) cells of the immune system because NK cell activation is regulated by MHC class I-specific inhibitory receptors. Therefore, safely reducing or eliminating expression of MHC class I has proven challenging. Genomic editing strategies to deplete MHC class II molecules have also proven difficult particularly in certain cell types for reasons including low editing efficiencies and low cell survival rates, preventing practical application as a cell therapy.

[0008] Thus, there exists a need for improved methods and compositions for modifying cells to overcome the problem of recipient immune rejection and the technical difficulties associated with the multiple genetic modifications required to produce a safer cell for transplant. The present disclosure provides genomic editing using *Neisseria meningitidis* CRISPR/Cas9 systems. NmeCas9 is smaller than *Streptococcus pyogenes* Cas9 (SpyCas9), allowing NmeCas9 to be suitable for messenger RNA (mRNA)-based delivery methods. NmeCas9 has an advantageous specificity and low off-target cleavage rates.

[0009] The engineered cell comprises a genetic modification in the HLA-A, TRAC, TRBC, or CIITA (class II major histocompatibility complex transactivator), which may be useful in cell therapy. The disclosure further provides compositions and methods to reduce or eliminate surface expression of endogenous T-cell receptor, MHC class I or II protein in a cell by genetically modifying the HLA-A, TRAC, TRBC, or CIITA gene.

[0010] In some embodiments, a method of reducing surface expression of HLA-A protein in an engineered cell relative to an unmodified cell, is provided, the method comprising contacting a cell with a composition of any of embodiments provided herein. In some embodiments, a method of reducing surface expression of TRAC protein in an engineered cell relative to an unmodified cell, is provided, the method comprising contacting a cell with a composition of any of embodiments provided herein. In some embodiments, a method of reducing surface expression of TRBC protein in an engineered cell relative to an unmodified cell, is provided, the method comprising contacting a cell with a composition of any of embodiments provided herein. In some embodiments, a method of reducing surface expression of MHC class II protein in an engineered cell relative to an unmodified cell, is provided, the method comprising contacting a cell with a composition of any of embodiments provided herein.

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. **1**A shows the distribution of percent of HLA-A2−, HLA-A3− T cells following editing with HLA-A targeting guides.

[0012] FIG. 1B shows the percent of HLA-A2− HLA-A3+ (HLA-A-), HLA-A+, HLA-A3+ (HLA-A3−) and HLA-A2−, HLA-A3− T cells following editing with select guides.

[0013] FIG. 1C shows percent HLA-B− T cells following editing with select HLA-A guides.

[0014] FIG. 2 shows percent HLA-A2 negative T cells following editing with a dilution series of HLA-A guides.

[0015] FIG. 3A shows the distribution of percent of CD3− T cells following editing with TRAC guides.

[0016] FIG. 3B shows percent editing at the TRAC locus.

[0017] FIG. 4 shows percent CD3 negative T cells following editing with a dilution series of TRAC guides.

[0018] FIG. 5A shows the distribution of percent of CD3− T cells following editing with TRBC guides.

[0019] FIG. 5B shows percent editing at the TRBC loci.

[0020] FIG. 6 shows percent CD3 negative T cells following editing with a dilution series of TRBC guides.

[0021] FIG. 7A shows the distribution of percent of HLA-DP, DQ, DR− T cells following editing with CIITA guides.

[0022] FIG. 7B shows percent editing at the CIITA locus.

[0023] FIG. 8 shows percentage of HLA II-DR, DP, DQ negative cells on the left vertical axis and mean percent editing on the right vertical axis following editing with CIITA guides.

[0024] FIG. 9A shows the percent of reads with C to T conversions at the CIITA locus following editing with a dilution series of CIITA guides.

[0025] FIG. 9B shows the percent of HLA-DP, DQ, DR negative T cells following editing with a dilution series of CIITA guides.

[0026] FIGS. 10A-10B show editing at the AAVS1 locus represented as indel frequency.

[0027] FIGS. 11A-11B show indel frequency at AAVS1 following editing with a dilution series of AAVS1 guides.

[0028] FIG. 12 shows mean editing frequency at various TRAC sgRNA concentrations.

[0029] FIG. 13 shows mean editing frequency at the CIITA locus.

[0030] FIG. 14 shows mean percentage of T cells negative for HLA-A2 surface expression.

[0031] FIG. 15 shows mean percentage editing at the CIITA locus.

[0032] FIGS. 16A-16L show mean percentage editing at a series of doses of the noted guide RNAs.

[0033] FIGS. 17A-17D show mean percentage editing of CD8+ T cells that are negative for respective expression markers at various sgRNA concentrations.

DETAILED DESCRIPTION

[0034] The present disclosure provides engineered cells, as well as methods and compositions for genetically modifying a cell to make an engineered cell and populations of engineered cells, that are useful, for example, for adoptive cell transfer (ACT) therapies. The disclosure provided herein overcomes certain hurdles of prior methods by providing methods and compositions for genetically modifying the HLA-A, TRAC, TRBC, CIITA, or AAVS1 locus to reduce expression of HLA-A, TRAC, TRBC, or MHC class II protein on the surface of a cell. In some embodiments, the disclosure provides engineered cells with reduced or eliminated surface expression of HLA-A, TRAC, TRBC, or MHC class II as a result of a genetic modification in the HLA-A, TRAC, TRBC, or CIITA gene. In some embodiments, the disclosure provides compositions and methods for reducing or eliminating expression of HLA-A, TRAC, TRBC, or MHC class II protein and compositions and methods to further reduce the cell's susceptibility to immune rejection. For example, in some embodiments, the methods and compositions comprise reducing or eliminating surface expression of HLA-A protein by genetically modifying the HLA-A gene. In some embodiments, the methods and compositions comprise reducing or eliminating surface expression of TRAC protein by genetically modifying the TRAC gene. In some embodiments, the methods and compositions comprise reducing or eliminating surface expression of TRBC1 protein by genetically modifying the TRBC1 gene. In some embodiments, the methods and compositions comprise reducing or eliminating surface expression of TRBC2 protein by genetically modifying the TRBC2 gene. In some embodiments, the methods and compositions comprise reducing or eliminating surface expression of MHC class II protein by genetically modifying CIITA. The engineered cell compositions produced by the methods disclosed herein have desirable properties, including e.g., reduced expression of MHC molecules, reduced immunogenicity in vitro and in vivo, increased survival, and increased genetic compatibility with greater subjects for transplant.

[0035] The term "about" or "approximately" means an acceptable error for a particular value as determined by one of ordinary skill in the art, which depends in part on how the value is measured or determined, or a degree of variation that does not substantially affect the properties of the described subject matter, or within the tolerances accepted in the art, e.g., within 10%, 5%, 2%, or 1%. Accordingly, unless indicated to the contrary, the numerical parameters set forth in the following specification and attached claims are approximations that may vary depending upon the desired properties sought to be obtained. At the very least, and not as an attempt to limit the application of the doctrine of equivalents to the scope of the claims, each numerical parameter should at least be construed in light of the number of reported significant digits and by applying ordinary rounding techniques.

[0036] Provided herein are the following numbered embodiments:

[0037] Embodiment 1 is an engineered cell, which has reduced or eliminated surface expression of HLA-A relative to an unmodified cell, comprising a genetic modification in the HLA-A gene, wherein the genetic modification comprises at least one nucleotide within the genomic coordinates chr6:29942540-29945459.

[0038] Embodiment 2 is the engineered cell of embodiment 1, which has reduced or eliminated surface expression of HLA-A relative to an unmodified cell, comprising a genetic modification in the HLA-A gene, wherein the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from any one of the genomic coordinates listed in Table 1 or wherein the genetic modification comprises at least one nucleotide within the genomic coordinates targeted by a guide RNA comprising a guide sequence of any one of SEQ ID NOs: 66, 61, 2-60, 62-65, 67-80.

[0039] Embodiment 3 is an engineered cell, which has reduced or eliminated surface expression of HLA-A relative to an unmodified cell, comprising a genetic modification in the HLA-A gene, wherein the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from: chr6:29942891-

29942915; chr6:29942609-29942633; chr6:29944266-29944290; chr6:29942889-29942913; chr6:29944471-29944495; and chr6:29944470-29944494.

[0040] Embodiment 4 is an engineered human cell, which has reduced or eliminated surface expression of HLA-A relative to an unmodified cell, comprising a genetic modification in the HLA-A gene, wherein the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from: chr6:29942891-29942915; chr6:29942609-29942633; chr6:29942889-29942913; chr6:29944471-29944495; chr6:29944266-29944290; chr6:29942785-29942809.

[0041] Embodiment 5 is an engineered human cell, which has reduced or eliminated surface expression of HLA-A relative to an unmodified cell, comprising a genetic modification in an HLA-A gene, wherein the genetic modification comprises an indel, a C to T substitution, or an A to G substitution within the genomic coordinates chosen from: chr6:29942891-29942915; chr6:29942609-29942633; chr6:29944266-29944290; chr6:29942889-29942913; chr6:29944471-29944495; and chr6:29944470-29944494.

[0042] Embodiment 6 is an engineered human cell, which has reduced or eliminated surface expression of HLA-A relative to an unmodified cell, comprising a genetic modification in an HLA-A gene, wherein the genetic modification comprises an indel, a C to T substitution, or an A to G substitution within the genomic coordinates chosen from: chr6:29942891-29942915; chr6:29942609-29942633; chr6:29942889-29942913; chr6:29944471-29944495; chr6:29944266-29944290; chr6:29942785-29942809.

[0043] Embodiment 7 is the engineered cell of any one of embodiments 1-6, wherein the HLA-A expression is reduced or eliminated by a genomic editing system that binds to an HLA-A genomic target sequence comprising at least 5 contiguous nucleotides within the genomic coordinates chosen from: chr6:29942891-29942915; chr6:29942609-29942633; chr6:29944266-29944290; chr6:29942889-29942913; chr6:29944471-29944495; and chr6:29944470-29944494 or chosen from chr6:29942891-29942915; chr6:29942609-29942633; chr6:29942889-29942913; chr6:29944471-29944495; chr6:29944266-29944290; chr6:29942785-29942809.

[0044] Embodiment 8 is the engineered cell of any one of embodiments 1-7, wherein the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from any one of the genomic coordinates listed in Table 1.

[0045] Embodiment 9 is the engineered cell of any one of embodiments 1-8, wherein the cell is homozygous for HLA-C.

[0046] Embodiment 10 is the engineered cell of any one of embodiments 1-9, wherein the cell is homozygous for HLA-B and homozygous for HLA-C.

[0047] Embodiment 11 is a composition comprising an HLA-A guide RNA and optionally an RNA-guided DNA binding agent or nucleic acid encoding an RNA-guided DNA binding agent, wherein the HLA-A guide RNA comprises: i. a guide sequence selected from SEQ ID NOs: 66, 61, 2-60, 62-65, 67-80; ii. at least 19, 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence selected from SEQ ID NOs: 66, 61, 2-60, 62-65, 67-80; or iii. a guide sequence that is at least 95%, 90%, or 85% identical to a sequence selected from SEQ ID NOs: 66, 61, 2-60, 62-65, 67-80; iv. a sequence that comprises 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 1; v. at least 20, 21, 22, 23, or 24, contiguous nucleotides of a sequence from (iv); or vi. a guide sequence that is at least 95%, 90%, or 85% identical to a sequence selected from (iv).

[0048] Embodiment 12 is a method of making an engineered human cell, which has reduced or eliminated surface expression of HLA-A protein relative to an unmodified cell, comprising contacting a cell with a composition comprising an HLA-A guide RNA and optionally an RNA-guided DNA binding agent or nucleic acid encoding an RNA-guided DNA binding agent, wherein the HLA-A guide RNA comprises: i. a guide sequence selected from SEQ ID NOs: 66, 61, 2-60, 62-65, 67-80; ii. at least 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence selected from SEQ ID NOs: 66, 61, 2-60, 62-65, 67-80; or iii. a guide sequence that is at least 95%, 90%, or 85% identical to a sequence selected from SEQ ID NOs: 66, 61, 2-60, 62-65, 67-80; iv. a sequence that comprises 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 1; v. at least 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence from (iv); or vi. a guide sequence that is at least 95%, 90%, or 85% identical to a sequence selected from (iv).

[0049] Embodiment 13 is a method of reducing surface expression of HLA-A protein in a human cell relative to an unmodified cell, comprising contacting a cell with a composition comprising an HLA-A guide RNA and optionally an RNA-guided DNA binding agent or nucleic acid encoding an RNA-guided DNA binding agent, wherein the HLA-A guide RNA comprises: i. a guide sequence selected from SEQ ID NOs: 66, 61, 2-60, 62-65, 67-80; ii. at least 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence selected from SEQ ID NOs: 66, 61, 2-60, 62-65, 67-80; or iii. a guide sequence that is at least 95%, 90%, or 85% identical to a sequence selected from SEQ ID NOs: 66, 61, 2-60, 62-65, 67-80; iv. a sequence that comprises 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 1; v. at least 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence from (iv); or vi. a guide sequence that is at least 95%, 90%, or 85% identical to a sequence selected from (iv).

[0050] Embodiment 14 is the composition or method of any one of embodiments 11-13, wherein the HLA-A guide RNA comprises a guide sequence of any one of SEQ ID NO: 66, 61, 13, 55, 70, and 71.

[0051] Embodiment 15 is the composition or method of any one of embodiments 11-13, wherein the HLA-A guide RNA comprises a guide sequence of any one of SEQ ID NOs: 61, 66, 13, 17, 55, and 70.

[0052] Embodiment 16 is the composition or method of any one of embodiments 11-13, wherein the HLA-A guide RNA comprises a guide sequence comprising a sequence of SEQ ID NO: 61.

[0053] Embodiment 17 is the composition or method of any one of embodiments 11-13, wherein the HLA-A guide RNA comprises a guide sequence comprising a sequence of SEQ ID NO: 66.

[0054] Embodiment 18 is a population of cells comprising the engineered cells of any one of embodiments 1-10, or the engineered cells produced by the method of any one of embodiments 12-17 or by use of the composition of embodiment 11.

[0055] Embodiment 19 is a pharmaceutical composition comprising (a) the engineered cells of any one of embodiments 1-10; the engineered cells produced by the method of any one of embodiments 12-17 or by use of the composition of embodiment 11; or (b) a population of cells of embodiment 18.

[0056] Embodiment 20 is an engineered human cell, which has reduced or eliminated surface expression of TRAC relative to an unmodified cell, comprising a genetic modification in the TRAC gene, wherein the genetic modification comprises at least one nucleotide within the genomic coordinates chr14:22547505-22551621 or chr14:22547462-22551621.

[0057] Embodiment 21 is the engineered cell of embodiment 20, wherein the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from any one of the genomic coordinates listed in Table 2 or wherein the genetic modification comprises at least one nucleotide within the genomic coordinates targeted by a guide RNA comprising a guide sequence of any one of SEQ ID NOs: 111, 107, 101-106, 108-110, and 112-120.

[0058] Embodiment 22 is the engineered cell of embodiment 20 or 21, wherein the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from: chr14:22550574-22550598; chr14:22550544-22550568; chr14:22547505-22547529; or chr14:22547525-22547549; chr14:22547674-22547698.

[0059] Embodiment 23 is the engineered cell of embodiment 20 or 21, wherein the genetic modification comprises at least one nucleotide within the genomic coordinates chr14:22547481-22547505.

[0060] Embodiment 24 is the engineered cell of embodiment 20 or 21, wherein the genetic modification comprises at least one nucleotide within the genomic coordinates chr14:22547471-22547495.

[0061] Embodiment 25 is the engineered cell of embodiment 20 or 21, wherein the genetic modification comprises at least one nucleotide within the genomic coordinates chr14:22547470-22547494.

[0062] Embodiment 26 is the engineered cell of embodiment 20 or 21, wherein the genetic modification comprises at least one nucleotide within the genomic coordinates chr14:22547462-22547486.

[0063] Embodiment 27 is an engineered human cell, which has reduced or eliminated expression of TRAC relative to an unmodified cell, comprising a genetic modification in the TRAC gene, wherein the genetic modification comprises an indel, a C to T substitution, or an A to G substitution within the genomic coordinates chosen from: chr14:22550574-22550598; chr14:22550544-22550568; chr14:22547505-22547529; chr14:22547525-22547549; or chr14:22547674-22547698.

[0064] Embodiment 28 is the engineered cell of any one of embodiments 20-27, wherein the TRAC expression is reduced or eliminated by a genomic editing system that binds to a TRAC target sequence comprising at least 5 contiguous nucleotides within the genomic coordinates chosen from: chr14:22550574-22550598; chr14:22550544-22550568; chr14:22547505-22547529; chr14:22547525-22547549; or chr14:22547674-22547698.

[0065] Embodiment 29 is a composition comprising: a) a TRAC guide RNA comprising a guide sequence that i) targets a TRAC genomic target sequence; or ii) directs an RNA-guided DNA binding agent to induce a double stranded break (DSB) or a single-stranded break (SSB) in a TRAC genomic target sequence comprising at least 10 contiguous nucleotides within the genomic coordinates chosen from: chr14:22550574-22550598; chr14:22550544-22550568; chr14:22547505-22547529; chr14:22547525-22547549; or chr14:22547674-22547698.

[0066] Embodiment 30 is a composition comprising: (a) a TRAC guide RNA and optionally (b) an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent, wherein the TRAC guide RNA comprises: i) a guide sequence selected from SEQ ID NOs: 111, 107, 101-106, 108-110, and 112-120; ii) at least 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence selected from SEQ ID NOs: 111, 107, 101-106, 108-110, and 112-120; or iii) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence selected from SEQ ID NOs: 111, 107, 101-106, 108-110, and 112-120; or iv) a sequence that comprises 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 2; v) at least 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence from (iv); or vi) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence selected from (iv).

[0067] Embodiment 31 is the composition of embodiment 30, for use in altering a DNA sequence within the TRAC locus in a cell.

[0068] Embodiment 32 is the composition of embodiment 30, for use in reducing or eliminating the expression of TRAC protein in a cell.

[0069] Embodiment 33 is a method of making an engineered human cell, which has reduced or eliminated surface expression of TRAC protein relative to an unmodified cell, comprising contacting a cell with a composition comprising: (a) a TRAC guide RNA and optionally (b) an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent, wherein the TRAC guide RNA comprises: i) a guide sequence selected from SEQ ID NOs: 111, 107, 101-106, 108-110, and 112-120; ii) at least 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence selected from SEQ ID NOs: 111, 107, 101-106, 108-110, and 112-120; or iii) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence selected from SEQ ID NOs: 111, 107, 101-106, 108-110, and 112-120; or iv) a sequence that comprises 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 2; v) at least 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence from (iv); or vi) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence selected from (iv).

[0070] Embodiment 34 is a method of reducing surface expression of TRAC protein in a human cell relative to an unmodified cell, comprising contacting a cell with a composition comprising: (a) a TRAC guide RNA and optionally (b) an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent, wherein the TRAC guide RNA comprises: i) a guide sequence selected from SEQ ID NOs: 111, 107, 101-106, 108-110, and 112-120; ii) at least 20, 21, 22, 23, 24, or 25 contiguous nucleotides of a sequence selected from SEQ ID NOs: 111, 107, 101-106, 108-110, and 112-120; or iii) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence selected from SEQ ID NOs: 111, 107, 101-106, 108-110, and 112-120; or iv) a sequence that comprises 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 2; v) at least 20, 21, 22, 23, 24, or 25 contiguous nucleotides of a sequence from (iv); or vi) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence selected from (iv).

[0071] Embodiment 35 is the composition or method of any one of embodiments 29-34, wherein the TRAC guide RNA comprises a guide sequence of any one of SEQ ID NO: 111, 107, 101, 102, and 103.

[0072] Embodiment 36 is the composition or method of any one of embodiments 29-34, wherein the TRAC guide RNA comprises a guide sequence comprising a sequence of SEQ ID NO: 107.

[0073] Embodiment 37 is the composition or method of any one of embodiments 29-34, wherein the TRAC guide RNA comprises a guide sequence comprising a sequence of SEQ ID NO: 111.

[0074] Embodiment 38 is the composition or method of any one of embodiments 29-34, wherein the TRAC guide RNA comprises a guide sequence comprising a sequence of SEQ ID NO: 117.

[0075] Embodiment 39 is the composition or method of any one of embodiments 29-34, wherein the TRAC guide RNA comprises a guide sequence comprising a sequence of SEQ ID NO: 118.

[0076] Embodiment 40 is the composition or method of any one of embodiments 29-34, wherein the TRAC guide RNA comprises a guide sequence comprising a sequence of SEQ ID NO: 119.

[0077] Embodiment 41 is the composition or method of any one of embodiments 29-34, wherein the TRAC guide RNA comprises a guide sequence comprising a sequence of SEQ ID NO: 120.

[0078] Embodiment 42 is a population of cells comprising the engineered cells of any one of embodiments 20-28 or the engineered cells produced by use of the composition of claim **29** or **30** or the method of any one of embodiments 33-41, wherein greater than about 50%, greater than about 55%, greater than about 60%, greater than about 65%, greater than about 70%, greater than about 75%, greater than about 80%, greater than about 85%, greater than about 90%, greater than about 95%, greater than about 98%, or greater than about 99% of the population of cells are CD3-cells.

[0079] Embodiment 43 is a population of cells comprising the engineered cells of any one of embodiments 20-28, or the engineered cells produced by use of the composition of embodiment 29 or 30 or by the method of any one of embodiments 33-41, or the population of cells of embodiment 42, wherein greater than about 50%, greater than about 55%, greater than about 60%, greater than about 65%, greater than about 70%, greater than about 75%, greater than about 80%, greater than about 85%, greater than about 90%, greater than about 95%, greater than about 98%, or greater than about 99% of the population lacks an endogenous T-cell receptor.

[0080] Embodiment 44 is a population of cells comprising the engineered cells of any one of embodiments 20-28, or the engineered cells produced by use of the composition of embodiment 29 or 30 or by the method of any one of embodiments 33-41, or the population of cells of embodiment 42 or 43, wherein the expression of the TRAC gene in the population has been reduced relative to an unaltered population of the same cell by at least about 50%, at least about 55%, by at least about 60%, at least about 65%, at least about 70%, by at least about 75%, at least about 80%, at least about 85%, by at least about 90%, at least about 95%, or at least about 98%, or at least about 99%.

[0081] Embodiment 45 is a pharmaceutical composition comprising the engineered cells of any one of embodiments 20-28, or the engineered cells produced by use of the composition of embodiment 29 or 30 or by the method of any one of embodiments 33-41, or the population of cells of any one of embodiments 42-44.

[0082] Embodiment 46 is an engineered cell, which has reduced or eliminated surface expression of TRBC relative to an unmodified cell, comprising a genetic modification in the TRBC locus, wherein the genetic modification comprises at least one nucleotide within the genomic coordinates chr7: 142791756-142802543.

[0083] Embodiment 47 is an engineered cell, which has reduced or eliminated surface expression of TRBC relative to an unmodified cell, comprising a genetic modification in the TRBC locus, wherein the genetic modification comprises at least one nucleotide within the genomic coordinates: (a) chr7:142791862-142793149; (b) chr7: 142791756-142792721; or (c) chr7:142801104-142802543; or wherein the genetic modification comprises at least one nucleotide within the genomic coordinates targeted by a guide RNA comprising a guide sequence of any one of SEQ ID NOs: 215, 201-214, and 216-265.

[0084] Embodiment 48 is the engineered cell of embodiments 46 or 47, wherein the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from any one of the genomic coordinates listed in Table 3.

[0085] Embodiment 49 is the engineered cell of embodiments 46 or 47, wherein the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from: (a) chr7: 142792690-142792714; or chr7:142792693-142792717; or (b) chr7:142791756-142791780; chr7: 142791761-142791785; chr7:142791820-142791844; chr7:142791939-142791963; chr7:142791940-142791964; or chr7:142792004-142792028; or (c) chr7:142801104-142801124; chr7:142802103-142802127; or chr7:142802106-142802130.

[0086] Embodiment 50 is the engineered cell of embodiment 46 or 47, wherein the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from: chr7:142792690-142792714; chr7:142802103-142802127; and chr7:142802106-14280213.

[0087] Embodiment 51 is an engineered cell, which has reduced or eliminated expression of TRBC relative to an unmodified cell, comprising a genetic modification in the human TRBC gene, wherein the genetic modification comprises an indel, a C to T substitution, or an A to G substitution within the genomic coordinates (a) chr7:142791862-142793149; (b) chr7: 142791756-142792721; or (c) chr7:142801104-142802543.

[0088] Embodiment 52 is an engineered human cell, which has reduced or eliminated expression of TRBC relative to an unmodified cell, comprising a genetic modification in the TRBC gene, wherein the genetic modification comprises an indel, a C to T substitution, or an A to G substitution within the genomic coordinates chosen from: (a) chr7:142792690-142792714 or chr7:142792693-142792717; or (b) chr7:142791756-142791780; chr7:142791761-142791785; chr7: 142791820-142791844; chr7:142791939-142791963; chr7:142791940-142791964; or chr7:142792004-142792028; or (c) chr7:142801104-142801124; chr7: 142802103-142802127; or chr7:142802106-142802130.

[0089] Embodiment 53 is an engineered human cell, which has reduced or eliminated expression of TRBC relative to an unmodified cell, comprising a genetic modification in the TRBC gene, wherein the genetic modification comprises an indel, a C to T substitution, or an A to G substitution within the genomic coordinates chosen from: chr7:142792690-142792714; chr7: 142802103-142802127; and chr7:142802106-14280213.

[0090] Embodiment 54 is the engineered cell of any one of embodiments 46-53, wherein the TRBC expression is reduced or eliminated by a genomic editing system that binds to a TRBC target sequence comprising at least 5 contiguous nucleotides within the genomic coordinates chosen from: (a) chr7:142792690-142792714 or chr7:142792693-142792717; or (b) chr7:142791756-142791780; chr7:142791761-142791785; chr7:142791820-142791844; chr7: 142791939-142791963; chr7:142791940-142791964; or chr7:142792004-142792028; or (c) chr7: 142801104-142801124; chr7:142802103-142802127; or chr7:142802106-142802130.

[0091] Embodiment 55 is the engineered cell of any one of embodiments 46-54, wherein the TRBC expression is reduced or eliminated by a genomic editing system that binds to a TRBC target sequence comprising at least 5 contiguous nucleotides within the genomic coordinates chosen from: chr7:142792690-142792714; chr7:142802103-142802127; and chr7:142802106-14280213.

[0092] Embodiment 56 is a composition comprising (a) a TRBC guide RNA and optionally (b) an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent, wherein the TRBC guide RNA comprises: i) a guide sequence selected from SEQ ID NOs: 215, 201-214, and 216-265; ii) at least 20, 21, 22, 23, 24, or 25 contiguous nucleotides of a sequence selected from SEQ ID NOs: 215, 201-214, and 216-265; iii) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence selected from SEQ ID NOs: 215, 201-214, and 216-265; iv) a sequence that comprises 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 3; v) at least 20, 21, 22, 23, 24, or 25 contiguous nucleotides of a sequence from (iv); or vi) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence selected from (iv).

[0093] Embodiment 57 is the composition of embodiment 56, for use in altering a DNA sequence within the TRBC locus in a cell.

[0094] Embodiment 58 is the composition of embodiment 56, for use in reducing or eliminating the expression of TRBC protein in a cell.

[0095] Embodiment 59 is a method of making an engineered human cell, which has reduced or eliminated surface expression of TRBC protein relative to an unmodified cell, comprising contacting a cell with: (a) a TRBC guide RNA and optionally (b) an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent, wherein the TRBC guide RNA comprises: i) a guide sequence selected from SEQ ID NOs: 215, 201-214, and 216-265; ii) at least 20, 21, 22, 23, 24, or 25 contiguous nucleotides of a sequence selected from SEQ ID NOs: 215, 201-214, and 216-265; iii) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence selected from SEQ ID NOs: 215, 201-214, and 216-265; iv) a sequence that comprises 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 3; v) at least 20, 21, 22, 23, 24, or 25 contiguous nucleotides of a sequence from (iv); or vi) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence selected from (iv).

[0096] Embodiment 60 is a method of reducing surface expression of TRBC protein in a human cell relative to an unmodified cell, comprising contacting a cell with: (a) a TRBC guide RNA and optionally (b) an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent, wherein the TRBC guide RNA comprises: i) a guide sequence selected from SEQ ID NOs: 215, 201-214, and 216-265; ii) at least 20, 21, 22, 23, 24, or 25 contiguous nucleotides of a sequence selected from SEQ ID NOs: 215, 201-214, and 216-265; iii) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence selected from SEQ ID NOs: 215, 201-214, and 216-265; iv) a sequence that comprises 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 3; v) at least 20, 21, 22, 23, 24, or 25 contiguous nucleotides of a sequence from (iv); or vi) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence selected from (iv).

[0097] Embodiment 61 is the method or composition of any one of embodiments 56-60, wherein the TRBC guide RNA comprises a guide sequence of any one of SEQ ID NO: 215, 216, 223, 224, 229, 230, 246, 259, and 260.

[0098] Embodiment 62 is the method or composition of any one of embodiments 56-61, wherein the TRBC guide RNA comprises a guide sequence of any one of SEQ ID NO: 215, 216, 224, 229, 246, 259, and 260.

[0099] Embodiment 63 is the method or composition of any one of embodiments 56-62, wherein the TRBC guide RNA comprises a guide sequence of any one of SEQ ID NOs: 215, 259, and 260.

[0100] Embodiment 64 is a population of cells comprising the engineered cells of any one of embodiments 46-55 or the engineered cells produced by use of the composition of embodiment 56 or by the method of any one of embodiments 59-63, wherein greater than about 50%, greater than about 55%, greater than about 60%, greater than about 65%, greater than about 70%, greater than about 75%, greater than about 80%, greater than about 85%, greater than about 90%, greater than about 95%, greater than about 98%, or greater than about 99% of the population of cells are CD3-cells.

[0101] Embodiment 65 is a population of cells comprising the engineered cells of any one of embodiments 46-55 or the engineered cells produced by use of the composition of embodiment 56 or by the method of any one of embodiments 59-63, wherein greater than about 50%, greater than about 55%, greater than about 60%, greater than about 65%, greater than about 70%, greater than about 75%, greater than about 80%, greater than about 85%, greater than about 90%, greater than about 95%, greater than about 98%, or greater than about 99% of the population lacks an endogenous T-cell receptor.

[0102] Embodiment 66 is a population of cells comprising the engineered cells of any one of embodiments 46-55 or the engineered cells produced by use of the composition of embodiment 56 or by the method of any one of embodiments 59-63, wherein the expression of the TRBC gene in the population has been reduced relative to an unaltered population of the same cell by at least about 50%, at least about 55%, by at least about 60%, at least about 65%, at least about 70%, by at least about 75%, at least about 80%, at least about 85%, by at least about 90%, at least about 95%, at least about 98%, or at least about 99%.

[0103] Embodiment 67 is a pharmaceutical composition comprising (a) the engineered cells of any one of embodiments 46-55, or the engineered cells produced by use of the composition of embodiment 56 or by the method of any one of embodiments 59-63, or (b) a population of cells of any one of embodiments 64-66.

[0104] Embodiment 68 is an engineered cell, which has reduced or eliminated surface expression of MHC class II relative to an unmodified cell, comprising a genetic modification in the CIITA gene, wherein the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from: (a) chr16:10877363-10907788 or (b) chr16:10906515-10908136.

[0105] Embodiment 69 is the engineered cell of embodiment 68, wherein the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from any one of the genomic coordinates listed in Table 4.

[0106] Embodiment 70 is the engineered cell of embodiment 68, wherein the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from: (a) chr16:10907504-10907528; chr16:10906643-10906667; chr16:10907508-10907532; chr16:10907539-10907559; chr16:10895658-10895682; chr16:10895668-10895692; chr16:10895750-10895774; chr16:10895753-10895777; chr16:10895754-10895778; chr16:10898684-10898708; chr16:10901529-10901553; chr16:10902121-10902145; chr16:10902701-10902725; chr16:10904726-10904750; chr16:10904760-10904784; chr16:10906493-10906517; chr16:10906515-10906539; chr16:10906631-10906655; chr16:10906636-10906660; chr16:10906770-10906794; chr16:10906788-10906812; chr16:10906789-10906813; chr16:10906816-10906840; chr16:10907148-10907172; chr16:10907254-10907278; chr16:10907331-10907355; chr16:10907477-10907501; chr16:10907497-10907521; chr16:10907503-10907527; chr16:10907574-10907598; or (b) chr16:10906889-10906913 or chr16:10907504-10907528.

[0107] Embodiment 71 is the engineered cell of embodiment 68, wherein the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from: chr16:10907504-10907528; chr16:10906643-10906667; chr16:10895658-10895682; chr16:10902701-10902725; chr16:10906493-10906517; chr16:10906631-10906655; chr16:10907477-10907501; chr16:10907497-10907521; or chr16:10907508-10907532.

[0108] Embodiment 72 is an engineered cell, which has reduced or eliminated surface expression of MHC class II relative to an unmodified cell, comprising a genetic modification in the CIITA locus, wherein the genetic modification comprises an indel, a C to T substitution, or an A to G substitution within the genomic coordinates chosen from: (a) chr16:10907504-10907528; chr16:10906643-10906667; chr16:10907508-10907532; chr16:10907539-10907559; chr16:10895658-10895682; chr16:10895668-10895692; chr16:10895750-10895774; chr16:10895753-10895777; chr16:10895754-10895778; chr16:10898684-10898708; chr16:10901529-10901553; chr16:10902121-10902145; chr16:10902701-10902725; chr16:10904726-10904750; chr16:10904760-10904784; chr16:10906493-10906517; chr16:10906515-10906539; chr16:10906631-10906655; chr16:10906636-10906660; chr16:10906770-10906794; chr16:10906788-10906812; chr16:10906789-10906813; chr16:10906816-10906840; chr16:10907148-10907172; chr16:10907254-10907278; chr16:10907331-10907355; chr16:10907477-10907501; chr16:10907497-10907521; chr16:10907503-10907527; chr16:10907574-10907598; or (b) chr16:10907504-10907528.

[0109] Embodiment 73 is an engineered cell, which has reduced or eliminated surface expression of MHC class II relative to an unmodified cell, comprising a genetic modification in the CIITA locus, wherein the genetic modification comprises an indel, a C to T substitution, or an A to G substitution within the genomic coordinates chosen from: chr16:10907504-10907528; chr16:10906643-10906667; chr16:10895658-10895682; chr16:10902701-10902725; chr16:10906493-10906517; chr16:10906631-10906655; chr16:10907477-10907501; and chr16:10907497-10907521; chr16:10907504-10907528; chr16:10907508-10907532.

[0110] Embodiment 74 is the engineered cell of any one of embodiments 68-73, wherein the MHC class II expression is reduced or eliminated by a genomic editing system that binds to a CIITA genomic target sequence comprising at least 5 contiguous nucleotides within the genomic coordinates chosen from (a) chr16: 10907504-10907528; chr16:10906643-10906667; chr16:10907508-10907532; chr16:10907539-10907559; chr16:10895658-10895682; chr16: 10895668-10895692; chr16:10895750-10895774; chr16:10895753-10895777; chr16:10895754-10895778; chr16:10898684-10898708; chr16: 10901529-10901553; chr16:10902121-10902145; chr16:10902701-10902725; chr16:10904726-10904750; chr16:10904760-10904784; chr16: 10906493-10906517; chr16:10906515-10906539; chr16:10906631-10906655; chr16:10906636-10906660; chr16:10906770-10906794; chr16: 10906788-10906812; chr16:10906789-10906813; chr16:10906816-10906840; chr16:10907148-10907172; chr16:10907254-10907278; chr16: 10907331-10907355; chr16:10907477-10907501; chr16:10907497-10907521; chr16:10907503-10907527; chr16:10907574-10907598; or chr16: 10907504-10907528.

[0111] Embodiment 75 is the engineered cell of any one of embodiments 68-74, wherein the MHC class II expression is reduced or eliminated by a genomic editing system that binds to a CIITA genomic target sequence comprising at least 5 contiguous nucleotides within the genomic coordinates chosen from: chr16: 10907504-10907528; chr16:10906643-10906667; chr16:10895658-10895682; chr16:10902701-10902725; chr16:10906493-10906517; chr16: 10906631-10906655; chr16:10907477-10907501; chr16:10907497-10907521; or chr16:10907508-10907532.

[0112] Embodiment 76 is a composition comprising (a) a CIITA guide RNA and optionally (b) an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent, wherein the CIITA guide RNA comprises: i) a guide sequence selected from SEQ ID NOs: 301, 422, 302-421, and 423-576; or ii) at least 20, 21, 22, 23, 24, or 25 contiguous nucleotides of a sequence selected from SEQ ID NOs: 301, 422, 302-421, and 423-576; or iii) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence selected from SEQ ID NOs: 301, 422, 302-421, and 423-576; or iv) a sequence that comprises 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 4; or v) at least 20, 21, 22, 23, 24, or 25 contiguous nucleotides of a sequence from (iv); or vi) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence selected from (iv).

[0113] Embodiment 77 is the composition of embodiment 76, for use in altering a DNA sequence within the CIITA gene in a cell.

[0114] Embodiment 78 is the composition of embodiment 76, for use in reducing or eliminating the expression of the CIITA in a cell.

[0115] Embodiment 79 is a method of making an engineered human cell, which has reduced or eliminated surface expression of MHC class II protein relative to an unmodified cell, comprising contacting a cell with: (a) a CIITA guide RNA and optionally (b) an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent, wherein the CIITA guide RNA comprises: i) a guide sequence selected from SEQ ID NOs: 301, 422, 302-421, and 423-576; or ii) at least 20, 21, 22, 23, 24, or 25 contiguous nucleotides of a sequence selected from SEQ ID NOs: 301, 422, 302-421, and 423-576; or iii) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence selected from SEQ ID NOs: 301, 422, 302-421, and 423-576; or iv) a sequence that comprises 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 4; or v) at least 20, 21, 22, 23, 24, or 25 contiguous nucleotides of a sequence from (iv); or vi) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence selected from (iv).

[0116] Embodiment 80 is a method of reducing surface expression of MHC class II protein in a human cell relative to an unmodified cell, comprising contacting a cell with: (a) a CIITA guide RNA and optionally (b) an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent, wherein the CIITA guide RNA comprises: i) a guide sequence selected from SEQ ID NOs: 301, 422, 302-421, and 423-576; or ii) at least 20, 21, 22, 23, 24, or 25 contiguous nucleotides of a sequence selected from SEQ ID NOs: 301, 422, 302-421, and 423-576; or iii) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence selected from SEQ ID NOs: 301, 422, 302-421, and 423-576; or iv) a sequence that comprises 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 4; or v) at least 20, 21, 22, 23, 24, or 25 contiguous nucleotides of a sequence from (iv); or vi) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence selected from (iv).

[0117] Embodiment 81 is the method or composition of any one of embodiments 76-80, wherein the CIITA guide RNA comprises a guide sequence of any one of SEQ ID NOs: 301, 422, 302, 320, 321, 324, 326, 327, 332, 354, 361, 372, 400, 408, 414, 415, 419, 420, 428, 431, 432, 434, 451, 455, 458, 462, 463, 464, 468.

[0118] Embodiment 82 is the method or composition of any one of embodiments 76-81, wherein the CIITA guide RNA comprises a guide sequence of any one of SEQ ID NO: 538.

[0119] Embodiment 83 is the method or composition of any one of embodiments 76-81, wherein the CIITA guide RNA comprises a guide sequence of any one of SEQ ID NOs: 301, 422, 302, 320, 372, 414, 419, 462, and 463.

[0120] Embodiment 84 is a population of cells comprising the engineered cells of any one of embodiments 68-75 or the engineered cells produced by use of the composition of embodiment 76 or by the method of any one of embodiments 79-83, wherein greater than about 50%, greater than about 55%, greater than about 60%, greater than about 65%, greater than about 70%, greater than about 75%, greater than about 80%, greater than about 85%, greater than about 90%, greater than about 95%, greater than about 98%, or greater than about 99% of the population of cells are MHC class II molecules negative as measured by flow cytometry.

[0121] Embodiment 85 is a population of cells comprising the engineered cells of any one of embodiments 68-75 or the engineered cells produced by use of the composition of embodiment 76 or by the method of any one of embodiments 79-83, wherein greater than about 50%, greater than about 55%, greater than about 60%, greater than about 65%, greater than about 70%, greater than about 75%, greater than about 80%, greater than about 85%, greater than about 90%, greater than about 95%, greater than about 98%, or greater than about 99% of the population of cells are negative for MHC class II molecules as measured by next generation sequencing (NGS).

[0122] Embodiment 86 is a population of cells comprising the engineered cells of any one of embodiments 68-75 or the engineered cells produced by use of the composition of embodiment 76 or by the method of any one of embodiments 79-83, wherein the expression of MHC Class II molecules in the population has been reduced relative to an unaltered population of the same cell by at least about 50%, at least about 55%, by at least about 60%, at least about 65%, at least about 70%, by at least about 75%, at least about 80%, at least about 85%, by at least about 90%, at least about 95%, at least about 98%, or at least about 99%.

[0123] Embodiment 87 is a pharmaceutical composition comprising (a) the engineered cells of any one of embodiments 68-75 or the engineered cells produced by use of the composition of embodiment 76 or by the method of any one of embodiments 79-83, or (b) a population of cells of any one of embodiments 84-86.

[0124] Embodiment 88 is an engineered cell comprising a genetic modification in the AAVS1 locus, wherein the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from: chr19:55115151-55116209.

[0125] Embodiment 89 is the engineered cell of embodiment 88, wherein the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from any one of the genomic coordinates listed in Table 5.

[0126] Embodiment 90 is the engineered cell of embodiment 88, wherein the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from: chr19:55115218-55115242; chr19:55115477-55115501; chr19:55115504-55115528; chr19:55115513-55115537; chr19:55115514-55115538; chr19:55115517-55115541; chr19:55115518-55115542; chr19:55115549-55115573; chr19:55115574-55115598; chr19:55115606-55115630; chr19:55115933-55115957; chr19:55116026-55116050; chr19:55116045-55116069; chr19:55116084-55116108; chr19:55115276-55115300; chr19:55115509-55115533; chr19:55115579-55115603; chr19:55115863-55115887; chr19:55115906-55115930; or chr19:55116006-55116030.

[0127] Embodiment 91 is an engineered cell comprising a genetic modification in the AAVS1 gene, wherein the genetic modification comprises an indel, a C to T substitution, or an A to G substitution within the genomic coordinates chosen from: chr19:55115218-55115242; chr19:55115477-55115501; chr19:55115504-55115528; chr19:55115513-55115537; chr19:55115514-55115538; chr19:55115517-55115541; chr19:55115518-55115542; chr19:55115549-55115573; chr19:55115574-55115598; chr19:55115606-55115630; chr19:55115933-55115957; chr19:55116026-55116050; chr19:55116045-55116069; chr19:55116084-55116108; chr19:55115276-55115300; chr19:55115509-55115533; chr19:55115579-55115603; chr19:55115863-55115887; chr19:55115906-55115930; or chr19:55116006-55116030.

[0128] Embodiment 92 is the engineered cell of any one of embodiments 88-91, wherein the genetic modification is induced by a genomic editing system that binds to an AAVS1 genomic target sequence comprising at least 5 contiguous nucleotides within the genomic coordinates chosen from chr19:55115218-55115242; chr19:55115477-55115501; chr19:55115504-55115528; chr19:55115513-55115537; chr19:55115514-55115538; chr19:55115517-55115541; chr19:55115518-55115542; chr19:55115549-55115573; chr19:55115574-55115598; chr19:55115606-55115630; chr19:55115933-55115957; chr19:55116026-55116050; chr19:55116045-55116069; chr19:55116084-55116108; chr19:55115276-55115300; chr19:55115509-55115533; chr19:55115579-55115603; chr19:55115863-55115887; chr19:55115906-55115930; or chr19:55116006-55116030.

[0129] Embodiment 93 is a composition comprising: a) an AAVS1 guide RNA comprising a guide sequence that i) targets an AAVS1 genomic target sequence; or ii) directs an RNA-guided DNA binding agent to induce a double stranded break (DSB) or a single-stranded break (SSB) in an AAVS1 genomic target sequence comprising at least 10 contiguous nucleotides within the genomic coordinates chosen from: chr19:55115218-55115242; chr19:55115477-55115501; chr19:55115504-55115528; chr19:55115513-55115537; chr19:55115514-55115538; chr19:55115517-55115541; chr19:55115518-55115542; chr19:55115549-55115573; chr19:55115574-55115598; chr19:55115606-55115630; chr19:55115933-55115957; chr19:55116026-55116050; chr19:55116045-55116069; chr19:55116084-55116108; chr19:55115276-55115300; chr19:55115509-55115533; chr19:55115579-55115603; chr19:55115863-55115887; chr19:55115906-55115930; or chr19:55116006-55116030.

[0130] Embodiment 94 is a composition comprising: a) an AAVS1 guide RNA (gRNA) and optionally (b) an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent, wherein the AAVS1 guide RNA comprises: i) a guide sequence selected from SEQ ID NOs: 601-774; ii) at least 20, 21, 22, 23, 24, or 25 contiguous nucleotides of a sequence selected from SEQ ID NOs: 601-774; iii) a guide sequence at least 95%, 90%, or 85% identical to a sequence selected from SEQ ID NOs: 601-774; iv) a sequence that comprises 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 5; v) at least 20, 21, 22, 23, 24, or 25 contiguous nucleotides of a sequence from (iv); or vi) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence selected from (iv).

[0131] Embodiment 95 is a method of making an engineered human cell comprising contacting a cell with: (a) an AAVS1 guide RNA and optionally (b) an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent, wherein the AAVS1 guide RNA comprises: i) a guide sequence selected from SEQ ID NOs: 601-774; ii) at least 20, 21, 22, 23, 24, or 25 contiguous nucleotides of a sequence selected from SEQ ID NOs: 601-774; iii) a guide sequence at least 95%, 90%, or 85% identical to a sequence selected from SEQ ID NOs: 601-774; iv) a sequence that comprises 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 5; v) at least 20, 21, 22, 23, or 24, contiguous nucleotides of a sequence from (iv); or vi) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence selected from (iv).

[0132] Embodiment 96 is the method or composition of any one of embodiments 93-95, wherein the AAVS1 guide RNA comprises a guide sequence of any one of SEQ ID NOs: 611, 620, 622, 626, 627, 628, 629, 632, 633, 634, 656, 659, 660, 661, 673, 691, 692, 730, 734, and 746.

[0133] Embodiment 97 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-96, wherein the genetic modification comprises at least 2, at least 3, at least 4, at least 5, at least 6, at least 7, at least 8, at least 9, or at least 10 contiguous nucleotides within the genomic coordinates.

[0134] Embodiment 98 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-97, wherein the genetic modification comprises at least 5, 6, 7, 8, 9, or 10 contiguous nucleotides within the genomic coordinates.

[0135] Embodiment 99 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-98, wherein the genomic target sequence comprises at least 10 contiguous nucleotides within the genomic coordinates.

[0136] Embodiment 100 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-99, wherein the genomic target sequence comprises at least 15 contiguous nucleotides within the genomic coordinates.

[0137] Embodiment 101 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-100, wherein the genomic target sequence comprises at least 17, 18, 19, 20, 21, 22, 23, or 24 contiguous nucleotides within the genomic coordinates.

[0138] Embodiment 102 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-101, wherein the genetic modification comprises an indel.

[0139] Embodiment 103 is the engineered cell of any one of embodiments 1-102, wherein the genetic modification comprises an insertion of a heterologous coding sequence.

[0140] Embodiment 104 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-103, wherein the genetic modification comprises at least one A to G substitution within the genomic coordinates.

[0141] Embodiment 105 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-104, wherein the genetic modification comprises at least one C to T substitution within the genomic coordinates.

[0142] Embodiment 106 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-105, wherein the cell has a genetic modification in the CIITA gene.

[0143] Embodiment 107 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-106, wherein the cell has reduced expression of TRAC protein on the surface of the cell.

[0144] Embodiment 108 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-107, wherein the cell has reduced expression of TRBC protein on the surface of the cell.

[0145] Embodiment 109 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-108, wherein the cell has reduced expression of MHC class II molecules on the surface of the cell.

[0146] Embodiment 110 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-109, wherein the engineered cell is an immune cell.

[0147] Embodiment 111 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-110, wherein the cell is a primary cell.

[0148] Embodiment 112 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-111, wherein the engineered cell is a monocyte, macrophage, mast cell, dendritic cell, or granulocyte.

[0149] Embodiment 113 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-112, wherein the engineered cell is a lymphocyte.

[0150] Embodiment 114 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-113, wherein the cell is a T cell.

[0151] Embodiment 115 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-114, wherein the cell is a CD8+ T cell.

[0152] Embodiment 116 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-114, wherein the cell is a CD4+ T cell.

[0153] Embodiment 117 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-113, wherein the cell is a natural killer (NK) cell.

[0154] Embodiment 118 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-113, wherein the cell is a macrophage.

[0155] Embodiment 119 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-113, wherein the cell is a B cell.

[0156] Embodiment 120 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-113, wherein the cell is a plasma B cell.

[0157] Embodiment 121 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-113, wherein the cell is memory B cell.

[0158] Embodiment 122 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-109, wherein the cell is a stem cell.

[0159] Embodiment 123 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-109, wherein the cell is a progenitor cell.

[0160] Embodiment 124 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-109, wherein the cell is an HSC.

[0161] Embodiment 125 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-109, wherein the cell is an iPSC.

[0162] Embodiment 126 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-121, wherein the cell is an activated cell.

[0163] Embodiment 127 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-121, wherein the cell is a non-activated cell.

[0164] Embodiment 128 is the population of embodiment 18 or pharmaceutical composition of embodiment 19, wherein the population of cells is at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% HLA-A negative as measured by flow cytometry.

[0165] Embodiment 129 is the population of any one of embodiments 42-44 and 64-66 or pharmaceutical composition of embodiment 45 or 67, wherein the population of cells is at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 99.5% endogenous TCR protein negative as measured by flow cytometry.

[0166] Embodiment 130 is the population of any one of embodiments 84-86 or pharmaceutical composition of embodiment 87, wherein the population of cells is at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 99.5% HLA-DP, DQ, DR negative as measured by flow cytometry.

[0167] Embodiment 131 is the population of embodiment 18 or pharmaceutical composition of embodiment 19, wherein at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 99.5% of the population of cells comprises the genetic modification in the HLA-A gene as measured by next-generation sequencing (NGS).

[0168] Embodiment 132 is the population of any one of embodiments 42-44 or pharmaceutical composition of embodiment 45, wherein at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% of the population of cells comprises the genetic modification in the TRAC gene as measured by next-generation sequencing (NGS).

[0169] Embodiment 133 is the population of any one of embodiments 64-66 or pharmaceutical composition of embodiment 67, wherein at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% of the population of cells comprises the genetic modification in the TRBC gene as measured by next-generation sequencing (NGS).

[0170] Embodiment 134 is the population of any one of embodiments 84-86 or pharmaceutical composition of embodiment 87, wherein at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% of the population of cells comprises the genetic modification in the CIITA gene as measured by next-generation sequencing (NGS).

[0171] Embodiment 135 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-134, wherein the cell is an allogeneic cell.

[0172] Embodiment 136 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-135, wherein the cell is a primary cell.

[0173] Embodiment 137 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of the embodiments 1-136, wherein the cell is a CD4+ T cell.

[0174] Embodiment 138 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-136, wherein the cell is a CD8+ T cell.

[0175] Embodiment 139 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-136, wherein the cell is a memory T cell.

[0176] Embodiment 140 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-136, wherein the cell is a B cell.

[0177] Embodiment 141 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-136, wherein the cell is a plasma B cell.

[0178] Embodiment 142 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-136, wherein the cell is a memory B cell.

[0179] Embodiment 143 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-136, wherein the cell is a natural killer (NK) cell.

[0180] Embodiment 144 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-136, wherein the cell is a macrophage.

[0181] Embodiment 145 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-136, wherein the cell is stem cell.

[0182] Embodiment 146 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-136, wherein the cell is a pluripotent stem cell (PSC).

[0183] Embodiment 147 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-136, wherein the cell is a hematopoietic stem cell (HSC).

[0184] Embodiment 148 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-136, wherein the cell is an induced pluripotent stem cell (iPSC).

[0185] Embodiment 149 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-136, wherein the cell is a mesenchymal stem cell (MSC).

[0186] Embodiment 150 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-136, wherein the cell is a neural stem cell (NSC).

[0187] Embodiment 151 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-136, wherein the cell is a limbal stem cell (LSC).

[0188] Embodiment 152 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-136, wherein the cell is a progenitor cell, e.g. an endothelial progenitor cell or a neural progenitor cell.

[0189] Embodiment 153 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-136, wherein the cell is a tissue-specific primary cell.

[0190] Embodiment 154 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-136, wherein the cell is a chosen from: chondrocyte, myocyte, and keratinocyte.

[0191] Embodiment 155 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-136, wherein the cell is an activated cell.

[0192] Embodiment 156 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-136, wherein the cell is a non-activated cell.

[0193] Embodiment 157 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-156, wherein the cells are engineered with a genomic editing system.

[0194] Embodiment 158 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 7-157, wherein the genomic editing system comprises an RNA-guided DNA-binding agent or a nucleic acid encoding an RNA-guided DNA binding agent.

[0195] Embodiment 159 is the engineered cell, population of cells, pharmaceutical composition, or method of embodiment 158, wherein the RNA-guided DNA-binding agent or the RNA-guided DNA-binding agent encoded by the nucleic acid is *N. meningitidis* Cas9 (NmeCas9).

[0196] Embodiment 160 is the engineered cell, population of cells, pharmaceutical composition, or method of embodiment 159, wherein the NmeCas9 is Nme1Cas9, Nme2Cas9, or Nme3Cas9.

[0197] Embodiment 161 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 158-160, wherein the RNA-guided DNA-binding agent or the RNA-guided DNA-binding agent encoded by the nucleic acid has double-stranded endonuclease activity.

[0198] Embodiment 162 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 158-161, wherein the RNA-guided DNA-binding agent or the RNA-guided DNA-binding agent encoded by the nucleic acid has nickase activity.

[0199] Embodiment 163 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 158-161, wherein the RNA-guided DNA-binding agent or the RNA-guided DNA-binding agent encoded by the nucleic acid comprises a dCas9 DNA binding domain.

[0200] Embodiment 164 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 158-160, wherein the RNA-guided DNA-binding agent or nucleic acid encoding the RNA-guided DNA binding agent is a A to G base editor.

[0201] Embodiment 165 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 158-160, wherein the RNA-guided DNA-binding agent or nucleic acid encoding the RNA-guided DNA binding agent is a C to T base editor.

[0202] Embodiment 166 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 158-165, wherein the RNA-guided DNA-binding agent or the RNA-guided DNA-binding agent encoded by the nucleic acid comprises a deaminase region.

[0203] Embodiment 167 is the engineered cell of embodiment 158, wherein the RNA-guided DNA-binding agent or the RNA-guided DNA-binding agent encoded by the nucleic acid comprises an APOBEC3A deaminase (A3A) and an *N. meningitidis* Cas9 nickase.

[0204] Embodiment 168 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-167, wherein the guide RNA is provided to the cell in a vector.

[0205] Embodiment 169 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-168, wherein the RNA-guided DNA binding agent is provided to the cell in a vector, optionally in the same vector as the guide RNA.

[0206] Embodiment 170 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-169, wherein the exogenous nucleic acid is provided to the cell in a vector.

[0207] Embodiment 171 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 168-170, wherein the vector is a viral vector.

[0208] Embodiment 172 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 168-170, wherein the vector is a non-viral vector.

[0209] Embodiment 173 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 168-171, wherein the vector is a lentiviral vector.

[0210] Embodiment 174 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 168-171, wherein the vector is a retroviral vector.

[0211] Embodiment 175 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 168-171, wherein the vector is an AAV.

[0212] Embodiment 176 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-175, wherein the guide RNA is provided to the cell in a lipid nucleic acid assembly composition, optionally in the same lipid nucleic acid assembly composition as an RNA-guided DNA binding agent.

[0213] Embodiment 177 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-176, wherein the exogenous nucleic acid is provided to the cell in a lipid nucleic acid assembly composition.

[0214] Embodiment 178 is the engineered cell, population of cells, pharmaceutical composition, or method of embodiment 176 or 177, wherein the lipid nucleic acid assembly composition is a lipid nanoparticle (LNP).

[0215] Embodiment 179 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-178, wherein the exogenous nucleic acid is integrated into the genome of the cell.

[0216] Embodiment 180 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-179, wherein the exogenous nucleic acid is integrated into the genome of the cell by homologous recombination (HR).

[0217] Embodiment 181 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-180, wherein the exogenous nucleic acid is integrated into a safe harbor locus in the genome of the cell.

[0218] Embodiment 182 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-181, wherein the guide RNA is a single guide RNA.

[0219] Embodiment 183 is the engineered cell, population of cells, pharmaceutical composition, or method of embodiment 182, wherein the single guide RNA comprises the nucleotide sequence of SEQ ID NO: 900 3' to the guide sequence.

[0220] Embodiment 184 is the engineered cell, population of cells, pharmaceutical composition, or method of embodiment 182 or 183, wherein the single guide RNA comprises a guide region and a conserved region, wherein the conserved region comprising one or more of: (a) a shortened repeat/anti-repeat region, wherein the shortened repeat/anti-repeat region lacks 2-24 nucleotides, wherein (i) one or more of nucleotides 37-48 and 53-64 is deleted and optionally one or more of nucleotides 37-64 is substituted relative to SEQ ID NO: 900; and (ii) nucleotide 36 is linked to nucleotide 65 by at least 2 nucleotides; or (b) a shortened hairpin 1 region, wherein the shortened hairpin 1 lacks 2-10, optionally 2-8 nucleotides, wherein (i) one or more of nucleotides 82-86 and 91-95 is deleted and optionally one or more of positions 82-96 is substituted relative to SEQ ID NO: 900; and (ii) nucleotide 81 is linked to nucleotide 96 by at least 4 nucleotides; or (c) a shortened hairpin 2 region, wherein the shortened hairpin 2 lacks 2-18, optionally 2-16 nucleotides, wherein (i) one or more of nucleotides 113-121 and 126-134 is deleted and optionally one or more of nucleotides 113-134 is substituted relative to SEQ ID NO: 900; and (ii) nucleotide 112 is linked to nucleotide 135 by at least 4 nucleotides; wherein one or both nucleotides 144-145 are optionally deleted relative to SEQ ID NO: 900; wherein at least 10 nucleotides are modified nucleotides.

[0221] Embodiment 185 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-184, wherein the guide RNA comprises at least one modification.

[0222] Embodiment 186 is the engineered cell, population of cells, pharmaceutical composition, or method of embodiment 185, wherein the modification comprises a modified nucleotide selected from a 2'—O-methyl (2'—OMe) modified nucleotide, 2'—O-(2-methoxyethyl) (2'—O-moe) modified nucleotide, a 2'-fluoro (2'—F) modified nucleotide, a phosphorothioate (PS) linkage between nucleotides, or an inverted abasic modified nucleotide.

[0223] Embodiment 187 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-186, wherein the gRNA comprises a 5' end modification, a modification in the repeat/anti-repeat region, a modification in the hairpin 1 region, a modification in the hairpin 2 region, or a 3' end modification.

[0224] Embodiment 188 is the engineered cell, population of cells, pharmaceutical composition, or method of embodiment 187, wherein the 5' end modification comprises at least one PS linkage, and wherein one or more of: i. there is one PS linkage, and the linkage is between the first and second nucleotides; ii. there are two PS linkages between the first three nucleotides; iii. there are PS linkages between any one or more of the first four nucleotides; and iv. there are PS linkages between any one or more of the first five nucleotides.

[0225] Embodiment 189 is the engineered cell, population of cells, pharmaceutical composition, or method of embodiment 187 or 188, wherein the 5' end modification further comprises at least one 2'—OMe, 2'—O-moe, inverted abasic, or 2'—F modified nucleotide.

[0226] Embodiment 190 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 187-189, wherein the 5' end modification comprises: i. a modification of one or more of the first 1-4 nucleotides, wherein the modification is a PS linkage, inverted abasic nucleotide, 2'—OMe, 2'—O-moe, or 2'—F; ii. a modification to the first nucleotide with 2'—OMe, 2'—O-moe, or 2'—F, and an optional one or two PS linkages to the next nucleotide or the first nucleotide of the 3' tail; iii. a modification to the first or second nucleotide with 2'—OMe, 2'—O-moe, or 2'—F, and optionally one or more PS linkages; iv. a modification to the first, second, or third nucleotides with 2'—OMe, 2'—O-moe, or 2'—F, and optionally one or more PS linkages; or v. a modification to the first, second, third, or forth nucleotides with 2'—OMe, 2'—O-moe, or 2'—F, and optionally one or more PS linkages.

[0227] Embodiment 191 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 187-180, wherein the 3' end modification comprises at least one PS linkage, and wherein one or more of: i. there is one PS linkage, and the linkage is between the last and second to last nucleotides; ii. there are two PS linkages between the last three nucleotides; and iii. there are PS linkages between any one or more of the last four nucleotides.

[0228] Embodiment 192 is the engineered cell, population of cells, pharmaceutical composition, or method of 191, wherein the 3' end modification further comprises at least one 2'-OMe, 2'—O-moe, inverted abasic, or 2'—F modified nucleotide.

[0229] Embodiment 193 is the engineered cell, population of cells, pharmaceutical composition, or method of 192, wherein the 3' end modification comprises: i. a modification of one or more of the last 1-4 nucleotides, wherein the modification is a PS linkage, inverted abasic nucleotide, 2'—OMe, 2'—O-moe, or 2'—F; ii. a modification to the last nucleotide with 2'—OMe, 2'-O-moe, or 2'—F, and an optional one or two PS linkages to the next nucleotide or the first nucleotide of the 3' tail; iii. a modification to the last or second to last nucleotide with 2'—OMe, 2'—O-moe, or 2'—F, and optionally one or more PS linkages; iv. a modification to the last, second to last, or third to last nucleotides with 2'—OMe, 2'—O-moe, or 2'—F, and optionally one or more PS linkages; or v. a modification to the last, second to last, third to last, or fourth to last nucleotides with 2'—OMe, 2'—O-moe, or 2'—F, and optionally one or more PS linkages.

[0230] Embodiment 194 is the engineered cell, population of cells, pharmaceutical composition, or method of 193, further comprising a 3' tail comprising a 2'—O-Me modified nucleotide.

[0231] Embodiment 195 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-194, wherein the guide RNA comprises a 5' end modification or a 3' end modification.

[0232] Embodiment 196 is the method or composition of any one of embodiments 1-195, wherein the guide RNA comprises:

[0233] the guide sequence, wherein the guide sequence comprises: 2'—O-Me modified nucleotides at the first four nucleotides 1-4; PS linkages between nucleotides 1-2, 2-3, and 3-4; and 2'—O-Me modified nucleotides at nucleotides 5, 8, 9, 11, 13, 18, and 22 of the guide sequence; a shortened repeat/anti-repeat region, wherein nucleotides 38-48 and 53-63 are deleted relative to SEQ ID NO: 900, comprising: 2'—O-Me modified nucleotides at nucleotides 25, 29, 30, 31, 32, 37, 49-52, 64, 65, 69, 70, and 73; a shortened hairpin 1 region, wherein nucleotides 86 and 91 are deleted relative to SEQ ID NO: 900, comprising: 2'—O-Me modified nucleotides at nucleotides 80, 81, 83, 84, 85, 87-90, 92-94, and 99; 2'—O-Me modified nucleotide at nucleotide 101 between the shortened hairpin 1 region and the shortened hairpin 2 region; a shortened hairpin 2 region, wherein nucleotides 112-120 and 127-135 are deleted relative to SEQ ID NO: 900, comprising: 2'—O-Me modified nucleotides at nucleotides 104, 110, 111, 122-125, 142, and 143, PS linkages between nucleotides 141-142 and 142-143, wherein one or both nucleotides 144-145 are optionally deleted relative to SEQ ID NO: 900.

[0234] Embodiment 197 is the method or composition of any one of embodiments 1-196, wherein the guide RNA comprises the modified nucleotides of any one of SEQ ID NOs: 904-909, 911, 995-997, and 1081-1089.

[0235] Embodiment 198 is the method or composition of any one of embodiments 1-197, wherein the guide RNA comprises the modified nucleotides of SEQ ID NO: 995.

[0236] Embodiment 199 is the method or composition of any one of embodiments 1-198, wherein the guide RNA comprises the modified nucleotides of SEQ ID NO: 1083.

[0237] Embodiment 200 is the method or composition of any one of embodiments 1-199, wherein the guide RNA is modified according to the pattern of any one of SEQ ID NOs: 904-909, 911, and 995-997, wherein each N in the pattern is any natural or non-natural nucleotide wherein the N's are collectively any one of the guide sequences of Tables 1-5.

[0238] Embodiment 201 is the method or composition of any one of embodiments 1-200, wherein the guide RNA is modified according to the pattern of SEQ ID NO: 995, wherein each N in the pattern is any natural or non-natural nucleotide wherein the N's are collectively any one of the guide sequences of Tables 1-5.

[0239] Embodiment 202 is a method of administering the engineered cell, population of cells, pharmaceutical composition of any one of embodiments 1-10, 18-28, 42-55, 64-75, 84-92, and 97-195 to a subject in need thereof.

[0240] Embodiment 203 is a method of administering the engineered cell, population of cells, or pharmaceutical composition of any one of embodiments 1-10, 18-28, 42-55, 64-75, 84-92, and 97-195 to a subject as an adoptive cell transfer (ACT) therapy.

[0241] Embodiment 204 is a method of treating a disease or disorder comprising administering the engineered cell, population of cells, or pharmaceutical composition of any one of embodiments 1-10, 18-28, 42-55, 64-75, 84-92, and 97-195 to a subject in need thereof.

[0242] Embodiment 205 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-10, 18-28, 42-55, 64-75, 84-92, and 97-195, for use in administering to a subject as an adoptive cell transfer (ACT) therapy.

[0243] Embodiment 206 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-10, 18-28, 42-55, 64-75, 84-92, and 97-195, for use in treating a subject with cancer.

[0244] Embodiment 207 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-10, 18-28, 42-55, 64-75, 84-92, and 97-195, for use in treating a subject with an infectious disease.

[0245] Embodiment 208 is the engineered cell, population of cells, pharmaceutical composition, or method of any one of embodiments 1-10, 18-28, 42-55, 64-75, 84-92, and 97-195, for use in treating a subject with an autoimmune disease.

I. Definitions

[0246] Unless stated otherwise, the following terms and phrases as used herein are intended to have the following meanings:

[0247] The term "or combinations thereof" as used herein refers to all permutations and combinations of the listed terms preceding the term. For example, "A, B, C, or combinations thereof" is intended to include at least one of: A, B, C, AB, AC, BC, or ABC, and if order is important in a particular context, also BA, CA, CB, ACB, CBA, BCA, BAC, or CAB. Continuing with this example, expressly included are combinations that contain repeats of one or more item or term, such as BB, AAA, AAB, BBC, CBBA, CABA, and so forth. The skilled artisan will understand that typically there is no limit on the number of items or terms in any combination, unless otherwise apparent from the context.

[0248] As used herein, the term "kit" refers to a packaged set of related components, such as one or more polynucleotides or compositions and one or more related materials such as delivery devices (e.g., syringes), solvents, solutions, buffers, instructions, or desiccants.

[0249] An "allogeneic" cell, as used herein, refers to a cell originating from a donor subject of the same species as a recipient subject, wherein the donor subject and recipient subject have genetic dissimilarity, e.g., genes at one or more loci that are not identical. Thus, e.g., a cell is allogeneic with respect to the subject to be administered the cell. As used herein, a cell that is removed or isolated from a donor, that will not be re-introduced into the original donor, is considered an allogeneic cell.

[0250] An "autologous" cell, as used herein, refers to a cell derived from the same subject to whom the material will later be re-introduced. Thus, e.g., a cell is considered autologous if it is removed from a subject and it will then be re-introduced into the same subject.

[0251] "CIITA" or "CIITA" or "C2TA," as used herein, refers to the nucleic acid sequence or protein sequence of "class II major histocompatibility complex transactivator;" the human gene has accession number NC_000016.10 (range 10866208 . . . 10941562), reference GRCh38.p13. The CIITA protein in the nucleus acts as a positive regulator of MHC class II gene transcription and is required for MHC class II protein expression.

[0252] As used herein, "MHC" or "MHC molecule(s)" or "MHC protein" or "MHC complex(es)," refers to a major histocompatibility complex molecule (or plural), and includes e.g., MHC class I and MHC class II molecules. In humans, MHC molecules are referred to as "human leukocyte antigen" complexes or "HLA molecules" or "HLA protein." The use of terms "MHC" and "HLA" are not meant to be limiting; as used herein, the term "MHC" may be used to refer to human MHC molecules, i.e., HLA molecules. Therefore, the terms "MHC" and "HLA" are used interchangeably herein.

[0253] The term "HLA-A," as used herein in the context of HLA-A protein, refers to the MHC class I protein molecule, which is a heterodimer consisting of a heavy chain (encoded by the HLA-A gene) and a light chain (i.e., beta-2 microglobulin). The term "HLA-A" or "HLA-A gene," as used herein in the context of nucleic acids refers to the gene encoding the heavy chain of the HLA-A protein molecule. The HLA-A gene is also referred to as "HLA class I histocompatibility, A alpha chain;" the human gene has accession number NC_000006.12 (29942532 . . . 29945870). The HLA-A gene is known to have thousands of different versions (also referred to as "alleles") across the population (and an individual may receive two different alleles of the HLA-A gene). A public database for HLA-A alleles, including sequence information, may be accessed at IPD-IMGT/HLA: https://www.ebi.ac.uk/ipd/imgt/hla/. All alleles of HLA-A are encompassed by the terms "HLA-A" and "HLA-A gene."

[0254] "HLA-B" as used herein in the context of nucleic acids refers to the gene encoding the heavy chain of the HLA-B protein molecule. The HLA-B is also referred to as "HLA class I histocompatibility, B alpha chain;" the human gene has accession number NC_000006.12 (31353875 . . . 31357179).

[0255] "HLA-C" as used herein in the context of nucleic acids refers to the gene encoding the heavy chain of the HLA-C protein molecule. The HLA-C is also referred to as "HLA class I histocompatibility, C alpha chain;" the human gene has accession number NC_000006.12 (31268749 . . . 31272092).

[0256] The term "TRAC," as used herein in the context of TRAC protein, refers to the T-cell receptor a-chain. "TRAC" as used herein in the context of nucleic acids refers to the gene encoding the T-cell receptor a-chain. A human wild-type TRAC sequence is available at NCBI Gene ID: 28755; Ensembl: ENSG00000277734. T Cell Receptor Alpha Constant, TCRA, IMD7, TRCA and TRA are gene synonyms for TRAC.

[0257] The term "TRBC" (or "TRBC1/2") is used to refer to the nucleic acid sequence or amino acid sequence of the "T-cell receptor 0-chain", e.g., TRBC1 and TRBC2. The terms "TRBC1" and "TRBC2," as used herein in the context of TRBC proteins, refer to two homologous proteins that comprise the T-cell receptor 0-chain. "TRBC1" and "TRBC2" as used herein in the context of nucleic acids refers to the genes encoding the T-cell receptor 0-chain. A human wild-type TRBC1 sequence is available at NCBI Gene ID: 28639; Ensembl: ENSG00000211751. T Cell Receptor Beta Constant, V_segment Translation Product, BV05S1J2.2, TCRBC1, and TCRB are gene synonyms for

TRBC1. A human wild-type TRBC2 sequence is available at NCBI Gene ID: 28638; Ensembl: ENSG00000211772. T Cell Receptor Beta Constant, V_segment Translation Product, and TCRBC2 are gene synonyms for TRBC2.

[0258] As used herein, the term "AAVS1" refers to the genomic location at chr19:50900000-58617616 according to hg38.

[0259] As used herein, the term "within the genomic coordinates" includes the boundaries of the genomic coordinate range given. For example, if chr6:29942854-chr6:29942913 is given, the coordinates chr6:29942854-chr6:29942913 are encompassed. Throughout this application, the referenced genomic coordinates are based on genomic annotations in the GRCh38 (also referred to as hg38) assembly of the human genome from the Genome Reference Consortium, available at the National Center for Biotechnology Information website. Tools and methods for converting genomic coordinates between one assembly and another are known in the art and can be used to convert the genomic coordinates provided herein to the corresponding coordinates in another assembly of the human genome, including conversion to an earlier assembly generated by the same institution or using the same algorithm (e.g., from GRCh38 to GRCh37), and conversion of an assembly generated by a different institution or algorithm (e.g., from GRCh38 to NCBI33, generated by the International Human Genome Sequencing Consortium). Available methods and tools known in the art include, but are not limited to, NCBI Genome Remapping Service, available at the National Center for Biotechnology Information website, UCSC LiftOver, available at the UCSC Genome Brower website, and Assembly Converter, available at the Ensembl.org website.

[0260] As used herein, the term "homozygous" refers to having two identical alleles of a particular gene.

[0261] As used herein, the term "subject" is intended to include living organisms in which an immune response can be elicited, including e.g., mammals, primates, humans.

[0262] "Polynucleotide" and "nucleic acid" are used herein to refer to a multimeric compound comprising nucleosides or nucleoside analogs which have nitrogenous heterocyclic bases or base analogs linked together along a backbone, including conventional RNA, DNA, mixed RNA-DNA, and polymers that are analogs thereof. A nucleic acid "backbone" can be made up of a variety of linkages, including one or more of sugar-phosphodiester linkages, peptide-nucleic acid bonds ("peptide nucleic acids" or PNA; PCT No. WO 95/32305), phosphorothioate linkages, methylphosphonate linkages, or combinations thereof. Sugar moieties of a nucleic acid can be ribose, deoxyribose, or similar compounds with substitutions, e.g., 2' methoxy or 2' halide substitutions. Nitrogenous bases can be conventional bases (A, G, C, T, U), analogs thereof (e.g., modified uridines such as 5-methoxyuridine, pseudouridine, or N1-methylpseudouridine, or others); inosine; derivatives of purines or pyrimidines (e.g., N^4-methyl deoxyguanosine, deaza- or aza-purines, deaza- or aza-pyrimidines, pyrimidine bases with substituent groups at the 5 or 6 position (e.g., 5-methylcytosine), purine bases with a substituent at the 2, 6, or 8 positions, 2-amino-6-methylaminopurine, O^6-methylguanine, 4-thio-pyrimidines, 4-amino-pyrimidines, 4-dimethylhydrazine-pyrimidines, and O^4-alkyl-pyrimidines; U.S. Pat. No. 5,378,825 and PCT No. WO 93/13121). For general discussion see *The Biochemistry of the Nucleic Acids* 5-36, Adams et al., ed., 11^{th} ed., 1992). Nucleic acids can include one or more "abasic" residues where the backbone includes no nitrogenous base for position(s) of the polymer (U.S. Pat. No. 5,585,481). A nucleic acid can comprise only conventional RNA or DNA sugars, bases and linkages, or can include both conventional components and substitutions (e.g., conventional bases with 2' methoxy linkages, or polymers containing both conventional bases and one or more base analogs). Nucleic acid includes "locked nucleic acid" (LNA), an analogue containing one or more LNA nucleotide monomers with a bicyclic furanose unit locked in an RNA mimicking sugar conformation, which enhance hybridization affinity toward complementary RNA and DNA sequences (Vester and Wengel, 2004, Biochemistry 43(42): 13233-41). RNA and DNA have different sugar moieties and can differ by the presence of uracil or analogs thereof in RNA and thymine or analogs thereof in DNA. [00263] "Guide RNA", "gRNA", and simply "guide" are used herein interchangeably to refer to, for example, the guide that directs an RNA-guided DNA binding agent to a target DNA and can be a single guide RNA, or the combination of a crRNA and a trRNA (also known as tracrRNA). Exemplary gRNAs include Class II Cas nuclease guide RNAs, in modified or unmodified forms. The crRNA and trRNA may be associated as a single RNA molecule (single guide RNA, sgRNA) or in two separate RNA strands (dual guide RNA, dgRNA). "Guide RNA" or "gRNA" refers to each type. The trRNA may be a naturally occurring sequence, or a trRNA sequence with modifications or variations compared to naturally-occurring sequences.

[0263] As used herein, a "guide sequence" refers to a sequence within a guide RNA that is complementary to a target sequence and functions to direct a guide RNA to a target sequence for binding or modification (e.g., cleavage) by an RNA-guided DNA binding agent. A "guide sequence" may also be referred to as a "targeting sequence," or a "spacer sequence." A guide sequence can be 19, 20, 21, 22, 23, or 24, or 25 nucleotides in length, e.g., in the case of *Neisseria meningitides*. In some embodiments, the Nme Cas9 guide sequence comprises at least 22, 23, or 24 contiguous nucleotides of a sequence selected from SEQ ID NOs: 2-80, 101-120, 201, 265, 301, 302, 304-576, or 601-774. In some embodiments, the target sequence is in a gene or on a chromosome, for example, and is complementary to the guide sequence. In some embodiments, the degree of complementarity or identity between a guide sequence and its corresponding target sequence is at least 80%, 85%, 90%, or 95%. For example, in some embodiments, the guide sequence comprises a sequence 24 contiguous nucleotides of a sequence selected from SEQ ID NO: 2-80, 101-120, 201, 265, 301, 302, 304-576, or 601-774. In some embodiments, the guide sequence and the target region may be 100% complementary or identical. In other embodiments, the guide sequence and the target region may contain at least one mismatch, i.e., one nucleotide that is not identical or not complementary, depending on the reference sequence. For example, the guide sequence and the target sequence may contain 1-2, preferably no more than 1 mismatch, where the total length of the target sequence is 19, 20, 21, 22, 23, or 24, nucleotides, or more. In some embodiments, the guide sequence and the target region may contain 1-2 mismatches where the guide sequence comprises at least 24 nucleotides, or more. In some embodiments, the guide sequence and the target region may contain 1-2 mismatches where the guide sequence comprises 24 nucleotides. That is, the guide sequence and the target region may form a duplex region having base pairs, or more. In certain embodiments, the duplex region may include 1-2 mismatches such that guide strand and target sequence are not fully complementary. Mismatch positions are known in the art as provided in, for example, PAM distal mismatches tend to be better tolerated than PAM proximal matches. Mismatch tolerances at other positions are known in the art (see, e.g., Edraki et al., 2019. Mol. Cell, 73:1-13).

[0264] Target sequences for RNA-guided DNA binding agents include both the positive and negative strands of genomic DNA (i.e., the sequence given and the sequence's reverse compliment), as a nucleic acid substrate for an RNA-guided DNA binding agent is a double stranded nucleic acid. Accordingly, where a guide sequence is said to be "complementary to a target sequence", it is to be understood that the guide sequence may direct a guide RNA to bind to the reverse complement of a target sequence. Thus, in some embodiments, where the guide sequence binds the reverse complement of a target sequence, the guide sequence is identical to certain nucleotides of the target sequence (e.g., the target sequence not including the PAM) except for the substitution of U for T in the guide sequence.

[0265] As used herein, an "RNA-guided DNA binding agent" means a polypeptide or complex of polypeptides having RNA and DNA binding activity, or a DNA-binding subunit of such a complex, wherein the DNA binding activity is sequence-specific and depends on the presence of a PAM and the sequence of the guide RNA. Exemplary RNA-guided DNA binding agents include Cas cleavases/nickases and inactivated forms thereof ("dCas DNA binding agents"). "Cas nuclease", also called "Cas protein" as used herein, encompasses Cas cleavases, Cas nickases, and dCas DNA binding agents. Cas cleavases/nickases and dCas DNA binding agents include a Csm or Cmr complex of a type III CRISPR system, the Cas10, Csm1, or Cmr2 subunit thereof, a Cascade complex of a type I CRISPR system, the Cas3 subunit thereof, and Class 2 Cas nucleases.

[0266] As used herein, a "Class 2 Cas nuclease" is a single-chain polypeptide with RNA-guided DNA binding activity. Class 2 Cas nucleases include Class 2 Cas cleavases/nickases (e.g., H840A, D10A, or N863A variants of Spy Cas9 and D16A and H588A of Nme Cas9, e.g., Nme2 Cas9), which further have RNA-guided DNA cleavases or nickase activity, and Class 2 dCas DNA binding agents, in which cleavase/nickase activity is inactivated. Class 2 Cas nucleases include, for example, Cas9, Cpf1, C2c1, C2c2, C2c3, HF Cas9 (e.g., N497A, R661A, Q695A, Q926A variants), HypaCas9 (e.g., N692A, M694A, Q695A, H698A variants), eSPCas9(1.0) (e.g., K810A, K1003A, R1060A variants), and eSPCas9(1.1) (e.g., K848A, K1003A, R1060A variants) proteins and modifications thereof. Cpf1 protein, Zetsche et al., Cell, 163: 1-13 (2015), is homologous to Cas9, and contains a RuvC-like nuclease domain. Cpf1 sequences of Zetsche are incorporated by reference in their entirety. See, e.g., Zetsche, Tables S1 and S3. See, e.g., Makarova et al., *Nat Rev Microbiol,* 13(11): 722-36 (2015); Shmakov et al., *Molecular Cell,* 60:385-397 (2015).

[0267] Several Cas9 orthologs have been obtained from *N. meningitidis* (Esvelt et al., NAT. METHODS, vol. 10, 2013, 1116-1121; Hou et al., PNAS, vol. 110, 2013, pages 15644-15649) (Nme1Cas9, Nme2Cas9, and Nme3Cas9). The Nme2Cas9 ortholog functions efficiently in mammalian cells, recognizes an N4CC PAM, and can be used for in vivo editing with cognate gRNAs (Ran et al., NATURE, vol. 520, 2015, pages 186-191; Kim et al., NAT. COMMUN., vol. 8, 2017, pages 14500). Nme2Cas9 can be specific and selective, e.g. capable of low off-target editing (Lee et al., MOL. THER., vol. 24, 2016, pages 645-654; Kim et al., 2017). See also e.g., WO/2020081568 (e.g., pages 28 and 42), describing an Nme2Cas9 D16A nickase, the contents of which are hereby incorporated by reference in its entirety. Throughout, "NmeCas9" or "Nme Cas9" is generic and encompasses any type of NmeCas9, including, Nme1Cas9, Nme2Cas9, and Nme3Cas9.

[0268] Exemplary nucleotide and polypeptide sequences of Cas9 molecules are provided in Table 7. Methods for identifying alternate nucleotide sequences encoding Cas9 polypeptide sequences, including alternate naturally occurring variants, are known in the art. Sequences with at least 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, or 99% identity to any of the Cas9 nucleic acid sequences, or nucleic acid sequences encoding the amino acid sequences provided herein are also contemplated.

[0269] As used herein, the term "editor" refers to an agent comprising a polypeptide that is capable of making a modification within a DNA sequence. In some embodiments, the editor is a cleavase, such as a Cas9 cleavase. In some embodiments, the editor is capable of deaminating a base within a DNA molecule, and it may be called a base editor. In some embodiments, the editor is capable of deaminating a cytosine (C) in DNA. In some embodiments, the editor is a fusion protein comprising an RNA-guided nickase fused to a cytidine deaminase. In some embodiments, the editor is a fusion protein comprising an RNA-guided nickase fused to an APOBEC3A deaminase (A3A). In some embodiments, the editor comprises a Cas9 nickase fused to an APOBEC3A deaminase (A3A). In some embodiments, the editor is a fusion protein comprising an RNA-guided nickase fused to a cytidine deaminase and a UGI. In some embodiments, the editor lacks a UGI.

[0270] As used herein, a "cytidine deaminase" means a polypeptide or complex of polypeptides that is capable of cytidine deaminase activity, that is catalyzing the hydrolytic deamination of cytidine or deoxycytidine, typically resulting in uridine or deoxyuridine. Cytidine deaminases encompass enzymes in the cytidine deaminase superfamily, and in particular, enzymes of the APOBEC family (APOBEC1, APOBEC2, APOBEC4, and APOBEC3 subgroups of enzymes), activation-induced cytidine deaminase (AID or AICDA) and CMP deaminases (see, e.g., Conticello et al., Mol. Biol. Evol. 22:367-77, 2005; Conticello, Genome Biol. 9:229, 2008; Muramatsu et al., J. Biol. Chem. 274: 18470-6, 1999); Carrington et al., Cells 9:1690 (2020)). In some embodiments, variants of any known cytidine deaminase or APOBEC protein are encompassed. Variants include proteins having a sequence that differs from wild-type protein by one or several mutations (i.e., substitutions, deletions, insertions), such as one or several single point substitutions. For instance, a shortened sequence could be used, e.g., by deleting N-terminal, C-terminal, or internal amino acids, preferably one to four amino acids at the C-terminus of the sequence. As used herein, the term "variant" refers to allelic variants, splicing variants, and natural or artificial mutants, which are homologous to a reference sequence. The variant is "functional" in that it shows a catalytic activity of DNA editing.

[0271] As used herein, the term "APOBEC3A" refers to a cytidine deaminase such as the protein expressed by the human A3A gene. The APOBEC3A may have catalytic DNA editing activity. An amino acid sequence of APOBEC3A has been described (UniPROT accession ID: p31941) and is included herein as SEQ ID NO: 827. In some embodiments, the APOBEC3A protein is a human APOBEC3A protein or a wild-type protein. Variants include proteins having a sequence that differs from wild-type APOBEC3A protein by one or several mutations (i.e., substitutions, deletions, insertions), such as one or several single point substitutions. For instance, a shortened APOBEC3A sequence could be used, e.g., by deleting N-terminal, C-terminal, or internal amino acids, preferably one to four amino acids at the C-terminus of the sequence. As used herein, the term "variant" refers to allelic variants, splicing variants, and natural or artificial mutants, which are homologous to an APOBEC3A reference sequence. The variant is "functional" in that it shows a catalytic activity of DNA editing. In some embodiments, an APOBEC3A (such as a human APOBEC3A) has a wild-type amino acid position 57 (as numbered in the wild-type sequence). In some embodiments, an APOBEC3A (such as a human APOBEC3A) has an asparagine at amino acid position 57 (as numbered in the wild-type sequence).

[0272] As used herein, a "nickase" is an enzyme that creates a single-strand break (also known as a "nick") in double strand DNA, i.e., cuts one strand but not the other of the DNA double helix. As used herein, an "RNA-guided DNA nickase" means a polypeptide or complex of polypeptides having DNA nickase activity, wherein the DNA nickase activity is sequence-specific and depends on the sequence of the RNA. Exemplary RNA-guided DNA nickases include Cas nickases. Class 2 Cas nickases include, polypeptides in which either the HNH or RuvC catalytic domain is inactivated, for example, Cas9 (e.g., H840A, D10A, or N863A variants of SpyCas9 or D16A variant of NmeCas9). Exemplary amino acid substitutions in the HNH or HNH-like nuclease domain or RuvC or RuvC-like domains for *N. meningitidis* include Nme2Cas9 D16A (HNH nickase) and Nme2Cas9 H588A (RuvC nickase). Cpf1, C2c1, C2c2, C2c3, HF Cas9 (e.g., N497A, R661A, Q695A, Q926A variants), HypaCas9 (e.g., N692A, M694A, Q695A, H698A variants), eSPCas9(1.0) (e.g., K810A, K1003A, R1060A variants), and eSPCas9(1.1) (e.g., K848A, K1003A, R1060A variants) proteins and modifications thereof. Cpf1 protein, Zetsche et al., Cell, 163: 1-13 (2015), is homologous to Cas9, and contains a RuvC-like protein domain. Cpf1 sequences of Zetsche are incorporated by reference in their entirety. See, e.g., Zetsche, Tables S1 and S3. "Cas9" encompasses *S. pyogenes* (Spy) Cas9, the variants of Cas9 listed herein, and equivalents thereof. See, e.g., Makarova et al., Nat Rev Microbiol, 13(11): 722-36 (2015); Shmakov et al., Molecular Cell, 60:385-397 (2015).

[0273] As used herein, the term "fusion protein" refers to a hybrid polypeptide which comprises protein domains from at least two different proteins. One protein may be located at the amino-terminal (N-terminal) portion of the fusion protein or at the carboxy-terminal (C-terminal) protein thus forming an "amino-terminal fusion protein" or a "carboxy-terminal fusion protein," respectively. Any of the proteins provided herein may be produced by any method known in the art. For example, the proteins provided herein may be produced via recombinant protein expression and purification, which is especially suited for fusion proteins comprising a peptide linker. Methods for recombinant protein expression and purification are well known, and include those described by Green and Sambrook, Molecular Cloning: A Laboratory Manual (4th ed., Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y. (2012)), the entire contents of which are incorporated herein by reference.

[0274] The term "linker," as used herein, refers to a chemical group or a molecule linking two adjacent molecules or moieties. Typically, the linker is positioned between, or flanked by, two groups, molecules, or other moieties and connected to each one via a covalent bond. In some embodiments, the linker is an amino acid or a plurality of amino acids (e.g., a peptide or protein) such as a 16-amino acid residue "XTEN" linker, or a variant thereof (See, e.g., the Examples; and Schellenberger et al. A recombinant polypeptide extends the in vivo half-life of peptides and proteins in a tunable manner. Nat. Biotechnol. 27, 1186-1190 (2009)). In some embodiments, the XTEN linker comprises the sequence SGSETPGTSESATPES (SEQ ID NO: 930), SGSETPGTSESA (SEQ ID NO: 931), or SGSETPGTSESATPEGGSGGS (SEQ ID NO: 932). In some embodiments, the linker is a peptide linker comprising one or more sequences selected from SEQ ID NOs: 933-994.

[0275] As used herein, the term "uracil glycosylase inhibitor" or "UGI" refers to a protein that is capable of inhibiting a uracil-DNA glycosylase (UDG) base-excision repair enzyme.

[0276] As used herein, "open reading frame" or "ORF" of a gene refers to a sequence consisting of a series of codons that specify the amino acid sequence of the protein that the gene codes for. The ORF begins with a start codon (e.g., ATG in DNA or AUG in RNA) and ends with a stop codon, e.g., TAA, TAG or TGA in DNA or UAA, UAG, or UGA in RNA.

[0277] As used herein, "ribonucleoprotein" (RNP) or "RNP complex" refers to a guide RNA together with an RNA-guided DNA binding agent, such as a Cas nuclease, e.g., a Cas cleavase, Cas nickase, or dCas DNA binding agent (e.g., Cas9). In some embodiments, the guide RNA guides the RNA-guided DNA binding agent such as Cas9 to a target sequence, and the guide RNA hybridizes with the target sequence and the agent binds to the target sequence; in cases where the agent is a cleavase or nickase, binding can be followed by cleaving or nicking.

[0278] As used herein, a first sequence is considered to "comprise a sequence with at least X % identity to" a second sequence if an alignment of the first sequence to the second sequence shows that X % or more of the positions of the second sequence in its entirety are matched by the first sequence. For example, the sequence AAGA comprises a sequence with 100% identity to the sequence AAG because an alignment would give 100% identity in that there are matches to all three positions of the second sequence. The differences between RNA and DNA (generally the exchange of uridine for thymidine or vice versa) and the presence of nucleoside analogs such as modified uridines do not contribute to differences in identity or complementarity among polynucleotides as long as the relevant nucleotides (such as thymidine, uridine, or modified uridine) have the same complement (e.g., adenosine for all of thymidine, uridine, or modified uridine; another example is cytosine and 5-methylcytosine, both of which have guanosine or modified guanosine as a complement). Thus, for example, the sequence 5'-AXG where X is any modified uridine, such as pseudouridine, N1-methyl pseudouridine, or 5-methoxyuridine, is considered 100% identical to AUG in that both are perfectly complementary to the same sequence (5'-CAU). Exemplary alignment algorithms are the Smith-Waterman and Needleman-Wunsch algorithms, which are well-known in the art. One skilled in the art will understand what choice of algorithm and parameter settings are appropriate for a given pair of sequences to be aligned; for sequences of generally similar length and expected identity >50% for amino acids or >75% for nucleotides, the Needleman-Wunsch algorithm with default settings of the Needleman-Wunsch algorithm interface provided by the EBI at the www.ebi.ac.uk web server is generally appropriate.

[0279] "Messenger RNA" or "mRNA" is used herein to refer to a polynucleotide and comprises an open reading frame that can be translated into a polypeptide (i.e., can serve as a substrate for translation by a ribosome and amino-acylated tRNAs). mRNA can comprise one or more modifications as provided below.

[0280] As used herein, a "genetic modification" is a change at the DNA level, e.g. induced by a CRISPR/Cas9 gRNA and Cas9 system. A genetic modification may comprise an insertion, deletion, or substitution (i.e., base sequence substitution, i.e., mutation), typically within a defined sequence or genomic locus. A genetic modification changes the nucleic acid sequence of the DNA. A genetic modification may be at a single nucleotide position. A genetic modification may be at multiple nucleotides, e.g., 2, 3, 4, 5 or more nucleotides, typically in close proximity to each other, e.g., contiguous nucleotides. A genetic modification can be in a coding sequence, e.g., an exon sequence. A genetic modification can be at a splice site, i.e., sufficiently close to a splice acceptor site or a splice donor site to disrupt splicing. A genetic modification can include insertion of a nucleotide sequence not endogenous to the genomic locus, e.g., insertion of a coding sequence of a heterologous open reading frame or gene. As used herein, a genetic modification can be used to prevent translation of an endogenous full-length protein having an amino acid sequence of the full-length protein prior to genetic modification of the genomic locus. Prevention of translation of an endogenous full-length protein or gene product includes prevention of translation of a protein or gene product of any length. Translation of an endogenous full-length protein can be prevented, for example, by a frameshift mutation that results in the generation of a premature stop codon or by generation of a nonsense mutation. Translation of an endogenous full-length protein can be prevented by disruption of splicing. Translation of an endogenous full-length protein can be prevented by the insertion of a heterologous coding sequence. Translation of an endogenous full-length protein, e.g., when the endogenous full-length protein contains an unwanted mutation, can be prevented by making a change at one or more positions to change an endogenous full-length protein coding sequence to provide a modified full-length coding sequence different from the endogenous sequence present in the cell, e.g., correction of a point mutation. Translation of an endogenous full-length protein can be prevented by altering the splicing of the endogenous full-length protein to produce a different protein by alternative splicing.

[0281] As used herein, "indel" refers to an insertion or deletion mutation consisting of a number of nucleotides that are either inserted, deleted, or inserted and deleted, e.g. at the site of double-stranded breaks (DSBs), in a target nucleic acid. As used herein, when indel formation results in an insertion, the insertion is a random insertion at the site of a DSB and may or may not be directed by or based on a template sequence.

[0282] As used herein, a "heterologous coding sequence" refers to a coding sequence that has been introduced as an exogenous source within a cell (e.g., inserted at a genomic locus such as a safe harbor locus including a TCR gene locus). That is, the introduced coding sequence is heterologous with respect to at least its insertion site. A polypeptide expressed from such heterologous coding sequence gene is referred to as a "heterologous polypeptide." The heterologous coding sequence can be naturally-occurring or engineered, and can be wild-type or a variant. The heterologous coding sequence may include nucleotide sequences other than the sequence that encodes the heterologous polypeptide (e.g., an internal ribosomal entry site). The heterologous coding sequence can be a coding sequence that occurs naturally in the genome, as a wild-type or a variant (e.g., mutant). For example, although the cell contains the coding sequence of interest (as a wild-type or as a variant), the same coding sequence or variant thereof can be introduced as an exogenous source for, e.g., expression at a locus that is highly expressed. The heterologous gcoding sequence can also be a coding sequence that is not naturally occurring in the genome, or that expresses a heterologous polypeptide that does not naturally occur in the genome. "Heterologous coding sequence", "exogenous coding sequence", and "transgene" are used interchangeably. In some embodiments, the heterologous coding sequence or transgene includes an exogenous nucleic acid sequence, e.g., a nucleic acid sequence is not endogenous to the recipient cell. In some embodiments, the heterologous coding sequence or transgene includes an exogenous nucleic acid sequence, e.g., a nucleic acid sequence that does not naturally occur in the recipient cell. For example, a heterologous coding sequence may be heterologous with respect to its insertion site and with respect to its recipient cell.

[0283] As used herein, "reduced or eliminated" expression of a protein on a cell refers to a partial or complete loss of expression of the protein relative to an unmodified cell. In some embodiments, the surface expression of a protein on a cell is measured by flow cytometry and has "reduced" or "eliminated" surface expression relative to an unmodified cell as evidenced by a reduction in fluorescence signal upon staining with the same antibody against the protein. A cell that has "reduced" or "eliminated" surface expression of a protein by flow cytometry relative to an unmodified cell may be referred to as "negative" for expression of that protein as evidenced by a fluorescence signal similar to a cell stained with an isotype control antibody. The "reduction" or "elimination" of protein expression can be measured by other known techniques in the field with appropriate controls known to those skilled in the art.

[0284] As used herein, "knockdown" refers to a decrease in expression of a particular gene product (e.g., protein, mRNA, or both), e.g., as compared to expression of an unedited target sequence. Knockdown of a protein can be measured by detecting total cellular amount of the protein from a sample, such as a tissue, fluid, or cell population of interest. It can also be measured by measuring a surrogate, marker, or activity for the protein. Methods for measuring knockdown of mRNA are known and include analyzing mRNA isolated from a sample of interest. In some embodiments, "knockdown" may refer to some loss of expression of a particular gene product, for example a decrease in the amount of mRNA transcribed or a decrease in the amount of protein expressed by a cell or population of cells (including in vivo populations such as those found in tissues).

[0285] As used herein, "knockout" refers to a loss of expression from a particular gene or of a particular protein in a cell. Knockout can result in a decrease in expression below the level of detection of the assay. Knockout can be measured either by detecting total cellular amount of a protein in a cell, a tissue or a population of cells.

[0286] As used herein, a "target sequence" or "genomic target sequence" refers to a sequence of nucleic acid in a target gene that has complementarity to the guide sequence of the gRNA. The interaction of the target sequence and the guide sequence directs an RNA-guided DNA binding agent to bind, and potentially nick or cleave (depending on the activity of the agent), within the target sequence.

[0287] As used herein, "treatment" refers to any administration or application of a therapeutic for disease or disorder in a subject, and includes inhibiting the disease, arresting its development, relieving one or more symptoms of the disease, curing the disease, or preventing one or more symptoms of the disease, including recurrence of the symptom.

[0288] Reference will now be made in detail to certain embodiments of the invention, examples of which are illustrated in the accompanying drawings. While the invention is described in conjunction with the illustrated embodiments, it will be understood that they are not intended to limit the invention to those embodiments. On the contrary, the invention is intended to cover all alternatives, modifications, and equivalents, which may be included within the invention as defined by the appended claims and included embodiments.

[0289] Before describing the present teachings in detail, it is to be understood that the disclosure is not limited to specific compositions or process steps, as such may vary. It should be noted that, as used in this specification and the appended claims, the singular form "a", "an" and "the" include plural references unless the context clearly dictates otherwise. Thus, for example, reference to "a conjugate" includes a plurality of conjugates and reference to "a cell" includes a plurality of cells and the like.

[0290] Numeric ranges are inclusive of the numbers defining the range. Measured and measurable values are understood to be approximate, taking into account significant digits and the error associated with the measurement. Also, the use of "comprise", "comprises", "comprising", "contain", "contains", "containing", "include", "includes", and "including" are not intended to be limiting. It is to be understood that both the foregoing general description and detailed description are exemplary and explanatory only and are not restrictive of the teachings.

[0291] Unless specifically noted in the specification, embodiments in the specification that recite "comprising" various components are also contemplated as "consisting of" or "consisting essentially of" the recited components; embodiments in the specification that recite "consisting of" various components are also contemplated as "comprising" or "consisting essentially of" the recited components; and embodiments in the specification that recite "consisting essentially of" various components are also contemplated as "consisting of" or "comprising" the recited components (this interchangeability does not apply to the use of these terms in the claims). The term "or" is used in an inclusive sense, i.e., equivalent to "and/or," unless the context clearly indicates otherwise.

[0292] The section headings used herein are for organizational purposes only and are not to be construed as limiting the desired subject matter in any way. In the event that any material incorporated by reference contradicts any term defined in this specification or any other express content of this specification, this specification controls. While the present teachings are described in conjunction with various embodiments, it is not intended that the present teachings be limited to such embodiments. On the contrary, the present teachings encompass various alternatives, modifications, and equivalents, as will be appreciated by those of skill in the art.

II. Genetically Modified Cells

A. Engineered Cell Compositions

[0293] The present disclosure provides engineered cell compositions which have reduced or eliminated surface expression of HLA-A, HLA-B, TRAC, TRBC, and/or MHC class II relative to an unmodified cell as disclosed herein. In some embodiments, the engineered cell composition comprises a genetic modification in the HLA-A, HLA-B, TRAC, TRBC, and/or CIITA gene. In some embodiments, the engineered cell composition comprises a genetic modification in each of the HLA-A, HLA-B, and CIITA genes. In some embodiments, the engineered cell is an allogeneic cell. In some embodiments, the engineered cell with reduced HLA-A, HLA-B, TRAC, TRBC, and/or MHC class II expression is useful for adoptive cell transfer therapies. In some embodiments, the engineered cell comprises additional genetic modifications in the genome of the cell to yield a cell that is desirable for allogeneic transplant purposes.

[0294] In some embodiments, an engineered cell which has reduced or eliminated surface expression of HLA-A relative to an unmodified cell is provided, comprising a genetic modification in the HLA-A gene, wherein the modification comprises at least one nucleotide from within the genomic coordinates chr6:29942540-29945459. In some embodiments, an engineered cell which has reduced or eliminated surface expression of HLA-A relative to an unmodified cell is provided, comprising a genetic modification in the HLA-A gene, wherein the modification comprises at least one nucleotide from within the genomic coordinates chosen from: chr6:29942891-29942915; chr6:29942609-29942633; chr6:29944266-29944290; chr6:29942889-29942913; chr6:29944471-29944495; and chr6:29944470-29944494.

[0295] In some embodiments, an engineered cell which has reduced or eliminated surface expression of HLA-A relative to an unmodified cell is provided, comprising a genetic modification in the HLA-A gene, wherein the modification comprises at least one nucleotide from within the genomic coordinates chosen from: chr6:29942891-29942915; chr6:29942609-29942633; chr6:29942889-29942913; chr6:29944471-29944495; chr6:29944266-29944290; chr6:29942785-29942809.

[0296] In some embodiments, an engineered cell which has reduced or eliminated surface expression of HLA-A relative to an unmodified cell is provided, comprising a genetic modification in the HLA-A gene, wherein the modification comprises at least 5 contiguous nucleotides within the genomic coordinates chr6:29942540-29945459. In some embodiments, the engineered cell which has reduced or eliminated surface expression of HLA-A relative to an unmodified cell, comprises a genetic modification in the HLA-A gene, wherein the modification comprises at least 6, 7, 8, 9, or 10 contiguous nucleotides within the genomic coordinates.

[0297] In some embodiments, an engineered cell which has reduced or eliminated surface expression of HLA-A relative to an unmodified cell is provided, comprising a genetic modification in the HLA-A gene, wherein the modification comprises at least 5 contiguous nucleotides within the genomic coordinates chosen from: chr6:29942891-29942915; chr6:29942609-29942633; chr6:29944266-29944290; chr6:29942889-29942913; chr6:29944471-29944495; and chr6:29944470-29944494. In some embodiments, the engineered cell which has reduced or eliminated surface expression of HLA-A relative to an unmodified cell, comprises a genetic modification in the HLA-A gene, wherein the modification comprises at least 6, 7, 8, 9, or 10 contiguous nucleotides within the genomic coordinates.

[0298] In some embodiments, an engineered cell which has reduced or eliminated surface expression of HLA-A relative to an unmodified cell is provided, comprising a genetic modification in the HLA-A gene, wherein the modification comprises at least 5 contiguous nucleotides within the genomic coordinates chosen from: chr6:29942891-29942915; chr6:29942609-29942633; chr6:29942889-29942913; chr6:29944471-29944495; chr6:29944266-29944290; chr6:29942785-29942809. In some embodiments, the engineered cell which has reduced or eliminated surface expression of HLA-A relative to an unmodified cell, comprises a genetic modification in the HLA-A gene, wherein the modification comprises at least 6, 7, 8, 9, or 10 contiguous nucleotides within the genomic coordinates.

[0299] In some embodiments, an engineered cell which has reduced or eliminated surface expression of HLA-A relative to an unmodified cell is provided, comprising a genetic modification in the HLA-A gene, wherein the modification comprises at least one C to T substitution or at least one A to G substitution within the genomic coordinates chr6:29942540-29945459.

[0300] In some embodiments, an engineered cell which has reduced or eliminated surface expression of HLA-A relative to an unmodified cell is provided, comprising a genetic modification in the HLA-A gene, wherein the modification comprises at least one C to T substitution or at least one A to G substitution within the genomic coordinates chosen from: chr6:29942891-29942915; chr6:29942609-29942633; chr6:29944266-29944290; chr6:29942889-29942913; chr6:29944471-29944495; and chr6:29944470-29944494.

[0301] In some embodiments, an engineered cell which has reduced or eliminated surface expression of HLA-A relative to an unmodified cell is provided, comprising a genetic modification in the HLA-A gene, wherein the modification comprises at least one C to T substitution or at least one A to G substitution within the genomic coordinates chosen from: chr6:29942891-29942915; chr6:29942609-29942633; chr6:29942889-29942913; chr6:29944471-29944495; chr6:29944266-29944290; chr6:29942785-29942809.

[0302] In some embodiments, an engineered cell, which has reduced or eliminated surface expression of HLA-A relative to an unmodified cell, comprises a genetic modification in the HLA-A gene, wherein the genetic modification comprises an indel, a C to T substitution, or an A to G substitution within the genomic coordinates chosen from: (a) chr6:29942891-29942915; chr6:29942609-29942633; chr6:29944266-29944290; chr6:29942889-29942913; chr6:29944471-29944495; and chr6:29944470-29944494 or (b) chr6:29942891-29942915; chr6:29942609-29942633; chr6:29942889-29942913; chr6:29944471-29944495; chr6:29944266-29944290; chr6:29942785-29942809. In some embodiments, the genetic modification comprises at least 5 contiguous nucleotides within the genomic coordinates. In some embodiments, the genetic modification comprises at least 6, 7, 8, 9, or 10 contiguous nucleotides within the genomic coordinates. In some embodiments, the genetic modification comprises at least one C to T substitution or at least one A to G substitution within the genomic coordinates.

[0303] In some embodiments, an engineered cell is provided that has reduced or eliminated surface expression of HLA-A by a genomic editing system that binds to an HLA-A genomic target sequence comprising at least 10 contiguous nucleotides within the genomic coordinates chosen from: (a) chr6:29942891-29942915; chr6:29942609-29942633; chr6:29944266-29944290; chr6:29942889-29942913; chr6:29944471-29944495; and chr6:29944470-29944494 or (b) chr6:29942891-29942915; chr6:29942609-29942633; chr6:29942889-29942913; chr6:29944471-29944495; chr6:29944266-29944290; chr6:29942785-29942809. In some embodiments, an engineered cell is provided that has reduced or eliminated surface expression of HLA-A by a genomic editing system that binds to an HLA-A genomic target sequence comprising at least 5 contiguous nucleotides within the genomic coordinates. In some embodiments, the HLA-A genomic target sequence comprises at least 15 contiguous nucleotides within the genomic coordinates. In some embodiments, the genomic editing system comprises an RNA-guided DNA-binding agent. In some embodiments, the RNA-guided DNA-binding agent comprises a Cas9 protein, such as an *N. meningitidis* Cas9.

[0304] In some embodiments, an engineered cell is provided that has reduced or eliminated surface expression of HLA-A by a genomic editing system that binds to an HLA-A genomic target sequence comprising at least 5 contiguous nucleotides within the genomic coordinates chosen from: (a) chr6:29942891-29942915; chr6:29942609-29942633; chr6:29944266-29944290; chr6:29942889-29942913; chr6:29944471-29944495; and chr6:29944470-29944494 or (b) chr6:29942891-29942915; chr6:29942609-29942633; chr6:29942889-29942913; chr6:29944471-29944495; chr6:29944266-29944290; chr6:29942785-29942809. In some embodiments, the HLA-A genomic target sequence comprises at least 10 contiguous nucleotides within the genomic coordinates. In some embodiments, the HLA-A genomic target sequence comprises at least 15 contiguous nucleotides within the genomic coordinates. In some embodiments, the genomic editing system comprises an RNA-guided DNA-binding agent. In some embodiments, the RNA-guided DNA-binding agent comprises a Cas9 protein, such as an N. *Meningitidis.*

[0305] In some embodiments, an engineered cell which has reduced or eliminated surface expression of TRAC relative to an unmodified cell is provided, comprising a genetic modification in the TRAC gene, wherein the modification comprises at least one nucleotide from within the genomic coordinates chr14:22547462-22551621. In some embodiments, an engineered cell which has reduced or eliminated surface expression of TRAC relative to an unmodified cell is provided, comprising a genetic modification in the TRAC gene, wherein the modification comprises at least one nucleotide from within the genomic coordinates chr14:22547505-22551621. In some embodiments, an engineered cell which has reduced or eliminated surface expression of TRAC relative to an unmodified cell is provided, comprising a genetic modification in the TRAC gene, wherein the modification comprises at least one nucleotide from within the genomic coordinates from: chr14:22547505-22547529; chr14:22547525-22547549; chr14:22547674-22547698; chr14:22550544-22550568; or chr14:22550574-22550598.

[0306] In some embodiments, an engineered cell which has reduced or eliminated surface expression of TRAC relative to an unmodified cell is provided, comprising a genetic modification in the TRAC gene, wherein the modification comprises at least 5 contiguous nucleotides within the genomic coordinates chr14:22547462-22551621. In some embodiments, an engineered cell which has reduced or eliminated surface expression of TRAC relative to an unmodified cell is provided, comprising a genetic modification in the TRAC gene, wherein the modification comprises at least 5 contiguous nucleotides within the genomic coordinates chr14:22547505-22551621. In some embodiments, the engineered cell which has reduced or eliminated surface expression of TRAC relative to an unmodified cell, comprises a genetic modification in the TRAC gene, wherein the modification comprises at least 6, 7, 8, 9, or 10 contiguous nucleotides within the genomic coordinates.

[0307] In some embodiments, an engineered cell which has reduced or eliminated surface expression of TRAC relative to an unmodified cell is provided, comprising a genetic modification in the TRAC gene, wherein the modification comprises at least 5 contiguous nucleotides within the genomic coordinates chosen from: chr14:22547505-22547529; chr14:22547525-22547549; chr14:22547674-22547698; chr14:22550544-22550568; or chr14:22550574-22550598. In some embodiments, the engineered cell which has reduced or eliminated surface expression of TRAC relative to an unmodified cell, comprises a genetic modification in the TRAC gene, wherein the modification comprises at least 6, 7, 8, 9, or 10 contiguous nucleotides within the genomic coordinates.

[0308] In some embodiments, an engineered cell which has reduced or eliminated surface expression of TRAC relative to an unmodified cell is provided, comprising a genetic modification in the TRAC gene, wherein the modification comprises at least one C to T substitution or at least one A to G substitution within the genomic coordinates chr14:22547505-22551621 or chr14:22547462-22551621.

[0309] In some embodiments, an engineered cell which has reduced or eliminated surface expression of TRAC relative to an unmodified cell is provided, comprising a genetic modification in the TRAC gene, wherein the modification comprises at least one C to T substitution or at least one A to G substitution within the genomic coordinates chosen from: chr14:22547505-22547529; chr14:22547525-22547549; chr14:22547674-22547698; chr14:22550544-22550568; or chr14:22550574-22550598.

[0310] In some embodiments, an engineered cell, which has reduced or eliminated surface expression of TRAC relative to an unmodified cell, comprises a genetic modification in the TRAC gene, wherein the genetic modification comprises an indel, a C to T substitution, or an A to G substitution within the genomic coordinates chosen from: (a) chr14:22547505-22547529; chr14:22547525-22547549; chr14:22547674-22547698; chr14:22550544-22550568; and chr14:22550574-22550598. In some embodiments, the genetic modification comprises at least 5 contiguous nucleotides within the genomic coordinates. In some embodiments, the genetic modification comprises at least 6, 7, 8, 9, or 10 contiguous nucleotides within the genomic coordinates. In some embodiments, the genetic modification comprises at least one C to T substitution or at least one A to G substitution within the genomic coordinates.

[0311] In some embodiments, an engineered cell is provided that has reduced or eliminated surface expression of TRAC by a genomic editing system that binds to a TRAC genomic target sequence comprising at least 10 contiguous nucleotides within the genomic coordinates chosen from: (a) chr14:22547505-22547529; chr14:22547525-22547549; chr14:22547674-22547698; chr14:22550544-22550568; and chr14:22550574-22550598. In some embodiments, an engineered cell is provided that has reduced or eliminated surface expression of TRAC by a genomic editing system that binds to a TRAC genomic target sequence comprising at least 5 contiguous nucleotides within the genomic coordinates. In some embodiments, the TRAC genomic target sequence comprises at least 15 contiguous nucleotides within the genomic coordinates. In some embodiments, the genomic editing system comprises an RNA-guided DNA-binding agent. In some embodiments, the RNA-guided DNA-binding agent comprises a Cas9 protein, such as an *N. meningitidis* Cas9.

[0312] In some embodiments, an engineered cell is provided that has reduced or eliminated surface expression of TRAC by a genomic editing system that binds to a TRAC genomic target sequence comprising at least 5 contiguous nucleotides within the genomic coordinates chosen from: (a) chr14:22547505-22547529; chr14:22547525-22547549; chr14:22547674-22547698; chr14:22550544-22550568; and chr14:22550574-22550598. In some embodiments, the TRAC genomic target sequence comprises at least 10 contiguous nucleotides within the genomic coordinates. In some embodiments, the TRAC genomic target sequence comprises at least 15 contiguous nucleotides within the genomic coordinates. In some embodiments, the genomic editing system comprises an RNA-guided DNA-binding agent. In some embodiments, the RNA-guided DNA-binding agent comprises a Cas9 protein, such as an N. *Meningitidis.*

[0313] In some embodiments, an engineered cell which has reduced or eliminated surface expression of TRBC relative to an unmodified cell is provided, comprising a genetic modification in the TRBC gene, wherein the modification comprises at least one nucleotide from within the genomic coordinates chosen from: chr7:142791756-142802543. In some embodiments, an engineered cell which has reduced or eliminated surface expression of TRBC relative to an unmodified cell is provided, comprising a genetic modification in the TRBC gene, wherein the modification comprises at least one nucleotide from within the genomic coordinates chosen from: (a) chr7:142791862-142793149; (b) chr7: 142791756-142792721; or (c) chr7: 142801104-142802543.

[0314] In some embodiments, an engineered cell which has reduced or eliminated surface expression of TRBC relative to an unmodified cell is provided, comprising a genetic modification in the TRBC gene, wherein the modification comprises at least one nucleotide from within the genomic coordinates from: (a) chr7:142792690-142792714; or chr7:142792693-142792717; or (b) chr7:142791756-142791780; chr7:142791761-142791785; chr7:142791820-142791844; chr7:142791939-142791963; chr7:142791940-142791964; or chr7:142792004-142792028; or (c) chr7: 142801104-142801124; chr7:142802103-142802127; or chr7:142802106-142802130.

[0315] In some embodiments, an engineered cell which has reduced or eliminated surface expression of TRBC relative to an unmodified cell is provided, comprising a genetic modification in the TRBC gene, wherein the modification comprises at least one nucleotide from within the genomic coordinates chosen from: chr7:142792690-142792714; chr7:142802103-142802127; and chr7: 142802106-14280213.

[0316] In some embodiments, an engineered cell which has reduced or eliminated surface expression of TRBC relative to an unmodified cell is provided, comprising a genetic modification in the TRBC gene, wherein the modification comprises at least 5 contiguous nucleotides within the genomic coordinates chosen from: (a) chr7:142791862-142793149; (b) chr7: 142791756-142792721; or (c) chr7: 142801104-142802543. In some embodiments, the engineered cell which has reduced or eliminated surface expression of TRBC relative to an unmodified cell, comprises a genetic modification in the TRBC gene, wherein the modification comprises at least 6, 7, 8, 9, or 10 contiguous nucleotides within the genomic coordinates.

[0317] In some embodiments, an engineered cell which has reduced or eliminated surface expression of TRBC relative to an unmodified cell is provided, comprising a genetic modification in the TRBC gene, wherein the modification comprises at least 5 contiguous nucleotides within the genomic coordinates chosen from: (a) chr7:142792690-142792714; or chr7:142792693-142792717; or (b) chr7: 142791756-142791780; chr7:142791761-142791785; chr7: 142791820-142791844; chr7:142791939-142791963; chr7: 142791940-142791964; or chr7:142792004-142792028; or (c) chr7:142801104-142801124; chr7:142802103-142802127; or chr7:142802106-142802130. In some embodiments, the engineered cell which has reduced or eliminated surface expression of TRBC relative to an unmodified cell, comprises a genetic modification in the TRBC gene, wherein the modification comprises at least 6, 7, 8, 9, or 10 contiguous nucleotides within the genomic coordinates.

[0318] In some embodiments, an engineered cell which has reduced or eliminated surface expression of TRBC relative to an unmodified cell is provided, comprising a genetic modification in the TRBC gene, wherein the modification comprises at least 5 contiguous nucleotides within the genomic coordinates chosen from: chr7:142792690-142792714; chr7:142802103-142802127; and chr7: 142802106-14280213. In some embodiments, the engineered cell which has reduced or eliminated surface expression of TRBC relative to an unmodified cell, comprises a genetic modification in the TRBC gene, wherein the modification comprises at least 6, 7, 8, 9, or 10 contiguous nucleotides within the genomic coordinates.

[0319] In some embodiments, an engineered cell which has reduced or eliminated surface expression of TRBC relative to an unmodified cell is provided, comprising a genetic modification in the TRBC gene, wherein the modification comprises at least one C to T substitution or at least one A to G substitution within the genomic coordinates chosen from:

[0320] In some embodiments, an engineered cell which has reduced or eliminated surface expression of TRBC relative to an unmodified cell is provided, comprising a genetic modification in the TRBC gene, wherein the modification comprises at least one C to T substitution or at least one A to G substitution within the genomic coordinates chosen from: (a) chr7:142791862-142793149; (b) chr7: 142791756-142792721; or (c) chr7:142801104-142802543.

[0321] In some embodiments, an engineered cell which has reduced or eliminated surface expression of TRBC relative to an unmodified cell is provided, comprising a genetic modification in the TRBC gene, wherein the modification comprises at least one C to T substitution or at least one A to G substitution within the genomic coordinates chosen from: (a) chr7:142792690-142792714; or chr7: 142792693-142792717; or (b) chr7:142791756-142791780; chr7:142791761-142791785; chr7:142791820-142791844; chr7:142791939-142791963; chr7:142791940-142791964; or chr7:142792004-142792028; or (c) chr7:142801104-142801124; chr7:142802103-142802127; or chr7: 142802106-142802130.

[0322] In some embodiments, an engineered cell which has reduced or eliminated surface expression of TRBC relative to an unmodified cell is provided, comprising a genetic modification in the TRBC gene, wherein the modification comprises at least one C to T substitution or at least one A to G substitution within the genomic coordinates chosen from: chr7:142792690-142792714; chr7: 142802103-142802127; and chr7:142802106-14280213.

[0323] In some embodiments, an engineered cell, which has reduced or eliminated surface expression of TRBC relative to an unmodified cell, comprises a genetic modification in the TRBC gene, wherein the genetic modification comprises an indel, a C to T substitution, or an A to G substitution within the genomic coordinates chosen from: (a) chr7:142792690-142792714; or chr7:142792693-142792717; or (b) chr7:142791756-142791780; chr7: 142791761-142791785; chr7:142791820-142791844; chr7: 142791939-142791963; chr7:142791940-142791964; or chr7:142792004-142792028; or (c) chr7:142801104-142801124; chr7:142802103-142802127; or chr7: 142802106-142802130. In some embodiments, the genetic modification comprises at least 5 contiguous nucleotides within the genomic coordinates. In some embodiments, the genetic modification comprises at least 6, 7, 8, 9, or 10 contiguous nucleotides within the genomic coordinates. In some embodiments, the genetic modification comprises at least one C to T substitution or at least one A to G substitution within the genomic coordinates.

[0324] In some embodiments, an engineered cell, which has reduced or eliminated surface expression of TRBC relative to an unmodified cell, comprises a genetic modification in the TRBC gene, wherein the genetic modification comprises an indel, a C to T substitution, or an A to G substitution within the genomic coordinates chosen from: chr7:142792690-142792714; chr7:142802103-142802127; and chr7:142802106-14280213. In some embodiments, the genetic modification comprises at least 5 contiguous nucleotides within the genomic coordinates. In some embodiments, the genetic modification comprises at least 6, 7, 8, 9, or 10 contiguous nucleotides within the genomic coordinates. In some embodiments, the genetic modification comprises at least one C to T substitution or at least one A to G substitution within the genomic coordinates.

[0325] In some embodiments, an engineered cell is provided that has reduced or eliminated surface expression of TRBC by a genomic editing system that binds to a TRBC genomic target sequence comprising at least 10 contiguous nucleotides within the genomic coordinates chosen from: (a) chr7:142792690-142792714; or chr7:142792693-142792717; or (b) chr7:142791756-142791780; chr7:142791761-142791785; chr7:142791820-142791844; chr7:142791939-142791963; chr7:142791940-142791964; or chr7:142792004-142792028; or (c) chr7:142801104-142801124; chr7:142802103-142802127; or chr7:142802106-142802130. In some embodiments, an engineered cell is provided that has reduced or eliminated surface expression of TRBC by a genomic editing system that binds to a TRBC genomic target sequence comprising at least 5 contiguous nucleotides within the genomic coordinates. In some embodiments, the TRBC genomic target sequence comprises at least 15 contiguous nucleotides within the genomic coordinates. In some embodiments, the genomic editing system comprises an RNA-guided DNA-binding agent. In some embodiments, the RNA-guided DNA-binding agent comprises a Cas9 protein, such as an *N. meningitidis* Cas9.

[0326] In some embodiments, an engineered cell is provided that has reduced or eliminated surface expression of TRBC by a genomic editing system that binds to a TRBC genomic target sequence comprising at least 10 contiguous nucleotides within the genomic coordinates chosen from: chr7:142792690-142792714; chr7:142802103-142802127; and chr7:142802106-14280213. In some embodiments, an engineered cell is provided that has reduced or eliminated surface expression of TRBC by a genomic editing system that binds to a TRBC genomic target sequence comprising at least 5 contiguous nucleotides within the genomic coordinates. In some embodiments, the TRBC genomic target sequence comprises at least 15 contiguous nucleotides within the genomic coordinates. In some embodiments, the genomic editing system comprises an RNA-guided DNA-binding agent. In some embodiments, the RNA-guided DNA-binding agent comprises a Cas9 protein, such as an *N. meningitidis* Cas9.

[0327] In some embodiments, an engineered cell is provided that has reduced or eliminated surface expression of TRBC by a genomic editing system that binds to a TRBC genomic target sequence comprising at least 5 contiguous nucleotides within the genomic coordinates chosen from: (a) chr7:142792690-142792714; or chr7:142792693-142792717; or (b) chr7:142791756-142791780; chr7:142791761-142791785; chr7:142791820-142791844; chr7:142791939-142791963; chr7:142791940-142791964; or chr7:142792004-142792028; or (c) chr7:142801104-142801124; chr7:142802103-142802127; or chr7:142802106-142802130. In some embodiments, the TRBC genomic target sequence comprises at least 10 contiguous nucleotides within the genomic coordinates. In some embodiments, the TRBC genomic target sequence comprises at least 15 contiguous nucleotides within the genomic coordinates. In some embodiments, the genomic editing system comprises an RNA-guided DNA-binding agent. In some embodiments, the RNA-guided DNA-binding agent comprises a Cas9 protein, such as an N. *Meningitidis.*

[0328] In some embodiments, an engineered cell is provided that has reduced or eliminated surface expression of TRBC by a genomic editing system that binds to a TRBC genomic target sequence comprising at least 5 contiguous nucleotides within the genomic coordinates chosen from: chr7:142792690-142792714; chr7:142802103-142802127; and chr7:142802106-14280213. In some embodiments, the TRBC genomic target sequence comprises at least 10 contiguous nucleotides within the genomic coordinates. In some embodiments, the TRBC genomic target sequence comprises at least 15 contiguous nucleotides within the genomic coordinates. In some embodiments, the genomic editing system comprises an RNA-guided DNA-binding agent. In some embodiments, the RNA-guided DNA-binding agent comprises a Cas9 protein, such as an N. *Meningitidis.*

[0329] In some embodiments, an engineered cell which has reduced or eliminated surface expression of MHC class II relative to an unmodified cell is provided, comprising a genetic modification in the CIITA locus, wherein the modification comprises at least one nucleotide of the genomic coordinates (a) chr16:10877363-10907788 or (b) chr16:10906515-10908136.

[0330] In some embodiments, an engineered cell which has reduced or eliminated surface expression of MHC class II relative to an unmodified cell is provided, comprising a genetic modification in the CIITA gene, wherein the modification comprises at least one nucleotide of the genomic coordinates chosen from: (a) chr16:10907504-10907528; chr16:10907508-10907532; chr16:10907539-10907559; chr16:10895658-10895682; chr16:10895668-10895692; chr16:10895750-10895774; chr16:10895753-10895777; chr16:10895754-10895778; chr16:10898684-10898708; chr16:10901529-10901553; chr16:10902121-10902145; chr16:10902701-10902725; chr16:10904726-10904750; chr16:10904760-10904784; chr16:10906493-10906517; chr16:10906515-10906539; chr16:10906631-10906655; chr16:10906636-10906660; chr16:10906643-10906667; chr16:10906770-10906794; chr16:10906788-10906812; chr16:10906789-10906813; chr16:10906816-10906840; chr16:10907148-10907172; chr16:10907254-10907278; chr16:10907331-10907355; chr16:10907477-10907501; chr16:10907497-10907521; chr16:10907503-10907527; and chr16:10907574-10907598; or (b) chr16:10906889-10906913; and chr16:10907504-10907528.

[0331] In some embodiments, an engineered cell which has reduced or eliminated surface expression of MHC class II relative to an unmodified cell is provided, comprising a genetic modification in the CIITA gene, wherein the modification comprises at least one nucleotide of the genomic coordinates chosen from: chr16:10895658-10895682; chr16:10902701-10902725; chr16:10906493-10906517; chr16:10906631-10906655; chr16:10906643-10906667;

chr16:10907477-10907501; and chr16:10907497-10907521; chr16:10907504-10907528; and chr16:10907508-10907532.

[0332] In some embodiments, an engineered cell which has reduced or eliminated surface expression of MHC class II relative to an unmodified cell is provided, comprising a genetic modification in the CIITA gene, wherein the modification comprises at least 5 contiguous nucleotides within the genomic coordinates chosen from: (a) chr16:10877363-10907788 or (b) chr16:10906515-10908136. In some embodiments, the engineered cell which has reduced or eliminated surface expression of MHC class II relative to an unmodified cell, comprises a genetic modification in the CIITA gene, wherein the modification comprises at least 6, 7, 8, 9, or 10 contiguous nucleotides within the genomic coordinates.

[0333] In some embodiments, an engineered cell which has reduced or eliminated surface expression of MHC class II relative to an unmodified cell is provided, comprising a genetic modification in the CIITA gene, wherein the modification comprises at least 5 contiguous nucleotides within the genomic coordinates chosen from: (a) chr16:10907504-10907528; chr16:10907508-10907532; chr16:10907539-10907559; chr16:10895658-10895682; chr16:10895668-10895692; chr16:10895750-10895774; chr16:10895753-10895777; chr16:10895754-10895778; chr16:10898684-10898708; chr16:10901529-10901553; chr16:10902121-10902145; chr16:10902701-10902725; chr16:10904726-10904750; chr16:10904760-10904784; chr16:10906493-10906517; chr16:10906515-10906539; chr16:10906631-10906655; chr16:10906636-10906660; chr16:10906643-10906667; chr16:10906770-10906794; chr16:10906788-10906812; chr16:10906789-10906813; chr16:10906816-10906840; chr16:10907148-10907172; chr16:10907254-10907278; chr16:10907331-10907355; chr16:10907477-10907501; chr16:10907497-10907521; chr16:10907503-10907527; chr16:10907574-10907598; or (b) chr16:10906889-10906913; or chr16:10907504-10907528. In some embodiments, the engineered cell which has reduced or eliminated surface expression of MHC class II relative to an unmodified cell, comprises a genetic modification in the CIITA gene, wherein the modification comprises at least 6, 7, 8, 9, or 10 contiguous nucleotides within the genomic coordinates.

[0334] In some embodiments, an engineered cell which has reduced or eliminated surface expression of MHC class II relative to an unmodified cell is provided, comprising a genetic modification in the CIITA gene, wherein the modification comprises at least 5 contiguous nucleotides within the genomic coordinates chosen from: chr16:10895658-10895682; chr16:10902701-10902725; chr16:10906493-10906517; chr16:10906631-10906655; chr16:10906643-10906667; chr16:10907477-10907501; and chr16:10907497-10907521; chr16:10907504-10907528; and chr16:10907508-10907532. In some embodiments, the engineered cell which has reduced or eliminated surface expression of MHC class II relative to an unmodified cell, comprises a genetic modification in the CIITA gene, wherein the modification comprises at least 6, 7, 8, 9, or 10 contiguous nucleotides within the genomic coordinates.

[0335] In some embodiments, an engineered cell which has reduced or eliminated surface expression of MHC class II relative to an unmodified cell is provided, comprising a genetic modification in the CIITA gene, wherein the modification comprises at least one C to T substitution or at least one A to G substitution within the genomic coordinates chosen from: (a) chr16:10877363-10907788 or (b) chr16:10906515-10908136.

[0336] In some embodiments, an engineered cell which has reduced or eliminated surface expression of MHC class II relative to an unmodified cell is provided, comprising a genetic modification in the CIITA gene, wherein the modification comprises at least one C to T substitution or at least one A to G substitution within the genomic coordinates chosen from: (a) chr16:10907504-10907528; chr16:10907508-10907532; chr16:10907539-10907559; chr16:10895658-10895682; chr16:10895668-10895692; chr16:10895750-10895774; chr16:10895753-10895777; chr16:10895754-10895778; chr16:10898684-10898708; chr16:10901529-10901553; chr16:10902121-10902145; chr16:10902701-10902725; chr16:10904726-10904750; chr16:10904760-10904784; chr16:10906493-10906517; chr16:10906515-10906539; chr16:10906631-10906655; chr16:10906636-10906660; chr16:10906643-10906667; chr16:10906770-10906794; chr16:10906788-10906812; chr16:10906789-10906813; chr16:10906816-10906840; chr16:10907148-10907172; chr16:10907254-10907278; chr16:10907331-10907355; chr16:10907477-10907501; chr16:10907497-10907521; chr16:10907503-10907527; chr16:10907574-10907598; or (b) chr16:10906889-10906913; or chr16:10907504-10907528.

[0337] In some embodiments, an engineered cell which has reduced or eliminated surface expression of MHC class II relative to an unmodified cell is provided, comprising a genetic modification in the CIITA gene, wherein the modification comprises at least one C to T substitution or at least one A to G substitution within the genomic coordinates chosen from: chr16:10895658-10895682; chr16:10902701-10902725; chr16:10906493-10906517; chr16:10906631-10906655; chr16:10906643-10906667; chr16:10907477-10907501; and chr16:10907497-10907521; chr16:10907504-10907528; and chr16:10907508-10907532.

[0338] In some embodiments, an engineered cell which has reduced or eliminated surface expression of MHC class II relative to an unmodified cell is provided, comprising a genetic modification in the CIITA gene, wherein the modification comprises at least one nucleotide within the genomic coordinates chosen from: (a) chr16:10877363-10907788 or (b) chr16:10906515-10908136.

[0339] In some embodiments, an engineered cell which has reduced or eliminated surface expression of MHC class II relative to an unmodified cell is provided, comprising a genetic modification in the CIITA gene, wherein the modification comprises at least one nucleotide within the genomic coordinates chosen from: (a) chr16:10907504-10907528; chr16:10907508-10907532; chr16:10907539-10907559; chr16:10895658-10895682; chr16:10895668-10895692; chr16:10895750-10895774; chr16:10895753-10895777; chr16:10895754-10895778; chr16:10898684-10898708; chr16:10901529-10901553; chr16:10902121-10902145; chr16:10902701-10902725; chr16:10904726-10904750; chr16:10904760-10904784; chr16:10906493-10906517; chr16:10906515-10906539; chr16:10906631-10906655; chr16:10906636-10906660; chr16:10906643-10906667; chr16:10906770-10906794; chr16:10906788-10906812; chr16:10906789-10906813; chr16:10906816-10906840; chr16:10907148-10907172; chr16:10907254-10907278; chr16:10907331-10907355; chr16:10907477-

10907501; chr16:10907497-10907521; chr16:10907503-10907527; chr16:10907574-10907598; or (b) chr16:10906889-10906913; or chr16:10907504-10907528.

[0340] In some embodiments, an engineered cell which has reduced or eliminated surface expression of MHC class II relative to an unmodified cell is provided, comprising a genetic modification in the CIITA gene, wherein the modification comprises at least one nucleotide within the genomic coordinates chosen from: chr16:10895658-10895682; chr16:10902701-10902725; chr16:10906493-10906517; chr16:10906631-10906655; chr16:10906643-10906667; chr16:10907477-10907501; and chr16:10907497-10907521; chr16:10907504-10907528; and chr16:10907508-10907532.

[0341] In some embodiments, an engineered cell, which has reduced or eliminated surface expression of MHC class II relative to an unmodified cell, comprises a genetic modification in the CIITA gene, the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from: (a) chr16:10877363-10907788 or (b) chr16:10906515-10908136.

[0342] In some embodiments, an engineered cell, which has reduced or eliminated surface expression of MHC class II relative to an unmodified cell, comprises a genetic modification in the CIITA gene, the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from: chr16:10907504-10907528; chr16:10907508-10907532; chr16:10907539-10907559; chr16:10895658-10895682; chr16:10895668-10895692; chr16:10895750-10895774; chr16:10895753-10895777; chr16:10895754-10895778; chr16:10898684-10898708; chr16:10901529-10901553; chr16:10902121-10902145; chr16:10902701-10902725; chr16:10904726-10904750; chr16:10904760-10904784; chr16:10906493-10906517; chr16:10906515-10906539; chr16:10906631-10906655; chr16:10906636-10906660; chr16:10906643-10906667; chr16:10906770-10906794; chr16:10906788-10906812; chr16:10906789-10906813; chr16:10906816-10906840; chr16:10907148-10907172; chr16:10907254-10907278; chr16:10907331-10907355; chr16:10907477-10907501; chr16:10907497-10907521; chr16:10907503-10907527; chr16:10907574-10907598; or (b) chr16:10906889-10906913; or chr16:10907504-10907528.

[0343] In some embodiments, an engineered cell, which has reduced or eliminated surface expression of MHC class II relative to an unmodified cell, comprises a genetic modification in the CIITA gene, the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from: chr16:10895658-10895682; chr16:10902701-10902725; chr16:10906493-10906517; chr16:10906631-10906655; chr16:10906643-10906667; chr16:10907477-10907501; and chr16:10907497-10907521; chr16:10907504-10907528; and chr16:10907508-10907532.

[0344] In some embodiments, an engineered cell, which has reduced or eliminated surface expression of MHC class II relative to an unmodified cell, comprises a genetic modification in the CIITA gene, wherein the genetic modification comprises an indel, a C to T substitution, or an A to G substitution within the genomic coordinates chosen from: (a) chr16:10907504-10907528; chr16:10907508-10907532; chr16:10907539-10907559; chr16:10895658-10895682; chr16:10895668-10895692; chr16:10895750-10895774; chr16:10895753-10895777; chr16:10895754-10895778; chr16:10898684-10898708; chr16:10901529-10901553; chr16:10902121-10902145; chr16:10902701-10902725; chr16:10904726-10904750; chr16:10904760-10904784; chr16:10906493-10906517; chr16:10906515-10906539; chr16:10906631-10906655; chr16:10906636-10906660; chr16:10906643-10906667; chr16:10906770-10906794; chr16:10906788-10906812; chr16:10906789-10906813; chr16:10906816-10906840; chr16:10907148-10907172; chr16:10907254-10907278; chr16:10907331-10907355; chr16:10907477-10907501; chr16:10907497-10907521; chr16:10907503-10907527; chr16:10907574-10907598; or (b) chr16:10906889-10906913; or chr16:10907504-10907528. In some embodiments, the genetic modification comprises at least 5 contiguous nucleotides within the genomic coordinates. In some embodiments, the genetic modification comprises at least 6, 7, 8, 9, or 10 contiguous nucleotides within the genomic coordinates. In some embodiments, the genetic modification comprises at least one C to T substitution or at least one A to G substitution within the genomic coordinates.

[0345] In some embodiments, an engineered cell, which has reduced or eliminated surface expression of MHC class II relative to an unmodified cell, comprises a genetic modification in the CIITA gene, wherein the genetic modification comprises an indel, a C to T substitution, or an A to G substitution within the genomic coordinates chosen from: chr16:10895658-10895682; chr16:10902701-10902725; chr16:10906493-10906517; chr16:10906631-10906655; chr16:10906643-10906667; chr16:10907477-10907501; and chr16:10907497-10907521; chr16:10907504-10907528; and chr16:10907508-10907532. In some embodiments, the genetic modification comprises at least 5 contiguous nucleotides within the genomic coordinates. In some embodiments, the genetic modification comprises at least 6, 7, 8, 9, or 10 contiguous nucleotides within the genomic coordinates. In some embodiments, the genetic modification comprises at least one C to T substitution or at least one A to G substitution within the genomic coordinates.

[0346] In some embodiments, an engineered cell is provided that has reduced or eliminated surface expression of MHC class II by a genomic editing system that binds to a CIITA genomic target sequence comprising at least 10 contiguous nucleotides within the genomic coordinates chosen from: (a) chr16:10907504-10907528; chr16:10907508-10907532; chr16:10907539-10907559; chr16:10895658-10895682; chr16:10895668-10895692; chr16:10895750-10895774; chr16:10895753-10895777; chr16:10895754-10895778; chr16:10898684-10898708; chr16:10901529-10901553; chr16:10902121-10902145; chr16:10902701-10902725; chr16:10904726-10904750; chr16:10904760-10904784; chr16:10906493-10906517; chr16:10906515-10906539; chr16:10906631-10906655; chr16:10906636-10906660; chr16:10906643-10906667; chr16:10906770-10906794; chr16:10906788-10906812; chr16:10906789-10906813; chr16:10906816-10906840; chr16:10907148-10907172; chr16:10907254-10907278; chr16:10907331-10907355; chr16:10907477-10907501; chr16:10907497-10907521; chr16:10907503-10907527; chr16:10907574-10907598; or (b) chr16:10906889-10906913; or chr16:10907504-10907528.

[0347] In some embodiments, an engineered cell is provided that has reduced or eliminated surface expression of MHC class II by a genomic editing system that binds to a CIITA genomic target sequence comprising at least 10 contiguous nucleotides within the genomic coordinates chosen from: (a) chr16:10895658-10895682; chr16:10902701-10902725; chr16:10906493-10906517; chr16:10906631-10906655; chr16:10906643-10906667; chr16:10907477-10907501; and chr16:10907497-10907521; chr16:10907504-10907528; and chr16:10907508-10907532. In some embodiments, an engineered cell is provided that has reduced or eliminated surface expression of MHC class II by a genomic editing system that binds to a CIITA genomic target sequence comprising at least 5 contiguous nucleotides within the genomic coordinates. In some embodiments, the CIITA genomic target sequence comprises at least 15 contiguous nucleotides within the genomic coordinates. In some embodiments, the genomic editing system comprises an RNA-guided DNA-binding agent. In some embodiments, the RNA-guided DNA-binding agent comprises a Cas9 protein, such as an *N. meningitidis* Cas9.

[0348] In some embodiments, an engineered cell is provided that has reduced or eliminated surface expression of MHC class II by a genomic editing system that binds to a CIITA genomic target sequence comprising at least 5 contiguous nucleotides within the genomic coordinates chosen from: (a) chr16:10907504-10907528; chr16:10907508-10907532; chr16:10907539-10907559; chr16:10895658-10895682; chr16:10895668-10895692; chr16:10895750-10895774; chr16:10895753-10895777; chr16:10895754-10895778; chr16:10898684-10898708; chr16:10901529-10901553; chr16:10902121-10902145; chr16:10902701-10902725; chr16:10904726-10904750; chr16:10904760-10904784; chr16:10906493-10906517; chr16:10906515-10906539; chr16:10906631-10906655; chr16:10906636-10906660; chr16:10906643-10906667; chr16:10906770-10906794; chr16:10906788-10906812; chr16:10906789-10906813; chr16:10906816-10906840; chr16:10907148-10907172; chr16:10907254-10907278; chr16:10907331-10907355; chr16:10907477-10907501; chr16:10907497-10907521; chr16:10907503-10907527; chr16:10907574-10907598; or (b) chr16:10906889-10906913; or chr16:10907504-10907528.

[0349] In some embodiments, an engineered cell is provided that has reduced or eliminated surface expression of MHC class II by a genomic editing system that binds to a CIITA genomic target sequence comprising at least 5 contiguous nucleotides within the genomic coordinates chosen from: chr16:10895658-10895682; chr16:10902701-10902725; chr16:10906493-10906517; chr16:10906631-10906655; chr16:10906643-10906667; chr16:10907477-10907501; and chr16:10907497-10907521; chr16:10907504-10907528; and chr16:10907508-10907532 . . . In some embodiments, the CIITA genomic target sequence comprises at least 10 contiguous nucleotides within the genomic coordinates. In some embodiments, the CIITA genomic target sequence comprises at least 15 contiguous nucleotides within the genomic coordinates. In some embodiments, the genomic editing system comprises an RNA-guided DNA-binding agent. In some embodiments, the RNA-guided DNA-binding agent comprises a Cas9 protein, such as an N. *Meningitidis.*

[0350] In some embodiments, an engineered cell is provided comprising a genetic modification in the AAVS1 gene, wherein the modification comprises at least one nucleotide from within the genomic coordinates chr19:55115151-55116209.

[0351] In some embodiments, an engineered cell is provided, comprising a genetic modification in the AAVS1 gene, wherein the modification comprises at least one nucleotide from within the genomic coordinates chosen from: chr19:55115218-55115242; chr19:55115477-55115501; chr19:55115504-55115528; chr19:55115513-55115537; chr19:55115514-55115538; chr19:55115517-55115541; chr19:55115518-55115542; chr19:55115549-55115573; chr19:55115574-55115598; chr19:55115606-55115630; chr19:55115933-55115957; chr19:55116026-55116050; chr19:55116045-55116069; chr19:55116084-55116108; chr19:55115276-55115300; chr19:55115509-55115533; chr19:55115579-55115603; chr19:55115863-55115887; chr19:55115906-55115930; orchr19:55116006-55116030.

[0352] In some embodiments, an engineered cell is provided, comprising a genetic modification in the AAVS1 gene, wherein the modification comprises at least 5 contiguous nucleotides within the genomic coordinates chr19:55115151-55116209. In some embodiments, the engineered cell comprises a genetic modification in the AAVS1 gene, wherein the modification comprises at least 6, 7, 8, 9, or 10 contiguous nucleotides within the genomic coordinates.

[0353] In some embodiments, an engineered cell is provided, comprising a genetic modification in the AAVS1 gene, wherein the modification comprises at least 5 contiguous nucleotides within the genomic coordinates chosen from: chr19:55115218-55115242; chr19:55115477-55115501; chr19:55115504-55115528; chr19:55115513-55115537; chr19:55115514-55115538; chr19:55115517-55115541; chr19:55115518-55115542; chr19:55115549-55115573; chr19:55115574-55115598; chr19:55115606-55115630; chr19:55115933-55115957; chr19:55116026-55116050; chr19:55116045-55116069; chr19:55116084-55116108; chr19:55115276-55115300; chr19:55115509-55115533; chr19:55115579-55115603; chr19:55115863-55115887; chr19:55115906-55115930; or chr19:55116006-55116030. In some embodiments, the engineered cell comprises a genetic modification in the AAVS1 gene, wherein the modification comprises at least 6, 7, 8, 9, or 10 contiguous nucleotides within the genomic coordinates.

[0354] In some embodiments, an engineered cell is provided, comprising a genetic modification in the AAVS1 gene, wherein the modification comprises at least one C to T substitution or at least one A to G substitution within the genomic coordinates chr19:55115151-55116209.

[0355] In some embodiments, an engineered cell is provided, comprising a genetic modification in the AAVS1 gene, wherein the modification comprises at least one C to T substitution or at least one A to G substitution within the genomic coordinates chosen from: chr19:55115218-55115242; chr19:55115477-55115501; chr19:55115504-55115528; chr19:55115513-55115537; chr19:55115514-55115538; chr19:55115517-55115541; chr19:55115518-55115542; chr19:55115549-55115573; chr19:55115574-55115598; chr19:55115606-55115630; chr19:55115933-55115957; chr19:55116026-55116050; chr19:55116045-55116069; chr19:55116084-55116108; chr19:55115276-55115300; chr19:55115509-55115533; chr19:55115579-55115603; chr19:55115863-55115887; chr19:55115906-55115930; or chr19:55116006-55116030.

[0356] In some embodiments, an engineered cell is provided that comprises a genetic modification in the AAVS1 gene, wherein the genetic modification comprises an indel, a C to T substitution, or an A to G substitution within the genomic coordinates chosen from: chr19:55115218-

55115242; chr19:55115477-55115501; chr19:55115504-55115528; chr19:55115513-55115537; chr19:55115514-55115538; chr19:55115517-55115541; chr19:55115518-55115542; chr19:55115549-55115573; chr19:55115574-55115598; chr19:55115606-55115630; chr19:55115933-55115957; chr19:55116026-55116050; chr19:55116045-55116069; chr19:55116084-55116108; chr19:55115276-55115300; chr19:55115509-55115533; chr19:55115579-55115603; chr19:55115863-55115887; chr19:55115906-55115930; or chr19:55116006-55116030. In some embodiments, the genetic modification comprises at least 5 contiguous nucleotides within the genomic coordinates. In some embodiments, the genetic modification comprises at least 6, 7, 8, 9, or 10 contiguous nucleotides within the genomic coordinates. In some embodiments, the genetic modification comprises at least one C to T substitution or at least one A to G substitution within the genomic coordinates.

[0357] In some embodiments, an engineered cell is provided that has a genetic modification in AAVS1 induced by a genomic editing system that binds to an AAVS1 genomic target sequence comprising at least 10 contiguous nucleotides within the genomic coordinates chosen from: chr19:55115218-55115242; chr19:55115477-55115501; chr19:55115504-55115528; chr19:55115513-55115537; chr19:55115514-55115538; chr19:55115517-55115541; chr19:55115518-55115542; chr19:55115549-55115573; chr19:55115574-55115598; chr19:55115606-55115630; chr19:55115933-55115957; chr19:55116026-55116050; chr19:55116045-55116069; chr19:55116084-55116108; chr19:55115276-55115300; chr19:55115509-55115533; chr19:55115579-55115603; chr19:55115863-55115887; chr19:55115906-55115930; or chr19:55116006-55116030. In some embodiments, an engineered cell is provided that has a genetic modification in AAVS1 induced by a genomic editing system that binds to an AAVS1 genomic target sequence comprising at least 5 contiguous nucleotides within the genomic coordinates. In some embodiments, the AAVS1 genomic target sequence comprises at least 15 contiguous nucleotides within the genomic coordinates. In some embodiments, the genomic editing system comprises an RNA-guided DNA-binding agent. In some embodiments, the RNA-guided DNA-binding agent comprises a Cas9 protein, such as an *N. meningitidis* Cas9.

[0358] In some embodiments, an engineered cell is provided that has a genetic modification in AAVS1 induced by a genomic editing system that binds to an AAVS1 genomic target sequence comprising at least 5 contiguous nucleotides within the genomic coordinates chosen from: chr19:55115218-55115242; chr19:55115477-55115501; chr19:55115504-55115528; chr19:55115513-55115537; chr19:55115514-55115538; chr19:55115517-55115541; chr19:55115518-55115542; chr19:55115549-55115573; chr19:55115574-55115598; chr19:55115606-55115630; chr19:55115933-55115957; chr19:55116026-55116050; chr19:55116045-55116069; chr19:55116084-55116108; chr19:55115276-55115300; chr19:55115509-55115533; chr19:55115579-55115603; chr19:55115863-55115887; chr19:55115906-55115930; or chr19:55116006-55116030. In some embodiments, the AAVS1 genomic target sequence comprises at least 10 contiguous nucleotides within the genomic coordinates. In some embodiments, the AAVS1 genomic target sequence comprises at least 15 contiguous nucleotides within the genomic coordinates. In some embodiments, the genomic editing system comprises an RNA-guided DNA-binding agent. In some embodiments, the RNA-guided DNA-binding agent comprises a Cas9 protein, such as an N. *Meningitidis.*

[0359] In some embodiments, for each given range of genomic coordinates, a range may encompass +/−10 nucleotides on either end of the specified coordinates. For each given range of genomic coordinates, the range may encompass +/−5 nucleotides on either end of the range. For example, if chr16:10923222-10923242 is given, in some embodiments the genomic target sequence or genetic modification may fall within chr16:10923212-10923252.

[0360] In some embodiments, a given range of genomic coordinates may comprise a target sequence on both strands of the DNA (i.e., the plus (+) strand and the minus (−) strand).

[0361] Genetic modifications in the HLA-A, TRAC, TRBC, CIITA, and AAVS1 genes are described further herein.

[0362] In some embodiments, a genetic modification in the HLA-A, TRAC, TRBC, CIITA, or AAVS1 locus comprises any one or more of an insertion, deletion, substitution, or deamination of at least one nucleotide in a target sequence. In some embodiments, the engineered cell which has reduced or eliminated surface expression of HLA-A relative to an unmodified cell is provided, comprising a genetic modification in the HLA-A gene. In some embodiments, the engineered cell which has reduced or eliminated surface expression of TRAC relative to an unmodified cell is provided, comprising a genetic modification in the TRAC gene. In some embodiments, the engineered cell which has reduced or eliminated surface expression of TRBC1 relative to an unmodified cell is provided, comprising a genetic modification in the TRBC1 gene. In some embodiments, the engineered cell which has reduced or eliminated surface expression of TRBC2 relative to an unmodified cell is provided, comprising a genetic modification in the TRBC2 gene. In some embodiments, the engineered cell which has reduced or eliminated surface expression of MHC class II relative to an unmodified cell is provided, comprising a genetic modification in the CIITA gene.

[0363] In some embodiments, the engineered cell which has reduced or eliminated surface expression of HLA-A, TRAC, TRBC, or MHC class II relative to an unmodified cell is provided, comprising a genetic modification in the HLA-A, TRAC, TRBC, or CIITA gene, wherein the cell further has reduced or eliminated expression of an endogenous TCR protein relative to an unmodified cell. In some embodiments, the engineered cell which has reduced or eliminated surface expression of HLA-A, TRAC, TRBC, or MHC class II relative to an unmodified cell is provided, comprising a genetic modification in the HLA-A, TRAC, TRBC, or CIITA gene, wherein the cell further comprises an exogenous nucleic acid, and further has reduced or eliminated expression of an endogenous TCR protein relative to an unmodified cell. In some embodiments, the engineered cell which has reduced or eliminated surface expression of HLA-A, TRAC, TRBC, or MHC class II relative to an unmodified cell is provided, comprising a genetic modification in the HLA-A, TRAC, TRBC, or CIITA gene, wherein the cell further has reduced or eliminated surface expression of MHC class I, and wherein the cell further has reduced or eliminated expression of an endogenous TCR protein relative to an unmodified cell.

[0364] In some embodiments, the engineered cell which has reduced or eliminated surface expression of HLA-A, TRAC, TRBC, or MHC class II relative to an unmodified cell is provided, comprising a genetic modification in the HLA-A, TRAC, TRBC, or CIITA gene, wherein the cell further comprises an exogenous nucleic acid, and wherein the cell further has reduced or eliminated surface expression of MHC class I, and wherein the cell further has reduced or eliminated expression of an endogenous TCR protein relative to an unmodified cell. In some embodiments, the engineered cell has reduced or eliminated expression of a TRAC protein relative to an unmodified cell. In some embodiments, the engineered cell has reduced or eliminated expression of a TRBC protein relative to an unmodified cell.

[0365] In some embodiments, the engineered cell which has reduced or eliminated surface expression of MHC class II relative to an unmodified cell is provided, comprising a genetic modification in the CIITA gene, wherein the modification comprises at least one nucleotide of an exon within the genomic coordinates chr16:10902662-chr16:10923285, and wherein the cell further comprises an exogenous nucleic acid, and wherein the cell further has reduced or eliminated surface expression of HLA-A, and wherein the cell further has reduced or eliminated expression of an endogenous TCR protein relative to an unmodified cell. In some embodiments, the engineered cell has reduced or eliminated expression of a TRAC protein relative to an unmodified cell. In some embodiments, the engineered cell has reduced or eliminated expression of a TRBC protein relative to an unmodified cell. In some embodiments, the engineered cell comprises a genetic modification in the HLA-A gene. In some embodiments, the engineered cell comprises a genetic modification that reduces expression of HLA-A protein on the surface of the engineered cell.

[0366] The engineered cell may be any of the exemplary cell types disclosed herein. In some embodiments, the engineered cell is an immune cell. In some embodiments, the engineered cell is a hematopoetic stem cell (HSC). In some embodiments, the engineered cell is an induced pluripotent stem cell (iPSC). In some embodiments, the engineered cell is a monocyte, macrophage, mast cell, dendritic cell, or granulocyte. In some embodiments, the engineered cell is monocyte. In some embodiments, the engineered cell is a macrophage. In some embodiments, the engineered cell is a mast cell. In some embodiments, the engineered cell is a dendritic cell.

[0367] In some embodiments, the engineered cell is a granulocyte. In some embodiments, the engineered cell is a lymphocyte. In some embodiments, the engineered cell is a T cell. In some embodiments, the engineered cell is a CD4+ T cell. In some embodiments, the engineered cell is a CD8+ T cell. In some embodiments, the engineered cell is a memory T cell. In some embodiments, the engineered cell is a B cell. In some embodiments, the engineered cell is a plasma B cell. In some embodiments, the engineered cell is a memory B cell.

[0368] In some embodiments, the disclosure provides a pharmaceutical composition comprising any one of the engineered cells disclosed herein. In some embodiments, the pharmaceutical composition comprises a population of any one of the engineered cells disclosed herein. In some embodiments, the population of cells is at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 99.5% negative for the surface antigen (e.g., HLA-A, MHC Class II (HLA-DP, DQ, DR), or endogenous TCR) as measured by flow cytometry. In some embodiments, the population of engineered cells that is at least 65% negative as measured by flow cytometry. In some embodiments, the population of engineered cells that is at least 70% negative as measured by flow cytometry. In some embodiments, the population of engineered cells is at least 80% negative as measured by flow cytometry. In some embodiments, the population of engineered cells is at least 90% negative as measured by flow cytometry. In some embodiments, the population of engineered cells is at least 91% negative as measured by flow cytometry. In some embodiments, the population of engineered cells is at least 92% negative as measured by flow cytometry. In some embodiments, the population of engineered cells is at least 93% negative as measured by flow cytometry. In some embodiments, the population of engineered cells is at least 94% negative as measured by flow cytometry. In some embodiments, the population of engineered cells is at least 95% endogenous TCR protein negative as measured by flow cytometry. In some embodiments, the population of engineered cells is at least 97% endogenous TCR protein negative as measured by flow cytometry. In some embodiments, the population of engineered cells is at least 98% endogenous TCR protein negative as measured by flow cytometry. In some embodiments, the population of engineered cells is at least 99% endogenous TCR protein negative as measured by flow cytometry.

[0369] In some embodiments, methods are provided for administering the engineered cells or pharmaceutical compositions disclosed herein to a subject in need thereof. In some embodiments, methods are provided for administering the engineered cells or pharmaceutical compositions disclosed herein to a subject as an ACT therapy. In some embodiments, methods are provided for administering the engineered cells or pharmaceutical compositions disclosed herein to a subject as a treatment for cancer. In some embodiments, methods are provided for administering the engineered cells or pharmaceutical compositions disclosed herein to a subject as a treatment for an autoimmune disease. In some embodiments, methods are provided for administering the engineered cells or pharmaceutical compositions disclosed herein to a subject as a treatment for an infectious disease.

B. Methods and Compositions for Reducing or Eliminating Surface Expression of HLA-A, TRAC, TRBC, and MHC Class II

[0370] The present disclosure provides methods and compositions for reducing or eliminating surface expression of HLA-A, TRAC, TRBC, or MHC class II protein on a cell relative to an unmodified cell by genetically modifying the HLA-A, TRAC, TRBC, or CIITA gene. The resultant genetically modified cell may also be referred to herein as an engineered cell. In some embodiments, an already-genetically modified (or engineered) cell may be the starting cell for further genetic modification using the methods or compositions provided herein. In some embodiments, the cell is an allogeneic cell. In some embodiments, a cell with reduced HLA-A, TRAC, TRBC, or MHC class II expression is useful for adoptive cell transfer therapies. In some embodiments, editing of the HLA-A, TRAC, TRBC, or CIITA gene is combined with additional genetic modifications to yield a cell that is desirable for allogeneic transplant purposes.

[0371] In some embodiments, the methods comprise reducing or eliminating surface expression of HLA-A protein on the surface of a cell comprising contacting a cell with a composition comprising an HLA-A guide RNA comprising a guide sequence that targets an HLA-A genomic target sequence comprising at least 10 contiguous nucleotides within the genomic coordinates chr6:29942540-29945459. In some embodiments, the methods further comprise contacting the cell with an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent. In some embodiments, the RNA-guided DNA binding agent is Cas9. In some embodiments, the RNA-guided DNA binding agent is *N. meningitidis* Cas9. In some embodiments, the HLA-A guide RNA is a *N. meningitidis* Cas9 guide RNA. In some embodiments, the RNA-guided DNA binding agent comprises a deaminase domain. In some embodiments the RNA-guided DNA binding agent comprises an APOBEC3A deaminase (A3A) and an RNA-guided nickase. In some embodiments, the expression of HLA-A protein on the surface of the cell (i.e., engineered cell) is thereby reduced. In some embodiments, the HLA-A guide RNA comprises a guide sequence selected from SEQ ID NO: 2-80. In some embodiments, the methods comprise making an engineered cell, which has reduced or eliminated surface expression of HLA-A protein relative to an unmodified cell, comprising contact the cell with a composition comprising an HLA-A guide RNA comprising a guide sequence that targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chr6:29942540-29945459. In some embodiments, the methods further comprise contacting the cell with an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent. In some embodiments, the RNA-guided DNA binding agent is Cas9. In some embodiments, the RNA-guided DNA binding agent is *N. meningitidis* Cas9. In some embodiments, the HLA-A guide RNA is a *N. meningitidis* Cas9 guide RNA. In some embodiments, the RNA-guided DNA binding agent comprises a deaminase region. In some embodiments the RNA-guided DNA binding agent comprises an APOBEC3A deaminase (A3A) and an RNA-guided nickase. In some embodiments, the expression of HLA-A protein on the surface of the cell (i.e., engineered cell) is thereby reduced. In some embodiments, the HLA-A guide RNA comprises a guide sequence selected from SEQ ID NO: 2-80.

[0372] In some embodiments, the methods comprise genetically modifying a cell to reduce or eliminate the surface expression of HLA-A protein comprising contacting the cell with a composition comprising an HLA-A guide RNA comprising a guide sequence that targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chr6:29942540-29945459. In some embodiments, the methods further comprise contacting the cell with an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent. In some embodiments, the RNA-guided DNA binding agent is Cas9. In some embodiments, the RNA-guided DNA binding agent is *N. meningitidis* Cas9. In some embodiments, the HLA-A guide RNA is a *N. meningitidis* Cas9 guide RNA. In some embodiments, the RNA-guided DNA binding agent comprises a deaminase region. In some embodiments the RNA-guided DNA binding agent comprises an APOBEC3A deaminase (A3A) and an RNA-guided nickase. In some embodiments, the expression of HLA-A protein on the surface of the cell (i.e., engineered cell) is thereby reduced. In some embodiments, the HLA-A guide RNA comprises a guide sequence selected from SEQ ID NO: 2-80.

[0373] In some embodiments, the methods comprise genetically modifying HLA-A comprising contacting a cell with a composition comprising an HLA-A guide RNA comprising a guide sequence that targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chr6:29942540-29945459. In some embodiments, the methods further comprise contacting the cell with an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent. In some embodiments, the RNA-guided DNA binding agent is Cas9. In some embodiments, the RNA-guided DNA binding agent is *N. meningitidis* Cas9. In some embodiments, the HLA-A guide RNA is a *N. meningitidis* Cas9 guide RNA. In some embodiments, the RNA-guided DNA binding agent comprises a deaminase region. In some embodiments the RNA-guided DNA binding agent comprises an APOBEC3A deaminase (A3A) and an RNA-guided nickase. In some embodiments, the expression of HLA-A protein on the surface of the cell (i.e., engineered cell) is thereby reduced. In some embodiments, the HLA-A guide RNA comprises a guide sequence selected from SEQ ID NO: 2-80.

[0374] In some embodiments, the methods comprise inducing a DSB or a single stranded break (SSB) in HLA-A comprising contacting a cell with a composition comprising an HLA-A guide RNA comprising a guide sequence targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chr6:29942540-29945459. In some embodiments, the methods further comprise contacting the cell with an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent. In some embodiments, the RNA-guided DNA binding agent is Cas9. In some embodiments, the RNA-guided DNA binding agent is *S. pyogenes* Cas9. In some embodiments, the HLA-A guide RNA is a *N. meningitidis* Cas9 guide RNA. In some embodiments, the RNA-guided DNA binding agent comprises a deaminase region. In some embodiments the RNA-guided DNA binding agent comprises an APOBEC3A deaminase (A3A) and an RNA-guided nickase. In some embodiments, the expression of HLA-A protein on the surface of the cell (i.e., engineered cell) is thereby reduced. In some embodiments, the HLA-A guide RNA comprises a guide sequence selected from SEQ ID NO: 2-80.

[0375] In some embodiments, the methods comprise reducing expression of the HLA-A protein in a cell comprising delivering a composition to a cell comprising contacting a cell with a composition comprising an HLA-A guide RNA comprising a guide sequence targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chr6:29942540-29945459. In some embodiments, the methods further comprise contacting the cell with an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent. In some embodiments, the RNA-guided DNA binding agent is Cas9. In some embodiments, the RNA-guided DNA binding agent is *N. meningitidis* Cas9. In some embodiments, the HLA-A guide RNA is a *N. meningitidis* Cas9 guide RNA. In some embodiments, the RNA-guided DNA binding agent comprises a deaminase region. In some embodiments the RNA-guided DNA binding agent comprises an APOBEC3A deaminase (A3A) and an RNA-guided nickase. In some embodiments, the expression of HLA-A protein on the surface of the cell (i.e., engineered cell) is thereby reduced. In some embodiments, the HLA-A guide RNA comprises a guide sequence selected from SEQ ID NO: 2-80.

[0376] In some embodiments, the methods of reducing expression of an HLA-A protein on the surface of a cell comprise contacting a cell with any one or more of the HLA-A guide RNAs disclosed herein. In some embodiments, the HLA-A guide RNA comprises a guide sequence selected from SEQ ID NO: 2-80.

[0377] In some embodiments, compositions are provided comprising an HLA-A guide RNA comprising a guide sequence that targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chr6:29942540-29945459. In some embodiments, the composition further comprises an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent. In some embodiments, the composition comprises an RNA-guided DNA binding agent that is Cas9. In some embodiments, the RNA-guided DNA binding agent is *N. meningitidis* Cas9. In some embodiments, the HLA-A guide RNA is a *N. meningitidis* Cas9 guide RNA. In some embodiments, the RNA-guided DNA binding agent comprises a deaminase region. In some embodiments the RNA-guided DNA binding agent comprises an APOBEC3A deaminase (A3A) and an RNA-guided nickase. In some embodiments, the HLA-A guide RNA comprises a guide sequence selected from SEQ ID NO: 2-80.

[0378] In some embodiments, a composition is provided, the composition comprising: a) an HLA-A guide RNA (gRNA) comprising i) a guide sequence selected from SEQ ID NOs: 2-80; or ii) at least 19, 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence selected from SEQ ID NOs: 2-80; or iii) a guide sequence at least 95%, 90%, or 85% identical to a sequence selected from SEQ ID NOs: 2-80; or iv) a sequence that comprises 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 1; or v) at least 19, 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence from (iv); or vi) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence selected from (iv). In some embodiments, the HLA-A guide RNA (gRNA) is a single guide RNA.

[0379] In some embodiments, a method of making an engineered cell, which has reduced or eliminated surface expression of HLA-A protein relative to an unmodified cell, is provided, the method comprising contacting a cell with a composition of any of the embodiments provided herein. In some embodiments, the composition comprises an HLA-A guide RNA, comprising a guide sequence of any one of: SEQ ID NOs: 1, 13, 55, 61, 66, 70, and 71. In some embodiments, the composition comprises an HLA-A guide RNA, comprising a guide sequence of any one of: SEQ ID NOs: 13, 55, 61, 66, 70, and 71. In some embodiments, the composition comprises an HLA-A guide RNA, comprising a guide sequence of any one of: SEQ ID NOs: 13, 17, 55, 61, 66, and 70.

[0380] In some embodiments, a method of reducing surface expression of HLA-A protein in an engineered cell relative to an unmodified cell, is provided, the method comprising contacting a cell with a composition of any of embodiments provided herein.

[0381] In some embodiments, the composition further comprises a uracil glycosylase inhibitor (UGI). In some embodiments, the composition comprises an RNA-guided DNA binding agent that the RNA-guided DNA binding agent generates a cytosine (C) to thymine (T) conversion with the HLA-A genomic target sequence. In some embodiments, the composition comprises an RNA-guided DNA binding agent that generates a adenosine (A) to guanine (G) conversion with the HLA-A genomic target sequence.

[0382] In some embodiments, an engineered cell produced by the methods described herein is provided. In some embodiments, the engineered cell produced by the methods and compositions described herein is an allogeneic cell. In some embodiments, the methods produce a composition comprising an engineered cell having reduced HLA-A expression. In some embodiments, the methods produce a composition comprising an engineered cell having reduced HLA-A protein expression. In some embodiments, the methods produce a composition comprising an engineered cell having reduced HLA-A levels in the cell nucleus. In some embodiments, the methods produce a composition comprising an engineered cell that expresses a truncated form of the HLA-A protein. In some embodiments, the methods produce a composition comprising an engineered cell that produces no detectable HLA-A protein. In some embodiments, the engineered cell has reduced HLA-A expression, reduced HLA-A protein, or reduced HLA-A levels in the cell nucleus as compared to an unmodified cell. In some embodiments, the engineered cell produced by the methods disclosed herein elicits a reduced response from CD4+ T cells as compared to an unmodified cell as measured in an in vitro cell culture assay containing CD4+ T cells.

[0383] In some embodiments, an engineered cell produced by the methods or compositions disclosed herein is provided wherein the cell has reduced or eliminated surface expression of HLA-A protein and wherein the cell comprises a genetic modification comprising at least 5 contiguous nucleotides within the genomic coordinates chr6:29942540-29945459. In some embodiments, an engineered cell produced by the methods or compositions disclosed herein is provided wherein the cell has reduced or eliminated surface expression of HLA-A protein and wherein the cell comprises a genetic modification comprising at least 10 contiguous nucleotides within the genomic coordinates chr6:29942540-29945459. In some embodiments, an engineered cell produced by the methods or compositions disclosed herein is provided wherein the cell has reduced or eliminated surface expression of HLA-A protein and wherein the cell comprises a genetic modification comprising at least one C to T substitution or at least one A to G substitution within the genomic coordinates chr6:29942540-29945459.

[0384] In some embodiments, the methods comprise reducing or eliminating surface expression of TRAC protein on the surface of a cell comprising contacting a cell with a composition comprising a TRAC guide RNA comprising a guide sequence that targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chr14:22547505-22551621 or chr14:22547462-22551621. In some embodiments, the methods further comprise contacting the cell with an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent. In some embodiments, the RNA-guided DNA binding agent is Cas9. In some embodiments, the RNA-guided DNA binding agent is *N. meningitidis* Cas9. In some embodiments, the TRAC guide RNA is a *N. meningitidis* Cas9 guide RNA. In some embodiments, the RNA-guided DNA binding agent comprises a deaminase domain. In some embodiments the RNA-guided DNA binding agent comprises an APOBEC3A deaminase (A3A) and an RNA-guided nickase. In some embodiments, the expression of TRAC protein on the surface of the cell (i.e., engineered cell) is thereby reduced. In some embodiments, the TRAC guide RNA comprises a guide sequence selected from SEQ ID NO: 101-120.

[0385] In some embodiments, the methods comprise making an engineered cell, which has reduced or eliminated surface expression of TRAC protein relative to an unmodified cell, comprising contact the cell with a composition comprising a TRAC guide RNA comprising a guide sequence that targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chr14:22547505-22551621 or chr14:22547462-22551621. In some embodiments, the methods further comprise contacting the cell with an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent. In some embodiments, the RNA-guided DNA binding agent is Cas9. In some embodiments, the RNA-guided DNA binding agent is *N. meningitidis* Cas9. In some embodiments, the TRAC guide RNA is a *N. meningitidis* Cas9 guide RNA. In some embodiments, the RNA-guided DNA binding agent comprises a deaminase region. In some embodiments the RNA-guided DNA binding agent comprises an APOBEC3A deaminase (A3A) and an RNA-guided nickase. In some embodiments, the expression of TRAC protein on the surface of the cell (i.e., engineered cell) is thereby reduced. In some embodiments, the TRAC guide RNA comprises a guide sequence selected from SEQ ID NO: 101-120.

[0386] In some embodiments, the methods comprise genetically modifying a cell to reduce or eliminate the surface expression of TRAC protein comprising contacting the cell with a composition comprising a TRAC guide RNA comprising a guide sequence that targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chr14:22547505-22551621 or chr14:22547462-22551621. In some embodiments, the methods further comprise contacting the cell with an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent. In some embodiments, the RNA-guided DNA binding agent is Cas9. In some embodiments, the RNA-guided DNA binding agent is *N. meningitidis* Cas9. In some embodiments, the TRAC guide RNA is a *N. meningitidis* Cas9 guide RNA. In some embodiments, the RNA-guided DNA binding agent comprises a deaminase region. In some embodiments the RNA-guided DNA binding agent comprises an APOBEC3A deaminase (A3A) and an RNA-guided nickase. In some embodiments, the expression of TRAC protein on the surface of the cell (i.e., engineered cell) is thereby reduced. In some embodiments, the TRAC guide RNA comprises a guide sequence selected from SEQ ID NO: 101-120.

[0387] In some embodiments, the methods comprise genetically modifying TRAC comprising contacting a cell with a composition comprising a TRAC guide RNA comprising a guide sequence that targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chr14:22547505-22551621 or chr14:22547462-22551621. In some embodiments, the methods further comprise contacting the cell with an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent. In some embodiments, the RNA-guided DNA binding agent is Cas9. In some embodiments, the RNA-guided DNA binding agent is *N. meningitidis* Cas9. In some embodiments, the TRAC guide RNA is a *N. meningitidis* Cas9 guide RNA. In some embodiments, the RNA-guided DNA binding agent comprises a deaminase region. In some embodiments the RNA-guided DNA binding agent comprises an APOBEC3A deaminase (A3A) and an RNA-guided nickase. In some embodiments, the expression of TRAC protein on the surface of the cell (i.e., engineered cell) is thereby reduced. In some embodiments, the TRAC guide RNA comprises a guide sequence selected from SEQ ID NO: 101-120.

[0388] In some embodiments, the methods comprise inducing a DSB or a single stranded break (SSB) in TRAC comprising contacting a cell with a composition comprising a TRAC guide RNA comprising a guide sequence targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chr14:22547505-22551621 or chr14:22547462-22551621. In some embodiments, the methods further comprise contacting the cell with an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent. In some embodiments, the RNA-guided DNA binding agent is Cas9. In some embodiments, the RNA-guided DNA binding agent is *S. pyogenes* Cas9. In some embodiments, the TRAC guide RNA is a *N. meningitidis* Cas9 guide RNA. In some embodiments, the RNA-guided DNA binding agent comprises a deaminase region. In some embodiments the RNA-guided DNA binding agent comprises an APOBEC3A deaminase (A3A) and an RNA-guided nickase. In some embodiments, the expression of TRAC protein on the surface of the cell (i.e., engineered cell) is thereby reduced. In some embodiments, the TRAC guide RNA comprises a guide sequence selected from SEQ ID NO: 101-120.

[0389] In some embodiments, the methods comprise reducing expression of the TRAC protein in a cell comprising delivering a composition to a cell comprising contacting a cell with a composition comprising a TRAC guide RNA comprising a guide sequence targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chr14:22547505-22551621 or chr14:22547462-22551621. In some embodiments, the methods further comprise contacting the cell with an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent. In some embodiments, the RNA-guided DNA binding agent is Cas9. In some embodiments, the RNA-guided DNA binding agent is *N. meningitidis* Cas9. In some embodiments, the TRAC guide RNA is a *N. meningitidis* Cas9 guide RNA. In some embodiments, the RNA-guided DNA binding agent comprises a deaminase region. In some embodiments the RNA-guided DNA binding agent comprises an APOBEC3A deaminase (A3A) and an RNA-guided nickase. In some embodiments, the expression of TRAC protein on the surface of the cell (i.e., engineered cell) is thereby reduced. In some embodiments, the TRAC guide RNA comprises a guide sequence selected from SEQ ID NO: 101-120.

[0390] In some embodiments, the methods of reducing expression of a TRAC protein on the surface of a cell comprise contacting a cell with any one or more of the

TRAC guide RNAs disclosed herein. In some embodiments, the TRAC guide RNA comprises a guide sequence selected from SEQ ID NO: 101-120.

[0391] In some embodiments, compositions are provided comprising a TRAC guide RNA comprising a guide sequence that targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chr14:22547505-22551621 or chr14:22547462-22551621. In some embodiments, the composition further comprises an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent. In some embodiments, the composition comprises an RNA-guided DNA binding agent that is Cas9. In some embodiments, the RNA-guided DNA binding agent is *N. meningitidis* Cas9. In some embodiments, the TRAC guide RNA is a *N. meningitidis* Cas9 guide RNA. In some embodiments, the RNA-guided DNA binding agent comprises a deaminase region. In some embodiments the RNA-guided DNA binding agent comprises an APOBEC3A deaminase (A3A) and an RNA-guided nickase. In some embodiments, the TRAC guide RNA comprises a guide sequence selected from SEQ ID NO: 101-120.

[0392] In some embodiments, a composition is provided, the composition comprising: a) a TRAC guide RNA (gRNA) comprising i) a guide sequence selected from SEQ ID NOs: 101-120; or ii) at least 19, 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence selected from SEQ ID NOs: 101-120; or iii) a guide sequence at least 95%, 90%, or 85% identical to a sequence selected from SEQ ID NOs: 101-120; or iv) a sequence that comprises 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 2; or v) at least 19, 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence from (iv); or vi) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence selected from (v). In some embodiments, the TRAC guide RNA (gRNA) is a single-guide RNA (sgRNA).

[0393] In some embodiments, a method of making an engineered cell, which has reduced or eliminated surface expression of TRAC protein relative to an unmodified cell, is provided, the method comprising contacting a cell with a composition of any of the embodiments provided herein. In some embodiments, the composition comprises a TRAC guide RNA, comprising a guide sequence of any one of: SEQ ID NO: 101, 102, 103, 105, 107, 109, 111, and 115. In some embodiments, the composition comprises a TRAC guide RNA comprising a guide sequence of any one of: SEQ ID NO: 101, 102, 103, 107, and 111.

[0394] In some embodiments, a method of reducing surface expression of TRAC protein in an engineered cell relative to an unmodified cell, is provided, the method comprising contacting a cell with a composition of any of embodiments provided herein.

[0395] embodiments, the composition further comprises a uracil glycosylase inhibitor (UGI). In some embodiments, the composition comprises an RNA-guided DNA binding agent that the RNA-guided DNA binding agent generates a cytosine (C) to thymine (T) conversion with the TRAC genomic target sequence. In some embodiments, the composition comprises an RNA-guided DNA binding agent that generates a adenosine (A) to guanine (G) conversion with the TRAC genomic target sequence.

[0396] In some embodiments, an engineered cell produced by the methods described herein is provided. In some embodiments, the engineered cell produced by the methods and compositions described herein is an allogeneic cell. In some embodiments, the methods produce a composition comprising an engineered cell having reduced TRAC expression. In some embodiments, the methods produce a composition comprising an engineered cell having reduced TRAC protein expression. In some embodiments, the methods produce a composition comprising an engineered cell having reduced TRAC levels in the cell nucleus. In some embodiments, the methods produce a composition comprising an engineered cell that expresses a truncated form of the TRAC protein. In some embodiments, the methods produce a composition comprising an engineered cell that produces no detectable TRAC protein. In some embodiments, the engineered cell has reduced TRAC expression, reduced TRAC protein, or reduced TRAC levels in the cell nucleus as compared to an unmodified cell. In some embodiments, the engineered cell produced by the methods disclosed herein elicits a reduced response from CD4+ T cells as compared to an unmodified cell as measured in an in vitro cell culture assay containing CD4+ T cells.

[0397] In some embodiments, an engineered cell produced by the methods or compositions disclosed herein is provided wherein the cell has reduced or eliminated surface expression of TRAC protein and wherein the cell comprises a genetic modification comprising at least 5 contiguous nucleotides within the genomic coordinates chr14: 22547505-22551621 or chr14:22547462-22551621. In some embodiments, an engineered cell produced by the methods or compositions disclosed herein is provided wherein the cell has reduced or eliminated surface expression of TRAC protein and wherein the cell comprises a genetic modification comprising at least 10 contiguous nucleotides within the genomic coordinates chr14: 22547505-22551621 or chr14:22547462-22551621. In some embodiments, an engineered cell produced by the methods or compositions disclosed herein is provided wherein the cell has reduced or eliminated surface expression of TRAC protein and wherein the cell comprises a genetic modification comprising at least one C to T substitution or at least one A to G substitution within the genomic coordinates chr14:22547505-22551621 or chr14:22547462-22551621.

[0398] In some embodiments, the methods comprise reducing or eliminating surface expression of TRBC protein on the surface of a cell comprising contacting a cell with a composition comprising a TRBC guide RNA comprising a guide sequence that targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chosen from chr7:142791756-142802543. In some embodiments, the methods comprise reducing or eliminating surface expression of TRBC protein on the surface of a cell comprising contacting a cell with a composition comprising a TRBC guide RNA comprising a guide sequence that targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chosen from: (a) chr7:142791862-142793149; (b) chr7: 142791756-142792721; or (c) chr7:142801104-142802543. In some embodiments, the methods further comprise contacting the cell with an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent. In some embodiments, the RNA-guided DNA binding agent is Cas9. In some embodiments, the RNA-guided DNA binding agent is *N. meningitidis* Cas9. In some embodiments, the TRBC guide RNA is a *N. meningitidis* Cas9 guide RNA. In some embodiments, the RNA-guided DNA binding agent comprises a deaminase domain. In some embodiments the RNA-guided DNA binding agent comprises an APOBEC3A deaminase (A3A) and an RNA-guided nickase. In some embodiments, the expression of TRBC protein on the surface of the cell (i.e., engineered cell) is thereby reduced. In some embodiments, the TRBC guide RNA comprises a guide sequence selected from SEQ ID NO: 201-265.

[0399] In some embodiments, the methods comprise making an engineered cell, which has reduced or eliminated surface expression of TRBC protein relative to an unmodified cell, comprising contact the cell with a composition comprising a TRBC guide RNA comprising a guide sequence that targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates coordinates chosen from: (a) chr7:142791862-142793149; (b) chr7: 142791756-142792721; or (c) chr7:142801104-142802543. In some embodiments, the methods further comprise contacting the cell with an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent. In some embodiments, the RNA-guided DNA binding agent is Cas9. In some embodiments, the RNA-guided DNA binding agent is *N. meningitidis* Cas9. In some embodiments, the TRBC guide RNA is a *N. meningitidis* Cas9 guide RNA. In some embodiments, the RNA-guided DNA binding agent comprises a deaminase region. In some embodiments the RNA-guided DNA binding agent comprises an APOBEC3A deaminase (A3A) and an RNA-guided nickase. In some embodiments, the expression of TRBC protein on the surface of the cell (i.e., engineered cell) is thereby reduced. In some embodiments, the TRBC guide RNA comprises a guide sequence selected from SEQ ID NO: 201-265.

[0400] In some embodiments, the methods comprise genetically modifying a cell to reduce or eliminate the surface expression of TRBC protein comprising contacting the cell with a composition comprising a TRBC guide RNA comprising a guide sequence that targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chosen from: (a) chr7:142791862-142793149; (b) chr7: 142791756-142792721; or (c) chr7:142801104-142802543. In some embodiments, the methods further comprise contacting the cell with an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent. In some embodiments, the RNA-guided DNA binding agent is Cas9. In some embodiments, the RNA-guided DNA binding agent is *N. meningitidis* Cas9. In some embodiments, the TRBC guide RNA is a *N. meningitidis* Cas9 guide RNA. In some embodiments, the RNA-guided DNA binding agent comprises a deaminase region. In some embodiments the RNA-guided DNA binding agent comprises an APOBEC3A deaminase (A3A) and an RNA-guided nickase. In some embodiments, the expression of TRBC protein on the surface of the cell (i.e., engineered cell) is thereby reduced. In some embodiments, the TRBC guide RNA comprises a guide sequence selected from SEQ ID NO: 201-265.

[0401] In some embodiments, the methods comprise genetically modifying TRBC comprising contacting a cell with a composition comprising a TRBC guide RNA comprising a guide sequence that targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates coordinates chosen from: (a) chr7:142791862-142793149; (b) chr7: 142791756-142792721; or (c) chr7:142801104-142802543. In some embodiments, the methods further comprise contacting the cell with an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent. In some embodiments, the RNA-guided DNA binding agent is Cas9. In some embodiments, the RNA-guided DNA binding agent is *N. meningitidis* Cas9. In some embodiments, the TRBC guide RNA is a *N. meningitidis* Cas9 guide RNA. In some embodiments, the RNA-guided DNA binding agent comprises a deaminase region. In some embodiments the RNA-guided DNA binding agent comprises an APOBEC3A deaminase (A3A) and an RNA-guided nickase. In some embodiments, the expression of TRBC protein on the surface of the cell (i.e., engineered cell) is thereby reduced. In some embodiments, the TRBC guide RNA comprises a guide sequence selected from SEQ ID NO: 201-265.

[0402] In some embodiments, the methods comprise inducing a DSB or a single stranded break (SSB) in TRBC comprising contacting a cell with a composition comprising a TRBC guide RNA comprising a guide sequence targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates coordinates chosen from: (a) chr7:142791862-142793149; (b) chr7: 142791756-142792721; or (c) chr7:142801104-142802543. In some embodiments, the methods further comprise contacting the cell with an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent. In some embodiments, the RNA-guided DNA binding agent is Cas9. In some embodiments, the RNA-guided DNA binding agent is *S. pyogenes* Cas9. In some embodiments, the TRBC guide RNA is a *N. meningitidis* Cas9 guide RNA. In some embodiments, the RNA-guided DNA binding agent comprises a deaminase region. In some embodiments the RNA-guided DNA binding agent comprises an APOBEC3A deaminase (A3A) and an RNA-guided nickase. In some embodiments, the expression of TRBC protein on the surface of the cell (i.e., engineered cell) is thereby reduced. In some embodiments, the TRBC guide RNA comprises a guide sequence selected from SEQ ID NO: 201-265.

[0403] In some embodiments, the methods comprise reducing expression of the TRBC protein in a cell comprising delivering a composition to a cell comprising contacting a cell with a composition comprising a TRBC guide RNA comprising a guide sequence targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates coordinates chosen from: (a) chr7: 142791862-142793149; (b) chr7: 142791756-142792721; or (c) chr7:142801104-142802543. In some embodiments, the methods further comprise contacting the cell with an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent. In some embodiments, the RNA-guided DNA binding agent is Cas9. In some embodiments, the RNA-guided DNA binding agent is *N. meningitidis* Cas9. In some embodiments, the TRBC guide RNA is a *N. meningitidis* Cas9 guide RNA. In some embodiments, the RNA-guided DNA binding agent comprises a deaminase region. In some embodiments the RNA-guided DNA binding agent comprises an APOBEC3A deaminase (A3A) and an RNA-guided nickase. In some embodiments, the expression of TRBC protein on the surface of the cell (i.e., engineered cell) is thereby reduced. In some embodiments, the TRBC guide RNA comprises a guide sequence selected from SEQ ID NO: 201-265.

[0404] In some embodiments, the methods of reducing expression of an TRBC protein on the surface of a cell comprise contacting a cell with any one or more of the TRBC guide RNAs disclosed herein. In some embodiments, the TRBC guide RNA comprises a guide sequence selected from SEQ ID NO: 201-265.

[0405] In some embodiments, compositions are provided comprising a TRBC guide RNA comprising a guide sequence that targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chosen from: (a) chr7:142791862-142793149; (b) chr7: 142791756-142792721; or (c) chr7:142801104-142802543. In some embodiments, the composition further comprises an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent. In some embodiments, the composition comprises an RNA-guided DNA binding agent that is Cas9. In some embodiments, the RNA-guided DNA binding agent is *N. meningitidis* Cas9. In some embodiments, the TRBC guide RNA is a *N. meningitidis* Cas9 guide RNA. In some embodiments, the RNA-guided DNA binding agent comprises a deaminase region. In some embodiments the RNA-guided DNA binding agent comprises an APOBEC3A deaminase (A3A) and an RNA-guided nickase. In some embodiments, the TRBC guide RNA comprises a guide sequence selected from SEQ ID NO: 201-265.

[0406] In some embodiments, a composition is provided, the composition comprising: a) a TRBC guide RNA (gRNA) comprising i) a guide sequence selected from SEQ ID NOs: 201-265; or ii) at least 19, 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence selected from SEQ ID NOs: 201-265; or iii) a guide sequence at least 95%, 90%, or 85% identical to a sequence selected from SEQ ID NOs: 201-265; or iv) a sequence that comprises 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 3; or v) at least 19, 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence from (iv); or vi) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence selected from (v). In some embodiments, the TRBC guide RNA that is a single-guide RNA (sgRNA).

[0407] In some embodiments, the composition further comprises a uracil glycosylase inhibitor (UGI). In some embodiments, the composition comprises an RNA-guided DNA binding agent that the RNA-guided DNA binding agent generates a cytosine (C) to thymine (T) conversion with the TRBC genomic target sequence. In some embodiments, the composition comprises an RNA-guided DNA binding agent that generates a adenosine (A) to guanine (G) conversion with the TRBC genomic target sequence.

[0408] In some embodiments, an engineered cell produced by the methods described herein is provided. In some embodiments, the engineered cell produced by the methods and compositions described herein is an allogeneic cell. In some embodiments, the methods produce a composition comprising an engineered cell having reduced TRBC expression. In some embodiments, the methods produce a composition comprising an engineered cell having reduced TRBC protein expression. In some embodiments, the methods produce a composition comprising an engineered cell having reduced TRBC levels in the cell nucleus. In some embodiments, the methods produce a composition comprising an engineered cell that expresses a truncated form of the TRBC protein. In some embodiments, the methods produce a composition comprising an engineered cell that produces no detectable TRBC protein. In some embodiments, the engineered cell has reduced TRBC expression, reduced TRBC protein, or reduced TRBC levels in the cell nucleus as compared to an unmodified cell. In some embodiments, the engineered cell produced by the methods disclosed herein elicits a reduced response from CD4+ T cells as compared to an unmodified cell as measured in an in vitro cell culture assay containing CD4+ T cells.

[0409] In some embodiments, an engineered cell produced by the methods or compositions disclosed herein is provided wherein the cell has reduced or eliminated surface expression of TRBC protein and wherein the cell comprises a genetic modification comprising at least 5 contiguous nucleotides within the genomic coordinates chosen from: (a) chr7:142791862-142793149; (b) chr7: 142791756-142792721; or (c) chr7:142801104-142802543. In some embodiments, an engineered cell produced by the methods or compositions disclosed herein is provided wherein the cell has reduced or eliminated surface expression of TRBC protein and wherein the cell comprises a genetic modification comprising at least 10 contiguous nucleotides within the genomic coordinates chosen from: (a) chr7:142791862-142793149; (b) chr7: 142791756-142792721; or (c) chr7: 142801104-142802543. In some embodiments, an engineered cell produced by the methods or compositions disclosed herein is provided wherein the cell has reduced or eliminated surface expression of TRBC protein and wherein the cell comprises a genetic modification comprising at least one C to T substitution or at least one A to G substitution within the genomic coordinates chosen from: (a) chr7: 142791862-142793149; (b) chr7: 142791756-142792721; or (c) chr7:142801104-142802543.

[0410] In some embodiments, an engineered cell, which has reduced or eliminated surface expression of TRBC relative to an unmodified cell, is provided, the engineered cell comprising a genetic modification in the TRBC gene, wherein the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from: (a) chr7:142792690-142792714; or chr7:142792693-142792717; or (b) chr7:142791756-142791780; chr7: 142791761-142791785; chr7:142791820-142791844; chr7: 142791939-142791963; chr7:142791940-142791964; or chr7:142792004-142792028; or (c) chr7:142801104-142801124; chr7:142802103-142802127; or chr7: 142802106-142802130.

[0411] In some embodiments, an engineered cell, which has reduced or eliminated surface expression of TRBC relative to an unmodified cell, is provided, the engineered cell comprising a genetic modification in the TRBC gene, wherein the genetic modification comprises an indel, a C to T substitution, or an A to G substitution within the genomic coordinates chosen from: (a) chr7:142792690-142792714; or chr7:142792693-142792717; or (b) chr7:142791756-142791780; chr7:142791761-142791785; chr7:142791820-142791844; chr7:142791939-142791963; chr7:142791940-142791964; or chr7:142792004-142792028; or (c) chr7: 142801104-142801124; chr7:142802103-142802127; or chr7:142802106-142802130.

[0412] In some embodiments, a method of making an engineered cell, which has reduced or eliminated surface expression of TRBC protein relative to an unmodified cell, is provided, the method comprising contacting a cell with a composition of any of the embodiments provided herein. In some embodiments, the composition comprises a TRBC guide RNA, comprising a guide sequence of any one of: SEQ ID NO: 215, 216, 223, 224, 229, 230, 246, 259, and 260. In some embodiments, the composition comprises a TRBC guide RNA, comprising a guide sequence of any one of: SEQ ID NO: 215, 216, 224, 229, 246, 259, and 260. In some embodiments, the composition comprises a TRBC guide RNA, comprising a guide sequence of any one of SEQ ID NOs: 215, 259, and 260.

[0413] In some embodiments, the methods comprise reducing or eliminating surface expression of MHC class II protein on the surface of a cell comprising contacting a cell with a composition comprising a CIITA guide RNA comprising a guide sequence that targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chosen from: (a) chr16:10877363-10907788 or (b) chr16:10906515-10908136. In some embodiments, the methods further comprise contacting the cell with an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent. In some embodiments, the RNA-guided DNA binding agent is Cas9. In some embodiments, the RNA-guided DNA binding agent is *N. meningitidis* Cas9. In some embodiments, the CIITA guide RNA is a *N. meningitidis* Cas9 guide RNA. In some embodiments, the RNA-guided DNA binding agent comprises a deaminase domain. In some embodiments the RNA-guided DNA binding agent comprises an APOBEC3A deaminase (A3A) and an RNA-guided nickase. In some embodiments, the expression of MHC class II protein on the surface of the cell (i.e., engineered cell) is thereby reduced. In some embodiments, the CIITA guide RNA comprises a guide sequence selected from SEQ ID NO: 301, 302, 304-576.

[0414] In some embodiments, the methods comprise making an engineered cell, which has reduced or eliminated surface expression of MHC class II protein relative to an unmodified cell, comprising contact the cell with a composition comprising a CIITA guide RNA comprising a guide sequence that targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chosen from: (a) chr16:10877363-10907788 or (b) chr16:10906515-10908136. In some embodiments, the methods further comprise contacting the cell with an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent. In some embodiments, the RNA-guided DNA binding agent is Cas9. In some embodiments, the RNA-guided DNA binding agent is *N. meningitidis* Cas9. In some embodiments, the CIITA guide RNA is a *N. meningitidis* Cas9 guide RNA. In some embodiments, the RNA-guided DNA binding agent comprises a deaminase region. In some embodiments the RNA-guided DNA binding agent comprises an APOBEC3A deaminase (A3A) and an RNA-guided nickase. In some embodiments, the expression of MHC class II protein on the surface of the cell (i.e., engineered cell) is thereby reduced. In some embodiments, the CIITA guide RNA comprises a guide sequence selected from SEQ ID NO: 301, 302, 304-576.

[0415] In some embodiments, the methods comprise genetically modifying a cell to reduce or eliminate the surface expression of MHC class II protein comprising contacting the cell with a composition comprising a CIITA guide RNA comprising a guide sequence that targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chosen from: (a) chr16:10877363-10907788 or (b) chr16:10906515-10908136. In some embodiments, the methods further comprise contacting the cell with an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent. In some embodiments, the RNA-guided DNA binding agent is Cas9. In some embodiments, the RNA-guided DNA binding agent is *N. meningitidis* Cas9. In some embodiments, the CIITA guide RNA is a *N. meningitidis* Cas9 guide RNA. In some embodiments, the RNA-guided DNA binding agent comprises a deaminase region. In some embodiments the RNA-guided DNA binding agent comprises an APOBEC3A deaminase (A3A) and an RNA-guided nickase. In some embodiments, the expression of MHC class II protein on the surface of the cell (i.e., engineered cell) is thereby reduced. In some embodiments, the CIITA guide RNA comprises a guide sequence selected from SEQ ID NO: 301, 302, 304-576.

[0416] In some embodiments, the methods comprise genetically modifying CIITA comprising contacting a cell with a composition comprising a CIITA guide RNA comprising a guide sequence that targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chosen from: (a) chr16:10877363-10907788 or (b) chr16:10906515-10908136. In some embodiments, the methods further comprise contacting the cell with an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent. In some embodiments, the RNA-guided DNA binding agent is Cas9. In some embodiments, the RNA-guided DNA binding agent is *N. meningitidis* Cas9. In some embodiments, the CIITA guide RNA is a *N. meningitidis* Cas9 guide RNA. In some embodiments, the RNA-guided DNA binding agent comprises a deaminase region. In some embodiments the RNA-guided DNA binding agent comprises an APOBEC3A deaminase (A3A) and an RNA-guided nickase. In some embodiments, the expression of MHC class II protein on the surface of the cell (i.e., engineered cell) is thereby reduced. In some embodiments, the CIITA guide RNA comprises a guide sequence selected from SEQ ID NO: 301, 302, 304-576.

[0417] In some embodiments, the methods comprise inducing a DSB or an single stranded break (SSB) in CIITA comprising contacting a cell with a composition comprising a CIITA guide RNA comprising a guide sequence targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chosen from: (a) chr16:10877363-10907788 or (b) chr16:10906515-10908136. In some embodiments, the methods further comprise contacting the cell with an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent. In some embodiments, the RNA-guided DNA binding agent is Cas9. In some embodiments, the RNA-guided DNA binding agent is *S. pyogenes* Cas9. In some embodiments, the CIITA guide RNA is a *N. meningitidis* Cas9 guide RNA. In some embodiments, the RNA-guided DNA binding agent comprises a deaminase region. In some embodiments the RNA-guided DNA binding agent comprises an APOBEC3A deaminase (A3A) and an RNA-guided nickase. In some embodiments, the expression of MHC class II protein on the surface of the cell (i.e., engineered cell) is thereby reduced. In some embodiments, the CIITA guide RNA comprises a guide sequence selected from SEQ ID NO: 301, 302, 304-576.

[0418] In some embodiments, the methods comprise reducing expression of the CIITA protein in a cell comprising delivering a composition to a cell comprising contacting a cell with a composition comprising a CIITA guide RNA comprising a guide sequence targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chosen from: (a) chr16:10877363-10907788 or (b) chr16:10906515-10908136. In some embodiments, the methods further comprise contacting the cell with an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent. In some embodiments, the RNA-guided DNA binding agent is Cas9. In some embodiments, the RNA-guided DNA binding agent is *N. meningitidis* Cas9. In some embodiments, the CIITA guide RNA is a *N. meningitidis* Cas9 guide RNA. In some embodiments, the RNA-guided DNA binding agent comprises a deaminase region. In some embodiments the RNA-guided DNA binding agent comprises an APOBEC3A deaminase (A3A) and an RNA-guided nickase. In some embodiments, the expression of MHC class II protein on the surface of the cell (i.e., engineered cell) is thereby reduced. In some embodiments, the CIITA guide RNA comprises a guide sequence selected from SEQ ID NO: 301, 302, 304-576.

[0419] In some embodiments, the methods of reducing expression of an MHC class II protein on the surface of a cell comprise contacting a cell with any one or more of the CIITA guide RNAs disclosed herein. In some embodiments, the CIITA guide RNA comprises a guide sequence selected from SEQ ID NO: 301, 302, 304-576.

[0420] In some embodiments, compositions are provided comprising a CIITA guide RNA comprising a guide sequence that targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chosen from: (a) chr16:10877363-10907788 or (b) chr16:10906515-10908136. In some embodiments, the composition further comprises an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent. In some embodiments, the composition comprises an RNA-guided DNA binding agent that is Cas9. In some embodiments, the RNA-guided DNA binding agent is *N. meningitidis* Cas9. In some embodiments, the CIITA guide RNA is a *N. meningitidis* Cas9 guide RNA. In some embodiments, the RNA-guided DNA binding agent comprises a deaminase region. In some embodiments the RNA-guided DNA binding agent comprises an APOBEC3A deaminase (A3A) and an RNA-guided nickase. In some embodiments, the CIITA guide RNA comprises a guide sequence selected from SEQ ID NO: 301, 302, 304-576.

[0421] In some embodiments, a composition is provided, the composition comprising: a) a CIITA guide RNA (gRNA) comprising i) a guide sequence selected from SEQ ID NOs: 301, 302, 304-576; or ii) at least 19, 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence selected from SEQ ID NOs: 301, 302, 304-576; or iii) a guide sequence at least 95%, 90%, or 85% identical to a sequence selected from SEQ ID NOs: 301, 302, 304-576; or iv) a sequence that comprises 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 4; or v) at least 19, 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence from (iv); or vi) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence selected from (v). In some embodiments, the CIITA guide RNA that is a single-guide RNA (sgRNA).

[0422] In some embodiments, a method of making an engineered cell, which has reduced or eliminated surface expression of MHC class II protein relative to an unmodified cell, is provided, the method comprising contacting a cell with a composition of any of the embodiments provided herein. In some embodiments, the composition comprises a CIITA guide RNA, comprising a guide sequence of any one of: SEQ ID NOs: 301-302, 320-321, 324, 326, 327, 332, 354, 361, 372, 400, 415, 419-420, 422, 428, 431, 432, 434, 451, 455, 458, 462-464, and 468. In some embodiments, the composition comprises a CIITA guide RNA, comprising a guide sequence of any one of: SEQ ID NOs: 301, 302, 320, 372, 414, 419, 422, and 462-463.

[0423] In some embodiments, the composition further comprises a uracil glycosylase inhibitor (UGI). In some embodiments, the composition comprises an RNA-guided DNA binding agent that the RNA-guided DNA binding agent generates a cytosine (C) to thymine (T) conversion with the CIITA genomic target sequence. In some embodiments, the composition comprises an RNA-guided DNA binding agent that generates a adenosine (A) to guanine (G) conversion with the CIITA genomic target sequence.

[0424] In some embodiments, an engineered cell produced by the methods described herein is provided. In some embodiments, the engineered cell produced by the methods and compositions described herein is an allogeneic cell. In some embodiments, the methods produce a composition comprising an engineered cell having reduced MHC class II expression. In some embodiments, the methods produce a composition comprising an engineered cell having reduced CIITA protein expression. In some embodiments, the methods produce a composition comprising an engineered cell having reduced CIITA levels in the cell nucleus. In some embodiments, the methods produce a composition comprising an engineered cell that expresses a truncated form of the CIITA protein. In some embodiments, the methods produce a composition comprising an engineered cell that produces no detectable CIITA protein. In some embodiments, the engineered cell has reduced MHC class II expression, reduced CIITA protein, or reduced CIITA levels in the cell nucleus as compared to an unmodified cell. In some embodiments, the engineered cell produced by the methods disclosed herein elicits a reduced response from CD4+ T cells as compared to an unmodified cell as measured in an in vitro cell culture assay containing CD4+ T cells.

[0425] In some embodiments, an engineered cell produced by the methods or compositions disclosed herein is provided wherein the cell has reduced or eliminated surface expression of MHC class II protein and wherein the cell comprises a genetic modification comprising at least 5 contiguous nucleotides within the genomic coordinates chosen from: (a) chr16:10877363-10907788 or (b) chr16:10906515-10908136. In some embodiments, an engineered cell produced by the methods or compositions disclosed herein is provided wherein the cell has reduced or eliminated surface expression of MHC class II protein and wherein the cell comprises a genetic modification comprising at least 10 contiguous nucleotides within the genomic coordinates chosen from: (a) chr16:10877363-10907788 or (b) chr16:10906515-10908136. In some embodiments, an engineered cell produced by the methods or compositions disclosed herein is provided wherein the cell has reduced or eliminated surface expression of MHC class II protein and wherein the cell comprises a genetic modification comprising at least one C to T substitution or at least one A to G substitution within the genomic coordinates chosen from: (a) chr16:10877363-10907788 or (b) chr16:10906515-10908136.

[0426] In some embodiments, the methods comprise making an engineered cell comprising contact the cell with a composition comprising an AAVS1 guide RNA comprising a guide sequence that targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chr19:55115151-55116209. In some embodiments, the methods further comprise contacting the cell with an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent. In some embodiments, the RNA-guided DNA binding agent is Cas9. In some embodiments, the RNA-guided DNA binding agent is *N. meningitidis* Cas9. In some embodiments, the AAVS1 guide RNA is a *N. meningitidis* Cas9 guide RNA. In some embodiments, the RNA-guided DNA binding agent comprises a deaminase region. In some embodiments the RNA-guided DNA binding agent comprises an APOBEC3A deaminase (A3A) and an RNA-guided nickase. In some embodiments, the AAVS1 guide RNA comprises a guide sequence selected from SEQ ID NO: 601-774.

[0427] In some embodiments, the methods comprise genetically modifying a cell comprising contacting the cell with a composition comprising an AAVS1 guide RNA comprising a guide sequence that targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chr19:55115151-55116209. In some embodiments, the methods further comprise contacting the cell with an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent. In some embodiments, the RNA-guided DNA binding agent is Cas9. In some embodiments, the RNA-guided DNA binding agent is *N. meningitidis* Cas9. In some embodiments, the AAVS1 guide RNA is a *N. meningitidis* Cas9 guide RNA. In some embodiments, the RNA-guided DNA binding agent comprises a deaminase region. In some embodiments the RNA-guided DNA binding agent comprises an APOBEC3A deaminase (A3A) and an RNA-guided nickase. In some embodiments, the AAVS1 guide RNA comprises a guide sequence selected from SEQ ID NO: 601-774.

[0428] In some embodiments, the methods comprise genetically modifying AAVS1 comprising contacting a cell with a composition comprising an AAVS1 guide RNA comprising a guide sequence that targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chr19:55115151-55116209. In some embodiments, the methods further comprise contacting the cell with an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent. In some embodiments, the RNA-guided DNA binding agent is Cas9. In some embodiments, the RNA-guided DNA binding agent is *N. meningitidis* Cas9. In some embodiments, the AAVS1 guide RNA is a *N. meningitidis* Cas9 guide RNA. In some embodiments, the RNA-guided DNA binding agent comprises a deaminase region. In some embodiments the RNA-guided DNA binding agent comprises an APOBEC3A deaminase (A3A) and an RNA-guided nickase. In some embodiments, the AAVS1 guide RNA comprises a guide sequence selected from SEQ ID NO: 601-774.

[0429] In some embodiments, the methods comprise inducing a DSB or a single stranded break (SSB) in AAVS1 comprising contacting a cell with a composition comprising an AAVS1 guide RNA comprising a guide sequence targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chr19:55115151-55116209. In some embodiments, the methods further comprise contacting the cell with an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent. In some embodiments, the RNA-guided DNA binding agent is Cas9. In some embodiments, the RNA-guided DNA binding agent is *S. pyogenes* Cas9. In some embodiments, the AAVS1 guide RNA is a *N. meningitidis* Cas9 guide RNA. In some embodiments, the RNA-guided DNA binding agent comprises a deaminase region. In some embodiments the RNA-guided DNA binding agent comprises an APOBEC3A deaminase (A3A) and an RNA-guided nickase. In some embodiments, the AAVS1 guide RNA comprises a guide sequence selected from SEQ ID NO: 601-774.

[0430] In some embodiments, the methods comprise delivering a composition to a cell comprising contacting a cell with a composition comprising an AAVS1 guide RNA comprising a guide sequence targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chr19:55115151-55116209. In some embodiments, the methods further comprise contacting the cell with an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent. In some embodiments, the RNA-guided DNA binding agent is Cas9. In some embodiments, the RNA-guided DNA binding agent is *N. meningitidis* Cas9. In some embodiments, the AAVS1 guide RNA is a *N. meningitidis* Cas9 guide RNA. In some embodiments, the RNA-guided DNA binding agent comprises a deaminase region. In some embodiments the RNA-guided DNA binding agent comprises an APOBEC3A deaminase (A3A) and an RNA-guided nickase. In some embodiments, the AAVS1 guide RNA comprises a guide sequence selected from SEQ ID NO: 601-774.

[0431] In some embodiments, the methods of genetically modifying the AAVS1 gene comprise contacting a cell with any one or more of the AAVS1 guide RNAs disclosed herein. In some embodiments, the AAVS1 guide RNA comprises a guide sequence selected from SEQ ID NO: 601-774.

[0432] In some embodiments, compositions are provided comprising an AAVS1 guide RNA comprising a guide sequence that targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chr19:55115151-55116209. In some embodiments, the composition further comprises an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent. In some embodiments, the composition comprises an RNA-guided DNA binding agent that is Cas9. In some embodiments, the RNA-guided DNA binding agent is *N. meningitidis* Cas9. In some embodiments, the AAVS1 guide RNA is a *N. meningitidis* Cas9 guide RNA. In some embodiments, the RNA-guided DNA binding agent comprises a deaminase region. In some embodiments the RNA-guided DNA binding agent comprises an APOBEC3A deaminase (A3A) and an RNA-guided nickase. In some embodiments, the AAVS1 guide RNA comprises a guide sequence selected from SEQ ID NO: 601-774.

[0433] In some embodiments, a composition is provided, the composition comprising: a) an AAVS1 guide RNA (gRNA) comprising i) a guide sequence selected from SEQ ID NOs: 601-774; or ii) at least 19, 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence selected from SEQ ID NOs: 601-774; or iii) a guide sequence at least 95%, 90%, or 85% identical to a sequence selected from SEQ ID NOs: 601-774; or iv) a sequence that comprises 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 5; or v) at least 19, 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence from (iv); or vi) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence selected from (v). In some embodiments, the AAVS1 guide RNA (gRNA) is a single guide RNA.

[0434] In some embodiments, a method of making an engineered cell is provided, the method comprising contacting a cell with a composition of any of the embodiments provided herein. In some embodiments, the composition comprises an AAVS1 guide RNA, comprising a guide sequence of any one of from: SEQ ID NOs: 611, 620, 622, 626, 627, 628, 629, 632, 633, 634, 656, 659, 660, 661, 673, 691, 692, 730, 734, and 746.

[0435] In some embodiments, the composition further comprises a uracil glycosylase inhibitor (UGI). In some embodiments, the composition comprises an RNA-guided DNA binding agent that the RNA-guided DNA binding agent generates a cytosine (C) to thymine (T) conversion with the AAVS1 genomic target sequence. In some embodiments, the composition comprises an RNA-guided DNA binding agent that generates a adenosine (A) to guanine (G) conversion with the AAVS1 genomic target sequence.

[0436] In some embodiments, an engineered cell produced by the methods described herein is provided. In some embodiments, the engineered cell produced by the methods and compositions described herein is an allogeneic cell. In some embodiments, the engineered cell produced by the methods disclosed herein elicits a reduced response from CD4+ T cells as compared to an unmodified cell as measured in an in vitro cell culture assay containing CD4+ T cells.

[0437] In some embodiments, an engineered cell produced by the methods or compositions disclosed herein is provided wherein the cell comprises a genetic modification comprising at least 5 contiguous nucleotides within the genomic coordinates chr19:55115151-55116209. In some embodiments, an engineered cell produced by the methods or compositions disclosed herein is provided wherein the cell comprises a genetic modification comprising at least 10 contiguous nucleotides within the genomic coordinates chr19:55115151-55116209. In some embodiments, an engineered cell produced by the methods or compositions disclosed herein is provided wherein the cell comprises a genetic modification comprising at least one C to T substitution or at least one A to G substitution within the genomic coordinates chr19:55115151-55116209. In some embodiments, the compositions disclosed herein further comprise a pharmaceutically acceptable carrier. In some embodiments, a cell produced by the compositions disclosed herein comprising a pharmaceutically acceptable carrier is provided. In some embodiments, compositions comprising the cells disclosed herein are provided.

C. HLA-A Guide RNAs

[0438] The methods and compositions provided herein disclose HLA-A guide RNAs useful for reducing the expression of HLA-A protein on the surface of a cell. In some embodiments, such guide RNAs direct an RNA-guided DNA binding agent to an HLA-A genomic target sequence and may be referred to herein as "HLA-A guide RNAs." In some embodiments, the HLA-A guide RNA directs an RNA-guided DNA binding agent to a human HLA-A genomic target sequence. In some embodiments, the HLA-A guide RNA comprises a guide sequence selected from SEQ ID NO: 2-80.

[0439] In some embodiments, the methods and compositions disclosed herein comprise an HLA-A guide RNA comprising a guide sequence that targets an HLA-A genomic target sequence comprising at least 10 nucleotides within the genomic coordinates chr6:29942540-29945459. In some embodiments, the method and composition disclosed herein comprise an HLA-A guide RNA comprising a guide sequence that targets an HLA-A genomic target sequence comprising at least one nucleotide within the genomic coordinates chr6:29942540-29945459.

[0440] In some embodiments, the methods and compositions disclosed herein comprise an HLA-A guide RNA comprising a guide sequence that directs an RNA-guided DNA binding agent to induce a double stranded break (DSB) or a single-stranded break (SSB) in an HLA-A gene, wherein the HLA-A guide RNA targets and HLA-A genomic target sequence comprising at least 10 contiguous nucleotides within the genomic coordinates chr6:29942540-29945459. In some embodiments, the methods and compositions disclosed herein comprise an HLA-A guide RNA comprising a guide sequence that directs an RNA-guided DNA binding agent to induce a double stranded break (DSB) or a single-stranded break (SSB) in an HLA-A gene, wherein the HLA-A guide RNA targets an HLA-A genomic target sequence comprising at least one nucleotide within the genomic coordinates chr6:29942540-29945459.

[0441] In some embodiments, the methods and compositions disclose an HLA-A guide RNA that directs an RNA-guided DNA binding agent to induce a double stranded break (DSB) or a single-stranded break (SSB) in an HLA-A genomic target sequence. In some embodiments, the methods and compositions disclose an HLA-A guide RNA that directs an RNA-guided DNA binding agent to make a cut in an HLA-A genomic target sequence. In embodiments wherein the RNA-guided DNA cutting agent is Cas9, the cut occurs at the third base from the protospacer adjacent motif (PAM) sequence.

[0442] In some embodiments, a composition is provided comprising an HLA-A guide RNA described herein and an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent.

[0443] In some embodiments, a composition is provided comprising an HLA-A single-guide RNA (sgRNA) comprising a guide sequence that targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chr6:29942540-29945459. In some embodiments, a composition is provided comprising an HLA-A sgRNA described herein and an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent.

[0444] In some embodiments, a composition is provided comprising an HLA-A dual-guide RNA (dgRNA) comprising a guide sequence that targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chr6:29942540-29945459. In some embodiments, a composition is provided comprising an HLA-A dgRNA described herein and an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent.

[0445] In some embodiments, the TRAC gRNA comprises a guide sequence selected from any one of SEQ ID NOs: 2-80. Exemplary HLA-A guide sequences are shown below in Table 1 (SEQ ID NOs: 2-80) with corresponding guide RNA sequences 2-80.

TABLE 1

Exemplary HLA-A guide sequences

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1002-1080)	Exemplary Guide RNA Modified Sequence (SEQ ID NOs: 2002-2080)	Genomic Coordinates (hg38)
G028854	2	CCUGGGUCUGGUCCUCCCCAUCCC	CCUGGGUCUGGUCCUCCCCAUCCCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mC*mU*mGmGGUmCmUGmGUmCCUCCmCCAUmCCCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29944223-29944247
G028855	3	GUGGAGACCAGGCCUGCAGGGGAU	GUGGAGACCAGGCCUGCAGGGGAUGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mG*mU*mG*mGmAGAmCmCAmGGmCCUGCmAGGGmGAUmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29944264-29944288
G028856	4	CCGUGUCCUGGGUCUGGUCCUCCC	CCGUGUCCUGGGUCUGGUCCUCCCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mC*mG*mUmGUCmCmUGmGGmUCUGGmUCCUmCCCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29944229-29944253
G028857	5	UCCGUGUCCUGGGUCUGGUCCUCC	UCCGUGUCCUGGGUCUGGUCCUCCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mU*mC*mC*mGmUGUmCmCUmGGmGUCUGmGUCCmUCCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29944230-29944254
G028858	6	CACAUGGCAGGUGUAUCUCUGCUC	CACAUGGCAGGUGUAUCUCUGCUCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mA*mC*mAmUGGmCmAGmGUmGUAUCmUCUGmCUCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29944327-29944351
G028859	7	UGCUGCACAUGGCAGGUGUAUCUC	UGCUGCACAUGGCAGGUGUAUCUCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGC	mU*mG*mC*mUmGCAmCmAUmGGmCAGGUmGUAUmCUCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmU	chr6: 29944332-29944356

TABLE 1-continued

Exemplary HLA-A guide sequences					
Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1002-1080)	Exemplary Guide RNA Modified Sequence (SEQ ID NOs: 2002-2080)	Genomic Coordinates (hg38)
			AACGCUCUGCCUUCUGGCAUCGUU	GUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	
G028860	8	CCGCACGAACUGCGUGUCGUCCAC	CCGCACGAACUGCGUGUCGUCCACGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mC*mG*mCmACGmAmACmUGmCGUGUmCGUCmCACmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29942830-29942854
G028861	9	CGCACGAACUGCGUGUCGUCCACG	CGCACGAACUGCGUGUCGUCCACGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mG*mC*mAmCGAmAmCUmGCmGUGUCmGUCCmACGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29942829-29942853
G028862	10	GUGCGCUGCAGCGUCUCCUUCCCG	GUGCGCUGCAGCGUCUCCUUCCCGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mG*mU*mG*mCmGCUmGmCAmGCmGUCUCmCUUCmCCGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29943511-29943535
G028863	11	CCGAGGAUGGCCGUCAUGGCGCCC	CCGAGGAUGGCCGUCAUGGCGCCCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mC*mG*mAmGGAmUmGGmCCmGUCAUmGGCGmCCCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29942547-29942571
G028864	12	CGAGGAUGGCCGUCAUGGCGCCCC	CGAGGAUGGCCGUCAUGGCGCCCCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mG*mA*mGmGAUmGmGCmCGmUCAUGmGCGCmCCCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29942548-29942572
G028865	13	AAGGUUCCAUCCCCUGCAGGCCUG	AAGGUUCCAUCCCCUGCAGGCCUGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCC	mA*mA*mG*mGmUUCmCmAUmCCmCCUGCmAGGCmCUGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCU	chr6: 29944266-29944290

TABLE 1-continued

Exemplary HLA-A guide sequences

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1002-1080)	Exemplary Guide RNA Modified Sequence (SEQ ID NOs: 2002-2080)	Genomic Coordinates (hg38)
			UUCUGGCAUCGUU	CUmGmCCmUmUmCmUGGCAUCG*mU*mU	
G028866	14	AGACCAGGCCUGCAGGGGAUGGAA	AGACCAGGCCUGCAGGGGAUGGAAGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mA*mG*mA*mCmCAGmGmCCmUGmCAGGGmGAUGmGAAmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29944268-29944292
G028867	15	ACUUCUGGAAGGUUCCAUCCCCUG	ACUUCUGGAAGGUUCCAUCCCCUGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mA*mC*mU*mUmCUGmGmAAmGGmUUCCAmUCCCmCUGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29944274-29944298
G028868	16	CGAUGAAGCGGGGCUCCCCGCGGC	CGAUGAAGCGGGGCUCCCCGCGGCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mG*mA*mUmGAAmGmCGmGGmGCUCCmCCGCmGGCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29942795-29942819
G028869	17	UCCGUGUCCCGGCCCGGCCGCGGG	UCCGUGUCCCGGCCCGGCCGCGGGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mU*mC*mC*mGmUGUmCmCCmGGmCCCGGmCCGCmGGGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29942785-29942809
G028870	18	CCGUGUCCCGGCCCGGCCGCGGGG	CCGUGUCCCGGCCCGGCCGCGGGGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mC*mG*mUmGUCmCmCGmGCmCCGGCmCGCGmGGGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29942786-29942810
G028871	19	CAGACGCCGAGGAUGGCCGUCAUG	CAGACGCCGAGGAUGGCCGUCAUGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCC	mC*mA*mG*mAmCGCmCmGAmGGmAUGGCmCGUCmAUGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCU	chr6: 29942541-29942565

TABLE 1-continued

Exemplary HLA-A guide sequences

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1002-1080)	Exemplary Guide RNA Modified Sequence (SEQ ID NOs: 2002-2080)	Genomic Coordinates (hg38)
			UUCUGGCAUCG UU	CUmGmCCmUmUmCmUG GCAUCG*mU*mU	
G028872	20	CCAGACGCCGA GGAUGGCCGUC AU	CCAGACGCCGA GGAUGGCCGUC AUGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mC*mA*mGmACGm CmCGmAGmGAUGGmCC GUmCAUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr6: 29942540-29942564
G028873	21	AGACGCCGAGG AUGGCCGUCAU GG	AGACGCCGAGG AUGGCCGUCAU GGGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mA*mG*mA*mCmGCCm GmAGmGAmUGGCCmGU CAmUGGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr6: 29942542-29942566
G028874	22	CUGUCGAACCG CACGAACUGCG UG	CUGUCGAACCG CACGAACUGCG UGGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mU*mG*mUmCGAm AmCCmGCmACGAAmCU GCmGUGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr6: 29942838-29942862
G028875	23	UCGUGCGGUUC GACAGCGACGC CG	UCGUGCGGUUC GACAGCGACGC CGGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mU*mC*mG*mUmGCGm GmUUmCGmACAGCmGA CGmCCGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr6: 29942852-29942876
G028876	24	CCAUCCCCAUC GUGGGCAUCAU UG	CCAUCCCCAUC GUGGGCAUCAU UGGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mC*mA*mUmCCCm CmAUmCGmUGGGCmAU CAmUUGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr6: 29944517-29944541
G028877	25	UCCUCUGGCUC GCGGCGUCGCU GU	UCCUCUGGCUC GCGGCGUCGCU GUGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mU*mC*mC*mUmCUGm GmCUmCGmCGGCGmUC GCmUGUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr6: 29942858-29942882

TABLE 1-continued

Exemplary HLA-A guide sequences					
Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1002-1080)	Exemplary Guide RNA Modified Sequence (SEQ ID NOs: 2002-2080)	Genomic Coordinates (hg38)
G028878	26	CUCUGGUUGUAGUAGCCGCGCAGG	CUCUGGUUGUAGUAGCCGCGCAGGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mU*mC*mUmGGUmUmGUmAGmUAGCCmGCGCmAGGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29942988-29943012
G028879	27	ACCUGGGGACCCUGCGCGGCUACU	ACCUGGGGACCCUGCGCGGCUACUGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mA*mC*mC*mUmGGGmGmACmCCmUGCGCmGGCUmACUmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29942984-29943008
G028880	28	CGCUCUGGUUGUAGUAGCCGCGCA	CGCUCUGGUUGUAGUAGCCGCGCAGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mG*mC*mUmCUGmGmUUmGUmAGUAGmCCGCmGCAmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29942990-29943014
G028881	29	GCUCUGGUUGUAGUAGCCGCGCAG	GCUCUGGUUGUAGUAGCCGCGCAGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mG*mC*mU*mCmUGGmUmUGmUAmGUAGCmCGCGmCAGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29942989-29943013
G028882	30	UAUUUUUUUCUAUAGUGUGAGACA	UAUUUUUUUCUAUAGUGUGAGACAGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mU*mA*mU*mUmUUUmUmUCmUAmUAGUGmUGAGmACAmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29945435-29945459
G028883	31	ACGCCUACGACGGCAAGGAUUACA	ACGCCUACGACGGCAAGGAUUACAGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mA*mC*mG*mCmCUAmCmGAmCGmGCAAGmGAUUmACAmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29943342-29943366

TABLE 1-continued

Exemplary HLA-A guide sequences

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1002-1080)	Exemplary Guide RNA Modified Sequence (SEQ ID NOs: 2002-2080)	Genomic Coordinates (hg38)
G028884	32	UUUGUUCUACCCCAGGCAGUGACA	UUUGUUCUACCCCAGGCAGUGACAGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mU*mU*mU*mGmUUCmUmACmCCmCAGGCmAGUGmACAmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29945218-29945242
G028885	33	UUUUGUUCUACCCCAGGCAGUGAC	UUUUGUUCUACCCCAGGCAGUGACGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mU*mU*mU*mUmGUUmCmUAmCCmCCAGGmCAGUmGACmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29945217-29945241
G028886	34	AGCGACCACAGCUCCAGUGAUCAC	AGCGACCACAGCUCCAGUGAUCACGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mA*mG*mC*mGmACCmAmCAmGCmUCCAGmUGAUmCACmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29944558-29944582
G028887	35	GAGCUCCGUGUCCUGGGUCUGGUC	GAGCUCCGUGUCCUGGGUCUGGUCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mG*mA*mG*mCmUCCmGmUGmUCmCUGGGmUCUGmGUCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29944234-29944258
G028888	36	UCCACGAGCUCCGUGUCCUGGGUC	UCCACGAGCUCCGUGUCCUGGGUCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mU*mC*mC*mAmCGAmGmCUmCCmGUGUCmCUGGmGUCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29944239-29944263
G028889	37	ACGAGCUCCGUGUCCUGGGUCUGG	ACGAGCUCCGUGUCCUGGGUCUGGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mA*mC*mG*mAmGCUmCmCGmUGmUCCUGmGGUCmUGGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29944236-29944260

TABLE 1-continued

Exemplary HLA-A guide sequences					
Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1002-1080)	Exemplary Guide RNA Modified Sequence (SEQ ID NOs: 2002-2080)	Genomic Coordinates (hg38)
G028890	38	CGAGCUCCGUGUCCUGGGUCUGGU	CGAGCUCCGUGUCCUGGGUCUGGUGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mG*mA*mGmCUCmCmGUmGUmCCUGGmGUCUmGGUmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29944235-29944259
G028891	39	GCAGGCCUGGUCUCCACGAGCUCC	GCAGGCCUGGUCUCCACGAGCUCCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mG*mC*mA*mGmGCCmUmGGmUCmUCCACmGAGCmUCCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29944251-29944275
G028892	40	UCCCCUGCAGGCCUGGUCUCCACG	UCCCCUGCAGGCCUGGUCUCCACGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mU*mC*mC*mCmCUGmCmAGmGCmCUGGUmCUCCmACGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29944257-29944281
G028893	41	AUCCGUGUCCCGGCCCGGCCGCGG	AUCCGUGUCCCGGCCCGGCCGCGGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mA*mU*mC*mCmGUGmUmCCmCGmGCCCGmGCCGmCGGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29942784-29942808
G028894	42	CCUUCCCGUUCUCCAGGUAUCUGC	CCUUCCCGUUCUCCAGGUAUCUGCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mC*mU*mUmCCCmGmUUmCUmCCAGGmUAUCmUGCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29943495-29943519
G028895	43	CCGUUCUCCAGGUAUCUGCGGAGC	CCGUUCUCCAGGUAUCUGCGGAGCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mC*mG*mUmUCUmCmCAmGGmUAUCUmGCGGmAGCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29943490-29943514

TABLE 1-continued

Exemplary HLA-A guide sequences

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1002-1080)	Exemplary Guide RNA Modified Sequence (SEQ ID NOs: 2002-2080)	Genomic Coordinates (hg38)
G028896	44	ACCUGCCAUGUGCAGCAUGAGGGU	ACCUGCCAUGUGCAGCAUGAGGGUGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mA*mC*mC*mUmGCCmAmUGmUGmCAGCAmUGAGmGGUmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29944345-29944369
G028897	45	CACCUGCCAUGUGCAGCAUGAGGG	CACCUGCCAUGUGCAGCAUGAGGGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mA*mC*mCmUGCmCmAUmGUmGCAGCmAUGAmGGGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29944344-29944368
G028898	46	ACCUGGCAGCGGGAUGGGGAGGAC	ACCUGGCAGCGGGAUGGGGAGGACGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mA*mC*mC*mUmGGCmAmGCmGGmGAUGGmGGAGmGACmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29944219-29944243
G028899	47	CACUGACCUGGCAGCGGGAUGGGG	CACUGACCUGGCAGCGGGAUGGGGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mA*mC*mUmGACmCmUGmGCmAGCGGmGAUGmGGGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29944214-29944238
G028900	48	CCUGGCAGCGGGAUGGGGAGGACC	CCUGGCAGCGGGAUGGGGAGGACCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mC*mU*mGmGCAmGmCGmGGmAUGGGmGAGGmACCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29944220-29944244
G028901	49	CCAUCUCAGGGUGAGGGGCUUGGG	CCAUCUCAGGGUGAGGGGCUUGGGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mC*mA*mUmCUCmAmGGmGUmGAGGGmGCUUmGGGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29944366-29944390

TABLE 1-continued

Exemplary HLA-A guide sequences

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1002-1080)	Exemplary Guide RNA Modified Sequence (SEQ ID NOs: 2002-2080)	Genomic Coordinates (hg38)
G028902	50	UGGCCGUCAUGGCGCCCCGAACCC	UGGCCGUCAUGGCGCCCCGAACCCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mU*mG*mG*mCmCGUmCmAUmGGmCGCCCmCGAAmCCCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29942554-29942578
G028903	51	AUGUCCGCCGCGGUCCAAGAGCGC	AUGUCCGCCGCGGUCCAAGAGCGCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mA*mU*mG*mUmCCGmCmCGmCGmGUCCAmAGAGmCGCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29943379-29943403
G028904	52	GGUACCCGUGCGCUGCAGCGUCUC	GGUACCCGUGCGCUGCAGCGUCUCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mG*mG*mU*mAmCCCmGmUGmCGmCUGCAmGCGUmCUCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29943518-29943542
G028905	53	GGGAAGGAGACGCUGCAGCGCACG	GGGAAGGAGACGCUGCAGCGCACGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mG*mG*mG*mAmAGGmAmGAmCGmCUGCAmGCGCmACGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29943518-29943542
G028906	54	GCGGGCGCCGUGGAUAGAGCAGGA	GCGGGCGCCGUGGAUAGAGCAGGAGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mG*mC*mG*mGmGCGmCmCGmUGmGAUAGmAGCAmGGAmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29942895-29942919
G028907	55	UCCUGCUCUAUCCACGGCGCCCGC	UCCUGCUCUAUCCACGGCGCCCGCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mU*mC*mC*mUmGCUmCmUAmUCmCACGGmCGCCmCGCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29942889-29942913

TABLE 1-continued

Exemplary HLA-A guide sequences

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1002-1080)	Exemplary Guide RNA Modified Sequence (SEQ ID NOs: 2002-2080)	Genomic Coordinates (hg38)
G028908	56	CCCCUGGUACCCGUGCGCUGCAGC	CCCCUGGUACCCGUGCGCUGCAGCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mC*mC*mCmUGGmUmACmCCmGUGCGmCUGCmAGCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29943523-29943547
G028909	57	UGGUACCCGUGCGCUGCAGCGUCU	UGGUACCCGUGCGCUGCAGCGUCUGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mU*mG*mG*mUmACCmCmGUmGCmGCUGCmAGCGmUCUmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29943519-29943543
G028910	58	CAUCUCAGGGUGAGGGGCUUGGGC	CAUCUCAGGGUGAGGGGCUUGGGCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mA*mU*mCmUCAmGmGGmUGmAGGGGmCUUGmGGCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29944365-29944389
G028911	59	ACGCAGUUCGUGCGGUUCGACAGC	ACGCAGUUCGUGCGGUUCGACAGCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mA*mC*mG*mCmAGUmUmCGmUGmCGGUUmCGACmAGCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29942845-29942869
G028912	60	CGCCCAGGUCUGGGUCAGGGCCAG	CGCCCAGGUCUGGGUCAGGGCCAGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mG*mC*mCmCAGmGmUCmUGmGGUCAmGGGCmCAGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29942595-29942619
G028913	61	CACUCACCCGCCCAGGUCUGGGUC	CACUCACCCGCCCAGGUCUGGGUCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mA*mC*mUmCACmCmCGmCCmCAGGUmCUGGmGUCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29942609-29942633

TABLE 1-continued

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1002-1080)	Exemplary Guide RNA Modified Sequence (SEQ ID NOs: 2002-2080)	Genomic Coordinates (hg38)
Exemplary HLA-A guide sequences					
G028914	62	CCGCCCAGGUCUGGGUCAGGGCCA	CCGCCCAGGUCUGGGUCAGGGCCAGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mC*mG*mCmCCAmGmGUmCUmGGGUCmAGGGmCCAmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29942596-29942620
G028915	63	CCCGCCCAGGUCUGGGUCAGGGCC	CCCGCCCAGGUCUGGGUCAGGGCCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mC*mC*mGmCCCmAmGGmUCmUGGGUmCAGGmGCCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29942597-29942621
G028916	64	CAGCGACGCCGCGAGCCAGAGGAU	CAGCGACGCCGCGAGCCAGAGGAUGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mA*mG*mCmGACmGmCCmGmGCmGAGCCmAGAGmGAUmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29942865-29942889
G028917	65	CGCGAGCCAGAGGAUGGAGCCGCG	CGCGAGCCAGAGGAUGGAGCCGCGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mG*mC*mGmAGCmCmAGmAGmGAUGGmAGCCmGCGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29942874-29942898
G028918	66	GCUCUAUCCACGGCGCCCGCGGCU	GCUCUAUCCACGGCGCCCGCGGCUGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mG*mC*mU*mCmUAUmCmCAmCGmGCGCCmCGCGmGCUmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29942891-29942915
G028919	67	CACAUCAGAGCCCUGGGCACUGUC	CACAUCAGAGCCCUGGGCACUGUCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mA*mC*mAmUCAmGmAGmCCmCUGGGmCACUmGUCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29945232-29945256

TABLE 1-continued

Exemplary HLA-A guide sequences

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1002-1080)	Exemplary Guide RNA Modified Sequence (SEQ ID NOs: 2002-2080)	Genomic Coordinates (hg38)
G028920	68	CAGACCCAGGACACGGAGCUCGUG	CAGACCCAGGACACGGAGCUCGUGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mA*mG*mAmCCCmAmGGmACmACGGAmGCUCmGUGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29944243-29944267
G028921	69	CCAGGACACGGAGCUCGUGGAGAC	CCAGGACACGGAGCUCGUGGAGACGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mC*mA*mGmGACmAmCGmGAmGCUCGmUGGAmGACmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29944248-29944272
G028922	70	CUCUGGGAAAAGAGGGGAAGGUGA	CUCUGGGAAAAGAGGGGAAGGUGAGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mU*mC*mUmGGGmAmAAmAGmAGGGGmAAGGmUGAmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29944471-29944495
G028923	71	UCUGGGAAAAGAGGGGAAGGUGAG	UCUGGGAAAAGAGGGGAAGGUGAGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mU*mC*mU*mGmGGAmAmAAmGAmGGGGAmAGGUmGAGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29944470-29944494
G028924	72	AGACGCUGCAGCGCACGGGUACCA	AGACGCUGCAGCGCACGGGUACCAGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mA*mG*mA*mCmGCUmGmCAmGCmGCACGmGGUAmCCAmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29943525-29943549
G028925	73	UCCUUUUCUAUCUGUGGGAAGAAA	UCCUUUUCUAUCUGUGGGAAGAAAGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mU*mC*mC*mUmUUUmCmUAmUCmUGUGGmGAAGmAAAmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	Unavailable. Imperfect alignment to human genome.

TABLE 1-continued

Exemplary HLA-A guide sequences					
Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1002-1080)	Exemplary Guide RNA Modified Sequence (SEQ ID NOs: 2002-2080)	Genomic Coordinates (hg38)
G028926	74	CACUGCCUGGGGUAGAACAAAAAC	CACUGCCUGGGGUAGAACAAAAACGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mA*mC*mUmGCCmUmGGmGGmUAGAAmCAAAmAACmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29945209-29945233
G028927	75	ACAACCAGAGCGAGGCCGGUGAGU	ACAACCAGAGCGAGGCCGGUGAGUGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mA*mC*mA*mAmCCAmGmAGmCGmAGGCCmGGUGmAGUmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29943008-29943032
G028928	76	CUACAACCAGAGCGAGGCCGGUGA	CUACAACCAGAGCGAGGCCGGUGAGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mU*mA*mCmAACmCmAGmAGmCGAGGmCCGGmUGAmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29943006-29943030
G028929	77	UACAACCAGAGCGAGGCCGGUGAG	UACAACCAGAGCGAGGCCGGUGAGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mU*mA*mC*mAmACCmAmGAmGCmGAGGCmCGGUmGAGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29943007-29943031
G028930	78	CCAGAGCGAGGCCGGUGAGUGACC	CCAGAGCGAGGCCGGUGAGUGACCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mC*mA*mGmAGCmGmAGmGCmCGGUGmAGUGmACCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29943012-29943036
G028933	79	CCAUCCCGCUGCCAGGUCAGUGUG	CCAUCCCGCUGCCAGGUCAGUGUGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mC*mA*mUmCCCmGmCUmGCmCAGGUmCAGUmGUGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29944206-29944230

TABLE 1-continued

Exemplary HLA-A guide sequences					
Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1002-1080)	Exemplary Guide RNA Modified Sequence (SEQ ID NOs: 2002-2080)	Genomic Coordinates (hg38)
G028934	80	GUAUCUGCGGAGCCACUCCACGCA	GUAUCUGCGGAGCCACUCCACGCAGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mG*mU*mA*mUmCUGmCmGGmAGmCCACUmCCACmGCAmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr6: 29943479-29943503

[0446] Throughout this application, the terms “mA,” “mC,” “mU,” or “mG” may be used to denote a nucleotide that has been modified with 2'—O-Me. Throughout this application, the terms A*, C*, U*, or G* may be used to denote a nucleotide that is linked to the next (e.g., 3') nucleotide with a phosphorothioate (PS) bond.

[0447] In some embodiments, the HLA-A guide RNA comprises a guide sequence selected from SEQ ID NOs: 2-80. In some embodiments, the HLA-A guide RNA comprises a guide sequence that is at least 19, 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence selected from SEQ ID NOs: 2-80. In some embodiments, the HLA-A guide RNA comprises a guide sequence that is at least 9500, 90%, or 85% identical to a sequence selected from SEQ ID NOs: 2-80. In some embodiments, the HLA-A guide RNA comprises a guide sequence that is at least 9500 identical to a sequence selected from SEQ ID NOs: 2-80.

[0448] In some embodiments, the HLA-A guide RNA comprises a guide sequence that comprises at least 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 1. As used herein, at least 10 contiguous nucleotides±10 nucleotides of a genomic coordinate means, for example, at least 10 contiguous nucleotides within the genomic coordinates wherein the genomic coordinates include 10 nucleotides in the 5' direction and 10 nucleotides in the 3' direction from the ranges listed in Table 1. For example, an HLA-A guide RNA may comprise 10 contiguous nucleotides within the genomic coordinates chosen from: chr6:29942891-29942915; chr6:29942609-29942633; chr6:29944266-29944290; chr6:29942889-29942913; chr6:29944471-29944495; and chr6:29944470-29944494; including the boundary nucleotides of these ranges. In some embodiments, the HLA-A guide RNA comprises a guide sequence that is at least 19, 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence that comprises 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 1. In some embodiments, the HLA-A guide RNA comprises a guide sequence that is at least 95%, 90%, or 85% identical to a sequence selected from a sequence that is 19, 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence that comprises 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 1.

[0449] In some embodiments, the HLA-A guide RNA comprises a guide sequence that comprises at least 15 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 1. In some embodiments, the HLA-A guide RNA comprises a guide sequence that comprises at least 24 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 1.

[0450] In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 2. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 3. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 4. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 5. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 6. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 7. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 8. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 9. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 10. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 11. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 12. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 13. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 14. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 15. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 16. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 17. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 18. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 19. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 20. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 21. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 22. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 23. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 24. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 25. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 26. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 27. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 28. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 29. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 30. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 31. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 32. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 33. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 34. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 35. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 36. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 37. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 38. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 39. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 40. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 41. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 42. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 43. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 44. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 45. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 46. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 47. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 48. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 49. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 50. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 51. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 52. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 53. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 54. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 55. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 56. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 57. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 58. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 59. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 60. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 61. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 62. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 63. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 64. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 65. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 66. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 67. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 68. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 69. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 70. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 71. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 72. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 73. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 74. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 75. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 76. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 77. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 78. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 79. In some embodiments, the HLA-A guide RNA comprises SEQ ID NO: 80.

[0451] In some embodiments, the HLA-A guide RNA comprises a guide sequence of any one of: SEQ ID NOs: 13, 55, 61, 66, and 70-71.

[0452] In some embodiments, the HLA-A guide RNA comprises a sequence listed in Table 1. In some embodiments, the HLA-A guide RNA comprises a sequence of any one of SEQ ID NO: 2-80. In some embodiments, the HLA-A guide RNA comprises a sequence of SEQ ID NO: 61 or 66. In some embodiments, the HLA-A guide RNA comprises a sequence of SEQ ID NO: 61. In some embodiments, the HLA-A guide RNA comprises a sequence of SEQ ID NO: 61. In some embodiments, the HLA-A guide RNA comprises a guide sequence comprising a sequence of any one of SEQ ID NO: 2-80. In some embodiments, the HLA-A guide RNA comprises a guide sequence comprising a sequence of SEQ ID NO: 61 or 66. In some embodiments, the HLA-A guide RNA comprises a guide sequence comprising a sequence of SEQ ID NO: 61. In some embodiments, the HLA-A guide RNA comprises a guide sequence comprising a sequence of SEQ ID NO: 66. In some embodiments, the HLA-A guide RNA comprises a sequence of any one of SEQ ID NOs: 1002-1080. In some embodiments, the HLA-A guide RNA comprises a sequence of SEQ ID NO: 1061 or 1066. In some embodiments, the HLA-A guide RNA comprises a sequence of SEQ ID NO: 1061. In some embodiments, the HLA-A guide RNA comprises a sequence of SEQ ID NO: 1066. In some embodiments, the HLA-A guide RNA comprises a sequence of any one of SEQ ID NOs: 2002-2080, 3001, and 3002. In some embodiments, the HLA-A guide RNA comprises a sequence of any one of SEQ ID NOs: 2061, 2066, 3001, and 3002. In some embodiments, the HLA-A guide RNA comprises a sequence of SEQ ID NO: 2061. In some embodiments, the HLA-A guide RNA comprises a sequence of SEQ ID NO: 2066. In some embodiments, the HLA-A guide RNA comprises a sequence of SEQ ID NO: 3001. In some embodiments, the HLA-A guide RNA comprises a sequence of SEQ ID NO: 3002.

[0453] In some embodiments, the HLA-A guide RNA is a single guide RNA (sgRNA) comprising a sequence of any one of the sgRNA sequences listed in Table 1.

[0454] Additional embodiments of HLA-A guide RNAs are provided herein, including e.g., exemplary modifications to the guide RNA.

1. Genetic Modifications to HLA-A

[0455] In some embodiments, the methods and compositions disclosed herein genetically modify at least one nucleotide of the HLA-A gene in a cell. In some embodiments, the genetic modification to HLA-A reduces or eliminates the expression of HLA-A protein on the surface of the genetically modified cell (or engineered cell). Genetic modifications encompass the population of modifications that results from contact with a genomic editing system (e.g., the population of edits that result from Cas9 and an HLA-A guide RNA, or the population of edits that result from BC22 and an HLA-A guide RNA).

[0456] In some embodiments, the genetic modification comprises at least one nucleotide within the genomic coordinates chr6:29942540-29945459. In some embodiments, the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from any of the genomic coordinates listed in Table 1.

[0457] In some embodiments, the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from: chr6:29942891-29942915; chr6:29942609-29942633; chr6:29944266-29944290; chr6:29942889-29942913; chr6:29944471-29944495; and chr6:29944470-29944494.

[0458] In some embodiments, the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from: chr6:29942891-29942915; chr6:29942609-29942633; chr6:29942889-29942913; chr6:29944471-29944495; chr6:29944266-29944290; chr6:29942785-29942809

[0459] In some embodiments, the genetic modification comprises at least 5 contiguous nucleotides within the genomic coordinates chosen from: chr6:29942891-29942915; chr6:29942609-29942633; chr6:29944266-29944290; chr6:29942889-29942913; chr6:29944471-29944495; and chr6:29944470-29944494. In some embodiments, the genetic modification comprises at least 10 contiguous nucleotides within the genomic coordinates chosen from: chr6:29942891-29942915; chr6:29942609-29942633; chr6:29944266-29944290; chr6:29942889-29942913; chr6:29944471-29944495; chr6:29944470-29944494. In some embodiments, the genetic modification comprises at least one indel, at least one C to T substitution, or at least one A to G substitution within the genomic coordinates chosen from: chr6:29942891-29942915; chr6:29942609-29942633; chr6:29944266-29944290; chr6:29942889-29942913; chr6:29944471-29944495; chr6:29944470-29944494.

[0460] In some embodiments, the genetic modification comprises at least 5 contiguous nucleotides within the genomic coordinates chosen from: chr6:29942891-29942915; chr6:29942609-29942633; chr6:29942889-29942913; chr6:29944471-29944495; chr6:29944266-29944290; chr6:29942785-29942809 In some embodiments, the genetic modification comprises at least 10 contiguous nucleotides within the genomic coordinates chosen from: chr6:29942891-29942915; chr6:29942609-29942633; chr6:29942889-29942913; chr6:29944471-29944495; chr6:29944266-29944290; chr6:29942785-29942809 In some embodiments, the genetic modification comprises at least one indel, at least one C to T substitution, or at least one A to G substitution within the genomic coordinates chosen from: chr6:29942891-29942915; chr6:29942609-29942633; chr6:29942889-29942913; chr6:29944471-29944495; chr6:29944266-29944290; chr6:29942785-29942809

[0461] In some embodiments, the modification to HLA-A comprises any one or more of an insertion, deletion, substitution, or deamination of at least one nucleotide in a target sequence. In some embodiments, the modification to HLA-A comprises an insertion of 1, 2, 3, 4 or 5 or more nucleotides in a target sequence. In some embodiments, the modification to HLA-A comprises a deletion of 1, 2, 3, 4 or 5 or more nucleotides in a target sequence. In other embodiments, the modification to HLA-A comprises an insertion of 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20 or 25 or more nucleotides in a target sequence. In other embodiments, the modification to HLA-A comprises a deletion of 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20 or 25 or more nucleotides in a target sequence. In some embodiments, the modification to HLA-A comprises an indel, which is generally defined in the art as an insertion or deletion of less than 1000 base pairs (bp). In some embodiments, the modification to HLA-A comprises an indel which results in a frameshift mutation in a target sequence. In some embodiments, the modification to HLA-A comprises a substitution of 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20 or 25 or more nucleotides in a target sequence. In some embodiments, the modification to HLA-A comprises one or more of an insertion, deletion, or substitution of nucleotides resulting from the incorporation of a template nucleic acid. In some embodiments, the modification to HLA-A is not transient.

[0462] In some embodiments, the methods and compositions disclosed herein modify the HLA-A gene in a cell using an RNA-guided DNA binding agent (e.g., a Cas enzyme). In some embodiments, the RNA-guided DNA binding agent is Cas9. In some embodiments, the RNA-guided DNA binding agent cuts within the HLA-A gene, wherein the HLA-A guide RNA targets an HLA-A genomic target sequence comprising at least 10 contiguous nucleotides within the genomic coordinates chr6:29942540-29945459.

[0463] In some embodiments, the genetic modification to HLA-A results in utilization of an out-of-frame stop codon. In some embodiments, the genetic modification to HLA-A results in exon skipping during splicing. In some embodiments, the genetic modification to HLA-A results in reduced HLA-A protein expression by the cell. In some embodiments, the modification to the HLA-A results in reduced or eliminated HLA-A protein expression on the surface of the cell.

[0464] In some embodiments, HLA-A expression on the surface of a cell is reduced as a result of the genetic modification to HLA-A. In some embodiments, HLA-A expression on the surface of a cell is absent as a result of the genetic modification to HLA-A.

2. Efficacy of HLA-A Guide RNAs

[0465] The efficacy of an HLA-A guide RNA may be determined by techniques available in the art that assess the editing efficiency of a guide RNA, the levels of HLA-A mRNA, or the levels of HLA-A protein in a target cell. In some embodiments, the reduction or elimination of HLA-A protein on the surface of a cell may be determined by comparison to an unmodified cell (or "relative to an unmodified cell"). An engineered cell or cell population may also be compared to a population of unmodified cells.

[0466] An "unmodified cell" (or "unmodified cells") refers to a control cell (or cells) of the same type of cell in an experiment or test, wherein the "unmodified" control cell has not been contacted with an HLA-A guide (i.e., a non-engineered cell). Therefore, an unmodified cell (or cells) may be a cell that has not been contacted with a guide RNA, or a cell that has been contacted with a guide RNA that does not target HLA-A.

[0467] In some embodiments, the efficacy of an HLA-A guide RNA is determined by measuring the reduction or elimination of HLA-A protein on the surface of the target cells). In some embodiments, HLA-A protein expression is measured by flow cytometry (e.g., with an antibody against HLA-A2/HLA-A3). In some embodiments, the population of cells is enriched (e.g., by FACS or MACS) and is at least 65%, 70%, 80%, 90%, 91%, 92%, 93%, or 94% HLA-A negative as measured by flow cytometry relative to a population of unmodified cells. In some embodiments, the population of cells is not enriched (e.g., by FACS or MACS) and is at least 65%, 70%, 80%, 90%, 91%, 92%, 93%, or 94% HLA-A negative as measured by flow cytometry relative to a population of unmodified cells.

[0468] In some embodiments, the population of cells is at least 65% HLA-A negative as measured by flow cytometry relative to a population of unmodified cells. In some embodiments, the population of cells is at least 70% HLA-A negative as measured by flow cytometry relative to a population of unmodified cells. In some embodiments, the population of cells is at least 80% HLA-A negative as measured by flow cytometry relative to a population of unmodified cells. In some embodiments, the population of cells is at least 90% HLA-A negative as measured by flow cytometry relative to a population of unmodified cells. In some embodiments, the population of cells is at least 91% HLA-A negative as measured by flow cytometry relative to a population of unmodified cells. In some embodiments, the population of cells is at least 92% HLA-A negative as measured by flow cytometry relative to a population of unmodified cells. In some embodiments, the population of cells is at least 93% HLA-A negative as measured by flow cytometry relative to a population of unmodified cells. In some embodiments, the population of cells is at least 94% HLA-A negative as measured by flow cytometry relative to a population of unmodified cells.

[0469] In some embodiments, an effective HLA-A guide RNA may be determined by measuring the response of immune cells in vitro or in vivo (e.g., CD8+ T cells) to the genetically modified target cell. For example, a reduced response from CD8+ T cells is indicative of an effective HLA-A guide RNA. A CD8+ T cell response may be evaluated by an assay that measures CD8+ T cell activation responses, e.g., CD8+ T cell proliferation, expression of activation markers, and/or cytokine production (IL-2, IFN-γ, TNF-α) (e.g., flow cytometry, ELISA). The CD8+ T cell response may be assessed in vitro or in vivo. In some embodiments, the CD8+ T cell response may be evaluated by co-culturing the genetically modified cell with CD8+ T cells in vitro. In some embodiments, CD8+ T cell activity may be evaluated in an in vivo model, e.g., a rodent model. In an in vivo model, e.g., genetically modified cells may be administered with CD8+ T cell; survival of the genetically modified cells is indicative of the ability to avoid CD8+ T cell lysis. In some embodiments, the methods produce a composition comprising a cell that survives in vivo in the presence of CD8+ T cells for greater than 1, 2, 3, 4, 5, or 6 weeks or more. In some embodiments, the methods produce a composition comprising a cell that survives in vivo in the presence of CD8+ T cells for at least one week to six weeks. In some embodiments, the methods produce a composition comprising a cell that survives in vivo in the presence of CD8+ T cells for at least two to four weeks. In some embodiments, the methods produce a composition comprising a cell that survives in vivo in the presence of CD8+ T cells for at least four to six weeks. In some embodiments, the methods produce a composition comprising a cell that survives in vivo in the presence of CD8+ T cells for more than six weeks.

[0470] The efficacy of an HLA-A guide RNA may also be assessed by the survival of the cell post-editing. In some embodiments, the cell survives post editing for at least one week to six weeks. In some embodiments, the cell survives post editing for at least two weeks. In some embodiments, the cell survives post editing for at least three weeks. In some embodiments, the cell survives post editing for at least four weeks. In some embodiments, the cell survives post editing for at least five weeks. In some embodiments, the cell survives post editing for at least six weeks. In some embodiments, the cell survives post editing for at least twelve weeks. The viability of a genetically modified cell may be measured using standard techniques, including e.g., by measures of cell death, by flow cytometry live/dead staining, or cell proliferation.

D. TRAC Guide RNAs

[0471] The methods and compositions provided herein disclose TRAC guide RNAs useful for reducing the expression of TRAC protein on the surface of a cell. In some embodiments, such guide RNAs direct an RNA-guided DNA binding agent to a TRAC genomic target sequence and may be referred to herein as "TRAC guide RNAs." In some embodiments, the TRAC guide RNA directs an RNA-guided DNA binding agent to a human TRAC genomic target sequence. In some embodiments, the TRAC guide RNA comprises a guide sequence selected from SEQ ID NO: 101-120.

[0472] In some embodiments, the methods and compositions disclosed herein comprise a TRAC guide RNA comprising a guide sequence that targets a TRAC genomic target sequence comprising at least 10 nucleotides within the genomic coordinates chr14:22547505-22551621 or chr14:22547462-22551621. In some embodiments, the method and composition disclosed herein comprise a TRAC guide RNA comprising a guide sequence that targets a TRAC genomic target sequence comprising at least one nucleotide within the genomic coordinates chr14:22547505-22551621 or chr14:22547462-22551621.

[0473] In some embodiments, the methods and compositions disclosed herein comprise a TRAC guide RNA comprising a guide sequence that directs an RNA-guided DNA binding agent to induce a double stranded break (DSB) or a single-stranded break (SSB) in a TRAC gene, wherein the TRAC guide RNA targets and TRAC genomic target sequence comprising at least 10 contiguous nucleotides within the genomic coordinates chr14:22547505-22551621 or chr14:22547462-22551621. In some embodiments, the methods and compositions disclosed herein comprise a TRAC guide RNA comprising a guide sequence that directs an RNA-guided DNA binding agent to induce a double stranded break (DSB) or a single-stranded break (SSB) in a TRAC gene, wherein the TRAC guide RNA targets a TRAC genomic target sequence comprising at least one nucleotide within the genomic coordinates chr14:22547505-22551621 or chr14:22547462-22551621.

[0474] In some embodiments, the methods and compositions disclose a TRAC guide RNA that directs an RNA-guided DNA binding agent to induce a double stranded break (DSB) or a single-stranded break (SSB) in a TRAC genomic target sequence. In some embodiments, the methods and compositions disclose a TRAC guide RNA that directs an RNA-guided DNA binding agent to make a cut in a TRAC genomic target sequence. In embodiments wherein the RNA-guided DNA cutting agent is Cas9, the cut or "cut site" occurs at the third base from the protospacer adjacent motif (PAM) sequence.

[0475] In some embodiments, a composition is provided comprising a TRAC guide RNA described herein and an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent.

[0476] In some embodiments, a composition is provided comprising a TRAC single-guide RNA (sgRNA) comprising a guide sequence that targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chr14:22547505-22551621 or chr14:22547462-22551621. In some embodiments, a composition is provided comprising a TRAC sgRNA described herein and an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent.

[0477] In some embodiments, a composition is provided comprising a TRAC dual-guide RNA (dgRNA) comprising a guide sequence that targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chr14:22547505-22551621 or chr14:22547462-22551621. In some embodiments, a composition is provided comprising a TRAC dgRNA described herein and an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent.

[0478] In some embodiments, the TRAC gRNA comprises a guide sequence selected from any one of SEQ ID NOs: 101-120. Exemplary TRAC guide sequences are shown below in Table 2 (SEQ ID NOs: 101-120) with corresponding guide RNA sequences 101-120.

TABLE 2

Exemplary TRAC guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID Nos: 1101-1120)	Exemplary Guide RNA Modified Sequence (SEQ ID NOs: 2101-2120)	Genomic Coordinates (hg38)
G021469	101	AUAUCCAGAACCCUGACCCUGCCG	AUAUCCAGAACCCUGACCCUGCCGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mA*mU*mA*mUmCCAmGmAAmCCmCUGACmCCUGmCCGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr14: 22547505-22547529
G021481	102	GCCGUGUACCAGCUGAGAGACUCU	GCCGUGUACCAGCUGAGAGACUCUGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mG*mC*mC*mGmUGUmAmCCmAGmCUGAGmAGACmUCUmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr14: 22547525-22547549
G028935	103	CAGGCCACAGCACUGUUGCUCUUG	CAGGCCACAGCACUGUUGCUCUUGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mA*mG*mGmCCAmCmAGmCAmCUGUUmGCUCmUUGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr14: 22547674-22547698
G028936	104	CCCAGCCCAGGUAAGGGCAGCUUU	CCCAGCCCAGGUAAGGGCAGCUUUGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mC*mC*mAmGCCmCmAGmGUmAAGGGmCAGCmUUUmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr14: 22547768-22547792
G028937	105	GUCUUACCUGUUUCAAAGCUUUUC	GUCUUACCUGUUUCAAAGCUUUUCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGC	mG*mU*mC*mUmUACmCmUGmUUmUCAAAmGCUUmUUCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmU	chr14: 22549665-22549689

TABLE 2-continued

Exemplary TRAC guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID Nos: 1101-1120)	Exemplary Guide RNA Modified Sequence (SEQ ID NOs: 2101-2120)	Genomic Coordinates (hg38)
			AACGCUCUGCC UUCUGGCAUCG UU	GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	
G028938	106	CUUUGAAACAG GUAAGACAGGG GU	CUUUGAAACAG GUAAGACAGGG GUGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mU*mU*mUmGAAm AmCAmGGmUAAGAmCA GGmGGUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr14: 22549671-22549695
G028939	107	UUAGGUUCGUA UCUGUAAAACC AA	UUAGGUUCGUA UCUGUAAAACC AAGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mU*mU*mA*mGmGUUm CmGUmAUmCUGUAmAA ACmCAAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr14: 22550544-22550568
G028940	108	UUACAGAUACG AACCUAAACUU UC	UUACAGAUACG AACCUAAACUU UCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mU*mU*mA*mCmAGAm UmACmGAmACCUAmAA CUmUUCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr14: 22550550-22550574
G028941	109	UUGAAAGUUUA GGUUCGUAUCU GU	UUGAAAGUUUA GGUUCGUAUCU GUGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mU*mU*mG*mAmAAGm UmUUmAGmGUUCGmUA UCmUGUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr14: 22550552-22550576
G028942	110	ACUUUCAAAAC CUGUCAGUGAU UG	ACUUUCAAAAC CUGUCAGUGAU UGGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mA*mC*mU*mUmUCAm AmAAmCCmUGUCAmGU GAmUUGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr14: 22550568-22550592
G028943	111	AAAACCUGUCA GUGAUUGGGUU CC	AAAACCUGUCA GUGAUUGGGUU CCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mA*mA*mA*mAmCCUm GmUCmAGmUGAUUmGG GUmUCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr14: 22550574-22550598

TABLE 2-continued

Exemplary TRAC guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID Nos: 1101-1120)	Exemplary Guide RNA Modified Sequence (SEQ ID NOs: 2101-2120)	Genomic Coordinates (hg38)
G028944	112	UGUCAGUGAUU GGGUUCCGAAU CC	UGUCAGUGAUU GGGUUCCGAAU CCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mU*mG*mU*mCmAGUm GmAUmUGmGGUUCmCG AAmUCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr14: 22550580-22550604
G028945	113	GGGUUCCGAAU CCUCCUCCUGA AA	GGGUUCCGAAU CCUCCUCCUGA AAGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mG*mG*mG*mUmUCCm GmAAmUCmCUCCUmCC UGmAAAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr14: 22550591-22550615
G028946	114	AAGGCCCCUCA CCUCAGCUGGA CC	AAGGCCCCUCA CCUCAGCUGGA CCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mA*mA*mG*mGmCCCm CmUCmACmCUCAGmCU GGmACCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr14: 22550652-22550676
G028947	115	AGCUUCAAGGC CCCUCACCUCA GC	AGCUUCAAGGC CCCUCACCUCA GCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mA*mG*mC*mUmUCAm AmGGmCCmCCUCAmCC UCmAGCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr14: 22550658-22550682
G028948	116	AUUCCAGAUCU GCAAGAUUGUA AG	AUUCCAGAUCU GCAAGAUUGUA AGGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mA*mU*mU*mCmCAGm AmUCmUGmCAAGAmUU GUmAAGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr14: 22551597-22551621
G021475	117	AACCCUGAUCC UCUUGUCCCAC AG	AACCCUGAUCC UCUUGUCCCAC AGGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mA*mA*mC*mCmCUGm AmUCmCUmCUUGUmCC CAmCAGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr14: 22547481-22547505
G021476	118	AUCAUGUCCUA ACCCUGAUCCU CU	AUCAUGUCCUA ACCCUGAUCCU CUGUUGUAGCU	mA*mU*mC*mAmUGUm CmCUmAAmCCCUGmAU CCmUCUmGUUGmUmAm	chr14: 22547471-22547495

TABLE 2-continued

Exemplary TRAC guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID Nos: 1101-1120)	Exemplary Guide RNA Modified Sequence (SEQ ID NOs: 2101-2120)	Genomic Coordinates (hg38)
			CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	
G021477	119	GAUCAUGUCCU AACCCUGAUCC UC	GAUCAUGUCCU AACCCUGAUCC UCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mG*mA*mU*mCmAUGm UmCCmUAmACCCUmGA UCmCUCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr14: 22547470-22547494
G021478	120	GGAAAUGAGAU CAUGUCCUAAC CC	GGAAAUGAGAU CAUGUCCUAAC CCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mG*mG*mA*mAmAUGm AmGAmUCmAUGUCmCU AAmCCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr14: 22547462-22547486

[0479] Throughout this application, the terms "mA," "MC", "mU," or "mG" may be used to denote a nucleotide that has been modified with 2'—O-Me. Throughout this application, the terms A*, C*, U*, or G* may be used to denote a nucleotide that is linked to the next (e.g., 3') nucleotide with a phosphorothioate (PS) bond.

[0480] In some embodiments, the TRAC guide RNA comprises a guide sequence selected from SEQ ID NOs: 101-120. In some embodiments, the TRAC guide RNA comprises a guide sequence that is at least 19, 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence selected from SEQ ID NOs: 101-120. In some embodiments, the TRAC guide RNA comprises a guide sequence that is at least 95%, 90%, or 85% identical to a sequence selected from SEQ ID NOs: 101-120. In some embodiments, the TRAC guide RNA comprises a guide sequence that is at least 95% identical to a sequence selected from SEQ ID NOs: 101-120.

[0481] In some embodiments, the TRAC guide RNA comprises a guide sequence that comprises at least 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 2. As used herein, at least 10 contiguous nucleotides±10 nucleotides of a genomic coordinate means, for example, at least 10 contiguous nucleotides within the genomic coordinates wherein the genomic coordinates include 10 nucleotides in the 5' direction and 10 nucleotides in the 3' direction from the ranges listed in Table 2. For example, a TRAC guide RNA may comprise 10 contiguous nucleotides within the genomic coordinates chosen from: chr14:22547505-22547529; chr14:22547525-22547549; chr14:22547674-22547698; chr14:22550544-22550568; or chr14:22550574-22550598; including the boundary nucleotides of these ranges. As another example, a TRAC guide RNA may comprise 10 contiguous nucleotides within the genomic coordinates chosen from: chr14: 22547481-22547505; chr14:22547471-22547495; chr14: 22547470-22547494; or chr14:22547462-22547486; including the boundary nucleotides of these ranges. In some embodiments, the TRAC guide RNA comprises a guide sequence that is at least 19, 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence that comprises 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 2. In some embodiments, the TRAC guide RNA comprises a guide sequence that is at least 95%, 90%, or 85% identical to a sequence selected from a sequence that is 19, 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence that comprises 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 2.

[0482] In some embodiments, the TRAC guide RNA comprises a guide sequence that comprises at least 15 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 2. In some embodiments, the TRAC guide RNA comprises a guide sequence that comprises at least 24 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 2.

[0483] In some embodiments, the TRAC guide RNA comprises SEQ ID NO: 101. In some embodiments, the TRAC guide RNA comprises SEQ ID NO: 102. In some embodiments, the TRAC guide RNA comprises SEQ ID NO: 103. In some embodiments, the TRAC guide RNA comprises SEQ ID NO: 104. In some embodiments, the

TRAC guide RNA comprises SEQ ID NO: 105. In some embodiments, the TRAC guide RNA comprises SEQ ID NO: 106. In some embodiments, the TRAC guide RNA comprises SEQ ID NO: 107. In some embodiments, the TRAC guide RNA comprises SEQ ID NO: 108. In some embodiments, the TRAC guide RNA comprises SEQ ID NO: 109. In some embodiments, the TRAC guide RNA comprises SEQ ID NO: 110. In some embodiments, the TRAC guide RNA comprises SEQ ID NO: 111. In some embodiments, the TRAC guide RNA comprises SEQ ID NO: 112. In some embodiments, the TRAC guide RNA comprises SEQ ID NO: 113. In some embodiments, the TRAC guide RNA comprises SEQ ID NO: 114. In some embodiments, the TRAC guide RNA comprises SEQ ID NO: 115. In some embodiments, the TRAC guide RNA comprises SEQ ID NO: 116. In some embodiments, the TRAC guide RNA comprises SEQ ID NO: 117. In some embodiments, the TRAC guide RNA comprises SEQ ID NO: 118. In some embodiments, the TRAC guide RNA comprises SEQ ID NO: 119. In some embodiments, the TRAC guide RNA comprises SEQ ID NO: 120.

[0484] In some embodiments, the TRAC guide RNA comprises a nucleotide chosen from: SEQ ID NOs: 101-103, 107, 111, 117, or 118.

[0485] In some embodiments, the TRAC guide RNA comprises a sequence listed in Table 2. In some embodiments, the TRAC guide RNA comprises a sequence of any one of SEQ ID NOs: 101-120. In some embodiments, the TRAC guide RNA comprises a sequence of any one of SEQ ID NOs: 107, 111, and 117-120. In some embodiments, the TRAC guide RNA comprises a sequence of SEQ ID NO: 107. In some embodiments, the TRAC guide RNA comprises a sequence of SEQ ID NO: 111. In some embodiments, the TRAC guide RNA comprises a sequence of SEQ ID NO: 117. In some embodiments, the TRAC guide RNA comprises a sequence of SEQ ID NO: 118. In some embodiments, the TRAC guide RNA comprises a sequence of SEQ ID NO: 119. In some embodiments, the TRAC guide RNA comprises a sequence of SEQ ID NO: 120. In some embodiments, the TRAC guide RNA comprises a guide sequence comprising a sequence of any one of SEQ ID NOs: 101-120. In some embodiments, the TRAC guide RNA comprises a guide sequence comprising a sequence of any one of SEQ ID NOs: 107, 111, and 117-120. In some embodiments, the TRAC guide RNA comprises a guide sequence comprising a sequence of SEQ ID NO: 107. In some embodiments, the TRAC guide RNA comprises a guide sequence comprising a sequence of SEQ ID NO: 111. In some embodiments, the TRAC guide RNA comprises a guide sequence comprising a sequence of SEQ ID NO: 117. In some embodiments, the TRAC guide RNA comprises a guide sequence comprising a sequence of SEQ ID NO: 118. In some embodiments, the TRAC guide RNA comprises a guide sequence comprising a sequence of SEQ ID NO: 119. In some embodiments, the TRAC guide RNA comprises a guide sequence comprising a sequence of SEQ ID NO: 120. In some embodiments, the TRAC guide RNA comprises a sequence of any one of SEQ ID NOs: 1101-1120. In some embodiments, the TRAC guide RNA comprises a sequence of any one of SEQ ID NOs: 1107, 1111, and 1117-1120. In some embodiments, the TRAC guide RNA comprises a sequence of SEQ ID NO: 1107. In some embodiments, the TRAC guide RNA comprises a sequence of SEQ ID NO: 1111. In some embodiments, the TRAC guide RNA comprises a sequence of SEQ ID NO: 1117. In some embodiments, the TRAC guide RNA comprises a sequence of SEQ ID NO: 1118. In some embodiments, the TRAC guide RNA comprises a sequence of SEQ ID NO: 1119. In some embodiments, the TRAC guide RNA comprises a sequence of SEQ ID NO: 1120. In some embodiments, the TRAC guide RNA comprises a sequence of any one of SEQ ID NOs: 2101-2120, 3003, and 3004. In some embodiments, the TRAC guide RNA comprises a sequence of any one of SEQ ID NOs: 2107, 2111, 2117-2120, 3003, and 3004. In some embodiments, the TRAC guide RNA comprises a sequence of SEQ ID NO: 2107. In some embodiments, the TRAC guide RNA comprises a sequence of SEQ ID NO: 2111. In some embodiments, the TRAC guide RNA comprises a sequence of SEQ ID NO: 2117. In some embodiments, the TRAC guide RNA comprises a sequence of SEQ ID NO: 2118. In some embodiments, the TRAC guide RNA comprises a sequence of SEQ ID NO: 2119. In some embodiments, the TRAC guide RNA comprises a sequence of SEQ ID NO: 2120. In some embodiments, the TRAC guide RNA comprises a sequence of SEQ ID NO: 3003. In some embodiments, the TRAC guide RNA comprises a sequence of SEQ ID NO: 3004.

[0486] In some embodiments, the TRAC guide RNA is a single guide RNA (sgRNA) comprising a sequence of any one of the sgRNA sequences listed in Table 2.

[0487] Additional embodiments of TRAC guide RNAs are provided herein, including e.g., exemplary modifications to the guide RNA. 1. Genetic modifications to TRAC

[0488] In some embodiments, the methods and compositions disclosed herein genetically modify at least one nucleotide of the TRAC gene in a cell. In some embodiments, the genetic modification to TRAC reduces or eliminates the expression of TRAC protein on the surface of the genetically modified cell (or engineered cell). Genetic modifications encompass the population of modifications that results from contact with a genomic editing system (e.g., the population of edits that result from Cas9 and a TRAC guide RNA, or the population of edits that result from BC22 and a TRAC guide RNA).

[0489] In some embodiments, the genetic modification comprises at least one nucleotide within the genomic coordinates chr14:22547505-22551621 or chr14:22547462-22551621. In some embodiments, the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from any of the genomic coordinates listed in Table 2.

[0490] In some embodiments, the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from: chr14:22547505-22547529; chr14:22547525-22547549; chr14:22547674-22547698; chr14:22550544-22550568; or chr14:22550574-22550598. In some embodiments, the genetic modification comprises at least 5 contiguous nucleotides within the genomic coordinates chosen from: chr14:22547505-22547529; chr14:22547525-22547549; chr14:22547674-22547698; chr14:22550544-22550568; or chr14:22550574-22550598. In some embodiments, the genetic modification comprises at least 10 contiguous nucleotides within the genomic coordinates chosen from: chr14:22547505-22547529; chr14:22547525-22547549; chr14:22547674-22547698; chr14:22550544-22550568; or chr14:22550574-22550598. In some embodiments, the genetic modification comprises at least 10 contiguous nucleotides within the genomic coordinates chosen from: chr14:22547481-22547505; chr14:22547471-22547495; chr14:22547470-22547494; or chr14:22547462-22547486.

[0491] In some embodiments, the genetic modification comprises at least one indel, at least one C to T substitution, or at least one A to G substitution within the genomic coordinates chosen from: chr14:22547505-22547529; chr14:22547525-22547549; chr14:22547674-22547698; chr14:22550544-22550568; or chr14:22550574-22550598. In some embodiments, the genetic modification comprises at least one indel, at least one C to T substitution, or at least one A to G substitution within the genomic coordinates chosen from: chr14:22547481-22547505; chr14:22547471-22547495; chr14:22547470-22547494; or chr14:22547462-22547486.

[0492] In some embodiments, the modification to TRAC comprises any one or more of an insertion, deletion, substitution, or deamination of at least one nucleotide in a target sequence. In some embodiments, the modification to TRAC comprises an insertion of 1, 2, 3, 4 or 5 or more nucleotides in a target sequence. In some embodiments, the modification to TRAC comprises a deletion of 1, 2, 3, 4 or 5 or more nucleotides in a target sequence. In other embodiments, the modification to TRAC comprises an insertion of 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20 or 25 or more nucleotides in a target sequence. In other embodiments, the modification to TRAC comprises a deletion of 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20 or 25 or more nucleotides in a target sequence. In some embodiments, the modification to TRAC comprises an indel, which is generally defined in the art as an insertion or deletion of less than 1000 base pairs (bp). In some embodiments, the modification to TRAC comprises an indel which results in a frameshift mutation in a target sequence. In some embodiments, the modification to TRAC comprises a substitution of 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20 or 25 or more nucleotides in a target sequence. In some embodiments, the modification to TRAC comprises one or more of an insertion, deletion, or substitution of nucleotides resulting from the incorporation of a template nucleic acid. In some embodiments, the modification to TRAC comprises an insertion of a donor nucleic acid in a target sequence. In some embodiments, the modification to TRAC is not transient.

[0493] In some embodiments, the methods and compositions disclosed herein modify the TRAC gene in a cell using an RNA-guided DNA binding agent (e.g., a Cas enzyme). In some embodiments, the RNA-guided DNA binding agent is Cas9. In some embodiments, the RNA-guided DNA binding agent cuts within the TRAC gene, wherein the TRAC guide RNA targets a TRAC genomic target sequence comprising at least 10 contiguous nucleotides chr14:22547505-22551621 or chr14:22547462-22551621.

[0494] In some embodiments, the genetic modification to TRAC results in utilization of an out-of-frame stop codon. In some embodiments, the genetic modification to TRAC results in exon skipping during splicing. In some embodiments, the genetic modification to TRAC results in reduced TRAC protein expression by the cell. In some embodiments, the modification to the TRAC gene results in reduced or eliminated TRAC protein expression on the surface of the cell.

[0495] In some embodiments, TRAC expression on the surface of a cell is reduced as a result of the genetic modification to TRAC. In some embodiments, TRAC expression on the surface of a cell is absent as a result of the genetic modification to TRAC.

2. Efficacy of TRAC Guide RNAs

[0496] In some embodiments, the efficacy of a TRAC gRNA is determined when delivered or expressed together with other components forming an RNP. In some embodiments, the TRAC gRNA is expressed together with an RNA-guided DNA binding agent, such as a Cas protein, e.g. Cas9. In some embodiments, the TRAC gRNA is delivered to or expressed in a cell line that already stably expresses an RNA-guided DNA nuclease, such as a Cas nuclease or nickase, e.g. Cas9 nuclease or nickase. In some embodiments the TRAC gRNA is delivered to a cell as part of a RNP. In some embodiments, the TRAC gRNA is delivered to a cell along with a mRNA encoding an RNA-guided DNA nuclease, such as a Cas nuclease or nickase, e.g. Cas9 nuclease or nickase.

[0497] As described herein, use of an RNA-guided DNA nuclease and a TRAC guide RNA disclosed herein can lead to double-stranded breaks in the DNA which can produce errors in the form of insertion/deletion (indel) mutations upon repair by cellular machinery. Many mutations due to indels alter the reading frame or introduce premature stop codons and, therefore, produce a non-functional protein.

[0498] In some embodiments, the efficacy of particular TRAC gRNAs is determined based on in vitro models. In some embodiments, the in vitro model is HEK293 cells stably expressing Cas9 (HEK293_Cas9). In some embodiments the in vitro model is a peripheral blood mononuclear cell (PBMC). In some embodiments, the in vitro model is a T cell, such as primary human T cells. With respect to using primary cells, commercially available primary cells can be used to provide greater consistency between experiments. In some embodiments, the number of off-target sites at which a deletion or insertion occurs in an in vitro model (e.g., in T cell) is determined, e.g., by analyzing genomic DNA from transfected cells in vitro with Cas9 mRNA and the guide RNA. In some embodiments, such a determination comprises analyzing genomic DNA from the cells transfected in vitro with Cas9 mRNA, the TRAC guide RNA, and a donor oligonucleotide. Exemplary procedures for such determinations are provided in the working examples in which HEK293 cells, PBMCs, and human CD3+ T cells are used.

[0499] In some embodiments, the efficacy of particular TRAC gRNAs is determined across multiple in vitro cell models for a gRNA selection process. In some embodiments, a cell line comparison of data with selected TRAC gRNAs is performed. In some embodiments, cross screening in multiple cell models is performed.

[0500] In some embodiments, the efficacy of a TRAC guide RNA is measured by percent indels of TRAC. In some embodiments, the percent editing of TRAC is compared to the percent indels necessary to achieve knockdown of the TRAC protein products

[0501] In some embodiments, the efficacy of a guide RNA is measured by reduced or eliminated expression of a component of the T-cell receptor (TCR). In embodiments, the reduced or eliminated expression of a component of the T-cell receptor (TCR) includes reduced or eliminated expression of TRAC. In some embodiments, said reduced or eliminated expression of said component of the TCR is the result of introduction of one or more, e.g., one or two, e.g., one TRAC gRNA molecule described herein to said component of the TCR into said cell. In embodiments, said reduced or eliminated expression of a component of the TCR is as measured by flow cytometry, e.g., as described herein.

[0502] In some embodiments, the efficacy of a TRAC guide RNA is measured by the number and/or frequency of indels at off-target sequences within the genome of the target cell type, such as a T cell. In some embodiments, efficacious guide RNAs are provided which produce indels at off target sites at very low frequencies (e.g., <5%) in a cell population and/or relative to the frequency of indel creation at the target site. Thus, the disclosure provides for guide RNAs which do not exhibit off-target indel formation in the target cell type (e.g., a T cell), or which produce a frequency of off-target indel formation of <5% in a cell population and/or relative to the frequency of indel creation at the target site. In some embodiments, the disclosure provides guide RNAs which do not exhibit any off target indel formation in the target cell type (e.g., T cell). In some embodiments, guide RNAs are provided which produce indels at less than 5 off-target sites, e.g., as evaluated by one or more methods described herein. In some embodiments, guide RNAs are provided which produce indels at less than or equal to 4, 3, 2, or 1 off-target site(s) e.g., as evaluated by one or more methods described herein. In some embodiments, the off-target site(s) does not occur in a protein coding region in the target cell (e.g., hepatocyte) genome.

[0503] In some embodiments, detecting gene editing events, such as the formation of insertion/deletion ("indel") mutations and homology directed repair (HDR) events in target DNA utilize linear amplification with a tagged primer and isolating the tagged amplification products (herein after referred to as "LAM-PCR," or "Linear Amplification (LA)" method).

[0504] In some embodiments, the efficacy of a guide RNA is measured by the levels of functional protein complexes comprising the expressed protein product of the gene. In some embodiments, the efficacy of a guide RNA is measured by flow cytometric analysis of TCR expression by which the live population of edited cells is analyzed for loss of the TCR.

E. TRBC Guide RNAs

[0505] The methods and compositions provided herein disclose TRBC guide RNAs useful for reducing the expression of TRBC1 protein or TRBC2 protein, or both, on the surface of a cell. In some embodiments, such guide RNAs direct an RNA-guided DNA binding agent to a TRBC1 genomic target sequence and may be referred herein as "TRBC1 guide RNAs." In some embodiments, the TRBC1 guide RNA directs an RNA-guided DNA binding agent to a human TRBC1 genomic target sequence. In some embodiments, such guide RNAs direct an RNA-guided DNA binding agent to a TRBC2 genomic target sequence and may be referred herein as "TRBC2 guide RNAs." In some embodiments, the TRBC2 guide RNA directs an RNA-guided DNA binding agent to a human TRBC2 genomic target sequence. In some embodiments, such guide RNAs direct an RNA-guided DNA binding agent to a TRBC1 genomic target sequence and to a TRBC2 genomic target sequence and may be referred herein as "TRBC1/2 guide RNAs." In some embodiments, the TRBC1/2 guide RNA directs an RNA-guided DNA binding agent to a human TRBC1 genomic target sequence and to a human TRBC2 genomic target sequence. In some embodiments, a TRBC1 guide RNA may contain a mismatch to a TRBC2 genomic target sequence, but the TRBC1 guide RNA may still direct an RNA-guided DNA binding agent to a TRBC2 genomic target sequence. In some embodiments, a TRBC2 guide RNA may contain a mismatch to a TRBC1 genomic target sequence, but the TRBC2 guide RNA may still direct an RNA-guided DNA binding agent to a TRBC1 genomic target sequence.

[0506] In some embodiments, the methods and compositions disclosed herein comprise a TRBC1 guide RNA comprising a guide sequence that targets a TRBC1 genomic target sequence comprising at least 10 nucleotides within the genomic coordinates chr7:142791862-142793149. In some embodiments, the method and composition disclosed herein comprise a TRBC1 guide RNA comprising a guide sequence that targets a TRBC1 genomic target sequence comprising at least one nucleotide within the genomic coordinates chr7: 142791862-142793149.

[0507] In some embodiments, the methods and compositions disclosed herein comprise a TRBC2 guide RNA comprising a guide sequence that targets a TRBC2 genomic target sequence comprising at least 10 nucleotides within the genomic coordinates chr7:142801104-142802543. In some embodiments, the method and composition disclosed herein comprise a TRBC2 guide RNA comprising a guide sequence that targets a TRBC2 genomic target sequence comprising at least one nucleotide within the genomic coordinates chr7: 142801104-142802543.

[0508] In some embodiments, the methods and compositions disclosed herein comprise a TRBC1 guide RNA comprising a guide sequence that directs an RNA-guided DNA binding agent to induce a double stranded break (DSB) or a single-stranded break (SSB) in a TRBC1 gene, wherein the TRBC1 guide RNA targets and TRBC1 genomic target sequence comprising at least 10 contiguous nucleotides within the genomic coordinates chr7:142791862-142793149. In some embodiments, the methods and compositions disclosed herein comprise a TRBC1 guide RNA comprising a guide sequence that directs an RNA-guided DNA binding agent to induce a double stranded break (DSB) or a single-stranded break (SSB) in a TRBC1 gene, wherein the TRBC1 guide RNA targets a TRBC1 genomic target sequence comprising at least one nucleotide within the genomic coordinates chr7:142791862-142793149.

[0509] In some embodiments, the methods and compositions disclosed herein comprise a TRBC2 guide RNA comprising a guide sequence that directs an RNA-guided DNA binding agent to induce a double stranded break (DSB) or a single-stranded break (SSB) in a TRBC2 gene, wherein the TRBC2 guide RNA targets and TRBC2 genomic target sequence comprising at least 10 contiguous nucleotides within the genomic coordinates chr7:142801104-142802543. In some embodiments, the methods and compositions disclosed herein comprise a TRBC2 guide RNA comprising a guide sequence that directs an RNA-guided DNA binding agent to induce a double stranded break (DSB) or a single-stranded break (SSB) in a TRBC2 gene, wherein the TRBC2 guide RNA targets a TRBC2 genomic target sequence comprising at least one nucleotide within the genomic coordinates chr7:142801104-142802543.

[0510] In some embodiments, the methods and compositions disclose a TRBC1 guide RNA that directs an RNA-guided DNA binding agent to induce a double stranded break (DSB) or a single-stranded break (SSB) in a TRBC1 or TRBC2 genomic target sequence. In some embodiments, the methods and compositions disclose a TRBC1 or TRBC2 guide RNA that directs an RNA-guided DNA binding agent to make a cut in a TRBC1 or TRBC2 genomic target sequence. In embodiments wherein the RNA-guided DNA cutting agent is Cas9, the cut or "cut site" occurs at the third base from the protospacer adjacent motif (PAM) sequence.

[0511] In some embodiments, a composition is provided comprising a TRBC guide RNA described herein and an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent.

[0512] In some embodiments, a composition is provided comprising a TRBC1 single-guide RNA (sgRNA) comprising a guide sequence that targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chr7:142791862-142793149. In some embodiments, a composition is provided comprising a TRBC1 sgRNA described herein and an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent.

[0513] In some embodiments, a composition is provided comprising a TRBC2 single-guide RNA (sgRNA) comprising a guide sequence that targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chr7:142801104-142802543. In some embodiments, a composition is provided comprising a TRBC2 sgRNA described herein and an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent.

[0514] In some embodiments, a composition is provided comprising a TRBC1 dual-guide RNA (dgRNA) comprising a guide sequence that targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chr7:142791862-142793149. In some embodiments, a composition is provided comprising a TRBC1 dgRNA described herein and an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent.

[0515] In some embodiments, a composition is provided comprising a TRBC2 dual-guide RNA (dgRNA) comprising a guide sequence that targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chr7:142801104-142802543. In some embodiments, a composition is provided comprising a TRBC2 dgRNA described herein and an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent.

[0516] In some embodiments, the TRBC gRNA comprises a guide sequence selected from any one of SEQ ID NOs: 201-265. Exemplary TRBC guide sequences are shown below in Table 3 (SEQ ID NOs: 201-265) with corresponding guide RNA sequences 201-265.

TABLE 3

Exemplary TRBC1 and TRBC2 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Target	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1201-1265)	Exemplary Guide RNA Modified Sequence (SEQ ID NOs: 2201-2265)	Genomic Coordinates (hg38)
G028952	201	TRBC1	CACGGACCCGCAGCCCCUCAAGGA	CACGGACCCGCAGCCCCUCAAGGAGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mA*mC*mGmGACmCmCGmCAmGCCCCmUCAAmGGAmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr7:142791862-142791886
G028953	202	TRBC1	ACGGACCCGCAGCCCCUCAAGGAG	ACGGACCCGCAGCCCCUCAAGGAGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mA*mC*mG*mGmACCmCmGCmAGmCCCCUmCAAGmGAGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr7:142791863-142791887

TABLE 3-continued

Exemplary TRBC1 and TRBC2 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Target	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1201-1265)	Exemplary Guide RNA Modified Sequence (SEQ ID NOs: 2201-2265)	Genomic Coordinates (hg38)
G028968	203	TRBC1	GACGAGUGGACCCAGGAUAGGGCC	GACGAGUGGACCCAGGAUAGGGCCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mG*mA*mC*mGmAGUmGmGAmCCmCAGGAmUAGGmGCCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr7: 142792010-142792034
G028969	204	TRBC1	UGGACCCAGGAUAGGGCCAAACCC	UGGACCCAGGAUAGGGCCAAACCCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mU*mG*mG*mAmCCCmAmGGmAUmAGGGCmCAAAmCCCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr7: 142792016-142792040
G028970	205	TRBC1	GGACCCAGGAUAGGGCCAAACCCG	GGACCCAGGAUAGGGCCAAACCCGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mG*mG*mA*mCmCCAmGmGAmUAmGGGCCmAAACmCCGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr7: 142792017-142792041
G028977	206	TRBC1	CCCUGCUUUCUUUCAGACUGUGGC	CCCUGCUUUCUUUCAGACUGUGGCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mC*mC*mUmGCUmUmUCmUUmUCAGAmCUGUmGGCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCA	chr7: 142792505-142792529

TABLE 3-continued

Exemplary TRBC1 and TRBC2 guide sequences.						
Guide ID	SEQ ID NO to the Guide Sequence	Target	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1201-1265)	Exemplary Guide RNA Modified Sequence (SEQ ID NOs: 2201-2265)	Genomic Coordinates (hg38)
					AU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	
G028978	207	TRBC1	GACUGUGGCUUUACCUCGGGUAAG	GACUGUGGCUUUACCUCGGGUAAGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mG*mA*mC*mUmGUGmGmCUmUUmACCUCmGGGUmAAGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr7: 142792520-142792544
G028979	208	TRBC1	ACUGUGGCUUUACCUCGGGUAAGU	ACUGUGGCUUUACCUCGGGUAAGUGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mA*mC*mU*mGmUGGmCmUUmUAmCCUCGmGGUAmAGUmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr7: 142792521-142792545
G028980	209	TRBC1	UGGCUUUACCUCGGGUAAGUAAGC	UGGCUUUACCUCGGGUAAGUAAGCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mU*mG*mG*mCmUUUmAmCCmUCmGGGUAmAGUAmAGCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr7: 142792525-142792549

TABLE 3-continued

Exemplary TRBC1 and TRBC2 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Target	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1201-1265)	Exemplary Guide RNA Modified Sequence (SEQ ID NOs: 2201-2265)	Genomic Coordinates (hg38)
G028981	210	TRBC1	UACCUCGGGUAAGUAAGCCCUUCC	UACCUCGGGUAAGUAAGCCCUUCCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mU*mA*mC*mCmUCGmGmGUmAAmGUAAGmCCCUmUCCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr7: 142792531-142792555
G028982	211	TRBC1	AGGAAGGGCUUACUUACCCGAGGU	AGGAAGGGCUUACUUACCCGAGGUGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mA*mG*mG*mAmAGGmGmCUmUAmCUUACmCCGAmGGUmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr7: 142792532-142792556
G028983	212	TRBC1	CGGGUAAGUAAGCCCUUCCUUUC	CGGGUAAGUAAGCCCUUCCUUUUCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mG*mG*mGmUAAmGmUAmAGmCCCUUmCCUUmUUCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr7: 142792536-142792560
G028984	213	TRBC1	GGGUAAGUAAGCCCUUCCUUUUCC	GGGUAAGUAAGCCCUUCCUUUUCCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mG*mG*mG*mUmAAGmUmAAmGCmCCUUCmCUUUmUCCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmG	chr7: 142792537-142792561

TABLE 3-continued

Exemplary TRBC1 and TRBC2 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Target	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1201-1265)	Exemplary Guide RNA Modified Sequence (SEQ ID NOs: 2201-2265)	Genomic Coordinates (hg38)
					mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	
G028985	214	TRBC1	UGCCAAC AGUGUCC UACCAGC AAG	UGCCAACAGUGUCCUACCA GCAAGGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mU*mG*mC* mCmAACmAm GUmGUmCCU ACmCAGCmA AGmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	chr7: 142792682-142792706
G028986	215	TRBC1	GUGUCCU ACCAGCA AGGGGUC CUG	GUGUCCUACCAGCAAGGGG UCCUGGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mG*mU*mG* mUmCCUmAm CCmAGmCAA GGmGGUCmC UGmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	chr7: 142792690-142792714
G028987	216	TRBC1	UCCUACC AGCAAGG GGUCCUG UCU	UCCUACCAGCAAGGGGUCC UGUCUGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mU*mC*mC* mUmACCmAm GCmAAmGGG GUmCCUGmU CUmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	chr7: 142792693-142792717
G028989	217	TRBC1	ACCAUGG CCAUCAA CACAAGG GCG	ACCAUGGCCAUCAACACAA GGGCGGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG	mA*mC*mC* mAmUGGmCm CAmUCmAAC ACmAAGGmG	chr7: 142792776-142792800

TABLE 3-continued

Exemplary TRBC1 and TRBC2 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Target	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1201-1265)	Exemplary Guide RNA Modified Sequence (SEQ ID NOs: 2201-2265)	Genomic Coordinates (hg38)
				CAACGCUCUGCCUUCUGGC AUCGUU	CGmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	
G028990	218	TRBC1	CCAUGGU AAGCAGG AGGGCAG GAU	CCAUGGUAAGCAGGAGGGC AGGAUGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mC*mC*mA* mUmGGUmAm AGmCAmGGA GGmGCAGmG AUmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	chr7: 142792793-142792817
G028991	219	TRBC1	GGUCAAG AGAAAGG AUUUCUG AAG	GGUCAAGAGAAAGGAUUUC UGAAGGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mG*mG*mU* mCmAAGmAm GAmAAmGGA UUmUCUGmA AGmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	chr7: 142793119-142793143
G028992	220	TRBC1	GUCAAGA GAAAGGA UUUCUGA AGG	GUCAAGAGAAAGGAUUUCU GAAGGGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mG*mU*mC* mAmAGAmGm AAmAGmGAU UUmCUGAmA GGmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC	chr7: 142793120-142793144

TABLE 3-continued

Exemplary TRBC1 and TRBC2 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Target	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1201-1265)	Exemplary Guide RNA Modified Sequence (SEQ ID NOs: 2201-2265)	Genomic Coordinates (hg38)
					mUmUmCmUG GCAUCG*mU *mU	
G028993	221	TRBC1	GGCUGCC UUCAGAA AUCCUUU CUC	GGCUGCCUUCAGAAAUCCU UUCUCGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mG*mG*mC* mUmGCCmUm UCmAGmAAA UCmCUUUmC UCmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	chr7: 142793125-142793149
G028949	222	TRBC1/2	CCCACAC CCAAAAG GCCACAC UGG	CCCACACCCAAAAGGCCAC ACUGGGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mC*mC*mC* mAmCACmCm CAmAAmAGG CCmACACmU GGmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	chr7: 142791756-142791780
G028950	223	TRBC1/2	ACCCAAA AGGCCAC ACUGGUG UGC	ACCCAAAAGGCCACACUGG UGUGCGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mA*mC*mC* mCmAAAmAm GGmCCmACA CUmGGUGmU GCmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	chr7: 142791761-142791785
G028951	224	TRBC1/2	UCCUUCC CAUUCAC CCACCAG CUC	UCCUUCCCAUUCACCCACC AGCUCGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mU*mC*mC* mUmUCCmCm AUmUCmACC CAmCCAGmC UCmGUUGmU mAmGmCUCC CmUmGmAmA	chr7: 142791820-142791844

TABLE 3-continued

Exemplary TRBC1 and TRBC2 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Target	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1201-1265)	Exemplary Guide RNA Modified Sequence (SEQ ID NOs: 2201-2265)	Genomic Coordinates (hg38)
					mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	
G028954	225	TRBC1/2	GACCCGC AGCCCCU CAAGGAG CAG	GACCCGCAGCCCCUCAAGG AGCAGGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mG*mA*mC* mCmCGCmAm GCmCCmCUC AAmGGAGmC AGmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	chr7: 142791866-142791890
G028955	226	TRBC1/2	ACCCGCA GCCCCUC AAGGAGC AGC	ACCCGCAGCCCCUCAAGGA GCAGCGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mA*mC*mC* mCmGCAmGm CCmCCmUCA AGmGAGCmA GCmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	chr7: 142791867-142791891
G028956	227	TRBC1/2	CUCAAGG AGCAGCC CGCCCUC AAU	CUCAAGGAGCAGCCCGCCC UCAAUGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mC*mU*mC* mAmAGGmAm GCmAGmCCC GCmCCUCmA AUmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	chr7: 142791878-142791902

TABLE 3-continued

Exemplary TRBC1 and TRBC2 guide sequences.						
Guide ID	SEQ ID NO to the Guide Sequence	Target	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1201-1265)	Exemplary Guide RNA Modified Sequence (SEQ ID NOs: 2201-2265)	Genomic Coordinates (hg38)
G028957	228	TRBC1/2	AGCCCGCCCUCAAUGACUCCAGAU	AGCCCGCCCUCAAUGACUCCAGAUGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mA*mG*mC*mCmCGCmCmCUmCAmAUGACmUCCAmGAUmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr7:142791888-142791912
G028958	229	TRBC1/2	CGGCCACCUUCUGGCAGAACCCC	CGGCCACCUUCUGGCAGAACCCCCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mG*mG*mCmCACmCmUUmCUmGGCAGmAACCmCCCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr7:142791939-142791963
G028959	230	TRBC1/2	GGGGGUUCUGCCAGAAGGUGCCG	GGGGGUUCUGCCAGAAGGUGGCCGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mG*mG*mG*mGmGUUmCmUGmCCmAGAAGmGUGGmCCGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr7:142791939-142791963
G028960	231	TRBC1/2	CGGGGGUUCUGCCAGAAGGUGGCC	CGGGGGUUCUGCCAGAAGGUGGCCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mG*mG*mGmGGUmUmUCUmGCmCAGAAmGGUGmGCCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGm	chr7:142791940-142791964

TABLE 3-continued

Exemplary TRBC1 and TRBC2 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Target	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1201-1265)	Exemplary Guide RNA Modified Sequence (SEQ ID NOs: 2201-2265)	Genomic Coordinates (hg38)
					CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	
G028961	232	TRBC1/2	CCUUCUG GCAGAAC CCCCGCA ACC	CCUUCUGGCAGAACCCCCG CAACCGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mC*mC*mU* mUmCUGmGm CAmGAmACC CCmCGCAmA CCmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	chr7: 142791945- 142791969
G028962	233	TRBC1/2	UGGUUGC GGGGGUU CUGCCAG AAG	UGGUUGCGGGGGUUCUGCC AGAAGGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mU*mG*mG* mUmUGCmGm GGmGGmUUC UGmCCAGmA AGmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	chr7: 142791946- 142791970
G028963	234	TRBC1/2	CUGGGUC CACUCGU CAUUCUC CGA	CUGGGUCCACUCGUCAUUC UCCGAGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mC*mU*mG* mGmGUCmCm ACmUCmGUC AUmUCUCmC GAmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	chr7: 142792001- 142792025
G028964	235	TRBC1/2	CCUGGGU CCACUCG	CCUGGGUCCACUCGUCAUU CUCCGGUUGUAGCUCCCUG	mC*mC*mU* mGmGGUmCm	chr7: 142792002-

TABLE 3-continued

Exemplary TRBC1 and TRBC2 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Target	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1201-1265)	Exemplary Guide RNA Modified Sequence (SEQ ID NOs: 2201-2265)	Genomic Coordinates (hg38)
			UCAUUCU CCG	AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	CAmCUmCGU CAmUUCUmC CGmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	142792026
G028965	236	TRBC1/2	GAGAAUG ACGAGUG GACCCAG GAU	GAGAAUGACGAGUGGACCC AGGAUGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mG*mA*mG* mAmAUGmAm CGmAGmUGG ACmCCAGmG AUmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	chr7: 142792004- 142792028
G028966	237	TRBC1/2	UGACGAG UGGACCC AGGAUAG GGC	UGACGAGUGGACCCAGGAU AGGGCGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mU*mG*mA* mCmGAGmUm GGmACmCCA GGmAUAGmG GCmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	chr7: 142792009- 142792033
G028967	238	TRBC1/2	GCCCUAU CCUGGGU CCACUCG UCA	GCCCUAUCCUGGGUCCACU CGUCAGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mG*mC*mC* mCmUAUmCm CUmGGmGUC CAmCUCGmU CAmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm	chr7: 142792009- 142792033

TABLE 3-continued

Exemplary TRBC1 and TRBC2 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Target	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1201-1265)	Exemplary Guide RNA Modified Sequence (SEQ ID NOs: 2201-2265)	Genomic Coordinates (hg38)
					CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	
G028971	239	TRBC1/2	AGGCCUG GGGUAGA GCAGGUG AGU	AGGCCUGGGGUAGAGCAGG UGAGUGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mA*mG*mG* mCmCUGmGm GGmUAmGAG CAmGGUGmA GUmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	chr7: 142792062-142792086
G028972	240	TRBC1/2	AGGCCCC ACUCACC UGCUCUA CCC	AGGCCCCACUCACCUGCUC UACCCGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mA*mG*mG* mCmCCCmAm CUmCAmCCU GCmUCUAmC CCmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	chr7: 142792069-142792093
G028973	241	TRBC1/2	GAGCAGG UGAGUGG GGCCUGG GGA	GAGCAGGUGAGUGGGGCCU GGGGAGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mG*mA*mG* mCmAGGmUm GAmGUmGGG GCmCUGGmG GAmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	chr7: 142792074-142792098
G028974	242	TRBC1/2	UCCCCAG GCCCCAC UCACCUG CUC	UCCCCAGGCCCCACUCACC UGCUCGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG	mU*mC*mC* mCmCAGmGm CCmCCmACU CAmCCUGmC	chr7: 142792074-142792098

TABLE 3-continued

Exemplary TRBC1 and TRBC2 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Target	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1201-1265)	Exemplary Guide RNA Modified Sequence (SEQ ID NOs: 2201-2265)	Genomic Coordinates (hg38)
				CAACGCUCUGCCUUCUGGC AUCGUU	UCmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	
G028975	243	TRBC1/2	CUCCCCA GGCCCCA CUCACCU GCU	CUCCCCAGGCCCCACUCAC CUGCUGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mC*mU*mC* mCmCCAmGm GCmCCmCAC UCmACCUmG CUmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	chr7: 142792075-142792099
G028976	244	TRBC1/2	UCUCCCC AGGCCCC ACUCACC UGC	UCUCCCCAGGCCCCACUCA CCUGCGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mU*mC*mU* mCmCCCmAm GGmCCmCCA CUmCACCmU GCmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	chr7: 142792076-142792100
G028988	245	TRBC1/2	ACCAGCA AGGGGUC CUGUCUG CCA	ACCAGCAAGGGGUCCUGUC UGCCAGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mA*mC*mC* mAmGCAmAm GGmGGmUCC UGmUCUGmC CAmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC	chr7: 142792697-142792721

TABLE 3-continued

Exemplary TRBC1 and TRBC2 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Target	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1201-1265)	Exemplary Guide RNA Modified Sequence (SEQ ID NOs: 2201-2265)	Genomic Coordinates (hg38)
					mUmUmCmUG GCAUCG*mU *mU	
G027904	246	TRBC2	CCACACC CAAAAGG CCACAC	CCACACCCAAAAGGCCACA CGUUUUAGAGCUAGAAAUA GCAAGUUAAAAUAAGGCUA GUCCGUUAUCACGAAAGGG CACCGAGUCGGUGCU	mC*mC*mA* CACCCAAAA GGCCACACG UUUUAGAmG mCmUmAmGm AmAmAmUmA mGmCAAGUU AAAAUAAGG CUAGUCCGU UAUCACGAA AGGGCACCG AGUCGGmU* mG*mC*mU	chr7: 142801104-142801124
G028994	247	TRBC2	CACAGAC CCGCAGC CCCUCAA GGA	CACAGACCCGCAGCCCCUC AAGGAGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mC*mA*mC* mAmGACmCm CGmCAmGCC CCmUCAAmG GAmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	chr7: 142801209-142801233
G028995	248	TRBC2	ACAGACC CGCAGCC CCUCAAG GAG	ACAGACCCGCAGCCCCUCA AGGAGGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mA*mC*mA* mGmACCmCm GCmAGmCCC CUmCAAGmG AGmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	chr7: 142801210-142801234
G028996	249	TRBC2	UGGACCC AGGAUAG GGCCAAA CCU	UGGACCCAGGAUAGGGCCA AACCUGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mU*mG*mG* mAmCCCmAm GGmAUmAGG GCmCAAAmC CUmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm	chr7: 142801363-142801387

TABLE 3-continued

Exemplary TRBC1 and TRBC2 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Target	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1201-1265)	Exemplary Guide RNA Modified Sequence (SEQ ID NOs: 2201-2265)	Genomic Coordinates (hg38)
					CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	
G028997	250	TRBC2	GGACCCA GGAUAGG GCCAAAC CUG	GGACCCAGGAUAGGGCCAA ACCUGGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mG*mG*mA* mCmCCAmGm GAmUAmGGG CCmAAACmC UGmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	chr7: 142801364-142801388
G028998	251	TRBC2	GCCAAAC CUGUCAC CCAGAUC GUC	GCCAAACCUGUCACCCAGA UCGUCGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mG*mC*mC* mAmAACmCm UGmUCmACC CAmGAUCmG UCmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	chr7: 142801378-142801402
G028999	252	TRBC2	CCUGUCA CCCAGAU CGUCAGC GCC	CCUGUCACCCAGAUCGUCA GCGCCGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mC*mC*mU* mGmUCAmCm CCmAGmAUC GUmCAGCmG CCmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	chr7: 142801384-142801408
G029000	253	TRBC2	CCCUGUU UUCUUUC AGACUGU GGC	CCCUGUUUUCUUUCAGACU GUGGCGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG	mC*mC*mC* mUmGUUmUm UCmUUmUCA GAmCUGUmG	chr7: 142801927-142801951

TABLE 3-continued

Exemplary TRBC1 and TRBC2 guide sequences.						
Guide ID	SEQ ID NO to the Guide Sequence	Target	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1201-1265)	Exemplary Guide RNA Modified Sequence (SEQ ID NOs: 2201-2265)	Genomic Coordinates (hg38)
				CAACGCUCUGCCUUCUGGC AUCGUU	GCmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	
G029001	254	TRBC2	UGUUUUC UUUCAGA CUGUGGC UUC	UGUUUUCUUUCAGACUGUG GCUUCGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mU*mG*mU* mUmUUCmUm UUmCAmGAC UGmUGGCmU UCmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	chr7: 142801930- 142801954
G029002	255	TRBC2	UGGCUUC ACCUCCG GUAAGUG AGU	UGGCUUCACCUCCGGUAAG UGAGUGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mU*mG*mG* mCmUUCmAm CCmUCmCGG UAmAGUGmA GUmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	chr7: 142801947- 142801971
G029003	256	TRBC2	AGGAGAG ACUCACU UACCGGA GGU	AGGAGAGACUCACUUACCG GAGGUGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mA*mG*mG* mAmGAGmAm CUmCAmCUU ACmCGGAmG GUmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC	chr7: 142801954- 142801978

TABLE 3-continued

Exemplary TRBC1 and TRBC2 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Target	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1201-1265)	Exemplary Guide RNA Modified Sequence (SEQ ID NOs: 2201-2265)	Genomic Coordinates (hg38)
					mUmUmCmUG GCAUCG*mU *mU	
G029004	257	TRBC2	CUGUUGA CAAGAAC AGAGCAU GUA	CUGUUGACAAGAACAGAGC AUGUAGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mC*mU*mG* mUmUGAmCm AAmGAmACA GAmGCAUmG UAmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	chr7: 142802080- 142802104
G029005	258	TRBC2	UGUCAAC AGAGUCU UACCAGC AAG	UGUCAACAGAGUCUUACCA GCAAGGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mU*mG*mU* mCmAACmAm GAmGUmCUU ACmCAGCmA AGmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	chr7: 142802095- 142802119
G029006	259	TRBC2	GAGUCUU ACCAGCA AGGGGUC CUG	GAGUCUUACCAGCAAGGGG UCCUGGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mG*mA*mG* mUmCUUmAm CCmAGmCAA GGmGGUCmC UGmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	chr7: 142802103- 142802127
G029007	260	TRBC2	UCUUACC AGCAAGG GGUCCUG UCU	UCUUACCAGCAAGGGGUCC UGUCUGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mU*mC*mU* mUmACCmAm GCmAAmGGG GUmCCUGmU CUmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU	chr7: 142802106- 142802130

TABLE 3-continued

Exemplary TRBC1 and TRBC2 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Target	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1201-1265)	Exemplary Guide RNA Modified Sequence (SEQ ID NOs: 2201-2265)	Genomic Coordinates (hg38)
					mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	
G029008	261	TRBC2	ACCAUGG CCAUCAG CACGAGG GCA	ACCAUGGCCAUCAGCACGA GGGCAGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mA*mC*mC* mAmUGGmCm CAmUCmAGC ACmGAGGmG CAmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	chr7: 142802189- 142802213
G029009	262	TRBC2	UCUCUUC CACAGGU CAAGAGA AAG	UCUCUUCCACAGGUCAAGA GAAAGGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mU*mC*mU* mCmUUCmCm ACmAGmGUC AAmGAGAmA AGmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	chr7: 142802490- 142802514
G029010	263	TRBC2	UCAAGAG AAAGGAU UCCAGAG GCU	UCAAGAGAAAGGAUUCCAG AGGCUGUUGUAGCUCCCUG AAACCGUUGCUACAAUAAG GCCGUCGAAAGAUGUGCCG CAACGCUCUGCCUUCUGGC AUCGUU	mU*mC*mA* mAmGAGmAm AAmGGmAUU CCmAGAGmG CUmGUUGmU mAmGmCUCC CmUmGmAmA mAmCmCGUU mGmCUAmCA AU*AAGmGm CCmGmUmCm GmAmAmAmG mAmUGUGCm CGmCAAmCG CUCUmGmCC mUmUmCmUG GCAUCG*mU *mU	chr7: 142802503- 142802527

TABLE 3-continued

Exemplary TRBC1 and TRBC2 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Target	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1201-1265)	Exemplary Guide RNA Modified Sequence (SEQ ID NOs: 2201-2265)	Genomic Coordinates (hg38)
G029011	264	TRBC2	AGCUAGCCUCUGGAAUCCUUUCUC	AGCUAGCCUCUGGAAUCCUUUCUCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mA*mG*mC*mUmAGCmCmUCmUGmGAAUCmCUUUmCUCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr7:142802507-142802531
G029012	265	TRBC2	GGAUGGUUUUGGAGCUAGCCUCUG	GGAUGGUUUUGGAGCUAGCCUCUGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mG*mG*mA*mUmGGUmUmUUmGGmAGCUAmGCCUmCUGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr7:142802519-142802543

[0517] The terms "mA," "mC," "mU," or "mG" may be used to denote a nucleotide that has been modified with 2'—O-Me. The terms A*, C*, U*, or G* may be used to denote a nucleotide that is linked to the next (e.g., 3') nucleotide with a phosphorothioate (PS) bond.

[0518] In some embodiments, the TRBC guide RNA comprises a guide sequence selected from SEQ ID NOs: 201-265. In some embodiments, the TRBC guide RNA comprises a guide sequence that is at least 19, 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence selected from SEQ ID NOs: 201-265. In some embodiments, the TRBC guide RNA comprises a guide sequence that is at least 95%, 90%, or 85% identical to a sequence selected from SEQ ID NOs: 201-265. In some embodiments, the TRBC guide RNA comprises a guide sequence that is at least 95% identical to a sequence selected from SEQ ID NOs: 201-265.

[0519] In some embodiments, the TRBC guide RNA comprises a guide sequence that comprises at least 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 3. As used herein, at least 10 contiguous nucleotides±10 nucleotides of a genomic coordinate means, for example, at least 10 contiguous nucleotides within the genomic coordinates wherein the genomic coordinates include 10 nucleotides in the 5' direction and 10 nucleotides in the 3' direction from the ranges listed in Table 3. For example, a TRBC guide RNA may comprise 10 contiguous nucleotides within the genomic coordinates chosen from: (a) chr7:142792690-142792714 and chr7:142792693-142792717; or (b) chr7:142791761-142791785; chr7:142791820-142791844; and chr7:142791939-142791963; chr7:142791756-142791780; or (c) chr7:142801104-142801124; chr7:142802103-142802127; and chr7:142802106-142802130; including the boundary nucleotides of these ranges. In some embodiments, the TRBC guide RNA comprises a guide sequence that is at least 19, 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence that comprises 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 3. In some embodiments, the TRBC guide RNA comprises a guide sequence that is at least 95%, 90%, or 85% identical to a sequence selected from a sequence that is 19, 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence that comprises 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 3.

[0520] In some embodiments, the TRBC guide RNA comprises a guide sequence that comprises at least 15 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 3. In some embodiments, the TRAC guide RNA comprises a guide sequence that comprises at least 24 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 3.

[0521] In some embodiments, the TRBC1 guide RNA comprises SEQ ID NO: 201. In some embodiments, the TRBC1 guide RNA comprises SEQ ID NO: 202. In some embodiments, the TRBC1 guide RNA comprises SEQ ID NO: 203. In some embodiments, the TRBC1 guide RNA comprises SEQ ID NO: 204. In some embodiments, the TRBC1 guide RNA comprises SEQ ID NO: 205. In some embodiments, the TRBC1 guide RNA comprises SEQ ID NO: 206. In some embodiments, the TRBC1 guide RNA comprises SEQ ID NO: 207. In some embodiments, the TRBC1 guide RNA comprises SEQ ID NO: 208. In some embodiments, the TRBC1 guide RNA comprises SEQ ID NO: 209. In some embodiments, the TRBC1 guide RNA comprises SEQ ID NO: 210. In some embodiments, the TRBC1 guide RNA comprises SEQ ID NO: 211. In some embodiments, the TRBC1 guide RNA comprises SEQ ID NO: 212. In some embodiments, the TRBC1 guide RNA comprises SEQ ID NO: 213. In some embodiments, the TRBC1 guide RNA comprises SEQ ID NO: 214. In some embodiments, the TRBC1 guide RNA comprises SEQ ID NO: 215. In some embodiments, the TRBC1 guide RNA comprises SEQ ID NO: 216. In some embodiments, the TRBC1 guide RNA comprises SEQ ID NO: 217. In some embodiments, the TRBC1 guide RNA comprises SEQ ID NO: 218. In some embodiments, the TRBC1 guide RNA comprises SEQ ID NO: 219. In some embodiments, the TRBC1 guide RNA comprises SEQ ID NO: 220. In some embodiments, the TRBC1 guide RNA comprises SEQ ID NO: 221.

[0522] In some embodiments, the TRBC guide RNA comprises SEQ ID NO. 222. In some embodiments, the TRBC guide RNA comprises SEQ ID NO. 223. In some embodiments, the TRBC guide RNA comprises SEQ ID NO. 224. In some embodiments, the TRBC guide RNA comprises SEQ ID NO. 225. In some embodiments, the TRBC guide RNA comprises SEQ ID NO. 226. In some embodiments, the TRBC guide RNA comprises SEQ ID NO. 227. In some embodiments, the TRBC guide RNA comprises SEQ ID NO. 228. In some embodiments, the TRBC guide RNA comprises SEQ ID NO. 229. In some embodiments, the TRBC guide RNA comprises SEQ ID NO. 230. In some embodiments, the TRBC guide RNA comprises SEQ ID NO. 231. In some embodiments, the TRBC guide RNA comprises SEQ ID NO. 232. In some embodiments, the TRBC guide RNA comprises SEQ ID NO. 233. In some embodiments, the TRBC guide RNA comprises SEQ ID NO. 234. In some embodiments, the TRBC guide RNA comprises SEQ ID NO. 235. In some embodiments, the TRBC guide RNA comprises SEQ ID NO. 236. In some embodiments, the TRBC guide RNA comprises SEQ ID NO. 237. In some embodiments, the TRBC guide RNA comprises SEQ ID NO. 238. In some embodiments, the TRBC guide RNA comprises SEQ ID NO. 239. In some embodiments, the TRBC guide RNA comprises SEQ ID NO. 240. In some embodiments, the TRBC guide RNA comprises SEQ ID NO. 241. In some embodiments, the TRBC guide RNA comprises SEQ ID NO. 242. In some embodiments, the TRBC guide RNA comprises SEQ ID NO. 243. In some embodiments, the TRBC guide RNA comprises SEQ ID NO. 244. In some embodiments, the TRBC guide RNA comprises SEQ ID NO. 245.

[0523] In some embodiments, the TRBC2 guide RNA comprises SEQ ID NO. 246. In some embodiments, the TRBC2 guide RNA comprises SEQ ID NO. 247. In some embodiments, the TRBC2 guide RNA comprises SEQ ID NO. 248. In some embodiments, the TRBC2 guide RNA comprises SEQ ID NO. 249. In some embodiments, the TRBC2 guide RNA comprises SEQ ID NO. 250. In some embodiments, the TRBC2 guide RNA comprises SEQ ID NO. 251. In some embodiments, the TRBC2 guide RNA comprises SEQ ID NO. 252. In some embodiments, the TRBC2 guide RNA comprises SEQ ID NO. 253. In some embodiments, the TRBC2 guide RNA comprises SEQ ID NO. 254. In some embodiments, the TRBC2 guide RNA comprises SEQ ID NO. 255. In some embodiments, the TRBC2 guide RNA comprises SEQ ID NO. 256. In some embodiments, the TRBC2 guide RNA comprises SEQ ID NO. 257. In some embodiments, the TRBC2 guide RNA comprises SEQ ID NO. 258. In some embodiments, the TRBC2 guide RNA comprises SEQ ID NO. 259. In some embodiments, the TRBC2 guide RNA comprises SEQ ID NO. 260. In some embodiments, the TRBC2 guide RNA comprises SEQ ID NO. 261. In some embodiments, the TRBC2 guide RNA comprises SEQ ID NO. 262. In some embodiments, the TRBC2 guide RNA comprises SEQ ID NO. 263. In some embodiments, the TRBC2 guide RNA comprises SEQ ID NO. 264. In some embodiments, the TRBC2 guide RNA comprises SEQ ID NO. 265.

[0524] In some embodiments, the TRBC1 guide RNA comprises a guide sequence of any one of: SEQ ID NOs: 215-216.

[0525] In some embodiments, the TRBC guide RNA comprises a nucleotide chosen form SEQ ID NOs: 223-224 and 229-230.

[0526] In some embodiments, the TRBC2 guide RNA comprises a guide sequence of any one of SEQ ID NOs: 246 and 259-260.

[0527] In some embodiments, the TRBC guide RNA comprises a guide sequence of any one of SEQ ID NOs: 215, 259, and 260.

[0528] In some embodiments, the TRBC guide RNA comprises a sequence listed in Table 3. In some embodiments, the TRBC guide RNA comprises a sequence of any one of SEQ ID NOs: 201-265. In some embodiments, the TRBC guide RNA comprises a sequence of SEQ ID NO: 215. In some embodiments, the TRBC guide RNA comprises a guide sequence comprising a sequence of any one of SEQ ID NOs: 201-265. In some embodiments, the TRBC guide RNA comprises a guide sequence comprising a sequence of SEQ ID NO: 215.

[0529] In some embodiments, the TRBC guide RNA comprises a sequence of any one of SEQ ID NOs: 1201-1265. In some embodiments, the TRBC guide RNA comprises a sequence of SEQ ID NO: 1215. In some embodiments, the TRBC guide RNA comprises a sequence of any one of SEQ ID NOs: 2201-2265 and 3005. In some embodiments, the TRBC guide RNA comprises a sequence of SEQ ID NO: 2215 or 3005. In some embodiments, the TRBC guide RNA comprises a sequence of SEQ ID NO: 2215. In some embodiments, the TRBC guide RNA comprises a sequence of SEQ ID NO: 3005.

[0530] In some embodiments, the TRBC guide RNA is a single guide RNA (sgRNA) comprising a sequence of any one of the sgRNA sequences listed in Table 3.

[0531] Additional embodiments of TRBC guide RNAs are provided herein, including e.g., exemplary modifications to the guide RNA.

1. Genetic Modifications to TRBC

[0532] In some embodiments, the methods and compositions disclosed herein genetically modify at least one nucleotide of the TRBC1 gene in a cell. In some embodiments, the genetic modification to TRBC1 reduces or eliminates the expression of TRBC1 protein on the surface of the genetically modified cell (or engineered cell). Genetic modifications encompass the population of modifications that results from contact with a genomic editing system (e.g., the population of edits that result from Cas9 and a TRBC1 guide RNA, or the population of edits that result from BC22 and a TRBC1 guide RNA).

[0533] In some embodiments, the methods and compositions disclosed herein genetically modify at least one nucleotide of the TRBC2 gene in a cell. In some embodiments, the genetic modification to TRBC2 reduces or eliminates the expression of TRBC2 protein on the surface of the genetically modified cell (or engineered cell). Genetic modifications encompass the population of modifications that results from contact with a genomic editing system (e.g., the population of edits that result from Cas9 and a TRBC2 guide RNA, or the population of edits that result from BC22 and a TRBC2 guide RNA).

[0534] In some embodiments, the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from: (a) chr7:142791862-142793149 or (b) chr7:142801104-142802543. In some embodiments, the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from any of the genomic coordinates listed in Table 3.

[0535] In some embodiments, the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from: chr7:142792690-142792714 and chr7:142792693-142792717. In some embodiments, the genetic modification comprises at least 5 contiguous nucleotides within the genomic coordinates chosen from: chr7:142792690-142792714 and chr7:142792693-142792717. In some embodiments, the genetic modification comprises at least 10 contiguous nucleotides within the genomic coordinates chosen from: chr7:142792690-142792714 and chr7:142792693-142792717.

[0536] In some embodiments, the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from: chr7:142791761-142791785; chr7:142791820-142791844; chr7:142791939-142791963; and chr7:142791756-142791780. In some embodiments, the genetic modification comprises at least 5 contiguous nucleotides within the genomic coordinates chosen from: chr7:142791761-142791785; chr7:142791820-142791844; chr7:142791939-142791963; and chr7:142791756-142791780. In some embodiments, the genetic modification comprises at least 10 contiguous nucleotides within the genomic coordinates chosen from: chr7:142791761-142791785; chr7:142791820-142791844; chr7:142791939-142791963; and chr7:142791756-142791780.

[0537] In some embodiments, the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from: chr7:142801104-142801124; chr7:142802103-142802127; and chr7:142802106-142802130. In some embodiments, the genetic modification comprises at least 5 contiguous nucleotides within the genomic coordinates chosen from: chr7:142801104-142801124; chr7:142802103-142802127; and chr7:142802106-142802130. In some embodiments, the genetic modification comprises at least 10 contiguous nucleotides within the genomic coordinates chosen from chr7:142801104-142801124; chr7:142802103-142802127; and chr7:142802106-142802130.

[0538] In some embodiments, the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from: chr7:142792690-142792714; chr7:142802103-142802127; and chr7:142802106-14280213. In some embodiments, the genetic modification comprises at least 5 contiguous nucleotides within the genomic coordinates chosen from: chr7:142792690-142792714; chr7:142802103-142802127; and chr7:142802106-14280213. In some embodiments, the genetic modification comprises at least 10 contiguous nucleotides within the genomic coordinates chosen from chr7:142792690-142792714; chr7:142802103-142802127; and chr7:142802106-14280213.

[0539] In some embodiments, the genetic modification comprises at least one indel, at least one C to T substitution, or at least one A to G substitution within the genomic coordinates chosen from: (a) chr7:142792690-142792714 and chr7:142792693-142792717; (b) chr7:142791761-142791785; chr7:142791820-142791844; chr7:142791939-142791963; and chr7:142791756-142791780; or (c) chr7:142801104-142801124; chr7:142802103-142802127; and chr7:142802106-142802130.

[0540] In some embodiments, the genetic modification comprises at least one indel, at least one C to T substitution, or at least one A to G substitution within the genomic coordinates chosen from: chr7:142792690-142792714; chr7:142802103-142802127; and chr7:142802106-14280213.

[0541] In some embodiments, the modification to TRBC comprises any one or more of an insertion, deletion, substitution, or deamination of at least one nucleotide in a target sequence. In some embodiments, the modification to TRBC comprises an insertion of 1, 2, 3, 4 or 5 or more nucleotides in a target sequence. In some embodiments, the modification to TRBC comprises a deletion of 1, 2, 3, 4 or 5 or more nucleotides in a target sequence. In other embodiments, the modification to TRBC comprises an insertion of 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20 or 25 or more nucleotides in a target sequence. In other embodiments, the modification to TRBC comprises a deletion of 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20 or 25 or more nucleotides in a target sequence. In some embodiments, the modification to TRBC comprises an indel, which is generally defined in the art as an insertion or deletion of less than 1000 base pairs (bp). In some embodiments, the modification to TRBC comprises an indel which results in a frameshift mutation in a target sequence. In some embodiments, the modification to TRBC comprises a substitution of 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20 or 25 or more nucleotides in a target sequence. In some embodiments, the modification to TRBC comprises one or more of an insertion, deletion, or substitution of nucleotides resulting from the incorporation of a template nucleic acid. In some embodiments, the modification to TRBC is not transient.

[0542] In some embodiments, the methods and compositions disclosed herein modify the TRBC gene in a cell using an RNA-guided DNA binding agent (e.g., a Cas enzyme). In some embodiments, the RNA-guided DNA binding agent is Cas9. In some embodiments, the RNA-guided DNA binding agent cuts within the TRBC1 gene, wherein the TRBC1 guide RNA targets a TRBC1 genomic target sequence comprising at least 10 contiguous nucleotides chr7:142791862-142793149. In some embodiments, the RNA-guided DNA binding agent cuts within the TRBC2 gene, wherein the TRBC3 guide RNA targets a TRBC2 genomic target sequence comprising at least 10 contiguous nucleotides within chr7:142801104-142802543.

[0543] In some embodiments, the genetic modification to TRBC results in utilization of an out-of-frame stop codon. In some embodiments, the genetic modification to TRBC results in exon skipping during splicing. In some embodiments, the genetic modification to TRBC1 results in reduced or eliminated TRBC1 protein expression by the cell. In some embodiments, the modification to the TRBC1 gene results in reduced or eliminated TRBC1 protein expression on the surface of the cell. In some embodiments, the genetic modification to TRBC2 results in reduced or eliminated TRBC2 protein expression by the cell. In some embodiments, the modification to the TRBC2 gene results in eliminated TRBC2 protein expression on the surface of the cell.

[0544] In some embodiments, TRBC1 expression on the surface of a cell is reduced as a result of the genetic modification to TRBC1. In some embodiments, TRBC1 expression on the surface of a cell is absent as a result of the genetic modification to TRBC1.

[0545] In some embodiments, TRBC2 expression on the surface of a cell is reduced as a result of the genetic modification to TRBC2. In some embodiments, TRBC2 expression on the surface of a cell is absent as a result of the genetic modification to TRBC2.

2. Efficacy of TRBC Guide RNAs

[0546] In some embodiments, the efficacy of a TRBC gRNA is determined when delivered or expressed together with other components forming an RNP. In some embodiments, the TRBC gRNA is expressed together with an RNA-guided DNA binding agent, such as a Cas protein, e.g. Cas9. In some embodiments, the TRBC gRNA is delivered to or expressed in a cell line that already stably expresses an RNA-guided DNA nuclease, such as a Cas nuclease or nickase, e.g. Cas9 nuclease or nickase. In some embodiments the TRBC gRNA is delivered to a cell as part of an RNP. In some embodiments, the TRBC gRNA is delivered to a cell along with a mRNA encoding an RNA-guided DNA nuclease, such as a Cas nuclease or nickase, e.g. Cas9 nuclease or nickase.

[0547] As described herein, use of an RNA-guided DNA nuclease and a TRBC guide RNA disclosed herein can lead to double-stranded breaks in the DNA which can produce errors in the form of insertion/deletion (indel) mutations upon repair by cellular machinery. Many mutations due to indels alter the reading frame or introduce premature stop codons and, therefore, produce a non-functional protein.

[0548] In some embodiments, the efficacy of particular TRBC gRNAs is determined based on in vitro models. In some embodiments, the in vitro model is HEK293 cells stably expressing Cas9 (HEK293_Cas9). In some embodiments the in vitro model is a peripheral blood mononuclear cell (PBMC). In some embodiments, the in vitro model is a T cell, such as primary human T cells. With respect to using primary cells, commercially available primary cells can be used to provide greater consistency between experiments. In some embodiments, the number of off-target sites at which a deletion or insertion occurs in an in vitro model (e.g., in T cell) is determined, e.g., by analyzing genomic DNA from transfected cells in vitro with Cas9 mRNA and the guide RNA. In some embodiments, such a determination comprises analyzing genomic DNA from the cells transfected in vitro with Cas9 mRNA, the TRBC guide RNA, and a donor oligonucleotide. Exemplary procedures for such determinations are provided in the working examples in which HEK293 cells, PBMCs, and human CD3+ T cells are used.

[0549] In some embodiments, the efficacy of particular TRBC gRNAs is determined across multiple in vitro cell models for a gRNA selection process. In some embodiments, a cell line comparison of data with selected TRBC gRNAs is performed. In some embodiments, cross screening in multiple cell models is performed.

[0550] In some embodiments, the efficacy of a TRBC guide RNA is measured by percent indels of TRBC. In some embodiments, the percent editing of TRBC is compared to the percent indels necessary to achieve knockdown of the TRBC protein products

[0551] In some embodiments, the efficacy of a guide RNA is measured by reduced or eliminated expression of a component of the T-cell receptor (TCR). In embodiments, the reduced or eliminated expression of a component of the T-cell receptor (TCR) includes reduced or eliminated expression of TRBC. In some embodiments, said reduced or eliminated expression of said component of the TCR is the result of introduction of one or more, e.g., one or two, e.g., one TRBC gRNA molecule described herein to said component of the TCR into said cell. In embodiments, said reduced or eliminated expression of a component of the TCR is as measured by flow cytometry, e.g., as described herein.

[0552] In some embodiments, the efficacy of a TRBC guide RNA is measured by the number and/or frequency of indels at off-target sequences within the genome of the target cell type, such as a T cell. In some embodiments, efficacious guide RNAs are provided which produce indels at off target sites at very low frequencies (e.g., <5%) in a cell population and/or relative to the frequency of indel creation at the target site. Thus, the disclosure provides for guide RNAs which do not exhibit off-target indel formation in the target cell type (e.g., a T cell), or which produce a frequency of off-target indel formation of <5% in a cell population and/or relative to the frequency of indel creation at the target site. In some embodiments, the disclosure provides guide RNAs which do not exhibit any off target indel formation in the target cell type (e.g., T cell). In some embodiments, guide RNAs are provided which produce indels at less than 5 off-target sites, e.g., as evaluated by one or more methods described herein. In some embodiments, guide RNAs are provided which produce indels at less than or equal to 4, 3, 2, or 1 off-target site(s) e.g., as evaluated by one or more methods described herein. In some embodiments, the off-target site(s) does not occur in a protein coding region in the target cell (e.g., hepatocyte) genome.

[0553] In some embodiments, detecting gene editing events, such as the formation of insertion/deletion ("indel") mutations and homology directed repair (HDR) events in target DNA utilize linear amplification with a tagged primer and isolating the tagged amplification products (herein after referred to as "LAM-PCR," or "Linear Amplification (LA)" method).

[0554] In some embodiments, the efficacy of a guide RNA is measured by the levels of functional protein complexes comprising the expressed protein product of the gene. In some embodiments, the efficacy of a guide RNA is measured by flow cytometric analysis of TCR expression by which the live population of edited cells is analyzed for loss of the TCR.

F. CIITA Guide RNAs

[0555] The methods and compositions provided herein disclose CIITA guide RNAs useful for reducing the expression of MHC class II protein on the surface of a cell. In some embodiments, such guide RNAs direct an RNA-guided DNA binding agent to a CIITA genomic target sequence and may be referred to herein as "CIITA guide RNAs." In some embodiments, the CIITA guide RNA directs an RNA-guided DNA binding agent to a human CIITA genomic target sequence. In some embodiments, the CIITA guide RNA comprises a guide sequence selected from SEQ ID NOs: 301, 302, 304-576.

[0556] In some embodiments, the methods and compositions disclosed herein comprise a CIITA guide RNA comprising a guide sequence that targets a CIITA genomic target sequence comprising at least 10 contiguous nucleotides within the genomic coordinates chosen from: (a) chr16:10877363-10907788 or (b) chr16:10906515-10908136. In some embodiments, the methods and compositions disclosed herein comprise a CIITA guide RNA comprising a guide sequence that targets a CIITA genomic target sequence comprising at least one nucleotide within the genomic coordinates chosen from: (a) chr16:10877363-10907788 or (b) chr16:10906515-10908136.

[0557] In some embodiments, the methods and compositions disclosed herein comprise a CIITA guide RNA comprising a guide sequence that directs an RNA-guided DNA binding agent to make cut in a CIITA gene, wherein the CIITA guide RNA targets a CIITA genomic target sequence comprising at least 10 contiguous nucleotides within the genomic coordinates chosen from: (a) chr16:10877363-10907788 or (b) chr16:10906515-10908136. In some embodiments, the methods and compositions disclosed herein comprise a CIITA guide RNA comprising a guide sequence that directs an RNA-guided DNA binding agent to make cut in a CIITA gene, wherein the CIITA guide RNA targets a CIITA genomic target sequence comprising at least one nucleotide within the genomic coordinates chosen from: (a) chr16:10877363-10907788 or (b) chr16:10906515-10908136.

[0558] In some embodiments, the methods and compositions disclose a CIITA guide RNA that directs an RNA-guided DNA binding agent to induce a double stranded break (DSB) or a single-stranded break (SSB) in a CIITA genomic target sequence. In some embodiments, the methods and compositions disclose a CIITA guide RNA that directs an RNA-guided DNA binding agent to make a cut in a CIITA genomic target sequence. In embodiments wherein the RNA-guided DNA cutting agent is Cas9, the cut or "cut site" occurs at the third base from the protospacer adjacent motif (PAM) sequence.

[0559] In some embodiments, a composition is provided comprising a CIITA guide RNA described herein and an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent.

[0560] In some embodiments, a composition is provided comprising a CIITA single-guide RNA (sgRNA) comprising a guide sequence that targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chosen from: (a) chr16:10877363-10907788 or (b) chr16:10906515-10908136. In some embodiments, a composition is provided comprising a CIITA sgRNA described herein and an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent.

[0561] In some embodiments, a composition is provided comprising a CIITA dual-guide RNA (dgRNA) comprising a guide sequence that targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chosen from: (a) chr16:10877363-10907788 or (b) chr16:10906515-10908136. In some embodiments, a composition is provided comprising a CIITA dgRNA described herein and an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent.

[0562] Exemplary CIITA guide sequences are shown below in Table 4.

TABLE 4

Exemplary CIITA guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
G026584	301	UCAAAGUACCCUACAGGAGGACCA	UCAAAGUACCCUACAGGAGGACCAGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mU*mC*mA*mAmAGUmAmCCmCUmACAGGmAGGAmCCAmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	2301	chr16: 10907504-10907528
G026585	302	AGUACCCUACAGGAGGACCAGUUC	AGUACCCUACAGGAGGACCAGUUCGUUGUAGCUCCCUGA	mA*mG*mU*mAmCCCmUmACmAGmGAGGAmCCAGmUUCmGUUGmUmAm	2302	chr16: 10907508-10907532

TABLE 4-continued

Exemplary CIITA guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
			AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU		
G029013	304	UACCUGUC AGAGCCCC AAGGUAAA	UACCUGUCAGAGC CCCAAGGUAAAGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mU*mA*mC*mCmUGUm CmAGmAGmCCCCAmAG GUmAAAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2304	chr16: 10877363-10877387
G029014	305	UUUACCUU GGGGCUCU GACAGGUA	UUUACCUUGGGGC UCUGACAGGUAGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mU*mU*mU*mAmCCUm UmGGmGGmCUCUGmAC AGmGUAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2305	chr16: 10877363-10877387
G029015	306	UUUUACCU UGGGGCUC UGACAGGU	UUUUACCUUGGGG CUCUGACAGGUGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mU*mU*mU*mUmACCm UmUGmGGmGCUCUmGA CAmGGUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2306	chr16: 10877364-10877388
G029016	307	GCCUGGGA GGGAAGAC AAUGCUCA	GCCUGGGAGGGAA GACAAUGCUCAGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mG*mC*mC*mUmGGGm AmGGmGAmAGACAmAU GCmUCAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2307	chr16: 10895259-10895283
G029017	308	UCUUCCCU CCCAGGCA GCUCACAG	UCUUCCCUCCCAG GCAGCUCACAGGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mU*mC*mU*mUmCCCm UmCCmCAmGGCAGmCU CAmCAGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2308	chr16: 10895268-10895292
G029018	309	UCCCUCCC AGGCAGCU CACAGUGU	UCCCUCCCAGGCA GCUCACAGUGUGU UGUAGCUCCCUGA AACCGUUGCUACA	mU*mC*mC*mCmUCCm CmAGmGCmAGCUCmAC AGmUGUmGUUGmUmAm GmCUCCCmUmGmAmAm	2309	chr16: 10895271-10895295

TABLE 4-continued

Exemplary CIITA guide sequences.						
Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
			AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU		
G029019	310	CUCACAGU GUGCCACC AUGGAGUU	CUCACAGUGUGCC ACCAUGGAGUUGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mU*mC*mAmCAGm UmGUmGCmCACCAmUG GAmGUUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2310	chr16: 10895285- 10895309
G029020	311	UCACAGUG UGCCACCA UGGAGUUG	UCACAGUGUGCCA CCAUGGAGUUGGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mU*mC*mA*mCmAGUm GmUGmCCmACCAUmGG AGmUUGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2311	chr16: 10895286- 10895310
G029021	312	CACAGUGU GCCACCAU GGAGUUGG	CACAGUGUGCCAC CAUGGAGUUGGGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mA*mC*mAmGUGm UmGCmCAmCCAUGmGA GUmUGGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2312	chr16: 10895287- 10895311
G029022	313	GAAGAGAU UGAGCUCU ACUCAGGU	GAAGAGAUUGAGC UCUACUCAGGUGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mG*mA*mA*mGmAGAm UmUGmAGmCUCUAmCU CAmGGUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2313	chr16: 10895406- 10895430
G029023	314	ACCUGAGU AGAGCUCA AUCUCUUC	ACCUGAGUAGAGC UCAAUCUCUUCGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mA*mC*mC*mUmGAGm UmAGmAGmCUCAAmUC UCmUUCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2314	chr16: 10895406- 10895430
G029024	315	GAGAUUGA GCUCUACU CAGGUGGG	GAGAUUGAGCUCU ACUCAGGUGGGGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA	mG*mA*mG*mAmUUGm AmGCmUCmUACUCmAG GUmGGGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU	2315	chr16: 10895409- 10895433

TABLE 4-continued

Exemplary CIITA guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
			ACGCUCUGCCUUC UGGCAUCGUU	mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU		
G029025	316	AUUGAGCU CUACUCAG GUGGGCCC	AUUGAGCUCUACU CAGGUGGGCCCGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mA*mU*mU*mGmAGCm UmCUmACmUCAGGmUG GGmCCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2316	chr16: 10895412- 10895436
G029026	317	UUGAGCUC UACUCAGG UGGGCCCU	UUGAGCUCUACUC AGGUGGGCCCUGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mU*mU*mG*mAmGCUm CmUAmCUmCAGGUmGG GCmCCUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2317	chr16: 10895413- 10895437
G029027	318	AGGUGGGC CCUCCUCC CUCUGGUC	AGGUGGGCCCUCC UCCCUCUGGUCGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mA*mG*mG*mUmGGGm CmCCmUCmCUCCCmUC UGmGUCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2318	chr16: 10895426- 10895450
G029028	319	CUUUUCCU CCCAGAAC CCGACACA	CUUUUCCUCCCAG AACCCGACACAGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mU*mU*mUmUCCm UmCCmCAmGAACCmCG ACmACAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2319	chr16: 10895655- 10895679
G029029	320	GUCUGUGU CGGGUUCU GGGAGGAA	GUCUGUGUCGGGU UCUGGGAGGAAGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mG*mU*mC*mUmGUGm UmCGmGGmUUCUGmGG AGmGAAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2320	chr16: 10895658- 10895682
G029030	321	AACCCGAC ACAGACAC CAUCAACU	AACCCGACACAGA CACCAUCAACUGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mA*mA*mC*mCmCGAm CmACmAGmACACCmAU CAmACUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU	2321	chr16: 10895668- 10895692

TABLE 4-continued

Exemplary CIITA guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
				CUmGmCCmUmUmCmUG GCAUCG*mU*mU		
G029031	322	GGCUUAUG CCAAUAUC GGUGAGGA	GGCUUAUGCCAAU AUCGGUGAGGAGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mG*mG*mC*mUmUAUm GmCCmAAmUAUCGmGU GAmGGAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2322	chr16: 10895747- 10895771
G029032	323	CUUCCUCA CCGAUAUU GGCAUAAG	CUUCCUCACCGAU AUUGGCAUAAGGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mU*mU*mCmCUCm AmCCmGAmUAUUGmGC AUmAAGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2323	chr16: 10895749- 10895773
G029033	324	GCUUCCUC ACCGAUAU UGGCAUAA	GCUUCCUCACCGA UAUUGGCAUAAGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mG*mC*mU*mUmCCUm CmACmCGmAUAUUmGG CAmUAAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2324	chr16: 10895750- 10895774
G029034	325	UGCCAAUA UCGGUGAG GAAGCACC	UGCCAAUAUCGGU GAGGAAGCACCGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mU*mG*mC*mCmAAUm AmUCmGGmUGAGGmAA GCmACCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2325	chr16: 10895753- 10895777
G029035	326	GGUGCUUC CUCACCGA UAUUGGCA	GGUGCUUCCUCAC CGAUAUUGGCAGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mG*mG*mU*mGmCUUm CmCUmCAmCCGAUmAU UGmGCAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2326	chr16: 10895753- 10895777
G029036	327	GCCAAUAU CGGUGAGG AAGCACCU	GCCAAUAUCGGUG AGGAAGCACCUGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mG*mC*mC*mAmAUAm UmCGmGUmGAGGAmAG CAmCCUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2327	chr16: 10895754- 10895778

TABLE 4-continued

Exemplary CIITA guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
G029037	328	UUUUCCUUGUCUGGGCAGCGGAAC	UUUUCCUUGUCUGGGCAGCGGAACGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mU*mU*mU*mUmCCUmUmGUmCUmGGGCAmGCGGmAACmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	2328	chr16: 10898651-10898675
G029038	329	GGGCAGCGGAACUGGACCAGUAUG	GGGCAGCGGAACUGGACCAGUAUGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mG*mG*mG*mCmAGCmGmGAmACmUGGACmCAGUmAUGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	2329	chr16: 10898663-10898687
G029039	330	GAACUGGACCAGUAUGUCUUCCAG	GAACUGGACCAGUAUGUCUUCCAGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mG*mA*mA*mCmUGGmAmCCmAGmUAUGUmCUUCmCAGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	2330	chr16: 10898671-10898695
G029040	331	AACUGGACCAGUAUGUCUUCCAGG	AACUGGACCAGUAUGUCUUCCAGGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mA*mA*mC*mUmGGAmCmCAmGUmAUGUCmUUCCmAGGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	2331	chr16: 10898672-10898696
G029041	332	AUGUCUUCCAGGACUCCCAGCUGG	AUGUCUUCCAGGACUCCCAGCUGGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mA*mU*mG*mUmCUUmCmCAmGGmACUCCmCAGCmUGGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	2332	chr16: 10898684-10898708
G029042	333	UUACUGAAAAUGUCCUUGCUCAGG	UUACUGAAAAUGUCCUUGCUCAGGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mU*mU*mA*mCmUGAmAmAAmUGmUCCUUmGCUCmAGGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	2333	chr16: 10898712-10898736

TABLE 4-continued

Exemplary CIITA guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
G029043	334	AACUUACUGAAAAUGUCCUUGCUC	AACUUACUGAAAAUGUCCUUGCUCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mA*mA*mC*mUmUACmUmGAmAAmAUGUCmCUUGmCUCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	2334	chr16: 10898715-10898739
G029044	335	AAACUUACUGAAAAUGUCCUUGCU	AAACUUACUGAAAAUGUCCUUGCUGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mA*mA*mA*mCmUUAmCmUGmAAmAAUGUmCCUUmGCUmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	2335	chr16: 10898716-10898740
G029045	336	CCACCCACCACAAACUUACUGAAA	CCACCCACCACAAACUUACUGAAAGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mC*mA*mCmCCAmCmCAmCAmAACUUmACUGmAAAmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	2336	chr16: 10898727-10898751
G029046	337	UGUGCUCUACUUUGAGAAAAACCA	UGUGCUCUACUUUGAGAAAAACCAGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mU*mG*mU*mGmCUCmUmACmUUmUGAGAmAAAAmCCAmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	2337	chr16: 10898906-10898930
G029047	338	GGUUUUUCUCAAAGUAGAGCACAU	GGUUUUUCUCAAAGUAGAGCACAUGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mG*mG*mU*mUmUUUmCmUCmAAmAGUAGmAGCAmCAUmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	2338	chr16: 10898907-10898931
G029048	339	GGUCCUAUGUGCUCUACUUUGAGA	GGUCCUAUGUGCUCUACUUUGAGAGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mG*mG*mU*mCmCUAmUmGUmGCmUCUACmUUUGmAGAmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	2339	chr16: 10898913-10898937
G029049	340	CAGAAAAGUCAGAAAAGACGUGAG	CAGAAAAGUCAGAAAAGACGUGAGGUUGUAGCUCCCUGA	mC*mA*mG*mAmAAAmGmUCmAGmAAAAGmACGUmGAGmGUUGmUmAm	2340	chr16: 10898983-10899007

TABLE 4-continued

Exemplary CIITA guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
			AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU		
G029050	341	AGAAAAGU CAGAAAAG ACGUGAGU	AGAAAAGUCAGAA AAGACGUGAGUGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mA*mG*mA*mAmAAGm UmCAmGAmAAAGAmCG UGmAGUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2341	chr16: 10898984-10899008
G029051	342	GAAAAGUC AGAAAAGA CGUGAGUG	GAAAAGUCAGAAA AGACGUGAGUGGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mG*mA*mA*mAmAGUm CmAGmAAmAAGACmGU GAmGUGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2342	chr16: 10898985-10899009
G029052	343	UCACUCAC GUCUUUUC UGACUUUU	UCACUCACGUCUU UUCUGACUUUUGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mU*mC*mA*mCmUCAm CmGUmCUmUUUCUmGA CUmUUUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2343	chr16: 10898986-10899010
G029053	344	CUCACUCA CGUCUUUU CUGACUUU	CUCACUCACGUCU UUUCUGACUUUGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mU*mC*mAmCUCm AmCGmUCmUUUUCmUG ACmUUUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2344	chr16: 10898987-10899011
G029054	345	AAGUCAGA AAAGACGU GAGUGAGC	AAGUCAGAAAAGA CGUGAGUGAGCGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mA*mA*mG*mUmCAGm AmAAmAGmACGUGmAG UGmAGCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2345	chr16: 10898988-10899012
G029055	346	AGUCAGAA AAGACGUG AGUGAGCC	AGUCAGAAAAGAC GUGAGUGAGCCGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA	mA*mG*mU*mCmAGAm AmAAmGAmCGUGAmGU GAmGCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC	2346	chr16: 10898989-10899013

TABLE 4-continued

Exemplary CIITA guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
			AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU		
G029056	347	AAAAGACG UGAGUGAG CCCCUCCC	AAAAGACGUGAGU GAGCCCCUCCCGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mA*mA*mA*mAmGACm GmUGmAGmUGAGCmCC CUmCCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2347	chr16: 10898995- 10899019
G029057	348	GACGUGAG UGAGCCCC UCCCUGAU	GACGUGAGUGAGC CCCUCCCUGAUGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mG*mA*mC*mGmUGAm GmUGmAGmCCCCUmCC CUmGAUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2348	chr16: 10898999- 10899023
G029058	349	CUGCAGAG AAAACAUG UGAUCAGC	CUGCAGAGAAAAC AUGUGAUCAGCGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mU*mG*mCmAGAm GmAAmAAmCAUGUmGA UCmAGCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2349	chr16: 10901489- 10901513
G029059	350	GCUGCAGA GAAAACAU GUGAUCAG	GCUGCAGAGAAAA CAUGUGAUCAGGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mG*mC*mU*mGmCAGm AmGAmAAmACAUGmUG AUmCAGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2350	chr16: 10901490- 10901514
G029060	351	GGCUGCAG AGAAAACA UGUGAUCA	GGCUGCAGAGAAA ACAUGUGAUCAGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mG*mG*mC*mUmGCAm GmAGmAAmAACAUmGU GAmUCAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2351	chr16: 10901491- 10901515
G029061	352	UUUCUCUG CAGCCUUC CCAGAGGA	UUUCUCUGCAGCC UUCCCAGAGGAGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA	mU*mU*mU*mCmUCUm GmCAmGCmCUUCCmCA GAmGGAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU	2352	chr16: 10901502- 10901526

TABLE 4-continued

Exemplary CIITA guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
			ACGCUCUGCCUUC UGGCAUCGUU	mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU		
G029062	353	CAGCCUUC CCAGAGGA GCUUCCGG	CAGCCUUCCCAGA GGAGCUUCCGGGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mA*mG*mCmCUUm CmCCmAGmAGGAGmCU UCmCGGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2353	chr16: 10901510- 10901534
G029063	354	UUCCAGUG CUUCAGGU CUGCCGGA	UUCCAGUGCUUCA GGUCUGCCGGAGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mU*mU*mC*mCmAGUm GmCUmUCmAGGUCmUG CCmGGAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2354	chr16: 10901529- 10901553
G029064	355	ACACCUGG CUUCCAGU GCUUCAGG	ACACCUGGCUUCC AGUGCUUCAGGGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mA*mC*mA*mCmCUGm GmCUmUCmCAGUGmCU UCmAGGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2355	chr16: 10901538- 10901562
G029065	356	CAGCCCAC CUGCCCUG CACACCUG	CAGCCCACCUGCC CUGCACACCUGGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mA*mG*mCmCCAm CmCUmGCmCCUGCmAC ACmCUGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2356	chr16: 10901555- 10901579
G029066	357	CACAGCCC ACUUCCUC ACAGCUGA	CACAGCCCACUUC CUCACAGCUGAGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mA*mC*mAmGCCm CmACmUUmCCUCAmCA GCmUGAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2357	chr16: 10902017- 10902041
G029067	358	ACAGCCCA CUUCCUCA CAGCUGAG	ACAGCCCACUUCC UCACAGCUGAGGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mA*mC*mA*mGmCCCm AmCUmUCmCUCACmAG CUmGAGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU	2358	chr16: 10902018- 10902042

TABLE 4-continued

Exemplary CIITA guide sequences.						
Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
				CUmGmCCmUmUmCmUG GCAUCG*mU*mU		
G029068	359	CUGCCUGC GCUGUUCA ACCAGGAG	CUGCCUGCGCUGU UCAACCAGGAGGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mU*mG*mCmCUGm CmGCmUGmUUCAAmCC AGmGAGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2359	chr16: 10902114-10902138
G029069	360	CCUGCGCU GUUCAACC AGGAGCCA	CCUGCGCUGUUCA ACCAGGAGCCAGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mC*mU*mGmCGCm UmGUmUCmAACCAmGG AGmCCAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2360	chr16: 10902117-10902141
G029070	361	CGCUGUUC AACCAGGA GCCAGCCU	CGCUGUUCAACCA GGAGCCAGCCUGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mG*mC*mUmGUUm CmAAmCCmAGGAGmCC AGmCCUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2361	chr16: 10902121-10902145
G029071	362	ACCAGGAG CCAGCCUC CGGCCAGA	ACCAGGAGCCAGC CUCCGGCCAGAGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mA*mC*mC*mAmGGAm GmCCmAGmCCUCCmGG CCmAGAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2362	chr16: 10902130-10902154
G029072	363	GCCUCCGG CCAGAUGC GCCUGGAG	GCCUCCGGCCAGA UGCGCCUGGAGGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mG*mC*mC*mUmCCGm GmCCmAGmAUGCGmCC UGmGAGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2363	chr16: 10902141-10902165
G029073	364	CCGGCCAG AUGCGCCU GGAGAAAA	CCGGCCAGAUGCG CCUGGAGAAAAGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mC*mG*mGmCCAm GmAUmGCmGCCUGmGA GAmAAAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2364	chr16: 10902145-10902169

TABLE 4-continued

Exemplary CIITA guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
G029074	365	AACAUACU GGGAAUCU GGUCGGUU	AACAUACUGGGAA UCUGGUCGGUUGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mA*mA*mC*mAmUACm UmGGmGAmAUCUGmGU CGmGUUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2365	chr16: 10902167- 10902191
G029075	366	CUGCCUUU GUCUCUUG CAGUGCCU	CUGCCUUUGUCUC UUGCAGUGCCUGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mU*mG*mCmCUUm UmGUmCUmCUUGCmAG UGmCCUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2366	chr16: 10902638- 10902662
G029076	367	UUGUCUCU UGCAGUGC CUUUCUCC	UUGUCUCUUGCAG UGCCUUUCUCCGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mU*mU*mG*mUmCUCm UmUGmCAmGUGCCmUU UCmUCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2367	chr16: 10902644- 10902668
G029077	368	UGCCUUUC UCCAGUUC CUCGUUGA	UGCCUUUCUCCAG UUCCUCGUUGAGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mU*mG*mC*mCmUUUm CmUCmCAmGUUCCmUC GUmUGAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2368	chr16: 10902657- 10902681
G029078	369	UCUCCCUG AGGGACCC AUCCAGUU	UCUCCCUGAGGGA CCCAUCCAGUUGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mU*mC*mU*mCmCCUm GmAGmGGmACCCAmUC CAmGUUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2369	chr16: 10902691- 10902715
G029079	370	CUCCCUGA GGGACCCA UCCAGUUU	CUCCCUGAGGGAC CCAUCCAGUUUGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mU*mC*mCmCUGm AmGGmGAmCCCAUmCC AGmUUUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2370	chr16: 10902692- 10902716
G029080	371	CCUGAGGG ACCCAUCC AGUUUGUC	CCUGAGGGACCCA UCCAGUUUGUCGU UGUAGCUCCCUGA	mC*mC*mU*mGmAGGm GmACmCCmAUCCAmGU UUmGUCmGUUGmUmAm	2371	chr16: 10902695- 10902719

TABLE 4-continued

Exemplary CIITA guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
			AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU		
G029081	372	GGACCCAU CCAGUUUG UCCCCACC	GGACCCAUCCAGU UUGUCCCCACCGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mG*mG*mA*mCmCCAm UmCCmAGmUUUGUmCC CCmACCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2372	chr16: 10902701- 10902725
G029082	373	CCAGUUUG UCCCCACC AUCUCCAC	CCAGUUUGUCCCC ACCAUCUCCACGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mC*mA*mGmUUUm GmUCmCCmCACCAmUC UCmCACmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2373	chr16: 10902709- 10902733
G029083	374	CAGUUUGU CCCCACCA UCUCCACU	CAGUUUGUCCCCA CCAUCUCCACUGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mA*mG*mUmUUGm UmCCmCCmACCAUmCU CCmACUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2374	chr16: 10902710- 10902734
G029084	375	AGUUUGUC CCCACCAU CUCCACUC	AGUUUGUCCCCAC CAUCUCCACUCGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mA*mG*mU*mUmUGUm CmCCmCAmCCAUCmUC CAmCUCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2375	chr16: 10902711- 10902735
G029085	376	CAAAUCUC UGAGGCUG GAACAGGG	CAAAUCUCUGAGG CUGGAACAGGGGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mA*mA*mAmUCUm CmUGmAGmGCUGGmAA CAmGGGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2376	chr16: 10902752- 10902776
G029086	377	ACCAUGGU AGAUGAAU AUACUGGA	ACCAUGGUAGAUG AAUAUACUGGAGU UGUAGCUCCCUGA AACCGUUGCUACA	mA*mC*mC*mAmUGGm UmAGmAUmGAAUAmUA CUmGGAmGUUGmUmAm GmCUCCCmUmGmAmAm	2377	chr16: 10902779- 10902803

TABLE 4-continued

Exemplary CIITA guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
			AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU		
G029087	378	CACCAUGG UAGAUGAA UAUACUGG	CACCAUGGUAGAU GAAUAUACUGGGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mA*mC*mCmAUGm GmUAmGAmUGAAUmAU ACmUGGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2378	chr16: 10902780- 10902804
G029088	379	UCACCAUG GUAGAUGA AUAUACUG	UCACCAUGGUAGA UGAAUAUACUGGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mU*mC*mA*mCmCAUm GmGUmAGmAUGAAmUA UAmCUGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2379	chr16: 10902781- 10902805
G029089	380	UCUACCAU GGUGAGUG CGGGGCCU	UCUACCAUGGUGA GUGCGGGGCCUGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mU*mC*mU*mAmCCAm UmGGmUGmAGUGCmGG GGmCCUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2380	chr16: 10902792- 10902816
G029090	381	CUACCAUG GUGAGUGC GGGGCCUG	CUACCAUGGUGAG UGCGGGGCCUGGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mU*mA*mCmCAUm GmGUmGAmGUGCGmGG GCmCUGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2381	chr16: 10902793- 10902817
G029091	382	UACCAUGG UGAGUGCG GGGCCUGG	UACCAUGGUGAGU GCGGGGCCUGGGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mU*mA*mC*mCmAUGm GmUGmAGmUGCGGmGG CCmUGGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2382	chr16: 10902794- 10902818
G029092	383	AUGGUGAG UGCGGGGC CUGGCUCC	AUGGUGAGUGCGG GGCCUGGCUCCGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA	mA*mU*mG*mGmUGAm GmUGmCGmGGGCCmUG GCmUCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU	2383	chr16: 10902798- 10902822

TABLE 4-continued

Exemplary CIITA guide sequences.						
Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
			ACGCUCUGCCUUC UGGCAUCGUU	mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU		
G029093	384	GUGAGUGC GGGGCCUG GCUCCCCG	GUGAGUGCGGGGC CUGGCUCCCCGGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mG*mU*mG*mAmGUGm CmGGmGGmCCUGGmCU CCmCCGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2384	chr16: 10902801-10902825
G029094	385	ACUCUCCA CCCCCAAU GUAGGUGA	ACUCUCCACCCCC AAUGUAGGUGAGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mA*mC*mU*mCmUCCm AmCCmCCmCAAUGmUA GGmUGAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2385	chr16: 10903710-10903734
G029095	386	CUCUCCAC CCCCAAUG UAGGUGAG	CUCUCCACCCCCA AUGUAGGUGAGGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mU*mC*mUmCCAm CmCCmCCmAAUGUmAG GUmGAGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2386	chr16: 10903711-10903735
G029096	387	UCUCCACC CCCAAUGU AGGUGAGG	UCUCCACCCCCAA UGUAGGUGAGGGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mU*mC*mU*mCmCACm CmCCmCAmAUGUAmGG UGmAGGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2387	chr16: 10903712-10903736
G029097	388	ACCCCCAA UGUAGGUG AGGUGCCC	ACCCCCAAUGUAG GUGAGGUGCCCGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mA*mC*mC*mCmCCAm AmUGmUAmGGUGAmGG UGmCCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2388	chr16: 10903717-10903741
G029098	389	CCAAUGUA GGUGAGGU GCCCCAGG	CCAAUGUAGGUGA GGUGCCCCAGGGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mC*mA*mAmUGUm AmGGmUGmAGGUGmCC CCmAGGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU	2389	chr16: 10903721-10903745

TABLE 4-continued

Exemplary CIITA guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
				CUmGmCCmUmUmCmUG GCAUCG*mU*mU		
G029099	390	AGGUGAGG UGCCCCAG GCCAGCCA	AGGUGAGGUGCCC CAGGCCAGCCAGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mA*mG*mG*mUmGAGm GmUGmCCmCCAGGmCC AGmCCAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2390	chr16: 10903728-10903752
G029100	391	GGUGAGGU GCCCCAGG CCAGCCAA	GGUGAGGUGCCCC AGGCCAGCCAAGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mG*mG*mU*mGmAGGm UmGCmCCmCAGGCmCA GCmCAAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2391	chr16: 10903729-10903753
G029101	392	GUGAGGUG CCCCAGGC CAGCCAAG	GUGAGGUGCCCCA GGCCAGCCAAGGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mG*mU*mG*mAmGGUm GmCCmCCmAGGCCmAG CCmAAGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2392	chr16: 10903730-10903754
G029102	393	UGAGGUGC CCCAGGCC AGCCAAGU	UGAGGUGCCCCAG GCCAGCCAAGUGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mU*mG*mA*mGmGUGm CmCCmCAmGGCCAmGC CAmAGUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2393	chr16: 10903731-10903755
G029103	394	GGUGCCCC AGGCCAGC CAAGUACC	GGUGCCCCAGGCC AGCCAAGUACCGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mG*mG*mU*mGmCCCm CmAGmGCmCAGCCmAA GUmACCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2394	chr16: 10903734-10903758
G029104	395	GUGCCCCA GGCCAGCC AAGUACC	GUGCCCCAGGCCA GCCAAGUACCCGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mG*mU*mG*mCmCCCm AmGGmCCmAGCCAmAG UAmCCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2395	chr16: 10903735-10903759

TABLE 4-continued

Exemplary CIITA guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
G029105	396	AAGUACCCCCUCCCAGUGGAUUCA	AAGUACCCCCUCCCAGUGGAUUCAGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mA*mA*mG*mUmACCmCmCCmUCmCCAGUmGGAUmUCAmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	2396	chr16: 10903751-10903775
G029106	397	CCUCCCGAGCAAACAUGACAGGUA	CCUCCCGAGCAAACAUGACAGGUAGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mC*mU*mCmCCGmAmGCmAAmACAUGmACAGmGUAmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	2397	chr16: 10903874-10903898
G029107	398	CUCCCGAGCAAACAUGACAGGUAA	CUCCCGAGCAAACAUGACAGGUAAGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mU*mC*mCmCGAmGmCAmAAmCAUGAmCAGGmUAAmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	2398	chr16: 10903875-10903899
G029108	399	CAAACAUGACAGGUAAGGACCCUU	CAAACAUGACAGGUAAGGACCCUUGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mA*mA*mAmCAUmGmACmAGmGUAAGmGACCmCUUmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	2399	chr16: 10903883-10903907
G029109	400	UUGUGCUCUGGAGAUGGAGAAGCA	UUGUGCUCUGGAGAUGGAGAAGCAGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mU*mU*mG*mUmGCUmCmUGmGAmGAUGGmAGAAmGCAmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	2400	chr16: 10904726-10904750
G029110	401	GCUUCUCCAUCUCCAGAGCACAAG	GCUUCUCCAUCUCCAGAGCACAAGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mG*mC*mU*mUmCUCmCmAUmCUmCCAGAmGCACmAAGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	2401	chr16: 10904727-10904751

TABLE 4-continued

Exemplary CIITA guide sequences.						
Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
G029111	402	CUUCUCCA UCUCCAGA GCACAAGA	CUUCUCCAUCUCC AGAGCACAAGAGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mU*mU*mCmUCCm AmUCmUCmCAGAGmCA CAmAGAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2402	chr16: 10904728- 10904752
G029112	403	UUCUCCAU CUCCAGAG CACAAGAC	UUCUCCAUCUCCA GAGCACAAGACGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mU*mU*mC*mUmCCAm UmCUmCCmAGAGCmAC AAmGACmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2403	chr16: 10904729- 10904753
G029113	404	UCUCCAUC UCCAGAGC ACAAGACG	UCUCCAUCUCCAG AGCACAAGACGGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mU*mC*mU*mCmCAUm CmUCmCAmGAGCAmCA AGmACGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2404	chr16: 10904730- 10904754
G029114	405	CCAUCUCC AGAGCACA AGACGUCC	CCAUCUCCAGAGC ACAAGACGUCCGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mC*mA*mUmCUCm CmAGmAGmCACAAmGA CGmUCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2405	chr16: 10904733- 10904757
G029115	406	CAUCUCCA GAGCACAA GACGUCCC	CAUCUCCAGAGCA CAAGACGUCCCGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mA*mU*mCmUCCm AmGAmGCmACAAGmAC GUmCCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2406	chr16: 10904734- 10904758
G029116	407	AGAGCACA AGACGUCC CCCACCCA	AGAGCACAAGACG UCCCCCACCCAGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mA*mG*mA*mGmCACm AmAGmACmGUCCCmCC ACmCCAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2407	chr16: 10904741- 10904765
G029117	408	ACCCAAUG CCCGGCAG CUGGAGAG	ACCCAAUGCCCGG CAGCUGGAGAGGU UGUAGCUCCCUGA	mA*mC*mC*mCmAAUm GmCCmCGmGCAGCmUG GAmGAGmGUUGmUmAm	2408	chr16: 10904760- 10904784

TABLE 4-continued

Exemplary CIITA guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
			AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU		
G029118	409	CCAUUUUG GAAGCUUG UUGGAGAC	CCAUUUUGGAAGC UUGUUGGAGACGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mC*mA*mUmUUUm GmGAmAGmCUUGUmUG GAmGACmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2409	chr16: 10904784- 10904808
G029119	410	CCAGGCCA UUUUGGAA GCUUGUUG	CCAGGCCAUUUUG GAAGCUUGUUGGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mC*mA*mGmGCCm AmUUmUUmGGAAGmCU UGmUUGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2410	chr16: 10904789- 10904813
G029120	411	GCCUGGUG AGUGAUGC GGGAUCUC	GCCUGGUGAGUGA UGCGGGAUCUCGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mG*mC*mC*mUmGGUm GmAGmUGmAUGCGmGG AUmCUCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2411	chr16: 10904807- 10904831
G029121	412	CCUGGUGA GUGAUGCG GGAUCUCU	CCUGGUGAGUGAU GCGGGAUCUCUGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mC*mU*mGmGUGm AmGUmGAmUGCGGmGA UCmUCUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2412	chr16: 10904808- 10904832
G029122	413	CCUUUGCA GAGCCGGU GGAGCAGU	CCUUUGCAGAGCC GGUGGAGCAGUGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mC*mU*mUmUGCm AmGAmGCmCGGUGmGA GCmAGUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2413	chr16: 10906489- 10906513
G029123	414	UAGAACUG CUCCACCG GCUCUGCA	UAGAACUGCUCCA CCGGCUCUGCAGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA	mU*mA*mG*mAmACUm GmCUmCCmACCGGmCU CUmGCAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC	2414	chr16: 10906493- 10906517

TABLE 4-continued

Exemplary CIITA guide sequences.						
Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
			AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU		
G029124	415	UACCGCUC ACUGCAGG ACACGUAU	UACCGCUCACUGC AGGACACGUAUGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mU*mA*mC*mCmGCUm CmACmUGmCAGGAmCA CGmUAUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2415	chr16: 10906515-10906539
G029125	416	CUCACUGC AGGACACG UAUGGUGC	CUCACUGCAGGAC ACGUAUGGUGCGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mU*mC*mAmCUGm CmAGmGAmCACGUmAU GGmUGCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2416	chr16: 10906520-10906544
G029126	417	UCACUGCA GGACACGU AUGGUGCC	UCACUGCAGGACA CGUAUGGUGCCGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mU*mC*mA*mCmUGCm AmGGmAmCmACGUAmUG GUmGCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2417	chr16: 10906521-10906545
G029127	418	AGGACACG UAUGGUGC CGAGCCCG	AGGACACGUAUGG UGCCGAGCCCGGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mA*mG*mG*mAmCACm GmUAmUGmGUGCCmGA GCmCCGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2418	chr16: 10906528-10906552
G029128	419	GCCCAGUC CGGGGUGG CCAGUUCC	GCCCAGUCCGGGG UGGCCAGUUCCGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mG*mC*mC*mCmAGUm CmCGmGGmGUGGCmCA GUmUCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2419	chr16: 10906631-10906655
G029129	420	GUUCUGCO CAGUCCGG GGUGGCCA	GUUCUGCCCAGUC CGGGGUGGCCAGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA	mG*mU*mU*mCmUGCm CmCAmGUmCCGGGmGU GGmCCAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU	2420	chr16: 10906636-10906660

TABLE 4-continued

Exemplary CIITA guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
			ACGCUCUGCCUUC UGGCAUCGUU	mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU		
G029130	421	CGUUCUGC CCAGUCCG GGGUGGCC	CGUUCUGCCCAGU CCGGGGUGGCCGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mG*mU*mUmCUGm CmCCmAGmUCCGGmGG UGmGCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2421	chr16: 10906637- 10906661
G029131	422	AGCUGCCG UUCUGCCC AGUCCGGG	AGCUGCCGUUCUG CCCAGUCCGGGGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mA*mG*mC*mUmGCCm GmUUmCUmGCCCAmGU CCmGGGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2422	chr16: 10906643- 10906667
G029132	423	GGGCAGAA CGGCAGCU GGCCCAAG	GGGCAGAACGGCA GCUGGCCCAAGGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mG*mG*mG*mCmAGAm AmCGmGCmAGCUGmGC CCmAAGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2423	chr16: 10906651- 10906675
G029133	424	CAAUAGCU CUUGCCCU GACCAGCU	CAAUAGCUCUUGC CCUGACCAGCUGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mA*mA*mUmAGCm UmCUmUGmCCCUGmAC CAmGCUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2424	chr16: 10906754- 10906778
G029134	425	CCAAUAGC UCUUGCCC UGACCAGC	CCAAUAGCUCUUG CCCUGACCAGCGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mC*mA*mAmUAGm CmUCmUUmGCCCUmGA CCmAGCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2425	chr16: 10906755- 10906779
G029135	426	UGCCCCAG CCCAAUAG CUCUUGCC	UGCCCCAGCCCAA UAGCUCUUGCCGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mU*mG*mC*mCmCCAm GmCCmCAmAUAGCmUC UUmGCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU	2426	chr16: 10906764- 10906788

TABLE 4-continued

Exemplary CIITA guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
				CUmGmCCmUmUmCmUG GCAUCG*mU*mU		
G029136	427	CUCACUGC CCCAGCCC AAUAGCUC	CUCACUGCCCCAG CCCAAUAGCUCGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mU*mC*mAmCUGm CmCCmCAmGCCCAmAU AGmCUCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2427	chr16: 10906769- 10906793
G029137	428	GCUCACUG CCCCAGCC CAAUAGCU	GCUCACUGCCCCA GCCCAAUAGCUGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mG*mC*mU*mCmACUm GmCCmCCmAGCCCmAA UAmGCUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2428	chr16: 10906770- 10906794
G029138	429	AAGCCCAG GCCCGGCU CACUGCCC	AAGCCCAGGCCCG GCUCACUGCCCGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mA*mA*mG*mCmCCAm GmGCmCCmGGCUCmAC UGmCCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2429	chr16: 91006783- 10906807
G029139	430	CAAGCCCA GGCCCGGC UCACUGCC	CAAGCCCAGGCCC GGCUCACUGCCGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mA*mA*mGmCCCm AmGGmCCmCGGCUmCA CUmGCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2430	chr16: 10906784- 10906808
G029140	431	GCCACAAG CCCAGGCC CGGCUCAC	GCCACAAGCCCAG GCCCGGCUCACGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mG*mC*mC*mAmCAAm GmCCmCAmGGCCCmGG CUmCACmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2431	chr16: 10906788- 10906812
G029141	432	GGCCACAA GCCCAGGC CCGGCUCA	GGCCACAAGCCCA GGCCCGGCUCAGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mG*mG*mC*mCmACAm AmGCmCCmAGGCCmCG GCmUCAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2432	chr16: 10906789- 10906813

TABLE 4-continued

Exemplary CIITA guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
G029142	433	CGGCCACAAGCCCAGGCCCGGCUC	CGGCCACAAGCCCAGGCCCGGCUCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mG*mG*mCmCACmAmAGmCCmCAGGCmCCGGmCUCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	2433	chr16:10906790-10906814
G029143	434	UUCCCCAGUACGACUUUGUCUUCU	UUCCCCAGUACGACUUUGUCUUCUGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mU*mU*mC*mCmCCAmGmUAmCGmACUUUmGUCUmUCUmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	2434	chr16:10906816-10906840
G029144	435	UCCCCAGUACGACUUUGUCUUCUC	UCCCCAGUACGACUUUGUCUUCUCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mU*mC*mC*mCmCAGmUmACmGAmCUUUGmUCUUmCUCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	2435	chr16:10906817-10906841
G029145	436	CCCCAGUACGACUUUGUCUUCUCU	CCCCAGUACGACUUUGUCUUCUCUGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mC*mC*mCmAGUmAmCGmACmUUUGUmCUUCmUCUmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	2436	chr16:10906818-10906842
G029146	437	AGUACGACUUUGUCUUCUCUGUCC	AGUACGACUUUGUCUUCUCUGUCCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mA*mG*mU*mAmCGAmCmUUmUGmUCUUCmUCUGmUCCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	2437	chr16:10906822-10906846
G029147	438	GCCUAUGGCCUGCAGGAUCUGCUC	GCCUAUGGCCUGCAGGAUCUGCUCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mG*mC*mC*mUmAUGmGmCCmUGmCAGGAmUCUGmCUCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	2438	chr16:10906875-10906899
G029148	439	CCUAUGGCCUGCAGGAUCUGCUCU	CCUAUGGCCUGCAGGAUCUGCUCUGUUGUAGCUCCCUGA	mC*mC*mU*mAmUGGmCmCUmGCmAGGAUmCUGCmUCUmGUUGmUmAm	2439	chr16:10906876-10906900

TABLE 4-continued

Exemplary CIITA guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
			AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU		
G029149	440	GCCUGCAG GAUCUGCU CUUCUCCC	GCCUGCAGGAUCU GCUCUUCUCCCGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mG*mC*mC*mUmGCAm GmGAmUCmUGCUCmUU CUmCCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2440	chr16: 10906882-10906906
G029150	441	CCUGCAGG AUCUGCUC UUCUCCCU	CCUGCAGGAUCUG CUCUUCUCCCUGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mC*mU*mGmCAGm GmAUmCUmGCUCUmUC UCmCCUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2441	chr16: 10906883-10906907
G029151	442	UCCCUGGG CCCACAGC CACUCGUG	UCCCUGGGCCCAC AGCCACUCGUGGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mU*mC*mC*mCmUGGm GmCCmCAmCAGCCmAC UCmGUGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2442	chr16: 10906902-10906926
G029152	443	UCGAGGAG CUGGAAGC GCAAGAUG	UCGAGGAGCUGGA AGCGCAAGAUGGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mU*mC*mG*mAmGGAm GmCUmGGmAAGCGmCA AGmAUGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2443	chr16: 10906993-10907017
G029153	444	UGGCCGGC CUUUUCCA GAAGAAGC	UGGCCGGCCUUUU CCAGAAGAAGCGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mU*mG*mG*mCmCGGm CmCUmUUmUCCAGmAA GAmAGCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2444	chr16: 10907077-10907101
G029154	445	UUCCAGAA GAAGCUGC UCCGAGGU	UUCCAGAAGAAGC UGCUCCGAGGUGU UGUAGCUCCCUGA AACCGUUGCUACA	mU*mU*mC*mCmAGAm AmGAmAGmCUGCUmCC GAmGGUmGUUGmUmAm GmCUCCCmUmGmAmAm	2445	chr16: 10907088-10907112

TABLE 4-continued

Exemplary CIITA guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
			AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU		
G029155	446	UCCAGAAG AAGCUGCU CCGAGGUU	UCCAGAAGAAGCU GCUCCGAGGUUGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mU*mC*mC*mAmGAAm GmAAmGCmUGCUCmCG AGmGUUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2446	chr16: 10907089- 10907113
G029156	447	AGAAGAAG CUGCUCCG AGGUUGCA	AGAAGAAGCUGCU CCGAGGUUGCAGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mA*mG*mA*mAmGAAm GmCUmGCmUCCGAmGG UUmGCAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2447	chr16: 10907092- 10907116
G029157	448	AGAAGCUG CUCCGAGG UUGCACCC	AGAAGCUGCUCCG AGGUUGCACCCGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mA*mG*mA*mAmGCUm GmCUmCCmGAGGUmUG CAmCCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2448	chr16: 10907095- 10907119
G029158	449	CUCCGAGG UUGCACCC UCCUCCUC	CUCCGAGGUUGCA CCCUCCUCCUCGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mU*mC*mCmGAGm GmUUmGCmACCCUmCC UCmCUCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2449	chr16: 10907103- 10907127
G029159	450	UCCGAGGU UGCACCCU CCUCCUCA	UCCGAGGUUGCAC CCUCCUCCUCAGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mU*mC*mC*mGmAGGm UmUGmCAmCCCUCmCU CCmUCAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2450	chr16: 10907104- 10907128
G029160	451	CUGGUCCA GAGCCUGA GCAAGGCC	CUGGUCCAGAGCC UGAGCAAGGCCGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA	mC*mU*mG*mGmUCCm AmGAmGCmCUGAGmCA AGmGCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU	2451	chr16: 10907148- 10907172

TABLE 4-continued

Exemplary CIITA guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
			ACGCUCUGCCUUC UGGCAUCGUU	mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU		
G029161	452	UGGUCCAG AGCCUGAG CAAGGCCG	UGGUCCAGAGCCU GAGCAAGGCCGGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mU*mG*mG*mUmCCAm GmAGmCCmUGAGCmAA GGmCCGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2452	chr16: 10907149-10907173
G029162	453	UCAGGGAU GACAGAGC ACCAAGAC	UCAGGGAUGACAG AGCACCAAGACGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mU*mC*mA*mGmGGAm UmGAmCAmGAGCAmCC AAmGACmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2453	chr16: 10907244-10907268
G029163	454	CAGGGAUG ACAGAGCA CCAAGACA	CAGGGAUGACAGA GCACCAAGACAGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mA*mG*mGmGAUm GmACmAGmAGCACmCA AGmACAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2454	chr16: 10907245-10907269
G029164	455	CAGAGCAC CAAGACAG AGCCCUGA	CAGAGCACCAAGA CAGAGCCCUGAGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mA*mG*mAmGCAm CmCAmAGmACAGAmGC CCmUGAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2455	chr16: 10907254-10907278
G029165	456	AGCACCAA GACAGAGC CCUGACGC	AGCACCAAGACAG AGCCCUGACGCGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mA*mG*mC*mAmCCAm AmGAmCAmGAGCCmCU GAmCGCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2456	chr16: 10907257-10907281
G029166	457	AAGACAGA GCCCUGAC GCUCCUCC	AAGACAGAGCCCU GACGCUCCUCCGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mA*mA*mG*mAmCAGm AmGCmCCmUGACGmCU CCmUCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU	2457	chr16: 10907263-10907287

TABLE 4-continued

Exemplary CIITA guide sequences.						
Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
				CUmGmCCmUmUmCmUG GCAUCG*mU*mU		
G029167	458	UGCCGGGC AGUGUGCC AGCUCUCA	UGCCGGGCAGUGU GCCAGCUCUCAGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mU*mG*mC*mCmGGGm CmAGmUGmUGCCAmGC UCmUCAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2458	chr16: 10907331-10907355
G029168	459	GCCGGGCA GUGUGCCA GCUCUCAG	GCCGGGCAGUGUG CCAGCUCUCAGGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mG*mC*mC*mGmGGCm AmGUmGUmGCCAGmCU CUmCAGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2459	chr16: 10907332-10907356
G029169	460	CAGGCCAG CUUGGCCA GCUCUGCC	CAGGCCAGCUUGG CCAGCUCUGCCGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mA*mG*mGmCCAm GmCUmUGmGCCAGmCU CUmGCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2460	chr16: 10907462-10907486
G029170	461	AGCUCCCA GGCCAGCU UGGCCAGC	AGCUCCCAGGCCA GCUUGGCCAGCGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mA*mG*mC*mUmCCCm AmGGmCCmAGCUUmGG CCmAGCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2461	chr16: 10907468-10907492
G029171	462	CUGCGGCC CAGCUCCC AGGCCAGC	CUGCGGCCCAGCU CCCAGGCCAGCGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mU*mG*mCmGGCm CmCAmGCmUCCCAmGG CCmAGCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2462	chr16: 10907477-10907501
G029172	463	GCAGACAU CAAAGUAC CCUACAGG	GCAGACAUCAAAG UACCCUACAGGGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mG*mC*mA*mGmACAm UmCAmAAmGUACCmCU ACmAGGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2463	chr16: 10907497-10907521

TABLE 4-continued

Exemplary CIITA guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
G029173	464	AUCAAAGU ACCCUACA GGAGGACC	AUCAAAGUACCCU ACAGGAGGACCGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mA*mU*mC*mAmAAGm UmACmCCmUACAGmGA GGmACCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2464	chr16: 10907503-10907527
G029174	465	GACCAGUU CCCAUCCG CAGACGUG	GACCAGUUCCCAU CCGCAGACGUGGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mG*mA*mC*mCmAGUm UmCCmCAmUCCGCmAG ACmGUGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2465	chr16: 10907523-10907547
G029175	466	GAUGGCCA AAGGCUUA GUCCAACA	GAUGGCCAAAGGC UUAGUCCAACAGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mG*mA*mU*mGmGCCm AmAAmGGmCUUAGmUC CAmACAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2466	chr16: 91007558-10907582
G029176	467	AAAGGCUU AGUCCAAC ACCCACCG	AAAGGCUUAGUCC AACACCCACCGGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mA*mA*mA*mGmGCUm UmAGmUCmCAACAmCC CAmCCGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2467	chr16: 10907565-10907589
G029177	468	GUCCAACA CCCACCGC GGGCCGCA	GUCCAACACCCAC CGCGGGCCGCAGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mG*mU*mC*mCmAACm AmCCmCAmCCGCGmGG CCmGCAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2468	chr16: 10907574-10907598
G029178	469	AGCUUCCU CCUGCAAU GCUUCCUG	AGCUUCCUCCUGC AAUGCUUCCUGGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mA*mG*mC*mUmUCCm UmCCmUGmCAAUGmCU UCmCUGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2469	chr16: 10907619-10907643

TABLE 4-continued

Exemplary CIITA guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
G029179	470	GCUUCCUC CUGCAAUG CUUCCUGG	GCUUCCUCCUGCA AUGCUUCCUGGGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mG*mC*mU*mUmCCUm CmCUmGCmAAUGCmUU CCmUGGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2470	chr16: 10907620-10907644
G029180	471	UCGCCACU CAGAGCCA GCCACAGG	UCGCCACUCAGAG CCAGCCACAGGGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mU*mC*mG*mCmCACm UmCAmGAmGCCAGmCC ACmAGGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2471	chr16: 10907648-10907672
G029181	472	UUCGCCAC UCAGAGCC AGCCACAG	UUCGCCACUCAGA GCCAGCCACAGGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mU*mU*mC*mGmCCAm CmUCmAGmAGCCAmGC CAmCAGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2472	chr16: 10907649-10907673
G029182	473	UUUCGCCA CUCAGAGC CAGCCACA	UUUCGCCACUCAG AGCCAGCCACAGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mU*mU*mU*mCmGCCm AmCUmCAmGAGCCmAG CCmACAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2473	chr16: 10907650-10907674
G029183	474	AUUUCGCC ACUCAGAG CCAGCCAC	AUUUCGCCACUCA GAGCCAGCCACGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mA*mU*mU*mUmCGCm CmACmUCmAGAGCmCA GCmCACmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2474	chr16: 10907651-10907675
G029184	475	AAGGAGCU CCCGCAGU ACCUAGCA	AAGGAGCUCCCGC AGUACCUAGCAGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mA*mA*mG*mGmAGCm UmCCmCGmCAGUAmCC UAmGCAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2475	chr16: 10907682-10907706
G029185	476	AGGAGCUC CCGCAGUA CCUAGCAU	AGGAGCUCCCGCA GUACCUAGCAUGU UGUAGCUCCCUGA	mA*mG*mG*mAmGCUm CmCCmGCmAGUACmCU AGmCAUmGUUGmUmAm	2476	chr16: 10907683-10907707

TABLE 4-continued

Exemplary CIITA guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
			AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU		
G029186	477	GGAGCUCC CGCAGUAC CUAGCAUU	GGAGCUCCCGCAG UACCUAGCAUUGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mG*mG*mA*mGmCUCm CmCGmCAmGUACCmUA GCmAUUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2477	chr16: 10907684- 10907708
G029187	478	CCAGCCAG UUGUCAUA GGGCCUCU	CCAGCCAGUUGUC AUAGGGCCUCUGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mC*mA*mGmCCAm GmUUmGUmCAUAGmGG CCmUCUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2478	chr16: 10907722- 10907746
G029188	479	GCACGCCC UCCAGCCA GUUGUCAU	GCACGCCCUCCAG CCAGUUGUCAUGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mG*mC*mA*mCmGCCm CmUCmCAmGCCAGmUU GUmCAUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2479	chr16: 10907731- 10907755
G029189	480	CUGGCUGG GCUGAUCU UCCAGCCU	CUGGCUGGGCUGA UCUUCCAGCCUGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mC*mU*mG*mGmCUGm GmGCmUGmAUCUUmCC AGmCCUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2480	chr16: 91007763- 10907787
G029190	481	UGGCUGGG CUGAUCUU CCAGCCUC	UGGCUGGGCUGAU CUUCCAGCCUCGU UGUAGCUCCCUGA AACCGUUGCUACA AUAAGGCCGUCGA AAGAUGUGCCGCA ACGCUCUGCCUUC UGGCAUCGUU	mU*mG*mG*mCmUGGm GmCUmGAmUCUUCmCA GCmCUCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	2481	chr16: 10907764- 10907788
G023421	482	UACCGCUC ACUGCAGG ACACGUAU	UACCGCUCACUGC AGGACACGUAUGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG	mU*mA*mC*mCmGCUm CmACmUGmCAGGAmCA CGmUAUmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU*	2482, 3016 and 3111	chr16: 10906515- 10906539

TABLE 4-continued

Exemplary CIITA guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
			CCGCAACGCUCUG CCGGCAUCGUU	AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU		
G023422	483	CUCACUGC AGGACACG UAUGGUGC	CUCACUGCAGGAC ACGUAUGGUGCGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mC*mU*mC*mAmCUGm CmAGmGAmCACGUmAU GGmUGCmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2483, 3017 and 3112	chr16: 10906520- 10906544
G023423	484	UCACUGCA GGACACGU AUGGUGCC	UCACUGCAGGACA CGUAUGGUGCCGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mU*mC*mA*mCmUGCm AmGGmACmACGUAmUG GUmGCCmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2484, 3018 and 3113	chr16: 10906521- 10906545
G023424	485	AGGACACG UAUGGUGC CGAGCCCG	AGGACACGUAUGG UGCCGAGCCCGGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mA*mG*mG*mAmCACm GmUAmUGmGUGCCmGA GCmCCGmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2485, 3019 and 3114	chr16: 10906528- 10906552
G023425	486	GGACACGU AUGGUGCC GAGCCCGC	GGACACGUAUGGU GCCGAGCCCGCGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mG*mG*mA*mCmACGm UmAUmGGmUGCCGmAG CCmCGCmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2486, 3020 and 3115	chr16: 10906529- 10906553
G023426	487	AUCCUAGU GGAGGUGG AUCUGGUG	AUCCUAGUGGAGG UGGAUCUGGUGGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mA*mU*mC*mCmUAGm UmGGmAGmGUGGAmUC UGmGUGmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2487, 3021 and 3116	chr16: 10906566- 10906590
G023427	488	GCAAGAGC CUGGAGCG GGAACUGG	GCAAGAGCCUGGA GCGGGAACUGGGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mG*mC*mA*mAmGAGm CmCUmGGmAGCGGmGA ACmUGGmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2488, 3022 and 3117	chr16: 10906615- 10906639
G023428	489	CAAGAGCC UGGAGCGG GAACUGGC	CAAGAGCCUGGAG CGGGAACUGGCGU UGUAGCUCCCUCC	mC*mA*mA*mGmAGCm CmUGmGAmGCGGGmAA CUmGGCmGUUGmUmAm	2489, 3023 and	chr16: 10906616- 10906640

TABLE 4-continued

Exemplary CIITA guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
			GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	3118	
G023429	490	GGGCAGAA CGGCAGCU GGCCCAAG	GGGCAGAACGGCA GCUGGCCCAAGGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mG*mG*mG*mCmAGAm AmCGmGCmAGCUGmGC CCmAAGmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2490, 3024 and 3119	chr16: 10906651-10906675
G023430	491	GCACAGCA AUCACUCG UGUCUCAC	GCACAGCAAUCAC UCGUGUCUCACGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mG*mC*mA*mCmAGCm AmAUmCAmCUCGUmGU CUmCACmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2491, 3025 and 3120	chr16: 10906723-10906747
G023431	492	CAAUAGCU CUUGCCCU GACCAGCU	CAAUAGCUCUUGC CCUGACCAGCUGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mC*mA*mA*mUmAGCm UmCUmUGmCCCUGmAC CAmGCUmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2492, 3026 and 3121	chr16: 10906754-10906778
G023432	493	CCAAUAGC UCUUGCCC UGACCAGC	CCAAUAGCUCUUG CCCUGACCAGCGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mC*mC*mA*mAmUAGm CmUCmUUmGCCCUmGA CCmAGCmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2493, 3027 and 3122	chr16: 10906755-10906779
G023433	494	AGCCGGGC CUGGGCUU GUGGCCGG	AGCCGGGCCUGGG CUUGUGGCCGGGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mA*mG*mC*mCmGGGm CmCUmGGmGCUUGmUG GCmCGGmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2494, 3028 and 3123	chr16: 10906791-10906815
G023434	495	CAGAGAAG ACAAAGUC GUACUGGG	CAGAGAAGACAAA GUCGUACUGGGGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mC*mA*mG*mAmGAAm GmACmAAmAGUCGmUA CUmGGGmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2495, 3029 and 3124	chr16: 10906819-10906843

TABLE 4-continued

Exemplary CIITA guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
G023435	496	AGUACGACUUUGUCUUCUCUGUCC	AGUACGACUUUGUCUUCUCUGUCCGUUGUAGCUCCCUCCGUUGCUACAAUAAGGCCGUCGAUGUGCCGCAACGCUCUGCCGGCAUCGUU	mA*mG*mU*mAmCGAmCmUUmUGmUCUUCmUCUGmUCCmGUUGmUmAmGmCUCCCmU(L1)mCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmC(L1)mGmAmUGUGCmCGmCAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2496, 3030 and 3125	chr16: 10906822-10906846
G023436	497	AGCAGAUCCUGCAGGCCAUAGGCA	AGCAGAUCCUGCAGGCCAUAGGCAGUUGUAGCUCCCUCCGUUGCUACAAUAAGGCCGUCGAUGUGCCGCAACGCUCUGCCGGCAUCGUU	mA*mG*mC*mAmGAUmCmCUmGCmAGGCCmAUAGmGCAmGUUGmUmAmGmCUCCCmU(L1)mCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmC(L1)mGmAmUGUGCmCGmCAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2497, 3031 and 3126	chr16: 10906874-10906898
G023437	498	GCCUAUGGCCUGCAGGAUCUGCUC	GCCUAUGGCCUGCAGGAUCUGCUCGUUGUAGCUCCCUCCGUUGCUACAAUAAGGCCGUCGAUGUGCCGCAACGCUCUGCCGGCAUCGUU	mG*mC*mC*mUmAUGmGmCCmUGmCAGGAmUCUGmCUCmGUUGmUmAmGmCUCCCmU(L1)mCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmC(L1)mGmAmUGUGCmCGmCAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2498, 3032 and 3127	chr16: 10906875-10906899
G023438	499	GAGCAGAUCCUGCAGGCCAUAGGC	GAGCAGAUCCUGCAGGCCAUAGGCGUUGUAGCUCCCUCCGUUGCUACAAUAAGGCCGUCGAUGUGCCGCAACGCUCUGCCGGCAUCGUU	mG*mA*mG*mCmAGAmUmCCmUGmCAGGCmCAUAmGGCmGUUGmUmAmGmCUCCCmU(L1)mCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmC(L1)mGmAmUGUGCmCGmCAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2499, 3033 and 3128	chr16: 10906875-10906899
G023439	500	CCUAUGGCCUGCAGGAUCUGCUCU	CCUAUGGCCUGCAGGAUCUGCUCUGUUGUAGCUCCCUCCGUUGCUACAAUAAGGCCGUCGAUGUGCCGCAACGCUCUGCCGGCAUCGUU	mC*mC*mU*mAmUGGmCmCUmGCmAGGAUmCUGCmUCUmGUUGmUmAmGmCUCCCmU(L1)mCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmC(L1)mGmAmUGUGCmCGmCAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2500, 3034 and 3129	chr16: 10906876-10906900
G023440	501	AGAGCAGAUCCUGCAGGCCAUAGG	AGAGCAGAUCCUGCAGGCCAUAGGGUUGUAGCUCCCUCCGUUGCUACAAUAAGGCCGUCGAUGUGCCGCAACGCUCUGCCGGCAUCGUU	mA*mG*mA*mGmCAGmAmUCmCUmGCAGGmCCAUmAGGmGUUGmUmAmGmCUCCCmU(L1)mCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmC(L1)mGmAmUGUGCmCGmCAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2501, 3035 and 3130	chr16: 10906876-10906900
G023441	502	AAGAGCAGAUCCUGCAGGCCAUAG	AAGAGCAGAUCCUGCAGGCCAUAGGUUGUAGCUCCCUCCGUUGCUACAAUAAGGCCGUCGAUGUGCCGCAACGCUCUGCCGGCAUCGUU	mA*mA*mG*mAmGCAmGmAUmCCmUGCAGmGCCAmUAGmGUUGmUmAmGmCUCCCmU(L1)mCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmC(L1)mGmAmUGUGCmCGmCAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2502, 3036 and 3131	chr16: 10906877-10906901

TABLE 4-continued

Exemplary CIITA guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
G023442	503	GCCUGCAG GAUCUGCU CUUCUCCC	GCCUGCAGGAUCU GCUCUUCUCCCGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mG*mC*mC*mUmGCAm GmGAmUCmUGCUCmUU CUmCCCmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2503, 3037 and 3132	chr16: 10906882-10906906
G023443	504	GGAUCUGC UCUUCUCC CUGGGCCC	GGAUCUGCUCUUC UCCCUGGGCCCGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mG*mG*mA*mUmCUGm CmUCmUUmCUCCCmUG GGmCCCmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2504, 3038 and 3133	chr16: 10906889-10906913
G023444	505	CACUCGUG GCGGCCGA UGAGGUUU	CACUCGUGGCGGC CGAUGAGGUUUGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mC*mA*mC*mUmCGUm GmGCmGGmCCGAUmGA GGmUUUmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2505, 3039 and 3134	chr16: 10906918-10906942
G023445	506	UGAGGUUU UCAGCCAC AUCUUGAA	UGAGGUUUUCAGC CACAUCUUGAAGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mU*mG*mA*mGmGUUm UmUCmAGmCCACAmUC UUmGAAmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2506, 3040 and 3135	chr16: 10906934-10906958
G023446	507	UGAAGAGA CCUGACCG CGUUCUGC	UGAAGAGACCUGA CCGCGUUCUGCGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mU*mG*mA*mAmGAGm AmCCmUGmACCGCmGU UCmUGCmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2507, 3041 and 3136	chr16: 10906954-10906978
G023447	508	UCGAGGAG CUGGAAGC GCAAGAUG	UCGAGGAGCUGGA AGCGCAAGAUGGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mU*mC*mG*mAmGGAm GmCUmGGmAAGCGmCA AGmAUGmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2508, 3042 and 3137	chr16: 10906993-10907017
G023448	509	AGAUGGCU UCCUGCAC AGCACGUG	AGAUGGCUUCCUG CACAGCACGUGGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mA*mG*mA*mUmGGCm UmUCmCUmGCACAmGC ACmGUGmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm	2509, 3043 and 3138	chr16: 10907012-10907036

TABLE 4-continued

Exemplary CIITA guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
				CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU		
G023449	510	UGGCCGGCCUUUUCCAGAAGAAGC	UGGCCGGCCUUUUCCAGAAGAAGCGUUGUAGCUCCCUCCGUUGCUACAAUAAGGCCGUCGAUGUGCCGCAACGCUCUGCCGGCAUCGUU	mU*mG*mG*mCmCGGmCmCUmUUmUCCAGmAAGAmAGCmGUUGmUmAmGmCUCCCmU(L1)mCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmC(L1)mGmAmUGUGCmCGmCAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2510, 3044 and 3139	chr16:10907077-10907101
G023450	511	UUCCAGAAGAAGCUGCUCCGAGGU	UUCCAGAAGAAGCUGCUCCGAGGUGUUGUAGCUCCCUCCGUUGCUACAAUAAGGCCGUCGAUGUGCCGCAACGCUCUGCCGGCAUCGUU	mU*mU*mC*mCmAGAmAmGAmAGmCUGCUmCCGAmGGUmGUUGmUmAmGmCUCCCmU(L1)mCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmC(L1)mGmAmUGUGCmCGmCAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2511, 3045 and 3140	chr16:10907088-10907112
G023451	512	UCCAGAAGAAGCUGCUCCGAGGUU	UCCAGAAGAAGCUGCUCCGAGGUUGUUGUAGCUCCCUCCGUUGCUACAAUAAGGCCGUCGAUGUGCCGCAACGCUCUGCCGGCAUCGUU	mU*mC*mC*mAmGAAmGmAAmGCmUGCUCmCGAGmGUUmGUUGmUmAmGmCUCCCmU(L1)mCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmC(L1)mGmAmUGUGCmCGmCAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2512, 3046 and 3141	chr16:10907089-10907113
G023452	513	AGAAGAAGCUGCUCCGAGGUUGCA	AGAAGAAGCUGCUCCGAGGUUGCAGUUGUAGCUCCCUCCGUUGCUACAAUAAGGCCGUCGAUGUGCCGCAACGCUCUGCCGGCAUCGUU	mA*mG*mA*mAmGAAmGmCUmGCmUCCGAmGGUUmGCAmGUUGmUmAmGmCUCCCmU(L1)mCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmC(L1)mGmAmUGUGCmCGmCAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2513, 3047 and 3142	chr16:10907092-10907116
G023453	514	AGAAGCUGCUCCGAGGUUGCACCC	AGAAGCUGCUCCGAGGUUGCACCCGUUGUAGCUCCCUCCGUUGCUACAAUAAGGCCGUCGAUGUGCCGCAACGCUCUGCCGGCAUCGUU	mA*mG*mA*mAmGCUmGmCUmCCmGAGGUmUGCAmCCCmGUUGmUmAmGmCUCCCmU(L1)mCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmC(L1)mGmAmUGUGCmCGmCAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2514, 3048 and 3143	chr16:10907095-10907119
G023454	515	CUCCGAGGUUGCACCCUCCUCCUC	CUCCGAGGUUGCACCCUCCUCCUCGUUGUAGCUCCCUCCGUUGCUACAAUAAGGCCGUCGAUGUGCCGCAACGCUCUGCCGGCAUCGUU	mC*mU*mC*mCmGAGmGmUUmGCmACCCUmCCUCmCUCmGUUGmUmAmGmCUCCCmU(L1)mCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmC(L1)mGmAmUGUGCmCGmCAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2515, 3049 and 3144	chr16:10907103-10907127
G023455	516	AGGUUGCACCCUCCUCCUCACAGC	AGGUUGCACCCUCCUCCUCACAGCGUUGUAGCUCCCUCCGUUGCUACAAUAAGGCCGUCGAUGUG	mA*mG*mG*mUmUGCmAmCCmCUmCCUCCmUCACmAGCmGUUGmUmAmGmCUCCCmU(L1)mCmCGUUmGmCUAmCAAU*	2516, 3050 and 3145	chr16:10907108-10907132

TABLE 4-continued

Exemplary CIITA guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
			CCGCAACGCUCUG CCGGCAUCGUU	AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU		
G023456	517	CUGGUCCA GAGCCUGA GCAAGGCC	CUGGUCCAGAGCC UGAGCAAGGCCGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mC*mU*mG*mGmUCCm AmGAmGCmCUGAGmCA AGmGCCmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2517, 3051 and 3146	chr16: 10907148- 10907172
G023457	518	UGGUCCAG AGCCUGAG CAAGGCCG	UGGUCCAGAGCCU GAGCAAGGCCGGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mU*mG*mG*mUmCCAm GmAGmCCmUGAGCmAA GGmCCGmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2518, 3052 and 3147	chr16: 10907149- 10907173
G023458	519	AGCAAGGC CGACGCCC UAUUUGAG	AGCAAGGCCGACG CCCUAUUUGAGGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mA*mG*mC*mAmAGGm CmCGmACmGCCCUmAU UUmGAGmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2519, 3053 and 3148	chr16: 10907163- 10907187
G023459	520	GACGCCCU AUUUGAGC UGUCCGGC	GACGCCCUAUUUG AGCUGUCCGGCGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mG*mA*mC*mGmCCCm UmAUmUUmGAGCUmGU CCmGGCmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2520, 3054 and 3149	chr16: 10907172- 10907196
G023460	521	AAGCCGGA CAGCUCAA AUAGGGCG	AAGCCGGACAGCU CAAAUAGGGCGGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mA*mA*mG*mCmCGGm AmCAmGCmUCAAAmUA GGmGCGmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2521, 3055 and 3150	chr16: 10907174- 10907198
G023461	522	GAGCUGUC CGGCUUCU CCAUGGAG	GAGCUGUCCGGCU UCUCCAUGGAGGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mG*mA*mG*mCmUGUm CmCGmGCmUUCUCmCA UGmGAGmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2522, 3056 and 3151	chr16: 10907184- 10907208
G023462	523	AGCUGUCC GGCUUCUC CAUGGAGC	AGCUGUCCGGCUU CUCCAUGGAGCGU UGUAGCUCCCUCC	mA*mG*mC*mUmGUCm CmGGmCUmUCUCCmAU GGmAGCmGUUGmUmAm	2523, 3057 and	chr16: 10907185- 10907209

TABLE 4-continued

Exemplary CIITA guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
			GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	3152	
G023463	524	UCAGGGAU GACAGAGC ACCAAGAC	UCAGGGAUGACAG AGCACCAAGACGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mU*mC*mA*mGmGGAm UmGAmCAmGAGCAmCC AAmGACmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2524, 3058 and 3153	chr16: 10907244- 10907268
G023464	525	CAGAGCAC CAAGACAG AGCCCUGA	CAGAGCACCAAGA CAGAGCCCUGAGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mC*mA*mG*mAmGCAm CmCAmAGmACAGAmGC CCmUGAmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2525, 3059 and 3154	chr16: 10907254- 10907278
G023465	526	AGCACCAA GACAGAGC CCUGACGC	AGCACCAAGACAG AGCCCUGACGCGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mA*mG*mC*mAmCCAm AmGAmCAmGAGCCmCU GAmCGCmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2526, 3060 and 3155	chr16: 10907257- 10907281
G023466	527	GGGACCGG CCACUUCU UCUCAGUC	GGGACCGGCCACU UCUUCUCAGUCGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mG*mG*mG*mAmCCGm GmCCmACmUUCUUmCU CAmGUCmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2527, 3061 and 3156	chr16: 10907287- 10907311
G023467	528	GGCCACUU CUUCUCAG UCACAGCC	GGCCACUUCUUCU CAGUCACAGCCGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mG*mG*mC*mCmACUm UmCUmUCmUCAGUmCA CAmGCCmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2528 3062 and 3157	chr16: 10907293- 10907317
G023468	529	ACAGCCCU ACUUUGUG CCGGGCAG	ACAGCCCUACUUU GUGCCGGGCAGGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mA*mC*mA*mGmCCCm UmACmUUmUGUGCmCG GGmCAGmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2529 3063 and 3158	chr16: 10907317- 10907341

TABLE 4-continued

Exemplary CIITA guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
G023469	530	UGCCGGGC AGUGUGCC AGCUCUCA	UGCCGGGCAGUGU GCCAGCUCUCAGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mU*mG*mC*mCmGGGm CmAGmUGmUGCCAmGC UCmUCAmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2530, 3064 and 3159	chr16: 10907331-10907355
G023470	531	GCCGGGCA GUGUGCCA GCUCUCAG	GCCGGGCAGUGUG CCAGCUCUCAGGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mG*mC*mC*mGmGGCm AmGUmGUmGCCAGmCU CUmCAGmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2531, 3065 and 3160	chr16: 10907332-10907356
G023471	532	CCUCCACG CUCACGGG ACUCUAUG	CCUCCACGCUCAC GGGACUCUAUGGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mC*mC*mU*mCmCACm GmCUmCAmCGGGAmCU CUmAUGmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2532, 3066 and 3161	chr16: 10907392-10907416
G023472	533	UCACGGGA CUCUAUGU CGGCCUGC	UCACGGGACUCUA UGUCGGCCUGCGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mU*mC*mA*mCmGGGm AmCUmCUmAUGUCmGG CCmUGCmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2533, 3067 and 3162	chr16: 10907401-10907425
G023473	534	UGGGAGCU GGGCCGCA GACAUCAA	UGGGAGCUGGGCC GCAGACAUCAAGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mU*mG*mG*mGmAGCm UmGGmGCmCGCAGmAC AUmCAAmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2534, 3068 and 3163	chr16: 10907484-10907508
G023474	535	GGGAGCUG GGCCGCAG ACAUCAAA	GGGAGCUGGGCCG CAGACAUCAAAGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mG*mG*mG*mAmGCUm GmGGmCCmGCAGAmCA UCmAAAmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2535, 3069 and 3164	chr16: 91007485-10907509
G023475	536	GCAGACAU CAAAGUAC CCUACAGG	GCAGACAUCAAAG UACCCUACAGGGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mG*mC*mA*mGmACAm UmCAmAAmGUACCmCU ACmAGGmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2536, 3070 and 3165	chr16: 10907497-10907521

TABLE 4-continued

Exemplary CIITA guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
G023476	537	AUCAAAGU ACCCUACA GGAGGACC	AUCAAAGUACCCU ACAGGAGGACCGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mA*mU*mC*mAmAAGm UmACmCCmUACAGmGA GGmACCmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2537, 3071 and 3166	chr16: 10907503- 10907527
G023477	538	UCAAAGUA CCCUACAG GAGGACCA	UCAAAGUACCCUA CAGGAGGACCAGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mU*mC*mA*mAmAGUm AmCCmCUmACAGGmAG GAmCCAmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2538, 3072 and 3167	chr16: 10907504- 10907528
G023478	539	AGUACCCU ACAGGAGG ACCAGUUC	AGUACCCUACAGG AGGACCAGUUCGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mA*mG*mU*mAmCCCm UmACmAGmGAGGAmCC AGmUUCmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2539, 3073 and 3168	chr16: 10907508- 10907532
G023479	540	UCCGCAGA CGUGAGGA CCUGGGCG	UCCGCAGACGUGA GGACCUGGGCGGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mU*mC*mC*mGmCAGm AmCGmUGmAGGACmCU GGmGCGmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2540, 3074 and 3169	chr16: 10907535- 10907559
G023480	541	GGACCUGG GCGAUGGC CAAAGGCU	GGACCUGGGCGAU GGCCAAAGGCUGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mG*mG*mA*mCmCUGm GmGCmGAmUGGCCmAA AGmGCUmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2541, 3075 and 3170	chr16: 10907548- 10907572
G023481	542	GGGCGAUG GCCAAAGG CUUAGUCC	GGGCGAUGGCCAA AGGCUUAGUCCGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mG*mG*mG*mCmGAUm GmGCmCAmAAGGCmUU AGmUCCmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2542, 3076 and 3171	chr16: 10907554- 10907578
G023482	543	GGCGAUGG CCAAAGGC UUAGUCCA	GGCGAUGGCCAAA GGCUUAGUCCAGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mG*mG*mC*mGmAUGm GmCCmAAmAGGCUmUA GUmCCAmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm	2543, 3077 and 3172	chr16: 10907555- 10907579

TABLE 4-continued

Exemplary CIITA guide sequences.						
Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
				CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU		
G023483	544	CGCGGGCCGCAGAGUCCGAGCUGG	CGCGGGCCGCAGAGUCCGAGCUGGGUUGUAGCUCCCUCCGUUGCUACAAUAAGGCCGUCGAUGUGCCGCAACGCUCUGCCGGCAUCGUU	mC*mG*mC*mGmGGCmCmGCmAGmAGUCCmGAGCmUGGmGUUGmUmAmGmCUCCCmU(L1)mCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmC(L1)mGmAmUGUGCmCGmCAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2544, 3078 and 3173	chr16: 10907587-10907611
G023484	545	GCGGGCCGCAGAGUCCGAGCUGGC	GCGGGCCGCAGAGUCCGAGCUGGCGUUGUAGCUCCCUCCGUUGCUACAAUAAGGCCGUCGAUGUGCCGCAACGCUCUGCCGGCAUCGUU	mG*mC*mG*mGmGCCmGmCAmGAmGUCCGmAGCUmGGCmGUUGmUmAmGmCUCCCmU(L1)mCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmC(L1)mGmAmUGUGCmCGmCAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2545, 3079 and 3174	chr16: 10907588-10907612
G023485	546	CGGGCCGCAGAGUCCGAGCUGGCC	CGGGCCGCAGAGUCCGAGCUGGCCGUUGUAGCUCCCUCCGUUGCUACAAUAAGGCCGUCGAUGUGCCGCAACGCUCUGCCGGCAUCGUU	mC*mG*mG*mGmCCGmCmAGmAGmUCCGAmGCUGmGCCmGUUGmUmAmGmCUCCCmU(L1)mCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmC(L1)mGmAmUGUGCmCGmCAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2546, 3080 and 3175	chr16: 10907589-10907613
G023486	547	UGAGUGGCGAAAUCAAGGACAAGG	UGAGUGGCGAAAUCAAGGACAAGGGUUGUAGCUCCCUCCGUUGCUACAAUAAGGCCGUCGAUGUGCCGCAACGCUCUGCCGGCAUCGUU	mU*mG*mA*mGmUGGmCmGAmAAmUCAAGmGACAmAGGmGUUGmUmAmGmCUCCCmU(L1)mCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmC(L1)mGmAmUGUGCmCGmCAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2547, 3081 and 3176	chr16: 10907662-10907686
G023487	548	GAGUGGCGAAAUCAAGGACAAGGA	GAGUGGCGAAAUCAAGGACAAGGAGUUGUAGCUCCCUCCGUUGCUACAAUAAGGCCGUCGAUGUGCCGCAACGCUCUGCCGGCAUCGUU	mG*mA*mG*mUmGGCmGmAAmAUmCAAGGmACAAmGGAmGUUGmUmAmGmCUCCCmU(L1)mCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmC(L1)mGmAmUGUGCmCGmCAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2548, 3082 and 3177	chr16: 10907663-10907687
G023488	549	AAAUCAAGGACAAGGAGCUCCCGC	AAAUCAAGGACAAGGAGCUCCCGCGUUGUAGCUCCCUCCGUUGCUACAAUAAGGCCGUCGAUGUGCCGCAACGCUCUGCCGGCAUCGUU	mA*mA*mA*mUmCAAmGmGAmCAmAGGAGmCUCCmCGCmGUUGmUmAmGmCUCCCmU(L1)mCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmC(L1)mGmAmUGUGCmCGmCAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2549, 3083 and 3178	chr16: 10907671-10907695
G023489	550	AAGGAGCUCCCGCAGUACCUAGCA	AAGGAGCUCCCGCAGUACCUAGCAGUUGUAGCUCCCUCCGUUGCUACAAUAAGGCCGUCGAUGUG	mA*mA*mG*mGmAGCmUmCCmCGmCAGUAmCCUAmGCAmGUUGmUmAmGmCUCCCmU(L1)mCmCGUUmGmCUAmCAAU*	2550, 3084 and 3179	chr16: 10907682-10907706

TABLE 4-continued

Exemplary CIITA guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
			CCGCAACGCUCUG CCGGCAUCGUU	AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU		
G023490	551	AGGAGCUC CCGCAGUA CCUAGCAU	AGGAGCUCCCGCA GUACCUAGCAUGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mA*mG*mG*mAmGCUm CmCCmGCmAGUACmCU AGmCAUmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2551, 3085 and 3180	chr16: 10907683- 10907707
G023491	552	GGAGCUCC CGCAGUAC CUAGCAUU	GGAGCUCCCGCAG UACCUAGCAUUGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mG*mG*mA*mGmCUCm CmCGmCAmGUACCmUA GCmAUUmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2552, 3086 and 3181	chr16: 10907684- 10907708
G023492	553	GCCCUAUG ACAACUGG CUGGAGGG	GCCCUAUGACAAC UGGCUGGAGGGGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mG*mC*mC*mCmUAUm GmACmAAmCUGGCmUG GAmGGGmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2553, 3087 and 3182	chr16: 10907726- 10907750
G023493	554	UGCCACGC UUUCUGGC UGGGCUGA	UGCCACGCUUUCU GGCUGGGCUGAGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mU*mG*mC*mCmACGm CmUUmUCmUGGCUmGG GCmUGAmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2554, 3088 and 3183	chr16: 10907752- 10907776
G023494	555	ACGCUUUC UGGCUGGG CUGAUCUU	ACGCUUUCUGGCU GGGCUGAUCUUGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mA*mC*mG*mCmUUUm CmUGmGCmUGGGCmUG AUmCUUmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2555, 3089 and 3184	chr16: 10907756- 10907780
G023495	556	CUUUCUGG CUGGGCUG AUCUUCCA	CUUUCUGGCUGGG CUGAUCUUCCAGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mC*mU*mU*mUmCUGm GmCUmGGmGCUGAmUC UUmCCAmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2556 3090 and 3185	chr16: 10907759- 10907783
G023496	557	CUGGCUGG GCUGAUCU UCCAGCCU	CUGGCUGGGCUGA UCUUCCAGCCUGU UGUAGCUCCCUCC	mC*mU*mG*mGmCUGm GmGCmUGmAUCUUmCC AGmCCUmGUUGmUmAm	2557, 3091 and	chr16: 10907763- 10907787

TABLE 4-continued

Exemplary CIITA guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
			GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	3186	
G023497	558	UGGCUGGG CUGAUCUU CCAGCCUC	UGGCUGGGCUGAU CUUCCAGCCUCGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mU*mG*mG*mCmUGGm GmCUmGAmUCUUCmCA GCmCUCmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2558, 3092 and 3187	chr16: 10907764-10907788
G023498	559	GGAGCCCU ACUCGGGC CAUCGGCG	GGAGCCCUACUCG GGCCAUCGGCGGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mG*mG*mA*mGmCCCm UmACmUCmGGGCCmAU CGmGCGmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2559, 3093 and 3188	chr16: 10907802-10907826
G023499	560	ACAGGAAG CAGAAGGU GCUUGCGA	ACAGGAAGCAGAA GGUGCUUGCGAGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mA*mC*mA*mGmGAAm GmCAmGAmAGGUGmCU UGmCGAmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2560, 3094 and 3189	chr16: 10907839-10907863
G023500	561	GCUUGCGA GGUACCUG AAGCGGCU	GCUUGCGAGGUAC CUGAAGCGGCUGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mG*mC*mU*mUmGCGm AmGGmUAmCCUGAmAG CGmGCUmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2561, 3095 and 3190	chr16: 10907855-10907879
G023501	562	GAAUUUGG CAGCACGU GGUACAGG	GAAUUUGGCAGCA CGUGGUACAGGGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mG*mA*mA*mUmUUGm GmCAmGCmACGUGmGU ACmAGGmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2562, 3096 and 3191	chr16: 10907950-10907974
G023502	563	AAUUUGGC AGCACGUG GUACAGGA	AAUUUGGCAGCAC GUGGUACAGGAGU UGUAGCUCCCUCC GUUGCUACAAUAA GGCCGUCGAUGUG CCGCAACGCUCUG CCGGCAUCGUU	mA*mA*mU*mUmUGGm CmAGmCAmCGUGGmUA CAmGGAmGUUGmUmAm GmCUCCCmU(L1)mCm CGUUmGmCUAmCAAU* AAGmGmCCmGmUmC(L 1)mGmAmUGUGCmCGm CAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2563, 3097 and 3192	chr16: 10907951-10907975

TABLE 4-continued

Exemplary CIITA guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
G023503	564	AUUUGGCAGCACGUGGUACAGGAG	AUUUGGCAGCACGUGGUACAGGAGGUUGUAGCUCCCUCCGUUGCUACAAUAAGGCCGUCGAUGUGCCGCAACGCUCUGCCGGCAUCGUU	mA*mU*mU*mUmGGCmAmGCmACmGUGGUmAcAGmGAGmGUUGmUmAmGmCUCCCmU(L1)mCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmC(L1)mGmAmUGUGCmCGmCAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2564, 3098 and 3193	chr16: 10907952-10907976
G023504	565	CCGGCCGCCUCUCUUUUCUGGGCA	CCGGCCGCCUCUCUUUUCUGGGCAGUUGUAGCUCCCUCCGUUGCUACAAUAAGGCCGUCGAUGUGCCGCAACGCUCUGCCGGCAUCGUU	mC*mC*mG*mGmCCGmCmCUmCUmCUUUUmCUGGmGCAmGUUGmUmAmGmCUCCCmU(L1)mCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmC(L1)mGmAmUGUGCmCGmCAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2565, 3099 and 3194	chr16: 10907980-10908004
G023505	566	CCUCUCUUUUCUGGGCACCCGCCU	CCUCUCUUUUCUGGGCACCCGCCUGUUGUAGCUCCCUCCGUUGCUACAAUAAGGCCGUCGAUGUGCCGCAACGCUCUGCCGGCAUCGUU	mC*mC*mU*mCmUCUmUmUUmCUmGGGCAmCCCGmCCUmGUUGmUmAmGmCUCCCmU(L1)mCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmC(L1)mGmAmUGUGCmCGmCAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2566, 3100 and 3195	chr16: 10907987-10908011
G023506	567	CCUCCUGAUGCACAUGUACUGGGC	CCUCCUGAUGCACAUGUACUGGGCGUUGUAGCUCCCUCCGUUGCUACAAUAAGGCCGUCGAUGUGCCGCAACGCUCUGCCGGCAUCGUU	mC*mC*mU*mCmCUGmAmUGmCAmCAUGUmACUGmGGCmGUUGmUmAmGmCUCCCmU(L1)mCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmC(L1)mGmAmUGUGCmCGmCAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2567, 3101 and 3196	chr16: 10908015-10908039
G023507	568	GCCUUGGAGGCGGCGGGCCAAGAC	GCCUUGGAGGCGGCGGGCCAAGACGUUGUAGCUCCCUCCGUUGCUACAAUAAGGCCGUCGAUGUGCCGCAACGCUCUGCCGGCAUCGUU	mG*mC*mC*mUmUGGmAmGGmCGmGCGGGmCCAAmGACmGUUGmUmAmGmCUCCCmU(L1)mCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmC(L1)mGmAmUGUGCmCGmCAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2568, 3102 and 3197	chr16: 10908042-10908066
G023508	569	CCUUGGAGGCGGCGGGCCAAGACU	CCUUGGAGGCGGCGGGCCAAGACUGUUGUAGCUCCCUCCGUUGCUACAAUAAGGCCGUCGAUGUGCCGCAACGCUCUGCCGGCAUCGUU	mC*mC*mU*mUmGGAmGmGCmGGmCGGGCmCAAGmACUmGUUGmUmAmGmCUCCCmU(L1)mCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmC(L1)mGmAmUGUGCmCGmCAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2569, 3103 and 3198	chr16: 10908043-10908067
G023509	570	AGGCGGCGGGCCAAGACUUCUCCC	AGGCGGCGGGCCAAGACUUCUCCCGUUGUAGCUCCCUCCGUUGCUACAAUAAGGCCGUCGAUGUGCCGCAACGCUCUGCCGGCAUCGUU	mA*mG*mG*mCmGGCmGmGGmCCmAAGACmUUCUmCCCmGUUGmUmAmGmCUCCCmU(L1)mCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmC(L1)mGmAmUGUGCmCGmCAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2570, 3104 and 3199	chr16: 10908049-10908073

TABLE 4-continued

Exemplary CIITA guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOS: 1301, 1302, 1304-1576)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2301, 2302, 2304-2576 and 3016-3205)	SEQ ID NOS	Genomic Coordinates (hg38)
G023510	571	CGGCGGGCCAAGACUUCUCCCUGG	CGGCGGGCCAAGACUUCUCCCUGGGUUGUAGCUCCCUCCGUUGCUACAAUAAGGCCGUCGAUGUGCCGCAACGCUCUGCCGGCAUCGUU	mC*mG*mG*mCmGGGmCmCAmAGmACUUCmUCCCmUGGmGUUGmUmAmGmCUCCCmU(L1)mCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmC(L1)mGmAmUGUGCmCGmCAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2571, 3105 and 3200	chr16: 10908052-10908076
G023511	572	CCCUGGACCUCCGCAGCACUGGCA	CCCUGGACCUCCGCAGCACUGGCAGUUGUAGCUCCCUCCGUUGCUACAAUAAGGCCGUCGAUGUGCCGCAACGCUCUGCCGGCAUCGUU	mC*mC*mC*mUmGGAmCmCUmCCmGCAGCmACUGmGCAmGUUGmUmAmGmCUCCCmU(L1)mCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmC(L1)mGmAmUGUGCmCGmCAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2572, 3106 and 3201	chr16: 10908070-10908094
G023512	573	CCUGGACCUCCGCAGCACUGGCAU	CCUGGACCUCCGCAGCACUGGCAUGUUGUAGCUCCCUCCGUUGCUACAAUAAGGCCGUCGAUGUGCCGCAACGCUCUGCCGGCAUCGUU	mC*mC*mU*mGmGACmCmUCmCGmCAGCAmCUGGmCAUmGUUGmUmAmGmCUCCCmU(L1)mCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmC(L1)mGmAmUGUGCmCGmCAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2573, 3107 and 3202	chr16: 10908071-10908095
G023513	574	CUGGACCUCCGCAGCACUGGCAUU	CUGGACCUCCGCAGCACUGGCAUUGUUGUAGCUCCCUCCGUUGCUACAAUAAGGCCGUCGAUGUGCCGCAACGCUCUGCCGGCAUCGUU	mC*mU*mG*mGmACCmUmCCmGCmAGCACmUGGCmAUUmGUUGmUmAmGmCUCCCmU(L1)mCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmC(L1)mGmAmUGUGCmCGmCAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2574, 3108 and 3203	chr16: 10908072-10908096
G023514	575	GGGAGCCUCGUGGGACUCAGCUGU	GGGAGCCUCGUGGGACUCAGCUGUGUUGUAGCUCCCUCCGUUGCUACAAUAAGGCCGUCGAUGUGCCGCAACGCUCUGCCGGCAUCGUU	mG*mG*mG*mAmGCCmUmCGmUGmGGACUmCAGCmUGUmGUUGmUmAmGmCUCCCmU(L1)mCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmC(L1)mGmAmUGUGCmCGmCAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2575, 3109 and 3204	chr16: 10908111-10908135
G023515	576	GGAGCCUCGUGGGACUCAGCUGUG	GGAGCCUCGUGGGACUCAGCUGUGGUUGUAGCUCCCUCCGUUGCUACAAUAAGGCCGUCGAUGUGCCGCAACGCUCUGCCGGCAUCGUU	mG*mG*mA*mGmCCUmCmGUmGGmGACUCmAGCUmGUGmGUUGmUmAmGmCUCCCmU(L1)mCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmC(L1)mGmAmUGUGCmCGmCAAmCGCUCUmGmCC(L1)GGCAUCGmU*mU	2576, 3110 and 3205	chr16: 10908112-10908136

[0563] The terms “mA,” “mC,” “mU,” or “mG” may be used to denote a nucleotide that has been modified with 2'—O-Me. The terms A*, C*, U*, or G* may be used to denote a nucleotide that is linked to the next (e.g., 3') nucleotide with a phosphorothioate (PS) bond. As used herein, the “(L1)” refers to an internal linker having a bridging length of about 15-21 atoms (e.g., about 18 atoms) as described below, e.g., see Table 28.

[0564] In some embodiments, the CIITA guide RNA comprises a guide sequence selected from SEQ TD NOs: 301, 302, 304-576. In some embodiments, the CIITA guide RNA comprises a guide sequence that is at least 19, 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence selected from SEQ ID NOs: 301, 302, 304-576. In some embodiments, the CIITA guide RNA comprises a guide sequence that is at least 95%, 90%, or 85% identical to a sequence selected from SEQ ID NOs: 301, 302, 304-576. In some embodiments, the CIITA guide RNA comprises a guide sequence that is at least 95% identical to a sequence selected from SEQ ID NOs: 301, 302, 304-576.

[0565] In some embodiments, the CIITA guide RNA comprises a guide sequence that comprises at least 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 4. As used herein, at least 10 contiguous nucleotides±10 nucleotides of a genomic coordinate means, for example, at least 10 contiguous nucleotides within the genomic coordinates wherein the genomic coordinates include 10 nucleotides in the 5' direction and 10 nucleotides in the 3' direction from the ranges listed in Table 4. For example, a CIITA guide RNA may comprise 10 contiguous nucleotides within the genomic coordinates chosen from: (a) chr16:10907504-10907528; chr16:10907508-10907532; chr16:10907539-10907559; chr16:10895658-10895682; chr16:10895668-10895692; chr16:10895750-10895774; chr16:10895753-10895777; chr16:10895754-10895778; chr16:10898684-10898708; chr16:10901529-10901553; chr16:10902121-10902145; chr16:10902701-10902725; chr16:10904726-10904750; chr16:10904760-10904784; chr16:10906493-10906517; chr16:10906515-10906539; chr16:10906631-10906655; chr16:10906636-10906660; chr16:10906643-10906667; chr16:10906770-10906794; chr16:10906788-10906812; chr16:10906789-10906813; chr16:10906816-10906840; chr16:10907148-10907172; chr16:10907254-10907278; chr16:10907331-10907355; chr16:10907477-10907501; chr16:10907497-10907521; chr16:10907503-10907527; and chr16:10907574-10907598; or (b) chr16:10906889-10906913; and chr16: 10907504-10907528; including the boundary nucleotides of these ranges. In some embodiments, the CIITA guide RNA comprises a guide sequence that is at least 19, 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence that comprises 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 4. In some embodiments, the CIITA guide RNA comprises a guide sequence that is at least 95%, 90%, or 85% identical to a sequence selected from a sequence that is 19, 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence that comprises 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 4.

[0566] In some embodiments, the CIITA guide RNA comprises a guide sequence that comprises at least 15 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 4. In some embodiments, the CIITA guide RNA comprises a guide sequence that comprises at least 24 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 4.

[0567] In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 301. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 302. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 304. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 305. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 306. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 307. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 308. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 309. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 310. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 311. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 312. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 313. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 314. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 315. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 316. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 317. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 318. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 319. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 320. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 321. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 322. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 323. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 324. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 325. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 326. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 327. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 328. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 329. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 330. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 331. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 332. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 333. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 334. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 335. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 336. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 337. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 338. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 339. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 340. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 341. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 342. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 343. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 344. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 345. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 346. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 347. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 348. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 349. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 350. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 351. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 352. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 353. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 354. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 355. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 356. In some embodiments, the CIITA guide RNA comprises SEQ ID NO:

357. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 358. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 359. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 360. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 361. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 362. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 363. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 364. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 365. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 366. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 367. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 368. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 369. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 370. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 371. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 372. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 373. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 374. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 375. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 376. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 377. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 378. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 379. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 380. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 381. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 382. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 383. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 384. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 385. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 386. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 387. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 388. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 389. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 390. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 391. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 392. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 393. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 394. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 395. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 396. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 397. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 398. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 399. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 400. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 401. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 402. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 403. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 404. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 405. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 406. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 407. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 408. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 409. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 410. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 411. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 412. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 413. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 414. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 415. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 416. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 417. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 418. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 419. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 420. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 421. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 422. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 423. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 424. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 425. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 426. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 427. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 428. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 429. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 430. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 431. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 432. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 433. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 434. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 435. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 436. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 437. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 438. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 439. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 440. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 441. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 442. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 443. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 444. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 445. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 446. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 447. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 448. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 449. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 450. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 451. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 452. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 453. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 454. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 455. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 456. In some embodiments, the CIITA

guide RNA comprises SEQ ID NO: 457. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 458. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 459. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 460. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 461. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 462. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 463. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 464. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 465. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 466. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 467. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 468. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 469. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 470. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 471. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 472. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 473. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 474. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 475. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 476. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 477. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 478. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 479. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 480. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 481. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 482. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 483. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 484. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 485. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 486. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 487. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 488. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 489. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 490. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 491. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 492. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 493. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 494. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 495. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 496. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 497. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 498. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 499. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 500. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 501. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 502. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 503. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 504. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 505. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 506. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 507. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 508. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 509. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 510. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 511. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 512. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 513. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 514. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 515. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 516. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 517. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 518. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 519. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 520. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 521. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 522. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 523. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 524. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 525. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 526. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 527. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 528. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 529. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 530. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 531. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 532. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 533. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 534. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 535. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 536. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 537. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 538. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 539. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 540. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 541. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 542. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 543. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 544. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 545. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 546. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 547. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 548. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 549. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 550. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 551. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 552. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 553. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 554. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 555. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 556. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 557. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 558. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 559. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 560. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 561. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 562. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 563. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 564. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 565. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 566. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 567. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 568. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 569. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 570. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 571. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 572. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 573. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 574. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 575. In some embodiments, the CIITA guide RNA comprises SEQ ID NO: 576.

[0568] In some embodiments, the CIITA guide RNA comprises a guide sequence of any one of: SEQ ID NOs: 301-302, 320-321, 324, 326-327, 332, 354, 361, 372, 400, 408, 414-415, 419-420, 422, 428, 431-432, 434, 451, 455, 458, 462-464, 468, 504, and 538.

[0569] In some embodiments, the CIITA guide RNA comprises a guide sequence of any one of SEQ ID NOs: 301-302, 320, 372, 414, 419, 422, and 462-463.

[0570] In some embodiments, the CIITA guide RNA comprises a sequence listed in Table 4. In some embodiments, the CIITA guide RNA comprises a sequence of any one of SEQ ID NOs: 301, 302, 304-576. In some embodiments, the CIITA guide RNA comprises a sequence of SEQ ID NO: 301 or 422. In some embodiments, the CIITA guide RNA comprises a sequence of SEQ ID NO: 301. In some embodiments, the CIITA guide RNA comprises a sequence of SEQ ID NO: 422. In some embodiments, the CIITA guide RNA comprises a guide sequence comprising a sequence of any one of SEQ ID NOs: 301, 302, 304-576. In some embodiments, the CIITA guide RNA comprises a guide sequence comprising a sequence of SEQ ID NO: 301 or 422. In some embodiments, the CIITA guide RNA comprises a guide sequence comprising a sequence of SEQ ID NO: 301. In some embodiments, the CIITA guide RNA comprises a guide sequence comprising a sequence of SEQ ID NO: 422. In some embodiments, the CIITA guide RNA comprises a sequence of any one of SEQ ID NOs: 1301, 1302, 1304-1576. In some embodiments, the CIITA guide RNA comprises a sequence of SEQ ID NO: 1301 or 1422. In some embodiments, the CIITA guide RNA comprises a sequence of SEQ ID NO: 1301. In some embodiments, the CIITA guide RNA comprises a sequence of SEQ ID NO: 1422. In some embodiments, the CIITA guide RNA comprises a sequence of any one of SEQ ID NOs: 2301, 2302, 2304-2576, 3006, and 3007. In some embodiments, the CIITA guide RNA comprises a sequence of any one of SEQ ID NOs: 2301, 2422, 3006, and 3007. In some embodiments, the CIITA guide RNA comprises a sequence of SEQ ID NO: 2301. In some embodiments, the CIITA guide RNA comprises a sequence of SEQ ID NO: 2422. In some embodiments, the CIITA guide RNA comprises a sequence of SEQ ID NO: 3006. In some embodiments, the CIITA guide RNA comprises a sequence of SEQ ID NO: 3007.

[0571] In some embodiments, the CIITA guide RNA is a single guide RNA (sgRNA) comprising a sequence of any one of the sgRNA sequences listed in Table 4.

[0572] Additional embodiments of CIITA guide RNAs are provided herein, including e.g., exemplary modifications to the guide RNA.

1. Genetic Modifications to CIITA

[0573] In some embodiments, the methods and compositions disclosed herein genetically modify at least one nucleotide of the CIITA locus in a cell. Because CIITA protein regulates expression of MHC class II, in some embodiments, the genetic modification to CIITA alters the production of CIITA protein, and thereby reduces the expression of MHC class II protein on the surface of the genetically modified cell (or engineered cell). Genetic modifications encompass the population of modifications that results from contact with a genomic editing system (e.g., the population of edits that result from Cas9 and a CIITA guide RNA, or the population of edits that result from BC22 and a CIITA guide RNA).

[0574] In some embodiments, the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from: (a) chr16:10877363-10907788 or (b) chr16:10906515-10908136. In some embodiments, the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from any of the genomic coordinates listed in Table 4.

[0575] In some embodiments, the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from: chr16:10907504-10907528; chr16:10907508-10907532; chr16:10907539-10907559; chr16:10895658-10895682; chr16:10895668-10895692; chr16:10895750-10895774; chr16:10895753-10895777; chr16:10895754-10895778; chr16:10898684-10898708; chr16:10901529-10901553; chr16:10902121-10902145; chr16:10902701-10902725; chr16:10904726-10904750; chr16:10904760-10904784; chr16:10906493-10906517; chr16:10906515-10906539; chr16:10906631-10906655; chr16:10906636-10906660; chr16:10906643-10906667; chr16:10906770-10906794; chr16:10906788-10906812; chr16:10906789-10906813; chr16:10906816-10906840; chr16:10907148-10907172; chr16:10907254-10907278; chr16:10907331-10907355; chr16:10907477-10907501; chr16:10907497-10907521; chr16:10907503-10907527; and chr16:10907574-10907598.

[0576] In some embodiments, the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from: chr16:10895658-10895682; chr16:10902701-10902725; chr16:10906493-10906517; chr16:10906631-10906655; chr16:10906643-10906667; chr16:10907477-10907501; and chr16:10907497-10907521; chr16:10907504-10907528; and chr16:10907508-10907532.

[0577] In some embodiments, the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from: chr16:10906889-10906913; and chr16:10907504-10907528.

[0578] In some embodiments, the genetic modification comprises at least 5 contiguous nucleotides within the genomic coordinates chosen from: chr16:10907504-10907528; chr16:10907508-10907532; chr16:10907539-10907559; chr16:10895658-10895682; chr16:10895668-10895692; chr16:10895750-10895774; chr16:10895753-10895777; chr16:10895754-10895778; chr16:10898684-10898708; chr16:10901529-10901553; chr16:10902121-10902145; chr16:10902701-10902725; chr16:10904726-10904750; chr16:10904760-10904784; chr16:10906493-10906517; chr16:10906515-10906539; chr16:10906631-10906655; chr16:10906636-10906660; chr16:10906643-10906667; chr16:10906770-10906794; chr16:10906788-10906812; chr16:10906789-10906813; chr16:10906816-10906840; chr16:10907148-10907172; chr16:10907254-10907278; chr16:10907331-10907355; chr16:10907477-10907501; chr16:10907497-10907521; chr16:10907503-10907527; and chr16:10907574-10907598. In some embodiments, the genetic modification comprises at least 10 contiguous nucleotides within the genomic coordinates chosen from: chr16:10907504-10907528; chr16:10907508-10907532; chr16:10907539-10907559; chr16:10895658-10895682; chr16:10895668-10895692; chr16:10895750-10895774; chr16:10895753-10895777; chr16:10895754-10895778; chr16:10898684-10898708; chr16:10901529-10901553; chr16:10902121-10902145; chr16:10902701-10902725; chr16:10904726-10904750; chr16:10904760-10904784; chr16:10906493-10906517; chr16:10906515-10906539; chr16:10906631-10906655; chr16:10906636-10906660; chr16:10906643-10906667; chr16:10906770-10906794; chr16:10906788-10906812; chr16:10906789-10906813; chr16:10906816-10906840; chr16:10907148-10907172; chr16:10907254-10907278; chr16:10907331-10907355; chr16:10907477-10907501; chr16:10907497-10907521; chr16:10907503-10907527; and chr16:10907574-10907598. In some embodiments, the genetic modification comprises at least one indel, at least one C to T substitution, or at least one A to G substitution within the genomic coordinates chosen from: chr16:10907504-10907528; chr16:10907508-10907532; chr16:10907539-10907559; chr16:10895658-10895682; chr16:10895668-10895692; chr16:10895750-10895774; chr16:10895753-10895777; chr16:10895754-10895778; chr16:10898684-10898708; chr16:10901529-10901553; chr16:10902121-10902145; chr16:10902701-10902725; chr16:10904726-10904750; chr16:10904760-10904784; chr16:10906493-10906517; chr16:10906515-10906539; chr16:10906631-10906655; chr16:10906636-10906660; chr16:10906643-10906667; chr16:10906770-10906794; chr16:10906788-10906812; chr16:10906789-10906813; chr16:10906816-10906840; chr16:10907148-10907172; chr16:10907254-10907278; chr16:10907331-10907355; chr16:10907477-10907501; chr16:10907497-10907521; chr16:10907503-10907527; and chr16:10907574-10907598.

[0579] In some embodiments, the genetic modification comprises at least 5 contiguous nucleotides within the genomic coordinates chosen from: chr16:10895658-10895682; chr16:10902701-10902725; chr16:10906493-10906517; chr16:10906631-10906655; chr16:10906643-10906667; chr16:10907477-10907501; and chr16:10907497-10907521; chr16:10907504-10907528; and chr16:10907508-10907532. In some embodiments, the genetic modification comprises at least 10 contiguous nucleotides within the genomic coordinates chosen from: chr16:10895658-10895682; chr16:10902701-10902725; chr16:10906493-10906517; chr16:10906631-10906655; chr16:10906643-10906667; chr16:10907477-10907501; and chr16:10907497-10907521; chr16:10907504-10907528; and chr16:10907508-10907532. In some embodiments, the genetic modification comprises at least one indel, at least one C to T substitution, or at least one A to G substitution within the genomic coordinates chosen from: chr16:10895658-10895682; chr16:10902701-10902725; chr16:10906493-10906517; chr16:10906631-10906655; chr16:10906643-10906667; chr16:10907477-10907501; and chr16:10907497-10907521; chr16:10907504-10907528; and chr16:10907508-10907532.

[0580] In some embodiments, the genetic modification comprises at least 5 contiguous nucleotides within the genomic coordinates chosen from: chr16:10906889-10906913; and chr16:10907504-10907528. In some embodiments, the genetic modification comprises at least 10 contiguous nucleotides within the genomic coordinates chosen from: chr16:10906889-10906913; and chr16:10907504-10907528. In some embodiments, the genetic modification comprises at least one indel, at least one C to T substitution, or at least one A to G substitution within the genomic coordinates chosen from: chr16:10906889-10906913; and chr16:10907504-10907528.

[0581] In some embodiments, the methods and compositions disclosed herein modify the CIITA locus in a cell, wherein the modification to CIITA comprises an insertion of an exogenous nucleic acid. In some embodiments, the exogenous nucleic acid is a protein-coding gene. The protein encoded by the exogenous nucleic acid may be expressed by the cell.

[0582] In some embodiments, the modification to CIITA comprises any one or more of an insertion, deletion, substitution or deamination of at least one nucleotide in a target sequence. In some embodiments, the modification to CIITA comprises an insertion of 1, 2, 3, 4 or 5 or more nucleotides in a target sequence. In some embodiments, the modification to CIITA comprises a deletion of 1, 2, 3, 4 or 5 or more nucleotides in a target sequence. In other embodiments, the modification to CIITA comprises an insertion of 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20 or 25 or more nucleotides in a target sequence. In other embodiments, the modification to CIITA comprises a deletion of 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20 or 25 or more nucleotides in a target sequence. In some embodiments, the modification to CIITA comprises an indel, which is generally defined in the art as an insertion or deletion of less than 1000 base pairs (bp). In some embodiments, the modification to CIITA comprises an indel which results in a frameshift mutation in a target sequence. In some embodiments, the modification to CIITA comprises a substitution of 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20 or 25 or more nucleotides in a target sequence. In some embodiments, the modification to CIITA comprises one or more of an insertion, deletion, or substitution of nucleotides resulting from the incorporation of a template nucleic acid. In some embodiments, the modification to CIITA comprises an insertion of a donor nucleic acid in a target sequence. In some embodiments, the modification to CIITA is not transient.

[0583] In some embodiments, the methods and compositions disclosed herein modify the CIITA locus in a cell using an RNA-guided DNA binding agent (e.g., a Cas enzyme). In some embodiments, the RNA-guided DNA binding agent is Cas9. In some embodiments, the RNA-guided DNA binding agent cuts within the CIITA gene, wherein the CIITA guide RNA targets a CIITA genomic target sequence comprising at least 10 contiguous nucleotides within the genomic coordinates chosen from: (a) chr16:10877363-10907788 and (b) chr16:10906515-10908136.

[0584] In some embodiments, the genetic modification to CIITA results in a truncated form of the CIITA protein. In some embodiments, the truncated CIITA protein does not bind to GTP. In some embodiments, the truncated CIITA protein does not localize to the nucleus. In some embodiments, the CIITA protein (e.g., a truncated form of the CIITA protein) has impaired activity as compared to the wildtype CIITA protein's activity relating to regulating MHC class II expression. In some embodiments, MHC class II expression on the surface of a cell is reduced as a result of impaired CIITA protein activity. In some embodiments, MHC class II expression on the surface of a cell is absent as a result of impaired CIITA protein activity.

2. Efficacy of CIITA Guide RNAs

[0585] The efficacy of a CIITA guide RNA may be determined by techniques available in the art that assess the editing efficiency of a guide RNA, the levels of CIITA protein or mRNA, or the levels of MHC class II in a target cell. In some embodiments, the reduction or elimination of MHC class II protein on the surface of a cell may be determined by comparison to an unmodified cell (or "relative to an unmodified cell"). An engineered cell or cell population may also be compared to a population of unmodified cells.

[0586] In some embodiments, the efficacy of a CIITA guide RNA is determined by measuring levels of CIITA protein in a cell. The levels of CIITA protein may be detected by, e.g., cell lysate and western blot with an anti-CIITA antibody. In some embodiments, the efficacy of a CIITA guide RNA is determined by measuring levels of CIITA protein in the cell nucleus. In some embodiments, the efficacy of a CIITA guide RNA is determined by measuring levels of CIITA mRNA in a cell. The levels of CIITA mRNA may be detected by e.g., RT-PCR. In some embodiments, a decrease in the levels CIITA protein or CIITA mRNA in the target cell as compared to an unmodified cell is indicative of an effective CIITA guide RNA.

[0587] An "unmodified cell" (or "unmodified cells") refers to a control cell (or cells) of the same type of cell in an experiment or test, wherein the "unmodified" control cell has not been contacted with a CIITA guide (i.e., a non-engineered cell). Therefore, an unmodified cell (or cells) may be a cell that has not been contacted with a guide RNA, or a cell that has been contacted with a guide RNA that does not target CIITA.

[0588] In some embodiments, the efficacy of a CIITA guide RNA is determined by measuring the reduction or elimination of MHC class II protein expression by the target cells. The CIITA protein functions as a transactivator, activating the MHC class II promoter, and is essential for the expression of MHC class II protein. In some embodiments, MHC class II protein expression may be detected on the surface of the target cells. In some embodiments, MHC class II protein expression is measured by flow cytometry. In some embodiments, an antibody against MHC class II protein (e.g., anti-HLA-DR, -DQ, -DP) may be used to detect MHC class II protein expression e.g., by flow cytometry. In some embodiments, a reduction or elimination in MHC class II protein on the surface of a cell (or population of cells) as compared to an unmodified cell (or population of unmodified cells) is indicative of an effective CIITA guide RNA. In some embodiments, a cell (or population of cells) that has been contacted with a particular CIITA guide RNA and RNA-guided DNA binding agent that is negative for MHC class II protein by flow cytometry is indicative of an effective CIITA guide RNA.

[0589] In some embodiments, the MHC class II protein expression is reduced or eliminated in a population of cells using the methods and compositions disclosed herein. In some embodiments, the population of cells is enriched (e.g., by FACS or MACS) and is at least 65%, 70%, 80%, 90%, 91%, 92%, 93%, or 94% MHC class II negative as measured by flow cytometry relative to a population of unmodified cells. In some embodiments, the population of cells is not enriched (e.g., by FACS or MACS) and is at least 65%, 70%, 80%, 90%, 91%, 92%, 93%, or 94% MHC class II negative as measured by flow cytometry relative to a population of unmodified cells.

[0590] In some embodiments, the population of cells is at least 65% MHC class II negative as measured by flow cytometry relative to a population of unmodified cells. In some embodiments, the population of cells is at least 70% MHC class II negative as measured by flow cytometry relative to a population of unmodified cells. In some embodiments, the population of cells is at least 80% MHC class II negative as measured by flow cytometry relative to a population of unmodified cells. In some embodiments, the population of cells is at least 90% MHC class II negative as measured by flow cytometry relative to a population of unmodified cells. In some embodiments, the population of cells is at least 91% MHC class II negative as measured by flow cytometry relative to a population of unmodified cells. In some embodiments, the population of cells is at least 92% MHC class II negative as measured by flow cytometry relative to a population of unmodified cells. In some embodiments, the population of cells is at least 93% MHC class II negative as measured by flow cytometry relative to a population of unmodified cells. In some embodiments, the population of cells is at least 94% MHC class II negative as measured by flow cytometry relative to a population of unmodified cells.

[0591] In some embodiments, an effective CIITA guide RNA may be determined by measuring the response of immune cells in vitro or in vivo (e.g., CD4+ T cells) to the genetically modified target cell. A CD4+ T cell response may be evaluated by an assay that measures the activation response of CD4+ T cells e.g., CD4+ T cell proliferation, expression of activation markers, or cytokine production (IL-2, IL-12, IFN-7) (e.g., flow cytometry, ELISA). The response of CD4+ T cells may be evaluated in in vitro cell culture assays in which the genetically modified cell is co-cultured with cells comprising CD4+ T cells. For example, the genetically modified cell may be co-cultured e.g., with PBMCs, purified CD3+ T cells comprising CD4+ T cells, purified CD4+ T cells, or a CD4+ T cell line. The CD4+ T cell response elicited from the genetically modified cell may be compared to the response elicited from an unmodified cell. A reduced response from CD4+ T cells is indicative of an effective CIITA guide RNA.

[0592] The efficacy of a CIITA guide RNA may also be assessed by the survival of the cell post-editing. In some embodiments, the cell survives post editing for at least one week to six weeks. In some embodiments, the cell survives post editing for at least one week to twelve weeks. In some embodiments, the cell survives post editing for at least two weeks. In some embodiments, the cell survives post editing for at least three weeks. In some embodiments, the cell survives post editing for at least four weeks. In some embodiments, the cell survives post editing for at least five weeks. In some embodiments, the cell survives post editing for at least six weeks. The viability of a genetically modified cell may be measured using standard techniques, including e.g., by measures of cell death, by flow cytometry live/dead staining, or cell proliferation.

G. AAVS1 Guide RNAs

[0593] The methods and compositions provided herein disclose AAVS1 guide RNAs useful for creating insertion sites within the AAVS1 locus. In some embodiments, such guide RNAs direct an RNA-guided DNA binding agent to am AAVS1 genomic target sequence and may be referred to herein as "AAVS1 guide RNAs." In some embodiments, the AAVS1 guide RNA directs an RNA-guided DNA binding agent to a human AAVS1 genomic target sequence. In some embodiments, the AAVS1 guide RNA comprises a guide sequence selected from SEQ ID NO: 601-774.

[0594] In some embodiments, the methods and compositions disclosed herein comprise an AAVS1 guide RNA comprising a guide sequence that targets an AAVS1 genomic target sequence comprising at least 10 contiguous nucleotides within the genomic coordinates chr19: 55115151-55116209. In some embodiments, the methods and compositions disclosed herein comprise a AAVS1 guide RNA comprising a guide sequence that targets an AAVS1 genomic target sequence comprising at least one nucleotide within the genomic coordinates chr19: 55115151-55116209.

[0595] In some embodiments, the methods and compositions disclosed herein comprise a AAVS1 guide RNA comprising a guide sequence that directs an RNA-guided DNA binding agent to make cut in an AAVS1 gene, wherein the AAVS1 guide RNA targets an AAVS1 genomic target sequence comprising at least 10 contiguous nucleotides within the genomic coordinates chr19: 55115151-55116209. In some embodiments, the methods and compositions disclosed herein comprise an AAVS1 guide RNA comprising a guide sequence that directs an RNA-guided DNA binding agent to make cut in an AAVS1 gene, wherein the AAVS1 guide RNA targets an AAVS1 genomic target sequence comprising at least one nucleotide within the genomic coordinates chr19: 55115151-55116209.

[0596] In some embodiments, the methods and compositions disclose an AAVS1 guide RNA that directs an RNA-guided DNA binding agent to induce a double stranded break (DSB) or a single-stranded break (SSB) in an AAVS1 genomic target sequence. In some embodiments, the methods and compositions disclose an AAVS1 guide RNA that directs an RNA-guided DNA binding agent to make a cut in an AAVS1 genomic target sequence. In embodiments wherein the RNA-guided DNA cutting agent is Cas9, the cut or "cut site" occurs at the third base from the protospacer adjacent motif (PAM) sequence.

[0597] In some embodiments, a composition is provided comprising an AAVS1 guide RNA described herein and an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent.

[0598] In some embodiments, a composition is provided comprising an AAVS1 single-guide RNA (sgRNA) comprising a guide sequence that targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chr19: 55115151-55116209. In some embodiments, a composition is provided comprising an AAVS1 sgRNA described herein and an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent.

[0599] In some embodiments, a composition is provided comprising an AAVS1 dual-guide RNA (dgRNA) comprising a guide sequence that targets a genomic target comprising at least 10 contiguous nucleotides within the genomic coordinates chr19: 55115151-55116209. In some embodiments, a composition is provided comprising an AAVS1 dgRNA described herein and an RNA-guided DNA binding agent or a nucleic acid encoding an RNA-guided DNA binding agent.

[0600] In some embodiments, the AAVS1 gRNA comprises a guide sequence selected from any one of SEQ ID NOs: 601-774. Exemplary AAVS1 guide sequences are shown below in Table 5.

TABLE 5

Exemplary AAVS1 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1601-1774)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2601-2774)	Genomic Coordinates (hg38)
G025808	601	AUAUCCAGAACCCUGACCCUGCCG	AUAUCCAGAACCCUGACCCUGCCGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mA*mU*mA*mUmCCAmGmAAmCCmCUGACmCCUGmCCGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr14: 22547505-22547529
G025809	602	CCCACAGAUAUCCAGAACCCUGAC	CCCACAGAUAUCCAGAACCCUGACGUUGUAGCUCCCUGAAACCG	mC*mC*mC*mAmCAGmAmUAmUCmCAGAAmCCCUmGACmGUUGmUmAmGmCUCCCmUmGmAmAm	chr14: 22547498-22547522

TABLE 5-continued

Exemplary AAVS1 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1601-1774)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2601-2774)	Genomic Coordinates (hg38)
			UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	
G025810	603	UUGUCCCACAG AUAUCCAGAAC CC	UUGUCCCACAG AUAUCCAGAAC CCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mU*mU*mG*mUmCCCm AmCAmGAmUAUCCmAG AAmCCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr14: 22547494-22547518
G025811	604	CUUGUCCCACA GAUAUCCAGAA CC	CUUGUCCCACA GAUAUCCAGAA CCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mU*mU*mGmUCCm CmACmAGmAUAUCmCA GAmACCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr14: 22547493-22547517
G025812	605	AUCCUCUUGUC CCACAGAUAUC CA	AUCCUCUUGUC CCACAGAUAUC CAGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mA*mU*mC*mCmUCUm UmGUmCCmCACAGmAU AUmCCAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr14: 22547488-22547512
G025813	606	GAUCCUCUUGU CCCACAGAUAU CC	GAUCCUCUUGU CCCACAGAUAU CCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mG*mA*mU*mCmCUCm UmUGmUCmCCACAmGA UAmUCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr14: 22547487-22547511
G025814	607	AACCCUGAUCC UCUUGUCCCAC AG	AACCCUGAUCC UCUUGUCCCAC AGGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mA*mA*mC*mCmCUGm AmUCmCUmCUUGUmCC CAmCAGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr14: 22547481-22547505
G025815	608	GCCGUGUACCA GCUGAGAGACU CU	GCCGUGUACCA GCUGAGAGACU CUGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC	mG*mC*mC*mGmUGUm AmCCmAGmCUGAGmAG ACmUCUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU	chr14: 22547525-22547549

TABLE 5-continued

Exemplary AAVS1 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1601-1774)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2601-2774)	Genomic Coordinates (hg38)
			AACGCUCUGCC UUCUGGCAUCG UU	GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	
G025816*	609	AAGAGAAAGGG AGUAGAGGCGG CC	AAGAGAAAGGG AGUAGAGGCGG CCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mA*mA*mG*mAmGAAm AmGGmGAmGUAGAmGG CGmGCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115151-55115175
G025817	610	AGUAGAGGCGG CCACGACCUGG UG	AGUAGAGGCGG CCACGACCUGG UGGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mA*mG*mU*mAmGAGm GmCGmGCmCACGAmCC UGmGUGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115162-55115186
G025818	611	CCACGACCUGG UGAACACCUAG GA	CCACGACCUGG UGAACACCUAG GAGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mC*mA*mCmGACm CmUGmGUmGAACAmCC UAmGGAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115173-55115197
G025819	612	GACGCACCAUU CUCACAAAGGG AG	GACGCACCAUU CUCACAAAGGG AGGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mG*mA*mC*mGmCACm CmAUmUCmUCACAmAA GGmGAGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115195-55115219
G025820	613	CUCACAAAGGG AGUUUUCCACA CG	CUCACAAAGGG AGUUUUCCACA CGGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mU*mC*mAmCAAm AmGGmGAmGUUUUmCC ACmACGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115206-55115230
G025821	614	UCACAAAGGGA GUUUUCCACAC GG	UCACAAAGGGA GUUUUCCACAC GGGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mU*mC*mA*mCmAAAm GmGGmAGmUUUUCmCA CAmCGGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115207-55115231

TABLE 5-continued

Exemplary AAVS1 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1601-1774)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2601-2774)	Genomic Coordinates (hg38)
G025822	615	CACAAAGGGAG UUUUCCACACG GA	CACAAAGGGAG UUUUCCACACG GAGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mA*mC*mAmAAGm GmGAmGUmUUUCCmAC ACmGGAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115208-55115232
G025823	616	ACAAAGGGAGU UUUCCACACGG AC	ACAAAGGGAGU UUUCCACACGG ACGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mA*mC*mA*mAmAGGm GmAGmUUmUUCCAmCA CGmGACmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115209-55115233
G025824	617	CAAAGGGAGUU UUCCACACGGA CA	CAAAGGGAGUU UUCCACACGGA CAGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mA*mA*mAmGGGm AmGUmUUmUCCACmAC GGmACAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115210-55115234
G025825	618	AGGGAGUUUUC CACACGGACAC CC	AGGGAGUUUUC CACACGGACAC CCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mA*mG*mG*mGmAGUm UmUUmCCmACACGmGA CAmCCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115213-55115237
G025826	619	GUUUUCCACAC GGACACCCCCC UC	GUUUUCCACAC GGACACCCCCC UCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mG*mU*mU*mUmUCCm AmCAmCGmGACACmCC CCmCUCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115218-55115242
G025827	620	CACACGGACAC CCCCCUCCUCA CC	CACACGGACAC CCCCCUCCUCA CCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mA*mC*mAmCGGm AmCAmCCmCCCCUmCC UCmACCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115224-55115248
G025828	621	ACACGGACACC CCCCUCCUCAC CA	ACACGGACACC CCCCUCCUCAC CAGUUGUAGCU	mA*mC*mA*mCmGGAm CmACmCCmCCCUCmCU CAmCCAmGUUGmUmAm	chr19: 55115225-55115249

TABLE 5-continued

Exemplary AAVS1 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1601-1774)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2601-2774)	Genomic Coordinates (hg38)
			CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	
G025829	622	GGACACCCCCC UCCUCACCACA GC	GGACACCCCCC UCCUCACCACA GCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mG*mG*mA*mCmACCm CmCCmCUmCCUCAmCC ACmAGCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115229-55115253
G025830	623	GCCCUGCCAGG ACGGGGCUGGC UA	GCCCUGCCAGG ACGGGGCUGGC UAGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mG*mC*mC*mCmUGCm CmAGmGAmCGGGGmCU GGmCUAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115251-55115275
G025831	624	GGUGAAGAGCC AAAGUUAGAAC UC	GGUGAAGAGCC AAAGUUAGAAC UCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mG*mG*mU*mGmAAGm AmGCmCAmAAGUUmAG AAmCUCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115337-55115361
G025832*	625	UCUUGGGAAGU GUAAGGAAGCU GC	UCUUGGGAAGU GUAAGGAAGCU GCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mU*mC*mU*mUmGGGm AmAGmUGmUAAGGmAA GCmUGCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115404-55115428
G025833	626	GCUGCAGCACC AGGAUCAGUGA AA	GCUGCAGCACC AGGAUCAGUGA AAGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mG*mC*mU*mGmCAGm CmACmCAmGGAUCmAG UGmAAAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115423-55115447
G025834	627	ACCAGGAUCAG UGAAACGCACC AG	ACCAGGAUCAG UGAAACGCACC AGGUUGUAGCU CCCUGAAACCG UUGCUACAAUA	mA*mC*mC*mAmGGAm UmCAmGUmGAAACmGC ACmCAGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC	chr19: 55115431-55115455

TABLE 5-continued

Exemplary AAVS1 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1601-1774)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2601-2774)	Genomic Coordinates (hg38)
			AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	
G025835	628	GGUUCUGGGAG AGGGUAGCGCA GG	GGUUCUGGGAG AGGGUAGCGCA GGGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mG*mG*mU*mUmCUGm GmGAmGAmGGGUAmGC GCmAGGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115477-55115501
G025836	629	GAGGGUAGCGC AGGGUGGCCAC UG	GAGGGUAGCGC AGGGUGGCCAC UGGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mG*mA*mG*mGmGUAm GmCGmCAmGGGUGmGC CAmCUGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115487-55115511
G025837	630	GCCACUGAGAA CCGGGCAGGUC AC	GCCACUGAGAA CCGGGCAGGUC ACGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mG*mC*mC*mAmCUGm AmGAmACmCGGGCmAG GUmCACmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115504-55115528
G025838	631	CCACUGAGAAC CGGGCAGGUCA CG	CCACUGAGAAC CGGGCAGGUCA CGGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mC*mA*mCmUGAm GmAAmCCmGGGCAmGG UCmACGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115505-55115529
G025839	632	CACUGAGAACC GGGCAGGUCAC GC	CACUGAGAACC GGGCAGGUCAC GCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mA*mC*mUmGAGm AmACmCGmGGCAGmGU CAmCGCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115506-55115530
G025840*	633	ACUGAGAACCG GGCAGGUCACG CA	ACUGAGAACCG GGCAGGUCACG CAGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC	mA*mC*mU*mGmAGAm AmCCmGGmGCAGGmUC ACmGCAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU	chr19: 55115507-55115531

TABLE 5-continued

Exemplary AAVS1 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1601-1774)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2601-2774)	Genomic Coordinates (hg38)
			UUCUGGCAUCG UU	CUmGmCCmUmUmCmUG GCAUCG*mU*mU	
G025841*	634	AACCGGGCAGG UCACGCAUCCC CC	AACCGGGCAGG UCACGCAUCCC CCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mA*mA*mC*mCmGGGm CmAGmGUmCACGCmAU CCmCCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115513-55115537
G025842	635	ACCGGGCAGGU CACGCAUCCCC CC	ACCGGGCAGGU CACGCAUCCCC CCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mA*mC*mC*mGmGGCm AmGGmUCmACGCAmUC CCmCCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115514-55115538
G025843	636	GGGCAGGUCAC GCAUCCCCCCC UU	GGGCAGGUCAC GCAUCCCCCCC UUGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mG*mG*mG*mCmAGGm UmCAmCGmCAUCCmCC CCmCUUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115517-55115541
G025844	637	GGCAGGUCACG CAUCCCCCCCU UC	GGCAGGUCACG CAUCCCCCCCU UCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mG*mG*mC*mAmGGUm CmACmGCmAUCCCmCC CCmUUCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115518-55115542
G025845	638	CCUUCCCUCCC ACCCCCUGCCA AG	CCUUCCCUCCC ACCCCCUGCCA AGGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mC*mU*mUmCCCm UmCCmCAmCCCCmUG CCmAAGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115537-55115561
G025846	639	CUUCCCUCCCA CCCCCUGCCAA GC	CUUCCCUCCCA CCCCCUGCCAA GCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mU*mU*mCmCCUm CmCCmACmCCCCUmGC CAmAGCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115538-55115562

TABLE 5-continued

Exemplary AAVS1 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1601-1774)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2601-2774)	Genomic Coordinates (hg38)
G025847	640	CCCCCUGCCAA GCUCUCCCUCC CA	CCCCCUGCCAA GCUCUCCCUCC CAGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mC*mC*mCmCUGm CmCAmAGmCUCUCmCC UCmCCAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115549-55115573
G025848	641	GAUCCUCUCUG GCUCCAUCGUA AG	GAUCCUCUCUG GCUCCAUCGUA AGGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mG*mA*mU*mCmCUCm UmCUmGGmCUCCAmUC GUmAAGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115574-55115598
G025849	642	AGAGGUUCUGG CAAGGAGAGAG AU	AGAGGUUCUGG CAAGGAGAGAG AUGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mA*mG*mA*mGmGUUm CmUGmGCmAAGGAmGA GAmGAUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115606-55115630
G025850	643	GGGGGUGUGUC ACCAGAUAAGG AA	GGGGGUGUGUC ACCAGAUAAGG AAGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mG*mG*mG*mGmGUGm UmGUmCAmCCAGAmUA AGmGAAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115643-55115667
G025851	644	CAAUAUCAGGA GACUAGGAAGG AG	CAAUAUCAGGA GACUAGGAAGG AGGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mA*mA*mUmAUCm AmGGmAGmACUAGmGA AGmGAGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115696-55115720
G025852	645	AUGGGGCUUUU CUGUCACCAAU CC	AUGGGGCUUUU CUGUCACCAAU CCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mA*mU*mG*mGmGGCm UmUUmUCmUGUCAmCC AAmUCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115731-55115755
G025853	646	UUUCUGUCACC AAUCCUGUCCC UA	UUUCUGUCACC AAUCCUGUCCC UAGUUGUAGCU	mU*mU*mU*mCmUGUm CmACmCAmAUCCUmGU CCmCUAmGUUGmUmAm	chr19: 55115739-55115763

TABLE 5-continued

Exemplary AAVS1 guide sequences.					
Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1601-1774)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2601-2774)	Genomic Coordinates (hg38)
			CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	
G025854	647	UUCUGUCACCA AUCCUGUCCCU AG	UUCUGUCACCA AUCCUGUCCCU AGGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mU*mU*mC*mUmGUCm AmCCmAAmUCCUGmUC CCmUAGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115740-55115764
G025855	648	UCUGUCACCAA UCCUGUCCCUA GU	UCUGUCACCAA UCCUGUCCCUA GUGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mU*mC*mU*mGmUCAm CmCAmAUmCCUGUmCC CUmAGUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115741-55115765
G025856	649	GUGGGGUGGAG GGGACAGAUAA AA	GUGGGGUGGAG GGGACAGAUAA AAGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mG*mU*mG*mGmGGUm GmGAmGGmGGACAmGA UAmAAAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115774-55115798
G025857	650	UGGAGGGGACA GAUAAAAGUAC CC	UGGAGGGGACA GAUAAAAGUAC CCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mU*mG*mG*mAmGGGm GmACmAGmAUAAAmAG UAmCCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19 : 55115780-55115804
G025858	651	GGACAGAUAAA AGUACCCAGAA CC	GGACAGAUAAA AGUACCCAGAA CCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mG*mG*mA*mCmAGAm UmAAmAAmGUACCmCA GAmACCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115786-55115810
G025859	652	AAAGUACCCAG AACCAGAGCCA CA	AAAGUACCCAG AACCAGAGCCA CAGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA	mA*mA*mA*mGmUACm CmCAmGAmACCAGmAG CCmACAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU	chr19: 55115795-55115819

TABLE 5-continued

Exemplary AAVS1 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1601-1774)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2601-2774)	Genomic Coordinates (hg38)
			AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	
G025860	653	UACCCAGAACC AGAGCCACAUU AA	UACCCAGAACC AGAGCCACAUU AAGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mU*mA*mC*mCmCAGm AmACmCAmGAGCCmAC AUmUAAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115799-55115823
G025861	654	ACCCAGAACCA GAGCCACAUUA AC	ACCCAGAACCA GAGCCACAUUA ACGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mA*mC*mC*mCmAGAm AmCCmAGmAGCCAmCA UUmAACmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115800-55115824
G025862	655	UUAACCGGCCC UGGGAAUAUAA GG	UUAACCGGCCC UGGGAAUAUAA GGGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mU*mU*mA*mAmCCGm GmCCmCUmGGGAAmUA UAmAGGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115819-55115843
G025863	656	UAACCGGCCCU GGGAAUAUAAG GU	UAACCGGCCCU GGGAAUAUAAG GUGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mU*mA*mA*mCmCGGm CmCCmUGmGGAAUmAU AAmGGUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115820-55115844
G025864	657	AGGUGGUCCCA GCUCGGGGACA CA	AGGUGGUCCCA GCUCGGGGACA CAGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mA*mG*mG*mUmGGUm CmCCmAGmCUCGGmGG ACmACAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115840-55115864
G025865	658	GGAUCCCUGGA GGCAGCAAACA UG	GGAUCCCUGGA GGCAGCAAACA UGGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC	mG*mG*mA*mUmCCCm UmGGmAGmGCAGCmAA ACmAUGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU	chr19: 55115864-55115888

TABLE 5-continued

Exemplary AAVS1 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1601-1774)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2601-2774)	Genomic Coordinates (hg38)
			UUCUGGCAUCG UU	CUmGmCCmUmUmCmUG GCAUCG*mU*mU	
G025866	659	ACAUGCUGUCC UGAAGUGGACA UA	ACAUGCUGUCC UGAAGUGGACA UAGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mA*mC*mA*mUmGCUm GmUCmCUmGAAGUmGG ACmAUAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115883-55115907
G025867	660	CAUGCUGUCCU GAAGUGGACAU AG	CAUGCUGUCCU GAAGUGGACAU AGGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mA*mU*mGmCUGm UmCCmUGmAAGUGmGA CAmUAGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115884-55115908
G025868	661	GAGGAAGAAGA CUAGCUGAGCU CU	GAGGAAGAAGA CUAGCUGAGCU CUGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mG*mA*mG*mGmAAGm AmAGmACmUAGCUmGA GCmUCUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115920-55115944
G025869	662	AGGAAGAAGAC UAGCUGAGCUC UC	AGGAAGAAGAC UAGCUGAGCUC UCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mA*mG*mG*mAmAGAm AmGAmCUmAGCUGmAG CUmCUCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115921-55115945
G025870	663	GGAAGAAGACU AGCUGAGCUCU CG	GGAAGAAGACU AGCUGAGCUCU CGGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mG*mG*mA*mAmGAAm GmACmUAmGCUGAmGC UCmUCGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115922-55115946
G025871	664	AGCUGAGCUCU CGGACCCCUGG AA	AGCUGAGCUCU CGGACCCCUGG AAGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mA*mG*mC*mUmGAGm CmUCmUCmGGACCmCC UGmGAAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115933-55115957

TABLE 5-continued

Exemplary AAVS1 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1601-1774)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2601-2774)	Genomic Coordinates (hg38)
G025872	665	CUGCCCAAAUG AAAGGAGUGAG AG	CUGCCCAAAUG AAAGGAGUGAG AGGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mU*mG*mCmCCAm AmAUmGAmAAGGAmGU GAmGAGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55116019-55116043
G025873	666	UGCCCAAAUGA AAGGAGUGAGA GG	UGCCCAAAUGA AAGGAGUGAGA GGGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mU*mG*mC*mCmCAAm AmUGmAAmAGGAGmUG AGmAGGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55116020-55116044
G025874	667	AAUGAAAGGAG UGAGAGGUGAC CC	AAUGAAAGGAG UGAGAGGUGAC CCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mA*mA*mU*mGmAAAm GmGAmGUmGAGAGmGU GAmCCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55116026-55116050
G025875	668	GACCCGAAUCC ACAGGAGAACG GG	GACCCGAAUCC ACAGGAGAACG GGGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mG*mA*mC*mCmCGAm AmUCmCAmCAGGAmGA ACmGGGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55116045-55116069
G025876*	669	AAGCAAGAGGA UGGAGAGGUGG CU	AAGCAAGAGGA UGGAGAGGUGG CUGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mA*mA*mG*mCmAAGm AmGGmAUmGGAGAmGG UGmGCUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55116084-55116108
G025877	670	UUGUCCAGAAA AACGGUGAUGA UG	UUGUCCAGAAA AACGGUGAUGA UGGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mU*mU*mG*mUmCCAm GmAAmAAmACGGUmGA UGmAUGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55116136-55116160
G025878	671	UGGCCGCCUCU ACUCCCUUUCU CU	UGGCCGCCUCU ACUCCCUUUCU CUGUUGUAGCU CCCUGAAACCG	mU*mG*mG*mCmCGCm CmUCmUAmCUCCCmUU UCmUCUmGUUGmUmAm GmCUCCCmUmGmAmAm	chr19: 55115152-55115176

TABLE 5-continued

Exemplary AAVS1 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1601-1774)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2601-2774)	Genomic Coordinates (hg38)
			UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	
G025879	672	GUUCACCAGGU CGUGGCCGCCU CU	GUUCACCAGGU CGUGGCCGCCU CUGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mG*mU*mU*mCmACCm AmGGmUCmGUGGCmCG CCmUCUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115165-55115189
G025880	673	UGUUCACCAGG UCGUGGCCGCC UC	UGUUCACCAGG UCGUGGCCGCC UCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mU*mG*mU*mUmCACm CmAGmGUmCGUGGmCC GCmCUCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115166-55115190
G025881	674	GUCCUAGGUGU UCACCAGGUCG UG	GUCCUAGGUGU UCACCAGGUCG UGGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mG*mU*mC*mCmUAGm GmUGmUUmCACCAmGG UCmGUGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115174-55115198
G025882	675	UGCGUCCUAGG UGUUCACCAGG UC	UGCGUCCUAGG UGUUCACCAGG UCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mU*mG*mC*mGmUCCm UmAGmGUmGUUCAmCC AGmGUCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115177-55115201
G025883	676	UGUGAGAAUGG UGCGUCCUAGG UG	UGUGAGAAUGG UGCGUCCUAGG UGGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mU*mG*mU*mGmAGAm AmUGmGUmGCGUCmCU AGmGUGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115188-55115212
G025884	677	GAAAACUCCCU UUGUGAGAAUG GU	GAAAACUCCCU UUGUGAGAAUG GUGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC	mG*mA*mA*mAmACUm CmCCmUUmUGUGAmGA AUmGGUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU	chr19: 55115200-55115224

TABLE 5-continued

Exemplary AAVS1 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1601-1774)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2601-2774)	Genomic Coordinates (hg38)
			AACGCUCUGCC UUCUGGCAUCG UU	GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	
G025885	678	ACCUGUGAGAU AAGGCCAGUAG CC	ACCUGUGAGAU AAGGCCAGUAG CCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mA*mC*mC*mUmGUGm AmGAmUAmAGGCCmAG UAmGCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115270-55115294
G025886	679	UACCUGUGAGA UAAGGCCAGUA GC	UACCUGUGAGA UAAGGCCAGUA GCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mU*mA*mC*mCmUGUm GmAGmAUmAAGGCmCA GUmAGCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115271-55115295
G025887	680	UUACCUGUGAG AUAAGGCCAGU AG	UUACCUGUGAG AUAAGGCCAGU AGGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mU*mU*mA*mCmCUGm UmGAmGAmUAAGGmCC AGmUAGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115272-55115296
G025888*	681	AGUUUUACCUG UGAGAUAAGGC CA	AGUUUUACCUG UGAGAUAAGGC CAGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mA*mG*mU*mUmUUAm CmCUmGUmGAGAUmAA GGmCCAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115276-55115300
G025889	682	GUGCGUCAGUU UUACCUGUGAG AU	GUGCGUCAGUU UUACCUGUGAG AUGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mG*mU*mG*mCmGUCm AmGUmUUmUACCUmGU GAmGAUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115283-55115307
G025890	683	UAUAUUGUUCC UCCGUGCGUCA GU	UAUAUUGUUCC UCCGUGCGUCA GUGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mU*mA*mU*mAmUUGm UmUCmCUmCCGUGmCG UCmAGUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115297-55115321

TABLE 5-continued

Exemplary AAVS1 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1601-1774)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2601-2774)	Genomic Coordinates (hg38)
G025891	684	UAACUUUGGCUCUUCACCUUUCUA	UAACUUUGGCUCUUCACCUUUCUAGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mU*mA*mA*mCmUUUmGmGCmUCmUUCACmCUUUmCUAmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr19: 55115331-55115355
G025892	685	CUAACUUUGGCUCUUCACCUUUCU	CUAACUUUGGCUCUUCACCUUUCUGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mU*mA*mAmCUUmUmGGmCUmCUUCAmCCUUmUCUmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr19: 55115332-55115356
G025893	686	UCUAACUUUGGCUCUUCACCUUUC	UCUAACUUUGGCUCUUCACCUUUCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mU*mC*mU*mAmACUmUmUGmGCmUCUUCmACCUmUUCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr19: 55115333-55115357
G025894	687	GGUCCUGAGUUCUAACUUUGGCUC	GGUCCUGAGUUCUAACUUUGGCUCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mG*mG*mU*mCmCUGmAmGUmUCmUAACUmUUGGmCUCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr19: 55115343-55115367
G025895	688	UCCUGGUGCUGCAGCUUCCUUACA	UCCUGGUGCUGCAGCUUCCUUACAGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mU*mC*mC*mUmGGUmGmCUmGCmAGCUUmCCUUmACAmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr19: 55115414-55115438
G021566	689	CAGCUAGUCUUCUUCCUCCAACCC	CAGCUAGUCUUCUUCCUCCAACCCGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mA*mG*mCmUAGmUmCUmUCmUUCCUmCCAAmCCCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr19: 55115914-55115938
G025896	690	GCGUGACCUGCCCGGUUCUCAGUG	GCGUGACCUGCCCGGUUCUCAGUGGUUGUAGCU	mG*mC*mG*mUmGACmCmUGmCCmCGGUUmCUCAmGUGmGUUGmUmAm	chr19: 55115506-55115530

TABLE 5-continued

Exemplary AAVS1 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1601-1774)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2601-2774)	Genomic Coordinates (hg38)
			CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	
G025897	691	GAUGCGUGACC UGCCCGGUUCU CA	GAUGCGUGACC UGCCCGGUUCU CAGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mG*mA*mU*mGmCGUm GmACmCUmGCCCGmGU UCmUCAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115509-55115533
G025898	692	GUUUGCUUACG AUGGAGCCAGA GA	GUUUGCUUACG AUGGAGCCAGA GAGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mG*mU*mU*mUmGCUm UmACmGAmUGGAGmCC AGmAGAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115579-55115603
G025899	693	GAACCUCUAAG GUUUGCUUACG AU	GAACCUCUAAG GUUUGCUUACG AUGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mG*mA*mA*mCmCUCm UmAAmGGmUUUGCmUU ACmGAUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115590-55115614
G025900*	694	CCUGGAGCCAU CUCUCUCCUUG CC	CCUGGAGCCAU CUCUCUCCUUG CCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mC*mU*mGmGAGm CmCAmUCmUCUCUmCC UUmGCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115615-55115639
G025901	695	CACCCCCAUUU CCUGGAGCCAU CU	CACCCCCAUUU CCUGGAGCCAU CUGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mA*mC*mCmCCCm AmUUmUCmCUGGAmGC CAmUCUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115626-55115650
G025902*	696	UCUGGUGACAC ACCCCCAUUUC CU	UCUGGUGACAC ACCCCCAUUUC CUGUUGUAGCU CCCUGAAACCG UUGCUACAAUA	mU*mC*mU*mGmGUGm AmCAmCAmCCCCCmAU UUmCCUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC	chr19: 55115636-55115660

TABLE 5-continued

Exemplary AAVS1 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1601-1774)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2601-2774)	Genomic Coordinates (hg38)
			AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	
G025903	697	UUCCUUAUCUG GUGACACACCC CC	UUCCUUAUCUG GUGACACACCC CCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mU*mU*mC*mCmUUAm UmCUmGGmUGACAmCA CCmCCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115643-55115667
G025904	698	GGCAGAUUCCU UAUCUGGUGAC AC	GGCAGAUUCCU UAUCUGGUGAC ACGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mG*mG*mC*mAmGAUm UmCCmUUmAUCUGmGU GAmCACmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115649-55115673
G025905	699	AGGCAGAUUCC UUAUCUGGUGA CA	AGGCAGAUUCC UUAUCUGGUGA CAGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mA*mG*mG*mCmAGAm UmUCmCUmUAUCUmGG UGmACAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115650-55115674
G025906	700	UAGGCAGAUUC CUUAUCUGGUG AC	UAGGCAGAUUC CUUAUCUGGUG ACGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mU*mA*mG*mGmCAGm AmUUmCCmUUAUCmUG GUmGACmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115651-55115675
G025907	701	UUAGGCAGAUU CCUUAUCUGGU GA	UUAGGCAGAUU CCUUAUCUGGU GAGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mU*mU*mA*mGmGCAm GmAUmUCmCUUAUmCU GGmUGAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115652-55115676
G025908	702	CUAACCCCCAC CUCCUGUUAGG CA	CUAACCCCCAC CUCCUGUUAGG CAGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC	mC*mU*mA*mAmCCCm CmCAmCCmUCCUGmUU AGmGCAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU	chr19: 55115669-55115693

TABLE 5-continued

Exemplary AAVS1 guide sequences.					
Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1601-1774)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2601-2774)	Genomic Coordinates (hg38)
			UUCUGGCAUCG UU	CUmGmCCmUmUmCmUG GCAUCG*mU*mU	
G025909	703	CUCCUGAUAUU GGGUCUAACCC CC	CUCCUGAUAUU GGGUCUAACCC CCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mU*mC*mCmUGAm UmAUmUGmGGUCUmAA CCmCCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115684-55115708
G025910	704	AGUCUCCUGAU AUUGGGUCUAA CC	AGUCUCCUGAU AUUGGGUCUAA CCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mA*mG*mU*mCmUCCm UmGAmUAmUUGGGmUC UAmACCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115687-55115711
G025911	705	CCUAGUCUCCU GAUAUUGGGUC UA	CCUAGUCUCCU GAUAUUGGGUC UAGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mC*mU*mAmGUCm UmCCmUGmAUAUUmGG GUmCUAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115690-55115714
G025912	706	UCCUAGUCUCC UGAUAUUGGGU CU	UCCUAGUCUCC UGAUAUUGGGU CUGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mU*mC*mC*mUmAGUm CmUCmCUmGAUAUmUG GGmUCUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115691-55115715
G025913	707	UUCCUAGUCUC CUGAUAUUGGG UC	UUCCUAGUCUC CUGAUAUUGGG UCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mU*mU*mC*mCmUAGm UmCUmCCmUGAUAmUU GGmGUCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115692-55115716
G025914	708	CUUCCUAGUCU CCUGAUAUUGG GU	CUUCCUAGUCU CCUGAUAUUGG GUGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mU*mU*mCmCUAm GmUCmUCmCUGAUmAU UGmGGUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115693-55115717

TABLE 5-continued

Exemplary AAVS1 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1601-1774)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2601-2774)	Genomic Coordinates (hg38)
G025915	709	CAUCCUUAGGC CUCCUCCUUCC UA	CAUCCUUAGGC CUCCUCCUUCC UAGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mA*mU*mCmCUUm AmGGmCCmUCCUCmCU UCmCUAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115710-55115734
G025916	710	AAAAGCCCCAU CCUUAGGCCUC CU	AAAAGCCCCAU CCUUAGGCCUC CUGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mA*mA*mA*mAmGCCm CmCAmUCmCUUAGmGC CUmCCUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115718-55115742
G025917	711	GUGACAGAAAA GCCCCAUCCUU AG	GUGACAGAAAA GCCCCAUCCUU AGGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mG*mU*mG*mAmCAGm AmAAmAGmCCCCAmUC CUmUAGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115725-55115749
G025918	712	UUGGUGACAGA AAAGCCCCAUC CU	UUGGUGACAGA AAAGCCCCAUC CUGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mU*mU*mG*mGmUGAm CmAGmAAmAAGCCmCC AUmCCUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115728-55115752
G025919	713	GACAGGAUUGG UGACAGAAAAG CC	GACAGGAUUGG UGACAGAAAAG CCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mG*mA*mC*mAmGGAm UmUGmGUmGACAGmAA AAmGCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115735-55115759
G025920	714	UAGGGACAGGA UUGGUGACAGA AA	UAGGGACAGGA UUGGUGACAGA AAGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mU*mA*mG*mGmGACm AmGGmAUmUGGUGmAC AGmAAAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115739-55115763
G025921	715	CUAGGGACAGG AUUGGUGACAG AA	CUAGGGACAGG AUUGGUGACAG AAGUUGUAGCU	mC*mU*mA*mGmGGAm CmAGmGAmUUGGUmGA CAmGAAmGUUGmUmAm	chr19: 55115740-55115764

TABLE 5-continued

Exemplary AAVS1 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1601-1774)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2601-2774)	Genomic Coordinates (hg38)
			CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	
G025922	716	ACUAGGGACAG GAUUGGUGACA GA	ACUAGGGACAG GAUUGGUGACA GAGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mA*mC*mU*mAmGGGm AmCAmGGmAUUGGmUG ACmAGAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115741-55115765
G025923	717	CUGGGUACUUU UAUCUGUCCCC UC	CUGGGUACUUU UAUCUGUCCCC UCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mU*mG*mGmGUAm CmUUmUUmAUCUGmUC CCmCUCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115782-55115806
G025924	718	UCUGGGUACUU UUAUCUGUCCC CU	UCUGGGUACUU UUAUCUGUCCC CUGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mU*mC*mU*mGmGGUm AmCUmUUmUAUCUmGU CCmCCUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115783-55115807
G025925	719	UUCUGGGUACU UUUAUCUGUCC CC	UUCUGGGUACU UUUAUCUGUCC CCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mU*mU*mC*mUmGGGm UmACmUUmUUAUCmUG UCmCCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115784-55115808
G025926	720	UGGUUCUGGGU ACUUUUAUCUG UC	UGGUUCUGGGU ACUUUUAUCUG UCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mU*mG*mG*mUmUCUm GmGGmUAmCUUUUmAU CUmGUCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115787-55115811
G025927	721	CUCUGGUUCUG GGUACUUUUAU CU	CUCUGGUUCUG GGUACUUUUAU CUGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA	mC*mU*mC*mUmGGUm UmCUmGGmGUACUmUU UAmUCUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU	chr19: 55115790-55115814

TABLE 5-continued

Exemplary AAVS1 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1601-1774)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2601-2774)	Genomic Coordinates (hg38)
			AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	
G025928	722	GCUCUGGUUCU GGGUACUUUUA UC	GCUCUGGUUCU GGGUACUUUUA UCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mG*mC*mU*mCmUGGm UmUCmUGmGGUACmUU UUmAUCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115791-55115815
G025929	723	GGCUCUGGUUC UGGGUACUUUU AU	GGCUCUGGUUC UGGGUACUUUU AUGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mG*mG*mC*mUmCUGm GmUUmCUmGGGUAmCU UUmUAUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115792-55115816
G025930	724	GAGCUGGGACC ACCUUAUAUUC CC	GAGCUGGGACC ACCUUAUAUUC CCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mG*mA*mG*mCmUGGm GmACmCAmCCUUAmUA UUmCCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115831-55115855
G025931	725	GUCCCCGAGCU GGGACCACCUU AU	GUCCCCGAGCU GGGACCACCUU AUGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mG*mU*mC*mCmCCGm AmGCmUGmGGACCmAC CUmUAUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115837-55115861
G025932	726	UGUCCCCGAGC UGGGACCACCU UA	UGUCCCCGAGC UGGGACCACCU UAGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mU*mG*mU*mCmCCCm GmAGmCUmGGGACmCA CCmUUAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115838-55115862
G025933	727	CCAGGGAUCCU GUGUCCCCGAG CU	CCAGGGAUCCU GUGUCCCCGAG CUGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC	mC*mC*mA*mGmGGAm UmCCmUGmUGUCCmCC GAmGCUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU	chr19: 55115850-55115874

TABLE 5-continued

Exemplary AAVS1 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1601-1774)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2601-2774)	Genomic Coordinates (hg38)
			UUCUGGCAUCG UU	CUmGmCCmUmUmCmUG GCAUCG*mU*mU	
G025934	728	GUUUGCUGCCU CCAGGGAUCCU GU	GUUUGCUGCCU CCAGGGAUCCU GUGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mG*mU*mU*mUmGCUm GmCCmUCmCAGGGmAU CCmUGUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115861-55115885
G025935	729	UGUUUGCUGCC UCCAGGGAUCC UG	UGUUUGCUGCC UCCAGGGAUCC UGGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mU*mG*mU*mUmUGCm UmGCmCUmCCAGGmGA UCmCUGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115862-55115886
G025936	730	AUGUUUGCUGC CUCCAGGGAUC CU	AUGUUUGCUGC CUCCAGGGAUC CUGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mA*mU*mG*mUmUUGm CmUGmCCmUCCAGmGG AUmCCUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115863-55115887
G025937	731	GGACAGCAUGU UUGCUGCCUCC AG	GGACAGCAUGU UUGCUGCCUCC AGGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mG*mG*mA*mCmAGCm AmUGmUUmUGCUGmCC UCmCAGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115870-55115894
G025938	732	CCACUUCAGGA CAGCAUGUUUG CU	CCACUUCAGGA CAGCAUGUUUG CUGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mC*mA*mCmUUCm AmGGmACmAGCAUmGU UUmGCUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115878-55115902
G025939	733	UGUCCACUUCA GGACAGCAUGU UU	UGUCCACUUCA GGACAGCAUGU UUGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mU*mG*mU*mCmCACm UmUCmAGmGACAGmCA UGmUUUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115881-55115905

TABLE 5-continued

Exemplary AAVS1 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1601-1774)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2601-2774)	Genomic Coordinates (hg38)
G025940	734	CUUCUUCCUCC AACCCGGGCCC CU	CUUCUUCCUCC AACCCGGGCCC CUGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mU*mU*mCmUUCm CmUCmCAmACCCGmGG CCmCCUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115906-55115930
G025941	735	UCAGCUAGUCU UCUUCCUCCAA CC	UCAGCUAGUCU UCUUCCUCCAA CCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mU*mC*mA*mGmCUAm GmUCmUUmCUUCCmUC CAmACCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115915-55115939
G025942	736	AGAGCUCAGCU AGUCUUCUUCC UC	AGAGCUCAGCU AGUCUUCUUCC UCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mA*mG*mA*mGmCUCm AmGCmUAmGUCUUmCU UCmCUCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115920-55115944
G025943	737	GAGAGCUCAGC UAGUCUUCUUC CU	GAGAGCUCAGC UAGUCUUCUUC CUGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mG*mA*mG*mAmGCUm CmAGmCUmAGUCUmUC UUmCCUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115921-55115945
G025944	738	GUCCGAGAGCU CAGCUAGUCUU CU	GUCCGAGAGCU CAGCUAGUCUU CUGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mG*mU*mC*mCmGAGm AmGCmUCmAGCUAmGU CUmUCUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115925-55115949
G025945	739	GCCCCCUGUCA UGGCAUCUUCC AG	GCCCCCUGUCA UGGCAUCUUCC AGGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mG*mC*mC*mCmCCUm GmUCmAUmGGCAUmCU UCmCAGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115951-55115975
G025946	740	CUCUUCCAGCC CCCUGUCAUGG CA	CUCUUCCAGCC CCCUGUCAUGG CAGUUGUAGCU CCCUGAAACCG	mC*mU*mC*mUmUCCm AmGCmCCmCCUGUmCA UGmGCAmGUUGmUmAm GmCUCCCmUmGmAmAm	chr19: 55115959-55115983

TABLE 5-continued

Exemplary AAVS1 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1601-1774)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2601-2774)	Genomic Coordinates (hg38)
			UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	
G025947	741	CUCUAGUCUGU GCUAGCUCUUC CA	CUCUAGUCUGU GCUAGCUCUUC CAGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mU*mC*mUmAGUm CmUGmUGmCUAGCmUC UUmCCAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115975-55115999
G025948	742	UCUCUAGUCUG UGCUAGCUCUU CC	UCUCUAGUCUG UGCUAGCUCUU CCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mU*mC*mU*mCmUAGm UmCUmGUmGCUAGmCU CUmUCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115976-55116000
G025949	743	CUCUCUAGUCU GUGCUAGCUCU UC	CUCUCUAGUCU GUGCUAGCUCU UCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mU*mC*mUmCUAm GmUCmUGmUGCUAmGC UCmUUCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115977-55116001
G025950	744	CCUCUCUAGUC UGUGCUAGCUC UU	CCUCUCUAGUC UGUGCUAGCUC UUGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mC*mU*mCmUCUm AmGUmCUmGUGCUmAG CUmCUUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115978-55116002
G025951	745	CUUACCUCUCU AGUCUGUGCUA GC	CUUACCUCUCU AGUCUGUGCUA GCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mU*mU*mAmCCUm CmUCmUAmGUCUGmUG CUmAGCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55115982-55116006
G025952	746	CAUUUGGGCAG CUCCCCUACCC CC	CAUUUGGGCAG CUCCCCUACCC CCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC	mC*mA*mU*mUmUGGm GmCAmGCmUCCCCmUA CCmCCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU	chr19: 55116006-55116030

TABLE 5-continued

Exemplary AAVS1 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1601-1774)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2601-2774)	Genomic Coordinates (hg38)
			AACGCUCUGCC UUCUGGCAUCG UU	GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	
G025953	747	CUCCUUUCAUU UGGGCAGCUCC CC	CUCCUUUCAUU UGGGCAGCUCC CCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mU*mC*mCmUUUm CmAUmUUmGGGCAmGC UCmCCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55116013-55116037
G025954	748	ACUCCUUUCAU UUGGGCAGCUC CC	ACUCCUUUCAU UUGGGCAGCUC CCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mA*mC*mU*mCmCUUm UmCAmUUmUGGGCmAG CUmCCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55116014-55116038
G025955	749	CACUCCUUUCA UUUGGGCAGCU CC	CACUCCUUUCA UUUGGGCAGCU CCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mA*mC*mUmCCUm UmUCmAUmUUGGGmCA GCmUCCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55116015-55116039
G025956	750	CUCUCACUCCU UUCAUUUGGGC AG	CUCUCACUCCU UUCAUUUGGGC AGGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mU*mC*mUmCACm UmCCmUUmUCAUUmUG GGmCAGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55116019-55116043
G025957	751	CCUCUCACUCC UUUCAUUUGGG CA	CCUCUCACUCC UUUCAUUUGGG CAGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mC*mU*mCmUCAm CmUCmCUmUUCAUmUU GGmGCAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55116020-55116044
G025958	752	ACCUCUCACUC CUUUCAUUUGG GC	ACCUCUCACUC CUUUCAUUUGG GCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mA*mC*mC*mUmCUCm AmCUmCCmUUUCAmUU UGmGGCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55116021-55116045

TABLE 5-continued

Exemplary AAVS1 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1601-1774)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2601-2774)	Genomic Coordinates (hg38)
G025959	753	CUCCUGUGGAUUCGGGUCACCUCU	CUCCUGUGGAUUCGGGUCACCUCUGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mU*mC*mCmUGUmGmGAmUUmCGGGUmCACCmUCUmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr19: 55116039-55116063
G025960	754	CACCCCGUUCUCCUGUGGAUUCGG	CACCCCGUUCUCCUGUGGAUUCGGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mA*mC*mCmCCGmUmUCmUCmCUGUGmGAUUmCGGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr19: 55116048-55116072
G025961	755	UGCUUUCUUUGCCUGGACACCCCG	UGCUUUCUUUGCCUGGACACCCCGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mU*mG*mC*mUmUUCmUmUUmGCmCUGGAmCACCmCCGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr19: 55116065-55116089
G025962	756	AUCCUCUUGCUUUCUUUGCCUGGA	AUCCUCUUGCUUUCUUUGCCUGGAGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mA*mU*mC*mCmUCUmUmGCmUUmUCUUUmGCCUmGGAmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr19: 55116072-55116096
G025963	757	CAUCCUCUUGCUUUCUUUGCCUGG	CAUCCUCUUGCUUUCUUUGCCUGGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mA*mU*mCmCUCmUmUGmCUmUUCUUmUGCCmUGGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr19: 55116073-55116097
G025964	758	CCAUCCUCUUGCUUUCUUUGCCUG	CCAUCCUCUUGCUUUCUUUGCCUGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mC*mA*mUmCCUmCmUUmGCmUUUCUmUUGCmCUGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr19: 55116074-55116098
G025965	759	CCACCUCUCCAUCCUCUUGCUUUC	CCACCUCUCCAUCCUCUUGCUUUCGUUGUAGCU	mC*mC*mA*mCmCUCmUmCCmAUmCCUCUmUGCUmUUCmGUUGmUmAm	chr19: 55116082-55116106

TABLE 5-continued

Exemplary AAVS1 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1601-1774)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2601-2774)	Genomic Coordinates (hg38)
			CCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	GmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	
G025966	760	GUCUCCCUGGCUUUAGCCACCUCU	GUCUCCCUGGCUUUAGCCACCUCUGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mG*mU*mC*mUmCCCmUmGGmCUmUUAGCmCACCmUCUmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr19: 55116098-55116122
G025967	76	CCCCGUCUCCCUGGCUUUAGCCAC	CCCCGUCUCCCUGGCUUUAGCCACGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mC*mC*mCmGUCmUmCCmCUmGGCUUmUAGCmCACmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr19: 55116102-55116126
G025968	762	AAGUACCCCGUCUCCCUGGCUUUA	AAGUACCCCGUCUCCCUGGCUUUAGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mA*mA*mG*mUmACCmCmCGmUCmUCCCUmGGCUmUUAmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr19: 55116107-55116131
G025969	763	CCAAAGUACCCCGUCUCCCUGGCU	CCAAAGUACCCCGUCUCCCUGGCUGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mC*mC*mA*mAmAGUmAmCCmCCmGUCUCmCCUGmGCUmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr19: 55116110-55116134
G025970	764	UCUGGACAACCCCAAAGUACCCCG	UCUGGACAACCCCAAAGUACCCCGGUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU	mU*mC*mU*mGmGACmAmACmCCmCAAAGmUACCmCCGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	chr19: 55116121-55116145
G025971	765	UUCUGGACAACCCCAAAGUACCCC	UUCUGGACAACCCCAAAGUACCCCGUUGUAGCUCCCUGAAACCGUUGCUACAAUA	mU*mU*mC*mUmGGAmCmAAmCCmCCAAAmGUACmCCCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmC	chr19: 55116122-55116146

TABLE 5-continued

Exemplary AAVS1 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1601-1774)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2601-2774)	Genomic Coordinates (hg38)
			AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	
G025972	766	CCGUUUUUCUG GACAACCCCAA AG	CCGUUUUUCUG GACAACCCCAA AGGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mC*mG*mUmUUUm UmCUmGGmACAACmCC CAmAAGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55116128-55116152
G025973	767	ACCGUUUUUCU GGACAACCCCA AA	ACCGUUUUUCU GGACAACCCCA AAGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mA*mC*mC*mGmUUUm UmUCmUGmGACAAmCC CCmAAAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55116129-55116153
G025974	768	CACCGUUUUUC UGGACAACCCC AA	CACCGUUUUUC UGGACAACCCC AAGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mA*mC*mCmGUUm UmUUmCUmGGACAmAC CCmCAAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55116130-55116154
G025975	769	UGCAUCAUCAC CGUUUUUCUGG AC	UGCAUCAUCAC CGUUUUUCUGG ACGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mU*mG*mC*mAmUCAm UmCAmCCmGUUUUmUC UGmGACmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55116138-55116162
G025976	770	CUGCAUCAUCA CCGUUUUUCUG GA	CUGCAUCAUCA CCGUUUUUCUG GAGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mU*mG*mCmAUCm AmUCmACmCGUUUmUU CUmGGAmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55116139-55116163
G025977	771	CCUGCAUCAUC ACCGUUUUUCU GG	CCUGCAUCAUC ACCGUUUUUCU GGGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC	mC*mC*mU*mGmCAUm CmAUmCAmCCGUUmUU UCmUGGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU	chr19: 55116140-55116164

TABLE 5-continued

Exemplary AAVS1 guide sequences.

Guide ID	SEQ ID NO to the Guide Sequence	Guide Sequence	Exemplary Guide RNA Full Sequence (SEQ ID NOs: 1601-1774)	Exemplary Guide RNA Modified Sequence (SEQ ID Nos: 2601-2774)	Genomic Coordinates (hg38)
			UUCUGGCAUCG UU	CUmGmCCmUmUmCmUG GCAUCG*mU*mU	
G025978	772	CUCCCCUUCUU GUAGGCCUGCA UC	CUCCCCUUCUU GUAGGCCUGCA UCGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mU*mC*mCmCCUm UmCUmUGmUAGGCmCU GCmAUCmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55116156-55116180
G025979	773	CUUGCGUCCCG CCUCCCCUUCU UG	CUUGCGUCCCG CCUCCCCUUCU UGGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mU*mU*mGmCGUm CmCCmGCmCUCCCmCU UCmUUGmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55116168-55116192
G025980	774	CUCCGACGGAU GUCUCCCUUGC GU	CUCCGACGGAU GUCUCCCUUGC GUGUUGUAGCU CCCUGAAACCG UUGCUACAAUA AGGCCGUCGAA AGAUGUGCCGC AACGCUCUGCC UUCUGGCAUCG UU	mC*mU*mC*mCmGACm GmGAmUGmUCUCCmCU UGmCGUmGUUGmUmAm GmCUCCCmUmGmAmAm AmCmCGUUmGmCUAmC AAU*AAGmGmCCmGmU mCmGmAmAmAmGmAmU GUGCmCGmCAAmCGCU CUmGmCCmUmUmCmUG GCAUCG*mU*mU	chr19: 55116185-55116209

[0601] The terms "mA," "mc," "mU," or "mG" may be used to denote a nucleotide that has been modified with 2'—O-Me.

[0602] In some embodiments, the AAVS1 guide RNA comprises a guide sequence selected from SEQ TD NOs: 601-774. In some embodiments, the AAVS1 guide RNA comprises a guide sequence that is at least 19, 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence selected from SEQ ID NOs: 601-774. In some embodiments, the AAVS1 guide RNA comprises a guide sequence that is at least 95%, 90%, or 85% identical to a sequence selected from SEQ ID NOs: 601-774. In some embodiments, the AAVS1 guide RNA comprises a guide sequence that is at least 95% identical to a sequence selected from SEQ ID NOs: 601-774.

[0603] In some embodiments, the AAVS1 guide RNA comprises a guide sequence that comprises at least 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 5. As used herein, at least 10 contiguous nucleotides±10 nucleotides of a genomic coordinate means, for example, at least 10 contiguous nucleotides within the genomic coordinates wherein the genomic coordinates include 10 nucleotides in the 5' direction and 10 nucleotides in the 3' direction from the ranges listed in Table 5. For example, an AAVS1 guide RNA may comprise 10 contiguous nucleotides within the genomic coordinates chosen from: chr19:55115218-55115242; chr19:55115477-55115501; chr19:55115504-55115528; chr19:55115513-55115537; chr19:55115514-55115538; chr19:55115517-55115541; chr19:55115518-55115542; chr19:55115549-55115573; chr19:55115574-55115598; chr19:55115606-55115630; chr19:55115933-55115957; chr19:55116026-55116050; chr19:55116045-55116069; chr19:55116084-55116108; chr19:55115276-55115300; chr19:55115509-55115533; chr19:55115579-55115603; chr19:55115863-55115887; chr19:55115906-55115930; or chr19:55116006-55116030; including the boundary nucleotides of these ranges. In some embodiments, the AAVS1 guide RNA comprises a guide sequence that is at least 19, 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence that comprises 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 5. In some embodiments, the AAVS1 guide RNA comprises a guide sequence that is at least 95%, 90%, or 85% identical to a sequence selected from a sequence that is 19, 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence that comprises 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 5.

[0604] In some embodiments, the AAVS1 guide RNA comprises a guide sequence that comprises at least 15 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 5. In some embodiments, the AAVS1 guide RNA comprises a guide sequence that comprises at least 24 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 5.

[0605] In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 601. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 602. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 603. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 604. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 605. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 606. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 607. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 608. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 609. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 610. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 611. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 612. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 613. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 614. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 615. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 616. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 617. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 618. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 619. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 620. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 621. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 622. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 623. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 624. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 625. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 626. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 627. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 628. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 629. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 630. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 631. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 632. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 633. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 634. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 635. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 636. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 637. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 638. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 639. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 640. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 641. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 642. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 643. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 644. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 645. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 646. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 647. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 648. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 649. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 650. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 651. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 652. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 653. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 654. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 655. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 656. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 657. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 658. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 659. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 660. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 661. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 662. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 663. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 664. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 665. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 666. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 667. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 668. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 669. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 670. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 671. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 672. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 673. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 674. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 675. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 676. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 677. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 678. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 679. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 680. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 681. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 682. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 683. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 684. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 685. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 686. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 687. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 688. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 689. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 690. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 691. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 692. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 693. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 694. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 695. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 696. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 697. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 698. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 699. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 700. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 701. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 702. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 703. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 704. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 705. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 706. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 707. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 708. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 709. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 710. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 711. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 712. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 713. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 714. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 715. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 716. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 717. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 718. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 719. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 720. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 721. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 722. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 723. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 724. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 725. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 726. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 727. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 728. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 729. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 730. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 731. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 732. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 733. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 734. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 735. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 736. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 737. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 738. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 739. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 740. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 741. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 742. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 743. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 744. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 745. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 746. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 747. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 748. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 749. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 750. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 751. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 752. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 753. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 754. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 755. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 756. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 757. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 758. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 759. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 760. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 761. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 762. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 763. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 764. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 765. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 766. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 767. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 768. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 769. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 770. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 771. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 772. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 773. In some embodiments, the AAVS1 guide RNA comprises SEQ ID NO: 774.

[0606] In some embodiments, the AAVS1 guide RNA comprises a guide sequence of any one of: SEQ ID NOs: 611, 620, 622, 626-629, 632-634, 656, 659-661, 673, 691-692, 730, 734, and 746.

[0607] In some embodiments, the AAVS1 guide RNA comprises a guide sequence of any one of: SEQ ID NOs: 611, 620, 622, 627-629, 636, 656, 659-661, and 673,

[0608] In some embodiments, the AAVS1 guide RNA comprises a guide sequence of any one of: SEQ ID NOs: 692, 709, 730, 734, 746, 748, 760-761, and 763.

[0609] Additional embodiments of AAVS1 guide RNAs are provided herein, including e.g., exemplary modifications to the guide RNA.

1. Genetic Modifications to AAVS1

[0610] In some embodiments, the methods and compositions disclosed herein genetically modify at least one nucleotide of the AAVS1 locus in a cell. Genetic modifications encompass the population of modifications that results from contact with a genomic editing system (e.g., the population of edits that result from Cas9 and an AAVS1 guide RNA, or the population of edits that result from BC22 and an AAVS1 guide RNA).

[0611] In some embodiments, the genetic modification comprises at least one nucleotide within the genomic coordinates chr19:55115151-55116209. In some embodiments, the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from any of the genomic coordinates listed in Table 5.

[0612] In some embodiments, the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from: chr19:55115218-55115242; chr19:55115477-55115501; chr19:55115504-55115528; chr19:55115513-55115537; chr19:55115514-55115538; chr19:55115517-55115541; chr19:55115518-55115542; chr19:55115549-55115573; chr19:55115574-55115598; chr19:55115606-55115630; chr19:55115933-55115957; chr19:55116026-55116050; chr19:55116045-55116069; chr19:55116084-55116108; chr19:55115276-55115300; chr19:55115509-55115533; chr19:55115579-55115603; chr19:55115863-55115887; chr19:55115906-55115930; and chr19:55116006-55116030. In some embodiments, the genetic modification comprises at least 5 contiguous nucleotides within the genomic coordinates chosen from: chr19:55115218-55115242; chr19:55115477-55115501; chr19:55115504-55115528; chr19:55115513-55115537; chr19:55115514-55115538; chr19:55115517-55115541; chr19:55115518-55115542; chr19:55115549-55115573; chr19:55115574-55115598; chr19:55115606-55115630; chr19:55115933-55115957; chr19:55116026-55116050; chr19:55116045-55116069; chr19:55116084-55116108; chr19:55115276-55115300; chr19:55115509-55115533; chr19:55115579-55115603; chr19:55115863-55115887; chr19:55115906-55115930; and chr19:55116006-55116030. In some embodiments, the genetic modification comprises at least 10 contiguous nucleotides within the genomic coordinates chosen from: chr19:55115218-55115242; chr19:55115477-55115501; chr19:55115504-55115528; chr19:55115513-55115537; chr19:55115514-55115538; chr19:55115517-55115541; chr19:55115518-55115542; chr19:55115549-55115573; chr19:55115574-55115598; chr19:55115606-55115630; chr19:55115933-55115957; chr19:55116026-55116050; chr19:55116045-55116069; chr19:55116084-55116108; chr19:55115276-55115300; chr19:55115509-55115533; chr19:55115579-55115603; chr19:55115863-55115887; chr19:55115906-55115930; and chr19:55116006-55116030.

[0613] In some embodiments, the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from: (a) chr19:55115518-55115542; chr19:55115517-55115541; chr19:55115504-55115528; chr19:55115514-55115538; chr19:55115477-55115501; chr19:55115276-55115300; chr19:55116026-55116050; chr19:55116084-55116108; chr19:55116045-55116069; chr19:55115933-55115957; chr19:55115218-55115242; and chr19:55115696-55115720; or (b) chr19:55115579-55115603; chr19:55116006-55116030; chr19:55115863-55115887; chr19:55116098-55116122; chr19:55115710-55115734; and chr19:55116014-55116038. In some embodiments, the genetic modification comprises at least 5 contiguous nucleotides within the genomic coordinates chosen from: (a) chr19:55115518-55115542; chr19:55115517-55115541; chr19:55115504-55115528; chr19:55115514-55115538; chr19:55115477-55115501; chr19:55115276-55115300; chr19:55116026-55116050; chr19:55116084-55116108; chr19:55116045-55116069; chr19:55115933-55115957; chr19:55115218-55115242; and chr19:55115696-55115720; or (b) chr19:55115579-55115603; chr19:55116006-55116030; chr19:55115863-55115887; chr19:55116098-55116122; chr19:55115710-55115734; and chr19:55116014-55116038. In some embodiments, the genetic modification comprises at least 10 contiguous nucleotides within the genomic coordinates chosen from: (a) chr19:55115518-55115542; chr19:55115517-55115541; chr19:55115504-55115528; chr19:55115514-55115538; chr19:55115477-55115501; chr19:55115276-55115300; chr19:55116026-55116050; chr19:55116084-55116108; chr19:55116045-55116069; chr19:55115933-55115957; chr19:55115218-55115242; and chr19:55115696-55115720; or (b) chr19:55115579-55115603; chr19:55116006-55116030; chr19:55115863-55115887; chr19:55116098-55116122; chr19:55115710-55115734; and chr19:55116014-55116038.

[0614] In some embodiments, the genetic modification comprises at least one indel, at least one C to T substitution, or at least one A to G substitution within the genomic coordinates chosen from: chr19:55115218-55115242; chr19:55115477-55115501; chr19:55115504-55115528; chr19:55115513-55115537; chr19:55115514-55115538; chr19:55115517-55115541; chr19:55115518-55115542; chr19:55115549-55115573; chr19:55115574-55115598; chr19:55115606-55115630; chr19:55115933-55115957; chr19:55116026-55116050; chr19:55116045-55116069; chr19:55116084-55116108; chr19:55115276-55115300; chr19:55115509-55115533; chr19:55115579-55115603; chr19:55115863-55115887; chr19:55115906-55115930; and chr19:55116006-55116030.

[0615] In some embodiments, the genetic modification comprises at least one indel, at least one C to T substitution, or at least one A to G substitution within the genomic coordinates chosen from: (a) chr19:55115518-55115542; chr19:55115517-55115541; chr19:55115504-55115528; chr19:55115514-55115538; chr19:55115477-55115501; chr19:55115276-55115300; chr19:55116026-55116050; chr19:55116084-55116108; chr19:55116045-55116069; chr19:55115933-55115957; chr19:55115218-55115242; and chr19:55115696-55115720; or (b) chr19:55115579-55115603; chr19:55116006-55116030; chr19:55115863-55115887; chr19:55116098-55116122; chr19:55115710-55115734; and chr19:55116014-55116038.

[0616] In some embodiments, the modification to AAVS1 comprises any one or more of an insertion, deletion, substitution, or deamination of at least one nucleotide in a target sequence. In some embodiments, the modification to AAVS1 comprises an insertion of 1, 2, 3, 4 or 5 or more nucleotides in a target sequence. In some embodiments, the modification to AAVS1 comprises a deletion of 1, 2, 3, 4 or 5 or more nucleotides in a target sequence. In other embodiments, the modification to AAVS1 comprises an insertion of 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20 or 25 or more nucleotides in a target sequence. In other embodiments, the modification to AAVS1 comprises a deletion of 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20 or 25 or more nucleotides in a target sequence. In some embodiments, the modification to AAVS1 comprises an indel, which is generally defined in the art as an insertion or deletion of less than 1000 base pairs (bp). In some embodiments, the modification to AAVS1 comprises an insertion of a donor nucleic acid in a target sequence. In some embodiments, the modification to AAVS1 is not transient.

[0617] In some embodiments, the methods and compositions disclosed herein modify the AAVS1 locus in a cell using an RNA-guided DNA binding agent (e.g., a Cas enzyme). In some embodiments, the RNA-guided DNA binding agent is Cas9. In some embodiments, the RNA-guided DNA binding agent cuts within the AAVS1 gene, wherein the AAVS1 guide RNA targets an AAVS1 genomic target sequence comprising at least 10 contiguous nucleotides chr19:55115151-55116209.

[0618] In some embodiments, the methods and compositions disclosed herein modify the AAVS1 locus in a cell, wherein the modification to AAVS1 comprises an insertion of an exogenous nucleic acid. In some embodiments, the exogenous nucleic acid is a protein-coding gene. The protein encoded by the exogenous nucleic acid may be expressed by the cell.

H. Exemplary Cell Types

[0619] In some embodiments, methods and compositions disclosed herein genetically modify a cell. In some embodiments, the cell is an allogeneic cell. In some embodiments the cell is a human cell. In some embodiments the genetically modified cell is referred to as an engineered cell. An engineered cell refers to a cell (or progeny of a cell) comprising an engineered genetic modification, e.g. that has been contacted with a genomic editing system and genetically modified by the genomic editing system. The terms "engineered cell" and "genetically modified cell" are used interchangeably throughout. The engineered cell may be any of the exemplary cell types disclosed herein.

[0620] In some embodiments, the cell is an immune cell. As used herein, "immune cell" refers to a cell of the immune system, including e.g., a lymphocyte (e.g., T cell, B cell, natural killer cell ("NK cell", and NKT cell, or iNKT cell)), monocyte, macrophage, mast cell, dendritic cell, or granulocyte (e.g., neutrophil, eosinophil, and basophil). In some embodiments, the cell is a primary immune cell. In some embodiments, the immune system cell may be selected from CD3+, CD4+ and CD8+ T cells, regulatory T cells (Tregs), B cells, NK cells, and dendritic cells (DC). In some embodiments, the immune cell is allogeneic.

[0621] In some embodiments, the cell is a lymphocyte. In some embodiments, the cell is an adaptive immune cell. In some embodiments, the cell is a T cell. In some embodiments, the cell is a B cell. In some embodiments, the cell is a NK cell. In some embodiments, the lymphocyte is allogeneic.

[0622] As used herein, a T cell can be defined as a cell that expresses a T cell receptor ("TCR" or "ap TCR" or "76 TCR"), however in some embodiments, the TCR of a T cell may be genetically modified to reduce its expression (e.g., by genetic modification to the TRAC or TRBC genes), therefore expression of the protein CD3 may be used as a marker to identify a T cell by standard flow cytometry methods. CD3 is a multi-subunit signaling complex that associates with the TCR. Thus, a T cell may be referred to as CD3+. In some embodiments, a T cell is a cell that expresses a CD3+ marker and either a CD4+ or CD8+ marker. In some embodiments, the T cell is allogeneic.

[0623] In some embodiments, the T cell expresses the glycoprotein CD8 and therefore is CD8+ by standard flow cytometry methods and may be referred to as a "cytotoxic" T cell. In some embodiments, the T cell expresses the glycoprotein CD4 and therefore is CD4+ by standard flow cytometry methods and may be referred to as a "helper" T cell. CD4+ T cells can differentiate into subsets and may be referred to as a Th1 cell, Th2 cell, Th9 cell, Th17 cell, Th22 cell, T regulatory ("Treg") cell, or T follicular helper cells ("Tfh"). Each CD4+ subset releases specific cytokines that can have either proinflammatory or anti-inflammatory functions, survival or protective functions. A T cell may be isolated from a subject by CD4+ or CD8+ selection methods.

[0624] In some embodiments, the T cell is a memory T cell. In the body, a memory T cell has encountered antigen. A memory T cell can be located in the secondary lymphoid organs (central memory T cells) or in recently infected tissue (effector memory T cells). A memory T cell may be a CD8+ T cell. A memory T cell may be a CD4+ T cell.

[0625] As used herein, a "central memory T cell" can be defined as an antigen-experienced T cell, and for example, may expresses CD62L and CD45RO. A central memory T cell may be detected as CD62L+ and CD45RO+ by Central memory T cells also express CCR7, therefore may be detected as CCR7+ by standard flow cytometry methods.

[0626] As used herein, an "early stem-cell memory T cell" (or "Tscm") can be defined as a T cell that expresses CD27 and CD45RA, and therefore is CD27+ and CD45RA+ by standard flow cytometry methods. A Tscm does not express the CD45 isoform CD45RO, therefore a Tscm will further be CD45RO− if stained for this isoform by standard flow cytometry methods. A CD45RO-CD27+ cell is therefore also an early stem-cell memory T cell. Tscm cells further express CD62L and CCR7, therefore may be detected as CD62L+ and CCR7+ by standard flow cytometry methods. Early stem-cell memory T cells have been shown to correlate with increased persistence and therapeutic efficacy of cell therapy products.

[0627] In some embodiments, the cell is a B cell. As used herein, a "B cell" can be defined as a cell that expresses CD19 or CD20, or B cell mature antigen ("BCMA"), and therefore a B cell is CD19+, or CD20+, or BCMA+ by standard flow cytometry methods. A B cell is further negative for CD3 and CD56 by standard flow cytometry methods. The B cell may be a plasma cell. The B cell may be a memory B cell. The B cell may be a naïve B cell. The B cell may be IgM+, or has a class-switched B cell receptor (e.g., IgG+, or IgA+). In some embodiments, the B cell is allogeneic.

[0628] In some embodiments, the cell is a mononuclear cell, such as from bone marrow or peripheral blood. In some embodiments, the cell is a peripheral blood mononuclear cell ("PBMC"). In some embodiments, the cell is a PBMC, e.g. a lymphocyte or monocyte. In some embodiments, the cell is a peripheral blood lymphocyte ("PBL"). In some embodiments, the mononuclear cell is allogeneic.

[0629] Cells used in ACT or tissue regenerative therapy are included, such as stem cells, progenitor cells, and primary cells. Stem cells, for example, include pluripotent stem cells (PSCs); induced pluripotent stem cells (iPSCs); embryonic stem cells (ESCs); mesenchymal stem cells (MSCs, e.g., isolated from bone marrow (BM), peripheral blood (PB), placenta, umbilical cord (UC) or adipose); hematopoietic stem cells (HSCs; e.g. isolated from BM or UC); neural stem cells (NSCs); tissue specific progenitor stem cells (TSPSCs); and limbal stem cells (LSCs). Progenitor and primary cells include mononuclear cells (MNCs, e.g., isolated from BM or PB); endothelial progenitor cells (EPCs, e.g. isolated from BM, PB, and UC); neural progenitor cells (NPCs); and tissue-specific primary cells or cells derived therefrom (TSCs) including chondrocytes, myocytes, and keratinocytes. Cells for organ or tissue transplantations such as islet cells, cardiomyocytes, thyroid cells, thymocytes, neuronal cells, skin cells, and retinal cells are also included.

[0630] In some embodiments, the cell is a human cell, such as a cell isolated from a human subject. In some embodiments, the cell is isolated from human donor PBMCs or leukopaks. In some embodiments, the cell is from a subject with a condition, disorder, or disease. In some embodiments, the cell is from a human donor with Epstein Barr Virus ("EBV").

[0631] In some embodiments, the methods are carried out ex vivo. As used herein, "ex vivo" refers to an in vitro method wherein the cell is capable of being transferred into a subject, e.g. as an ACT therapy. In some embodiments, an ex vivo method is an in vitro method involving an ACT therapy cell or cell population.

[0632] In some embodiments, the cell is from a cell line. In some embodiments, the cell line is derived from a human subject. In some embodiments, the cell line is a lymphoblastoid cell line ("LCL"). The cell may be cryopreserved and thawed. The cell may not have been previously cryopreserved.

[0633] In some embodiments, the cell is from a cell bank. In some embodiments, the cell is genetically modified and then transferred into a cell bank. In some embodiments the cell is removed from a subject, genetically modified ex vivo, and transferred into a cell bank. In some embodiments, a genetically modified population of cells is transferred into a cell bank. In some embodiments, a genetically modified population of immune cells is transferred into a cell bank. In some embodiments, a genetically modified population of immune cells comprising a first and second subpopulations, wherein the first and second sub-populations have at least one common genetic modification and at least one different genetic modification are transferred into a cell bank.

III. Details of the Genomic Editing Systems

[0634] In some embodiments, the genomic editing system is a CRISPR/Cas system, including e.g., a CRISPR guide RNA comprising a guide sequence and RNA-guided DNA binding agent, and described further herein. Further description of the CRISPR/Cas system methods and compositions for use therein are known in the art. See e.g., PCT/US2021/062922 filed Dec. 10, 2021, and U.S. Provisional Application No. 63/275,425 filed Nov. 3, 2021, the contents of each of which are hereby incorporated in their entireties.

A. CRISPR Guide RNA

[0635] Provided herein are guide sequences useful for modifying a target sequence, e.g., using a guide RNA comprising a disclosed guide sequence with an RNA-guided DNA binding agent (e.g., a CRISPR/Cas system). Guide sequences targeting the HLA-A, TRAC, TRBC, and CIITA are shown in Tables 1-5 (SEQ ID NOs: 2-80, 101-120, 201-265, 301, 302, 304-576, and 601-774), as are the genomic coordinates that these guide RNAs target.

[0636] In some embodiments, a gRNA provided herein comprises a guide region (guide sequence) and a conserved region comprising a repeat/anti-repeat region, a hairpin 1 region, and a hairpin 2 region, wherein one or more of the repeat/anti-repeat region, the hairpin 1 region, and the hairpin 2 region are shortened. In some embodiments, the gRNA is an *N. meningitidis* Cas9 (NmeCas9) gRNA.

[0637] In some embodiments, the guide RNA comprises a modified sgRNA. In some embodiments, the sgRNA comprises any one of the modification pattern of the modified sgRNA sequences provided in Tables 1-5, 6, 7, and 7A-7B. In some embodiments, the conserved region comprises any one of modified conserved region Nme guide RNA motifs in Table 7B, and wherein the conserved region is 3' of the guide region (guide sequence). In some embodiments, the conserved region comprises a modified sequence comprising any one of SEQ ID NOs: 1081-1089, and wherein the conserved region is 3' of the guide region (guide sequence). In some embodiments, the guide RNA comprises a nucleotide sequence selected from any one of SEQ ID NOs: 904-909, 911, and 995-997, where the N's represent collectively any guide sequence disclosed herein, including the guide sequences provided in Tables 1-5. In certain embodiments, the N's represent collectively a guide sequence that is at least 80%, 85%, 90%, 95%, or 100% identical to or complementary to any one of the guide sequences provided in Tables 1-5. In certain embodiments, the N's represent collectively any one of the guide sequences provided in Tables 1-5. In certain embodiments, when the N's represent collectively a guide sequence, within $(N)_{20-25}$, each N of the $(N)_{20-25}$ may be independently modified, e.g., modified with a 2'—OMe modification, optionally further with a PS modification, particularly at 1, 2, or 3 terminal nucleotides. In certain embodiments, the $(N)_{20-25}$ has the following sequence and modification pattern: mN*mN*mN*mNmNNNmNmNNmNNmNNNNNmNNN NmNNN.

[0638] An exemplary conserved region of an NmeCas9 single guide RNA (Nme sgRNA) is shown in Table 6 (SEQ ID NO: 900). The first row shows the numbering of the nucleotides; the second row shows an exemplary sequence; and the third (and fourth) rows show the regions. "Shortened" with respect to an sgRNA means that its conserved region lacks at least one nucleotide shown in Table 6, as discussed in detail below.

[0639] In some embodiments, the guide RNA is a Nme sgRNA comprising a conserved portion comprising a repeat/anti-repeat region, a hairpin 1 region, and a hairpin 2 region, wherein one or more of the repeat/anti-repeat region, the hairpin 1 region, and the hairpin 2 region are shortened. In some embodiments, the sgRNA described herein further comprises a guide region and a conserved region, wherein the conserved region comprises one or more of: (a) a shortened repeat/anti-repeat region, wherein the shortened repeat/anti-repeat region lacks 2-24 nucleotides, wherein (i) one or more of nucleotides 37-48 and 53-64 is deleted and optionally one or more of nucleotides 37-64 is substituted relative to SEQ ID NO: 900; and (ii) nucleotide 36 is linked to nucleotide 65 by at least 2 nucleotides; or (b) a shortened hairpin 1 region, wherein the shortened hairpin 1 lacks 2-10, optionally 2-8 nucleotides, wherein one or more of nucleotides 82-86 and 91-95 is deleted and optionally one or more of positions 82-96 is substituted relative to SEQ ID NO: 900; and nucleotide 81 is linked to nucleotide 96 by at least 4 nucleotides; or (c) a shortened hairpin 2 region, wherein the shortened hairpin 2 lacks 2-18, optionally 2-16 nucleotides, wherein (i) one or more of nucleotides 113-121 and 126-134 is deleted and optionally one or more of nucleotides 113-134 is substituted relative to SEQ ID NO: 900; and (ii) nucleotide 112 is linked to nucleotide 135 by at least 4 nucleotides; wherein one or both nucleotides 144-145 are optionally deleted relative to SEQ ID NO: 900; and wherein at least 10 nucleotides are modified nucleotides.

[0640] In some embodiment, the gRNA disclosed herein is a sgRNA.

[0641] In some embodiments, in the guide sequence, nucleotides 1-4 are modified nucleotides. In some embodiments, in the guide sequence, nucleotides 5, 8, 9, 11, 13, 18, and 22 are modified nucleotides. In some embodiments, in the guide sequence, nucleotides 1-5, 8, 9, 11, 13, 18, and 22 are modified nucleotides. In some embodiments, the modified nucleotides are 2'—O-methyl (2'—O-Me) modified nucleotides. In some embodiments, in the guide sequence, nucleotide 1 is linked to nucleotide 2 by a phosphorothioate (PS) linkage, nucleotide 2 is linked to nucleotide 3 by a PS linkage, and/or nucleotide 3 is linked to nucleotide 4 by a PS linkage.

[0642] In some embodiments, one or both nucleotides 144-145 are deleted relative to SEQ ID NO: 900.

[0643] In some embodiments, at least 10 nucleotides of the conserved region are modified nucleotides.

[0644] In some embodiments, a repeat/anti-repeat region of a gRNA is a shortened repeat/anti-repeat region lacking 2-24 nucleotides, e.g., any of the repeat/anti-repeat regions indicated in the numbered embodiments above or Tables 1-6 or described elsewhere herein, which may be combined with any of the shortened hairpin 1 region or hairpin 2 region described herein, including but not limited to combinations indicated in the numbered embodiments above and represented in the sequences of Tables 1-6 or described elsewhere herein. In some embodiments, one or more of positions 49-52, 87-90, or 122-125 is substituted relative to SEQ ID NO: 900. In some embodiments, all of positions 49-52, 87-90, or 122-125 are substituted relative to SEQ ID NO: 900. In some embodiments, the 3' tail provided in Tables 1-6 or described herein is deleted.

[0645] In some embodiments, the shortened repeat/anti-repeat region of the gRNA lacks 18 nucleotides. In some embodiments, the shortened repeat/anti-repeat region of the gRNA lacks 22 nucleotides.

[0646] In some embodiments, in the shortened repeat/anti-repeat region of the gRNA, nucleotide 36 is linked to nucleotide 65 by 6 nucleotides. In some embodiments, in the shortened repeat/anti-repeat region of the gRNA, nucleotide 36 is linked to nucleotide 65 by 7 nucleotides. In some embodiments, in the shortened repeat/anti-repeat region of the gRNA, nucleotide 36 is linked to nucleotide 65 by 8 nucleotides. In some embodiments, in the shortened repeat/anti-repeat region of the gRNA, nucleotide 36 is linked to nucleotide 65 by 9 nucleotides. In some embodiments, in the shortened repeat/anti-repeat region of the gRNA, nucleotide 36 is linked to nucleotide 65 by 10 nucleotides.

[0647] In some embodiments, in the shortened repeat/anti-repeat region of the gRNA, nucleotides 38-48 and 53-63 are deleted relative to SEQ ID NO: 900. In some embodiments, in the shortened repeat/anti-repeat region of the gRNA, nucleotides 38, 41-48, 53-60, and 63 are deleted relative to SEQ ID NO: 900.

[0648] In some embodiments, in the shortened repeat/anti-repeat region of the gRNA, nucleotide 36 is linked to nucleotide 65 by 6 nucleotides. In some embodiments, in the shortened repeat/anti-repeat region of the gRNA, nucleotides 38-48 and 53-63 are deleted relative to SEQ ID NO: 700, and nucleotide 36 is linked to nucleotide 65 by nucleotides 37, 49-52, and 64.

[0649] In some embodiments, in the shortened repeat/anti-repeat region of the gRNA, nucleotide 36 is linked to nucleotide 65 by 10 nucleotides. In some embodiments, in the shortened repeat/anti-repeat region of the gRNA, nucleotides 38, 41-48, 53-60, and 63 are deleted relative to SEQ ID NO: 900, and nucleotide 36 is linked to nucleotide 65 by nucleotides 37, 39, 40, 49-52, 61, 62, and 64.

[0650] In some embodiments, all of nucleotides 38-48 and nucleotides 53-63 of the upper stem of the shortened repeat/anti-repeat region are deleted relative to SEQ ID NO: 900.

[0651] In some embodiments, all of nucleotides 39-48 and nucleotides 53-62 of the upper stem of the shortened repeat/anti-repeat region are deleted relative to SEQ ID NO: 900, and nucleotides 38 and 63 is substituted.

[0652] In some embodiments, the shortened repeat/anti-repeat region has 14 modified nucleotides. In some embodiments, the shortened repeat/anti-repeat region has 15 modified nucleotides. In some embodiments, the shortened repeat/anti-repeat region has 16 modified nucleotides. In some embodiments, the shortened repeat/anti-repeat region has 17 modified nucleotides. In some embodiments, the shortened repeat/anti-repeat region has 18 modified nucleotides. In some embodiments, the shortened repeat/anti-repeat region has 19 modified nucleotides. In some embodiments, the shortened repeat/anti-repeat region has 20 modified nucleotides. In some embodiments, in the shortened repeat/anti-repeat region, nucleotides 25, 29, 30, 31, 32, 37, 49-52, 64, 65, 69, 70, and 73 are modified nucleotides. In some embodiments, the modified nucleotides are 2'—O-Me modified nucleotides.

[0653] In some embodiments, between the shortened repeat/anti-repeat region and the shortened hairpin 1 region, nucleotide 76 is linked to nucleotide 77 by a PS linkage.

[0654] In some embodiments, the shortened hairpin 1 region lacks 2 nucleotides. In some embodiments, the shortened hairpin 1 region lacks 21 nucleotides. In some embodiments, the shortened hairpin 1 region lacks 2 nucleotides, and nucleotides 86 and 91 are deleted relative to SEQ ID NO: 900. In some embodiments, the shortened hairpin 1 region lacks 2 nucleotides, and nucleotides 85 and 92 are deleted relative to SEQ ID NO: 900. In some embodiments, in the shortened hairpin 1 region, nucleotide 81 is linked to nucleotide 96 by 12 nucleotides. In some embodiments, in the shortened hairpin 1 region, nucleotide 81 is linked to nucleotide 96 by 12 nucleotides. In some embodiments, in the shortened hairpin 1 region, nucleotides 86 and 91 are deleted relative to SEQ ID NO: 900, and nucleotide 81 is linked to nucleotide 96 by nucleotides 82-85, 87-90, and 92-95. In some embodiments, in the shortened hairpin 1 region, nucleotides 85 and 92 are deleted relative to SEQ ID NO: 900, and nucleotide 81 is linked to nucleotide 96 by nucleotides 82-84, 86-91, and 93-95.

[0655] In some embodiments, the shortened hairpin 1 region has a duplex portion of 7 base paired nucleotides in length. In some embodiments, the shortened hairpin 1 region has a duplex portion of 8 base paired nucleotides in length.

[0656] In the stem of the shortened hairpin 1 region is seven base paired nucleotides in length. In some embodiments, nucleotides 85-86 and nucleotides 91-92 of the shortened hairpin 1 region are deleted.

[0657] In some embodiments, the shortened hairpin 1 region has 13 modified nucleotides. In some embodiments, in the shortened hairpin 1 region, nucleotides 80, 81, 83, 84, 85, 87-90, 92-94, and 99 are modified nucleotides. In some embodiments, the modified nucleotides are 2'—O-Me modified nucleotides.

[0658] In some embodiments, between the shortened hairpin 1 region and the shortened hairpin 2 region, nucleotide 101 is a modified nucleotide. In some embodiments, the modified nucleotide is a 2'—O-Me modified nucleotide.

[0659] In some embodiments, the shortened hairpin 2 lacks 18 nucleotides. In some embodiments, the shortened hairpin 2 has 24 nucleotides. In some embodiments, in the shortened hairpin 2 nucleotides 113-121 and 126-134 are deleted relative to SEQ ID NO: 900. In some embodiments, the shortened hairpin 2 lacks 18 nucleotides, and nucleotides 113-121 and 126-134 are deleted relative to SEQ ID NO: 900. In some embodiments, in the shortened hairpin 2 region, nucleotide 112 is linked to nucleotide 135 by 4 nucleotides. In some embodiments, in the shortened hairpin 2 region, nucleotides 113-121 and 126-134 are deleted relative to SEQ ID NO: 900 and nucleotide 112 is linked to nucleotide 135 by nucleotides 122-125. In some embodiments, in the shortened hairpin 2 region, nucleotides 112-120 and 127-135 are deleted relative to SEQ ID NO: 900. In some embodiments, the shortened hairpin 2 region lacks 18 nucleotides, and nucleotides 112-120 and 127-135 are deleted relative to SEQ ID NO: 900.

[0660] In some embodiments, the shortened repeat/anti-repeat region has a length of 28 nucleotides. In some embodiments, the shortened repeat/anti-repeat region has a length of 32 nucleotides.

[0661] In some embodiments, the upper stem of the shortened repeat/anti-repeat region comprises no more than one base pair. In some embodiments, the upper stem of the shortened repeat/anti-repeat region comprises no more than three base pairs.

[0662] In some embodiments, the shortened hairpin 2 region has 8 modified nucleotides. In some embodiments, the shortened hairpin 2 region has 9 modified nucleotides. In some embodiments, the shortened hairpin 2 region has 13 modified nucleotides. In some embodiments, in the shortened hairpin 2 region, nucleotides 104, 110, 111, 122-125, 142, and 143 are modified nucleotides. In some embodiments, in the shortened hairpin 2 region, nucleotides 104, 106-111, 122-125, 142, and 143 are modified nucleotides. In some embodiments, the modified nucleotides are 2'—O-Me modified nucleotides.

[0663] In some embodiments, in the shortened hairpin 2 region, nucleotide 141 is linked to nucleotide 142 by a PS linkage, and/or nucleotide 142 is linked to nucleotide 143 by a PS linkage.

[0664] In some embodiments, a guide RNA (gRNA) comprises a guide region and a conserved region, the conserved region comprising:

[0665] (a) a shortened repeat/anti-repeat region, wherein the shortened repeat/anti-repeat region lacks 18-22 nucleotides relative to SEQ ID NO: 900, wherein

[0666] (i) nucleotides 38-48 and 53-63 are deleted; and

[0667] (ii) nucleotide 36 is linked to nucleotide 65 by 6-10 nucleotides;

[0668] (b) a shortened hairpin 1 region, wherein the shortened hairpin 1 lacks 2 nucleotides, wherein nucleotides 86 and 91 are deleted or nucleotides 85 and 92 are deleted relative to SEQ ID NO: 900; and

[0669] (c) a shortened hairpin 2 region, wherein the shortened hairpin 2 lacks 18 nucleotides, wherein nucleotides 113-121 and 126-134 are deleted relative to SEQ ID NO: 900; and wherein nucleotides 144-145 are deleted relative to SEQ ID NO: 900; wherein at least 10 nucleotides are modified nucleotides.

[0670] In some embodiments, a guide RNA (gRNA) comprises a guide region and a conserved region, the conserved region comprising:

[0671] (a) a shortened repeat/anti-repeat region, wherein the shortened repeat/anti-repeat region lacks 18-22 nucleotides relative to SEQ ID NO: 900, wherein

[0672] (i) nucleotides 38, 41-48, 53-60, and 63 are deleted; and

[0673] (ii) nucleotide 36 is linked to nucleotide 65 by 6-10 nucleotides;

[0674] (b) a shortened hairpin 1 region, wherein the shortened hairpin 1 lacks 2 nucleotides, wherein nucleotides 86 and 91 are deleted or nucleotides 85 and 92 are deleted relative to SEQ ID NO: 700;

[0675] (c) a shortened hairpin 2 region, wherein the shortened hairpin 2 lacks 18 nucleotides, wherein nucleotides 113-121 and 126-134 are deleted relative to SEQ ID NO: 900; and wherein nucleotides 144-145 are deleted relative to SEQ ID NO: 900; wherein at least 10 nucleotides are modified nucleotides.

[0676] In some embodiments, a guide RNA (gRNA) is provided, the gRNA comprising a guide region and a conserved region, the conserved region comprising one or more of:

[0677] (a) a shortened repeat/anti-repeat region, wherein the shortened repeat/anti-repeat region lacks 18-22 nucleotides relative to SEQ ID NO: 900, wherein

[0678] (i) nucleotides 37-48 and 53-64 are deleted; and

[0679] (ii) nucleotide 36 is linked to nucleotide 65 by 6-10 nucleotides; or

[0680] (b) a shortened hairpin 1 region, wherein the shortened hairpin 1 lacks 2 nucleotides, wherein nucleotides 86 and 91 are deleted or nucleotides 85 and 92 are deleted relative to SEQ ID NO: 900; or

[0681] (c) a shortened hairpin 2 region, wherein the shortened hairpin 2 lacks 18 nucleotides, wherein nucleotides 113-121 and 126-134 are deleted relative to SEQ ID NO: 900; and

[0682] wherein nucleotides 144-145 are deleted relative to SEQ ID NO: 900;

[0683] wherein at least 10 nucleotides are modified nucleotides.

[0684] In further embodiments, the shortened repeat/anti-repeat region, wherein the shortened repeat/anti-repeat region lacks 22 nucleotides relative to SEQ ID NO: 900. In further embodiments, nucleotide 36 is linked to nucleotide 65 by a sequence comprising the nucleotide sequence UGAAAC. In further embodiments, the nucleotide 36 is linked to nucleotide 65 by 10 nucleotides. In further embodiments, the nucleotide 36 is linked to nucleotide 65 by a sequence comprising the nucleotide sequence UUCGAAAGAC (SEQ ID NO: 3011).

[0685] In some embodiments, the guide RNA (gRNA) of the previous embodiment comprising a guide region and a conserved region, the conserved region comprising:

[0686] (a) a shortened repeat/anti-repeat region, wherein the shortened repeat/anti-repeat region lacks 18-22 nucleotides, wherein

[0687] (i) nucleotides 37-48 and 53-64 are deleted relative to SEQ ID NO: 900; and

[0688] (ii) nucleotide 36 is linked to nucleotide 65 by 6-10 nucleotides;

[0689] (b) a shortened hairpin 1 region, wherein the shortened hairpin 1 lacks 2 nucleotides relative to SEQ ID NO: 900, wherein nucleotides 86 and 91 are deleted or nucleotides 85 and 92 are deleted;

[0690] (c) a shortened hairpin 2 region, wherein the shortened hairpin 2 lacks 18 nucleotides, wherein nucleotides 113-121 and 126-134 are deleted relative to SEQ ID NO: 900; and

[0691] (d) wherein nucleotides 144-145 are deleted relative to SEQ ID NO: 900; wherein at least 10 nucleotides are modified nucleotides.

[0692] In further embodiments, the shortened repeat/anti-repeat region, wherein the shortened repeat/anti-repeat region lacks 22 nucleotides relative to SEQ ID NO: 900. In further embodiments, nucleotide 36 is linked to nucleotide 65 by a sequence comprising the nucleotide sequence UGAAAC. In further embodiments, the nucleotide 36 is linked to nucleotide 65 by 10 nucleotides. In further embodiments, the nucleotide 36 is linked to nucleotide 65 by a sequence comprising the nucleotide sequence UUCGAAAGAC (SEQ ID NO: 3011).

[0693] In some embodiments, a guide RNA (gRNA) is provided, the gRNA comprising: a guide sequence comprising:

[0694] 2'—O-Me modified nucleotides at the first four nucleotides 1-4;

[0695] PS linkages between nucleotides 1-2, 2-3, and 3-4; and

[0696] 2'—O-Me modified nucleotides at nucleotides 5, 8, 9, 11, 13, 18, and 22 of the guide sequence;

a shortened repeat/anti-repeat region, wherein nucleotides 38-48 and 53-63 are deleted relative to SEQ ID NO: 900, comprising:

[0697] 2'—O-Me modified nucleotides at nucleotides 25, 29, 30, 31, 32, 37, 49-52, 64, 65, 69, 70, and 73;

a PS linkage between nucleotides 76-77 between the shortened repeat/anti-repeat region and the shortened hairpin 1 region;

a shortened hairpin 1 region, wherein nucleotides 86 and 91 are deleted relative to SEQ ID NO: 900, comprising:

[0698] 2'—O-Me modified nucleotides at nucleotides 80, 81, 83, 84, 85, 87-90, 92-94, and 99; 2'—O-Me modified nucleotide at nucleotide 101 between the shortened hairpin 1 region and the shortened hairpin 2 region;

a shortened hairpin 2 region, wherein nucleotides 112-120 and 127-135 are deleted relative to SEQ ID NO: 900, comprising:

[0699] 2'—O-Me modified nucleotides at nucleotides 104, 110, 111, 122-125, 142, and 143,

[0700] PS linkages between nucleotides 141-142 and 142-143,

wherein one or both nucleotides 144-145 are optionally deleted relative to SEQ ID NO: 900.

[0701] In some embodiments, a guide RNA (gRNA) is provided, the gRNA comprising: a guide sequence comprising:

[0702] 2'—O-Me modified nucleotides at the first four nucleotides 1-4;

[0703] PS linkages between nucleotides 1-2, 2-3, and 3-4; and

[0704] 2'—O-Me modified nucleotides at nucleotides 5, 8, 9, 11, 13, 18, and 22 of the guide sequence;

a shortened repeat/anti-repeat region, wherein nucleotides 38-48 and 53-63 are deleted relative to SEQ ID NO: 900, comprising:

[0705] 2'—O-Me modified nucleotides at nucleotides 25, 29, 30, 31, 32, 37, 49-52, 64, 65, 69, 70, and 73;

a shortened hairpin 1 region, wherein nucleotides 86 and 91 are deleted relative to SEQ ID NO: 900, comprising:

[0706] 2'—O-Me modified nucleotides at nucleotides 80, 81, 83, 84, 85, 87-90, 92-94, and 99;

2'—O-Me modified nucleotide at nucleotide 101 between the shortened hairpin 1 region and the shortened hairpin 2 region;

a shortened hairpin 2 region, wherein nucleotides 112-120 and 127-135 are deleted relative to SEQ ID NO: 900, comprising:

[0707] 2'—O-Me modified nucleotides at nucleotides 104, 110, 111, 122-125, 142, and 143,

[0708] PS linkages between nucleotides 141-142 and 142-143,

wherein one or both nucleotides 144-145 are optionally deleted relative to SEQ ID NO: 900.

[0709] In some embodiments, a guide RNA (gRNA) is provided, the gRNA comprising: a guide sequence comprising:

[0710] 2'—O-Me modified nucleotides at the first four nucleotides 1-4;

[0711] PS linkages between nucleotides 1-2, 2-3, and 3-4; and

[0712] 2'—O-Me modified nucleotides at nucleotides 5, 8, 9, 11, 13, 18, and 22 of the guide sequence;

a shortened repeat/anti-repeat region, wherein nucleotides 38-48 and 53-63 are deleted relative to SEQ ID NO: 900, comprising:

[0713] 2'—O-Me modified nucleotides at nucleotides 25, 29, 30, 31, 32, 37, 49-52, 64, 65, 69, 70, and 73;

a PS linkage between nucleotides 76-77 between the shortened repeat/anti-repeat region and the shortened hairpin 1 region;

a shortened hairpin 1 region, wherein nucleotides 86 and 91 are deleted relative to SEQ ID NO: 900, comprising:

[0714] 2'—O-Me modified nucleotides at nucleotides 80, 81, 83, 84, 85, 87-90, 92-94, and 99;

2'—O-Me modified nucleotide at nucleotide 101 between the shortened hairpin 1 region and the shortened hairpin 2 region;

a shortened hairpin 2 region, wherein nucleotides 112-120 and 127-135 are deleted relative to SEQ ID NO: 900, comprising:

[0715] 2'—O-Me modified nucleotides at nucleotides 104, 106-111, 122-125, 142, and 143,

[0716] PS linkages between nucleotides 141-142 and 142-143,

wherein one or both nucleotides 144-145 are optionally deleted relative to SEQ ID NO: 900.

[0717] In some embodiments, a guide RNA (gRNA) is provided, the gRNA comprising: a guide sequence comprising:

[0718] 2'—O-Me modified nucleotides at the first four nucleotides 1-4;

[0719] PS linkages between nucleotides 1-2, 2-3, and 3-4; and

[0720] 2'—O-Me modified nucleotides at nucleotides 5, 8, 9, 11, 13, 18, and 22 of the guide sequence;

a shortened repeat/anti-repeat region, wherein nucleotides 38-48 and 53-63 are deleted relative to SEQ ID NO: 900, comprising:

[0721] 2'—O-Me modified nucleotides at nucleotides 25, 29, 30, 31, 32, 37, 49-52, 64, 65, 69, 70, and 73;

a shortened hairpin 1 region, wherein nucleotides 86 and 91 are deleted relative to SEQ ID NO: 900, comprising:

[0722] 2'—O-Me modified nucleotides at nucleotides 80, 81, 83, 84, 85, 87-90, 92-94, and 99;

2'—O-Me modified nucleotide at nucleotide 101 between the shortened hairpin 1 region and the shortened hairpin 2 region;

a shortened hairpin 2 region, wherein nucleotides 112-120 and 127-135 are deleted relative to SEQ ID NO: 900, comprising:

[0723] 2'—O-Me modified nucleotides at nucleotides 104, 106-111, 122-125, 142, and 143,

[0724] PS linkages between nucleotides 141-142 and 142-143,

wherein one or both nucleotides 144-145 are optionally deleted relative to SEQ ID NO: 900.

[0725] In some embodiments, the NmeCas9 sgRNA comprises any one of the Nme Cas9 guide sequences disclosed herein and additional nucleotides to form a crRNA, e.g., with the following exemplary scaffold nucleotide sequence following the guide sequence at its 3' end:

(SEQ ID NO: 899)
GUUGUAGCUCCCUUUCUCAUUUCGGAAACGAAAUGAGAACCGUUGCUACA

AUAAGGCCGUCUGAAAAGAUGUGCCGCAACGCUCUGCCCCUUAAAGCUUC

UGCUUUAAGGGGCAUCGUUUA.

[0726] In some embodiments, the NmeCas9 sgRNA comprises any one of the guide sequences disclosed herein and additional nucleotides to form a crRNA with the following nucleotide sequence following the guide sequence at its 3' end:

(SEQ ID NO: 901)
$(N)_{20-25}$GUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAA
AAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU;

(SEQ ID NO: 902)
$(N)_{20-25}$GUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAA
AGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUUUAUU;

(SEQ ID NO: 903)
$(N)_{20-25}$GUUGUAGCUCCCUGGAAACCCGUUGCUACAAUAAGGCCGUCG
AAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUUUAUU

where A, C, G, U, and N are adenine, cytosine, guanine, uracil, and any ribonucleotide, respectively, unless otherwise indicated. In some embodiments, N equals 24. In some embodiments, N equals 25.

[0727] In some embodiments, the sgRNA comprises a conserved region comprising one of the following sequences in 5' to 3' orientation:

(SEQ ID NO: 906)
GUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmG
mCCmGmUmCmGmAmAmAmGmAmUGUGCmCGCmAmAmCmGCUCUmGmCCmU
mUmCmUGmGCmAmUC*mG*mU*mU;
or

(SEQ ID NO: 1082)
mGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGm
GmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUm
UmCmUGGCAUCG*mU*mU;

or any one of SEQ ID NOs: 1081-1089, where A, C, G, U, and N are adenine, cytosine, guanine, uracil, and any ribonucleotide, respectively, unless otherwise indicated. An m is indicative of a 2'O-methyl modification, and an * is indicative of a phosphorothioate linkage between the nucleotides.

[0728] In certain embodiments, the guide sequence is 20-25 nucleotides in length ((N)20-25), wherein each nucleotide may be independently modified. In certain embodiments, each of nucleotides 1-3 of the 5' end of the guide is independently modified. In certain embodiments, each of nucleotides 1-3 of the 5' end of the guide is independently modified with a 2'—OMe modification. In certain embodiments, each of nucleotides 1-3 of the 5' end of the guide is independently modified with a phosphorothioate linkage to the adjacent nucleotide residue. In certain embodiments, each of nucleotides 1-3 of the 5' end of the guide is independently modified with a 2'—OMe modification and a phosphorothioate linkage to the adjacent nucleotide residue.

[0729] In certain embodiments, sgRNA, such as an sgRNA comprising Exemplary NmeCas9 sgRNA, further includes a 3' tail, e.g., a 3' tail of 1, 2, 3, 4, or more nucleotides. In certain embodiments, the tail includes one or more modified nucleotides. In certain embodiments, the modified nucleotide is selected from a 2'—O-methyl (2'—OMe) modified nucleotide, a 2'—O-(2-methoxyethyl) (2'—O-moe) modified nucleotide, a 2'-fluoro (2'-F) modified nucleotide, a phosphorothioate (PS) linkage between nucleotides, and an inverted abasic modified nucleotide, or a combination thereof. In certain embodiments, the modified nucleotide includes a 2'—OMe modified nucleotide. In certain embodiments, the modified nucleotide includes a PS linkage between nucleotides. In certain embodiments, the modified nucleotide includes a 2'—OMe modified nucleotide and a PS linkage between nucleotides.

[0730] In some embodiments, the guide RNA is a chemically modified guide RNA. In some embodiments, the guide RNA is a chemically modified single guide RNA. The chemically modified guide RNAs may comprise one or more of the modifications as shown in Tables 1-5. The chemically modified guide RNAs may comprise one or more of modified nucleotides of any one of SEQ ID NOs: 904-909, 911, 995-997, and 1081-1089.

[0731] In some embodiments, the guide RNA is a sgRNA comprising the modification pattern shown in any one of SEQ ID NO: 904-909, 911, 995-997, and 1081-1089.

[0732] In some embodiments, the guide RNA comprises a sgRNA comprising the modification pattern shown in SEQ ID NO: 907. In some embodiments, the guide RNA comprises a sgRNA comprising the modified nucleotides of SEQ ID NO: 907, including a guide sequence disclosed herein. In some embodiments, the guide RNA is a sgRNA comprising a sequence of SEQ ID NO: 907 or a sequence that is at least 99%, 98%, 97%, 96%, 95%, 94%, 93%, 92%, 91%, or 90% identical to SEQ ID NO: 907. In some embodiments, the guide RNA comprises a sgRNA comprising the modification pattern shown in SEQ ID NO: 995. In some embodiments, the guide RNA comprises a sgRNA comprising the modified nucleotides of SEQ ID NO: 995, including a guide sequence disclosed herein. In some embodiments, the guide RNA is a sgRNA comprising a sequence of SEQ ID NO: 995 or a sequence that is at least 99%, 98%, 97%, 96%, 95%, 94%, 93%, 92%, 91%, or 90% identical to SEQ ID NO: 995. In some embodiments, the guide RNA comprises a sgRNA comprising the modification pattern shown in SEQ ID NO: 996. In some embodiments, the guide RNA comprises a sgRNA comprising the modified nucleotides of SEQ ID NO: 996, including a guide sequence disclosed herein. In some embodiments, the guide RNA is a sgRNA comprising a sequence of SEQ ID NO: 996 or a sequence that is at least 99%, 98%, 97%, 96%, 95%, 94%, 93%, 92%, 91%, or 90% identical to SEQ ID NO: 996. In some embodiments, the guide RNA comprises a sgRNA comprising the modification pattern shown in SEQ ID NO: 997. In some embodiments, the guide RNA comprises a sgRNA comprising the modified nucleotides of SEQ ID NO: 997, including a guide sequence disclosed herein. In some embodiments, the guide RNA is a sgRNA comprising a sequence of SEQ ID NO: 997 or a sequence that is at least 99%, 98%, 97%, 96%, 95%, 94%, 93%, 92%, 91%, or 90% identical to SEQ ID NO: 997.

[0733] In some embodiments, the guide RNA comprises a sgRNA comprising the modification pattern shown in SEQ ID NO: 1082. In some embodiments, the guide RNA comprises a sgRNA comprising the modified nucleotides of SEQ ID NO: 1082, including a guide sequence disclosed herein. In some embodiments, the guide RNA is a sgRNA comprising a sequence of SEQ ID NO: 1082 or a sequence that is at least 99%, 98%, 97%, 96%, 95%, 94%, 93%, 92%, 91%, or 90% identical to SEQ ID NO: 1082. In some embodiments, the guide RNA comprises a sgRNA comprising the modification pattern shown in SEQ ID NO: 1083. In some embodiments, the guide RNA comprises a sgRNA comprising the modified nucleotides of SEQ ID NO: 1083, including a guide sequence disclosed herein. In some embodiments, the guide RNA is a sgRNA comprising a sequence of SEQ ID NO: 1083 or a sequence that is at least 99%, 98%, 97%, 96%, 95%, 94%, 93%, 92%, 91%, or 90% identical to SEQ ID NO: 1083. In some embodiments, the guide RNA comprises a sgRNA comprising the modification pattern shown in SEQ ID NO: 1084. In some embodiments, the guide RNA comprises a sgRNA comprising the modified nucleotides of SEQ ID NO: 1084, including a guide sequence disclosed herein. In some embodiments, the guide RNA is a sgRNA comprising a sequence of SEQ ID NO: 1084 or a sequence that is at least 99%, 98%, 97%, 96%, 95%, 94%, 93%, 92%, 91%, or 90% identical to SEQ ID NO: 1084. In some embodiments, the guide RNA comprises a sgRNA comprising the modification pattern shown in SEQ ID NO: 1085. In some embodiments, the guide RNA comprises a sgRNA comprising the modified nucleotides of SEQ ID NO: 1085, including a guide sequence disclosed herein. In some embodiments, the guide RNA is a sgRNA comprising a sequence of SEQ ID NO: 1085 or a sequence that is at least 99%, 98%, 97%, 96%, 95%, 94%, 93%, 92%, 91%, or 90% identical to SEQ ID NO: 1085.

[0734] The guide RNA may further comprise a trRNA. In each composition and method embodiment described herein, the crRNA and trRNA may be associated as a single RNA (sgRNA) or may be on separate RNAs (dgRNA). In the context of sgRNAs, the crRNA and trRNA components may be covalently linked, e.g., via a phosphodiester bond or other covalent bond. In some embodiments, a crRNA or trRNA sequence may be referred to as a "scaffold" or "conserved portion" of a guide RNA.

[0735] In each of the compositions, use, and method embodiments described herein, the guide RNA may comprise two RNA molecules as a "dual guide RNA" or "dgRNA." The dgRNA comprises a first RNA molecule comprising a crRNA comprising, e.g., a guide sequence shown in Tables 1-5, and a second RNA molecule comprising a trRNA. The first and second RNA molecules may not be covalently linked, but may form an RNA duplex via the base pairing between portions of the crRNA and the trRNA.

[0736] In each of the composition, use, and method embodiments described herein, the guide RNA may comprise a single RNA molecule as a "single guide RNA" or "sgRNA". The sgRNA may comprise a crRNA (or a portion thereof) comprising a guide sequence shown in Table 1, covalently linked to a trRNA. The sgRNA may comprise 20, 21, 22, 23, or 24 contiguous nucleotides of a guide sequence shown in Tables 1-5. In some embodiments, the crRNA and the trRNA are covalently linked via a linker. In some embodiments, the sgRNA forms a stem-loop structure via the base pairing between portions of the crRNA and the trRNA. In some embodiments, the crRNA and the trRNA are covalently linked via one or more bonds that are not a phosphodiester bond.

[0737] In some embodiments, the trRNA may comprise all or a portion of a trRNA sequence derived from a naturally-occurring CRISPR/Cas system. In some embodiments, the trRNA comprises a truncated or modified wild type trRNA. The length of the trRNA depends on the CRISPR/Cas system used. In some embodiments, the trRNA comprises or consists of 55, 60, 65, 70, 75, 80, 85, 90, 100, or more than 100 nucleotides. In some embodiments, the trRNA may comprise certain secondary structures, such as, for example, one or more hairpin or stem-loop structures, or one or more bulge structures.

[0738] In some embodiments, a composition comprising one or more guide RNAs comprising a guide sequence of any one in Tables 1-5 is provided.

[0739] In one aspect, a composition comprising a guide RNA that comprises a guide sequence that is at least 90% or 95% identical to any of the nucleic acids in Tables 1-5 is provided.

[0740] In other embodiments, a composition is provided that comprises at least one, e.g., at least two gRNAs comprising guide sequences selected from any two or more of the guide sequences shown in Tables 1-5. In some embodiments, the composition comprises at least two gRNAs that each comprise a guide sequence at least 90% or 95% identical to any of the nucleic acids shown in Tables 1-5.

[0741] In some embodiments, the guide RNA compositions of the present invention are designed to recognize (e.g., hybridize to) a target sequence in HLA-A. For example, the HLA-A target sequence may be recognized and cleaved by a provided Cas cleavase comprising a guide RNA. In some embodiments, an RNA-guided DNA binding agent, such as a Cas cleavase, may be directed by a guide RNA to a target sequence in HLA-A, where the guide sequence of the guide RNA hybridizes with the target sequence and the RNA-guided DNA binding agent, such as a Cas cleavase, cleaves the target sequence.

[0742] In some embodiments, the guide RNA compositions of the present invention are designed to recognize (or hybridize to) a target sequence in TRAC. For example, the TRAC target sequence may be recognized and cleaved by a provided Cas cleavase comprising a guide RNA. In some embodiments, an RNA-guided DNA binding agent, such as a Cas cleavase, may be directed by a guide RNA to a target sequence in TRAC, where the guide sequence of the guide RNA hybridizes with the target sequence and the RNA-guided DNA binding agent, such as a Cas cleavase, cleaves the target sequence.

[0743] In some embodiments, the guide RNA compositions of the present invention are designed to recognize (e.g., hybridize to) a target sequence in TRBC1. For example, the TRBC1 target sequence may be recognized and cleaved by a provided Cas cleavase comprising a guide RNA. In some embodiments, an RNA-guided DNA binding agent, such as a Cas cleavase, may be directed by a guide RNA to a target sequence in TRBC1, where the guide sequence of the guide RNA hybridizes with the target sequence and the RNA-guided DNA binding agent, such as a Cas cleavase, cleaves the target sequence.

[0744] In some embodiments, the guide RNA compositions of the present invention are designed to recognize (e.g., hybridize to) a target sequence in TRBC2. For example, the TRBC2 target sequence may be recognized and cleaved by a provided Cas cleavase comprising a guide RNA. In some embodiments, an RNA-guided DNA binding agent, such as a Cas cleavase, may be directed by a guide RNA to a target sequence in TRBC2, where the guide sequence of the guide RNA hybridizes with the target sequence and the RNA-guided DNA binding agent, such as a Cas cleavase, cleaves the target sequence.

[0745] In some embodiments, the guide RNA compositions of the present invention are designed to recognize (e.g., hybridize to) a target sequence in CIITA. For example, the CIITA target sequence may be recognized and cleaved by a provided Cas cleavase comprising a guide RNA. In some embodiments, an RNA-guided DNA binding agent, such as a Cas cleavase, may be directed by a guide RNA to a target sequence in CIITA, where the guide sequence of the guide RNA hybridizes with the target sequence and the RNA-guided DNA binding agent, such as a Cas cleavase, cleaves the target sequence.

[0746] In some embodiments, the guide RNA compositions of the present invention are designed to recognize (e.g., hybridize to) a target sequence in AAVS1. For example, the AAVS1 target sequence may be recognized and cleaved by a provided Cas cleavase comprising a guide RNA. In some embodiments, an RNA-guided DNA binding agent, such as a Cas cleavase, may be directed by a guide RNA to a target sequence in AAVS1, where the guide sequence of the guide RNA hybridizes with the target sequence and the RNA-guided DNA binding agent, such as a Cas cleavase, cleaves the target sequence.

[0747] In some embodiments, the selection of the one or more guide RNAs is determined based on target sequences within HLA-A, TRAC, TRBC, CIITA, or AAVS1. In some embodiments, the compositions comprising one or more guide sequences comprise a guide sequence that is complementary to the corresponding genomic region shown in Tables 1-5, according to coordinates from human reference genome hg38. Guide sequences of further embodiments may be complementary to sequences in the close vicinity of the genomic coordinate listed in any of the Tables 1-5 within HLA-A, TRAC, TRBC, CIITA, or AAVS1. For example, guide sequences of further embodiments may be complementary to sequences that comprise 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Tables 1-5.

[0748] Without being bound by any particular theory, modifications (e.g., frameshift mutations resulting from indels occurring as a result of a nuclease-mediated DSB) in certain regions of the target gene may be less tolerable than mutations in other regions, thus the location of a DSB is an important factor in the amount or type of protein knockdown that may result. In some embodiments, a gRNA complementary or having complementarity to a target sequence within the target gene used to direct an RNA-guided DNA binding agent to a particular location in the target gene.

[0749] In some embodiments, the Nme guide sequence is at least 90% or 95% or 100% identical to the reverse complement of a target sequence present in an HLA-A, TRAC, TRBC, CIITA, or AAVS1 gene. In some embodiments, the target sequence may be complementary to the guide sequence of the guide RNA. In some embodiments, the degree of complementarity or identity between a guide sequence of an Nme guide RNA and its corresponding target sequence is at least 80%, 85%, 90% or 95%; or 100%. In some embodiments, the target sequence and the guide sequence of the gRNA may be 100% complementary or identical.

[0750] In some embodiments, the target sequence and the guide sequence of the Nme gRNA may contain at least one mismatch. For example, the target sequence and the guide sequence of the gRNA may contain 1 or 2, less preferably 3, or 4 mismatches, where the total length of the guide sequence is 24 nucleotides. In some embodiments, the target sequence and the guide sequence of the gRNA may contain 1-2 mismatches, where the guide sequence is 24 nucleotides.

[0751] In some embodiments, the Nme guide sequence comprises a sequence of at least 21, 22, 23 or 24 contiguous nucleotides of a sequence selected from SEQ ID NOs: 2-80, 101-120, 201-265, 301, 302, 304-576, and 601-774.

[0752] In some embodiments, a composition or formulation disclosed herein comprises an mRNA comprising an open reading frame (ORF) encoding an RNA-guided DNA binding agent, such as a Cas nuclease as described herein. In some embodiments, an mRNA comprising an ORF encoding an RNA-guided DNA binding agent, such as a Cas nuclease, is provided, used, or administered.

[0753] In some embodiments, the gRNA is chemically modified. A gRNA comprising one or more modified nucleosides or nucleotides is called a "modified" gRNA or "chemically modified" gRNA, to describe the presence of one or more non-naturally or naturally occurring components or configurations that are used instead of or in addition to the canonical A, G, C, and U residues. Modified nucleosides and nucleotides can include one or more of: (i) alteration, e.g., replacement, of one or both of the non-linking phosphate oxygens or of one or more of the linking phosphate oxygens in the phosphodiester backbone linkage (an exemplary backbone modification); (ii) alteration, e.g., replacement, of a constituent of the ribose sugar, e.g., of the 2' hydroxyl on the ribose sugar (an exemplary sugar modification); (iii) wholesale replacement of the phosphate moiety with "dephospho" linkers (an exemplary backbone modification); (iv) modification or replacement of a naturally occurring nucleobase, including with a non-canonical nucleobase (an exemplary base modification); (v) replacement or modification of the ribose-phosphate backbone (an exemplary backbone modification); (vi) modification of the 3' end or 5' end of the oligonucleotide, e.g., removal, modification or replacement of a terminal phosphate group or conjugation of a moiety, cap or linker (such 3' or 5' cap modifications may comprise a sugar or backbone modification); and (vii) modification or replacement of the sugar (an exemplary sugar modification).

[0754] Chemical modifications such as those listed above can be combined to provide modified gRNAs comprising nucleosides and nucleotides (collectively "residues") that can have two, three, four, or more modifications. For example, a modified residue can have a modified sugar and a modified nucleobase. In some embodiments, every base of a gRNA is modified, e.g., all bases have a modified phosphate group, such as a phosphorothioate group. In certain embodiments, all, or substantially all, of the phosphate groups of an gRNA molecule are replaced with phosphorothioate groups. In some embodiments, modified gRNAs comprise at least one modified residue at or near the 5' end of the RNA. In some embodiments, modified gRNAs comprise at least one modified residue at or near the 3' end of the RNA.

[0755] In some embodiments, the gRNA comprises one, two, three or more modified residues. In some embodiments, at least 5% (e.g., at least 5%, at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, or 100%) of the positions in a modified gRNA are modified nucleosides or nucleotides.

[0756] In some embodiments of a backbone modification, the phosphate group of a modified residue can be modified by replacing one or more of the oxygens with a different substituent. Further, the modified residue, e.g., modified residue present in a modified nucleic acid, can include the wholesale replacement of an unmodified phosphate moiety with a modified phosphate group as described herein. In some embodiments, the backbone modification of the phosphate backbone can include alterations that result in either an uncharged linker or a charged linker with unsymmetrical charge distribution.

[0757] Examples of modified phosphate groups include phosphorothioate, phosphoroselenates, borano phosphates, borano phosphate esters, hydrogen phosphonates, phosphoroamidates, alkyl or aryl phosphonates and phosphotriesters.

[0758] Scaffolds that can mimic nucleic acids can also be constructed wherein the phosphate linker and ribose sugar are replaced by nuclease resistant nucleoside or nucleotide surrogates. Such modifications may comprise backbone and sugar modifications. In some embodiments, the nucleobases can be tethered by a surrogate backbone. Examples can include, without limitation, the morpholino, cyclobutyl, pyrrolidine and peptide nucleic acid (PNA) nucleoside surrogates. The modified nucleosides and modified nucleotides can include one or more modifications to the sugar group, i.e. at sugar modification. For example, the 2' hydroxyl group (OH) can be modified, e.g. replaced with a number of different "oxy" or "deoxy" substituents. In some embodiments, modifications to the 2' hydroxyl group can enhance the stability of the nucleic acid since the hydroxyl can no longer be deprotonated to form a 2'-alkoxide ion. Examples of 2' hydroxyl group modifications can include alkoxy or aryloxy (OR, wherein "R" can be, e.g., alkyl, cycloalkyl, aryl, aralkyl, heteroaryl or a sugar); polyethyleneglycols (PEG), $O(CH_2CH_2O)_nCH_2CH_2OR$ wherein R can be, e.g., H or optionally substituted alkyl, and n can be an integer from 0 to 20. In some embodiments, the 2' hydroxyl group modification can be 2'—O-Me. In some embodiments, the 2' hydroxyl group modification can be a 2'-fluoro modification, which replaces the 2' hydroxyl group with a fluoride. In some embodiments, the 2' hydroxyl group modification can include "locked" nucleic acids (LNA) in which the 2' hydroxyl can be connected, e.g., by a C_{1-6} alkylene or C_{1-6} heteroalkylene bridge, to the 4' carbon of the same ribose sugar, where exemplary bridges can include methylene, propylene, ether, or amino bridges. In some embodiments, the 2' hydroxyl group modification can included "unlocked" nucleic acids (UNA) in which the ribose ring lacks the C2'—C3' bond. In some embodiments, the 2' hydroxyl group modification can include the methoxyethyl group (MOE), ($OCH_2CH_2OCH_3$, e.g., a PEG derivative). [00708]"Deoxy" 2' modifications can include hydrogen (i.e. deoxyribose sugars, e.g., at the overhang portions of partially dsRNA); halo (e.g., bromo, chloro, fluoro, or iodo); amino (wherein amino can be, e.g., NH_2; alkylamino, dialkylamino, heterocyclyl, arylamino, diarylamino, heteroarylamino, diheteroarylamino, or amino acid); $NH(CH_2CH_2NH)_nCH_2CH_2$— amino (wherein amino can be, e.g., as described herein), —NHC(O)R (wherein R can be, e.g., alkyl, cycloalkyl, aryl, aralkyl, heteroaryl or sugar), cyano; mercapto; alkyl-thio-alkyl; thioalkoxy; and alkyl, cycloalkyl, aryl, alkenyl and alkynyl, which may be optionally substituted with e.g., an amino as described herein.

[0759] The sugar modification can comprise a sugar group which may also contain one or more carbons that possess the opposite stereochemical configuration than that of the corresponding carbon in ribose. Thus, a modified nucleic acid can include nucleotides containing e.g., arabinose, as the sugar. The modified nucleic acids can also include abasic sugars. These abasic sugars can also be further modified at one or more of the constituent sugar atoms. The modified nucleic acids can also include one or more sugars that are in the L form, e.g. L- nucleosides.

[0760] The modified nucleosides and modified nucleotides described herein, which can be incorporated into a modified nucleic acid, can include a modified base, also called a nucleobase. Examples of nucleobases include, but are not limited to, adenine (A), guanine (G), cytosine (C), and uracil (U). These nucleobases can be modified or wholly replaced to provide modified residues that can be incorporated into modified nucleic acids. The nucleobase of the nucleotide can be independently selected from a purine, a pyrimidine, a purine analog, or pyrimidine analog. In some embodiments, the nucleobase can include, for example, naturally-occurring and synthetic derivatives of a base.

[0761] In embodiments employing a dual guide RNA, each of the crRNA and the tracr RNA can contain modifications. Such modifications may be at one or both ends of the crRNA or tracr RNA. In embodiments comprising an sgRNA, one or more residues at one or both ends of the sgRNA may be chemically modified, or the entire sgRNA may be chemically modified. Certain embodiments comprise a 5' end modification. Certain embodiments comprise a 3' end modification. In certain embodiments, one or more or all of the nucleotides in single stranded overhang of a gRNA molecule are deoxynucleotides.

[0762] In some embodiments, the gRNAs disclosed herein comprise one of the modification patterns disclosed in WO2018/107028 A1, published Jun. 14, 2018 the contents of which are hereby incorporated by reference in their entirety.

[0763] The terms "mA," "mC," "mU," or "mG" may be used to denote a nucleotide that has been modified with 2'—O-Me. The terms "fA," "fC," "fU," or "fG" may be used to denote a nucleotide that has been substituted with 2'-F. A "*" may be used to depict a PS modification. The terms A*, C*, U*, or G* may be used to denote a nucleotide that is linked to the next (e.g., 3') nucleotide with a PS bond. The terms "mA*," "mC*,", "mU*," or "mG*" may be used to denote a nucleotide that has been substituted with 2'—O-Me and that is linked to the next (e.g., 3') nucleotide with a PS bond.

[0764] In some embodiments, the gRNAs disclosed herein comprise one or more internal linkers. As used herein, "internal linker" describes a non-nucleotide segment joining two nucleotides within a guide RNA. If the gRNA contains a spacer region, the internal linker is located outside of the spacer region (e.g., in the scaffold or conserved region of the gRNA). The length of an internal linker may be dependent on, for example, the number of nucleotides replaced by the linker and the position of the linker in the gRNA. Exemplary linker-containing gRNAs are disclosed in WO 2022/261292 A1, published Dec. 15, 2022, the content of which is hereby incorporated by reference in its entirety.

[0765] gRNAs disclosed herein may comprise an internal linker. In general, any internal linker compatible with the function of the gRNA may be used. It may be desirable for the linker to have a degree of flexibility. In some embodiments, the internal linker comprises at least two, three, four, five, six, or more on-pathway single bonds. A bond is on-pathway if it is part of the shortest path of bonds between the two nucleotides whose 5' and 3' positions are connected to the linker.

[0766] As used herein the length of the internal linker can be defined by its bridging length. The "bridging length" of an internal linker as used herein refers to the distance or number of atoms in the shortest chain of atoms on the pathway from the first atom of the linker (bound to a 3' substituent, such as an oxygen or phosphate, of the preceding nucleotide to the last atom of the linker (bound to a 5' substituent, such as an oxygen or phosphate) of the following nucleotide) (e.g., from ~ to #in the structure of Formula (I) described below). Approximate predicted bridging lengths for various linkers are provided in the table below.

[0767] Exemplary predicted linker lengths by number of atoms, number of ethylene glycol units, approximate linker length in Angstroms on the assumption that an ethylene glycol monomer is about 3.7 Angstroms, and suitable location for substitution of at least the entire loop portion of a hairpin structure are provided in the Table 28 below. Substitution of two nucleotides requires a linker length of at least about 11 Angstroms. Substitution of at least 3 nucleotides requires a linker length of at least about 16 Angstroms.

TABLE 28

Number of atoms	Number of Ethylene Glycol units	Approximate length in Angstroms	Suitable location for complete loop substitution
3	1	3.7	Repeat-anti-repeat (for both loop and stem when no stem present)
6	2	7.4	Repeat-anti-repeat (for both loop and stem when no stem present)
9	3	11.1	Repeat-anti-repeat (for both loop and stem when no stem present), Nexus
12	4	14.8	Nexus
15	5	18.5	Repeat-anti-repeat, hairpin 1, hairpin 2
18	6	22.2	Repeat-anti-repeat, hairpin 1, hairpin 2
21	7	25.9	Repeat-anti-repeat, hairpin 1, hairpin 2
24	8	29.6	Repeat-anti-repeat, hairpin 1, hairpin 2
27	9	33.3	Repeat-anti-repeat, hairpin 1, hairpin 2
30	10	37	Repeat-anti-repeat, hairpin 1, hairpin 2

[0768] In some embodiments, the internal linker comprises a structure of formula (I):

$$\sim L0\text{-}L1\text{-}L2\text{-}\# \qquad (I)$$

[0769] wherein:

[0770] ~ indicates a bond to a 3' substituent of the preceding nucleotide;

[0771] #indicates a bond to a 5' substituent of the following nucleotide;

[0772] L0 is null or C_{1-3} aliphatic;

[0773] L1 is $-[E^1-(R^1)]_m-$, where

[0774] each R^1 is independently a C_{1-5} aliphatic group, optionally substituted with 1 or 2 E^2,

[0775] each E^1 and E^2 are independently a hydrogen bond acceptor, or are each independently chosen from cyclic hydrocarbons, and heterocyclic hydrocarbons, and

[0776] each m is 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10; and

[0777] L2 is null, C_{1-3} aliphatic, or is a hydrogen bond acceptor.

[0778] In some embodiments, L1 comprises one or more —CH_2CH_2O—, —CH_2OCH_2—, or —OCH_2CH_2— units ("ethylene glycol subunits"). In some embodiments, the number of —CH_2CH_2O—, —CH_2OCH_2—, or —OCH_2CH_2— units is in the range of 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10.

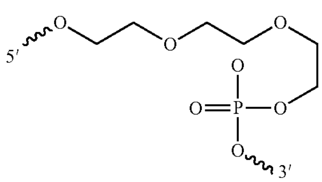
[0779] In some embodiments, m is 1, 2, 3, 4 or 5. In some embodiments, m is 1, 2, or 3. In some embodiments, m is 6, 7, 8, 9, or 10.

[0780] In some embodiments, L0 is null. In some embodiments, L0 is —CH_2— or —CH_2CH_2—.

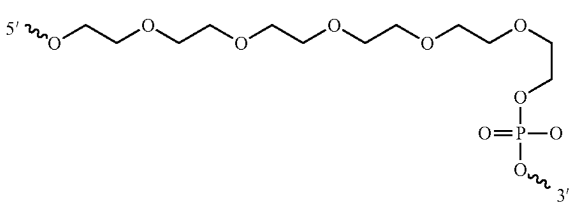
[0781] In some embodiments, L2 is null. In some embodiments, L2 is —O—, —S—, or C_{1-3} aliphatic. In some embodiments, L2 is —O—. In some embodiments, L2 is —S—. In some embodiments, L2 is —CH_2— or —CH_2CH_2—.

[0782] In the tables herein, L1 and L2, are optionally, C9 and C18, respectively as follows:

C-9 (~11 A)



C-18 (~23 A)



[0783] In certain embodiments, the internal linker has a bridging length of about 3-30 atoms, optionally 12-21 atoms, and the linker substitutes for at least 2 nucleotides of the gRNA. In certain embodiments, the internal linker has a bridging length of about 6-18 atoms, optionally about 6-12 atoms, and the linker substitutes for at least 2 nucleotides of the gRNA. In certain embodiments, the internal linker substitutes for 2-12 nucleotides.

[0784] In some embodiments, the guide RNA comprises a nucleic acid sequence of SEQ ID NO: 900, including modifications disclosed elsewhere herein. Table 29 shows various embodiments of the gRNA structures and species with possible number of internal linkers and positions.

TABLE 29

gRNA structures	Type	# internal linkers	Positions of internal linkers
Repeat/anti-R; Hp1; Hp2	Nme	3	All of repeat/anti-R; Hp1; Hp2
Repeat/anti-R; Hp1; Hp2	Nme	2	Any two of repeat/anti-R; Hp1; Hp2
Repeat/anti-R; Hp1; Hp2	Nme	1	Any one of repeat/anti-R; Hp1; Hp2

[0785] In certain embodiments, the internal linker is in a repeat-anti-repeat region of the gRNA. In certain embodiments, the internal linker substitutes for at least 4 nucleotides of the repeat-anti-repeat region of the gRNA. In certain embodiments, the internal linker substitutes for the loop in the repeat-anti-repeat region of an Nme Cas9 gRNA, corresponding to nucleotides 49-52 in SEQ ID NO: 900.

[0786] In certain embodiments, the internal linker is in a hairpin region of the gRNA. In certain embodiments, the internal linker substitutes for at least 4 nucleotides of the hairpin region of the gRNA. In certain embodiments, the internal linker substitutes for the loop in the hairpin 1 region of an Nme Cas9 gRNA, corresponding to nucleotides 87-90 in SEQ ID NO: 900. In certain embodiments, the internal linker substitutes for at least 4 nucleotides the loop in the hairpin 2 region of an Nme Cas9 gRNA, corresponding to nucleotides 122-125 in SEQ ID NO: 900. In certain embodiments, the internal linker substitutes for the loop in the hairpin 1 region of an Nme Cas9 gRNA, corresponding to nucleotides 87-90 in SEQ ID NO: 900 and for at least 4 nucleotides the loop in the hairpin 2 region of an Nme Cas9 gRNA, corresponding to nucleotides 122-125 in SEQ ID NO: 900.

TABLE 6

exemplary NmeCas9 sgRNA(SEQ ID NO: 900)

1-24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48
NNNNNNNNNNNNNNNNNNNNNNNN	G	U	U	G	U	A	G	C	U	C	C	C	U	U	U	C	U	C	A	U	U	U	C	G
					Lower stem																Upper stem			
Guide region													Repeat/Anti-Repeat region											

49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77
G	A	A	A	C	G	A	A	A	U	G	A	G	A	A	C	C	G	U	U	G	C	U	A	C	A	A	U	A
	Loop								Upper stem												Lower stem							
														Repeat/Anti-Repeat region														

78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108
A	G	G	C	C	G	U	C	U	G	A	A	A	A	G	A	U	G	U	G	C	C	G	C	A	A	C	G	C	U	C
				Stem						Loop								Stem (96: unpaired)							Lower stem					Bulge
												Hairpin 1																	Hairpin 2	

109	110	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130	131	132	133	134
U	G	C	C	C	C	U	U	A	A	A	G	C	U	U	C	U	G	C	U	U	U	A	A	G	G
						Upper Stem								Loop							Upper Stem				
													Hairpin 2												

135	136	137	138	139	140	141	142	143	144	145
G	G	C	A	U	C	G	U	U	U	A
	Upper Stem			Bulge		Lower Stem				
				Hairpin 2					Tail	

B. Ribonucleoprotein Complex

[0787] In some embodiments, the disclosure provides compositions comprising one or more gRNAs comprising one or more guide sequences from Tables 1-5 and an RNA-guided DNA binding agent, e.g., a nuclease, such as a Cas nuclease, such as Cas9. In some embodiments, the RNA-guided DNA-binding agent has cleavase activity, which can also be referred to as double-strand endonuclease activity. In some embodiments, the RNA-guided DNA-binding agent comprises a Cas nuclease. Examples of Cas9 nucleases include those of the type II CRISPR systems of *N. meningitidis* and other prokaryotes known in the art, and modified (e.g., engineered or mutant) versions thereof. In some embodiments, the RNA-guided DNA-binding agent comprises a Cas nickase. In some embodiments, the RNA-guided nickase is modified or derived from a Cas protein, such as a Class 2 Cas nuclease (which may be, e.g., a Cas nuclease of Type II). Class 2 Cas nuclease include, for example, Cas9 proteins and modifications thereof.

[0788] In some embodiments, the Cas nuclease is the Cas9 nuclease from *Neisseria meningitidis.*

[0789] In some embodiments, the Cas nickase is a nickase form of the Cas9 nuclease from *Neisseria meningitidis*. See e.g., WO/2020081568, describing an Nme2Cas9 D16A nickase fusion protein.

[0790] In some embodiments, the gRNA together with an RNA-guided DNA binding agent is called a ribonucleoprotein complex (RNP). In some embodiments, the RNA-guided DNA binding agent is a Cas nuclease. In some embodiments, the gRNA together with a Cas nuclease is called a Cas RNP. In some embodiments, the RNP comprises Type-II components. In some embodiments, the Cas nuclease is the Cas9 protein from the Type-II CRISPR/Cas system. In some embodiment, the gRNA together with Cas9 is called a Cas9 RNP.

[0791] Wild type Cas9 has two nuclease domains: RuvC and HNH. The RuvC domain cleaves the non-target DNA strand, and the HNH domain cleaves the target strand of DNA. In some embodiments, the Cas9 protein comprises more than one RuvC domain or more than one HNH domain. In some embodiments, the Cas9 protein is a wild type Cas9. In each of the composition, use, and method embodiments, the Cas induces a double strand break in target DNA.

[0792] In some embodiments, chimeric Cas nucleases are used, where one domain or region of the protein is replaced by a portion of a different protein. In some embodiments, a Cas nuclease domain may be replaced with a domain from a different nuclease such as FokI. In some embodiments, a Cas nuclease may be a modified nuclease.

[0793] In some embodiments, the RNA-guided DNA-binding agent has single-strand nickase activity, i.e., can cut one DNA strand to produce a single-strand break, also known as a "nick." In some embodiments, the RNA-guided DNA-binding agent comprises a Cas nickase. A nickase is an enzyme that creates a nick in dsDNA, i.e., cuts one strand but not the other of the DNA double helix. In some embodiments, a Cas nickase is a version of a Cas nuclease (e.g., a Cas nuclease discussed above) in which an endonucleolytic active site is inactivated, e.g., by one or more alterations (e.g., point mutations) in a catalytic domain. See e.g., U.S. Pat. No. 8,889,356 for discussion of Cas nickases and exemplary catalytic domain alterations. In some embodiments, a Cas nickase such as a Cas9 nickase has an inactivated RuvC or HNH domain.

[0794] In some embodiments, the RNA-guided DNA-binding agent is modified to contain only one functional nuclease domain. For example, the agent protein may be modified such that one of the nuclease domains is mutated or fully or partially deleted to reduce its nucleic acid cleavage activity. In some embodiments, a nickase is used having a RuvC domain with reduced activity. In some embodiments, a nickase is used having an inactive RuvC domain. In some embodiments, a nickase is used having an HNH domain with reduced activity. In some embodiments, a nickase is used having an inactive HNH domain.

[0795] In some embodiments, a conserved amino acid within a Cas protein nuclease domain is substituted to reduce or alter nuclease activity. In some embodiments, a Cas nuclease may comprise an amino acid substitution in the RuvC or RuvC-like nuclease domain. Exemplary amino acid substitutions in the HNH or HNH-like nuclease domain or RuvC or RuvC-like domains for *N. meningitidis* include Nme2Cas9D16A (HNH nickase) and Nme2Cas9H588A (RuvC nickase).

[0796] In some embodiments, an mRNA encoding a nickase is provided in combination with a pair of guide RNAs that are complementary to the sense and antisense strands of the target sequence, respectively. In this embodiment, the guide RNAs direct the nickase to a target sequence and introduce a DSB by generating a nick on opposite strands of the target sequence (i.e., double nicking). In some embodiments, use of double nicking may improve specificity and reduce off-target effects. In some embodiments, a nickase is used together with two separate guide RNAs targeting opposite strands of DNA to produce a double nick in the target DNA. In some embodiments, a nickase is used together with two separate guide RNAs that are selected to be in close proximity to produce a double nick in the target DNA.

[0797] In some embodiments, the RNA-guided DNA-binding agent lacks cleavase and nickase activity. In some embodiments, the RNA-guided DNA-binding agent comprises a dCas DNA-binding polypeptide. A dCas polypeptide has DNA-binding activity while essentially lacking catalytic (cleavase/nickase) activity. In some embodiments, the dCas polypeptide is a dCas9 polypeptide. In some embodiments, the RNA-guided DNA-binding agent lacking cleavase and nickase activity or the dCas DNA-binding polypeptide is a version of a Cas nuclease (e.g., a Cas nuclease discussed above) in which its endonucleolytic active sites are inactivated, e.g., by one or more alterations (e.g., point mutations) in its catalytic domains. See, e.g., US 2014/0186958 A1; US 2015/0166980 A1.

[0798] In some embodiments, the RNA-guided DNA binding agent comprises one or more heterologous functional domains (e.g., is or comprises a fusion polypeptide).

[0799] In some embodiments, the RNA-guided DNA binding agent comprises a APOBEC3 deaminase. In some embodiments, a APOBEC3 deaminase is a APOBEC3A (A3A). In some embodiments, the A3A is a human A3A. In some embodiments, the A3A is a wild-type A3A.

[0800] In some embodiments, the RNA-guided DNA binding agent comprises a deaminase and an RNA-guided nickase. In some embodiments, the mRNA further comprises a linker to link the sequencing encoding A3A to the sequence sequencing encoding RNA-guided nickase. In some embodiments, the linker is an organic molecule, group, polymer, or chemical moiety. In some embodiments, the linker is a peptide linker. In some embodiments, the peptide linker is any stretch of amino acids having at least 1, at least 2, at least 3, at least 4, at least 5, at least 6, at least 7, at least 8, at least 9, at least 10, at least 15, at least 20, at least 25, at least 30, at least 40, at least 50, or more amino acids. In some embodiments, the peptide linker is the 16 residue "XTEN" linker, or a variant thereof (See, e.g., the Examples; and Schellenberger et al. A recombinant polypeptide extends the in vivo half-life of peptides and proteins in a tunable manner. Nat. Biotechnol. 27, 1186-1190 (2009)).

[0801] In some embodiments, the peptide linker is the 16 residue "XTEN" linker, or a variant thereof (See, e.g., the Examples; and Schellenberger et al. A recombinant polypeptide extends the in vivo half-life of peptides and proteins in a tunable manner. Nat. Biotechnol. 27, 1186-1190 (2009)). In some embodiments, the XTEN linker comprises the sequence SGSETPGTSESATPES (SEQ ID NO: 930), SGSETPGTSESA (SEQ ID NO: 931), or SGSETPGTSESATPEGGSGGS (SEQ ID NO: 932).

[0802] In some embodiments, the peptide linker comprises a $(GGGGS)_n$(SEQ ID NO: 998), a $(G)_n$, an $(EAAAK)_n$(SEQ ID NO: 935), a $(GGS)_n$, an SGSETPGTSESATPES (SEQ ID NO: 930) motif (see, e.g., Guilinger J P, Thompson D B, Liu D R. Fusion of catalytically inactive Cas9 to FokI nuclease improves the specificity of genome modification. *Nat. Biotechnol.* 2014; 32(6): 577-82; the entire contents are incorporated herein by reference), or an $(XP)_n$ motif, or a combination of any of these, wherein n is independently an integer between 1 and 30 (SEQ ID NO: 3012). See, WO2015089406, e.g., paragraph [0012], the entire content of which is incorporated herein by reference.

[0803] In some embodiments, the peptide linker comprises one or more sequences selected from SEQ ID NOs: 930-994.

[0804] In some embodiments, the heterologous functional domain may facilitate transport of the RNA-guided DNA-binding agent into the nucleus of a cell. For example, the heterologous functional domain may be a nuclear localization signal (NLS). In some embodiments, the RNA-guided DNA-binding agent may be fused with 1-5 NLS(s). In some embodiments, the RNA-guided DNA-binding agent may be fused with 2, 3, or 4 NLS(s). In some embodiments, the RNA-guided DNA-binding agent may be fused with two NLS(s). In some embodiments, the RNA-guided DNA-binding agent may be fused with one NLS. Where one NLS is used, the NLS may be linked at the N-terminus or the C-terminus of the RNA-guided DNA-binding agent sequence. In some embodiments, the NLS is not linked to the C-terminus. It may also be inserted within the RNA-guided DNA binding agent sequence. In certain circumstances, at least the two NLSs are the same (e.g., two SV40 NLSs). In certain embodiments, at least two different NLSs are present the RNA-guided DNA binding agent. In some embodiments, the RNA-guided DNA-binding agent is fused to two SV40 NLS sequences linked at the carboxy terminus. In some embodiments, the RNA-guided DNA-binding agent may be fused with two NLSs, one linked at the N-terminus and one at the C-terminus. In some embodiments, the RNA-guided DNA-binding agent may be fused with 3 NLSs. In some embodiments, the RNA-guided DNA-binding agent may be fused with no NLS. In some embodiments, the NLS may be a monopartite sequence, such as, e.g., the SV40 NLS, PKKKRKV (SEQ ID NO: 3013) or PKKKRRV (SEQ ID NO: 923). In some embodiments, the NLS may be a bipartite sequence, such as the NLS of nucleoplasmin, KRPAATKKAGQAKKKK (SEQ ID NO: 924). In a specific embodiment, a single PKKKRKV (SEQ ID NO: 3013) NLS may be linked at the C-terminus of the RNA-guided DNA-binding agent. One or more linkers are optionally included at the fusion site. In some embodiments, the NLS comprises one or more sequences selected from SEQ ID NOs: 912-924.

[0805] In some embodiments, the RNA-guided DNA binding agent comprises an editor. An exemplary editor is BC22n which includes a *H. sapiens* APOBEC3A fused to Nme D16A Cas9 nickase by an XTEN linker, and mRNA encoding BC22n. An mRNA encoding NmeCas9 BC22n is provided (SEQ ID NO: 822).

[0806] In some embodiments, the heterologous functional domain may be capable of modifying the intracellular half-life of the RNA-guided DNA binding agent. In some embodiments, the half-life of the RNA-guided DNA binding agent may be increased. In some embodiments, the half-life of the RNA-guided DNA-binding agent may be reduced. In some embodiments, the heterologous functional domain may be capable of increasing the stability of the RNA-guided DNA-binding agent. In some embodiments, the heterologous functional domain may be capable of reducing the stability of the RNA-guided DNA-binding agent. In some embodiments, the heterologous functional domain may act as a signal peptide for protein degradation. In some embodiments, the protein degradation may be mediated by proteolytic enzymes, such as, for example, proteasomes, lysosomal proteases, or calpain proteases. In some embodiments, the heterologous functional domain may comprise a PEST sequence. In some embodiments, the RNA-guided DNA-binding agent may be modified by addition of ubiquitin or a polyubiquitin chain. In some embodiments, the ubiquitin may be a ubiquitin-like protein (UBL). Non-limiting examples of ubiquitin-like proteins include small ubiquitin-like modifier (SUMO), ubiquitin cross-reactive protein (UCRP, also known as interferon-stimulated gene-15 (ISG15)), ubiquitin-related modifier-1 (URM1), neuronal-precursor-cell-expressed developmentally downregulated protein-8 (NEDD8, also called Rubl in *S. cerevisiae*), human leukocyte antigen F-associated (FAT10), autophagy-8 (ATG8) and -12 (ATG12), Fau ubiquitin-like protein (FUB1), membrane-anchored UBL (MUB), ubiquitin fold-modifier-1 (UFM1), and ubiquitin-like protein-5 (UBL5).

[0807] In some embodiments, the heterologous functional domain may be a marker domain. Non-limiting examples of marker domains include fluorescent proteins, purification tags, epitope tags, and reporter gene sequences. In some embodiments, the marker domain may be a fluorescent protein. Non-limiting examples of suitable fluorescent proteins include green fluorescent proteins (e.g., GFP, GFP-2, tagGFP, turboGFP, sfGFP, EGFP, Emerald, Azami Green, Monomeric Azami Green, CopGFP, AceGFP, ZsGreen1), yellow fluorescent proteins (e.g., YFP, EYFP, Citrine, Venus, YPet, PhiYFP, ZsYellow1), blue fluorescent proteins (e.g., EBFP, EBFP2, Azurite, mKalamal, GFPuv, Sapphire, T-sapphire), cyan fluorescent proteins (e.g., ECFP, Cerulean, CyPet, AmCyani, Midoriishi-Cyan), red fluorescent proteins (e.g., mKate, mKate2, mPlum, DsRed monomer, mCherry, mRFP1, DsRed-Express, DsRed2, DsRed-Monomer, HcRed-Tandem, HcRed1, AsRed2, eqFP611, mRasberry, mStrawberry, Jred), and orange fluorescent proteins (mOrange, mKO, Kusabira-Orange, Monomeric Kusabira-Orange, mTangerine, tdTomato) or any other suitable fluorescent protein. In other embodiments, the marker domain may be a purification tag or an epitope tag. Non-limiting exemplary tags include glutathione-S-transferase (GST), chitin binding protein (CBP), maltose binding protein (MBP), thioredoxin (TRX), poly(NANP), tandem affinity purification (TAP) tag, myc, AcV5, AU1, AU5, E, ECS, E2, FLAG, HA, nus, Softag 1, Softag 3, Strep, SBP, Glu-Glu, HSV, KT3, S, S1, T7, V5, VSV-G, 6×His (SEQ ID NO: 3014), 8×His (SEQ ID NO: 3015), biotin carboxyl carrier protein (BCCP), poly-His, and calmodulin. Non-limiting exemplary reporter genes include glutathione-S-transferase (GST), horseradish peroxidase (HRP), chloramphenicol acetyltransferase (CAT), beta-galactosidase, beta-glucuronidase, luciferase, or fluorescent proteins.

[0808] In additional embodiments, the heterologous functional domain may target the RNA-guided DNA-binding agent to a specific organelle, cell type, tissue, or organ. In some embodiments, the heterologous functional domain may target the RNA-guided DNA-binding agent to mitochondria.

[0809] In further embodiments, the heterologous functional domain may be an effector domain such as an editor domain. When the RNA-guided DNA-binding agent is directed to its target sequence, e.g., when a Cas nuclease is directed to a target sequence by a gRNA, the effector such as an editor domain may modify or affect the target sequence. In some embodiments, the effector such as an editor domain may be chosen from a nucleic acid binding domain, a nuclease domain (e.g., a non-Cas nuclease domain), an epigenetic modification domain, a transcriptional activation domain, or a transcriptional repressor domain. In some embodiments, the heterologous functional domain is a nuclease, such as a FokI nuclease. See, e.g., U.S. Pat. No. 9,023,649. In some embodiments, the heterologous functional domain is a transcriptional activator or repressor. See, e.g., Qi et al., "Repurposing CRISPR as an RNA-guided platform for sequence-specific control of gene expression," Cell 152:1173-83 (2013); Perez-Pinera et al., "RNA-guided gene activation by CRISPR-Cas9-based transcription factors," *Nat. Methods* 10:973-6 (2013); Mali et al., "CAS9 transcriptional activators for target specificity screening and paired nickases for cooperative genome engineering," *Nat. Biotechnol.* 31:833-8 (2013); Gilbert et al., "CRISPR-mediated modular RNA-guided regulation of transcription in eukaryotes," Cell 154:442-51 (2013). As such, the RNA-guided DNA-binding agent essentially becomes a transcription factor that can be directed to bind a desired target sequence using a guide RNA.

C. Determination of Efficacy of Guide RNAs

[0810] In some embodiments, the efficacy of a guide RNA is determined when delivered or expressed together with other components (e.g., an RNA-guided DNA binding agent) forming an RNP. In some embodiments, the guide RNA is expressed together with an RNA-guided DNA binding agent, such as a Cas protein, e.g., Cas9. In some embodiments, the guide RNA is delivered to or expressed in a cell line that already stably expresses an RNA-guided DNA nuclease, such as a Cas nuclease or nickase, e.g., Cas9 nuclease or nickase. In some embodiments the guide RNA is delivered to a cell as part of a RNP. In some embodiments, the guide RNA is delivered to a cell along with a mRNA encoding an RNA-guided DNA nuclease, such as a Cas nuclease or nickase, e.g., Cas9 nuclease or nickase.

[0811] As described herein, use of an RNA-guided DNA nuclease and a guide RNA disclosed herein can lead to DSBs, SSBs, or site-specific binding that results in nucleic acid modification in the DNA or pre-mRNA which can produce errors in the form of insertion/deletion (indel) mutations upon repair by cellular machinery. Many mutations due to indels alter the reading frame, introduce premature stop codons, or induce exon skipping and, therefore, produce a non-functional protein.

[0812] In some embodiments, the efficacy of particular guide RNAs is determined based on in vitro models. In some embodiments, the in vitro model is T cell line. In some embodiments, the in vitro model is HEK293 T cells. In some embodiments, the in vitro model is HEK293 cells stably expressing Cas9 (HEK293_Cas9). In some embodiments, the in vitro model is a lymphoblastoid cell line. In some embodiments, the in vitro model is primary human T cells. In some embodiments, the in vitro model is primary human B cells. In some embodiments, the in vitro model is primary human peripheral blood lymphocytes. In some embodiments, the in vitro model is primary human peripheral blood mononuclear cells.

[0813] In some embodiments, the number of off-target sites at which a deletion or insertion occurs in an in vitro model is determined, e.g., by analyzing genomic DNA from the cells transfected in vitro with Cas9 mRNA and the guide RNA. In some embodiments, such a determination comprises analyzing genomic DNA from cells transfected in vitro with Cas9 mRNA, the guide RNA, and a donor oligonucleotide. Exemplary procedures for such determinations are provided in the working examples below.

[0814] In some embodiments, the efficacy of particular gRNAs is determined across multiple in vitro cell models for a guide RNA selection process. In some embodiments, a cell line comparison of data with selected guide RNAs is performed. In some embodiments, cross screening in multiple cell models is performed.

[0815] In some embodiments, the efficacy of particular guide RNAs is determined based on in vivo models. In some embodiments, the in vivo model is a rodent model. In some embodiments, the rodent model is a mouse which expresses the target gene. In some embodiments, the rodent model is a mouse which expresses an HLA-A gene. In some embodiments, the rodent model is a mouse which expresses a human HLA-A gene. In some embodiments, the rodent model is a mouse which expresses an HLA-B gene. In some embodiments, the rodent model is a mouse which expresses a human HLA-B gene. In some embodiments, the rodent model is a mouse which expresses a TRAC gene. In some embodiments, the rodent model is a mouse which expresses a human TRAC gene. In some embodiments, the rodent model is a mouse which expresses a TRBC1 gene. In some embodiments, the rodent model is a mouse which expresses a human TRBC1 gene. In some embodiments, the rodent model is a mouse which expresses a human TRBC2 gene. In some embodiments, the rodent model is a mouse which expresses a TRBC2 gene. In some embodiments, the rodent model is a mouse which expresses a CIITA gene. In some embodiments, the rodent model is a mouse which expresses a human CIITA gene. In some embodiments, the in vivo model is a non-human primate, for example cynomolgus monkey.

[0816] In some embodiments, the efficacy of a guide RNA is evaluated by on target cleavage efficiency. In some embodiments, the efficacy of a guide RNA is measured by percent editing at the target location, e.g., HLA-A, TRAC, TRBC, CIITA, or AAVS1. In some embodiments, deep sequencing may be utilized to identify the presence of modifications (e.g., insertions, deletions) introduced by genomic editing. Indel percentage can be calculated from next generation sequencing "NGS."

[0817] In some embodiments, the efficacy of a guide RNA is measured by the number or frequency of indels at off-target sequences within the genome of the target cell type. In some embodiments, efficacious guide RNAs are provided which produce indels at off target sites at very low frequencies (e.g., <5%) in a cell population or relative to the frequency of indel creation at the target site. Thus, the disclosure provides for guide RNAs which do not exhibit off-target indel formation in the target cell type (e.g., T cells or B cells), or which produce a frequency of off-target indel formation of <5% in a cell population or relative to the frequency of indel creation at the target site. In some embodiments, the disclosure provides guide RNAs which do not exhibit any off target indel formation in the target cell type (e.g., T cells or B cells). In some embodiments, guide RNAs are provided which produce indels at less than 5 off-target sites, e.g., as evaluated by one or more methods described herein. In some embodiments, guide RNAs are provided which produce indels at less than or equal to 4, 3, 2, or 1 off-target site(s) e.g., as evaluated by one or more methods described herein. In some embodiments, the off-target site(s) does not occur in a protein coding region in the target cell (e.g., T cells or B cells) genome.

[0818] In some embodiments, linear amplification is used to detect genomic editing events, such as the formation of insertion/deletion ("indel") mutations, translocations, and homology directed repair (HDR) events in target DNA. For example, linear amplification with a unique sequence-tagged primer and isolating the tagged amplification products (herein after referred to as "UnIT," or "Unique Identifier Tagmentation" method) may be used.

[0819] In some embodiments, the efficacy of a guide RNA is measured by the number of chromosomal rearrangements within the target cell type. Kromatid dGH assay may used to detect chromosomal rearrangements, including e.g., translocations, reciprocal translocations, translocations to off-target chromosomes, deletions (i.e., chromosomal rearrangements where fragments were lost during the cell replication cycle due to the editing event). In some embodiments, the target cell type has less than 10, less than 8, less than 5, less than 4, less than 3, less than 2, or less than 1 chromosomal rearrangement. In some embodiments, the target cell type has no chromosomal rearrangements.

D. Delivery of gRNA Compositions

[0820] Lipid nanoparticles (LNPs) are a well-known means for delivery of nucleotide and protein cargo and may be used for delivery of the guide RNAs, compositions, or pharmaceutical formulations disclosed herein. In some embodiments, the LNPs deliver nucleic acid, protein, or nucleic acid together with protein.

[0821] In some embodiments, the invention comprises a method for delivering any one of the gRNAs disclosed herein to a subject, wherein the gRNA is formulated as an LNP. In some embodiments, the LNP comprises the gRNA and a Cas9 or an mRNA encoding Cas9.

[0822] In some embodiments, the invention comprises a composition comprising any one of the gRNAs disclosed and an LNP. In some embodiments, the composition further comprises a Cas9 or an mRNA encoding Cas9.

[0823] In some embodiments, the LNPs comprise cationic lipids. In some embodiments, the LNPs comprise (9Z,12Z)-3-((4,4-bis(octyloxy)butanoyl)oxy)-2-((((3-(diethylamino)propoxy)carbonyl)oxy)methyl)propyl octadeca-9,12-dienoate, also called 3-((4,4-bis(octyloxy)butanoyl)oxy)-2-((((3-(diethylamino)propoxy)carbonyl)oxy)methyl)propyl (9Z,12Z)-octadeca-9,12-dienoate) (Lipid A) or another ionizable lipid. See, e.g., lipids of WO/2017/173054 and references described therein. In some embodiments, the LNPs comprise molar ratios of a cationic lipid amine to RNA phosphate (N:P) of about 4.5, 5.0, 5.5, 6.0, or 6.5. In some embodiments, the term cationic and ionizable in the context of LNP lipids is interchangeable, e.g., wherein ionizable lipids are cationic depending on the pH.

[0824] In some embodiments, the LNP comprises a lipid component, and the lipid component comprises: about 35 mol % Lipid A; about 15 mol % neutral lipid (e.g., distearoylphosphatidylcholine (DSPC)); about 47.5 mol % helper lipid (e.g., cholesterol); and about 2.5 mol % stealth lipid (e.g., 1,2-dimyristoyl-rac-glycero-3-methylpolyoxyethylene glycol 2000 (PEG2k-DMG)), and wherein the N/P ratio of the LNP composition is about 3-7.

[0825] In some embodiments, the gRNAs disclosed herein are formulated as LNPs for use in preparing a medicament for treating a disease or disorder.

[0826] Electroporation is a well-known means for delivery of cargo, and any electroporation methodology may be used for delivery of any one of the gRNAs disclosed herein. In some embodiments, electroporation may be used to deliver any one of the gRNAs disclosed herein and Cas9 or an mRNA encoding Cas9.

[0827] In some embodiments, the invention comprises a method for delivering any one of the gRNAs disclosed herein to an ex vivo cell, wherein the gRNA is formulated as an LNP or not formulated as an LNP. In some embodiments, the LNP comprises the gRNA and a Cas9 or an mRNA encoding Cas9.

[0828] In some embodiments, the guide RNA compositions described herein, alone or encoded on one or more vectors, are formulated in or administered via a lipid nanoparticle; see e.g., WO/2017/173054 and WO 2019/067992, the contents of which are hereby incorporated by reference in their entirety.

[0829] In certain embodiments, the invention comprises DNA or RNA vectors encoding any of the guide RNAs comprising any one or more of the guide sequences described herein. In some embodiments, in addition to guide RNA sequences, the vectors further comprise nucleic acids that do not encode guide RNAs. Nucleic acids that do not encode guide RNA include, but are not limited to, promoters, enhancers, regulatory sequences, and nucleic acids encoding an RNA-guided DNA nuclease, which can be a nuclease such as Cas9. In some embodiments, the vector comprises one or more nucleotide sequence(s) encoding a crRNA, a trRNA, or a crRNA and trRNA. In some embodiments, the vector comprises one or more nucleotide sequence(s) encoding a sgRNA and an mRNA encoding an RNA-guided DNA nuclease, which can be a Cas nuclease, such as Cas9. In some embodiments, the vector comprises one or more nucleotide sequence(s) encoding a crRNA, a trRNA, and an mRNA encoding an RNA-guided DNA nuclease, which can be a Cas protein, such as, Cas9. In one embodiment, the Cas9 is from *N. meningitidis* (i.e., Nine Cas9). In some embodiments, the nucleotide sequence encoding the crRNA, trRNA, or crRNA and trRNA (which may be a sgRNA) comprises or consists of a guide sequence flanked by all or a portion of a repeat sequence from a naturally-occurring CRISPR/Cas system. The nucleic acid comprising or consisting of the crRNA, trRNA, or crRNA and trRNA may further comprise a vector sequence wherein the vector sequence comprises or consists of nucleic acids that are not naturally found together with the crRNA, trRNA, or crRNA and trRNA.

IV. Therapeutic Methods and Uses

[0830] Any of the engineered cells and compositions described herein can be used in a method of treating a variety of diseases and disorders, as described herein. In some embodiments, the genetically modified cell (engineered cell) or population of genetically modified cells (engineered cells) and compositions may be used in methods of treating a variety of diseases and disorders. In some embodiments, a method of treating any one of the diseases or disorders described herein is encompassed, comprising administering any one or more composition described herein.

[0831] In some embodiments, the methods and compositions described herein may be used to treat diseases or disorders in need of delivery of a therapeutic agent. In some embodiments, the invention provides a method of providing an immunotherapy in a subject, the method including administering to the subject an effective amount of an engineered cell (or population of engineered cells) as described herein, for example, a cell of any of the aforementioned cell aspects and embodiments.

[0832] In some embodiments, the methods comprise administering to a subject a composition comprising an engineered cell described herein as an adoptive cell transfer therapy. In some embodiments, the engineered cell is an allogeneic cell.

[0833] In some embodiments of the methods, the method includes administering a lymphodepleting agent or immunosuppressant prior to administering to the subject an effective amount of the engineered cell (or engineered cells) as described herein, for example, a cell of any of the aforementioned cell aspects and embodiments. In another aspect, the invention provides a method of preparing engineered cells (e.g., a population of engineered cells).

[0834] Immunotherapy is the treatment of disease by activating or suppressing the immune system. Immunotherapies designed to elicit or amplify an immune response are classified as activation immunotherapies. Cell-based immunotherapies have been demonstrated to be effective in the treatment of some cancers. Immune effector cells such as lymphocytes, macrophages, dendritic cells, natural killer cells, cytotoxic T lymphocytes (CTLs), T helper cells, B cells, or their progenitors such as hematopoietic stem cells (HSC) or induced pluripotent stem cells (iPSC) can be programmed to act in response to abnormal antigens expressed on the surface of tumor cells. Thus, cancer immunotherapy allows components of the immune system to destroy tumors or other cancerous cells. Cell-based immunotherapies have also been demonstrated to be effective in the treatment of autoimmune diseases or transplant rejection. Immune effector cells such as regulatory T cells (Tregs) or mesenchymal stem cells can be programmed to act in response to autoantigens or transplant antigens expressed on the surface of normal tissues.

[0835] In some embodiments, the invention provides a method of preparing engineered cells (e.g., a population of engineered cells). The population of engineered cells may be used for immunotherapy.

[0836] In some embodiments, the invention provides a method of treating a subject in need thereof that includes administering engineered cells prepared by a method of preparing cells described herein, for example, a method of any of the aforementioned aspects and embodiments of methods of preparing cells.

[0837] In some embodiments, the engineered cells can be used to treat cancer, infectious diseases, inflammatory diseases, autoimmune diseases, cardiovascular diseases, neurological diseases, ophthalmologic diseases, renal diseases, liver diseases, musculoskeletal diseases, red blood cell diseases, or transplant rejections. In some embodiments, the engineered cells can be used in cell transplant, e.g., to the heart, liver, lung, kidney, pancreas, skin, or brain. (See e.g., Deuse et al., Nature Biotechnology 37:252-258 (2019).)

[0838] In some embodiments, the engineered cells can be used as a cell therapy comprising an allogeneic stem cell therapy. In some embodiments, the cell therapy comprises induced pluripotent stem cells (iPSCs). iPSCs may be induced to differentiate into other cell types including e.g., beta islet cells, neurons, and blood cells. In some embodiments, the cell therapy comprises hematopoietic stem cells. In some embodiments, the stem cells comprise mesenchymal stem cells that can develop into bone, cartilage, muscle, and fat cells. In some embodiments, the stem cells comprise ocular stem cells. In some embodiments, the allogeneic stem cell transplant comprises allogeneic bone marrow transplant. In some embodiments, the stem cells comprise pluripotent stem cells (PSCs). In some embodiments, the stem cells comprise induced embryonic stem cells (ESCs).

[0839] In some embodiments, the methods provide for administering the engineered cells to a subject, wherein the administration is an injection. In some embodiments, the methods provide for administering the engineered cells to a subject, wherein the administration is an intravascular injection or infusion. In some embodiments, the methods provide for administering the engineered cells to a subject, wherein the administration is a single dose.

[0840] In some embodiments, the methods provide for reducing a sign or symptom associated of a subject's disease treated with a composition disclosed herein. In some embodiments, the subject has a response to treatment with a composition disclosed herein that lasts more than one week. In some embodiments, the subject has a response to treatment with a composition disclosed herein that lasts more than two weeks. In some embodiments, the subject has a response to treatment with a composition disclosed herein that lasts more than three weeks. In some embodiments, the subject has a response to treatment with a composition disclosed herein that lasts more than one month.

[0841] In some embodiments, the methods provide for administering the engineered cells to a subject, and wherein the subject has a response to the administered cell that comprises a reduction in a sign or symptom associated with the disease treated by the cell therapy. In some embodiments, the subject has a response that lasts more than one week. In some embodiments, the subject has a response that lasts more than one month. In some embodiments, the subject has a response that lasts for at least 1-6 weeks.

TABLE7

V. Additional Sequences

Description	SEQ ID NO	Sequence
HLA-A *S. pyogenes* guide G000529	801	mG*mG*mC*CACGGAGCGAGACAUCUGUUUUAGAmGmCmUmAmGmAmAmAmUmAmGmCAAGUUAAAAUAAGGCUAGUCCGUUAUCAmAmCmUmUmGmAmAmAmAmAmGmUmGmGmCmAmCmCmGmAmGmUmCmGmGmUmGmCmU*mU*mU*mU
HLA-A control guide G018995	802	mA*mC*mA*GCGACGCCGCGAGCCAGGUUUUAGAmGmCmUmAmGmAmAmAmUmAmGmCAAGUUAAAAUAAGGCUAGUCCGUUAUCAmAmCmUmUmGmAmAmAmAmAmGmUmGmGmCmAmCmCmGmAmGmUmCmGmGmUmGmCmU*mU*mU*mU
TRAC control guide (G027891)	803	mU*mU*mC*AAAACCUGUCAGUGAUUGUUUUAGAmGmCmUmAmGmAmAmAmUmAmGmCAAGUUAAAAUAAGGUAGUCCGUUAUCACGAAAGGGCACCGAGUCGGmU*mG*mC*mU
TRAC control guide (G023516)	804, 3206, and 3207	MG*MC*MC*MGMUGUMAMCCMAGMCUGAGMAGACMUCUMGUUGMUMAMGMCUCCCMU(L1)MCMCGUUMGMCUAMCAAU*AAGMGMCCMGMUMC(L1)MGMAMUGUGCMCGMCAAMCGCUCUMGMCC(L1)GGCAUCGMU*MU
TRAC control guide (G021470)	805	MC*MC*MC*MAMCAGMAMUAMUCMCAGAAMCCCUMGACMGUUGMUMAMGMCUCCCMUMGMAMAMAMCMCGUUMGMCUAMCAAU*AAGMGMCCMGMUMCMGMAMAMAMGMAMUGUGCMCGMCAAMCGCUCUMGMCCMUMUMCMUGGCAUCG*MU*MU
TRAC control guide (G021471)	806	MU*MU*MG*MUMCCCMAMCAMGAMUAUCCMAGAAMCCCMGUUGMUMAMGMCUCCCMUMGMAMAMAMCMCGUUMGMCUAMCAAU*AAGMGMCCMGMUMCMGMAMAMAMGMAMUGUGCMCGMCAAMCGCUCUMGMCCMUMUMCMUGGCAUCG*MU*MU
TRAC control guide (G021472)	807	mC*mU*mU*mGmUCCmCmACmAGmAUAUCmCAGAmACCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU
TRAC control guide (G021473)	808	mA*mU*mC*mCmUCUmUmGUmCCmCACAGmAUAUmCCAmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU
TRAC control guide (G021474)	809	mG*mA*mU*mCmCUCmUmUGmUCmCCACAmGAUAmUCCmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU
TRAC control guide (G021475)	810	mA*mA*mC*mCmCUGmAmUCmCUmCUUGUmCCCAmCAGmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU
TRAC control guide (G021481)	811	mG*mC*mC*mGmUGUmAmCCmAGmCUGAGmAGACmUCUmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU
ORF encoding	812	ATGgaggccTcccccgccTccggcccccggcaccTgaTggacccccacaTcTTcaccTccAACTTCAACAACggcATCggccggCACAAGaccTACCTGTGCTACgaggTggagcggCTGGACAACggcaccTccgTgAAGATGGACCAGCACcggggcTTCCTGCACAACCAGgccAAGAACCTG

TABLE7-continued

V. Additional Sequences

Description	SEQ ID NO	Sequence
Sp. Cas9		CTGTGCggcTTCTACggccggCACgccgagCTGcggTTCCTGGACCTGgTgcccTccCTGCAGCTGGACcccgccCAGATCTACcgggTgacc TGGTTCATCTCCTGGTCCCCCTGCTTCTccTGGggcTGCgccggcgaggTgcgggccTTCCTGCAGgagAACaccCACgTgcggCTGcggATC TTCgccgcccggATCTACGACTACGACccccTGTACAAGgaggccCTGCAGATGCTGcggGACgccggcgcccCAGgTgTccATCATGaccTAC GACgagTTCAAGCACTGCTGGGACaccTTCgTgGACCACCAGggcTGCcccTTCCAGcccTGGGACggcCTGGACgagCACTccCAGgccCTG TccggccggCTGcgggccATCCTGCAGAACCAGggcAACTccggcTccgagaccccggcaccTccgagTccgccacccccgagTccgacaag aagTacTccaTcggccTggCcaTcggcaccaacTccgTgggcTgggccgTgaTcaccgacgagTacaaggTgcccTccaagaagTTcaaggTg cTgggcaacaccgaccggcacTccaTcaagaagaaccTgaTcggcgcccTgcTgTTcgacTccggcgagaccgccgaggccacccggcTgaag cggaccgcccggcggcggTacacccggcggaagaaccggaTcTgcTaccTgcaggagaTcTTcTccaacgagaTggccaaggTggacgacTcc TTCTTccaccggcTggaggagTccTTccTggTggaggaggacaagaagcacgagcggcaccccaTcTTcggcaacaTcgTggacgaggTggcc TaccacgagaagTaccccaccaTcTaccaccTgcggaagaagcTggTggacTccaccgacaaggccgaccTgcggcTgaTcTaccTggcccTg gcccacaTgaTcaagTTccggggccacTTccTgaTcgagggcgaccTgaaccccgacaacTccgacgTggacaagcTgTTcaTccagcTggTg cagaccTacaaccagcTgTTcgaggagaaccccaTcaacgccTccggcgTggacgccaaggccaTccTgTccgcccggcTgTccaagTcccgg cggcTggagaaccTgaTcgcccagcTgcccggcgagaagaagaacggccTgTTcggcaaccTgaTcgcccTgTcccTgggccTgacccccaac TTcaagTccaacTTcgaccTggccgaggacgccaagcTgcagcTgTccaaggacaccTacgacgacgaccTggacaaccTgcTggcccagaTc ggcgaccagTacgccgaccTgTTccTggccgccaagaaccTgTccgacgccaTccTgcTgTccgacaTccTgcgggTgaacaccgagaTcacc aaggccccccTgTccgccTccaTgaTcaagcggTacgacgagcaccaccaggaccTgacccTgcTgaaggcccTggTgcggcagcagcTgccc gagaagTacaaggagaTcTTcTTcgaccagTccaagaacggcTacgccggcTacaTcgacggcggcgccTcccaggaggagTTcTacaagTTc aTcaagcccaTccTggagaagaTggacggcaccgaggagcTgcTggTgaagcTgaaccgggaggaccTgcTgcggaagcagcggaccTTcgac aacggcTccaTccccccaccagaTccaccTgggcgagcTgcacgccaTccTgcggcggcaggaggacTTcTaccccTTccTgaaggacaaccgg gagaagaTcgagaagaTccTgaccTTccggaTccccTacTacgTgggcccccTggcccggggcaacTcccggTTcgccTggaTgacccggaag TccgaggagaccaTcaccccccTggaacTTcgaggaggTggTggacaagggcgccTccgcccagTccTTcaTcgagcggaTgaccaacTTcgac aagaaccTgcccaacgagaaggTgcTgcccaagcacTcccTgcTgTacgagTacTTcaccgTgTacaacgagcTgaccaaggTgaagTacgTg accgagggcaTgcggaagcccgccTTccTgTccggcgagcagaagaaggccaTcgTggaccTgcTgTTcaagaccaaccggaaggTgaccgTg aagcagcTgaaggaggacTacTTcaagaagaTcgagTgcTTcgacTccgTggagaTcTccggcgTggaggaccggTTcaacgccTcccTgggc accTaccacgaccTgcTgaagaTcaTcaaggacaaggacTTccTggacaacgaggagaacgaggacaTccTggaggacaTcgTgcTgacccTg acccTgTTcgaggaccgggagaTgaTcgaggagcggcTgaagaccTacgcccaccTgTTcgacgacaaggTgaTgaagcagcTgaagcggcgg cggTacaccggcTggggccggcTgTcccggaagcTgaTcaacggcaTccgggacaagcagTccggcaagaccaTccTggacTTccTgaagTcc gacggcTTcgccaaccggaacTTcaTgcagcTgaTccacgacgacTcccTgaccTTcaaggaggacaTccagaaggcccaggTgTccggccag ggcgacTcccTgcacgagcacaTcgccaaccTggccggcTcccccgccaTcaagaagggcaTccTgcagaccgTgaaggTggTggacgagcTg gTgaaggTgaTgggccggcacaagcccgagaacaTcgTgaTcgagaTggcccgggagaaccagaccacccagaagggccagaagaacTcccgg gagcggaTgaagcggaTcgaggagggcaTcaaggagcTgggcTcccagaTccTgaaggagcaccccgTggagaacacccagcTgcagaacgag aagcTgTaccTgTacTaccTgcagaacggccgggacaTgTacgTggaccaggagcTggacaTcaaccggcTgTccgacTacgacgTggaccac aTcgTgccccagTccTTccTgaaggacgacTccaTcgacaacaaggTgcTgacccggTccgacaagaaccggggcaagTccgacaacgTgccc TccgaggaggTggTgaagaagaTgaagaacTacTggcggcagcTgcTgaacgccaagcTgaTcacccagcggaagTTcgacaaccTgaccaag gccgagcggggcggccTgTccgagcTggacaaggccggcTTcaTcaagcggcagcTggTggagacccggcagaTcaccaagcacgTggcccag aTccTggacTcccggaTgaacaccaagTacgacgagaacgacaagcTgaTccgggaggTgaaggTgaTcacccTgaagTccaagcTggTgTcc gacTTccggaaggacTTccagTTcTacaaggTgcgggagaTcaacaacTaccaccacgcccacgacgccTaccTgaacgccgTggTgggcacc gcccTgaTcaagaagTaccccaagcTggagTccgagTTcgTgTacggcgacTacaaggTgTacgacgTgcggaagaTgaTcgccaagTccgag caggagaTcggcaaggccaccgccaagTacTTcTTcTacTccaacaTcaTgaacTTcTTcaagaccgagaTcacccTggccaacggcgagaTc cggaagcggccccTgaTcgagaccaacggcgagaccggcgagaTcgTgTgggacaagggccgggacTTcgccaccgTgcggaaggTgcTgTcc aTgccccaggTgaacaTcgTgaagaagaccgaggTgcagaccggcggcTTcTccaaggagTccaTccTgcccaagcggaacTccgacaagcTg aTcgcccggaagaaggacTgggaccccaagaagTacggcggcTTcgacTcccccaccgTggccTacTccgTgcTggTggTggccaaggTggag aagggcaagTccaagaagcTgaagTccgTgaaggagcTgcTgggcaTcaccaTcaTggagcggTccTccTTcgagaagaaccccaTcgacTTc cTggaggccaagggcTacaaggaggTgaagaaggaccTgaTcaTcaagcTgcccaagTacTcccTgTTcgagcTggagaacggccggaagcgg aTgcTggccTccgccggcgagcTgcagaagggcaacgagcTggcccTgcccTccaagTacgTgaacTTccTgTaccTggccTcccacTacgag aagcTgaagggcTcccccgaggacaacgagcagaagcagcTgTTcgTggagcagcacaagcacTaccTggacgagaTcaTcgagcagaTcTcc gagTTcTccaagcgggTgaTccTggccgacgccaaccTggacaaggTgcTgTccgccTacaacaagcaccgggacaagcccaTccgggagcag gccgagaacaTcaTccaccTgTTcacccTgaccaacCTgggcgcccccgccgccTTcaagTacTTcgacaccaccaTcgaccggaagcggTac

TABLE7-continued

V. Additional Sequences

Description	SEQ ID NO	Sequence
		accTccaccaaggaggTgcTggacgccacccTgaTccaccagTccaTcaccggccTgTacgagacccggaTcgaccTgTcccagcTgggcggc gacggcggcggcTcccccaagaagaagcggaaggTgTgA
ORF encoding Sp. Cas9	813	ATGGACAAGAAGTACAGCATCGGACTGGACATCGGAACAAACAGCGTCGGATGGGCAGTCATCACAGACGAATACAAGGTCCCGAGCAAGAAG TTCAAGGTCCTGGGAAACACAGACAGACACAGCATCAAGAAGAACCTGATCGGAGCACTGCTGTTCGACAGCGGAGAAACAGCAGAAGCAACA AGACTGAAGAGAACAGCAAGAAGAAGATACACAAGAAGAAAGAACAGAATCTGCTACCTGCAGGAAATCTTCAGCAACGAAATGGCAAAGGTC GACGACAGCTTCTTCCACAGACTGGAAGAAAGCTTCCTGGTCGAAGAAGACAAGAAGCACGAAAGACACCCGATCTTCGGAAACATCGTCGAC GAAGTCGCATACCACGAAAAGTACCCGACAATCTACCACCTGAGAAAGAAGCTGGTCGACAGCACAGACAAGGCAGACCTGAGACTGATCTAC CTGGCACTGGCACACATGATCAAGTTCAGAGGACACTTCCTGATCGAAGGAGACCTGAACCCGGACAACAGCGACGTCGACAAGCTGTTCATC CAGCTGGTCCAGACATACAACCAGCTGTTCGAAGAAAACCCGATCAACGCAAGCGGAGTCGACGCAAAGGCAATCCTGAGCGCAAGACTGAGC AAGAGCAGAAGACTGGAAAACCTGATCGCACAGCTGCCGGGAGAAAAGAAGAACGGACTGTTCGGAAACCTGATCGCACTGAGCCTGGGACTG ACACCGAACTTCAAGAGCAACTTCGACCTGGCAGAAGACGCAAAGCTGCAGCTGAGCAAGGACACATACGACGACGACCTGGACAACCTGCTG GCACAGATCGGAGACCAGTACGCAGACCTGTTCCTGGCAGCAAAGAACCTGAGCGACGCAATCCTGCTGAGCGACATCCTGAGAGTCAACACA GAAATCACAAAGGCACCGCTGAGCGCAAGCATGATCAAGAGATACGACGAACACCACCAGGACCTGACACTGCTGAAGGCACTGGTCAGACAG CAGCTGCCGGAAAAGTACAAGGAAATCTTCTTCGACCAGAGCAAGAACGGATACGCAGGATACATCGACGGAGGAGCAAGCCAGGAAGAATTC TACAAGTTCATCAAGCCGATCCTGGAAAAGATGGACGGAACAGAAGAACTGCTGGTCAAGCTGAACAGAGAAGACCTGCTGAGAAAGCAGAGA ACATTCGACAACGGAAGCATCCCGCACCAGATCCACCTGGGAGAACTGCACGCAATCCTGAGAAGACAGGAAGACTTCTACCCGTTCCTGAAG GACAACAGAGAAAAGATCGAAAAGATCCTGACATTCAGAATCCCGTACTACGTCGGACCGCTGGCAAGAGGAAACAGCAGATTCGCATGGATG ACAAGAAAGAGCGAAGAAACAATCACACCGTGGAACTTCGAAGAAGTCGTCGACAAGGGAGCAAGCGCACAGAGCTTCATCGAAAGAATGACA AACTTCGACAAGAACCTGCCGAACGAAAAGGTCCTGCCGAAGCACAGCCTGCTGTACGAATACTTCACAGTCTACAACGAACTGACAAAGGTC AAGTACGTCACAGAAGGAATGAGAAAGCCGGCATTCCTGAGCGGAGAACAGAAGAAGGCAATCGTCGACCTGCTGTTCAAGACAAACAGAAAG GTCACAGTCAAGCAGCTGAAGGAAGACTACTTCAAGAAGATCGAATGCTTCGACAGCGTCGAAATCAGCGGAGTCGAAGACAGATTCAACGCA AGCCTGGGAACATACCACGACCTGCTGAAGATCATCAAGGACAAGGACTTCCTGGACAACGAAGAAAACGAAGACATCCTGGAAGACATCGTC CTGACACTGACACTGTTCGAAGACAGAGAAATGATCGAAGAAAGACTGAAGACATACGCACACCTGTTCGACGACAAGGTCATGAAGCAGCTG AAGAGAAGAAGATACACAGGATGGGGAAGACTGAGCAGAAAGCTGATCAACGGAATCAGAGACAAGCAGAGCGGAAAGACAATCCTGGACTTC CTGAAGAGCGACGGATTCGCAAACAGAAACTTCATGCAGCTGATCCACGACGACAGCCTGACATTCAAGGAAGACATCCAGAAGGCACAGGTC AGCGGACAGGGAGACAGCCTGCACGAACACATCGCAAACCTGGCAGGAAGCCCGGCAATCAAGAAGGGAATCCTGCAGACAGTCAAGGTCGTC GACGAACTGGTCAAGGTCATGGGAAGACACAAGCCGGAAAACATCGTCATCGAAATGGCAAGAGAAAACCAGACAACACAGAAGGGACAGAAG AACAGCAGAGAAAGAATGAAGAGAATCGAAGAAGGAATCAAGGAACTGGGAAGCCAGATCCTGAAGGAACACCCGGTCGAAAACACACAGCTG CAGAACGAAAAGCTGTACCTGTACTACCTGCAGAACGGAAGAGACATGTACGTCGACCAGGAACTGGACATCAACAGACTGAGCGACTACGAC GTCGACCACATCGTCCCGCAGAGCTTCCTGAAGGACGACAGCATCGACAACAAGGTCCTGACAAGAAGCGACAAGAACAGAGGAAAGAGCGAC AACGTCCCGAGCGAAGAAGTCGTCAAGAAGATGAAGAACTACTGGAGACAGCTGCTGAACGCAAAGCTGATCACACAGAGAAAGTTCGACAAC CTGACAAAGGCAGAGAGAGGAGGACTGAGCGAACTGGACAAGGCAGGATTCATCAAGAGACAGCTGGTCGAAACAAGACAGATCACAAAGCAC GTCGCACAGATCCTGGACAGCAGAATGAACACAAAGTACGACGAAAACGACAAGCTGATCAGAGAAGTCAAGGTCATCACACTGAAGAGCAAG CTGGTCAGCGACTTCAGAAAGGACTTCCAGTTCTACAAGGTCAGAGAAATCAACAACTACCACCACGCACACGACGCATACCTGAACGCAGTC GTCGGAACAGCACTGATCAAGAAGTACCCGAAGCTGGAAAGCGAATTCGTCTACGGAGACTACAAGGTCTACGACGTCAGAAAGATGATCGCA AAGAGCGAACAGGAAATCGGAAAGGCAACAGCAAAGTACTTCTTCTACAGCAACATCATGAACTTCTTCAAGACAGAAATCACACTGGCAAAC GGAGAAATCAGAAAGAGACCGCTGATCGAAACAAACGGAGAAACAGGAGAAATCGTCTGGGACAAGGGAAGAGACTTCGCAACAGTCAGAAAG GTCCTGAGCATGCCGCAGGTCAACATCGTCAAGAAGACAGAAGTCCAGACAGGAGGATTCAGCAAGGAAAGCATCCTGCCGAAGAGAAACAGC GACAAGCTGATCGCAAGAAAGAAGGACTGGGACCCGAAGAAGTACGGAGGATTCGACAGCCCGACAGTCGCATACAGCGTCCTGGTCGTCGCA AAGGTCGAAAAGGGAAAGAGCAAGAAGCTGAAGAGCGTCAAGGAACTGCTGGGAATCACAATCATGGAAAGAAGCAGCTTCGAAAAGAACCCG ATCGACTTCCTGGAAGCAAAGGGATACAAGGAAGTCAAGAAGGACCTGATCATCAAGCTGCCGAAGTACAGCCTGTTCGAACTGGAAAACGGA AGAAAGAGAATGCTGGCAAGCGCAGGAGAACTGCAGAAGGGAAACGAACTGGCACTGCCGAGCAAGTACGTCAACTTCCTGTACCTGGCAAGC CACTACGAAAAGCTGAAGGGAAGCCCGGAAGACAACGAACAGAAGCAGCTGTTCGTCGAACAGCACAAGCACTACCTGGACGAAATCATCGAA CAGATCAGCGAATTCAGCAAGAGAGTCATCCTGGCAGACGCAAACCTGGACAAGGTCCTGAGCGCATACAACAAGCACAGAGACAAGCCGATC AGAGAACAGGCAGAAAACATCATCCACCTGTTCACACTGACAAACCTGGGAGCACCGGCAGCATTCAAGTACTTCGACACAACAATCGACAGA AAGAGATACACAAGCACAAAGGAAGTCCTGGACGCAACACTGATCCACCAGAGCATCACAGGACTGTACGAAACAAGAATCGACCTGAGCCAG CTGGGAGGAGACGGAGGAGGAAGCCCGAAGAAGAAGAGAAAGGTCTAG

TABLE7-continued

V. Additional Sequences

Description	SEQ ID NO	Sequence
ORF encoding Sp. Cas9	814	ATGGACAAGAAGTACTCCATCGGCCTGGACATCGGCACCAACTCCGTGGGCTGGGCCGTGATCACCGACGAGTACAAGGTGCCCTCCAAGAAG TTCAAGGTGCTGGGCAACACCGACCGGCACTCCATCAAGAAGAACCTGATCGGCGCCCTGCTGTTCGACTCCGGCGAGACCGCCGAGGCCACC CGGCTGAAGCGGACCGCCCGGCGGCGGTACACCCGGCGGAAGAACCGGATCTGCTACCTGCAGGAGATCTTCTCCAACGAGATGGCCAAGGTG GACGACTCCTTCTTCCACCGGCTGGAGGAGTCCTTCCTGGTGGAGGAGGACAAGAAGCACGAGCGGCACCCCATCTTCGGCAACATCGTGGAC GAGGTGGCCTACCACGAGAAGTACCCCACCATCTACCACCTGCGGAAGAAGCTGGTGGACTCCACCGACAAGGCCGACCTGCGGCTGATCTAC CTGGCCCTGGCCCACATGATCAAGTTCCGGGGCCACTTCCTGATCGAGGGCGACCTGAACCCCGACAACTCCGACGTGGACAAGCTGTTCATC CAGCTGGTGCAGACCTACAACCAGCTGTTCGAGGAGAACCCCATCAACGCCTCCGGCGTGGACGCCAAGGCCATCCTGTCCGCCCGGCTGTCC AAGTCCCGGCGGCTGGAGAACCTGATCGCCCAGCTGCCCGGCGAGAAGAAGAACGGCCTGTTCGGCAACCTGATCGCCCTGTCCCTGGGCCTG ACCCCCAACTTCAAGTCCAACTTCGACCTGGCCGAGGACGCCAAGCTGCAGCTGTCCAAGGACACCTACGACGACGACCTGGACAACCTGCTG GCCCAGATCGGCGACCAGTACGCCGACCTGTTCCTGGCCGCCAAGAACCTGTCCGACGCCATCCTGCTGTCCGACATCCTGCGGGTGAACACC GAGATCACCAAGGCCCCCCTGTCCGCCTCCATGATCAAGCGGTACGACGAGCACCACCAGGACCTGACCCTGCTGAAGGCCCTGGTGCGGCAG CAGCTGCCCGAGAAGTACAAGGAGATCTTCTTCGACCAGTCCAAGAACGGCTACGCCGGCTACATCGACGGCGGCGCCTCCCAGGAGGAGTTC TACAAGTTCATCAAGCCCATCCTGGAGAAGATGGACGGCACCGAGGAGCTGCTGGTGAAGCTGAACCGGGAGGACCTGCTGCGGAAGCAGCGG ACCTTCGACAACGGCTCCATCCCCCACCAGATCCACCTGGGCGAGCTGCACGCCATCCTGCGGCGGCAGGAGGACTTCTACCCCTTCCTGAAG GACAACCGGGAGAAGATCGAGAAGATCCTGACCTTCCGGATCCCCTACTACGTGGGCCCCCTGGCCCGGGGCAACTCCCGGTTCGCCTGGATG ACCCGGAAGTCCGAGGAGACCATCACCCCCTGGAACTTCGAGGAGGTGGTGGACAAGGGCGCCTCCGCCCAGTCCTTCATCGAGCGGATGACC AACTTCGACAAGAACCTGCCCAACGAGAAGGTGCTGCCCAAGCACTCCCTGCTGTACGAGTACTTCACCGTGTACAACGAGCTGACCAAGGTG AAGTACGTGACCGAGGGCATGCGGAAGCCCGCCTTCCTGTCCGGCGAGCAGAAGAAGGCCATCGTGGACCTGCTGTTCAAGACCAACCGGAAG GTGACCGTGAAGCAGCTGAAGGAGGACTACTTCAAGAAGATCGAGTGCTTCGACTCCGTGGAGATCTCCGGCGTGGAGGACCGGTTCAACGCC TCCCTGGGCACCTACCACGACCTGCTGAAGATCATCAAGGACAAGGACTTCCTGGACAACGAGGAGAACGAGGACATCCTGGAGGACATCGTG CTGACCCTGACCCTGTTCGAGGACCGGGAGATGATCGAGGAGCGGCTGAAGACCTACGCCCACCTGTTCGACGACAAGGTGATGAAGCAGCTG AAGCGGCGGCGGTACACCGGCTGGGGCCGGCTGTCCCGGAAGCTGATCAACGGCATCCGGGACAAGCAGTCCGGCAAGACCATCCTGGACTTC CTGAAGTCCGACGGCTTCGCCAACCGGAACTTCATGCAGCTGATCCACGACGACTCCCTGACCTTCAAGGAGGACATCCAGAAGGCCCAGGTG TCCGGCCAGGGCGACTCCCTGCACGAGCACATCGCCAACCTGGCCGGCTCCCCCGCCATCAAGAAGGGCATCCTGCAGACCGTGAAGGTGGTG GACGAGCTGGTGAAGGTGATGGGCCGGCACAAGCCCGAGAACATCGTGATCGAGATGGCCCGGGAGAACCAGACCACCCAGAAGGGCCAGAAG AACTCCCGGGAGCGGATGAAGCGGATCGAGGAGGGCATCAAGGAGCTGGGCTCCCAGATCCTGAAGGAGCACCCCGTGGAGAACACCCAGCTG CAGAACGAGAAGCTGTACCTGTACTACCTGCAGAACGGCCGGGACATGTACGTGGACCAGGAGCTGGACATCAACCGGCTGTCCGACTACGAC GTGGACCACATCGTGCCCCAGTCCTTCCTGAAGGACGACTCCATCGACAACAAGGTGCTGACCCGGTCCGACAAGAACCGGGGCAAGTCCGAC AACGTGCCCTCCGAGGAGGTGGTGAAGAAGATGAAGAACTACTGGCGGCAGCTGCTGAACGCCAAGCTGATCACCCAGCGGAAGTTCGACAAC CTGACCAAGGCCGAGCGGGGCGGCCTGTCCGAGCTGGACAAGGCCGGCTTCATCAAGCGGCAGCTGGTGGAGACCCGGCAGATCACCAAGCAC GTGGCCCAGATCCTGGACTCCCGGATGAACACCAAGTACGACGAGAACGACAAGCTGATCCGGGAGGTGAAGGTGATCACCCTGAAGTCCAAG CTGGTGTCCGACTTCCGGAAGGACTTCCAGTTCTACAAGGTGCGGGAGATCAACAACTACCACCACGCCCACGACGCCTACCTGAACGCCGTG GTGGGCACCGCCCTGATCAAGAAGTACCCCAAGCTGGAGTCCGAGTTCGTGTACGGCGACTACAAGGTGTACGACGTGCGGAAGATGATCGCC AAGTCCGAGCAGGAGATCGGCAAGGCCACCGCCAAGTACTTCTTCTACTCCAACATCATGAACTTCTTCAAGACCGAGATCACCCTGGCCAAC GGCGAGATCCGGAAGCGGCCCCTGATCGAGACCAACGGCGAGACCGGCGAGATCGTGTGGGACAAGGGCCGGGACTTCGCCACCGTGCGGAAG GTGCTGTCCATGCCCCAGGTGAACATCGTGAAGAAGACCGAGGTGCAGACCGGCGGCTTCTCCAAGGAGTCCATCCTGCCCAAGCGGAACTCC GACAAGCTGATCGCCCGGAAGAAGGACTGGGACCCCAAGAAGTACGGCGGCTTCGACTCCCCCACCGTGGCCTACTCCGTGCTGGTGGTGGCC AAGGTGGAGAAGGGCAAGTCCAAGAAGCTGAAGTCCGTGAAGGAGCTGCTGGGCATCACCATCATGGAGCGGTCCTCCTTCGAGAAGAACCCC ATCGACTTCCTGGAGGCCAAGGGCTACAAGGAGGTGAAGAAGGACCTGATCATCAAGCTGCCCAAGTACTCCCTGTTCGAGCTGGAGAACGGC CGGAAGCGGATGCTGGCCTCCGCCGGCGAGCTGCAGAAGGGCAACGAGCTGGCCCTGCCCTCCAAGTACGTGAACTTCCTGTACCTGGCCTCC CACTACGAGAAGCTGAAGGGCTCCCCCGAGGACAACGAGCAGAAGCAGCTGTTCGTGGAGCAGCACAAGCACTACCTGGACGAGATCATCGAG CAGATCTCCGAGTTCTCCAAGCGGGTGATCCTGGCCGACGCCAACCTGGACAAGGTGCTGTCCGCCTACAACAAGCACCGGGACAAGCCCATC CGGGAGCAGGCCGAGAACATCATCCACCTGTTCACCCTGACCAACCTGGGCGCCCCCGCCGCCTTCAAGTACTTCGACACCACCATCGACCGG AAGCGGTACACCTCCACCAAGGAGGTGCTGGACGCCACCCTGATCCACCAGTCCATCACCGGCCTGTACGAGACCCGGATCGACCTGTCCCAG CTGGGCGGCGACGGCGGCGGCTCCCCCAAGAAGAAGCGGAAGGTGTGA
ORF encoding Sp. Cas9	815	AUGGACAAGAAGUACAGCAUCGGCCUGGACAUCGGCACGAACAGCGUUGGCUGGGCUGUGAUCACGGACGAGUACAAGGUUCCCUCAAAGAAG UUCAAGGUGCUGGGCAACACGGACCGGCACAGCAUCAAGAAGAAUCUCAUCGGUGCACUGCUGUUCGACAGCGGUGAGACGGCCGAAGCCACG CGGCUGAAGCGGACGGCCCGCCGGCGGUACACGCGGCGGAAGAACCGGAUCUGCUACCUGCAGGAGAUCUUCAGCAACGAGAUGGCCAAGGUG GACGACAGCUUCUUCCACCGGCUGGAGGAGAGCUUCCUGGUGGAGGAGGACAAGAAGCACGAGCGGCACCCCAUCUUCGGCAACAUCGUGGAC

TABLE7-continued

V. Additional Sequences

Description	SEQ ID NO	Sequence
		GAAGUCGCCUACCACGAGAAGUACCCCACCAUCUACCACCUGCGGAAGAAGCUGGUGGACUCGACUGACAAGGCCGACCUGCGGCUGAUCUAC CUGGCACUGGCCCACAUGAUAAAGUUCCGGGGCCACUUCCUGAUCGAGGGCGACCUGAACCCUGACAACAGCGACGUGGACAAGCUGUUCAUC CAGCUGGUGCAGACCUACAACCAGCUGUUCGAGGAGAACCCCAUCAACGCCAGCGGCGUGGACGCCAAGGCCAUCCUCAGCGCCCGCCUCAGC AAGAGCCGGCGGCUGGAGAAUCUCAUCGCCCAGCUUCCAGGUGAGAAGAAGAAUGGGCUGUUCGGCAAUCUCAUCGCACUCAGCCUGGGCCUG ACUCCCAACUUCAAGAGCAACUUCGACCUGGCCGAGGACGCCAAGCUGCAGCUCAGCAAGGACACCUACGACGACGACCUGGACAAUCUCCUG GCCCAGAUCGGCGACCAGUACGCCGACCUGUUCCUGGCUGCCAAGAAUCUCAGCGACGCCAUCCUGCUCAGCGACAUCCUGCGGGUGAACACA GAGAUCACGAAGGCCCCCCUCAGCGCCAGCAUGAUAAAGCGGUACGACGAGCACCACCAGGACCUGACGCUGCUGAAGGCACUGGUGCGGCAG CAGCUUCCAGAGAAGUACAAGGAGAUCUUCUUCGACCAGAGCAAGAAUGGGUACGCCGGGUACAUCGACGGUGGUGCCAGCCAGGAGGAGUUC UACAAGUUCAUCAAGCCCAUCCUGGAGAAGAUGGACGGCACAGAGGAGCUGCUGGUGAAGCUGAACAGGGAGGACCUGCUGCGGAAGCAGCGG ACGUUCGACAAUGGGAGCAUCCCCCACCAGAUCCACCUGGGUGAGCUGCACGCCAUCCUGCGGCGGCAGGAGGACUUCUACCCCUUCCUGAAG GACAACAGGGAGAAGAUCGAGAAGAUCCUGACGUUCCGGAUCCCCUACUACGUUGGCCCCCUGGCCCGCGGCAACAGCCGGUUCGCCUGGAUG ACGCGGAAGAGCGAGGAGACGAUCACUCCCUGGAACUUCGAGGAAGUCGUGGACAAGGGUGCCAGCGCCCAGAGCUUCAUCGAGCGGAUGACG AACUUCGACAAGAAUCUUCCAAACGAGAAGGUGCUUCCAAAGCACAGCCUGCUGUACGAGUACUUCACGGUGUACAACGAGCUGACGAAGGUG AAGUACGUGACAGAGGGCAUGCGGAAGCCCGCCUUCCUCAGCGGUGAGCAGAAGAAGGCCAUCGUGGACCUGCUGUUCAAGACGAACCGGAAG GUGACGGUGAAGCAGCUGAAGGAGGACUACUUCAAGAAGAUCGAGUGCUUCGACAGCGUGGAGAUCAGCGGCGUGGAGGACCGGUUCAACGCC AGCCUGGGCACCUACCACGACCUGCUGAAGAUCAUCAAGGACAAGGACUUCCUGGACAACGAGGAGAACGAGGACAUCCUGGAGGACAUCGUG CUGACGCUGACGCUGUUCGAGGACAGGGAGAUGAUAGAGGAGCGGCUGAAGACCUACGCCCACCUGUUCGACGACAAGGUGAUGAAGCAGCUG AAGCGGCGGCGGUACACGGGCUGGGGCCGGCUCAGCCGGAAGCUGAUCAAUGGGAUCCGAGACAAGCAGAGCGGCAAGACGAUCCUGGACUUC CUGAAGAGCGACGGCUUCGCCAACCGGAACUUCAUGCAGCUGAUCCACGACGACAGCCUGACGUUCAAGGAGGACAUCCAGAAGGCCCAGGUC AGCGGCCAGGGCGACAGCCUGCACGAGCACAUCGCCAAUCUCGCCGGGAGCCCCGCCAUCAAGAAGGGGAUCCUGCAGACGGUGAAGGUGGUG GACGAGCUGGUGAAGGUGAUGGGCCGGCACAAGCCAGAGAACAUCGUGAUCGAGAUGGCCAGGGAGAACCAGACGACUCAAAAGGGGCAGAAG AACAGCAGGGAGCGGAUGAAGCGGAUCGAGGAGGGCAUCAAGGAGCUGGGCAGCCAGAUCCUGAAGGAGCACCCCGUGGAGAACACUCAACUG CAGAACGAGAAGCUGUACCUGUACUACCUGCAGAAUGGGCGAGACAUGUACGUGGACCAGGAGCUGGACAUCAACCGGCUCAGCGACUACGAC GUGGACCACAUCGUUCCCCAGAGCUUCCUGAAGGACGACAGCAUCGACAACAAGGUGCUGACGCGGAGCGACAAGAACCGGGGCAAGAGCGAC AACGUUCCCUCAGAGGAAGUCGUGAAGAAGAUGAAGAACUACUGGCGGCAGCUGCUGAACGCCAAGCUGAUCACUCAACGGAAGUUCGACAAU CUCACGAAGGCCGAGCGGGGUGGCCUCAGCGAGCUGGACAAGGCCGGGUUCAUCAAGCGGCAGCUGGUGGAGACGCGGCAGAUCACGAAGCAC GUGGCCCAGAUCCUGGACAGCCGGAUGAACACGAAGUACGACGAGAACGACAAGCUGAUCAGGGAAGUCAAGGUGAUCACGCUGAAGAGCAAG CUGGUCAGCGACUUCCGGAAGGACUUCCAGUUCUACAAGGUGAGGGAGAUCAACAACUACCACCACGCCCACGACGCCUACCUGAACGCUGUG GUUGGCACGGCACUGAUCAAGAAGUACCCCAAGCUGGAGAGCGAGUUCGUGUACGGCGACUACAAGGUGUACGACGUGCGGAAGAUGAUAGCC AAGAGCGAGCAGGAGAUCGGCAAGGCCACGGCCAAGUACUUCUUCUACAGCAACAUCAUGAACUUCUUCAAGACAGAGAUCACGCUGGCCAAU GGUGAGAUCCGGAAGCGGCCCCUGAUCGAGACGAAUGGUGAGACGGGUGAGAUCGUGUGGGACAAGGGGCGAGACUUCGCCACGGUGCGGAAG GUGCUCAGCAUGCCCCAGGUGAACAUCGUGAAGAAGACAGAAGUCCAGACGGGUGGCUUCAGCAAGGAGAGCAUCCUUCCAAAGCGGAACAGC GACAAGCUGAUCGCCCGCAAGAAGGACUGGGACCCCAAGAAGUACGGUGGCUUCGACAGCCCCACCGUGGCCUACAGCGUGCUGGUGGUGGCC AAGGUGGAGAAGGGGAAGAGCAAGAAGCUGAAGAGCGUGAAGGAGCUGCUGGGCAUCACGAUCAUGGAGCGGAGCAGCUUCGAGAAGAACCCC AUCGACUUCCUGGAAGCCAAGGGGUACAAGGAAGUCAAGAAGGACCUGAUCAUCAAGCUUCCAAAGUACAGCCUGUUCGAGCUGGAGAAUGGG CGGAAGCGGAUGCUGGCCAGCGCCGGUGAGCUGCAGAAGGGGAACGAGCUGGCACUUCCCUCAAAGUACGUGAACUUCCUGUACCUGGCCAGC CACUACGAGAAGCUGAAGGGGAGCCCAGAGGACAACGAGCAGAAGCAGCUGUUCGUGGAGCAGCACAAGCACUACCUGGACGAGAUCAUCGAG CAGAUCAGCGAGUUCAGCAAGCGGGUGAUCCUGGCCGACGCCAAUCUCGACAAGGUGCUCAGCGCCUACAACAAGCACCGAGACAAGCCCAUC AGGGAGCAGGCCGAGAACAUCAUCCACCUGUUCACGCUGACGAAUCUCGGUGCCCCCGCUGCCUUCAAGUACUUCGACACGACGAUCGACCGG AAGCGGUACACGUCGACUAAGGAAGUCCUGGACGCCACGCUGAUCCACCAGAGCAUCACGGGCCUGUACGAGACGCGGAUCGACCUCAGCCAG CUGGGUGGCGACGGUGGUGGCAGCCCCAAGAAGAAGCGGAAGGUGUAG
ORF encoding Sp. Cas9	816	AUGGACAAGAAGUACAGCAUCGGCCUCGACAUCGGCACCAACAGCGUCGGCUGGGCCGUCAUCACCGACGAGUACAAGGUCCCCAGCAAGAAG UUCAAGGUCCUCGGCAACACCGACCGCCACAGCAUCAAGAAGAACCUCAUCGGCGCCCUCCUCUUCGACAGCGGCGAGACCGCCGAGGCCACC CGCCUCAAGCGCACCGCCCGCCGCCGCUACACCCGCCGCAAGAACCGCAUCUGCUACCUCCAGGAGAUCUUCAGCAACGAGAUGGCCAAGGUC GACGACAGCUUCUUCCACCGCCUCGAGGAGAGCUUCCUCGUCGAGGAGGACAAGAAGCACGAGCGCCACCCCAUCUUCGGCAACAUCGUCGAC GAGGUCGCCUACCACGAGAAGUACCCCACCAUCUACCACCUCCGCAAGAAGCUCGUCGACAGCACCGACAAGGCCGACCUCCGCCUCAUCUAC CUCGCCCUCGCCCACAUGAUCAAGUUCCGCGGCCACUUCCUCAUCGAGGGCGACCUCAACCCCGACAACAGCGACGUCGACAAGCUCUUCAUC CAGCUCGUCCAGACCUACAACCAGCUCUUCGAGGAGAACCCCAUCAACGCCAGCGGCGUCGACGCCAAGGCCAUCCUCAGCGCCCGCCUCAGC

TABLE7-continued

V. Additional Sequences

Description	SEQ ID NO	Sequence
		AAGAGCCGCCGCCUCGAGAACCUCAUCGCCCAGCUCCCCGGCGAGAAGAAGAACGGCCUCUUCGGCAACCUCAUCGCCCUCAGCCUCGGCCUC ACCCCCAACUUCAAGAGCAACUUCGACCUCGCCGAGGACGCCAAGCUCCAGCUCAGCAAGGACACCUACGACGACGACCUCGACAACCUCCUC GCCCAGAUCGGCGACCAGUACGCCGACCUCUUCCUCGCCGCCAAGAACCUCAGCGACGCCAUCCUCCUCAGCGACAUCCUCCGCGUCAACACC GAGAUCACCAAGGCCCCCCUCAGCGCCAGCAUGAUCAAGCGCUACGACGAGCACCACCAGGACCUCACCCUCCUCAAGGCCCUCGUCCGCCAG CAGCUCCCCGAGAAGUACAAGGAGAUCUUCUUCGACCAGAGCAAGAACGGCUACGCCGGCUACAUCGACGGCGGCGCCAGCCAGGAGGAGUUC UACAAGUUCAUCAAGCCCAUCCUCGAGAAGAUGGACGGCACCGAGGAGCUCCUCGUCAAGCUCAACCGCGAGGACCUCCUCCGCAAGCAGCGC ACCUUCGACAACGGCAGCAUCCCCCACCAGAUCCACCUCGGCGAGCUCCACGCCAUCCUCCGCCGCCAGGAGGACUUCUACCCCUUCCUCAAG GACAACCGCGAGAAGAUCGAGAAGAUCCUCACCUUCCGCAUCCCCUACUACGUCGGCCCCCUCGCCCGCGGCAACAGCCGCUUCGCCUGGAUG ACCCGCAAGAGCGAGGAGACCAUCACCCCCUGGAACUUCGAGGAGGUCGUCGACAAGGGCGCCAGCGCCCAGAGCUUCAUCGAGCGCAUGACC AACUUCGACAAGAACCUCCCCAACGAGAAGGUCCUCCCCAAGCACAGCCUCCUCUACGAGUACUUCACCGUCUACAACGAGCUCACCAAGGUC AAGUACGUCACCGAGGGCAUGCGCAAGCCCGCCUUCCUCAGCGGCGAGCAGAAGAAGGCCAUCGUCGACCUCCUCUUCAAGACCAACCGCAAG GUCACCGUCAAGCAGCUCAAGGAGGACUACUUCAAGAAGAUCGAGUGCUUCGACAGCGUCGAGAUCAGCGGCGUCGAGGACCGCUUCAACGCC AGCCUCGGCACCUACCACGACCUCCUCAAGAUCAUCAAGGACAAGGACUUCCUCGACAACGAGGAGAACGAGGACAUCCUCGAGGACAUCGUC CUCACCCUCACCCUCUUCGAGGACCGCGAGAUGAUCGAGGAGCGCCUCAAGACCUACGCCCACCUCUUCGACGACAAGGUCAUGAAGCAGCUC AAGCGCCGCCGCUACACCGGCUGGGGCCGCCUCAGCCGCAAGCUCAUCAACGGCAUCCGCGACAAGCAGAGCGGCAAGACCAUCCUCGACUUC CUCAAGAGCGACGGCUUCGCCAACCGCAACUUCAUGCAGCUCAUCCACGACGACAGCCUCACCUUCAAGGAGGACAUCCAGAAGGCCCAGGUC AGCGGCCAGGGCGACAGCCUCCACGAGCACAUCGCCAACCUCGCCGGCAGCCCCGCCAUCAAGAAGGGCAUCCUCCAGACCGUCAAGGUCGUC GACGAGCUCGUCAAGGUCAUGGGCCGCCACAAGCCCGAGAACAUCGUCAUCGAGAUGGCCCGCGAGAACCAGACCACCCAGAAGGGCCAGAAG AACAGCCGCGAGCGCAUGAAGCGCAUCGAGGAGGGCAUCAAGGAGCUCGGCAGCCAGAUCCUCAAGGAGCACCCCGUCGAGAACACCCAGCUC CAGAACGAGAAGCUCUACCUCUACUACCUCCAGAACGGCCGCGACAUGUACGUCGACCAGGAGCUCGACAUCAACCGCCUCAGCGACUACGAC GUCGACCACAUCGUCCCCCAGAGCUUCCUCAAGGACGACAGCAUCGACAACAAGGUCCUCACCCGCAGCGACAAGAACCGCGGCAAGAGCGAC AACGUCCCCAGCGAGGAGGUCGUCAAGAAGAUGAAGAACUACUGGCGCCAGCUCCUCAACGCCAAGCUCAUCACCCAGCGCAAGUUCGACAAC CUCACCAAGGCCGAGCGCGGCGGCCUCAGCGAGCUCGACAAGGCCGGCUUCAUCAAGCGCCAGCUCGUCGAGACCCGCCAGAUCACCAAGCAC GUCGCCCAGAUCCUCGACAGCCGCAUGAACACCAAGUACGACGAGAACGACAAGCUCAUCCGCGAGGUCAAGGUCAUCACCCUCAAGAGCAAG CUCGUCAGCGACUUCCGCAAGGACUUCCAGUUCUACAAGGUCCGCGAGAUCAACAACUACCACCACGCCCACGACGCCUACCUCAACGCCGUC GUCGGCACCGCCCUCAUCAAGAAGUACCCCAAGCUCGAGAGCGAGUUCGUCUACGGCGACUACAAGGUCUACGACGUCCGCAAGAUGAUCGCC AAGAGCGAGCAGGAGAUCGGCAAGGCCACCGCCAAGUACUUCUUCUACAGCAACAUCAUGAACUUCUUCAAGACCGAGAUCACCCUCGCCAAC GGCGAGAUCCGCAAGCGCCCCCUCAUCGAGACCAACGGCGAGACCGGCGAGAUCGUCUGGGACAAGGGCCGCGACUUCGCCACCGUCCGCAAG GUCCUCAGCAUGCCCCAGGUCAACAUCGUCAAGAAGACCGAGGUCCAGACCGGCGGCUUCAGCAAGGAGAGCAUCCUCCCCAAGCGCAACAGC GACAAGCUCAUCGCCCGCAAGAAGGACUGGGACCCCAAGAAGUACGGCGGCUUCGACAGCCCCACCGUCGCCUACAGCGUCCUCGUCGUCGCC AAGGUCGAGAAGGGCAAGAGCAAGAAGCUCAAGAGCGUCAAGGAGCUCCUCGGCAUCACCAUCAUGGAGCGCAGCAGCUUCGAGAAGAACCCC AUCGACUUCCUCGAGGCCAAGGGCUACAAGGAGGUCAAGAAGGACCUCAUCAUCAAGCUCCCCAAGUACAGCCUCUUCGAGCUCGAGAACGGC CGCAAGCGCAUGCUCGCCAGCGCCGGCGAGCUCCAGAAGGGCAACGAGCUCGCCCUCCCCAGCAAGUACGUCAACUUCCUCUACCUCGCCAGC CACUACGAGAAGCUCAAGGGCAGCCCCGAGGACAACGAGCAGAAGCAGCUCUUCGUCGAGCAGCACAAGCACUACCUCGACGAGAUCAUCGAG CAGAUCAGCGAGUUCAGCAAGCGCGUCAUCCUCGCCGACGCCAACCUCGACAAGGUCCUCAGCGCCUACAACAAGCACCGCGACAAGCCCAUC CGCGAGCAGGCCGAGAACAUCAUCCACCUCUUCACCCUCACCAACCUCGGCGCCCCCGCCGCCUUCAAGUACUUCGACACCACCAUCGACCGC AAGCGCUACACCAGCACCAAGGAGGUCCUCGACGCCACCCUCAUCCACCAGAGCAUCACCGGCCUCUACGAGACCCGCAUCGACCUCAGCCAG CUCGGCGGCGACGGCGGCGGCAGCCCCAAGAAGAAGCGCAAGGUCUAG
ORF encoding *Sp.* Cas9	817	ATGGACAAGAAGTACAGCATCGGCCTGGACATCGGCACCAACAGCGTGGGCTGGGCCGTGATCACCGACGAGTACAAGGTGCCCAGCAAGAAG TTCAAGGTGCTGGGCAACACCGACCGGCACAGCATCAAGAAGAACCTGATCGGCGCCCTGCTGTTCGACAGCGGCGAGACCGCCGAGGCCACC CGGCTGAAGCGGACCGCCCGGCGGCGGTACACCCGGCGGAAGAACCGGATCTGCTACCTGCAGGAGATCTTCAGCAACGAGATGGCCAAGGTG GACGACAGCTTCTTCCACCGGCTGGAGGAGAGCTTCCTGGTGGAGGAGGACAAGAAGCACGAGCGGCACCCCATCTTCGGCAACATCGTGGAC GAGGTGGCCTACCACGAGAAGTACCCCACCATCTACCACCTGCGGAAGAAGCTGGTGGACAGCACCGACAAGGCCGACCTGCGGCTGATCTAC CTGGCCCTGGCCCACATGATCAAGTTCCGGGGCCACTTCCTGATCGAGGGCGACCTGAACCCCGACAACAGCGACGTGGACAAGCTGTTCATC CAGCTGGTGCAGACCTACAACCAGCTGTTCGAGGAGAACCCCATCAACGCCAGCGGCGTGGACGCCAAGGCCATCCTGAGCGCCCGGCTGAGC AAGAGCCGGCGGCTGGAGAACCTGATCGCCCAGCTGCCCGGCGAGAAGAAGAACGGCCTGTTCGGCAACCTGATCGCCCTGAGCCTGGGCCTG ACCCCCAACTTCAAGAGCAACTTCGACCTGGCCGAGGACGCCAAGCTGCAGCTGAGCAAGGACACCTACGACGACGACCTGGACAACCTGCTG GCCCAGATCGGCGACCAGTACGCCGACCTGTTCCTGGCCGCCAAGAACCTGAGCGACGCCATCCTGCTGAGCGACATCCTGCGGGTGAACACC

TABLE7-continued

V. Additional Sequences		
Description	SEQ ID NO	Sequence
		GAGATCACCAAGGCCCCCCTGAGCGCCAGCATGATCAAGCGGTACGACGAGCACCACCAGGACCTGACCCTGCTGAAGGCCCTGGTGCGGCAG CAGCTGCCCGAGAAGTACAAGGAGATCTTCTTCGACCAGAGCAAGAACGGCTACGCCGGCTACATCGACGGCGGCGCCAGCCAGGAGGAGTTC TACAAGTTCATCAAGCCCATCCTGGAGAAGATGGACGGCACCGAGGAGCTGCTGGTGAAGCTGAACCGGGAGGACCTGCTGCGGAAGCAGCGG ACCTTCGACAACGGCAGCATCCCCCACCAGATCCACCTGGGCGAGCTGCACGCCATCCTGCGGCGGCAGGAGGACTTCTACCCCTTCCTGAAG GACAACCGGGAGAAGATCGAGAAGATCCTGACCTTCCGGATCCCCTACTACGTGGGCCCCCTGGCCCGGGGCAACAGCCGGTTCGCCTGGATG ACCCGGAAGAGCGAGGAGACCATCACCCCCTGGAACTTCGAGGAGGTGGTGGACAAGGGCGCCAGCGCCCAGAGCTTCATCGAGCGGATGACC AACTTCGACAAGAACCTGCCCAACGAGAAGGTGCTGCCCAAGCACAGCCTGCTGTACGAGTACTTCACCGTGTACAACGAGCTGACCAAGGTG AAGTACGTGACCGAGGGCATGCGGAAGCCCGCCTTCCTGAGCGGCGAGCAGAAGAAGGCCATCGTGGACCTGCTGTTCAAGACCAACCGGAAG GTGACCGTGAAGCAGCTGAAGGAGGACTACTTCAAGAAGATCGAGTGCTTCGACAGCGTGGAGATCAGCGGCGTGGAGGACCGGTTCAACGCC AGCCTGGGCACCTACCACGACCTGCTGAAGATCATCAAGGACAAGGACTTCCTGGACAACGAGGAGAACGAGGACATCCTGGAGGACATCGTG CTGACCCTGACCCTGTTCGAGGACCGGGAGATGATCGAGGAGCGGCTGAAGACCTACGCCCACCTGTTCGACGACAAGGTGATGAAGCAGCTG AAGCGGCGGCGGTACACCGGCTGGGGCCGGCTGAGCCGGAAGCTGATCAACGGCATCCGGGACAAGCAGAGCGGCAAGACCATCCTGGACTTC CTGAAGAGCGACGGCTTCGCCAACCGGAACTTCATGCAGCTGATCCACGACGACAGCCTGACCTTCAAGGAGGACATCCAGAAGGCCCAGGTG AGCGGCCAGGGCGACAGCCTGCACGAGCACATCGCCAACCTGGCCGGCAGCCCCGCCATCAAGAAGGGCATCCTGCAGACCGTGAAGGTGGTG GACGAGCTGGTGAAGGTGATGGGCCGGCACAAGCCCGAGAACATCGTGATCGAGATGGCCCGGGAGAACCAGACCACCCAGAAGGGCCAGAAG AACAGCCGGGAGCGGATGAAGCGGATCGAGGAGGGCATCAAGGAGCTGGGCAGCCAGATCCTGAAGGAGCACCCCGTGGAGAACACCCAGCTG CAGAACGAGAAGCTGTACCTGTACTACCTGCAGAACGGCCGGGACATGTACGTGGACCAGGAGCTGGACATCAACCGGCTGAGCGACTACGAC GTGGACCACATCGTGCCCCAGAGCTTCCTGAAGGACGACAGCATCGACAACAAGGTGCTGACCCGGAGCGACAAGAACCGGGGCAAGAGCGAC AACGTGCCCAGCGAGGAGGTGGTGAAGAAGATGAAGAACTACTGGCGGCAGCTGCTGAACGCCAAGCTGATCACCCAGCGGAAGTTCGACAAC CTGACCAAGGCCGAGCGGGGCGGCCTGAGCGAGCTGGACAAGGCCGGCTTCATCAAGCGGCAGCTGGTGGAGACCCGGCAGATCACCAAGCAC GTGGCCCAGATCCTGGACAGCCGGATGAACACCAAGTACGACGAGAACGACAAGCTGATCCGGGAGGTGAAGGTGATCACCCTGAAGAGCAAG CTGGTGAGCGACTTCCGGAAGGACTTCCAGTTCTACAAGGTGCGGGAGATCAACAACTACCACCACGCCCACGACGCCTACCTGAACGCCGTG GTGGGCACCGCCCTGATCAAGAAGTACCCCAAGCTGGAGAGCGAGTTCGTGTACGGCGACTACAAGGTGTACGACGTGCGGAAGATGATCGCC AAGAGCGAGCAGGAGATCGGCAAGGCCACCGCCAAGTACTTCTTCTACAGCAACATCATGAACTTCTTCAAGACCGAGATCACCCTGGCCAAC GGCGAGATCCGGAAGCGGCCCCTGATCGAGACCAACGGCGAGACCGGCGAGATCGTGTGGGACAAGGGCCGGGACTTCGCCACCGTGCGGAAG GTGCTGAGCATGCCCCAGGTGAACATCGTGAAGAAGACCGAGGTGCAGACCGGCGGCTTCAGCAAGGAGAGCATCCTGCCCAAGCGGAACAGC GACAAGCTGATCGCCCGGAAGAAGGACTGGGACCCCAAGAAGTACGGCGGCTTCGACAGCCCCACCGTGGCCTACAGCGTGCTGGTGGTGGCC AAGGTGGAGAAGGGCAAGAGCAAGAAGCTGAAGAGCGTGAAGGAGCTGCTGGGCATCACCATCATGGAGCGGAGCAGCTTCGAGAAGAACCCC ATCGACTTCCTGGAGGCCAAGGGCTACAAGGAGGTGAAGAAGGACCTGATCATCAAGCTGCCCAAGTACAGCCTGTTCGAGCTGGAGAACGGC CGGAAGCGGATGCTGGCCAGCGCCGGCGAGCTGCAGAAGGGCAACGAGCTGGCCCTGCCCAGCAAGTACGTGAACTTCCTGTACCTGGCCAGC CACTACGAGAAGCTGAAGGGCAGCCCCGAGGACAACGAGCAGAAGCAGCTGTTCGTGGAGCAGCACAAGCACTACCTGGACGAGATCATCGAG CAGATCAGCGAGTTCAGCAAGCGGGTGATCCTGGCCGACGCCAACCTGGACAAGGTGCTGAGCGCCTACAACAAGCACCGGGACAAGCCCATC CGGGAGCAGGCCGAGAACATCATCCACCTGTTCACCCTGACCAACCTGGGCGCCCCCGCCGCCTTCAAGTACTTCGACACCACCATCGACCGG AAGCGGTACACCAGCACCAAGGAGGTGCTGGACGCCACCCTGATCCACCAGAGCATCACCGGCCTGTACGAGACCCGGATCGACCTGAGCCAG CTGGGCGGCGACGGCGGCGGCAGCCCCAAGAAGAAGCGGAAGGTGTGA
amino acid sequence for Sp. Cas9	818	MDKKYSIGLDIGTNSVGWAVITDEYKVPSKKFKVLGNTDRHSIKKNLIGALLFDSGETAEATRLKRTARRRYTRRKNRICYLQEIFSNEMAKV DDSFFHRLEESFLVEEDKKHERHPIFGNIVDEVAYHEKYPTIYHLRKKLVDSTDKADLRLIYLALAHMIKFRGHFLIEGDLNPDNSDVDKLFI QLVQTYNQLFEENPINASGVDAKAILSARLSKSRRLENLIAQLPGEKKNGLFGNLIALSLGLTPNFKSNFDLAEDAKLQLSKDTYDDDLDNLL AQIGDQYADLFLAAKNLSDAILLSDILRVNTEITKAPLSASMIKRYDEHHQDLTLLKALVRQQLPEKYKEIFFDQSKNGYAGYIDGGASQEEF YKFIKPILEKMDGTEELLVKLNREDLLRKQRTFDNGSIPHQIHLGELHAILRRQEDFYPFLKDNREKIEKILTFRIPYYVGPLARGNSRFAWM TRKSEETITPWNFEEVVDKGASAQSFIERMTNFDKNLPNEKVLPKHSLLYEYFTVYNELTKVKYVTEGMRKPAFLSGEQKKAIVDLLFKTNRK VTVKQLKEDYFKKIECFDSVEISGVEDRFNASLGTYHDLLKIIKDKDFLDNEENEDILEDIVLTLTLFEDREMIEERLKTYAHLFDDKVMKQL KRRRYTGWGRLSRKLINGIRDKQSGKTILDFLKSDGFANRNFMQLIHDDSLTFKEDIQKAQVSGQGDSLHEHIANLAGSPAIKKGILQTVKVV DELVKVMGRHKPENIVIEMARENQTTQKGQKNSRERMKRIEEGIKELGSQILKEHPVENTQLQNEKLYLYYLQNGRDMYVDQELDINRLSDYD VDHIVPQSFLKDDSIDNKVLTRSDKNRGKSDNVPSEEVVKKMKNYWRQLLNAKLITQRKFDNLTKAERGGLSELDKAGFIKRQLVETRQITKH VAQILDSRMNTKYDENDKLIREVKVITLKSKLVSDFRKDFQFYKVREINNYHHAHDAYLNAVVGTALIKKYPKLESEFVYGDYKVYDVRKMIA KSEQEIGKATAKYFFYSNIMNFFKTEITLANGEIRKRPLIETNGETGEIVWDKGRDFATVRKVLSMPQVNIVKKTEVQTGGFSKESILPKRNS DKLIARKKDWDPKKYGGFDSPTVAYSVLVVAKVEKGKSKKLKSVKELLGITIMERSSFEKNPIDFLEAKGYKEVKKDLIIKLPKYSLFELENG

TABLE7-continued

V. Additional Sequences

Description	SEQ ID NO	Sequence
		RKRMLASAGELQKGNELALPSKYVNFLYLASHYEKLKGSPEDNEQKQLFVEQHKHYLDEIIEQISEFSKRVILADANLDKVLSAYNKHRDKPI REQAENIIHLFTLTNLGAPAAFKYFDTTIDRKRYTSTKEVLDATLIHQSITGLYETRIDLSQLGGDGGGSPKKKRKV
Open reading frame for Cas9 with Hibit tag	819	AUGGACAAGAAGUACUCCAUCGGCCUGGACAUCGGCACCAACUCCGUGGGCUGGGCCGUGAUCACCGACGAGUACAAGGUGCCCUCCAAGAAG UUCAAGGUGCUGGGCAACACCGACCGGCACUCCAUCAAGAAGAACCUGAUCGGCGCCCUGCUGUUCGACUCCGGCGAGACCGCCGAGGCCACC CGGCUGAAGCGGACCGCCCGGCGGCGGUACACCCGGCGGAAGAACCGGAUCUGCUACCUGCAGGAGAUCUUCUCCAACGAGAUGGCCAAGGUG GACGACUCCUUCUUCCACCGGCUGGAGGAGUCCUUCCUGGUGGAGGAGGACAAGAAGCACGAGCGGCACCCCAUCUUCGGCAACAUCGUGGAC GAGGUGGCCUACCACGAGAAGUACCCCACCAUCUACCACCUGCGGAAGAAGCUGGUGGACUCCACCGACAAGGCCGACCUGCGGCUGAUCUAC CUGGCCCUGGCCCACAUGAUCAAGUUCCGGGGCCACUUCCUGAUCGAGGGCGACCUGAACCCCGACAACUCCGACGUGGACAAGCUGUUCAUC CAGCUGGUGCAGACCUACAACCAGCUGUUCGAGGAGAACCCCAUCAACGCCUCCGGCGUGGACGCCAAGGCCAUCCUGUCCGCCCGGCUGUCC AAGUCCCGGCGGCUGGAGAACCUGAUCGCCCAGCUGCCCGGCGAGAAGAAGAACGGCCUGUUCGGCAACCUGAUCGCCCUGUCCCUGGGCCUG ACCCCCAACUUCAAGUCCAACUUCGACCUGGCCGAGGACGCCAAGCUGCAGCUGUCCAAGGACACCUACGACGACGACCUGGACAACCUGCUG GCCCAGAUCGGCGACCAGUACGCCGACCUGUUCCUGGCCGCCAAGAACCUGUCCGACGCCAUCCUGCUGUCCGACAUCCUGCGGGUGAACACC GAGAUCACCAAGGCCCCCCUGUCCGCCUCCAUGAUCAAGCGGUACGACGAGCACCACCAGGACCUGACCCUGCUGAAGGCCCUGGUGCGGCAG CAGCUGCCCGAGAAGUACAAGGAGAUCUUCUUCGACCAGUCCAAGAACGGCUACGCCGGCUACAUCGACGGCGGCGCCUCCCAGGAGGAGUUC UACAAGUUCAUCAAGCCCAUCCUGGAGAAGAUGGACGGCACCGAGGAGCUGCUGGUGAAGCUGAACCGGGAGGACCUGCUGCGGAAGCAGCGG ACCUUCGACAACGGCUCCAUCCCCCACCAGAUCCACCUGGGCGAGCUGCACGCCAUCCUGCGGCGGCAGGAGGACUUCUACCCCUUCCUGAAG GACAACCGGGAGAAGAUCGAGAAGAUCCUGACCUUCCGGAUCCCCUACUACGUGGGCCCCCUGGCCCGGGGCAACUCCCGGUUCGCCUGGAUG ACCCGGAAGUCCGAGGAGACCAUCACCCCCUGGAACUUCGAGGAGGUGGUGGACAAGGGCGCCUCCGCCCAGUCCUUCAUCGAGCGGAUGACC AACUUCGACAAGAACCUGCCCAACGAGAAGGUGCUGCCCAAGCACUCCCUGCUGUACGAGUACUUCACCGUGUACAACGAGCUGACCAAGGUG AAGUACGUGACCGAGGGCAUGCGGAAGCCCGCCUUCCUGUCCGGCGAGCAGAAGAAGGCCAUCGUGGACCUGCUGUUCAAGACCAACCGGAAG GUGACCGUGAAGCAGCUGAAGGAGGACUACUUCAAGAAGAUCGAGUGCUUCGACUCCGUGGAGAUCUCCGGCGUGGAGGACCGGUUCAACGCC UCCCUGGGCACCUACCACGACCUGCUGAAGAUCAUCAAGGACAAGGACUUCCUGGACAACGAGGAGAACGAGGACAUCCUGGAGGACAUCGUG CUGACCCUGACCCUGUUCGAGGACCGGGAGAUGAUCGAGGAGCGGCUGAAGACCUACGCCCACCUGUUCGACGACAAGGUGAUGAAGCAGCUG AAGCGGCGGCGGUACACCGGCUGGGGCCGGCUGUCCCGGAAGCUGAUCAACGGCAUCCGGGACAAGCAGUCCGGCAAGACCAUCCUGGACUUC CUGAAGUCCGACGGCUUCGCCAACCGGAACUUCAUGCAGCUGAUCCACGACGACUCCCUGACCUUCAAGGAGGACAUCCAGAAGGCCCAGGUG UCCGGCCAGGGCGACUCCCUGCACGAGCACAUCGCCAACCUGGCCGGCUCCCCCGCCAUCAAGAAGGGCAUCCUGCAGACCGUGAAGGUGGUG GACGAGCUGGUGAAGGUGAUGGGCCGGCACAAGCCCGAGAACAUCGUGAUCGAGAUGGCCCGGGAGAACCAGACCACCCAGAAGGGCCAGAAG AACUCCCGGGAGCGGAUGAAGCGGAUCGAGGAGGGCAUCAAGGAGCUGGGCUCCCAGAUCCUGAAGGAGCACCCCGUGGAGAACACCCAGCUG CAGAACGAGAAGCUGUACCUGUACUACCUGCAGAACGGCCGGGACAUGUACGUGGACCAGGAGCUGGACAUCAACCGGCUGUCCGACUACGAC GUGGACCACAUCGUGCCCCAGUCCUUCCUGAAGGACGACUCCAUCGACAACAAGGUGCUGACCCGGUCCGACAAGAACCGGGGCAAGUCCGAC AACGUGCCCUCCGAGGAGGUGGUGAAGAAGAUGAAGAACUACUGGCGGCAGCUGCUGAACGCCAAGCUGAUCACCCAGCGGAAGUUCGACAAC CUGACCAAGGCCGAGCGGGGCGGCCUGUCCGAGCUGGACAAGGCCGGCUUCAUCAAGCGGCAGCUGGUGGAGACCCGGCAGAUCACCAAGCAC GUGGCCCAGAUCCUGGACUCCCGGAUGAACACCAAGUACGACGAGAACGACAAGCUGAUCCGGGAGGUGAAGGUGAUCACCCUGAAGUCCAAG CUGGUGUCCGACUUCCGGAAGGACUUCCAGUUCUACAAGGUGCGGGAGAUCAACAACUACCACCACGCCCACGACGCCUACCUGAACGCCGUG GUGGGCACCGCCCUGAUCAAGAAGUACCCCAAGCUGGAGUCCGAGUUCGUGUACGGCGACUACAAGGUGUACGACGUGCGGAAGAUGAUCGCC AAGUCCGAGCAGGAGAUCGGCAAGGCCACCGCCAAGUACUUCUUCUACUCCAACAUCAUGAACUUCUUCAAGACCGAGAUCACCCUGGCCAAC GGCGAGAUCCGGAAGCGGCCCCUGAUCGAGACCAACGGCGAGACCGGCGAGAUCGUGUGGGACAAGGGCCGGGACUUCGCCACCGUGCGGAAG GUGCUGUCCAUGCCCCAGGUGAACAUCGUGAAGAAGACCGAGGUGCAGACCGGCGGCUUCUCCAAGGAGUCCAUCCUGCCCAAGCGGAACUCC GACAAGCUGAUCGCCCGGAAGAAGGACUGGGACCCCAAGAAGUACGGCGGCUUCGACUCCCCCACCGUGGCCUACUCCGUGCUGGUGGUGGCC AAGGUGGAGAAGGGCAAGUCCAAGAAGCUGAAGUCCGUGAAGGAGCUGCUGGGCAUCACCAUCAUGGAGCGGUCCUCCUUCGAGAAGAACCCC AUCGACUUCCUGGAGGCCAAGGGCUACAAGGAGGUGAAGAAGGACCUGAUCAUCAAGCUGCCCAAGUACUCCCUGUUCGAGCUGGAGAACGGC CGGAAGCGGAUGCUGGCCUCCGCCGGCGAGCUGCAGAAGGGCAACGAGCUGGCCCUGCCCUCCAAGUACGUGAACUUCCUGUACCUGGCCUCC CACUACGAGAAGCUGAAGGGCUCCCCCGAGGACAACGAGCAGAAGCAGCUGUUCGUGGAGCAGCACAAGCACUACCUGGACGAGAUCAUCGAG CAGAUCUCCGAGUUCUCCAAGCGGGUGAUCCUGGCCGACGCCAACCUGGACAAGGUGCUGUCCGCCUACAACAAGCACCGGGACAAGCCCAUC CGGGAGCAGGCCGAGAACAUCAUCCACCUGUUCACCCUGACCAACCUGGGCGCCCCCGCCGCCUUCAAGUACUUCGACACCACCAUCGACCGG AAGCGGUACACCUCCACCAAGGAGGUGCUGGACGCCACCCUGAUCCACCAGUCCAUCACCGGCCUGUACGAGACCCGGAUCGACCUGUCCCAG CUGGGCGGCGACGGCGGCGGCUCCCCCAAGAAGAAGCGGAAGGUGUCCGAGUCCGCCACCCCCGAGUCCGUGUCCGGCUGGCGGCUGUUCAAG AAGAUCUCCUGA

TABLE7-continued

V. Additional Sequences

Description	SEQ ID NO	Sequence
Amino acid sequence for Cas9 with Hibit tag	820	MDKKYSIGLDIGTNSVGWAVITDEYKVPSKKFKVLGNTDRHSIKKNLIGALLFDSGETAEATRLKRTARRRYTRRKNRICYLQEIFSNEMAKV DDSFFHRLEESFLVEEDKKHERHPIFGNIVDEVAYHEKYPTIYHLRKKLVDSTDKADLRLIYLALAHMIKFRGHFLIEGDLNPDNSDVDKLFI QLVQTYNQLFEENPINASGVDAKAILSARLSKSRRLENLIAQLPGEKKNGLFGNLIALSLGLTPNFKSNFDLAEDAKLQLSKDTYDDDLDNLL AQIGDQYADLFLAAKNLSDAILLSDILRVNTEITKAPLSASMIKRYDEHHQDLTLLKALVRQQLPEKYKEIFFDQSKNGYAGYIDGGASQEEF YKFIKPILEKMDGTEELLVKLNREDLLRKQRTFDNGSIPHQIHLGELHAILRRQEDFYPFLKDNREKIEKILTFRIPYYVGPLARGNSRFAWM TRKSEETITPWNFEEVVDKGASAQSFIERMTNFDKNLPNEKVLPKHSLLYEYFTVYNELTKVKYVTEGMRKPAFLSGEQKKAIVDLLFKTNRK VTVKQLKEDYFKKIECFDSVEISGVEDRFNASLGTYHDLLKIIKDKDFLDNEENEDILEDIVLTLTLFEDREMIEERLKTYAHLFDDKVMKQL KRRRYTGWGRLSRKLINGIRDKQSGKTILDFLKSDGFANRNFMQLIHDDSLTFKEDIQKAQVSGQGDSLHEHIANLAGSPAIKKGILQTVKVV DELVKVMGRHKPENIVIEMARENQTTQKGQKNSRERMKRIEEGIKELGSQILKEHPVENTQLQNEKLYLYYLQNGRDMYVDQELDINRLSDYD VDHIVPQSFLKDDSIDNKVLTRSDKNRGKSDNVPSEEVVKKMKNYWRQLLNAKLITQRKFDNLTKAERGGLSELDKAGFIKRQLVETRQITKH VAQILDSRMNTKYDENDKLIREVKVITLKSKLVSDFRKDFQFYKVREINNYHHAHDAYLNAVVGTALIKKYPKLESEFVYGDYKVYDVRKMIA KSEQEIGKATAKYFFYSNIMNFFKTEITLANGEIRKRPLIETNGETGEIVWDKGRDFATVRKVLSMPQVNIVKKTEVQTGGFSKESILPKRNS DKLIARKKDWDPKKYGGFDSPTVAYSVLVVAKVEKGKSKKLKSVKELLGITIMERSSFEKNPIDFLEAKGYKEVKKDLIIKLPKYSLFELENG RKRMLASAGELQKGNELALPSKYVNFLYLASHYEKLKGSPEDNEQKQLFVEQHKHYLDEIIEQISEFSKRVILADANLDKVLSAYNKHRDKPI REQAENIIHLFTLTNLGAPAAFKYFDTTIDRKRYTSTKEVLDATLIHQSITGLYETRIDLSQLGGDGGGSPKKKRKVSESATPESVSGWRLFK KIS
mRNA encoding UGI	821	GGGAAGCUCAGAAUAAACGCUCAACUUUGGCCGGAUCUGCCACCAUGACCAACCUGUCCGACAUCAUCGAGAAGGAGACCGGCAAGCAGCUGG UGAUCCAGGAGUCCAUCCUGAUGCUGCCCGAGGAGGUGGAGGAGGUGAUCGGCAACAAGCCCGAGUCCGACAUCCUGGUGCACACCGCCUACG ACGAGUCCACCGACGAGAACGUGAUGCUGCUGACCUCCGACGCCCCCGAGUACAAGCCCUGGGCCCUGGUGAUCCAGGACUCCAACGGCGAGA ACAAGAUCAAGAUGCUGUCCGGCGGCUCCAAGCGGACCGCCGACGGCUCCGAGUUCGAGUCCCCCAAGAAGAAGCGGAAGGUGGAGUGAUAGC UAGCACCAGCCUCAAGAACACCCGAAUGGAGUCUCUAAGCUACAUAAUACCAACUUACACUUUACAAAAUGUUGUCCCCCAAAAUGUAGCCAU UCGUAUCUGCUCCUAAUAAAAAGAAAGUUUCUUCACAUUCUCUCGAGAAAAAAAAAAAAUGGAAAAAAAAAAAACGGAAAAAAAAAAAAGGUA AAAAAAAAAAAUAUAAAAAAAAAAAACAUAAAAAAAAAAAACGAAAAAAAAAAAACGUAAAAAAAAAAAACUCAAAAAAAAAAAAGAUAAAAA AAAAAAACCUAAAAAAAAAAAAUGUAAAAAAAAAAAAGGAAAAAAAAAAAACGCAAAAAAAAAAAACACAAAAAAAAAAAAUGCAAAAAAAAA AAAAUCGAAAAAAAAAAAAUCUAAAAAAAAAAAACGAAAAAAAAAAAACCCAAAAAAAAAAAAGACAAAAAAAAAAAAUAGAAAAAAAAAAAA GUUAAAAAAAAAAAACUGAAAAAAAAAAAAUUUAAAAAAAAAAAAUCUAG
mRNA encoding Nme2Cas9 base editor	822	GGGAAGCUCAGAAUAAACGCUCAACUUUGGCCGGAUCUGCCACCAUGGACGGCUCCGGCGGCGGCUCCCCCAAGAAGAAGCGGAAGGUGGAGG ACAAGCGGCCCGCCGCCACCAAGAAGGCCGGCCAGGCCAAGAAGAAGAAGGGCGGCUCCGGCGGCGGCGAGGCCUCCCCCGCCUCCGGCCCCC GGCACCUGAUGGACCCCCACAUCUUCACCUCCAACUUCAACAACGGCAUCGGCCGGCACAAGACCUACCUGUGCUACGAGGUGGAGCGGCUGG ACAACGGCACCUCCGUGAAGAUGGACCAGCACCGGGGCUUCCUGCACAACCAGGCCAAGAACCUGCUGUGCGGCUUCUACGGCCGGCACGCCG AGCUGCGGUUCCUGGACCUGGUGCCCUCCCUGCAGCUGGACCCCGCCCAGAUCUACCGGGUGACCUGGUUCAUCUCCUGGUCCCCCUGCUUCU CCUGGGGCUGCGCCGGCGAGGUGCGGGCCUUCCUGCAGGAGAACACCCACGUGCGGCUGCGGAUCUUCGCCGCCCGGAUCUACGACUACGACC CCCUGUACAAGGAGGCCCUGCAGAUGCUGCGGGACGCCGGCGCCCAGGUGUCCAUCAUGACCUACGACGAGUUCAAGCACUGCUGGGACACCU UCGUGGACCACCAGGGCUGCCCCUUCCAGCCCUGGGACGGCCUGGACGAGCACUCCCAGGCCCUGUCCGGCCGGCUGCGGGCCAUCCUGCAGA ACCAGGGCAACUCCGGCUCCGAGACCCCCGGCACCUCCGAGUCCGCCACCCCCGAGUCCGCAGCGUUCAAACCAAAUCCCAUCAACUACAUCC UGGGCCUGGCCAUCGGCAUCGCCUCCGUGGGCUGGGCCAUGGUGGAGAUCGACGAGGAGGAGAACCCCAUCCGGCUGAUCGACCUGGGCGUGC GGGUGUUCGAGCGGGCCGAGGUGCCCAAGACCGGCGACUCCCUGGCCAUGGCCCGGCGGCUGGCCCGGUCCGUGCGGCGGCUGACCCGGCGGC GGGCCCACCGGCUGCUGCGGGCCCGGCGGCUGCUGAAGCGGGAGGGCGUGCUGCAGGCCGCCGACUUCGACGAGAACGGCCUGAUCAAGUCCC UGCCCAACACCCCCUGGCAGCUGCGGGCCGCCGCCCUGGACCGGAAGCUGACCCCCCUGGAGUGGUCCGCCGUGCUGCUGCACCUGAUCAAGC ACCGGGGCUACCUGUCCCAGCGGAAGAACGAGGGCGAGACCGCCGACAAGGAGCUGGGCGCCCUGCUGAAGGGCGUGGCCAACAACGCCCACG CCCUGCAGACCGGCGACUUCCGGACCCCCGCCGAGCUGGCCCUGAACAAGUUCGAGAAGGAGUCCGGCCACAUCCGGAACCAGCGGGGCGACU ACUCCCACACCUUCUCCCGGAAGGACCUGCAGGCCGAGCUGAUCCUGCUGUUCGAGAAGCAGAAGGAGUUCGGCAACCCCCACGUGUCCGGCG GCCUGAAGGAGGGCAUCGAGACCCUGCUGAUGACCCAGCGGCCCGCCCUGUCCGGCGACGCCGUGCAGAAGAUGCUGGGCCACUGCACCUUCG AGCCCGCCGAGCCCAAGGCCGCCAAGAACACCUACACCGCCGAGCGGUUCAUCUGGCUGACCAAGCUGAACAACCUGCGGAUCCUGGAGCAGG GCUCCGAGCGGCCCCUGACCGACACCGAGCGGGCCACCCUGAUGGACGAGCCCUACCGGAAGUCCAAGCUGACCUACGCCCAGGCCCGGAAGC UGCUGGGCCUGGAGGACACCGCCUUCUUCAAGGGCCUGCGGUACGGCAAGGACAACGCCGAGGCCUCCACCCUGAUGGAGAUGAAGGCCUACC

TABLE7-continued

V. Additional Sequences

Description	SEQ ID NO	Sequence
		ACGCCAUCUCCCGGGCCCUGGAGAAGGAGGGCCUGAAGGACAAGAAGUCCCCCCUGAACCUGUCCUCCGAGCUGCAGGACGAGAUCGGCACCG CCUUCUCCCUGUUCAAGACCGACGAGGACAUCACCGGCCGGCUGAAGGACCGGGUGCAGCCCGAGAUCCUGGAGGCCCUGCUGAAGCACAUCU CCUUCGACAAGUUCGUGCAGAUCUCCCUGAAGGCCCUGCGGCGGAUCGUGCCCCUGAUGGAGCAGGGCAAGCGGUACGACGAGGCCUGCGCCG AGAUCUACGGCGACCACUACGGCAAGAAGAACACCGAGGAGAAGAUCUACCUGCCCCCCAUCCCCGCCGACGAGAUCCGGAACCCCGUGGUGC UGCGGGCCCUGUCCCAGGCCCGGAAGGUGAUCAACGGCGUGGUGCGGCGGUACGGCUCCCCCGCCCGGAUCCACAUCGAGACCGCCCGGGAGG UGGGCAAGUCCUUCAAGGACCGGAAGGAGAUCGAGAAGCGGCAGGAGGAGAACCGGAAGGACCGGGAGAAGGCCGCCGCCAAGUUCCGGGAGU ACUUCCCCAACUUCGUGGGCGAGCCCAAGUCCAAGGACAUCCUGAAGCUGCGGCUGUACGAGCAGCAGCACGGCAAGUGCCUGUACUCCGGCA AGGAGAUCAACCUGGUGCGGCUGAACGAGAAGGGCUACGUGGAGAUCGACCACGCCCUGCCCUUCUCCCGGACCUGGGACGACUCCUUCAACA ACAAGGUGCUGGUGCUGGGCUCCGAGAACCAGAACAAGGGCAACCAGACCCCCUACGAGUACUUCAACGGCAAGGACAACUCCCGGGAGUGGC AGGAGUUCAAGGCCCGGGUGGAGACCUCCCGGUUCCCCCGGUCCAAGAAGCAGCGGAUCCUGCUGCAGAAGUUCGACGAGGACGGCUUCAAGG AGUGCAACCUGAACGACACCCGGUACGUGAACCGCUUCCUGUGCCAGUUCGUGGCCGACCACAUCCUGCUGACCGGCAAGGGCAAGCGGCGGG UGUUCGCCUCCAACGGCCAGAUCACCAACCUGCUGCGGGGCUUCUGGGGCCUGCGGAAGGUGCGGGCCGAGAACGACCGGCACCACGCCCUGG ACGCCGUGGUGGUGGCCUGCUCCACCGUGGCCAUGCAGCAGAAGAUCACCCGGUUCGUGCGGUACAAGGAGAUGAACGCCUUCGACGGCAAGA CCAUCGACAAGGAGACCGGCAAGGUGCUGCACCAGAAGACCCACUUCCCCCAGCCCUGGGAGUUCUUCGCCCAGGAGGUGAUGAUCCGGGUGU UCGGCAAGCCCGACGGCAAGCCCGAGUUCGAGGAGGCCGACACCCCCGAGAAGCUGCGGACCCUGCUGGCCGAGAAGCUGUCCUCCCGGCCCG AGGCCGUGCACGAGUACGUGACCCCCCUGUUCGUGUCCCGGGCCCCCAACCGGAAGAUGUCCGGCGCCCACAAGGACACCCUGCGGUCCGCCA AGCGGUUCGUGAAGCACAACGAGAAGAUCUCCGUGAAGCGGGUGUGGCUGACCGAGAUCAAGCUGGCCGACCUGGAGAACAUGGUGAACUACA AGAACGGCCGGGAGAUCGAGCUGUACGAGGCCCUGAAGGCCCGGCUGGAGGCCUACGGCGGCAACGCCAAGCAGGCCUUCGACCCCAAGGACA ACCCCUUCUACAAGAAGGGCGGCCAGCUGGUGAAGGCCGUGCGGGUGGAGAAGACCCAGGAGUCCGGCGUGCUGCUGAACAAGAAGAACGCCU ACACCAUCGCCGACAACGGCGACAUGGUGCGGGUGGACGUGUUCUGCAAGGUGGACAAGAAGGGCAAGAACCAGUACUUCAUCGUGCCCAUCU ACGCCUGGCAGGUGGCCGAGAACAUCCUGCCCGACAUCGACUGCAAGGGCUACCGGAUCGACGACUCCUACACCUUCUGCUUCUCCCUGCACA AGUACGACCUGAUCGCCUUCCAGAAGGACGAGAAGUCCAAGGUGGAGUUCGCCUACUACAUCAACUGCGACUCCUCCAACGGCCGGUUCUACC UGGCCUGGCACGACAAGGGCUCCAAGGAGCAGCAGUUCCGGAUCUCCACCCAGAACCUGGUGCUGAUCCAGAAGUACCAGGUGAACGAGCUGG GCAAGGAGAUCCGGCCCUGCCGGCUGAAGAAGCGGCCCCCCGUGCGGUAGUGACUAGCACCAGCCUCAAGAACACCCGAAUGGAGUCUCUAAG CUACAUAAUACCAACUUACACUUUACAAAAUGUUGUCCCCCAAAAUGUAGCCAUUCGUAUCUGCUCCUAAUAAAAAGAAAGUUUCUUCACAUU CUCUCGAGAAAAAAAAAAAAUGGAAAAAAAAAAAACGGAAAAAAAAAAAGGUAAAAAAAAAAAAUAUAAAAAAAAAAACAUAAAAAAAAAAAA CGAAAAAAAAAAAACGUAAAAAAAAAAAACUCAAAAAAAAAAAGAUAAAAAAAAAAAACCUAAAAAAAAAAAAUGUAAAAAAAAAAAAGGGAA AAAAAAAAACGCAAAAAAAAAAAACACAAAAAAAAAAAAUGCAAAAAAAAAAAAUCGAAAAAAAAAAAAUCUAAAAAAAAAAAACGAAAAAAA AAAAACCCAAAAAAAAAAAAGACAAAAAAAAAAAAUAGAAAAAAAAAAAGUUAAAAAAAAAAAACUGAAAAAAAAAAAAUUUAAAAAAAAAAA AUCUAG
Open reading frame for UGI encoded by SEQ ID NO: 821	824	ATGACCAACCTGTCCGACATCATCGAGAAGGAGACCGGCAAGCAGCTGGTGATCCAGGAGTCCATCCTGATGCTGCCCGAGGAGGTGGAGGAG GTGATCGGCAACAAGCCCGAGTCCGACATCCTGGTGCACACCGCCTACGACGAGTCCACCGACGAGAACGTGATGCTGCTGACCTCCGACGCC CCCGAGTACAAGCCCTGGGCCCTGGTGATCCAGGACTCCAACGGCGAGAACAAGATCAAGATGCTGTCCGGCGGCTCCAAGCGGACCGCCGAC GGCTCCGAGTTCGAGTCCCCCAAGAAGAAGCGGAAGGTGGAGTGATAG
MRNA encoding Nme2 Cas9	825	GGGAAGCUCAGAAUAAACGCUCAACUUUGGCCGGAUCUGCCACCAUGGACGGCUCCGGCGGCGGCUCCCCCAAGAAGAAGCGGAAGGUGGAGG ACAAGCGGCCCGCCGCCACCAAGAAGGCCGGCCAGGCCAAGAAGAAGAAGGGCGGCUCCGGCGGCGGCGCCGCCUUCAAGCCCAACCCCAUCA ACUACAUCCUGGGCCUGGACAUCGGCAUCGCCUCCGUGGGCUGGGCCAUGGUGGAGAUCGACGAGGAGGAGAACCCCAUCCGGCUGAUCGACC UGGGCGUGCGGGUGUUCGAGCGGGCCGAGGUGCCCAAGACCGGCGACUCCCUGGCCAUGGCCCGGCGGCUGGCCCGGUCCGUGCGGCGGCUGA CCCGGCGGCGGGCCCACCGGCUGCUGCGGGCCCGGCGGCUGCUGAAGCGGGAGGGCGUGCUGCAGGCCGCCGACUUCGACGAGAACGGCCUGA UCAAGUCCCUGCCCAACACCCCCUGGCAGCUGCGGGCCGCCGCCCUGGACCGGAAGCUGACCCCCCUGGAGUGGUCCGCCGUGCUGCUGCACC UGAUCAAGCACCGGGGCUACCUGUCCCAGCGGAAGAACGAGGGCGAGACCGCCGACAAGGAGCUGGGCGCCCUGCUGAAGGGCGUGGCCAACA ACGCCCACGCCCUGCAGACCGGCGACUUCCGGACCCCCGCCGAGCUGGCCCUGAACAAGUUCGAGAAGGAGUCCGGCCACAUCCGGAACCAGC GGGGCGACUACUCCCACACCUUCUCCCGGAAGGACCUGCAGGCCGAGCUGAUCCUGCUGUUCGAGAAGCAGAAGGAGUUCGGCAACCCCCACG UGUCCGGCGGCCUGAAGGAGGGCAUCGAGACCCUGCUGAUGACCCAGCGGCCCGCCCUGUCCGGCGACGCCGUGCAGAAGAUGCUGGGCCACU GCACCUUCGAGCCCGCCGAGCCCAAGGCCGCCAAGAACACCUACACCGCCGAGCGGUUCAUCUGGCUGACCAAGCUGAACAACCUGCGGAUCC UGGAGCAGGGCUCCGAGCGGCCCCUGACCGACACCGAGCGGGCCACCCUGAUGGACGAGCCCUACCGGAAGUCCAAGCUGACCUACGCCCAGG CCCGGAAGCUGCUGGGCCUGGAGGACACCGCCUUCUUCAAGGCCUGCGGUACGGCAAGGACAACGCCGAGGCCUCCACCCUGAUGGAGAUGA

TABLE7-continued

V. Additional Sequences

Description	SEQ ID NO	Sequence
		AGGCCUACCACGCCAUCUCCCGGGCCCUGGAGAAGGAGGGCCUGAAGGACAAGAAGUCCCCCCUGAACCUGUCCUCCGAGCUGCAGGACGAGA UCGGCACCGCCUUCUCCCUGUUCAAGACCGACGAGGACAUCACCGGCCGGCUGAAGGACCGGGUGCAGCCCGAGAUCCUGGAGGCCCUGCUGA AGCACAUCUCCUUCGACAAGUUCGUGCAGAUCUCCCUGAAGGCCCUGCGGCGGAUCGUGCCCCUGAUGGAGCAGGGCAAGCGGUACGACGAGG CCUGCGCCGAGAUCUACGGCGACCACUACGGCAAGAAGAACACCGAGGAGAAGAUCUACCUGCCCCCCAUCCCCGCCGACGAGAUCCGGAACC CCGUGGUGCUGCGGGCCCUGUCCCAGGCCCGGAAGGUGAUCAACGGCGUGGUGCGGCGGUACGGCUCCCCCGCCCGGAUCCACAUCGAGACCG CCCGGGAGGUGGGCAAGUCCUUCAAGGACCGGAAGGAGAUCGAGAAGCGGCAGGAGGAGAACCGGAAGGACCGGGAGAAGGCCGCCGCCAAGU UCCGGGAGUACUUCCCCAACUUCGUGGGCGAGCCCAAGUCCAAGGACAUCCUGAAGCUGCGGCUGUACGAGCAGCAGCACGGCAAGUGCCUGU ACUCCGGCAAGGAGAUCAACCUGGUGCGGCUGAACGAGAAGGGCUACGUGGAGAUCGACCACGCCCUGCCCUUCUCCCGGACCUGGGACGACU CCUUCAACAACAAGGUGCUGGUGCUGGGCUCCGAGAACCAGAACAAGGGCAACCAGACCCCCUACGAGUACUUCAACGGCAAGGACAACUCCC GGGAGUGGCAGGAGUUCAAGGCCCGGGUGGAGACCUCCCGGUUCCCCCGGUCCAAGAAGCAGCGGAUCCUGCUGCAGAAGUUCGACGAGGACG GCUUCAAGGAGUGCAACCUGAACGACACCCGGUACGUGAACCGGUUCCUGUGCCAGUUCGUGGCCGACCACAUCCUGCUGACCGGCAAGGGCA AGCGGCGGGUGUUCGCCUCCAACGGCCAGAUCACCAACCUGCUGCGGGGCUUCUGGGGCCUGCGGAAGGUGCGGGCCGAGAACGACCGGCACC ACGCCCUGGACGCCGUGGUGGUGGCCUGCUCCACCGUGGCCAUGCAGCAGAAGAUCACCCGGUUCGUGCGGUACAAGGAGAUGAACGCCUUCG ACGGCAAGACCAUCGACAAGGAGACCGGCAAGGUGCUGCACCAGAAGACCCACUUCCCCCAGCCCUGGGAGUUCUUCGCCCAGGAGGUGAUGA UCCGGGUGUUCGGCAAGCCCGACGGCAAGCCCGAGUUCGAGGAGGCCGACACCCCCGAGAAGCUGCGGACCCUGCUGGCCGAGAAGCUGUCCU CCCGGCCCGAGGCCGUGCACGAGUACGUGACCCCCCUGUUCGUGUCCCGGGCCCCCAACCGGAAGAUGUCCGGCGCCCACAAGGACACCCUGC GGUCCGCCAAGCGGUUCGUGAAGCACAACGAGAAGAUCUCCGUGAAGCGGGUGUGGCUGACCGAGAUCAAGCUGGCCGACCUGGAGAACAUGG UGAACUACAAGAACGGCCGGGAGAUCGAGCUGUACGAGGCCCUGAAGGCCCGGCUGGAGGCCUACGGCGGCAACGCCAAGCAGGCCUUCGACC CCAAGGACAACCCCUUCUACAAGAAGGGCGGCCAGCUGGUGAAGGCCGUGCGGGUGGAGAAGACCCAGGAGUCCGGCGUGCUGCUGAACAAGA AGAACGCCUACACCAUCGCCGACAACGGCGACAUGGUGCGGGUGGACGUGUUCUGCAAGGUGGACAAGAAGGGCAAGAACCAGUACUUCAUCG UGCCCAUCUACGCCUGGCAGGUGGCCGAGAACAUCCUGCCCGACAUCGACUGCAAGGGCUACCGGAUCGACGACUCCUACACCUUCUGCUUCU CCCUGCACAAGUACGACCUGAUCGCCUUCCAGAAGGACGAGAAGUCCAAGGUGGAGUUCGCCUACUACAUCAACUGCGACUCCUCCAACGGCC GGUUCUACCUGGCCUGGCACGACAAGGGCUCCAAGGAGCAGCAGUUCCGGAUCUCCACCCAGAACCUGGUGCUGAUCCAGAAGUACCAGGUGA ACGAGCUGGGCAAGGAGAUCCGGCCCUGCCGGCUGAAGAAGCGGCCCCCCGUGCGGUAGCUAGCACCAGCCUCAAGAACACCCGAAUGGAGUC UCUAAGCUACAUAAUACCAACUUACACUUUACAAAAUGUUGUCCCCCAAAAUGUAGCCAUUCGUAUCUGCUCCUAAUAAAAAGAAAGUUUCUU CACAUUCUCUCGAGAAAAAAAAAAAAAUGGAAAAAAAAAAAAACGGAAAAAAAAAAAAAGGUAAAAAAAAAAAAAUAUAAAAAAAAAAAAACAUAAAA AAAAAAAACGAAAAAAAAAAAAACGUAAAAAAAAAAAAACUCAAAAAAAAAAAAGAUAAAAAAAAAAAAACCUAAAAAAAAAAAAAUGUAAAAAAAAA AAAGGGAAAAAAAAAAAAACGCAAAAAAAAAAAAACACAAAAAAAAAAAAAUGCAAAAAAAAAAAAAUCGAAAAAAAAAAAAAUCUAAAAAAAAAAAA CGAAAAAAAAAAAAACCCAAAAAAAAAAAAAGACAAAAAAAAAAAAAUAGAAAAAAAAAAAAGUUAAAAAAAAAAAAACUGAAAAAAAAAAAAAAUUUAA AAAAAAAAAAUCUAG
mRNA encoding Nme2 Cas9 with Hibit tag	826	GGGAAGCUCAGAAUAAACGCUCAACUUUGGCCGGAUCUGCCACCAUGGACGGCUCCGGCGGCGGCUCCCCCAAGAAGAAGCGGAAGGUGGAGG ACAAGCGGCCCGCCGCCACCAAGAAGGCCGGCCAGGCCAAGAAGAAGAAGGGCGGCUCCGGCGGCGGCGCCGCCUUCAAGCCCAACCCCAUCA ACUACAUCCUGGGCCUGGACAUCGGCAUCGCCUCCGUGGGCUGGGCCAUGGUGGAGAUCGACGAGGAGGAGAACCCCAUCCGGCUGAUCGACC UGGGCGUGCGGGUGUUCGAGCGGGCCGAGGUGCCCAAGACCGGCGACUCCCUGGCCAUGGCCCGGCGGCUGGCCCGGUCCGUGCGGCGGCUGA CCCGGCGGCGGGCCCACCGGCUGCUGCGGGCCCGGCGGCUGCUGAAGCGGGAGGGCGUGCUGCAGGCCGCCGACUUCGACGAGAACGGCCUGA UCAAGUCCCUGCCCAACACCCCCUGGCAGCUGCGGGCCGCCGCCCUGGACCGGAAGCUGACCCCCCUGGAGUGGUCCGCCGUGCUGCUGCACC UGAUCAAGCACCGGGGCUACCUGUCCCAGCGGAAGAACGAGGGCGAGACCGCCGACAAGGAGCUGGGCGCCCUGCUGAAGGGCGUGGCCAACA ACGCCCACGCCCUGCAGACCGGCGACUUCCGGACCCCCGCCGAGCUGGCCCUGAACAAGUUCGAGAAGGAGUCCGGCCACAUCCGGAACCAGC GGGGCGACUACUCCCACACCUUCUCCCGGAAGGACCUGCAGGCCGAGCUGAUCCUGCUGUUCGAGAAGCAGAAGGAGUUCGGCAACCCCCACG UGUCCGGCGGCCUGAAGGAGGGCAUCGAGACCCUGCUGAUGACCCAGCGGCCCGCCCUGUCCGGCGACGCCGUGCAGAAGAUGCUGGGCCACU GCACCUUCGAGCCCGCCGAGCCCAAGGCCGCCAAGAACACCUACACCGCCGAGCGGUUCAUCUGGCUGACCAAGCUGAACAACCUGCGGAUCC UGGAGCAGGGCUCCGAGCGGCCCCUGACCGACACCGAGCGGGCCACCCUGAUGGACGAGCCCUACCGGAAGUCCAAGCUGACCUACGCCCAGG CCCGGAAGCUGCUGGGCCUGGAGGACACCGCCUUCUUCAAGGGCCUGCGGUACGGCAAGGACAACGCCGAGGCCUCCACCCUGAUGGAGAUGA AGGCCUACCACGCCAUCUCCCGGGCCCUGGAGAAGGAGGGCCUGAAGGACAAGAAGUCCCCCCUGAACCUGUCCUCCGAGCUGCAGGACGAGA UCGGCACCGCCUUCUCCCUGUUCAAGACCGACGAGGACAUCACCGGCCGGCUGAAGGACCGGGUGCAGCCCGAGAUCCUGGAGGCCCUGCUGA AGCACAUCUCCUUCGACAAGUUCGUGCAGAUCUCCCUGAAGGCCCUGCGGCGGAUCGUGCCCCUGAUGGAGCAGGGCAAGCGGUACGACGAGG CCUGCGCCGAGAUCUACGGCGACCACUACGGCAAGAAGAACACCGAGGAGAAGAUCUACCUGCCCCCCAUCCCCGCCGACGAGAUCCGGAACC CCGUGGUGCUGCGGGCCCUGUCCCAGGCCCGGAAGGUGAUCAACGGCGUGGUGCGGCGGUACGGCUCCCCCGCCCGGAUCCACAUCGAGACCG

TABLE7-continued

V. Additional Sequences

Description	SEQ ID NO	Sequence
		CCCGGGAGGUGGGCAAGUCCUUCAAGGACCGGAAGGAGAUCGAGAAGCGGCAGGAGGAGAACCGGAAGGACCGGGAGAAGGCCGCCGCCAAGU UCCGGGAGUACUUCCCCAACUUCGUGGGCGAGCCCAAGUCCAAGGACAUCCUGAAGCUGCGGCUGUACGAGCAGCAGCACGGCAAGUGCCUGU ACUCCGGCAAGGAGAUCAACCUGGUGCGGCUGAACGAGAAGGGCUACGUGGAGAUCGACCACGCCCUGCCCUUCUCCCGGACCUGGGACGACU CCUUCAACAACAAGGUGCUGGUGCUGGGCUCCGAGAACCAGAACAAGGGCAACCAGACCCCCUACGAGUACUUCAACGGCAAGGACAACUCCC GGGAGUGGCAGGAGUUCAAGGCCCGGGUGGAGACCUCCCGGUUCCCCCGGUCCAAGAAGCAGCGGAUCCUGCUGCAGAAGUUCGACGAGGACG GCUUCAAGGAGUGCAACCUGAACGACACCCGGUACGUGAACCGGUUCCUGUGCCAGUUCGUGGCCGACCACAUCCUGCUGACCGGCAAGGGCA AGCGGCGGGUGUUCGCCUCCAACGGCCAGAUCACCAACCUGCUGCGGGGCUUCUGGGGCCUGCGGAAGGUGCGGGCCGAGAACGACCGGCACC ACGCCCUGGACGCCGUGGUGGUGGCCUGCUCCACCGUGGCCAUGCAGCAGAAGAUCACCCGGUUCGUGCGGUACAAGGAGAUGAACGCCUUCG ACGGCAAGACCAUCGACAAGGAGACCGGCAAGGUGCUGCACCAGAAGACCCACUUCCCCCAGCCCUGGGAGUUCUUCGCCCAGGAGGUGAUGA UCCGGGUGUUCGGCAAGCCCGACGGCAAGCCCGAGUUCGAGGAGGCCGACACCCCCGAGAAGCUGCGGACCCUGCUGGCCGAGAAGCUGUCCU CCCGGCCCGAGGCCGUGCACGAGUACGUGACCCCCCUGUUCGUGUCCCGGGCCCCCAACCGGAAGAUGUCCGGCGCCCACAAGGACACCCUGC GGUCCGCCAAGCGGUUCGUGAAGCACAACGAGAAGAUCUCCGUGAAGCGGGUGUGGCUGACCGAGAUCAAGCUGGCCGACCUGGAGAACAUGG UGAACUACAAGAACGGCCGGGAGAUCGAGCUGUACGAGGCCCUGAAGGCCCGGCUGGAGGCCUACGGCGGCAACGCCAAGCAGGCCUUCGACC CCAAGGACAACCCCUUCUACAAGAAGGGCGGCCAGCUGGUGAAGGCCGUGCGGGUGGAGAAGACCCAGGAGUCCGGCGUGCUGCUGAACAAGA AGAACGCCUACACCAUCGCCGACAACGGCGACAUGGUGCGGGUGGACGUGUUCUGCAAGGUGGACAAGAAGGGCAAGAACCAGUACUUCAUCG UGCCCAUCUACGCCUGGCAGGUGGCCGAGAACAUCCUGCCCGACAUCGACUGCAAGGGCUACCGGAUCGACGACUCCUACACCUUCUGCUUCU CCCUGCACAAGUACGACCUGAUCGCCUUCCAGAAGGACGAGAAGUCCAAGGUGGAGUUCGCCUACUACAUCAACUGCGACUCCUCCAACGGCC GGUUCUACCUGGCCUGGCACGACAAGGGCUCCAAGGAGCAGCAGUUCCGGAUCUCCACCCAGAACCUGGUGCUGAUCCAGAAGUACCAGGUGA ACGAGCUGGGCAAGGAGAUCCGGCCCUGCCGGCUGAAGAAGCGGCCCCCGUGCGGUCCGAGUCCGCCACCCCCGAGUCCGUGUCCGGCUGGC GGCUGUUCAAGAAGAUCUCCUAGCUAGCACCAGCCUCAAGAACACCCGAAUGGAGUCUCUAAGCUACAUAAUACCAACUUACACUUUACAAAA UGUUGUCCCCCAAAAUGUAGCCAUUCGUAUCUGCUCCUAAUAAAAAGAAAGUUUCUUCACAUUCUCUCGAGAAAAAAAAAAAAAUGGAAAAAAA AAAAACGGAAAAAAAAAAAAAGGUAAAAAAAAAAAAAUAUAAAAAAAAAAAAACAUAAAAAAAAAAAAACGAAAAAAAAAAAAACGUAAAAAAAAAAAA ACUCAAAAAAAAAAAAAGAUAAAAAAAAAAAAACCUAAAAAAAAAAAAAUGUAAAAAAAAAAAAAGGAAAAAAAAAAAAACGCAAAAAAAAAAAAACA CAAAAAAAAAAAAAUGCAAAAAAAAAAAAAUCGAAAAAAAAAAAAAUCUAAAAAAAAAAAAACGAAAAAAAAAAAAACCCAAAAAAAAAAAAAGACAAA AAAAAAAAAUAGAAAAAAAAAAAAAGUUAAAAAAAAAAAAACUGAAAAAAAAAAAAAUUUAAAAAAAAAAAAAUCUAG
Wild-type NmeCas9 guide RNA-conserved portion only	899	GUUGUAGCUCCCUUUCUCAUUUCGGAAACGAAAUGAGAACCGUUGCUACAAUAAGGCCGUCUGAAAAGAUGUGCCGCAACGCUCUGCCCCUUA AAGCUUCUGCUUUAAGGGGCAUCGUUUA
Wild-type NmeCas9 guide RNA	900	NNNNNNNNNNNNNNNNNNNNNNNNGUUGUAGCUCCCUUUCUCAUUUCGGAAACGAAAUGAGAACCGUUGCUACAAUAAGGCCGUCUGAAAAGA UGUGCCGCAACGCUCUGCCCCUUAAAGCUUCUGCUUUAAGGGGCAUCGUUUA
Shortened/unmodified NmeCas9 guide RNA motif	901	$(N)_{20-25}$ GUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU
Shortened/unmodified NmeCas9 guide RNA motif	902	$(N)_{20-25}$ GUUGUAGCUCCCUGAAACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUUUAUU
Shortened/unmodified NmeCas9 guide RNA motif	903	$(N)_{20-25}$ GUUGUAGCUCCCUGGAAACCCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUUUAUU
Shortened/modified NmeCas9 guide RNA motif	904	mN*mNNNNNNNNmNNNmNNNNNNNNNNNNmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmA mAmAmGmAmUGUGCmCGCmAmAmCmGCUCUmGmCCmUmUmCmUGmGCmAmUC*mG*mU*mU

TABLE7-continued

V. Additional Sequences		
Description	SEQ ID NO	Sequence
(101-mer)		
Shortened/modified NmeCas9 guide RNA motif	905	$(N)_{20-25}$GUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGCmAmAmCmGCUCUmGmCCmUmUmCmUGmGCmAmUC*mG*mU*mU
Shortened/modified NmeCas9 guide RNA motif	906	GUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGCmAmAmCmGCUCUmGmCCmUmUmCmUGmGCmAmUC*mG*mU*mU
Shortened/modified NmeCas9 guide RNA motif (101-mer)	907	mN*mN*mN*mNmNNNmNmNNmNNmNNNNNmNNNNmNNNmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU
Shortened/modified NmeCas9 guide RNA motif	908	$(N)_{20-25}$ GUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU
Shortened/modified NmeCas9 guide RNA motif	909	GUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU
Shortened/unmodified NmeCas9 guide RNA motif	910	$(N)_{20-25}$GUUGUAGCUCCCUUCGAAAGACCGUUGCUACAAUAAGGCCGUCGAAAGAUGUGCCGCAACGCUCUGCCUUCUGGCAUCGUU
Shortened/modified NmeCas9 guide RNA motif (105-mer)	911	mN*mN*mN*mNmNNNmNmNNmNNmNNNNNmNNNNmNNNmGUUGmUmAmGmCUCCCmUmUmCmGmAmAmAmGmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU
Exemplary NLS1	912	LAAKRSRTT
Exemplary NLS2	913	QAAKRSRTT
Exemplary NLS3	914	PAPAKRERTT
Exemplary NLS4	915	QAAKRPRTT
Exemplary NLS5	916	RAAKRPRTT
Exemplary NLS6	917	AAAKRSWSMAA
Exemplary NLS7	918	AAAKRVWSMAF
Exemplary NLS8	919	AAAKRSWSMAF

TABLE7-continued

V. Additional Sequences		
Description	SEQ ID NO	Sequence
Exemplary NLS9	920	AAAKRKYFAA
Exemplary NLS10	921	RAAKRKAFAA
Exemplary NLS11	922	RAAKRKYFAV
Altern-ative SV40 NLS	923	PKKKRRV
Nucleo-plasmin MLS	924	KRPAATKKAGQAKKKK
exemplary XTEN	930	SGSETPGTSESATPES
exemplary XTEN	931	SGSETPGTSESA
exemplary XTEN	932	SGSETPGTSESATPEGGSGGS
amino acid sequence for exemplary linker	933	GGS
amino acid sequence for exemplary linker	934	GGGGS
amino acid sequence for exemplary linker	935	EAAAK
amino acid sequence for exemplary linker	936	SEGSA
amino acid sequence for exemplary linker	937	SEGSAGTST
amino acid sequence for exemplary linker	938	GGGGSGGGGS
amino acid sequence for exemplary linker	939	GGGGSEAAAK

TABLE7-continued

V. Additional Sequences

Description	SEQ ID NO	Sequence
amino acid sequence for exemplary linker	940	EAAAKGGGGS
amino acid sequence for exemplary linker	941	EAAAKEAAAK
amino acid sequence for exemplary linker	942	SEGSAGTSTESEGSA
amino acid sequence for exemplary linker	943	GGGGSGGGGSGGGGS
amino acid sequence for exemplary linker	944	GGGGSGGGGSEAAAK
amino acid sequence for exemplary linker	945	GGGGSEAAAKGGGGS
amino acid sequence for exemplary linker	946	EAAAKGGGGSEAAAK
amino acid sequence for exemplary linker	947	EAAAKEAAAKGGGGS
amino acid sequence for exemplary linker	948	SEGSAGTSTESEGSAGTSTE
amino acid sequence for exemplary linker	949	GGGGSGGGGSGGGGSEAAAK
amino acid sequence for exemplary linker	950	GGGGSGGGGSEAAAKGGGGS
amino acid sequence for exemplary linker	951	GGGGSEAAAKGGGGSEAAAK
amino acid sequence for exemplary linker	952	GGGGSEAAAKEAAAKGGGGS
amino acid sequence for exemplary linker	953	GGGGSEAAAKEAAAKEAAAK
amino acid sequence for exemplary linker	954	EAAAKGGGGSGGGGSGGGGS
amino acid sequence for exemplary linker	955	EAAAKGGGGSGGGGSEAAAK

TABLE7-continued

Description	SEQ ID NO	Sequence
V. Additional Sequences		
amino acid sequence for exemplary linker	956	EAAAKGGGGSEAAAKGGGGS
amino acid sequence for exemplary linker	957	EAAAKGGGGSEAAAKEAAAK
amino acid sequence for exemplary linker	958	EAAAKEAAAKGGGGSGGGGS
amino acid sequence for exemplary linker	959	EAAAKEAAAKGGGGSEAAAK
amino acid sequence for exemplary linker	960	EAAAKEAAAKEAAAKGGGGS
amino acid sequence for exemplary linker	961	SEGSAGTSTESEGSAGTSTESEGSA
amino acid sequence for exemplary linker	962	GGGGSGGGGSGGGGSGGGGSGGGGS
amino acid sequence for exemplary linker	963	GGGGGGGGSGGGGSGGGGSEAAAK
amino acid sequence for exemplary linker	964	GGGGSGGGGSGGGGSEAAAKGGGGS
amino acid sequence for exemplary linker	965	GGGGSGGGGSGGGGSEAAAKEAAAK
amino acid sequence for exemplary linker	966	GGGGSGGGGSEAAAKGGGGSGGGGS
amino acid sequence for exemplary linker	967	GGGGSGGGGSEAAAKGGGGSEAAAK
amino acid sequence for exemplary linker	968	GGGGSGGGGSEAAAKEAAAKGGGGS
amino acid sequence for exemplary linker	969	GGGGSGGGGSEAAAKEAAAKEAAAK
amino acid sequence for exemplary linker	970	GGGGSEAAAKGGGGSGGGGSGGGGS
amino acid sequence for exemplary linker	971	GGGGSEAAAKGGGGSGGGGSEAAAK
amino acid sequence for exemplary linker	972	GGGGSEAAAKGGGGSEAAAKGGGGS

TABLE7-continued

Description	SEQ ID NO	Sequence
V. Additional Sequences		
amino acid sequence for exemplary linker	973	GGGGSEAAAKGGGGSEAAAKEAAAK
amino acid sequence for exemplary linker	974	GGGGSEAAAKEAAAKGGGGSGGGGS
amino acid sequence for exemplary linker	975	GGGGSEAAAKEAAAKEAAAKGGGGS
amino acid sequence for exemplary linker	976	GGGGSEAAAKEAAAKEAAAKEAAAK
amino acid sequence for exemplary linker	977	EAAAKGGGGSGGGGSGGGGSGGGGS
amino acid sequence for exemplary linker	978	EAAAKGGGGSGGGGSGGGGSEAAAK
amino acid sequence for exemplary linker	979	EAAAKGGGGSGGGGSEAAAKGGGGS
amino acid sequence for exemplary linker	980	EAAAKGGGGSGGGGSEAAAKEAAAK
amino acid sequence for exemplary linker	981	EAAAKGGGGSEAAAKGGGGSGGGGS
amino acid sequence for exemplary linker	982	EAAAKGGGGSEAAAKGGGGSEAAAK
amino acid sequence for exemplary linker	983	EAAAKGGGGSEAAAKEAAAKGGGGS
amino acid sequence for exemplary linker	984	EAAAKGGGGSEAAAKEAAAKEAAAK
amino acid sequence for exemplary linker	985	EAAAKEAAAKGGGGSEAAAKGGGGS
amino acid sequence for exemplary linker	986	EAAAKEAAAKGGGGSEAAAKEAAAK
amino acid sequence for exemplary linker	987	EAAAKEAAAKEAAAKGGGGSEAAAK
amino acid sequence for exemplary linker	988	EAAAKEAAAKEAAAKEAAAKGGGGS

TABLE7-continued

V. Additional Sequences		
Description	SEQ ID NO	Sequence
amino acid sequence for exemplary linker	989	EAAAKEAAAKEAAAKEAAAKEAAAK
amino acid sequence for exemplary linker	990	GTKDSTKDIPETPSKD
amino acid sequence for exemplary linker	991	GRDVRQPEVKEEKPES
amino acid sequence for exemplary linker	992	EGKSSGSGSESKSTAG
amino acid sequence for exemplary linker	993	TPGSPAGSPTSTEEGT
amino acid sequence for exemplary linker	994	GSEPATSGSETPGTST
Exemplary modified Nme guide sgRNA	995	mN*mN*mN*mNmNNNmNmNNmNNmNNNNNmNNNNmNNNmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAUAAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU
Exemplary modified Nme guide sgRNA	996	mN*mN*mN*mNmNNNmNmNNmNNmNNNNNmNNNNmNNNmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGmCmUmCmUmGmCCmUmUmCmUGGCAUCG*mU*mU
Exemplary modified Nme guide sgRNA	997	mN*mN*mN*mNmNNNmNmNNmNNmNNNNNmNNNNmNNNmGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAAUAAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGmCmUmCmUmGmCCmUmUmCmUGGCAUCG*mU*mU

*The guide sequence disclosed in this Table may be unmodified, modified with the exemplary modification pattern shown in the Table, or modified with a different modification pattern disclosed herein or available in the art.

TABLE 7A

VI. Additional Exemplary Guide RNAs

Guide ID	Target	Guide Sequence	Exemplary Guide RNA Modified Sequence	Genomic Coordinates (hg38)
G034202	HLA-A	GCUCUAUC CACGGCGC CCGCGGCU (SEQ ID NO: 66)	mG*mC*mU*mCmUAUmCmCAmCGmGCGCCmCGCGmGCUmGUUGmUmAmGmCU CCCmUmGmAmAmAmCmCGUUmGmCUAmCAAUAAGmGmCCmGmUmCmGmAmA mAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU (SEQ ID NO: 3001)	chr6: 29942891-29942915
G034617	HLA-A	CACUCACC CGCCCAGG UCUGGGUC (SEQ ID NO: 61)	mC*mA*mC*mUmCACmCmCGmCCmCAGGUmCUGGmGUCmGUUGmUmAmGmCU CCCmUmGmAmAmAmCmCGUUmGmCUAmCAAUAAGmGmCCmGmUmCmGmAmA mAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU (SEQ ID NO: 3002)	chr6: 29942609-29942633
G034982	TRAC	AAAACCUG UCAGUGA UUGGGUU CC (SEQ ID NO: 111)	mA*mA*mA*mAmCCUmGmUCmAGmUGAUUmGGGUmUCCmGUUGmUmAmGmCU CCCmUmGmAmAmAmCmCGUUmGmCUAmCAAUAAGmGmCCmGmUmCmGmAmA mAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU (SEQ ID NO: 3003)	chr14: 22550574-22550598
G034981	TRAC	UUAGGUU CGUAUCUG UAAAACCA A (SEQ ID NO: 107)	mU*mU*mA*mGmGUUmCmGUmAUmCUGUAmAAACmCAAmGUUGmUmAmGmCU CCCmUmGmAmAmAmCmCGUUmGmCUAmCAAUAAGmGmCCmGmUmCmGmAmA mAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU (SEQ ID NO: 3004)	chr14: 22550544-22550568
G034618	TRBC1	GUGUCCUA CCAGCAAG GGGUCCUG (SEQ ID NO: 215)	mG*mU*mG*mUmCCUmAmCCmAGmCAAGGmGGUCmCUGmGUUGmUmAmGmCU CCCmUmGmAmAmAmCmCGUUmGmCUAmCAAUAAGmGmCCmGmUmCmGmAmA mAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU (SEQ ID NO: 3005)	chr7: 142792690-142792714
G034201	CIITA	UCAAAGU ACCCUACA GGAGGACC A (SEQ ID NO: 301)	mU*mC*mA*mAmAGUmAmCCmCUmACAGGmAGGAmCCAmGUUGmUmAmGmCU CCCmUmGmAmAmAmCmCGUUmGmCUAmCAAUAAGmGmCCmGmUmCmGmAmA mAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU (SEQ ID NO: 3006)	chr16: 10907504-10907528
G034619	CIITA	AGCUGCCG UUCUGCCC AGUCCGGG (SEQ ID NO: 422)	mA*mG*mC*mUmGCCmGmUUmCUmGCCCAmGUCCmGGGmGUUGmUmAmGmCU CCCmUmGmAmAmAmCmCGUUmGmCUAmCAAUAAGmGmCCmGmUmCmGmAmA mAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU (SEQ ID NO: 3007)	chr16: 10906643-10906667
G032794	HLA-B	UCUGGGA AAGGAGG GGAAGAU GAG (SEQ ID NO: 828)	mU*mC*mU*mGmGGAmAmAGmGAmGGGGAmAGAUmGAGmGUUGmUmAmGmCU CCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAm AmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*m U (SEQ ID NO: 2828)	chr6: 31355222-31355246
G034208	HLA-B	UCUGGGA AAGGAGG GGAAGAU GAG (SEQ ID NO: 828)	mU*mC*mU*mGmGGAmAmAGmGAmGGGGAmAGAUmGAGmGUUGmUmAmGmCU CCCmUmGmAmAmAmCmCGUUmGmCUAmCAAUAAGmGmCCmGmUmCmGmAmA mAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU (SEQ ID NO: 3008)	chr6: 31355222-31355246
G032795	HLA-B	CUCUGGGA AAGGAGG GGAAGAU GA (SEQ ID NO: 829)	mC*mU*mC*mUmGGGmAmAAmGGmAGGGGmAAGAmUGAmGUUGmUmAmGmCU CCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAm AmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*m U (SEQ ID NO: 2829)	chr6: 31355221-31355245
G034209	HLA-B	CUCUGGGA AAGGAGG GGAAGAU GA (SEQ ID NO: 829)	mC*mU*mC*mUmGGGmAmAAmGGmAGGGGmAAGAmUGAmGUUGmUmAmGmCU CCCmUmGmAmAmAmCmCGUUmGmCUAmCAAUAAGmGmCCmGmUmCmGmAmA mAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU (SEQ ID NO: 3009)	chr6: 31355221-31355245

TABLE 7A-continued

VI. Additional Exemplary Guide RNAs

Guide ID	Target	Guide Sequence	Exemplary Guide RNA Modified Sequence	Genomic Coordinates (hg38)
G032806	HLA-B	UCCCAGAG CCGUCUUC CCAGUCCA (SEQ ID NO: 830)	mU*mC*mC*mCmAGAmGmCCmGUmCUUCCmCAGUmCCAmGUUGmUmAmGmCU CCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAm AmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*m U (SEQ ID NO: 2830)	chr6: 31355205-31355229
G034211	HLA-B	UCCCAGAG CCGUCUUC CCAGUCCA (SEQ ID NO: 830)	mU*mC*mC*mCmAGAmGmCCmGUmCUUCCmCAGUmCCAmGUUGmUmAmGmCU CCCmUmGmAmAmAmCmCGUUmGmCUAmCAAUAAGmGmCCmGmUmCmGmAmA mAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU (SEQ ID NO: 3010)	chr6: 31355205-31355229
G021557	VEGFA	GCAUGGGC AGGGGCU GGGGUGC AC (SEQ ID NO: 831)	mG*mC*mA*mUmGGGmCmAGmGGmGCUGGmGGUGmCACmGUUGmUmAmGmCU CCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAm AmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*m U (SEQ ID NO: 2831)	chr6: 43774288-43774312
G021558	VEGFA	GAAUGGC AGGCGGA GGUUGUA CUG (SEQ ID NO: 832)	mG*mA*mA*mUmGGCmAmGGmCGmGAGGUmUGUAmCUGmGUUGmUmAmGmCU CCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAm AmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*m U (SEQ ID NO: 2832)	chr6: 43780852-43780876
G021567	VEGFA	GUGAGCA GGCACCUG UGCCAACA U (SEQ ID NO: 833)	mG*mU*mG*mAmGCAmGmGCmACmCUGUGmCCAAmCAUmGUUGmUmAmGmCU CCCmUmGmAmAmAmCmCGUUmGmCUAmCAAU*AAGmGmCCmGmUmCmGmAm AmAmGmAmUGUGCmCGmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*m U (SEQ ID NO: 2833)	chr6: 43781113-43781137

TABLE 7B

VII. Exemplary Modified Conserved Region Nme Guide RNA Motifs

#mer	Modified nucleotide sequence of conserved portion	SEQ ID NO
101	mGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAA U*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGCmAmAmC mGCUCUmGmCCmUmUmCmUGmGCmAmUC*mG*mU*mU	1081
101	mGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAA U*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCG CUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	1082
101	mGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAA UAAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGC UCUmGmCCmUmUmCmUGGCAUCG*mU*mU	1083
101	mGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAA U*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCG mCmUmCmUmGmCCmUmUmCmUGGCAUCG*mU*mU	1084
101	mGUUGmUmAmGmCUCCCmUmGmAmAmAmCmCGUUmGmCUAmCAA UAAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCGmCAAmCGm CmUmCmUmGmCCmUmUmCmUGGCAUCG*mU*mU	1085
105	mGUUGmUmAmGmCUCCCmUmUmCmGmAmAmAmGmAmCmCGUUmG mCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmC GmCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	1086
105	mGUUGmUmAmGmCUCCCmUmUmCmGmAmAmAmGmAmCmCGUUmG mCUAmCAAUAAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCG mCAAmCGCUCUmGmCCmUmUmCmUGGCAUCG*mU*mU	1087

TABLE 7B-continued

VII. Exemplary Modified Conserved Region Nme Guide RNA Motifs		
#mer	Modified nucleotide sequence of conserved portion	SEQ ID NO
105	mGUUGmUmAmGmCUCCCmUmUmCmGmAmAmAmGmAmCmCGUUmG mCUAmCAAU*AAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmC GmCAAmCGmCmUmCmUmGmCCmUmUmCmUGGCAUCG*mU*mU	1088
105	mGUUGmUmAmGmCUCCCmUmUmCmGmAmAmAmGmAmCmCGUUmG mCUAmCAAUAAGmGmCCmGmUmCmGmAmAmAmGmAmUGUGCmCG mCAAmCGmCmUmCmUmGmCCmUmUmCmUGGCAUCG*mU*mU	1089

EXAMPLES

[0842] The following examples are provided to illustrate certain disclosed embodiments and are not to be construed as limiting the scope of this disclosure in any way.

Example 1: General Methods

[0843] Generally, unless otherwise indicated, guide RNAs used throughout the Examples identified as "GXXXXXX" refer to 89-nt or 101-nt modified sgRNA format such as those shown in the Tables provided herein.

1.1 Next-Generation Sequencing ("NGS") and Analysis for On-Target Cleavage Efficiency.

[0844] Genomic DNA was extracted using QuickExtract™ DNA Extraction Solution (Lucigen, Cat. No. QE09050) according to manufacturer's protocol.

[0845] To quantitatively determine the efficiency of editing at the target location in the genome, deep sequencing was utilized to identify the presence of insertions, deletions, and substitution introduced by gene editing. PCR primers were designed around the target site within the gene of interest (e.g., HLA-A) and the genomic area of interest was amplified. Primer sequence design was done as is standard in the field.

[0846] Additional PCR was performed according to the manufacturer's protocols (Illumina) to add chemistry for sequencing. The amplicons were sequenced on an Illumina MiSeq or NextSeq instrument. The reads were aligned to the human reference genome (e.g., hg38) after eliminating those having low quality scores. Reads that overlapped the target region of interest were re-aligned to the local genome sequence to improve the alignment. Then the number of wild type reads versus the number of reads which contain C-to-T mutations, C-to-A/G mutations or indels was calculated. Insertions and deletions were scored in a 20 bp region centered on the predicted Cas9 cleavage site. Indel percentage is defined as the total number of sequencing reads with one or more base inserted or deleted within the 20 bp scoring region divided by the total number of sequencing reads, including wild type. C-to-T mutations or C-to-A/G mutations were scored in a 40 bp region including 10 bp upstream and 10 bp downstream of the 20 bp sgRNA target sequence. The C-to-T editing percentage is defined as the total number of sequencing reads with either one or more C-to-T mutations within the 40 bp region divided by the total number of sequencing reads, including wild type. The percentage of C-to-A/G mutations are calculated similarly.

1.2 In Vitro Transcription ("IVT") of mRNA

[0847] Capped and polyadenylated mRNA containing N1-methyl pseudo-U was generated by in vitro transcription using a linearized plasmid DNA template and T7 RNA polymerase. Plasmid DNA containing a T7 promoter, a sequence for transcription, and a polyadenylation sequence was linearized by incubating at 37° C. for 2 hours with XbaI with the following conditions: 200 ng/μL plasmid, 2 U/μL XbaI (NEB), and 1× reaction buffer. The XbaI was inactivated by heating the reaction at 65° C. for 20 min. The linearized plasmid was purified from enzyme and buffer salts. The IVT reaction to generate modified mRNA was performed by incubating at 37° C. for 1.5-4 hours in the following conditions: 50 ng/μL linearized plasmid; 2-5 mM each of GTP, ATP, CTP, and N1-methyl pseudo-UTP (Trilink); 10-25 mM ARCA (Trilink); 5 U/μL T7 RNA polymerase (NEB); 1 U/μL Murine RNase inhibitor (NEB); 0.004 U/μL Inorganic *E. coli* pyrophosphatase (NEB); and 1× reaction buffer. TURBO DNase (ThermoFisher) was added to a final concentration of 0.01 U/μL, and the reaction was incubated for an additional 30 minutes to remove the DNA template. The mRNA was purified using a MegaClear Transcription Clean-up kit (ThermoFisher) or a RNeasy Maxi kit (Qiagen) per the manufacturers' protocols. Alternatively, the mRNA was purified through a precipitation protocol, which in some cases was followed by HPLC-based purification. Briefly, after the DNase digestion, mRNA is purified using LiCl precipitation, ammonium acetate precipitation and sodium acetate precipitation. For HPLC purified mRNA, after the LiCl precipitation and reconstitution, the mRNA was purified by RP-IP HPLC (see, e.g., Kariko, et al. Nucleic Acids Research, 2011, Vol. 39, No. 21 e142). The fractions chosen for pooling were combined and desalted by sodium acetate/ethanol precipitation as described above. In a further alternative method, mRNA was purified with a LiCl precipitation method followed by further purification by tangential flow filtration. RNA concentrations were determined by measuring the light absorbance at 260 nm (Nanodrop), and transcripts were analyzed by capillary electrophoresis by Bioanlayzer (Agilent).

[0848] *Streptococcus pyogenes* ("Spy") Cas9 mRNA was generated from plasmid DNA encoding an open reading frame according to SEQ ID NOs: 812-817 (see sequences in Table 7). When SEQ ID NOs: 812-817 are referred to below with respect to RNAs, it is understood that Ts should be replaced with Us (which were N1-methyl pseudouridines as described above). Messenger RNAs used in the Examples include a 5' cap and a 3' polyadenylation region, e.g., up to 100 nts, and are identified by the SEQ ID NOs: 812-817 in Table 7.

1.3 T Cell Preparation

[0849] Healthy human donor apheresis was obtained commercially (Hemacare), and cells were washed and resuspended in in CliniMACS® PBS/EDTA buffer (Miltenyi Biotec Cat. 130-070-525) and processed in a MultiMACS™ Cell 24 Separator Plus device (Miltenyi Biotec). T cells were isolated via positive selection using a Straight from Leukopak® CD4/CD8 MicroBead kit, human (Miltenyi Biotec Cat. 130-122-352). T cells were aliquoted and cryopreserved for future use in Cryostor® CS10 (StemCell Technologies Cat. 07930). Upon thaw, T cells were plated at a density of 1.0×10^6 cells/mL in T cell growth media (TCGM) composed of CTS OpTmizer T Cell Expansion SFM and T Cell Expansion Supplement (ThermoFisher Cat. A1048501) containing 2.5% or 5% human AB serum (GeminiBio, Cat. 100-512), 1× Penicillin-Streptomycin, 1× Glutamax, 10 mM HEPES, further supplemented with 200 U/mL recombinant human interleukin-2 (Peprotech, Cat. 200-02), 5 ng/ml recombinant human interleukin 7 (Peprotech, Cat. 200-07), and 5 ng/ml recombinant human interleukin 15 (Peprotech, Cat. 200-15). T cells were rested in the T cell growth media for 24 hours at which time they were activated with TransAct™ (1:100 dilution, Miltenyi Biotec, Cat. 130-111-160). T cells were activated for 48 hours prior to electroporation.

1.4 RNA Electroporation

[0850] Guide RNAs targeting a specific gene were removed from their storage plates and denatured for 2 minutes at 95° C. before incubating at room temperature for 5 minutes. Forty-eight hours post T cell activation, T cells were harvested, centrifuged at 500 g for 5 minutes, and resuspended at a concentration of 12.5×10^6 T cells/mL in P3 electroporation buffer (Lonza Catalog #V4SP-3960). BC22n electroporation mix was prepared with 1×10^5 T cells, 20 ng/μL of BC22n mRNAs, 20 ng/μL of UGI mRNA and 20 pmols of sgRNA in a final volume of 20 μL of P3 electroporation buffer. The mixture was transferred to the corresponding wells of a 96-well Nucleofector™ plate (Lonza Catalog #V4SP-3960). Cells were electroporated in duplicate using Lonza shuttle 96w using manufacturer's pulse code. Immediately post electroporation, cells were recovered in 80 μL of TCGM without cytokines at 37° C. for 15 minutes. Electroporated T cells were subsequently cultured in TCGM further supplemented with 2× cytokines (200 U/mL recombinant human interleukin-2 (Peprotech, Cat. 200-02), 5 ng/ml recombinant human interleukin 7 (Peprotech, Cat. 200-07), and 5 ng/ml recombinant human interleukin 15 (Peprotech, Cat. 200-15) per well. The plates were incubated at 37° C. On day 4 post-electroporation, edited T cells were collected for NGS analysis. Fresh CTS Optimizer media (ThermoFisher Cat. A1048501) supplemented with 1× cytokines (200 U/mL recombinant human interleukin-2 (Peprotech, Cat. 200-02), 5 ng/ml recombinant human interleukin 7 (Peprotech, Cat. 200-07), and 5 ng/ml recombinant human interleukin 15 (Peprotech, Cat. 200-15) was added to the remaining cells and incubated at 37° C. degrees. On day 7 post-electroporation, edited T cells were collected for flow cytometry analysis.

1.5 RNA Electroporation of an sgRNA Dilution Series

[0851] Solutions containing sgRNAs targeting a gene of interest and solutions containing mRNA encoding Nme2 BC22n (SEQ ID No: 822) and mRNA encoding UGI (SEQ ID No. 821) were prepared in P3 electroporation buffer. Single guide RNAs were removed from their storage plates, denatured for 2 minutes at 95° C. and incubated at room temperature for 5 minutes. Forty-eight hours post-activation, T cells were harvested, centrifuged, and resuspended at a concentration of 12.5×10^6 T cells/mL in P3 electroporation buffer (Lonza Catalog #V4SP-3960). Each sgRNA was serially diluted 1:2 (v/v) in P3 electroporation buffer starting from a final concentration of 5 μM. Subsequently, 1×10^5 T cells, 20 ng/μL of BC22n mRNA and 20 ng/μL of UGI mRNA were added to serially diluted sgRNAs in a final volume of 20 μL of P3 electroporation buffer. The resulting mix was transferred in duplicate to 96-well Nucleofector™ plates. Cells were electroporated using the manufacturer's pulse code. Immediately post electroporation, cells were recovered in 80 μL of TCGM without cytokines at 37° C. for 15 minutes. After recovery, 80 μL of electroporated T cells were transferred to 96-well plates containing 80 μL of TCGM supplemented with 400 U/mL recombinant human interleukin-2 (Peprotech, Cat. 200-02), 10 ng/ml recombinant human interleukin 7 (Peprotech, Cat. 200-07), and 10 ng/ml recombinant human interleukin 15 (Peprotech, Cat. 200-15) per well. Plates were incubated at 37° C. for 4 days. On day 4 post-editing, T cells were sub-cultured 1:4 (v/v) and on day 7 post-editing, T cells were collected for flow cytometry analysis.

1.6 Lipid Nanoparticle Formulation

[0852] In general, the lipid nanoparticle (LNP) components were dissolved in 100% ethanol at various molar ratios. The RNA cargos (e.g., Cas9 or base editor mRNA and sgRNA) were dissolved in 25 mM citrate, 100 mM NaCl, pH 5.0, resulting in a concentration of RNA cargo of approximately 0.45 mg/mL. The LNPs used contained ionizable lipid ((9Z,12Z)-3-((4,4-bis(octyloxy)butanoyl)oxy)-2-((((3-(diethylamino)propoxy)carbonyl)oxy)methyl)propyl octadeca-9,12-dienoate, also called 3-((4,4-bis(octyloxy)butanoyl)oxy)-2-((((3-(diethylamino)propoxy)carbonyl)oxy) methyl)propyl(9Z,12Z)-octadeca-9,12-dienoate), also called herein Lipid A, cholesterol, distearoylphosphatidylcholine (DSPC), and 1,2-dimyristoyl-rac-glycero-3-methoxypolyethylene glycol-2000 (PEG2K-DMG) (catalog #GM-020 from NOF, Tokyo, Japan) in a molar ratio of 35 Lipid A: 47.5 cholesterol: 15 DSPC: 2.5 PEG2k-DMG. The LNPs were formulated with a lipid amine to RNA phosphate (N:P) molar ratio of about 6. In some cases, LNPs were prepared with a single RNA species such as a mRNA or a gRNA. In some cases, LNPs were prepared with a mixture of mRNA and a guide RNA.

[0853] The LNPs were prepared using a cross-flow technique utilizing impinging jet mixing of the lipid in ethanol with two volumes of RNA solution and one volume of water. First, the lipid in ethanol was mixed through a mixing cross with the two volumes of RNA solution. Then, a fourth stream of water was mixed with the outlet stream of the cross through an inline tee (See WO 2016/010840, FIG. **2**). The LNPs were held for at least 1 hour at room temperature, and further diluted with water (approximately 1:1 v/v). Diluted LNPs were buffer exchanged into 50 mM Tris, 45 mM NaCl, 5% (w/v) sucrose, pH 7.5 (TSS) and concentrated as needed by methods known in the art. The resulting mixture was then filtered using a 0.2 m sterile filter. The final LNPs were characterized to determine the encapsulation efficiency, polydispersity index, and average particle size. The final LNP was stored at 4° C. or −80° C. until further use.

Example 2: Screening of HLA-A Guide RNAs with Nme2 BC22n

[0854] Guide RNAs designed for the disruption of the HLA-A gene were screened for editing efficacy in T cells by assessing loss of two allelic versions of the MHC I surface protein, HLA-A2 and HLA-A3. The donor had a heterozygous HLA-A phenotype of A*02:01:01G and 03:01:01G. The percentage of T cells double negative for HLA-A2 and A3 ("% A2-/A3-") was determined by flow cytometry following editing at the HLA-A locus by electroporation with Nme2 base editor system and each test guide.

2.1 T Cell Preparation and Editing with RNA Electroporation

[0855] T cells from a single donor (no. 808) were prepared and activated as described in Example 1. T cells were electroporated with sgRNAs targeting HLA-A, mRNA encoding Nme2 BC22n SEQ ID NO: 822), and mRNA encoding UGI (SEQ ID NO: 821) as described in Example 1 and herein.

2.2 Flow Cytometry

[0856] On day 7 post-electroporation, edited T cells were phenotyped by flow cytometry to determine HLA-A2 and HLA-A3 protein expression. Briefly, T cells were incubated for 30 min at 4° C. with a mixture of antibodies against CD3 (BioLegend, Cat. No. 316314), CD4 (BioLegend, Cat. No. 317434), CD8 (BioLegend, Cat. No. 301046), Viakrome (Immunotech, Cat. No. C36628), HLA A2 (BioLegend, Cat. No. 343306), and HLA A3 (Fisher Scientific, Cat. No. 501122330) diluted at 1:100, and HLA B7 (Milteny Biotech, Cat. No. 130-120-234) diluted at 1:200 in cell staining buffer. Cells were subsequently washed and resuspended in 100 μL of cell staining buffer. Cells were then processed on a Cytoflex flow cytometer (Beckman Coulter) and analyzed using the FlowJo™ software package. T cells were gated based on size, shape, viability, and CD8 positivity.

[0857] Table 8 and FIG. 1A show the mean percentage of cells double negative for HLA-A2 and HLA-A3 following editing at the HLA-A locus. The percentage of cells that are double or single negative for HLA-A2 and HLA-B3 for select guides, and guide specificity of the select guides to HLA-A over HLA-B are shown in FIGS. 1B and 1C.

TABLE 8

Mean percentage of T cells HLA-A negative (double negative for HLA-A2 and HLA-A3) following editing at the HLA-A locus

Guide	% HLA A2/A3 −ve		
ID	Mean	SD	N
G028854	0.63	0.47	2
G028855	14.15	0.64	2
G028856	0.62	0.01	2
G028857	1.32	0.35	2
G028858	5.17	0.45	2
G028859	4.51	0.59	2
G028860	1.49	0.3	2
G028861	4.26	0.18	2
G028862	0.21	0.04	2
G028863	0.08	0	2
G028864	0.16	0.05	2
G028865	75.45	0.78	2
G028866	42.6	1.7	2
G028867	11.6	1.41	2
G028868	0.84	0.27	2
G028869	38.3	2.4	2
G028870	2.59	0.28	2

TABLE 8-continued

Mean percentage of T cells HLA-A negative (double negative for HLA-A2 and HLA-A3) following editing at the HLA-A locus

Guide	% HLA A2/A3 −ve		
ID	Mean	SD	N
G028871	0.19	0.03	2
G028872	0.07	0.03	2
G028873	0.03	0.01	2
G028874	28.65	1.48	2
G028875	0.37	0.14	2
G028876	0.57	0.05	2
G028877	10.9	0.57	2
G028878	3.59	0.01	2
G028879	1.73	0.23	2
G028880	0.41	0.01	2
G028881	0.06	0.03	2
G028882	0.04	0.01	2
G028883	0.27	0.01	2
G028884	0.01	0.01	2
G028885	0	0.01	2
G028886	2.83	1.07	2
G028887	0.84	0.62	2
G028888	11.25	0.21	2
G028889	0.91	0.07	2
G028890	1.15	0.23	2
G028891	25.7	1.84	2
G028892	5.53	0.28	2
G028893	1.57	0.18	2
G028894	60.6	4.38	2
G028895	1.02	0.1	2
G028896	5.39	0.61	2
G028897	0.37	0	2
G028898	0.28	0.15	2
G028899	52.5	5.09	2
G028900	19.5	2.12	2
G028901	27.45	0.49	2
G028902	22.85	0.64	2
G028903	55.5	3.54	2
G028904	1.92	0.02	2
G028905	6.86	0.88	2
G028906	5.88	0.64	2
G028907	91.1	2.26	2
G028908	8.82	0.08	2
G028909	1.05	0.23	2
G028910	2.43	1.03	2
G028911	0.02	0.03	2
G028912	18.3	4.67	2
G028913	91.15	2.76	2
G028914	30.25	2.62	2
G028915	8.47	1.39	2
G028916	0.59	0.21	2
G028917	0.01	0.01	2
G028918	81.4	2.69	2
G028919	0.49	0.44	2
G028920	22.4	2.69	2
G028921	2.12	0.31	2
G028922	87.5	6.51	2
G028923	67.45	13.51	2
G028924	46.2	3.39	2
G028925	0.07	0.06	2
G028926	0.07	0.03	2
G028927	0.86	0.06	2
G028928	8.48	0.93	2
G028929	2.9	0.32	2
G028930	0.08	0.11	2
G028933	8.31	1.34	2
G028934	27.7	1.7	2

Example 3: Dose Response Curves (DRC) for Select HLA-A Nme2 Guides

[0858] A titration experiment was conducted to evaluate the editing efficacy of sgRNAs designed for the disruption of the HLA-A locus by assessing loss of one allelic version of the MHC I surface protein, HLA-A2. An eight-point dose response curve was generated for each sgRNA by titrating each sgRNA with a fixed concentration of mRNA encoding Nme2 BC22n (SEQ ID NO:822) and mRNA encoding UGI (SEQ ID NO: 821) for electroporation in T cells. T cells were then expanded and phenotyped by flow cytometry to determine the editing efficiency of each sgRNA tested.

[0859] T cells were prepared as described in Example 1. T cells were edited with serially diluted sgRNAs by the mRNA electroporation method described in Example 1. On day 7 post-editing, edited T cells were phenotyped by flow cytometry to determine HLA-A2 protein expression. Briefly, T cells were transferred to U-bottom plates, spun at 500 g for 5 minutes and resuspended in 100 μL of FACs buffer (PBS with 2% FBS and 2 mM EDTA) containing the following staining reagents: PerCP/Cy5.5 anti-human CD3 (BioLegend, Cat. 317434) diluted 1:100 (v/v), BV421 anti-human CD4 (BioLegend, Cat. 317434) diluted 1:100 (v/v), BV785 anti-human CD8 (BioLegend, Cat. 301046) diluted 1:100 (v/v), PE anti-human HLA-A2 (BioLegend Inc., Cat. 343306) diluted 1:100 (v/v), FITC anti-human-HLA-B7 (Miltenyi Biotec Inc., Cat. 130-120-234) diluted 1:200 and Viakrome 808 (Beckman Coulter Cat. C36628) diluted 1:100. T cells were incubated in staining solution for 30 minutes at 4° C., washed and resuspended using 100 μL of FACS buffer. Then, T cells were processed in a Beckman Coulter Cytoflex LX flow cytometer, and analyzed using the FlowJo™ software package. T cells were gated based on size, singularity, viability and CD8 positivity. Table 9 and FIG. **2** show the percentage of CD8+ T cells that are negative for the expression of HLA-A2.

TABLE 9

Mean percentage of $CD8^+$ $HLA\text{-}A2^-$ T cells following base editing

Guide sgRNA (μg/mL)	G028907 Mean % $CD8^+$ Cells	G028907 SD % $CD8^+$ Cells	N	G028913 Mean % $CD8^+$ Cells	G028913 SD % $CD8^+$ Cells	N	G028869 Mean % $CD8^+$ Cells	G028869 SD % $CD8^+$ Cells	N
5	98.65	0.07	2	97.20	0.42	2	70.55	0.49	2
2.5	98.25	0.21	2	96.15	0.07	2	61.75	0.92	2
1.25	97.40	0.14	2	93.30	0.00	2	43.75	1.91	2
0.625	95.70	0.42	2	90.85	0.92	2	25.45	0.92	2
0.313	92.40	0.00	2	85.55	1.77	2	14.55	0.92	2
0.078	80.20	3.25	2	68.25	3.75	2	6.20	0.33	2
0	54.45	2.05	2	42.70	1.13	2	3.19	0.40	2
EC50	0.07038			0.08998			0.9213		

Guide sgRNA (μg/mL)	G028922 Mean % $CD8^+$ Cells	G028922 SD % $CD8^+$ Cells	N	G028918 Mean % $CD8^+$ Cells	G028918 SD % $CD8^+$ Cells	N	G028865 Mean % $CD8^+$ Cells	G028865 SD % $CD8^+$ Cells	N
5	98.65	0.21	2	92.15	0.07	2	90.95	1.06	2
2.5	95.00	0.99	2	89.45	1.20	2	90.20	1.27	2
1.25	74.90	2.12	2	87.25	1.06	2	87.20	0.42	2
0.625	54.20	0.42	2	83.40	0.71	2	81.35	1.34	2
0.313	36.80	0.85	2	72.05	0.64	2	75.45	1.06	2
0.078	18.60	0.42	2	48.80	0.85	2	55.60	2.26	2
0	8.73	0.02	2	24.65	1.20	2	34.30	1.56	2
EC50	0.6028			0.1429			0.09631		

Example 4: Screening of TRAC Guide RNAs with Nme2 Base Editing System

[0860] TRAC guide RNAs were screened for Nme2 base editing efficacy in T cells by assessing loss of CD3 cell surface expression by flow cytometry and editing frequency by NGS. CD3 is a cell-surface component of the T cell receptor complex and its presence at the cell surface is used as a surrogate marker for TRAC protein expression.

4.1 T Cell Preparation and Editing with RNA Electroporation

[0861] T cells were prepared and activated as described in Example 1. T cells were electroporated with sgRNAs targeting TRAC, mRNA encoding Nme2 BC22n (SEQ ID NO: 822), and mRNA encoding UGI (SEQ ID NO: 821) as described in Example 1 and herein.

4.2 Flow Cytometry and NGS Sequencing

[0862] T cells from a single donor (no. 3786) were prepared as described in Example 1. T cells were edited with sgRNA targeting the TRAC locus using mRNA electroporation as described in Example 1. On day 4 post-electroporation, edited T cell samples were subjected to PCR and NGS analysis as described in Example 1. On day 7 post-electroporation, T cells were phenotyped by flow cytometry. Briefly, T cells were incubated for 30 min at 4° C. with a mixture of antibodies against CD3 (BioLegend, Cat. No. 316314), CD4 (BioLegend, Cat. No. 317434), CD8 (BioLegend, Cat. No. 301046), Viakrome (Immunotech, Cat. No. C36628) diluted at 1:100 in cell staining buffer (BioLegend, Cat. No. 420201). Cells were subsequently washed and resuspended in 100 μL of cell staining buffer. Cells were then processed on a Cytoflex flow cytometer (Beckman Coulter) and analyzed using the FlowJo™ software package. T cells were gated based on size, shape, viability, CD8 and CD3 expression. Table 10 and FIG. **3**A show the mean percentage of CD3 negative T cells. Mean percent editing in is shown in Table 10 and FIG. **3**B.

TABLE 10

Mean percentage of CD3 negative T cells; Mean percent editing of the TRAC loci as a percentage of total NGS reads. N = 4 represents two technical replicates each with two primer sets.

	% CD3 −ve			% C to T			% C to A/G			% Indels		
Guide ID	Mean	SD	N	Mean	SD	N	Mean	SD	N	Mean	SD	N
G021469	93.25	1.20	2	94.59	0.81	4	2.83	0.64	4	0.72	0.09	4
G021481	95.50	1.56	2	93.52	0.31	4	2.59	0.14	4	1.75	0.22	4
G028935	93.75	1.20	2	95.22	0.34	4	1.67	0.35	4	1.65	0.14	4
G028936	2.66	0.73	2	50.63	5.08	4	1.26	0.37	4	2.48	0.35	4
G028937	46.25	8.13	2	88.13	2.82	4	2.04	1.03	4	0.87	0.57	4
G028938	9.93	0.24	2	26.77	3.16	4	0.91	0.06	4	0.35	0.05	4
G028939	93.10	0.99	2	95.12	0.97	4	1.28	0.18	4	0.55	0.08	4
G028940	1.28	0.01	2	7.76	0.85	4	1.22	0.10	4	0.21	0.06	4
G028941	11.95	0.21	2	95.88	1.30	4	0.92	0.19	4	0.39	0.07	4
G028942	30.75	5.73	2	50.37	4.35	4	1.47	0.24	4	0.21	0.08	4
G028943	93.65	5.44	2	92.43	3.16	4	1.94	0.62	4	1.80	0.33	4
G028944	1.25	0.19	2	1.42	0.10	4	0.73	0.11	4	0.12	0.04	4
G028945	9.89	1.86	2	46.15	5.94	4	1.01	0.07	4	0.58	0.11	4
G028946	4.29	0.11	2	41.35	2.67	4	0.88	0.45	4	0.66	0.10	4
G028947	1.05	0.08	2	95.68	1.72	4	1.83	1.65	4	0.63	0.05	4
G028948	0.27	0.19	2	1.83	0.14	4	0.56	0.07	4	0.20	0.09	4

Example 5: Dose Response Curves (DRC) for Select TRAC Nme2 Guides

[0863] A titration experiment was conducted to evaluate the editing efficacy of select sgRNAs that target the TRAC locus by assessing loss of CD3 cell surface expression. CD3 is a cell-surface component of the T cell receptor complex and its presence at the cell surface is used as a surrogate marker for TRAC protein expression. An eight-point dose response curve was generated for each sgRNA by titrating each sgRNA with a fixed concentration of Nme2 BC22n (SEQ ID NO:822) and UGI mRNA132 (SEQ ID NO: 821) in T cells using electroporation. T cells were then expanded and phenotyped by flow cytometry to determine the editing efficiency of each sgRNA tested.

[0864] T cells were prepared as described in Example 1. T cells were edited with serially diluted sgRNAs that target TRAC by mRNA electroporation as described in Example 1. On day 7 post-editing, edited T cells were phenotyped by flow cytometry to determine CD3 expression as described in Example 4. T cells were gated based on size, singularity, viability and CD8 positivity. Tables 11 and FIG. **4** show the percentage of CD8+ T cells that are negative for the expression of CD3.

TABLE 11

Mean percent CD3 negative T cells following base editing at the TRAC locus

G021469				G021481			
sgRNA	% CD3− of CD8+			sgRNA	% CD3− of CD8+		
(uM)	Ave	SD	N	(uM)	Ave	SD	N
5.00	95.40	0.00	2.00	5.00	97.35	0.21	2
2.50	92.20	0.85	2.00	2.50	94.15	0.21	2
1.25	88.45	2.90	2.00	1.25	85.6	0.42	2
0.63	78.55	3.46	2.00	0.63	74.75	1.48	2
0.31	64.80	1.84	2.00	0.31	54.75	4.45	2
0.16	47.90	0.42	2.00	0.16	39.9	1.27	2
0.08	25.40	5.37	2.00	0.08	22.05	0.64	2
0.00	0.47	0.34	2.00	0.00	0.375	0.06	2

TABLE 11-continued

Mean percent CD3 negative T cells following base editing at the TRAC locus

G028935				G028939			
sgRNA	% CD3− of CD8+			sgRNA	% CD3− of CD8+		
(uM)	Ave	SD	N	(uM)	Ave	SD	N
5.00	94.60	0.14	2.00	5.00	97.7	0.42	2
2.50	92.10	0.14	2.00	2.50	92.65	1.20	2
1.25	90.05	0.07	2.00	1.25	88.35	0.49	2
0.63	88.15	1.20	2.00	0.63	79.55	1.06	2
0.31	87.75	0.78	2.00	0.31	73.35	3.04	2
0.16	79.90	1.27	2.00	0.16	51.8	3.39	2
0.08	67.20	1.13	2.00	0.08	36.05	5.02	2
0.00	0.46	0.23	2.00	0.00	0.22	0.03	2

G028943			
sgRNA	% CD3− of CD8+		
(uM)	Ave	SD	N
5.00	97.50	0.71	2
2.50	95.55	0.64	2
1.25	91.95	3.46	2
0.63	91.10	2.97	2
0.31	85.40	0.71	2
0.16	70.05	2.33	2
0.08	50.10	0.28	2
0.00	0.40	0.04	2

Example 6: Screening of TRBC 1 and 2 Guide RNAs with Nme2 BC22n

[0865] Guide RNAs targeting TRBC 1 and/or TRBC 2 were screened for editing efficacy in T cells by assessing loss of CD3 cell surface expression by flow cytometry and for editing efficacy by NGS, following TRBC editing by mRNA delivery. CD3 is a cell-surface component of the T cell receptor complex and its presence at the cell surface is used as a surrogate marker for TRBC protein expression.

6.1 T Cell Preparation and Editing with RNA Electroporation

[0866] T cells were prepared and activated as described in Example 1. T cells were electroporated with sgRNAs targeting TRBC, mRNA encoding Nme2 BC22n (SEQ ID NO:822), and mRNA encoding UGI (SEQ ID NO: 821) as described in Example 1 and herein.

6.2 Flow Cytometry and NGS Sequencing

[0867] T cells from a single donor (no. 3786) were prepared as described in Example 1. T cells were edited with sgRNA targeting the TRBC1 and/or TRBC 2 locus using mRNA electroporation as described in Example 1. On day 4 post-electroporation, edited T cell samples were subjected PCR and NGS analysis as described in Example 1. On day 7 post-electroporation, T cells were phenotyped by flow cytometry. Briefly, T cells were incubated for 30 min at 4° C. with a mixture of antibodies against CD3 (BioLegend, Cat. No. 316314), CD4 (BioLegend, Cat. No. 317434), CD8 (BioLegend, Cat. No. 301046), and Viakrome (Immunotech, Cat. No. C36628) diluted at 1:100 in cell staining buffer (BioLegend, Cat. No. 420201). Cells were subsequently washed and resuspended in 100 μL of cell staining buffer. Cells were then processed on a Cytoflex flow cytometer (Beckman Coulter) and analyzed using the FlowJo™ software package. T cells were gated based on size, shape, viability, CD8 and CD3 expression.

[0868] Table 12 and FIG. **5**A show the mean percentage of CD3− T cells. Percent editing at the TRBC 1 and/or 2 locus is shown in Table 12 and in FIG. **5**B.

TABLE 12

Mean percentage of CD3 negative T cells; Mean percent editing of the TRBC loci as a percentage of total NGS reads.

	% CD3 -ve			% C to T			% C to A/G			% Indels			Gene
Guide ID	Mean	SD	N	Mean	SD	N	Mean	SD	N	Mean	SD	N	target
G028952	9.65	1.06	2	14.08	1.30	4	0.84	0.50	4	0.36	0.06	4	TRBC1
G028953	15.30	0.99	2	46.47	4.15	4	1.82	0.48	4	0.61	0.12	4	TRBC1
G028968	19.45	2.05	2	38.11	8.64	4	1.29	0.21	4	0.74	0.04	4	TRBC1
G028969	3.41	0.49	2	17.85	9.69	4	0.63	0.39	4	0.15	0.05	4	TRBC1
G028970	29.30	3.54	2	71.73	5.73	4	1.59	0.33	4	0.54	0.08	4	TRBC1
G028977	0.99	0.07	2	23.09	16.10	4	0.90	1.04	4	0.49	0.59	4	TRBC1
G028978	1.67	0.25	2	43.93	5.74	4	1.46	0.09	4	0.82	0.24	4	TRBC1
G028979	5.32	0.29	2	90.62	2.52	4	2.11	0.18	4	2.32	0.28	4	TRBC1
G028980	2.41	0.00	2	37.84	1.93	4	1.57	0.28	4	0.36	0.10	4	TRBC1
G028981	1.08	ND	1	65.21	5.04	4	1.99	0.24	4	0.55	0.03	4	TRBC1
G028982	15.60	0.14	2	89.82	1.24	4	3.90	0.33	4	3.30	0.57	4	TRBC1
G028983	1.42	0.35	2	65.07	2.82	4	4.05	0.28	4	5.47	0.59	4	TRBC1
G028984	0.22	0.08	2	18.19	1.46	4	1.92	0.32	4	1.14	0.17	4	TRBC1
G028985	5.78	0.12	2	47.94	1.02	4	1.77	0.22	4	0.45	0.15	4	TRBC1
G028986	92.30	4.53	2	91.48	1.28	4	3.58	0.49	4	1.83	0.17	4	TRBC1
G028987	68.85	4.45	2	82.47	2.97	4	2.65	0.14	4	0.59	0.16	4	TRBC1
G028989	19.00	0.57	2	92.81	0.70	4	3.76	0.30	4	1.43	0.16	4	TRBC1
G028990	0.73	0.25	2	79.63	2.77	4	1.72	0.21	4	0.75	0.19	4	TRBC1
G028991	0.31	0.26	2	9.78	0.55	4	0.58	0.21	4	0.10	0.03	4	TRBC1
G028992	0.19	0.08	2	3.26	0.29	4	0.45	0.30	4	0.10	0.04	4	TRBC1
G028993	0.16	0.16	2	7.36	0.89	4	1.57	1.38	4	0.56	0.42	4	TRBC1
G028949	62.95	6.43	2	65.32	3.28	4	2.23	0.26	4	0.98	0.14	4	TRBC1/2
G028950	65.60	4.38	2	63.75	44.44	4	1.13	1.30	4	0.40	0.47	4	TRBC1/2
G028951	70.90	1.27	2	69.08	46.09	4	1.25	0.83	4	1.87	1.25	4	TRBC1/2
G028954	1.48	ND	1	7.84	0.83	4	1.25	0.75	4	0.16	0.06	4	TRBC1/2
G028955	0.50	0.13	2	2.35	0.43	4	1.05	0.57	4	0.11	0.05	4	TRBC1/2
G028956	20.00	1.98	2	80.04	3.44	4	1.82	0.32	4	1.04	0.19	4	TRBC1/2
G028957	2.09	0.39	2	13.49	0.63	4	1.17	0.31	4	0.23	0.06	4	TRBC1/2
G028958	69.40	0.14	2	89.79	0.26	4	2.25	0.16	4	2.67	0.31	4	TRBC1/2
G028959	2.86	0.49	2	5.62	0.62	4	1.32	0.30	4	0.12	0.05	4	TRBC1/2
G028960	42.50	4.24	2	42.76	0.36	4	1.74	0.44	4	0.25	0.08	4	TRBC1/2
G028961	1.27	0.04	2	1.64	0.43	4	0.46	0.08	4	0.12	0.04	4	TRBC1/2
G028962	0.65	0.20	2	6.58	2.20	4	1.12	0.57	4	0.14	0.06	4	TRBC1/2
G028963	11.65	2.19	2	11.55	1.59	4	0.97	0.49	4	0.25	0.05	4	TRBC1/2
G028964	23.10	1.84	2	21.45	0.91	4	1.09	0.59	4	0.32	0.05	4	TRBC1/2
G028965	40.35	5.59	2	61.42	3.81	4	2.10	0.15	4	4.12	0.40	4	TRBC1/2
G028966	18.00	4.67	2	26.83	8.93	4	1.00	0.11	4	0.78	0.46	4	TRBC1/2
G028967	21.30	2.55	2	19.82	0.85	4	0.63	0.17	4	0.17	0.09	4	TRBC1/2
G028971	1.14	0.01	2	25.06	3.88	4	0.49	0.06	4	0.42	0.14	4	TRBC1/2
G028972	16.85	2.05	2	67.14	4.73	4	2.15	1.09	4	1.63	0.27	4	TRBC1/2
G028973	4.88	0.30	2	90.36	2.03	4	1.35	0.19	4	0.98	0.14	4	TRBC1/2
G028974	3.05	0.15	2	83.09	1.63	4	2.11	0.75	4	1.58	0.12	4	TRBC1/2
G028975	1.01	0.04	2	18.72	2.39	4	1.49	0.34	4	0.60	0.12	4	TRBC1/2
G028976	0.97	0.18	2	31.70	1.75	4	1.66	0.39	4	0.86	0.20	4	TRBC1/2
G028988	9.14	0.85	2	87.16	3.83	4	2.33	0.27	4	0.51	0.08	4	TRBC1/2
G028994	17.70	3.82	2	30.45	3.67	4	1.30	0.53	4	0.54	0.06	4	TRBC2
G028995	25.45	2.47	2	49.38	1.71	4	1.61	0.50	4	0.64	0.03	4	TRBC2
G028996	1.64	0.28	2	5.61	2.70	4	0.41	0.14	4	0.19	0.06	4	TRBC2
G028997	17.00	0.85	2	57.64	25.79	4	1.16	0.35	4	0.54	0.17	4	TRBC2
G028998	4.13	1.70	2	20.02	4.05	4	2.11	0.71	4	0.24	0.08	4	TRBC2

TABLE 12-continued

Mean percentage of CD3 negative T cells; Mean percent editing of the TRBC loci as a percentage of total NGS reads.

	% CD3 -ve			% C to T			% C to A/G			% Indels			Gene
Guide ID	Mean	SD	N	Mean	SD	N	Mean	SD	N	Mean	SD	N	target
G028999	1.51	0.12	2	5.33	0.63	4	1.88	2.00	4	0.16	0.03	4	TRBC2
G029000	0.58	0.06	2	4.08	0.41	4	0.79	0.25	4	0.32	0.08	4	TRBC2
G029001	0.25	0.01	2	0.60	0.05	4	0.48	0.04	4	0.11	0.03	4	TRBC2
G029002	8.26	0.19	2	73.39	0.87	4	1.13	0.26	4	0.73	0.05	4	TRBC2
G029003	35.00	0.28	2	93.20	0.57	4	1.43	0.17	4	3.78	0.38	4	TRBC2
G029004	34.20	2.12	2	44.18	0.76	4	1.66	0.18	4	0.35	0.07	4	TRBC2
G029005	5.40	0.40	2	40.16	2.70	4	1.34	0.18	4	0.48	0.06	4	TRBC2
G029006	88.15	2.76	2	89.41	0.89	4	3.49	0.58	4	4.54	0.26	4	TRBC2
G029007	60.55	0.64	2	88.74	2.32	4	2.40	0.77	4	0.84	0.23	4	TRBC2
G029008	48.55	1.48	2	95.20	0.44	4	1.56	0.26	4	2.05	0.23	4	TRBC2
G029009	0.95	0.22	2	68.27	2.76	4	2.26	0.91	4	12.78	0.16	4	TRBC2
G029010	1.42	0.25	2	61.68	1.86	4	1.78	0.12	4	1.13	0.10	4	TRBC2
G029011	0.25	0.01	2	12.47	1.05	4	2.16	0.42	4	0.18	0.02	4	TRBC2
G029012	0.17	0.01	2	47.87	2.46	4	2.07	0.15	4	0.22	0.05	4	TRBC2

"ND" indicates no data.

N = 4 represents two technical replicates each with two primer sets.

Example 7: Dose Response Curves (DRC) for Select TRBC Nme2 Guides

[0869] A titration experiment was conducted to evaluate the editing efficacy of select sgRNAs designed for the disruption of one or both TRBC genes by assessing loss of CD3 cell surface expression. CD3 is a cell-surface component of the T cell receptor complex and its presence at the cell surface is used as a surrogate marker for TRBC protein expression. An eight-point dose response curve was generated for each sgRNA by titrating each sgRNA with a fixed concentration of mRNA encoding Nme2 BC22n (SEQ ID NO:822) and mRNA encoding UGI (SEQ ID NO: 1821 in T cells using electroporation. T cells were then expanded and phenotyped by flow cytometry to determine the editing efficiency of each sgRNA tested.

[0870] T cells were prepared as described in Example 1. T cells were edited with serially diluted TRBC targeting sgRNAs by mRNA electroporation as described in Example 1 and Table 13. On day 7 post-editing, edited T cells were phenotyped by flow cytometry to determine loss of CD3 expression as described in Example 4. T cells were gated based on size, singularity, viability and CD8 positivity. Table 13 and FIG. **6** show the percentage of CD8+ T cells that are negative for the expression of CD3.

[0871] T cells were prepared as described in Example 1. T cells were edited with serially diluted TRBC targeting sgRNAs by mRNA electroporation as described in Example 1. On day 7 post-editing, edited T cells were phenotyped by flow cytometry to determine loss of CD3 expression as described in Example 3. T cells were gated based on size, singularity, viability and CD8 positivity. Tables 13 and FIG. **6** show the percentage of CD8+ T cells that are negative for the expression of CD3.

TABLE 13

Percentage of CD8 positive cells that are CD3 negative at various doses of TRBC sgRNA

	% CD3− of CD8+		
sgRNA (uM)	Ave	SD	N
G028986			
5.00	96.75	0.07	2
2.50	88.15	0.35	2
1.25	70.85	1.63	2
0.63	56	3.25	2
0.31	34.35	1.48	2
0.16	20	0.28	2
0.08	10.16	1.05	2
0.00	0.36	0.06	2
G029006			
5.00	95.05	0.35	2
2.50	82.75	0.21	2
1.25	68.35	8.27	2
0.63	55.05	4.60	2
0.31	37.55	2.33	2
0.16	19.7	1.13	2
0.08	13.1	1.13	2
0.00	0.43	0.00	2
G029007			
5.00	65.65	0.07	2
2.50	52.55	1.34	2
1.25	36.3	0.57	2
0.63	22.6	0.71	2
0.31	10.4	0.42	2
0.16	4.565	0.49	2
0.08	3.21	1.09	2
0.00	0.335	0.05	2

Example 8: Screening of CIITA Guide RNAs with Nme2 BC22n and Nme2 Cas9

8.1 Screen of CIITA sgRNA with Nme2 BC22n

[0872] CIITA guide RNAs were screened for editing efficacy in T cells by assessing loss of HLA DP, DQ, DR cell surface expression and editing by NGS, following CIITA editing by mRNA delivery. HLA-DP, DR and DQ are cell surface proteins whose expression requires the transcription factor CIITA and thus their presence at the cell surface is a surrogate marker for CIITA protein expression.

8.1.1 T Cell Preparation and Editing with RNA Electroporation

[0873] T cells from a single donor (no. 3786) were prepared and activated as described in Example 1. T cells were electroporated with sgRNAs targeting CIITA, mRNA encoding Nme2 BC22n (SEQ ID NO: 822), and mRNA encoding UGI (SEQ ID NO: 821) as described in Example 1 and herein.

8.1.2 Flow Cytometry and NGS Sequencing

[0874] T cells were prepared as described in Example 1. T cells were edited with sgRNA targeting the CIITA locus using mRNA electroporation as described in Example 1. On day 4 post-electroporation, edited T cell samples were subjected PCR and NGS analysis as described in Example 1. On day 7 post-electroporation, T cells were phenotyped by flow cytometry as described in Example 1 except the mixture of antibodies used were against CD3 (BioLegend, Cat. No. 316314), CD4 (BioLegend, Cat. No. 317434), CD8 (BioLegend, Cat. No. 301046), Viakrome (Immunotech, Cat. No. C36628) diluted at 1:100, and HLA II-DR, DP, DQ (BioLegend, Cat. No. 361706) diluted at 1:50 in cell staining buffer (BioLegend, Cat. No. 420201). T cells were gated based on size, shape, viability, CD8 and HLA II-DR, DP, DQ, expression.

[0875] Table 14 and FIG. **7**A show the mean percentage of CD3 T cells with loss of the HLA-DP, DQ, DR expression. Percent editing at the CIITA locus is shown in Table 14 and FIG. **7**B.

TABLE 14

Mean percentage of HLA II-DP, DR, DQ negative cells; Mean percent editing of the CIITA locus as a percentage of total NGS reads

	% HLA DP, DR, DQ -ve			% C to T		% C to A/G		% Indels		
Guide ID	Mean	SD	N	Mean	SD	Mean	SD	Mean	SD	N
G029013	34.40	4.95	2	18.75	1.90	0.60	0.11	0.13	0.07	3
G029014	54.60	5.80	2	96.18	0.32	1.71	0.05	0.78	0.15	3
G029015	44.85	0.07	2	88.66	1.75	1.40	0.08	0.42	0.09	3
G029016	37.05	7.99	2	3.20	0.50	0.23	0.17	0.02	0.02	3
G029017	41.75	12.66	2	90.90	1.36	0.55	0.01	7.28	1.08	3
G029018	45.05	4.74	2	46.49	0.66	0.41	0.09	3.13	0.17	3
G029019	44.70	4.81	2	59.06	0.13	0.42	0.04	0.30	0.05	3
G029020	37.85	3.46	2	40.66	0.14	0.52	0.15	0.31	0.04	3
G029021	31.35	1.91	2	31.25	1.13	0.38	0.06	0.17	0.04	3
G029022	38.60	0.42	2	5.70	1.01	0.44	0.36	0.21	0.27	3
G029023	38.90	3.68	2	14.55	0.94	0.77	0.11	0.03	0.04	3
G029024	29.75	4.17	2	12.24	0.41	0.23	0.04	0.06	0.03	3
G029025	41.05	5.30	2	12.60	0.73	0.32	0.06	0.35	0.15	3
G029026	49.35	1.34	2	5.75	0.66	0.24	0.06	0.04	0.03	3
G029027	54.25	2.90	2	1.45	0.07	0.30	0.06	0.75	0.14	3
G029028	50.50	0.71	2	68.54	3.78	2.29	0.09	6.12	0.59	3
G029029	88.00	2.69	2	78.07	3.53	0.97	0.36	0.18	0.10	3
G029030	77.85	1.48	2	87.88	0.43	0.57	0.23	0.60	0.03	3
G029031	62.70	0.42	2	75.45	1.67	0.15	0.06	0.05	0.00	3
G029032	49.60	2.12	2	2.41	0.30	0.17	0.12	0.17	0.23	3
G029033	69.60	1.84	2	58.57	3.90	0.49	0.14	1.09	0.58	3
G029034	50.90	6.08	1	16.00	2.53	0.38	0.28	0.10	0.06	3
G029035	75.00	8.91	2	69.61	3.67	0.40	0.15	1.75	0.31	3
G029036	71.65	14.78	2	0.00	0.00	0.00	0.00	0.00	0.00	3
G029037	41.85	8.27	2	33.36	3.21	0.58	0.18	0.93	0.12	3
G029038	56.00	2.55	2	6.63	3.25	0.31	0.13	0.01	0.02	3
G029039	50.15	1.20	2	12.85	18.84	1.01	1.16	0.00	0.00	3
G029040	55.50	0.57	2	36.13	12.71	0.30	0.24	0.01	0.02	3
G029041	70.00	2.97	2	29.51	1.69	0.51	0.16	0.11	0.02	3
G029042	52.10	0.85	2	14.95	2.44	0.45	0.04	0.03	0.00	3
G029043	52.65	4.45	2	8.31	0.36	0.43	0.18	0.05	0.02	3
G029044	54.40	1.13	2	3.50	0.04	0.47	0.13	0.07	0.05	3
G029045	48.30	1.56	2	5.11	0.72	0.24	0.09	0.04	0.03	3
G029046	60.60	2.83	2	85.08	2.95	1.26	0.24	0.58	0.11	3
G029047	48.00	5.80	2	91.28	2.37	1.44	0.43	2.26	0.66	3
G029048	56.35	3.32	2	77.44	0.57	1.31	0.68	1.76	0.63	3
G029049	41.15	8.56	2	7.42	0.23	0.09	0.04	1.55	2.65	3
G029050	59.15	4.74	2	19.53	0.87	0.17	0.07	0.02	0.02	3
G029051	51.60	5.66	2	9.63	0.80	0.15	0.05	0.07	0.03	3
G029052	52.60	1.98	2	5.81	0.21	0.29	0.04	0.20	0.06	3
G029053	49.25	1.48	2	0.37	0.07	0.20	0.06	0.16	0.23	3
G029054	47.60	11.03	2	3.23	0.11	0.14	0.03	0.04	0.04	3
G029055	50.10	8.77	2	2.06	0.27	0.10	0.06	0.02	0.01	3

TABLE 14-continued

Mean percentage of HLA II-DP, DR, DQ negative cells; Mean percent editing of the CIITA locus as a percentage of total NGS reads

	% HLA DP, DR, DQ -ve			% C to T		% C to A/G		% Indels		
Guide ID	Mean	SD	N	Mean	SD	Mean	SD	Mean	SD	N
G029056	62.45	2.76	2	79.87	3.36	0.79	0.08	1.87	0.15	3
G029057	53.20	3.82	2	4.50	0.24	0.26	0.12	0.03	0.01	3
G029058	51.95	5.16	2	0.91	0.15	0.12	0.03	0.26	0.18	3
G029059	52.55	6.29	2	1.25	0.11	0.21	0.06	0.29	0.23	3
G029060	43.25	3.46	2	0.45	0.11	0.18	0.10	0.22	0.17	3
G029061	41.30	8.63	1	27.96	2.63	0.33	0.15	0.55	0.11	3
G029062	57.55	0.21	2	65.48	5.49	1.08	0.15	4.29	2.51	3
G029063	72.90	0.71	2	63.82	6.02	1.43	0.25	0.19	0.07	3
G029064	50.15	4.31	2	8.25	1.52	0.38	0.07	0.60	0.30	3
G029065	45.90	2.12	2	19.00	1.71	1.61	0.46	0.41	0.37	3
G029066	43.80	7.07	2	1.93	0.50	0.95	0.24	0.08	0.03	3
G029067	45.75	0.78	2	2.32	0.48	0.78	0.27	0.06	0.02	3
G029068	44.55	0.35	2	3.58	0.11	0.14	0.05	0.04	0.03	3
G029069	52.75	3.46	2	32.35	0.63	0.35	0.02	0.10	0.05	3
G029070	73.30	4.38	2	92.11	0.53	1.28	0.22	0.67	0.14	3
G029071	50.65	1.06	2	35.16	2.51	0.29	0.08	0.19	0.10	3
G029072	42.35	2.33	2	1.47	0.07	0.27	0.03	0.05	0.04	3
G029073	41.30	9.48	2	46.89	2.53	0.66	0.23	0.21	0.15	3
G029074	52.20	0.14	2	48.77	3.02	0.61	0.04	0.19	0.08	3
G029075	49.30	1.13	2	12.47	1.16	0.69	0.19	0.76	0.82	3
G029076	43.45	3.04	2	17.10	0.40	0.47	0.05	0.79	0.97	3
G029077	48.20	2.55	2	6.55	1.34	0.16	0.12	0.14	0.03	3
G029078	43.30	0.42	2	9.78	0.71	1.18	0.12	0.07	0.07	3
G029079	58.65	0.35	2	56.55	3.15	1.49	0.10	0.41	0.04	3
G029080	50.35	0.21	2	11.06	0.91	1.24	0.14	0.07	0.03	3
G029081	85.40	1.84	2	69.63	4.27	1.26	0.21	0.57	0.31	3
G029082	51.85	4.17	2	9.09	2.32	1.87	0.85	0.04	0.04	3
G029083	46.55	0.35	2	6.81	2.37	1.21	0.07	0.03	0.03	3
G029084	46.60	12.16	2	35.64	9.40	0.94	0.55	0.40	0.14	3
G029085	38.40	7.50	2	92.18	1.18	1.09	0.03	1.98	0.05	3
G029086	44.85	2.47	2	7.25	0.44	0.34	0.08	0.73	0.16	3
G029087	47.30	0.28	2	17.14	1.34	0.43	0.07	0.75	0.17	3
G029088	49.85	4.17	2	30.15	3.86	0.20	0.34	0.22	0.38	3
G029089	45.00	0.28	2	2.98	0.25	0.66	0.38	0.10	0.05	3
G029090	39.10	2.26	2	8.97	0.53	0.39	0.20	0.34	0.13	3
G029091	40.05	3.18	2	25.28	4.23	3.02	2.63	0.31	0.28	3
G029092	49.50	0.99	2	90.70	4.31	3.27	2.57	3.66	0.69	3
G029093	46.70	7.07	2	0.39	0.05	2.79	2.28	0.08	0.07	3
G029094	46.40	4.81	2	11.75	0.73	0.98	0.08	0.33	0.10	3
G029095	42.75	7.57	2	3.37	0.46	0.89	0.17	0.11	0.05	3
G029096	33.95	4.60	2	19.19	4.61	0.94	0.18	0.33	0.19	3
G029097	33.90	1.70	2	85.22	0.86	1.66	0.16	6.17	0.23	3
G029098	45.70	1.41	2	55.36	3.68	0.96	0.14	0.35	0.16	3
G029099	42.75	0.64	2	11.94	1.85	1.17	0.08	0.14	0.06	3
G029100	34.55	0.49	2	4.72	0.45	1.52	0.56	0.13	0.15	3
G029101	40.95	0.21	2	1.72	0.44	1.16	0.39	0.07	0.06	3
G029102	35.40	1.98	2	0.44	0.13	0.50	0.05	0.08	0.05	3
G029103	32.20	0.99	2	2.30	0.43	0.35	0.24	0.10	0.13	4
G029104	50.40	5.52	2	34.26	1.13	0.94	0.21	0.38	0.18	4
G029105	38.05	1.34	2	93.46	0.85	1.80	0.13	3.54	0.79	4
G029106	43.05	3.18	2	24.88	0.99	0.31	0.07	0.09	0.06	4
G029107	49.40	2.97	2	39.56	2.80	0.35	0.05	0.28	0.02	4
G029108	39.05	9.69	2	31.33	5.11	0.27	0.02	0.15	0.05	4
G029109	71.15	0.92	2	89.12	9.14	8.05	8.87	0.79	0.20	4
G029110	37.25	0.78	2	53.06	3.72	1.41	0.60	0.45	0.15	4
G029111	35.20	2.40	2	43.21	4.33	1.02	0.22	0.47	0.25	4
G029112	41.90	0.42	2	4.38	0.37	0.67	0.21	0.22	0.20	4
G029113	41.30	9.19	2	11.35	0.91	0.66	0.04	0.45	0.16	4
G029114	34.00	3.96	2	8.69	0.44	0.63	0.08	0.27	0.03	4
G029115	45.35	11.67	2	50.53	1.40	0.99	0.19	2.22	0.77	4
G029116	59.55	4.88	2	31.29	2.75	0.81	0.09	0.49	0.24	4
G029117	81.45	0.35	2	91.14	1.77	2.68	1.60	0.41	0.10	4
G029118	51.90	1.27	2	16.93	0.68	1.79	0.61	0.89	0.47	4
G029119	63.15	5.30	2	46.34	0.58	1.54	0.39	0.65	0.16	4
G029120	55.30	2.69	2	12.79	0.80	0.47	0.14	0.13	0.06	4
G029121	56.15	3.32	2	16.62	0.64	0.62	0.17	0.07	0.05	4
G029122	55.30	8.34	2	3.08	1.96	0.20	0.27	0.10	0.17	4
G029123	90.80	0.99	2	89.28	0.71	0.63	0.94	0.17	0.22	4
G029124	65.90	3.39	2	43.34	6.94	0.19	0.13	0.42	0.28	4

TABLE 14-continued

Mean percentage of HLA II-DP, DR, DQ negative cells; Mean percent editing of the CIITA locus as a percentage of total NGS reads

	% HLA DP, DR, DQ -ve			% C to T		% C to A/G		% Indels		
Guide ID	Mean	SD	N	Mean	SD	Mean	SD	Mean	SD	N
G029125	54.35	4.31	2	11.68	3.17	0.23	0.27	0.06	0.07	4
G029126	37.75	0.49	2	12.13	1.22	0.58	0.19	0.47	0.40	4
G029127	45.20	7.07	2	5.05	1.26	0.28	0.32	0.06	0.08	4
G029128	81.55	2.05	2	88.85	2.04	0.82	0.29	2.03	0.33	4
G029129	75.35	5.02	2	51.88	0.94	0.49	0.04	0.86	0.13	4
G029130	58.20	8.77	2	18.50	1.66	0.40	0.15	1.43	0.40	4
G029131	93.15	1.91	2	93.47	1.53	0.78	0.37	1.63	0.67	4
G029132	53.50	8.20	2	4.52	0.15	0.53	0.11	0.05	0.03	4
G029133	63.55	1.91	2	44.52	1.48	1.33	0.30	0.99	0.51	4
G029134	47.70	1.98	2	7.17	1.32	1.32	0.54	0.34	0.31	4
G029135	43.25	61.16	2	85.48	1.21	1.52	0.31	10.28	1.03	4
G029136	52.75	2.47	2	10.14	0.23	1.00	0.46	0.13	0.06	4
G029137	65.90	0.57	2	50.89	1.82	1.30	0.57	0.27	0.07	4
G029138	57.60	6.79	2	38.26	1.45	2.03	0.49	0.25	0.05	4
G029139	48.30	1.27	2	16.01	0.82	1.37	1.10	0.06	0.06	4
G029140	75.15	0.07	2	47.78	3.71	1.10	0.64	0.17	0.06	4
G029141	69.90	3.82	2	34.65	1.70	0.93	0.61	0.32	0.39	4
G029142	60.95	3.75	2	15.07	1.14	0.89	0.68	0.06	0.06	4
G029143	67.65	1.06	2	36.84	1.93	0.63	0.32	0.18	0.09	4
G029144	56.70	2.55	2	5.40	0.54	0.44	0.31	0.04	0.02	4
G029145	54.40	9.48	2	0.78	0.04	0.68	0.53	0.18	0.17	4
G029146	63.20	8.63	2	63.58	3.19	0.93	0.80	1.93	0.28	4
G029147	56.05	1.20	2	10.86	0.91	0.66	0.41	0.05	0.04	4
G029148	51.95	0.49	2	1.50	0.36	0.35	0.20	0.04	0.03	4
G029149	59.30	2.69	2	3.23	0.18	0.49	0.15	0.14	0.08	4
G029150	49.40	1.70	2	46.90	1.43	1.09	0.23	0.89	0.15	4
G029151	52.95	9.83	2	9.70	0.57	0.59	0.42	0.89	0.12	4
G029152	56.90	3.11	2	1.80	0.23	1.06	1.52	3.20	3.20	4
G029153	52.90	4.95	2	0.23	0.10	0.65	0.27	0.29	0.20	4
G029154	54.45	4.88	2	1.15	0.38	0.49	0.12	0.85	0.61	4
G029155	56.70	0.28	2	14.72	1.08	0.67	0.12	0.44	0.42	4
G029156	53.75	0.35	2	6.87	0.62	0.73	0.14	0.13	0.11	4
G029157	56.25	.06	2	3.13	0.37	0.78	0.10	0.20	0.09	4
G029158	51.60	3.96	2	7.70	0.81	1.02	0.02	0.10	0.03	4
G029159	56.45	1.20	2	8.61	0.59	1.16	0.19	0.18	0.03	4
G029160	72.15	5.02	2	50.61	2.24	3.06	2.44	1.28	0.34	4
G029161	59.45	2.62	2	1.81	0.22	0.39	0.18	0.07	0.04	4
G029162	50.50	2.12	2	17.38	1.22	0.93	0.50	3.06	2.91	4
G029163	46.55	9.97	2	11.23	0.99	0.57	0.18	0.04	0.03	4
G029164	67.15	2.33	2	58.31	3.24	0.96	0.32	0.92	0.73	4
G029165	53.65	4.17	2	0.71	0.24	0.19	0.14	0.42	0.49	4
G029166	53.45	0.35	2	29.13	1.23	0.48	0.08	0.27	0.20	4
G029167	66.60	1.84	2	32.25	1.39	1.11	0.13	0.15	0.23	4
G029168	50.20	6.93	2	1.25	0.27	0.93	0.53	0.13	0.20	4
G029169	49.80	6.51	2	0.89	1.05	0.47	0.55	0.05	0.07	4
G029170	49.35	5.4	2	0.00	0.00	0.00	0.00	0.00	0.00	
G029171	84.10	0.00	2	0.00	0.00	0.00	0.00	0.00	0.00	
G029172	95.20	0.42	2	90.18	1.06	1.15	0.09	1.95	0.62	4
G029173	72.50	4.95	2	45.16	1.85	0.93	0.03	1.85	1.06	4
G029174	59.55	5.59	2	6.11	0.81	0.58	0.06	0.10	0.05	4
G029175	49.80	3.25	2	0.28	0.10	0.29	0.19	5.18	4.22	4
G029176	48.35	4.88	2	3.15	0.46	0.23	0.16	4.47	2.55	4
G029177	79.80	1.41	2	83.07	2.06	0.76	0.11	2.00	1.95	4
G029178	53.80	0.85	2	34.52	8.46	0.41	0.17	2.52	2.17	4
G029179	51.45	7.14	2	22.28	2.43	0.24	0.04	1.25	1.31	4
G029180	51.75	4.88	2	58.42	2.55	1.21	0.13	0.21	0.05	4
G029181	51.15	4.88	2	9.33	0.23	0.67	0.30	0.14	0.14	4
G029182	49.95	2.76	2	7.53	0.57	0.83	0.09	0.29	0.24	4
G029183	49.20	2.12	2	27.27	1.48	1.14	0.21	0.45	0.36	4
G029184	36.70	2.83	2	23.25	2.15	0.36	0.20	0.11	0.05	4
G029185	42.00	4.81	2	11.98	2.17	0.33	0.14	0.11	0.09	4
G029186	55.05	2.76	2	19.73	0.53	0.41	0.18	0.15	0.05	4
G029187	59.60	1.56	2	12.22	1.40	0.79	0.13	0.04	0.04	4
G029188	57.50	7.21	2	6.53	0.36	0.41	0.35	0.20	0.16	4
G029189	49.20	4.38	2	0.13	0.04	0.65	0.22	0.04	0.05	4
G029190	44.05	2.76	2	0.40	0.09	0.54	0.19	0.04	0.04	4

8.2. Screening of CIITA Guide RNAs with Nme2 Cas9

[0876] Additional sgRNAs targeting CIITA were screened for efficacy in T cells by assessing loss of HLA-DP/DQ/DR cell surface expression by flow cytometry and editing by NGS, following CIITA editing by mRNA delivery.

8.2.1 T Cell Preparation and Editing with RNA Electroporation

[0877] T cells were prepared as described in Example 1 using 2.5% o human AB serum (GeminiBio, Cat. 100-512) in the T Cell growth media. T cells were electroporated with sgRNA targeting the CIITA gene using mRNA electroporation as described in Example 1 except for the following differences. This study used mRNA encoding Nme2 Cas9 (SEQ ID No: 826) instead of mRNAs encoding Nme2 BC22n base editor and UGI, respectively.

[0878] Cas9 electroporation mix was prepared with 1×10^5 T cells, 30 ng/L of Nme2 Cas9 mRNA and 40 pmols of sgRNA in a final volume of 20 μL of P3 electroporation buffer.

8.2.2 Flow Cytometry and NGS Sequencing

[0879] On day 7 post-electroporation, edited T cell samples were collected and subjected to PCR and NGS analysis as described in Example 1. On day 13 post-editing, cells were phenotyped by flow cytometry to determine HLA II-DR, DP, DQ protein expression. Briefly, T cells were incubated for 30 min at 4° C. with a mixture of antibodies against CD3 (BioLegend, Cat. No. 317338), CD4 (BioLegend, Cat. No. 317434), CD8 (BioLegend, Cat. No. 301046), Viakrome (Beckman Coulter, Cat. No. C36628) diluted at 1:100, and HLA II-DR, DP, DQ (BioLegend, Cat. No. 361706) diluted at 1:50 in cell staining buffer (BioLegend, Cat. No. 420201) to determine HLA-DP, DQ, DR protein expression. T cells were gated based on size, shape, viability, CD3, CD8, and HLA II-DR, DP, DQ, expression.

[0880] Table 15 shows the percentage of T cells with loss of HLA II and mean percent editing of the CIITA gene. FIG. **8** shows percentage of HLA II-DR, DP, DQ negative cells edited with Nme2 Cas9 on the left vertical axis and mean percent editing on the right vertical axis.

TABLE 15

Mean percentage of HLA II-DP, DR, DQ negative cells; Mean percent indels oatthe CIITA locus as a percentage of total NGS reads

Guide	% INdels		CD8+, HLA-DP, DQ, DR−		
ID	Mean	SD	Mean	SD	N
Untreated			40.25	0.05	2
G023421	2.55	0.25	45.2	8.1	2
G023422	1.1	0	46.4	5.1	2
G023423	1.7	0.4	47.7	1.7	2
G023424	0.35	0.05	44.45	1.25	2
G023425	0.65	0.55	46.25	2.95	2
G023426	7.7	2.1	50.15	1.55	2
G023427	0.5	0.1	46.15	4.35	2
G023428	0.15	0.05	50.4	0.4	2
G023429	0.9	0.1	48.15	2.35	2
G023430	0.1	0	50.3	1.5	2
G023431	1.55	0.35	50.8	1.6	2
G023432	0.15	0.05	52.45	0.85	2
G023433	0.15	0.05	46.25	0.65	2
G023434	2.1	0.2	49.85	0.05	2
G023435	0.65	0.05	52.05	0.85	2
G023436	0.35	0.05	53.5	1.4	2
G023437	0.9	0.1	52.1	0.1	2

TABLE 15-continued

Mean percentage of HLA II-DP, DR, DQ negative cells; Mean percent indels oatthe CIITA locus as a percentage of total NGS reads

Guide	% INdels		CD8+, HLA-DP, DQ, DR−		
ID	Mean	SD	Mean	SD	N
G023438	0.15	0.15	53.15	1.35	2
G023439	0.1	0	51.05	0.25	2
G023440	0.35	0.15	54.35	1.05	2
G023441	25.4	1.1	59.7	4.3	2
G023442	0.2	0	57.75	1.55	2
G023443	18.65	0.25	69.15	15.05	2
G023444	28.15	6.05	58.65	1.85	2
G023445	2.45	0.35	55.25	0.15	2
G023446	0.9	0.3	49.15	0.55	2
G023447	0.35	0.15	54.45	3.75	2
G023448	0.2	0	51.5	0.6	2
G023449	0	0	53.6	3	2
G023450	0.1	0	54	2.3	2
G023451	0.15	0.05	51.05	2.75	2
G023452	0.55	0.15	52.75	2.45	2
G023453	0.1	0	52.85	3.55	2
G023454	0.35	0.15	55.65	4.35	2
G023455	0.25	0.05	53.3	4.2	2
G023456	12.8	2	56	3	2
G023457	0.35	0.05	55.85	3.75	2
G023458	1	0.3	52	3.2	2
G023459	1.3	0.4	51.15	3.05	2
G023460	0.45	0.05	53.75	1.35	2
G023461	0.45	0.05	53.25	1.65	2
G023462	1.05	0.15	52.2	4	2
G023463	0.2	0	51.15	3.95	2
G023464	6.6	1	54.1	1	2
G023465	0.05	0.05	52.45	0.45	2
G023466	0.2	0.2	51.2	2.7	2
G023467	0.2	0	54.75	2.45	2
G023468	3.25	0.15	58.3	3.9	2
G023469	1.6	0.1	56.85	4.55	2
G023470	0.1	0	54.25	2.95	2
G023471	0.55	0.25	53.65	4.05	2
G023472	0.1	0	53.2	1.7	2
G023473	2.5	1	57.2	2.3	2
G023474	0.6	0.1	57.65	1.75	2
G023475	12.75	1.05	58.65	2.25	2
G023476	10.9	1.3	58.25	1.05	2
G023477	93.8	1.1	80.45	3.95	2
G023478	52.75	5.95	64.9	2.2	2
G023479	13.75	2.05	58.85	0.25	2
G023480	0.1	0	55.5	1.2	2
G023481	0.2	0	54.8	1.4	2
G023482	0.5	0.4	52.8	2.1	2
G023483	0.1	0	53.35	2.65	2
G023484	0.1	0	59.7	8.4	2
G023485	0.25	0.05	50.2	0.2	2
G023486	4.5	0.2	52.45	0.65	2
G023487	0.4	0.1	52.85	3.35	2
G023488	1.4	0.2	52	2.7	2
G023489	0.15	0.05	52.85	2.35	2
G023490	0.15	0.15	53.2	1.8	2
G023491	1	0.2	53.45	0.95	2
G023492	40.25	12.05	59.6	0.8	2
G023493	0.8	0.4	56.7	0.9	2
G023494	0.3	0.2	51.55	2.35	2
G023495	1.5	1.3	53.25	1.05	2
G023496	0.1	0	52.05	3.15	2
G023497	0.05	0.05	52.85	0.35	2
G023498	0.1	0	53	0.9	2
G023499	0.4	0	53.65	4.05	2
G023500	4.2	0.3	53.7	3	2
G023501	3.1	0.1	51.95	2.35	2
G023502	3.95	0.35	53.95	0.95	2
G023503	2.95	0.05	53.5	3.5	2
G023504	0.25	0.05	51.7	1.6	2
G023505	4.1	0.1	52.9	1	2
G023506	1.95	0.15	49.8	0.9	2

TABLE 15-continued

Mean percentage of HLA II-DP, DR, DQ negative cells; Mean percent indels oatthe CIITA locus as a percentage of total NGS reads

Guide	% INdels		CD8+, HLA-DP, DQ, DR−		
ID	Mean	SD	Mean	SD	N
G023507	1.3	1	53.2	3.4	2
G023508	0.1	0	53.15	5.35	2
G023509	0.15	0.05	51.85	0.85	2
G023510	0.15	0.05	53.8	0.3	2
G023511	0.25	0.05	51.75	0.25	2
G023512	0.1	0	51.75	2.65	2
G023513	0.1	0	53.85	2.35	2
G023514	0.35	0.05	53.1	2.1	2
G023515	0.1	0	51.95	0.95	2

Example 9: Dose Response Curves (DRC) for Select CIITA Nme2 Guides

[0881] A titration experiment was conducted to evaluate the editing efficacy of select sgRNAs designed for the disruption of the CIITA gene by assessing loss of HLA DP, DQ, DR cell surface expression by flow cytometry and editing frequency at CIITA by NGS. HLA-DP, DR and DQ are cell surface proteins whose expression requires the transcription factor CIITA and thus their presence at the cell surface is a surrogate marker for CIITA protein expression.

[0882] An eight-point dose response curve was generated for each sgRNA by titrating each sgRNA with a fixed concentration of mRNA encoding Nme2 BC22n (SEQ ID NO:822) and mRNA encoding UGI (SEQ ID NO: 821) in T cells using electroporation. T cells were then expanded and phenotyped by flow cytometry to determine the editing efficiency of each sgRNA tested.

[0883] T cells were prepared single donor (no. 808) as described in Example 1. T cells were edited with serially diluted CIITA targeting sgRNAs (0, 0.94, 1.88, 3.75, 7.5, 15, 30, or 60 pmols) by mRNA electroporation as described in Example 1. On day 4 post-editing, T cell samples were collected and subjected to PCR and NGS analysis as described in Example 1. On day 7 post-editing, T cells were phenotyped by flow cytometry to determine HLA-DP, DQ, DR expression as described in Example 8. T cells were gated based on size, singularity, viability and CD8 positivity. Table 16 and FIG. **9**A show the NGS editing results. Table 16 and FIG. **9**B show the percentage of CD8+ T cells that are negative for HLA-DP, DQ, DR expression

TABLE 16

sgRNA	C > T			C > A/G			Indels			% HLA II-DP, DQ, DR- of CD8+		
(pmols)	Ave	SD	N	Ave	SD	N	Ave	SD	N	Ave	SD	N
G029029												
60	90.74%	1.38%	2	1.62%	0.27%	2	0.43%	0.08%	2	91.95	1.77	2
30	42.49%	0.45%	2	1.70%	0.22%	2	0.15%	0.04%	2	69.45	1.20	2
15	25.50%	2.00%	2	1.64%	0.17%	2	0.11%	0.05%	2	58.1	3.25	2
7.5	13.63%	0.29%	2	1.96%	0.12%	2	0.05%	0.00%	2	52.7	0.99	2
3.75	7.08%	0.37%	2	1.58%	0.16%	2	0.07%	0.00%	2	50.35	2.33	2
1.88	3.08%	0.13%	2	1.49%	0.20%	2	0.04%	0.00%	2	45.2	2.40	2
0.94	1.46%	0.34%	2	1.42%	0.19%	2	0.03%	0.01%	2	47.15	1.48	2
0	0.04%	0.00%	2	2.51%	1.11%	2	0.03%	0.02%	2	45.3	4.81	2
G029081												
60	87.77%	0.42%	2	1.29%	0.03%	2	1.05%	0.22%	2	92.15	1.48	2
30	36.59%	1.71%	2	1.06%	0.11%	2	0.42%	0.03%	2	66.85	2.33	2
15	21.90%	1.00%	2	0.86%	0.11%	2	0.19%	0.03%	2	56.2	2.55	2
7.5	10.56%	0.50%	2	0.90%	0.02%	2	0.17%	0.01%	2	52.8	0.85	2
3.75	5.17%	0.57%	2	1.04%	0.04%	2	0.06%	0.01%	2	50.7	1.70	2
1.88	2.39%	0.07%	2	0.86%	0.01%	2	0.04%	0.00%	2	49.15	1.77	2
0.94	1.01%	0.06%	2	0.76%	0.10%	2	0.02%	0.01%	2	46.35	0.92	2
0	0.10%	0.01%	2	0.90%	0.06%	2	0.02%	0.01%	2	50.6	2.55	2
G029123												
60	97.68%	0.04%	2	0.47%	0.66%	2	0.46%	0.01%	2	95.45	0.07	2
30	53.86%	0.40%	2	0.19%	0.27%	2	0.00%	0.00%	2	72.5	1.70	2
15	31.51%	2.00%	2	0.23%	0.33%	2	0.00%	0.00%	2	62.8	3.25	2
7.5	16.83%	2.14%	2	0.00%	0.00%	2	0.00%	0.00%	2	56.25	0.35	2
3.75	7.72%	2.36%	2	0.56%	0.21%	2	0.18%	0.25%	2	52.35	0.07	2
1.88	3.37%	0.94%	2	0.34%	0.48%	2	0.00%	0.00%	2	48.4	0.14	2
0.94	0.39%	0.05%	2	0.18%	0.25%	2	0.00%	0.00%	2	48.9	0.85	2
0	0.00%	0.00%	2	0.00%	0.00%	2	0.00%	0.00%	2	47.3	3.25	2
G029128												
60	94.41%	0.63%	2	0.50%	0.08%	2	2.61%	0.21%	2	86.6	1.98	2
30	61.36%	0.04%	2	0.47%	0.07%	2	1.57%	0.13%	2	67.6	1.98	2
15	41.10%	1.11%	2	0.26%	0.01%	2	1.11%	0.11%	2	64.3	3.39	2
7.5	24.54%	0.03%	2	0.23%	0.06%	2	1.05%	0.08%	2	56.4	1.70	2
3.75	11.39%	0.16%	2	0.17%	0.04%	2	0.72%	0.03%	2	52.85	0.49	2
1.88	4.88%	0.50%	2	0.17%	0.02%	2	0.67%	0.08%	2	50.1	0.99	2
0.94	1.90%	0.02%	2	0.15%	0.00%	2	0.63%	0.06%	2	49.85	0.21	2
0	0.20%	0.01%	2	0.16%	0.07%	2	0.60%	0.06%	2	46.45	1.20	2

TABLE 16-continued

sgRNA	C > T			C > A/G			Indels			% HLA II-DP, DQ, DR- of CD8+		
(pmols)	Ave	SD	N	Ave	SD	N	Ave	SD	N	Ave	SD	N
G029131												
60	94.92%	0.02%	2	1.48%	0.09%	2	1.86%	0.22%	2	95.3	0.99	2
30	79.21%	2.08%	2	1.49%	0.07%	2	1.24%	0.02%	2	86.75	4.74	2
15	64.70%	2.53%	2	1.03%	0.15%	2	0.65%	0.32%	2	79.85	0.21	2
7.5	41.12%	2.09%	2	1.32%	0.35%	2	0.38%	0.02%	2	70.3	0.57	2
3.75	23.58%	0.18%	2	1.20%	0.06%	2	0.27%	0.05%	2	60.1	1.70	2
1.88	11.88%	0.04%	2	1.20%	0.14%	2	0.23%	0.09%	2	55.15	1.06	2
0.94	4.51%	0.14%	2	1.05%	0.11%	2	0.15%	0.01%	2	50.2	1.56	2
0	0.26%	0.04%	2	1.07%	0.15%	2	0.10%	0.02%	2	44.95	0.49	2
G029171												
60	81.24%	3.07%	2	1.68%	0.07%	2	0.80%	0.15%	2	94.4	1.98	2
30	26.12%	2.63%	2	1.70%	0.16%	2	0.21%	0.08%	2	63.25	3.46	2
15	15.27%	1.25%	2	1.81%	0.12%	2	0.29%	0.07%	2	56.95	0.35	2
7.5	8.44%	0.58%	2	1.43%	0.01%	2	0.21%	0.08%	2	53.6	0.28	2
3.75	4.16%	0.26%	2	1.78%	0.11%	2	0.17%	0.05%	2	53.9	2.12	2
1.88	2.45%	0.10%	2	1.67%	0.03%	2	0.20%	0.00%	2	51	4.10	2
0.94	1.49%	0.23%	2	2.86%	2.01%	2	0.78%	0.82%	2	50.75	2.76	2
0	0.16%	0.02%	2	1.51%	0.14%	2	0.18%	0.01%	2	47.15	1.06	2
G029172												
60	92.73%	0.76%	2	1.29%	0.12%	2	2.73%	0.33%	2	97.2	0.28	2
30	50.99%	1.65%	2	0.94%	0.22%	2	6.70%	0.11%	2	75.45	2.19	2
15	28.49%	1.14%	2	0.96%	0.02%	2	8.41%	0.22%	2	64.8	2.83	2
7.5	15.76%	0.07%	2	0.69%	0.07%	2	7.58%	0.46%	2	59.55	0.35	2
3.75	7.75%	0.38%	2	0.77%	0.21%	2	7.21%	0.59%	2	55.95	0.78	2
1.88	3.11%	0.18%	2	0.75%	0.04%	2	8.28%	0.66%	2	50.45	5.44	2
0.94	0.88%	0.05%	2	1.26%	0.75%	2	24.29%	23.82%	2	49.45	3.61	2
0	0.08%	0.02%	2	0.60%	0.19%	2	7.52%	0.26%	2	46.2	0.99	2
G026584												
60	95.17%	0.99%	2	1.26%	0.25%	2	2.17%	0.03%	2	99.05	0.07	2
30	84.51%	1.04%	2	1.59%	0.03%	2	1.86%	0.12%	2	93.45	0.07	2
15	70.25%	0.04%	2	1.26%	0.05%	2	1.92%	0.01%	2	86.35	2.47	2
7.5	49.19%	2.18%	2	1.19%	0.32%	2	1.29%	0.07%	2	76.45	2.05	2
3.75	27.90%	2.71%	2	0.84%	0.15%	2	0.92%	0.24%	2	66.15	0.92	2
1.88	10.85%	0.07%	2	1.01%	0.14%	2	0.93%	0.19%	2	52.75	7.00	2
0.94	3.17%	0.16%	2	1.11%	0.06%	2	1.21%	0.03%	2	49.4	1.56	2
0	0.08%	0.03%	2	1.01%	0.26%	2	1.35%	0.53%	2	43	2.83	2
G026585												
60	86.56%	0.77%	2	1.17%	0.01%	2	9.62%	0.42%	2	96.35	1.06	2
30	49.55%	0.28%	2	1.23%	0.02%	2	5.08%	0.37%	2	74.3	3.25	2
15	29.91%	0.35%	2	1.13%	0.27%	2	3.10%	0.05%	2	65.15	3.61	2
7.5	16.67%	0.14%	2	1.08%	0.03%	2	1.72%	0.11%	2	57.2	1.13	2
3.75	8.49%	0.35%	2	1.01%	0.09%	2	1.10%	0.05%	2	51.85	6.43	2
1.88	3.16%	0.60%	2	1.07%	0.09%	2	0.64%	0.19%	2	51.15	4.31	2
0.94	1.04%	0.15%	2	0.95%	0.02%	2	0.52%	0.11%	2	45.95	2.19	2
0	0.05%	0.05%	2	0.95%	0.11%	2	0.40%	0.05%	2	47.9	2.69	2

Example 10: Screening of AAVS1 Guides with Nme2Cas9

[0884] Guide RNAs targeting AAVS1 were screened for editing efficacy in T cells by assessing editing by NGS, following AAVS1 editing by mRNA delivery.

T Cell Preparation and Editing with RNA Electroporation

[0885] T cells single donor (no. 613) were prepared as described in Example 1 using 2.5% human AB serum (GeminiBio, Cat. 100-512) in the T Cell growth media. T cells were electroporated with sgRNA targeting the AAVS1 gene using mRNA electroporation as described in Example 1 with the following exceptions.

[0886] This study used mRNA encoding Nme2 Cas9 (SEQ ID No: 825) instead of mRNAs encoding Nme2 BC22n base editor and UGI, respectively. Nme2 Cas9 electroporation mix was prepared with 1×10^5 T cells, 30 ng/μL of Nme2Cas9 mRNA and 40 pmols of sgRNA in a final volume of 20 μL of P3 electroporation buffer. On day 7 post-electroporation, DNA samples were subjected to PCR and subsequent NGS analysis as described in Example 1.

[0887] Table 17 shows mean percent indels at the AAVS1 gene with Nme2Cas9 guides. FIGS. **10**A-**10**B show editing at the AAVS1 gene represented as indel frequency.

TABLE 17

Mean percent indels at the AAVS1 locus as a percentage of total NGS reads						
Guide ID	Mean	SD	Guide ID	Mean	SD	N
G025808	0.6	0	G025894	95.9	4.1	2
G025809	0.25	0.05	G025895	0.15	0.05	2
G025810	2.3	0.3	G025896	20.1	10.3	2
G025811	4.25	0.45	G025897	99.75	0.05	2
G025812	0.2	0	G025898	99.55	0.05	2
G025813	0.3	0.1	G025899	39.5	9.4	2
G025814	0.45	0.05	G025900	54.1	41.9	2
G025815	0.3	0	G025901	16.3	13.4	2
G025816	0	0	G02502	52.75	1.55	2
G025817	0.9	0.1	G025903	2.7	0.2	2
G025818	65.55	2.25	G025904	2.7	1.1	2
G025819	0.35	0.05	G025905	0.5	0	2
G025820	1.1	0	G025906	0.4	0	2
G025821	3.8	0.3	G025907	0.1	0	2
G025822	41.7	3.8	G025908	16.9	3.1	2
G025823	0.35	0.05	G025909	2.95	0.35	2
G025824	11.25	3.95	G025910	1.45	0.15	2
G025825	3.95	1.55	G025911	12.3	0.5	2
G025826	1.25	0.65	G025912	2.75	0.35	2
G025827	89.3	2.6	G025913	1.85	0.15	2
G025828	32.05	2.95	G025914	3.7	0.4	2
G025829	98.1	1	G025915	24.75	2.25	2
G025830	48.45	5.65	G025916	1.4	0.2	2
G025831	39	3.9	G025917	3.25	0.65	2
G025832	52.9	5.1	G025918	6.15	0.85	2
G025833	82.15	3.55	G025919	4.7	0.4	2
G025834	89.75	0.75	G025920	0.25	0.05	2
G025835	92	0.8	G025921	1.15	0.35	2
G025836	98.55	0.05	G025922	0.75	0.25	2
G025837	37.35	0.15	G025923	10.9	1.1	2
G025838	1.7	0.7	G025924	1.05	0.05	2
G025839	99.2	0.3	G025925	0.2	0	2
G025840	68.55	11.45	G025926	2.5	0.2	2
G025841	97.45	0.45	G025927	0.2	0	2
G025842	3.45	0.05	G025928	0.2	0	2
G025843	43	2.7	G025929	39.1	5	2
G025844	0.1	0	G025930	0.8	0.1	2
G025845	13.5	0.6	G025931	1.1	0.2	2
G025846	0.9	0.1	G025932	9.7	1.2	2
G025847	0.55	0.15	G025933	29.6	3.3	2
G025848	3.9	0.3	G025934	0.5	0	2
G025849	38.4	0.3	G025935	4.15	0.75	2
G025850	6.7	0.2	G025936	85.1	1.2	2
G025851	1	0	G025937	15.5	0.6	2
G025852	0.15	0.05	G025938	1.25	0.35	2
G025853	0.25	0.05	G025939	2.75	0.55	2
G025854	15.85	0.05	G025940	98.3	0.6	2
G025855	27.4	2.3	G025941	0.1	0	2
G025856	0.95	0.05	G025942	0.25	0.05	2
G025857	4.9	0.2	G025943	0.25	0.05	2
G025858	0.3	0	G025944	0.15	0.05	2
G025859	1.55	0.15	G025945	2.2	0.3	2
G025860	0.2	0	G025946	1.6	0.2	2
G025861	0.25	0.05	G025947	0.2	0	2
G025862	0.15	0.05	G025948	1.1	0.1	2
G025863	88.6	1.4	G025949	3.45	0.55	2
G025864	10.55	0.55	G025950	0.2	0	2
G025865	2.7	0.1	G025951	0.35	0.05	2
G025866	98.35	0.05	G025952	82.2	1.9	2
G025867	98.75	0.05	G025953	9.25	0.65	2
G025868	89.85	0.05	G025954	19.2	3.3	2
G025869	3.1	0.1	G025955	7.5	1.1	2
G025870	15.45	2.15	G025956	0.55	0.05	2
G025871	0.3	0	G025957	1	0.1	2
G025872	4.15	0.05	G025958	0.5	0	2
G025873	9.1	0.4	G025959	3.05	0.45	2

TABLE 17-continued

Mean percent indels at the AAVS1 locus as a percentage of total NGS reads						
Guide ID	Mean	SD	Guide ID	Mean	SD	N
G025874	7.15	1.15	G025960	0.4	0.1	2
G025875	20.6	1.1	G025961	0.95	0.05	2
G025876	25.95	4.65	G025962	0.4	0.1	2
G025877	2.2	0.2	G025963	0.1	0	2
G025878	0.55	0.05	G025964	0.15	0.05	2
G025879	8.25	0.25	G025965	0.35	0.05	2
G025880	96.45	0.85	G025966	54.5	5.4	2
G025881	1.2	0	G025967	32.95	2.55	2
G025882	0.2	0	G025968	1.05	0.05	2
G025883	0.6	0	G025969	43.9	4	2
G025884	0.3	0	G025970	0.9	0.3	2
G025885	1.35	0.05	G025971	14	2	2
G025886	0.1	0	G025972	1.55	0.05	2
G025887	0.25	0.05	G025973	0.95	0.05	2
G025888	0.45	0.45	G025974	0.45	0.05	2
G025889	0.3	0.1	G025975	0.25	0.05	2
G025890	0.25	0.05	G025976	0.65	0.05	2
G025891	48.8	3.1	G025977	0.4	0	2
G025892	25.25	5.05	G025978	1	0.3	2
G025893	7.3	1.1	G025979	15.95	1.05	2
G025894	41.85	19.35	G025980	0.1	0	2
G025895	0.35	0.05				

Example 11: Dose Response of AAVS1 Guides with Nme2Cas9

[0888] A dilution series of AAVS1 guide RNAs were screened for editing efficacy by dose response in T cells by assessing editing by NGS following AAV S1 editing by mRNA delivery. A four-point dose response curve was generated for each sgRNA by titrating each sgRNA with a fixed concentration of mRNA encoding Nme2 Cas9 (SEQ ID NO: 825) in T cells using mRNA electroporation.

[0889] T cells single donor (no. 613) were prepared and activated as described in Example 1 using 2.5% o human AB serum (GeminiBio, Cat. 100-512) in T Cell growth media (TCGM). T cells were electroporated with serially diluted sgRNA targeting the AAVS1 gene using mRNA electroporation as described in Example 1 with the following exceptions. This study used mRNA encoding Nme2 Cas9 instead of mRNAs encoding Nme2 BC22n base editor and UGI, respectively. Cas9 electroporation mix was prepared with 1×10^5 T cells, 30 ng/L of Nme2Cas9 mRNA and 40, 13.3, 4.4 and 1.5 pmols of sgRNA in a final volume of 20 μL of P3 electroporation buffer. On day 3 post-editing, T cells were harvested and subjected to PCR and NGS sequencing as described in Example 1.

[0890] Table 18 and FIGS. **11**A-**11**B show the mean indel frequency at various AASV1 sgRNA concentrations.

TABLE 18

Mean indel frequency at AAVS1

sgRNA (uM)	G025836 Mean	SD	N	G025835 Mean	SD	N	G025829 Mean	SD	N	G025834 Mean	SD	N
2.00	1.00	0.00	2	0.91	0.01	2	1.00	0.00	2	0.88	0.03	2
0.67	0.93	0.02	2	0.49	0.07	2	0.87	0.03	2	0.44	0.03	2
0.22	0.67	0.16	2	0.18	0.00	2	0.25	0.02	2	0.07	0.02	2
0.07	0.44	0.24	2	0.06	0.03	2	0.12	0.04	2	0.02	0.01	2

sgRNA (uM)	G025827 Mean % editing	SD	N	G025880 Mean % editing	SD	N	G025866 Mean % editing	SD	N	G025868 Mean % editing	SD	N
2.00	0.99	0.00	2	0.97	0.00	2	0.97	0.00	2	0.97	0.00	2
0.67	0.90	0.03	2	0.93	0.04	2	0.97	0.00	2	0.78	0.09	2
0.22	0.14	0.00	2	0.16	0.05	2	0.85	0.05	2	0.10	0.02	2
0.07	0.03	0.00	2	0.03	0.01	2	0.39	0.06	2	0.02	0.01	2

sgRNA (uM)	G025867 Mean % editing	SD	N	G025863 Mean % editing	SD	N	G025818 Mean % editing	SD	N	G025843 Mean % editing	SD	N
2.00	0.97	0.01	2	0.97	0.00	2	0.97	0.00	2	0.95	0.00	2
0.67	0.97	0.01	2	0.92	0.02	2	0.75	0.08	2	0.52	0.08	2
0.22	0.73	0.08	2	0.19	0.04	2	0.05	0.01	2	0.05	0.02	2
0.07	0.26	0.05	2	0.04	0.01	2	0.01	0.00	2	0.01	0.00	2

Example 12. Screening of Insertion Guide RNAs with Nme2Cas9 or SpyCas9

[0891] A select sgRNA targeting CIITA (G023477) was redesigned for AAV insertion and evaluated for insertion efficacy in T cells. Cells were edited using Nme2Cas9 and an AAV encoding an insertion template including a GFP reporter gene flanked by homology arms to the guide cut site. Insertion efficiency was assessed by measuring luminescence and the absence of HLA-DP, DQ, DR expression by flow cytometry and editing at AAVS1 was assessed by NGS.

12.1 mRNA Electroporation and AAV Transduction of T Cells

[0892] T cells from a single donor (no. 613) were prepared and activated as described in Example 1. T cells were electroporated with sgRNA targeting the CIITA gene using mRNA electroporation as described in Example 1 except for the following differences. This study used mRNA encoding Nme2 Cas9 (SEQ ID No: 825) instead of mRNAs encoding Nme2 BC22n base editor and UGI, respectively. Guide G028533 targeting targeting CIITA was editing using mRNA encoding SpCas9 (SEQ ID No: 816). Nme2 Cas9 electroporation mix was prepared with 1×10^5 T cells in P3 buffer (Lonza), 600 ng of mRNA encoding Cas9 and 20 pmoles of sgRNA. Fifteen minutes after electroporation, T cells were transduced with AAV. Briefly, AAV encoding green fluorescent protein flanked by homology arms bracketing the guide cut site was added to 80 ul of CTS Optimizer T cell growth media supplemented with in new flat-bottom 96-well plates with final multiplicity of infection (MOI) of 300,000. Electroporated T cells were added to the resulting plates and incubated at 37° C. Twenty-four hours post-electroporation and transduction, cells were split 1:2 in 2 U-bottom plates replenished with CTS Optimizer media supplemented with cytokines.

12.2 Flow Cytometry and NGS Sequencing

[0893] On day 7 post-electroporation, edited T cell samples were subjected to PCR and NGS analysis as described in Example 1.

[0894] Edited T cells were phenotyped by flow cytometry 10 days post electroporation to determine HLA II-DR, DP, DQ protein expression. Briefly, T cells were incubated for 30 min at 4° C. with a mixture of antibodies against CD3 (BioLegend, Cat. No. 317338), CD4 (BioLegend, Cat. No. 317434), CD8 (BioLegend, Cat. No. 301046), Viakrome (Beckman Coulter, Cat. No. C36628) diluted at 1:100 and HLA II-DR, DP, DQ (BioLegend, Cat. No. 361711) diluted at 1:50 in cell staining buffer (BioLegend, Cat. No. 420201). Cells were subsequently washed and resuspended in 100 μL of cell staining buffer. Cells were then processed on a Cytoflex flow cytometer (Beckman Coulter) and analyzed using the FlowJo™ software package. T cells were gated based on size, shape, viability, as well as CD8, HLA II-DR, DP, DQ, and GFP expression. Table 19 shows the percentage of T cells with HLA II-DR, DP, DQ negative cells, HLA II-DR, DP, DQ negative cells with GFP, and mean percent indel frequency.

TABLE 19

Mean percentages of HLA II-DP, DR, DQ negative cells and HLA-DP, DQ, DR negative, GFP positive cells.; Mean percent indel frequency of the CIITA loci as a percentage of total NGS reads

	% Indel Frequency			% MHC Class II negative			% MHC Class II negative, GFP+		
Guide	Mean	SD	n	Mean	SD	n	Mean	SD	n
Untreated	0	n/a	1	46.5	n/a	1	0	0	2
AAV only	0	n/a	1	47.20	0.24	2	5.05	0.50	3

TABLE 19-continued

Mean percentages of HLA II-DP, DR, DQ negative cells and HLA-DP, DQ, DR negative, GFP positive cells.; Mean percent indel frequency of the CIITA loci as a percentage of total NGS reads

	% Indel Frequency			% MHC Class II negative			% MHC Class II negative, GFP+		
Guide	Mean	SD	n	Mean	SD	n	Mean	SD	n
G026584	0.86	0.02	3	87.10	0.82	2	59.47	2.30	3
G026588	0.98	0.002	3	75.40	0.82	2	14.57	14.57	3

Example 13. Dose Dependence of Additional TRAC Guide RNAs with Nme2Cas9

[0895] Guide RNAs targeting TRAC were screened for editing efficacy in T cells. TRAC editing was assessed by NGS for each concentration of sgRNA. T cells from a single donor were prepared and activated as described in Example 1 using 2.5% human AB serum (GeminiBio, Cat. 100-512) in T Cell growth media (TCGM). T cells were electroporated as described in Example 1 with sgRNAs targeting the TRAC locus at the concentration listed in Table 20 with the following exceptions. This study used 30 ng/μL mRNA encoding Nme2 Cas9 instead of mRNAs encoding Nme2 BC22n base editor and UGI.

[0896] On day 3 post-electroporation, T cells were harvested and subjected to PCR and subsequent NGS analysis as described in Example 1. Table 20 and FIG. **12** show the mean editing frequency at various TRAC sgRNA concentrations.

TABLE 20

Mean editing frequency at the TRAC locus

sgRNA	G021475			G021476			G021477			G021478		
(uM)	Mean	SD	N	Mean	SD	N	Mean	SD	N	Mean	SD	N
2.00	0.95	0.01	2	0.48	0.06	2	0.90	0.01	2	0.66	0.08	2
0.40	0.59	0.05	2	0.08	0.01	2	0.48	0.04	2	0.10	0.03	2
0.08	0.13	0.03	2	0.01	0.00	2	0.09	0.02	2	0.02	0.01	2
0.02	0.02	0.00	2	0.00	0.00	2	0.01	0.01	2	0.01	0.00	2

TABLE 21

Mean editing frequency at the CIITA locus.

Guide	sgRNA (uM)	Mean	SD	N
G023443	2	0.17	0.01	4
	0.4	0.01	0.00	4
	0.08	0.00	0.00	4
	0.016	0.00	0.00	4
	0.003	0.00	0.00	4
G023444	2	0.19	0.01	4
	0.4	0.02	0.01	4
	0.08	0.00	0.00	4
	0.016	0.00	0.00	4
	0.003	0.00	0.00	4
G023477	2	0.67	0.07	4
	0.4	0.16	0.06	4
	0.08	0.03	0.00	4
	0.016	0.00	0.00	4
	0.003	0.00	0.00	4
G023478	2	0.26	0.05	4
	0.4	0.03	0.01	4
	0.08	0.01	0.00	4
	0.016	0.00	0.00	4
	0.003	0.00	0.00	4
G023492	2	0.23	0.05	4
	0.4	0.02	0.01	4
	0.08	0.00	0.00	4
	0.016	0.00	0.00	4
	0.003	0.00	0.00	4

Example 14. Dose Dependence for Select CIITA Guide RNAs with Nme2Cas9

[0897] Guide RNAs targeting CIITA were screened for editing efficacy by dose response in T cells. CIITA editing was assessed by NGS for each concentration of sgRNA. A five-point dilution series was generated for each sgRNA by titrating each sgRNA with a fixed concentration of mRNA encoding Nme2 Cas9 in T cells using electroporation.

[0898] T cells were prepared from single donor as described in Example 1. T cells were electroporated with mRNA encoding Nme2 Cas9 and CIITA-targeting sgRNAs at the concentrations listed in Table 21 as described in Example 1. On day 3 post-editing, T cell samples were collected and subjected to PCR and NGS analysis as described in Example 1. Table 21 and FIG. **13** show the NGS editing results.

Example 15. Dose Responsiveness Analysis in Non-Activated T Cells

[0899] Guide RNAs were screened for editing efficacy in non-activated T cells by assessing editing frequency by NGS or by flow cytometry following lipid nanoparticle (LNP) delivery.

15.1 T Cell Preparation

[0900] Isolated, cryopreserved T cells from 3 donors were thawed in a water bath on Day 0 and plated at a density of 1×10^6 cells/mL in TCAM media containing CTS OpTmizer T Cell Expansion SFM and T Cell Expansion Supplement (ThermoFisher Cat. A1048501), 1× Penicillin-Streptomycin, 1× Glutamax, 10 mM HEPES, 200 U/mL recombinant human interleukin-2 (Peprotech, Cat. 200-02), 5 ng/mL recombinant human interleukin 7 (Peprotech, Cat. 200-07), 5 ng/mL recombinant human interleukin 15 (Peprotech, Cat. 200-15), 2.5% human AB serum (GeminiBio, Cat. 100-512).

15.2 T Cell Engineering

[0901] LNPs were generally prepared as described in Example 1. Lipid nanoparticles in this example were prepared with molar ratios of 35 Lipid A: 47.5 cholesterol: 15 DSPC: 2.5 PEG2k-DMG. LNPs were made with a lipid amine to RNA phosphate (N:P) molar ratio of about 6. LNPs were formulated with a single RNA species. LNPs were delivered to T cells in TCAM media containing ApoE3 (Peprotech, Cat. 350-02).

[0902] On Day 1 (about 24 hours after thaw), T cells were centrifuged, resuspended, and plated at 100,000 cells/well in 50 μL/well TCGM with 2.5% human AB serum. LNPs were applied at 1 μg/mL total RNA cargo of Nme2 base editor mRNA and 0.5 μg/mL total RNA cargo of UGI mRNA along with 10 μg/mL of ApoE3. An LNP formulated with CIITA G026584 and an LNP formulated with HLA-A G028918 were mixed and applied in an eight point 2-fold serial dilution series starting with a high dose of 1.7 μg/mL CIITA G026584 and of 0.7 μg/mL HLA-A G028918. No LNPs were applied to "untreated" samples.

[0903] On Day 3 or Day 4, cells were washed and activated with 1:100 dilution of Trans Act (Miltenyi Biotec). Cells were incubated at 37° C. with regular cell splitting and the addition of fresh media. On Day 9, cells were harvested for analysis by flow cytometry and NGS analysis. For flow cytometric analysis, cells were washed in FACS buffer (PBS+2% FBS+2 mM EDTA). Cells were incubated in a cocktail of antibodies targeting CD4 (Biolegend 317434), CD8 (Biolegend 301046), CD3 (Biolegend 317336), HLA-A2 (Biolegend 343320), HLA-A3 (eBioscience, 11-5754-42), HLA-DR, DP, DQ (Biolegend 361712), and ViaKrome 808 Fixable Viability Dye (Beckman Coulter, C36628). T cells were subsequently washed and analyzed on a Cytoflex instrument (Beckman Coulter). Data analysis was performed using FlowJo software package (v.10.6.1). T cells were gated on size, viability, CD4 or CD8 expression, and expression of markers indicated in Table 22. Flow cytometry data for a representative donor that received about 48 hours of LNP exposure are shown in Table 22 and FIG. **14**. Similar results were seen in all three donors and for both the 48- and 72-hour LNP exposure periods.

TABLE 22

Mean percent cells negative for HLA-A2 surface expression for a representative donor.

G028918	Treated			Untreated		
(μg/mL)	Mean	SD	n	Mean	SD	n
0.700	91.9	0.6	3	1.8	0.0	1
0.350	92.3	0.7	3	2.8	0.0	1
0.175	90.5	0.4	3	1.4	0.0	1
0.088	85.3	0.6	3	2.6	0.0	1
0.044	75.1	1.0	3	1.6	0.0	1
0.022	61.1	2.9	3	1.9	0.0	1
0.011	44.7	1.6	3	2.5	0.0	1
0.005	33.3	2.4	3	3.8	0.0	1

[0904] NGS analysis was performed on samples using 0.85 μg/mL G026584 and 0.35 μg/mL G028918 with 48-hour LNP exposure. NGS results are shown in Table 23 and FIG. **15**.

TABLE 23

Mean percent editing at the CIITA locus.

	C to T			C to A/G			Indel		
Sample	Mean	SD	n	Mean	SD	n	Mean	SD	n
Donor 1	50.9	2.2	3	8.6	1.5	3	7.5	0.5	3
Donor 2	64.5	1.2	3	7.9	0.6	3	5.8	0.9	3
Donor 3	52.6	0.3	2	8.4	0.8	2	6.8	0.2	2
Untreated	0.3	0.0	2	1.3	0.1	2	0.2	0.0	2

Example 16. Dose Response Analysis of Select Guide RNAs with LNP

[0905] Select HLA-A, TRAC, TRBC1 & TRBC2, and CIITA guide RNAs were screened for editing efficacy in T cells by assessing editing frequency by NGS and by flow cytometry following lipid nanoparticle (LNP) delivery.

16.1 T Cell Preparation & Activation

[0906] Isolated, cryopreserved T cells from 2 donors were thawed in a water bath on Day 0 and plated at a density of 1.5×10^6 cells/mL in TCAM media containing CTS OpTmizer T Cell Expansion SFM and T Cell Expansion Supplement (Gibco Cat. A3705001), 1× Penicillin-Streptomycin, 1× Glutamax, 10 mM HEPES, 200 U/mL recombinant human interleukin-2 (Peprotech, Cat. 200-02), 5 ng/mL recombinant human interleukin 7 (Peprotech, Cat. 200-07), and 5 ng/mL recombinant human interleukin 15 (Peprotech, Cat. 200-15) and 2.5% human AB serum (GeminiBio, Cat. 100-512). On Day 1 (24 hours post thaw) cells were washed and activated with TransAct™ (1:100 dilution, Miltenyi Biotec).

16.2 LNP Formulation

[0907] LNPs were generally prepared as described in Example 1.6. Lipid nanoparticles in this example were prepared with molar ratios of 35 Lipid A: 47.5 cholesterol: 15 DSPC: 2.5 PEG2k-DMG. LNPs were made with a lipid amine to RNA phosphate (N:P) molar ratio of about 6. LNPs were formulated with a single RNA cargo of Nme2BC22 mRNA, UGI mRNA, or a sgRNA as listed in Table 24 and Table 25. LNPs were delivered to T cells in TCAM media containing ApoE3 (Peprotech, Cat. 350-02).

16.3 T Cell Engineering

[0908] On Day 3, T cells were centrifuged, resuspended, and plated at 100,000 cells/well in 100 μL/well of TCAM with 2.5% human AB serum to which 100 μL of corresponding LNP formulation were added. LNP formulation containing gRNA was added to T cells in at concentrations listed in Table 24 along with 1 μg/mL total RNA cargo of Nme2 base editor mRNA and 0.5 μg/mL total RNA cargo of UGI mRNA. No LNPs were applied to "untreated" samples.

[0909] Beginning on Day 4, cells were split, and media refreshed regularly. On Day 7, a portion of cells were harvested for sequencing analysis at TRAC, TRBC1, TRBC2 and CIITA loci. NGS analysis was performed as described in Example 1. Table 24 and FIGS. **16**A-**16**L show mean percent editing for these cells. For each guide, data from one representative donor is shown in Table 24.

TABLE 24

Mean percent editing.

gRNA	sgRNA dose (μg/mL)	C to T Mean	C to T SD	C to T n	C to A/G Mean	C to A/G SD	C to A/G n	Indels Mean	Indels SD	Indels n
G021469	0.0003	2.27	0.28	3	0.86	0.05	3	0.33	0.08	3
(TRAC)	0.0012	7.70	0.24	3	0.88	0.02	3	0.24	0.05	3
	0.0049	25.76	1.03	3	0.87	0.03	3	0.30	0.07	3
	0.0195	62.68	1.69	3	0.92	0.02	3	0.27	0.07	3
	0.0781	91.88	0.16	3	1.12	0.03	3	0.52	0.08	3
	0.3125	96.90	0.59	3	1.12	0.07	3	0.69	0.15	3
	1.25	97.42	0.39	3	1.24	0.02	3	0.88	0.19	3
	5	97.01	0.40	3	1.71	0.10	3	0.92	0.08	3
G021481	0.00008	0.83	0.15	3	0.55	0.11	3	0.22	0.07	3
(TRAC)	0.00031	2.57	0.49	3	0.68	0.10	3	0.14	0.06	3
	0.00122	8.56	0.76	3	0.51	0.10	3	0.19	0.08	3
	0.00488	29.59	1.22	3	0.56	0.09	3	0.38	0.17	3
	0.01953	67.99	0.88	3	0.89	0.25	3	0.42	0.06	3
	0.07813	92.54	1.51	3	0.66	0.06	3	0.58	0.18	3
	0.3125	97.06	0.30	3	0.71	0.02	3	0.99	0.03	3
	1.25	96.85	0.44	3	0.93	0.07	3	1.71	0.10	3
G028935	0.0003	10.56	0.30	3	0.83	0.06	3	0.12	0.01	3
(TRAC)	0.0012	33.70	1.30	3	1.05	0.06	3	10.19	0.04	3
	0.0049	72.47	1.20	3	1.16	0.13	3	0.33	0.07	3
	0.0195	94.40	0.52	3	1.43	0.20	3	0.42	0.10	3
	0.0781	96.53	0.68	3	1.78	0.25	3	0.52	0.09	3
	0.3125	96.50	0.86	3	2.02	0.24	3	0.74	0.28	3
	1.25	96.32	0.28	3	2.27	0.08	3	1.02	0.05	3
	5	95.22	0.44	3	2.84	0.13	3	1.49	0.12	3
G028939	0.0003	2.63	0.49	2	1.37	0.41	2	10.20	0.02	2
(TRAC)	0.0012	9.65	0.16	2	1.21	0.90	2	0.13	0.02	2
	0.0049	29.60	1.03	2	1.42	0.69	2	0.19	0.00	1
	0.0195	66.33	2.98	2	1.68	0.47	2	0.33	0.04	2
	0.0781	192.37	0.69	2	1.93	0.34	2	0.60	0.30	2
	0.3125	94.59	0.32	2	1.65	0.91	2	0.39	0.30	2
	1.25				n.d.					
	5	94.89	4.17	2	1.87	0.03	2	0.21	0.08	2
G028943	0.0003	14.74	0.27	3	0.67	0.03	3	0.38	0.01	3
(TRAC)	0.0012	44.84	0.69	3	0.73	0.03	3	0.50	0.04	3
	0.0049	83.57	0.21	3	0.67	0.06	3	0.61	0.10	3
	0.0195	97.05	0.26	3	0.72	0.12	3	0.79	0.11	3
	0.0781	97.65	0.48	3	0.79	0.21	3	0.93	0.21	3
	0.3125	197.32	0.24	3	0.97	0.13	3	1.26	0.22	3
	1.25	96.51	0.35	3	1.37	0.09	3	1.82	0.34	3
	5	95.41	0.32	3	1.92	0.24	3	2.26	0.17	3
G028986	0.0003	2.31	0.41	3	1.60	0.54	3	0.14	0.01	3
(TRBC1)	0.0012	8.20	0.37	3	1.48	0.16	3	10.15	0.03	3
	0.0049	26.27	1.69	3	1.69	0.31	3	0.26	0.08	3
	0.0195	62.21	0.95	3	1.73	0.13	3	0.52	0.22	3
	0.0781	92.08	0.61	3	1.33	0.09	3	10.65	0.20	3
	0.3125	96.37	0.34	3	1.48	0.06	3	1.12	0.46	3
	1.25	97.18	0.23	3	1.52	0.21	3	0.98	0.11	3
	5	96.58	0.35	3	1.77	0.14	3	1.30	0.32	3
G029006	0.0003	3.00	0.73	3	0.55	0.02	3	0.25	0.03	3
(TRBC2)	0.0012	9.04	0.86	3	0.57	0.07	3	0.34	0.03	3
	0.0049	28.73	0.77	3	0.59	0.06	3	0.89	0.03	3
	0.0195	67.16	0.57	3	0.77	0.14	3	2.01	0.25	3
	0.0781	91.90	0.42	3	0.84	0.02	3	3.50	0.36	3
	0.3125	93.22	10.44	3	0.95	0.08	3	5.04	0.49	3
	1.25	91.67	10.70	3	1.42	0.13	3	6.27	0.41	3
	5	91.91	0.59	3	1.73	0.16	3	5.85	0.54	3
G029007	0.0003	1.38	0.35	3	0.61	0.01	3	0.09	0.00	3
(TRBC2)	0.0012	3.96	0.46	3	0.69	0.05	3	0.11	0.01	3
	0.0049	12.87	0.75	3	0.68	0.13	3	0.16	0.03	3
	0.0195	38.66	10.91	3	0.95	0.32	3	0.25	0.01	3
	0.0781	79.87	0.93	3	0.74	0.07	3	0.43	0.08	3
	0.3125	96.82	0.18	3	0.75	0.03	3	0.62	0.06	3
	1.25	97.63	0.22	3	1.01	0.26	3	0.73	0.09	3
	5	97.44	0.25	3	10.93	0.12	3	0.86	0.08	3
G026584	0.0003	1.30	0.06	3	0.66	0.03	3	0.08	0.03	3
(CIITA)	0.0012	3.97	0.40	3	0.60	0.04	3	0.10	0.01	3
	0.0049	13.87	0.18	3	0.61	0.07	3	0.10	0.04	3
	0.0195	42.35	0.82	3	0.59	0.04	3	0.17	0.04	3
	0.0781	82.44	0.99	3	0.64	0.05	3	0.29	0.08	3
	0.3125	97.03	0.60	3	0.77	0.32	3	0.43	0.08	3
	1.25	98.01	0.10	3	0.87	0.12	3	0.66	0.08	3
	5	97.83	0.23	3	0.97	0.03	3	0.68	0.06	3

TABLE 24-continued

Mean percent editing.										
	sgRNA	C to T			C to A/G			Indels		
gRNA	dose (μg/mL)	Mean	SD	n	Mean	SD	n	Mean	SD	n
G029123	0.0003	0.68	0.10	3	10.27	0.04	3	0.12	0.01	3
(CIITA)	0.0012	2.19	0.28	3	0.31	0.06	3	0.15	0.08	3
	0.0049	7.60	0.52	3	0.31	0.05	3	0.17	0.02	3
	0.0195	27.13	1.24	3	0.32	0.11	3	0.15	0.04	3
	0.0781	65.16	0.17	3	10.27	0.02	3	0.18	0.05	3
	0.3125	94.17	0.27	3	0.30	0.05	3	0.18	0.05	3
	1.25	98.54	0.52	3	0.41	0.13	3	0.25	0.01	3
	5	98.37	0.22	3	0.46	0.07	3	0.17	0.04	3
G029131	0.0003	2.43	0.08	3	0.25	0.03	3	0.15	0.03	3
(CIITA)	0.0012	7.66	0.42	3	0.24	0.03	3	0.19	0.07	3
	0.0049	24.67	0.25	3	0.34	0.08	3	0.29	0.08	3
	0.0195	60.40	0.99	3	0.32	0.07	3	0.58	0.04	3
	0.0781	91.11	0.39	3	0.46	0.04	3	0.95	0.08	3
	0.3125	97.56	0.12	3	0.42	0.10	3	1.27	0.06	3
	1.25	97.70	0.10	3	0.52	0.01	3	1.53	0.09	3
	5	97.59	0.29	3	0.71	0.02	3	1.46	0.35	3
	0.0003	0.46	0.07	3	0.39	0.02	3	0.15	0.03	3
	0.0012	1.42	0.25	3	0.39	0.03	3	0.14	0.07	3
	0.0049	5.11	0.21	3	0.43	0.03	3	0.12	0.01	3
G029172	0.0195	17.07	0.41	3	0.59	0.21	3	0.22	0.04	3
(CIITA)	0.0781	46.68	1.19	3	0.45	0.09	3	0.43	0.12	3
	0.3125	82.18	0.17	3	0.54	0.07	3	0.52	0.09	3
	1.25	95.09	0.61	3	0.55	0.10	3	0.54	0.08	3
	5	94.17	0.07	3	0.89	0.30	3	0.62	0.20	3

"n.d." indicates no data.

16.4 Flow Cytometry

[0910] On Day 10, cells were harvested for analysis by flow cytometry. For flow cytometric analysis, cells were washed in FACS buffer (PBS+2% FBS+2 mM EDTA). Cells were incubated in a cocktail of antibodies targeting CD3 (Biolegend 317336), CD4 (Biolegend 300538), CD8 (Biolegend 301051), HLA-A2 (Biolegend 343320), HLA-A3 (Invitrogen, 17-5754-42), HLA-B7 (Miltenyi Biotec, 130-120-234), HLA-DR, DP, DQ (Biolegend 361708), and ViaKrome 808 Fixable Viability Dye (Beckman Coulter, C36628). T cells were subsequently washed and analyzed on a Cytoflex instrument (Beckman Coulter). Data analysis was performed using FlowJo software package (v.10.6.1). T cells were gated based on size, viability, CD8 positivity, and expression of markers indicated in Table 25. Flow cytometry data for CD8+ T cells are shown in Table 25 and FIGS. **17**A-**17**D. CD3-, HLA-A2-/HLA-A3-, and HLA-DR, DP, DQ-cell populations indicate efficient disruption of TRAC, TRBC1 or TRBC2; HLA-A; or CIITA, respectively. Data from one representative donor is shown in Table 25. Similar results were seen in both donors.

TABLE 25

Mean percent CD8+ T cells expressing surface markers in representative donor. CIITA guides are reported as mean percent HLA-DP, DQ, DR- cells. HLA-A guides are reported as mean percent HLA-A2/A3- cells. TRAC and TRBC guides are reported as mean percent CD3− cells.

Guide	sgRNA dose (ug/ml)	Mean	SD	n	Guide	sgRNA dose (ug/ml)	Mean	SD	n
G026584	0.0003	14.8	2.5	3	G029123	0.0003	14.3	3.4	3
CIITA	0.0012	16.3	3.7	3	CIITA	0.0012	16.6	3.9	3
	0.0049	20.5	4.2	3		0.0049	18.4	3.6	3
	0.0195	36.5	6.5	2		0.0195	28.1	3.4	3
	0.0781	76.5	2.6	3		0.0781	50.6	2.7	3
	0.3125	96.3	0.2	3		0.3125	81.4	1.9	3
	1.25	98.8	0.3	3		1.25	93.2	1.1	3
	5	98.7	0.2	3		5	92.6	0.3	3
G029131	0.0003	16.2	1.7	3	G029172	0.0003	13.9	0.7	3
CIITA	0.0012	17.5	5.0	3	CIITA	0.0012	13.8	1.0	3
	0.0049	25.2	3.6	3		0.0049	15.5	1.4	3
	0.0195	46.9	4.0	3		0.0195	19.5	5.2	3
	0.0781	73.7	3.3	3		0.0781	45.4	2.3	3
	0.3125	88.5	1.3	3		0.3125	82.1	2.8	3
	1.25	94.6	0.7	3		1.25	95.8	0.4	3
	5	94.4	1.5	3		5	93.1	1.5	3

TABLE 25-continued

Mean percent CD8+ T cells expressing surface markers in representative donor. CIITA guides are reported as mean percent HLA-DP, DQ, DR- cells. HLA-A guides are reported as mean percent HLA-A2/A3- cells. TRAC and TRBC guides are reported as mean percent CD3− cells.

Guide	sgRNA dose (ug/ml)	Mean	SD	n	Guide	sgRNA dose (ug/ml)	Mean	SD	n
G028922	0.0003	2.6	0.4	3	G028865	0.0003	4.7	0.2	3
HLA-A	0.0012	2.7	0.9	3	HLA-A	0.0012	10.2	0.5	3
	0.0049	7.4	0.7	3		0.0049	26.1	1.1	3
	0.0195	22.6	0.8	3		0.0195	56.6	1.7	3
	0.0781	58.1	1.0	3		0.0781	77.1	1.7	3
	0.3125	91.9	0.9	3		0.3125	82.5	1.1	3
	1.25	99.1	0.1	3		1.25	87.5	1.1	3
	5	98.3	0.6	3		5	89.0	0.8	3
G028869	0.0003	2.4	0.8	3	G028907	0.0003	5.8	0.3	3
HLA-A	0.0012	2.1	1.3	3	HLA-A	0.0012	15.4	0.2	3
	0.0049	2.7	0.2	3		0.0049	39.5	2.9	3
	0.0195	7.1	0.5	3		0.0195	76.5	0.5	3
	0.0781	19.7	2.3	3		0.0781	92.4	0.4	3
	0.3125	47.9	1.4	3		0.3125	95.7	0.7	3
	1.25	68.7	0.6	3		1.25	97.7	0.5	3
	5	66.9	0.6	3		5	97.0	0.2	3
G028913	0.0003	4.3	0.7	3	G028918	0.0003	4.8	0.4	3
HLA-A	0.0012	11.8	0.7	3	HLA-A	0.0012	10.0	1.2	3
	0.0049	33.7	0.5	3		0.0049	31.0	1.0	3
	0.0195	71.5	0.9	3		0.0195	63.4	0.2	3
	0.0781	90.8	0.9	3		0.0781	88.4	1.0	3
	0.3125	95.4	0.5	3		0.3125	93.5	0.4	3
	1.25	97.3	0.6	3		1.25	94.3	0.5	3
	5	97.4	0.3	3		5	94.5	0.1	3
G029006	0.0003	4.2	0.8	3	G028986	0.0003	3.8	1.2	3
TRBC	0.0012	9.0	1.3	3	TRBC	0.0012	7.3	0.2	3
	0.0049	25.8	0.9	3		0.0049	22.2	0.3	3
	0.0195	54.7	1.5	3		0.0195	55.1	1.1	3
	0.0781	84.8	0.7	3		0.0781	89.7	1.3	3
	0.3125	96.6	0.2	3		0.3125	97.8	0.2	3
	1.25	99.1	0.2	3		1.25	99.1	0.1	3
	5	99.0	0.2	3		5	98.8	0.1	3
G029007	0.0003	3.3	0.4	3	G028943	0.0003	17.97	6.36	3
TRBC	0.0012	4.7	0.7	3	TRAC	0.0012	36.57	0.49	3
	0.0049	12.1	1.5	3		0.0049	73.73	1.19	3
	0.0195	29.4	1.7	3		0.0195	95.03	0.32	3
	0.0781	53.4	2.0	3		0.0781	99.00	0.44	3
	0.3125	61.9	0.8	3		0.3125	99.27	0.06	3
	1.25	65.5	1.1	3		1.25	99.33	0.25	3
	5	69.0	1.4	3		5	98.97	0.42	3
G021469	0.0003	7.04	3.60	3	G021481	0.00008	2.43	0.47	3
TRAC	0.0012	8.84	0.27	3	TRAC	0.00031	3.58	0.77	3
	0.0049	24.50	1.04	3		0.00122	8.21	1.05	3
	0.0195	57.13	1.20	3		0.00488	26.03	0.35	3
	0.0781	85.10	1.35	3		0.01953	62.40	2.12	3
	0.3125	92.30	2.12	3		0.07813	92.70	0.70	3
	1.25	96.90	0.26	3		0.3125	99.00	0.10	3
	5	97.57	0.40	3		1.25	99.47	0.21	3
G028935	0.0003	15.60	4.65	3	G028939	0.0003	7.87	3.58	3
TRAC	0.0012	34.63	0.83	3	TRAC	0.0012	15.70	0.10	3
	0.0049	70.97	2.47	3		0.0049	40.90	2.88	3
	0.0195	90.27	0.35	3		0.0195	75.90	1.28	3
	0.0781	94.17	1.00	3		0.0781	92.10	0.66	3
	0.3125	95.57	0.76	3		0.3125	98.43	0.06	3
	1.25	96.63	0.25	3		1.25	99.70	0.10	3
	5	96.47	0.06	3		5	99.53	0.06	3

Example 17. Off-Target Analysis

17.1 Biochemical Off-Target Analysis

[0911] A biochemical method (See, e.g., Cameron et al., Nature Methods. 6, 600-606; 2017) was used to determine potential off-target genomic sites cleaved by Nme2Cas9 using specific guides targeting HLA-A, TRAC, TRBC1 & TRBC2, and CIITA, respectively. Guide RNAs shown in Table 26 were screened using NA24385 genomic DNA (Coriell Institute) alongside control guides. The number of on target and potential off-target cleavage sites were detected using a guide concentration of 192 nM gRNA and 64 nM Nme2Cas9 protein in the biochemical assay for which results are shown in Table 26.

[0912] Digenome sequencing was also performed by methods known in the art for discovery of potential off-target sites for select guides using Nme2 base editor. Data not shown.

TABLE 26

Biochemical Off-Target Analysis		
Guide ID	Target Gene	Sites
G028865	HLA-A	18
G028869	HLA-A	11
G028907	HLA-A	7
G028913	HLA-A	13
G028918	HLA-A	4
G028922	HLA-A	14
G028935	TRAC	4
G028939	TRAC	4
G028943	TRAC	2
G028986	TRBC1	8
G029006	TRBC2	9
G029007	TRBC2	7
G026584	CIITA	6
G029123	CIITA	1
G029131	CIITA	2
G029172	CIITA	4
G021557	VEGFA	5
G021558	VEGFA	1
G021567	VEGFA	2

17.2 Targeted Sequencing at Potential Off-Target Sites

[0913] A two-phase process was applied to identify and then confirm potential off-target editing. In phase one, a computational off-target prediction method Cas-OFFinder (Bae et al., 2014) was combined with the biochemical off-target discovery assay described above to identify potential off-target editing sites. The final phase confirms editing at the potential off-target loci through the detection of C-to-T mutations using NGS in genome-edited cells. A multiplex PCR based rhAMPSeq (RNase H2 dependent PCR Amplification for Next Generation Sequencing) assay or NGS as described in Example 1 was employed to characterize potential off-target editing in cells.

[0914] Samples were prepared in triplicate for two T cell donors. T cells were prepared as described in Example 1. Cells were treated simultaneously with 3 LNPs, each formulated with a single RNA cargo of Nme2BC22 mRNA, UGI mRNA, or a select sgRNA as listed in Table 27. LNPs were generally prepared as described in Example 1 with lipid molar ratio of 35 Lipid A:47.5 cholesterol:15 DSPC:2.5 PEG. LNPs were pre-incubated in 10 μg/mL of human ApoE3. Approximately 50,000 T cells were treated with LNPs measured by RNA weight as follows: 100 ng of Nme2BC22 mRNA, 50 ng of UGI mRNA, and sgRNA at the dose shown in Table 27. Cells were incubated at 37° C. for about 24 hours then resuspended in fresh media for further growth. Approximately 96 hours (4 days) after LNP treatment, cells were harvested and NGS analysis was performed generally as described in Example 1 or via rhAmpSeq CRISPR Analysis System (IDT) by the manufacturer's protocol using cell lysate. NGS analysis used primers designed to identify percent C-to-T mutations at predicted off-target sites. Results of the potential off target site sequencing analysis from two donors are summarized in Table 27. Potential off target sites were considered to have confirmed base editing when the mean % C to T editing was statistically significant (P value of 0.05 or less) compared to donor-matched untreated controls. Edited sequence reads were manually inspected for potential off target sites reaching these criteria to confirm edit-relevant C to T repair structures.

TABLE 27

Editing confirmation at potential off target sites by sequencing				
Guide	Target	LNP Dose (ng)	Loci characterized	Loci with confirmed base editing
G028907	HLA-A	23.8	7	3
G028913	HLA-A	31.7	6	2
G028918	HLA-A	28.6	4	3
G028939	TRAC	21.5	3	0
G028943	TRAC	5.8	1	0
G028986	TRBC	37.7	7	1
G029006	TRBC	55.0	7	3
G026584	CIITA	70.6	7	2
G029131	CIITA	96.8	1	0

SEQUENCE LISTING

The patent application contains a lengthy sequence listing. A copy of the sequence listing is available in electronic form from the USPTO web site (https://seqdata.uspto.gov/?pageRequest=docDetail&DocID=US20250276017A1). An electronic copy of the sequence listing will also be available from the USPTO upon request and payment of the fee set forth in 37 CFR 1.19(b)(3).

1. An engineered cell, which has reduced or eliminated surface expression of HLA-A relative to an unmodified cell, comprising a genetic modification in the HLA-A gene, wherein the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from: a) chr6:29942540-29945459: b) chr6:29942891-29942915: chr6:29942609-29942633: chr6:29944266-29944290: chr6:29942889-29942913: chr6:29944471-29944495; and chr6:29944470-29944494; and c) chr6:29942785-29942809.

2. The engineered cell of claim 1, wherein the genetic modification comprises:

a) at least one nucleotide within the genomic coordinates chosen from the genomic coordinates listed in Table 1;

b) at least one nucleotide within the genomic coordinates targeted by a guide RNA comprising a guide sequence of any one of SEQ ID NOs: 66, 61, 2-60, 62-65, 67-80; and/or
c) an indel, a C to T substitution, or an A to G substitution within the genomic coordinates.

3. (canceled)

4. (canceled)

5. (canceled)

6. (canceled)

7. (canceled)

8. (canceled)

9. The engineered cell of claim **1**, wherein the cell is: a) homozygous for HLA-C or b) homozygous for HLA-B and homozygous for HLA-C.

10. (canceled)

11. A composition comprising an HLA-A guide RNA, wherein the HLA-A guide RNA comprises

i) a guide sequence chosen from SEQ ID NOs: 66, 61, 2-60, 62-65, and 67-80;
ii) at least 19, 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence chosen from SEQ ID NOs: 66, 61, 2-60, 62-65, and 67-80; or
iii) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence chosen from SEQ ID NOs: 66, 61, 2-60, 62-65, and 67-80;
iv) a sequence that comprises 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 1;
v) at least 20, 21, 22, 23, or 24, contiguous nucleotides of a sequence chosen from iv); or
vi) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence chosen from iv).

12. A method of a) making an engineered human cell, which has reduced or eliminated surface expression of HLA-A protein relative to an unmodified cell, or b) reducing surface expression of HLA-A protein in a human cell relative to an unmodified cell, the method comprising contacting a cell with a composition comprising an HLA-A guide RNA, wherein the HLA-A guide RNA comprises

i) a guide sequence chosen from SEQ ID NOs: 66, 61, 2-60, 62-65, and 67-80;
ii) at least 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence chosen from SEQ ID NOs: 66, 61, 2-60, 62-65, and 67-80; or
iii) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence chosen from SEQ ID NOs: 66, 61, 2-60, 62-65, and 67-80;
iv) a sequence that comprises 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 1;
v) at least 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence chosen from iv); or
vi) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence chosen from iv).

13. (canceled)

14. The composition of claim **11**, wherein the HLA-A guide RNA comprises:

a) a guide sequence chosen from SEQ ID NOs: 66, 61, 13, 55, 70, and 71,
b) a guide sequence chosen from SEQ ID NOs: 61, 66, 13, 17, 55, and 70;
c) a guide sequence comprising the sequence of SEQ ID NO: 61; or
d) a guide sequence comprising the sequence of SEQ ID NO: 66.

15. (canceled)

16. (canceled)

17. (canceled)

18. A population of cells comprising the engineered cells of claim **1**.

19. A pharmaceutical composition comprising the engineered cell of claim **1**.

20. An engineered human cell, which has reduced or eliminated surface expression of TRAC relative to an unmodified cell, comprising a genetic modification in the TRAC gene, wherein the genetic modification comprises: a) at least one nucleotide within the genomic coordinates chr14:22547505-22551621 or chr14:22547462-22551621; or b) an indel, a C to T substitution, or an A to G substitution within the genomic coordinates chosen from: chr14:22550574-22550598: chr14:22550544-22550568: chr14:22547505-22547529; chr14:22547525-22547549; and chr14:22547674-22547698.

21. The engineered cell of claim **20**, wherein the genetic modification comprises: a) at least one nucleotide within the genomic coordinates chosen from the genomic coordinates listed in Table 2 or at least one nucleotide within the genomic coordinates targeted by a guide RNA comprising a guide sequence chosen from SEQ ID NOs: 111, 107, 101-106, 108-110, and 112-120; and/or

b) at least one nucleotide within the genomic coordinates chosen from: chr14:22550574-22550598: chr14:22550544-22550568: chr14:22547505-22547529; chr14:22547525-22547549; and chr14:22547674-22547698.

22. (canceled)

23. (canceled)

24. (canceled)

25. A composition comprising:

a) a TRAC guide RNA comprising a guide sequence that i) targets a TRAC genomic target sequence; or ii) directs an RNA-guided DNA binding agent to induce a double stranded break (DSB) or a single-stranded break (SSB) in a TRAC genomic target sequence comprising at least 10 contiguous nucleotides within the genomic coordinates chosen from: chr14:22550574-22550598; chr14:22550544-22550568; chr14:22547505-22547529; chr14:22547525-22547549; and chr14:22547674-22547698; or
b) a TRAC guide RNA, wherein the TRAC guide RNA comprises:
    i) a guide sequence chosen from SEQ ID NOs: 111, 107, 101-106, 108-110, and 112-120;
    ii) at least 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence chosen from SEQ ID NOs: 111, 107, 101-106, 108-110, and 112-120; or
    iii) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence chosen from SEQ ID NOs: 111, 107, 101-106, 108-110, and 112-120;
    iv) a sequence that comprises 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 2;
    v) at least 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence chosen from iv); or
    vi) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence chosen from iv).

26. (canceled)

27. (canceled)

28. (canceled)

29. A method of I) making an engineered human cell, which has reduced or eliminated surface expression of TRAC protein relative to an unmodified cell, or II) reducing surface expression of TRAC protein in a human cell relative to an unmodified cell, the method comprising contacting a cell with a composition comprising a TRAC guide RNA, wherein the TRAC guide RNA comprises:

i) a guide sequence chosen from SEQ ID NOs: 111, 107, 101-106, 108-110, and 112-120;

ii) at least 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence chosen from SEQ ID NOs: 111, 107, 101-106, 108-110, and 112-120; or

iii) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence chosen from SEQ ID NOs: 111, 107, 101-106, 108-110, and 112-120; or

iv) a sequence that comprises 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 2;

v) at least 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence chosen from iv); or

vi) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence chosen from iv).

30. (canceled)

31. The composition of claim **25**,

wherein the TRAC guide RNA comprises:

a) a guide sequence chosen from SEQ ID NOs: 111, 107, 101, 102, and 103;

b) a guide sequence comprising the sequence of SEQ ID NO: 107; or

c) a guide sequence comprising the sequence of SEQ ID NO: 111.

32. (canceled)

33. (canceled)

34. A pharmaceutical composition comprising the engineered cell of claim **20**.

35. An engineered cell, which has reduced or eliminated surface expression of TRBC relative to an unmodified cell, comprising a genetic modification in the TRBC gene, wherein the genetic modification comprises:

I) at least one nucleotide within the genomic coordinates (a)_chr7:142791756-142802543; (b) chr7:142791862-142793149; (c) chr7: 142791756-142792721; or (d) chr7:142801104-142802543; or wherein the genetic modification comprises at least one nucleotide within the genomic coordinates targeted by a guide RNA comprising a guide sequence chosen from SEQ ID NOs: 215, 201-214, and 216-265; or

II) an indel, a C to T substitution, or an A to G substitution within the genomic coordinates chosen from: (a) chr7: 142791862-142793149; (b) chr7:142791756-142792721; (c) chr7:142801104-142802543; (d) chr7: 142792690-142792714 and chr7:142792693-142792717; (e) chr7:142791756-142791780; chr7: 142791761-142791785; chr7:142791820-142791844; chr7:142791939-142791963; chr7:142791940-142791964; and chr7:142792004-142792028; (f) chr7: 142801104-142801124; chr7:142802103-142802127; and chr7:142802106-142802130; and (g) chr7: 142802103-142802127.

36. (canceled)

37. The engineered cell of claim **35**, wherein the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from:

a) chr7:142792690-142792714; and chr7:142792693-142792717;

b) chr7:142791756-142791780; chr7:142791761-142791785; chr7:142791820-142791844; chr7: 142791939-142791963; chr7:142791940-142791964; and chr7:142792004-142792028;

c) chr7:142801104-142801124; chr7:142802103-142802127; and chr7:142802106-142802130; and

d) chr7:142802103-142802127; and chr7:142802106-14280213; and

e) the genomic coordinates listed in Table 3.

38. (canceled)

39. (canceled)

40. (canceled)

41. (canceled)

42. (canceled)

43. (canceled)

44. (canceled)

45. A composition comprising (a) a TRBC guide RNA, wherein the TRBC guide RNA comprises:

i) a guide sequence chosen from SEQ ID NOs: 215, 201-214, and 216-265;

ii) at least 20, 21, 22, 23, or 24, contiguous nucleotides of a sequence chosen from SEQ ID NOs: 215, 201-214, and 216-265;

iii) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence chosen from SEQ ID NOs: 215, 201-214, and 216-265;

iv) a sequence that comprises 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 3;

v) at least 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence chosen from iv); or

vi) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence chosen from iv).

46. (canceled)

47. (canceled)

48. A method of I) making an engineered human cell, which has reduced or eliminated surface expression of TRBC protein relative to an unmodified cell, or II) reducing surface expression of TRBC protein in a human cell relative to an unmodified cell, the method comprising contacting a cell with TRBC guide RNA and, wherein the TRBC guide RNA comprises:

i) a guide sequence chosen from SEQ ID NOs: 215, 201-214, and 216-265;

ii) at least 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence chosen from SEQ ID NOs: 215, 201-214, and 216-265;

iii) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence chosen from SEQ ID NOs: 215, 201-214, and 216-265;

iv) a sequence that comprises 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 3;

v) at least 20, 21, 22, 23, or 24, contiguous nucleotides of a sequence chosen from iv); or

vi) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence chosen from iv).

49. (canceled)

50. The composition of claim **45**, wherein the TRBC guide RNA comprises a guide sequence of:

a) any one of SEQ ID NOs: 215, 216, 223, 224, 229, 230, 246, 259, and 260,

b) any one of SEQ ID NOs: 215, 216, 224, 229, 246, 259, and 260; or

c) any one of SEQ ID NOs: 215, 259, and 260.

51. (canceled)

52. (canceled)

53. A pharmaceutical composition comprising the engineered cell of claim **35**.

54. An engineered cell, which has reduced or eliminated surface expression of MHC class II relative to an unmodified cell, comprising:

I) a genetic modification in the CIITA gene, wherein the genetic modification comprises at least one nucleotide within the genomic coordinates chosen from: (a) chr16:10877363-10907788; and (b) chr16:10906515-10908136;

II) a genetic modification in the CIITA gene, wherein the genetic modification comprises an indel, a C to T substitution, or an A to G substitution within the genomic coordinates chosen from: (a) chr16:10906643-10906667: chr16:10907504-10907528: chr16:10907508-10907532: chr16:10907539-10907559; chr16:10895658-10895682; chr16:10895668-10895692; chr16:10895750-10895774; chr16:10895753-10895777; chr16:10895754-10895778; chr16:10898684-10898708; chr16:10901529-10901553; chr16:10902121-10902145; chr16:10902701-10902725; chr16:10904726-10904750; chr16:10904760-10904784; chr16:10906493-10906517; chr16:10906515-10906539; chr16:10906631-10906655; chr16:10906636-10906660; chr16:10906770-10906794; chr16:10906788-10906812; chr16:10906789-10906813; chr16:10906816-10906840; chr16:10907148-10907172; chr16:10907254-10907278; chr16:10907331-10907355; chr16:10907477-10907501; chr16:10907497-10907521; chr16:10907503-10907527; and chr16:10907574-10907598; and (b) chr16:10907504-10907528; or

III) a genetic modification in the CIITA gene, wherein the genetic modification comprises an indel, a C to T substitution, or an A to G substitution within the genomic coordinates chosen from: a) chr16:10906643-10906667: chr16:10907504-10907528: chr16:10895658-10895682: chr16:10902701-10902725; chr16:10906493-10906517; chr16:10906631-10906655; chr16:10907477-10907501; chr16:10907497-10907521; chr16:10907504-10907528; and chr16:10907508-10907532; and

b) chr16:10906889-10906913.

55. (canceled)

56. (canceled)

57. (canceled)

58. (canceled)

59. (canceled)

60. (canceled)

61. (canceled)

62. A composition comprising a CIITA guide RNA, wherein the CIITA guide RNA comprises:

i) a guide sequence chosen from SEQ ID NOs: 422, 301-421, and 423-576; or

ii) at least 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence chosen from SEQ ID NOs: 422, 301-421, and 423-576; or

iii) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence chosen from SEQ ID NOs: 422, 301-421, and 423-576; or

iv) a sequence that comprises 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 4; or

v) at least 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence chosen from iv); or

vi) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence chosen from iv).

63. (canceled)

64. (canceled)

65. A method of I) making an engineered human cell, which has reduced or eliminated surface expression of MHC class II protein relative to an unmodified cell, or II) reducing surface expression of MHC class II protein in a human cell relative to an unmodified cell, the method comprising contacting a cell with a CIITA guide RNA, wherein the CIITA guide RNA comprises:

i) a guide sequence chosen from SEQ ID NOs: 422, 301-421, and 423-576; or

ii) at least 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence chosen from SEQ ID NOs: 422, 301-421, and 423-576; or

iii) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence chosen from SEQ ID NOs: 422, 301-421, and 423-576; or

iv) a sequence that comprises 10 contiguous nucleotides±10 nucleotides of a genomic coordinate listed in Table 4; or

v) at least 20, 21, 22, 23, or 24 contiguous nucleotides of a sequence chosen from iv); or

vi) a guide sequence that is at least 95%, 90%, or 85% identical to a sequence chosen from iv).

66. (canceled)

67. The composition of claim **62**, wherein the CIITA guide RNA comprises a guide sequence of chosen from SEQ ID NOs: 422, 301, 302, 320, 321, 324, 326, 327, 332, 354, 361, 372, 400, 408, 414, 415, 419, 420, 428, 431, 432, 434, 451, 455, 458, 462, 463, 464, and 468.

68. (canceled)

69. A pharmaceutical composition comprising the engineered cell of claim **54**.

70. (canceled)

71. (canceled)

72. (canceled)

73. (canceled)

74. (canceled)

75. (canceled)

76. The engineered cell of claim **54**,

wherein the cell:

a) has reduced expression of TRAC protein on the surface of the cell;

b) has reduced expression of TRBC protein on the surface of the cell;

c) is an immune cell;

d) is a stem cell;

e) is a primary cell; or

f) is engineered with a genomic editing system.

77. (canceled)

78. (canceled)

79. (canceled)
80. (canceled)
81. (canceled)
82. (canceled)
83. (canceled)
84. (canceled)
85. (canceled)
86. (canceled)
87. (canceled)
88. (canceled)
89. (canceled)
90. (canceled)
91. (canceled)
92. (canceled)
93. (canceled)
94. (canceled)
95. (canceled)
96. (canceled)
97. (canceled)
98. (canceled)
99. (canceled)
100. (canceled)
101. (canceled)
102. (canceled)
103. (canceled)

104. A method of treating a disease or disorder comprising administering the engineered cell, of claim **54** to a subject in need thereof.

105. (canceled)
106. (canceled)
107. (canceled)
108. (canceled)

* * * * *